Galois Cohomology

***EYNTKA* Series**

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Introduction

Group cohomology assigns to a group G acting on a module M a sequence of abelian groups $H^n(G, M)$ that measure the failure of exactness, the obstructions to lifting, and the extensions of M by G. When G is a Galois group $\mathrm{Gal}(L/K)$ and M is an arithmetic object (the multiplicative group of a field, the points of an algebraic variety, the idèle class group of a number field), these cohomology groups encode the deepest structural information in algebraic number theory. The Brauer group of a field is H^2. The Hilbert 90 theorem is the vanishing of H^1. The local and global reciprocity laws of class field theory are consequences of the cup product structure on Tate cohomology. The Selmer group controlling the Mordell-Weil rank of an elliptic curve is a subgroup of H^1 defined by local conditions.

This book develops the theory of group cohomology for profinite groups and applies it systematically to the Galois groups of local and global fields. The treatment is self-contained given the prerequisites listed below, and proceeds from the abstract foundations (Part I) through the structural theory of cohomological dimension and the Brauer group (Part II) to the arithmetic applications: local and global Tate duality, the Poitou-Tate exact sequence, class field theory via class formations, Selmer groups, descent and non-abelian cohomology (Part III).

Prerequisites

The reader is assumed to be familiar with the material in the following books from the *Everything You Need To Know About* series:

- [9]: groups, rings, modules, tensor products, categories. In particular: the structure of finitely generated modules over PIDs.
- [10]: the Artin-Wedderburn theorem for semisimple algebras and the basic theory of central simple algebras.
- [6]: Dedekind domains, prime splitting, the class group, the Dirichlet unit theorem, and the ideal-theoretic formulation of class field theory (the Artin symbol, ray class groups). The

cohomological reformulation of class field theory is developed in this book (chapters 8 and 10) and used in [24].

- [14]: places and completions of number fields, the structure of non-archimedean local fields (valuation rings, ramification, the local norm), the product formula, and strong approximation.
- [13]: chain complexes, derived functors (Ext, Tor), spectral sequences (Grothendieck, Hochschild-Serre), and the universal coefficient theorem.

Class field theory ([24]) is *not* a prerequisite. On the contrary, this book provides the cohomological tools that [24] will use: the cohomological formalism of local and global reciprocity is developed in chapters 8 and 10. The arithmetic inputs that this formalism needs and that the book does not prove, among them the two fundamental inequalities of global class field theory and the existence theorems, are cited from the literature at the places where they are used.

Structure

The book has three parts. Part I (chapters 1 to 4) develops the abstract theory: profinite groups, cohomology of discrete and profinite groups, Tate cohomology, and cup products. No fields or arithmetic appear. The results are purely algebraic and topological. Part II (chapters 5 to 7) develops the structural theory: cohomological dimension, pro-p-groups and the Golod-Shafarevich inequality, and the Brauer group. These chapters bridge the abstract theory of Part I and the arithmetic of Part III. Part III (chapters 8 to 11) applies the machinery to local and global fields: Tate duality, the adèles and idèles, the Poitou-Tate sequence, class formations, reciprocity, Selmer groups, descent and non-abelian cohomology. The closing chapter, chapter 12, points to where the subject continues.

Conventions

All groups are written multiplicatively unless otherwise stated. Profinite groups carry the profinite topology, and all homomorphisms between profinite groups are continuous. For a profinite group G and a discrete abelian group M, $\mathrm{Hom}(G, M)$ denotes the group $\mathrm{Hom}_{\mathrm{cts}}(G, M)$ of continuous homomorphisms. Modules over profinite groups are discrete (every element has an open stabilizer) unless explicitly marked as profinite. The notation $\varprojlim$ denotes the (categorical) limit and $\varinjlim$ the colimit; "inverse limit" and "direct limit" are avoided. For a number field K: the ring of integers is $\mathcal{O}_K$, the absolute Galois group is $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, and the completion at a place v is K_v. Cross-references use `\cref` throughout. Internal references to other books in the series are by citation code (e.g., [9]).

Part I

Foundation

The cohomology of a group G with coefficients in a G-module M is a sequence of functors $H^n(G, -)$ that extend the fixed-point functor $H^0(G, M) = M^G$ to higher degrees. The theory has two sources: the algebraic, rooted in the homological algebra of group rings (resolutions, derived functors, Ext), and the topological, rooted in the cohomology of classifying spaces and the structure of profinite groups as limits of finite groups. Part I develops both perspectives and fuses them into a single framework.

Chapter 1 constructs the category of profinite groups and establishes the foundational results: the characterization as compact, Hausdorff, totally disconnected topological groups, Sylow theory, and the profinite completion. Chapter 2 develops cohomology for discrete groups via resolutions, establishing the long exact sequence, the Hochschild-Serre spectral sequence, and the interpretation of H^1 and H^2 in terms of extensions and outer actions. Chapter 3 passes to the profinite setting, where cohomology is computed as a colimit of finite-group cohomology, and Shapiro's lemma and inflation-restriction become the primary computational tools. Chapter 4 introduces Tate cohomology (extending the theory to negative degrees for finite groups) and the cup product, the multiplicative structure that will drive the duality theorems and reciprocity laws of Part III.

No fields or number-theoretic objects appear in Part I. The results are purely algebraic and topological, applicable to any profinite group acting on any discrete module. The arithmetic content begins in Part II, where the abstract machine acquires the characteristic shapes (cohomological dimension, pro-p-structure, Brauer classes) that govern its behavior over fields.

1

Profinite Groups

The absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ is the central object of modern algebraic number theory. For a finite extension L/K, the Galois group $\mathrm{Gal}(L/K)$ is a finite group whose structure is studied in [9]. For the separable closure, G_K is an infinite group carrying a natural topology (the Krull topology) inherited from the finite quotients $\mathrm{Gal}(L/K)$ as L ranges over finite Galois extensions of K. Extracting cohomological information from G_K requires first understanding this topology.

This chapter develops the theory of profinite groups: topological groups arising as limits of finite discrete groups. The central result is that a topological group is profinite if and only if it is compact, Hausdorff, and totally disconnected (theorem 1.1.10). From there, the structure of profinite groups (open and closed subgroups, Sylow theory, supernatural numbers) is developed, and the absolute Galois groups of many common fields are examined as profinite groups, establishing the concrete objects on which the cohomological machinery of the rest of the book will act.

1.1 Limits and Profinite Groups

Galois theory produces, for each number field K, a tower of finite groups $\mathrm{Gal}(L/K)$ indexed by the finite Galois extensions L/K, connected by restriction maps. The "total" Galois group G_K should assemble from this tower: an element of G_K should be a family of elements, one in each $\mathrm{Gal}(L/K)$, compatible with the restriction maps. Making this precise requires the formalism of limits over cofiltered diagrams.

Throughout this section, all topological groups are assumed Hausdorff unless stated otherwise. The reader is assumed familiar with the basic theory of topological groups from [17].

Definition 1.1.1: Cofiltered System of Topological Groups

A *cofiltered system* of topological groups consists of:

1. a nonempty directed partially ordered set $(I, \leq)$,
2. for each $i \in I$, a topological group G_i, and
3. for each pair $i \leq j$ in I, a continuous group homomorphism $\varphi_{ji} : G_j \to G_i$,

subject to $\varphi_{ii} = \mathrm{id}_{G_i}$ for all i, and $\varphi_{ki} = \varphi_{ji} \circ \varphi_{kj}$ for all $i \leq j \leq k$.

The directedness condition on I (every pair of elements has an upper bound) ensures that the system is "connected": any two groups in the system can be compared via a common third.

1.1.2 Remark: Categorical Vocabulary This text uses modern categorical terminology throughout. What many classical references (including [26] and [30]) call an "inverse limit" or "projective limit" is a *limit*, specifically a limit over a cofiltered index category. Dually, a "direct limit" or "injective limit" is a *colimit* (a filtered colimit). The notation $\varprojlim$ for limits and $\varinjlim$ for colimits is standard and retained here.

Definition 1.1.3: Limit of a Cofiltered System

Let $(G_i, \varphi_{ji})_{i \in I}$ be a cofiltered system of topological groups. The *limit* of the system is the topological group:

$$\varprojlim_{i \in I} G_i = \left\{ (g_i)_{i \in I} \in \prod_{i \in I} G_i \;\middle|\; \varphi_{ji}(g_j) = g_i \text{ for all } i \leq j \right\}$$

equipped with the subspace topology from $\prod_{i \in I} G_i$, where each G_i carries its given topology and the product carries the product topology.

An element of $\varprojlim G_i$ is a *compatible sequence*, a choice of element $g_i \in G_i$ for each i, such that the transition maps carry later choices to earlier ones. The limit is a closed subgroup of the product, being the intersection of the closed sets $\{(g_i) \mid \varphi_{ji}(g_j) = g_i\}$ as (i, j) ranges over pairs with $i \leq j$. Each such set is closed as the map $(g_i) \mapsto \varphi_{ji}(g_j) g_i^{-1}$ is continuous (built from two projections, φ_{ji}, inversion and multiplication in G_i) and $\{e\}$ is closed in the Hausdorff group G_i.

Proposition 1.1.4: Limits of Compact Groups

If $(G_i, \varphi_{ji})_{i \in I}$ is a cofiltered system of compact Hausdorff topological groups, then $\varprojlim G_i$ is a compact Hausdorff topological group.

***Proof*:**
The product $\prod_{i \in I} G_i$ is compact by Tychonoff's theorem (see [17, Tychonoff's Theorem]) and Hausdorff as a product of Hausdorff spaces. Since $\varprojlim G_i$ is a closed subgroup of this product, it is compact (a closed subset of a compact space is compact) and Hausdorff (a subspace of a

Hausdorff space is Hausdorff).

With the notion of limit in hand, the central definition of this chapter can be stated.

Definition 1.1.5: Profinite Group

A *profinite group* is a topological group isomorphic to the limit of a cofiltered system of finite groups, where each finite group carries the discrete topology.

Since every finite discrete group is compact and Hausdorff, proposition 1.1.4 shows that every profinite group is compact and Hausdorff. The converse question (which compact Hausdorff groups are profinite) is answered by theorem 1.1.10 below.

Any group gives rise to a profinite group through the following construction.

Definition 1.1.6: Profinite Completion

Let G be a group. The *profinite completion* of G is the limit:

$$\widehat{G} = \varprojlim_{N \trianglelefteq_{\text{fin}} G} G/N$$

where N ranges over all normal subgroups of finite index in G, ordered by reverse inclusion ($N_1 \leq N_2$ in the poset when $N_1 \supseteq N_2$), and the transition maps are the natural projections $G/N_1 \to G/N_2$ for $N_1 \subseteq N_2$.

The natural map $\iota\colon G \to \widehat{G}$, sending g to the tuple of cosets $(gN)_N$, is a group homomorphism. Its kernel is $\bigcap_N N$, the intersection of all finite-index normal subgroups. A group for which this intersection is trivial (so that ι is injective) is called *residually finite*.

Example 1.1.7: Profinite Groups and Completions

1. Every finite group G, equipped with the discrete topology, is profinite: it is the limit of the constant system consisting of G alone. The profinite completion satisfies $\widehat{G} \cong G$.

2. The profinite completion of $\mathbb{Z}$ is:

$$\widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

   where $\mathbb{Z}_p = \varprojlim_{k \geq 1} \mathbb{Z}/p^k\mathbb{Z}$ is the ring of p-adic integers. The isomorphism follows from the Chinese Remainder Theorem: for $n = \prod p_i^{a_i}$, the canonical map $\mathbb{Z}/n\mathbb{Z} \xrightarrow{\sim} \prod \mathbb{Z}/p_i^{a_i}\mathbb{Z}$ is an isomorphism, and passing to the limit over n gives the decomposition. The group $\mathbb{Z}$ is residually finite (the intersection $\bigcap_{n \geq 1} n\mathbb{Z} = \{0\}$) and embeds into $\widehat{\mathbb{Z}}$ with dense image.

3. The additive group $\mathbb{Q}$ has no proper subgroups of finite index: if $H \leq \mathbb{Q}$ satisfies $[\mathbb{Q} : H] = n$, then $n\mathbb{Q} \subseteq H$, and $n\mathbb{Q} = \mathbb{Q}$ forces $H = \mathbb{Q}$. The profinite completion is therefore trivial: $\widehat{\mathbb{Q}} = \{0\}$.

4. The p-adic integers $\mathbb{Z}_p$ form a profinite group by construction. The open subgroups are precisely $p^k\mathbb{Z}_p$ for $k \geq 0$, and the finite quotients $\mathbb{Z}_p/p^k\mathbb{Z}_p \cong \mathbb{Z}/p^k\mathbb{Z}$ are all p-groups. This is the prototypical example of a *pro-p-group* (section 1.3).

5. (If the reader has read [4]) For a connected scheme X with geometric point $\overline{x}$, the étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x})$ is a profinite group: it is the limit of the automorphism groups of finite étale covers of X, ordered by refinement. When $X = \operatorname{Spec} K$ for a field K, this recovers the absolute Galois group G_K.

The definition of a profinite group presents it as a limit of some system of finite groups, but there is a canonical such system: the finite quotients of the group itself. The following proposition shows that every profinite group is determined by its own open normal subgroups.

Proposition 1.1.8: Profinite Groups as Limits of Finite Quotients

Let G be a profinite group. Then the open normal subgroups of G form a cofiltered system (ordered by reverse inclusion), every such subgroup has finite index, every open neighbourhood of the identity contains an open normal subgroup, and the natural map:

$$G \xrightarrow{\sim} \varprojlim_{U \trianglelefteq_{\text{open}} G} G/U$$

is an isomorphism of topological groups. If $G = \varprojlim_{i \in I} G_i$ for a cofiltered system of finite discrete groups with projections $\pi_i \colon G \to G_i$, then every open neighbourhood of the identity contains $\ker \pi_j$ for some $j \in I$.

***Proof*:**
Write $G = \varprojlim_{i \in I} G_i$ with each G_i finite and discrete, and let $\pi_i \colon G \to G_i$ denote the projection. Each $\ker \pi_i$ is open in G (the preimage of $\{e\}$ under the continuous map π_i to a discrete space) and normal with $G/\ker \pi_i$ finite (it embeds into G_i). Conversely, any open normal subgroup U of G has finite index: the cosets of U form an open cover of the compact space G, so finitely many suffice, giving $[G : U] < \infty$. If U and V are open normal, so is $U \cap V$, so the open normal subgroups form a cofiltered system.

Let W be an open neighbourhood of e. The topology of G is the subspace topology from $\prod_i G_i$, and each G_i is discrete, so W contains a basic neighbourhood $\{g \in G \mid \pi_{i_1}(g) = e, \dots, \pi_{i_k}(g) = e\}$ of e for some $i_1, \dots, i_k \in I$. Directedness gives $j \in I$ with $j \geq i_l$ for every l. Then $\pi_{i_l} = \varphi_{j i_l} \circ \pi_j$, so $\ker \pi_j \subseteq \ker \pi_{i_l}$ for every l, and $\ker \pi_j \subseteq W$. This proves the last sentence of the statement, and since $\ker \pi_j$ is open and normal, every open neighbourhood of e contains an open normal subgroup.

The natural homomorphism $\alpha \colon G \to \varprojlim_U G/U$ sends g to the tuple $(gU)_U$. It is continuous, because each projection $G \to G/U$ is continuous: the preimage of a point of the discrete group G/U is a coset of the open subgroup U. It is injective: if $g \in \bigcap_U U$, then $g \in \ker \pi_i$ for every i, hence $g_i = e$ for all i, so $g = e$ in $G \subseteq \prod G_i$. It is surjective. Let $(g_U U)_U \in \varprojlim_U G/U$, and put $h_i = \pi_i(g_{\ker \pi_i}) \in G_i$. For $i \leq j$ the inclusion $\ker \pi_j \subseteq \ker \pi_i$ and the compatibility of the thread give $g_{\ker \pi_j} \ker \pi_i = g_{\ker \pi_i} \ker \pi_i$, so

$$\varphi_{ji}(h_j) = \pi_i(g_{\ker \pi_j}) = \pi_i(g_{\ker \pi_i}) = h_i$$

Hence $h = (h_i)_i$ lies in $\varprojlim_i G_i = G$. Let U be open normal, and choose j with $\ker \pi_j \subseteq U$, as in the previous paragraph. Since $\pi_j(h) = h_j = \pi_j(g_{\ker \pi_j})$, the elements h and $g_{\ker \pi_j}$ have the same coset modulo $\ker \pi_j$, hence modulo U, and compatibility of the thread gives

$hU = g_{\ker \pi_j} U = g_U U$. So $\alpha(h) = (g_U U)_U$. Finally, α is a continuous bijection from the compact space G to the Hausdorff space $\varprojlim_U G/U$, hence a homeomorphism.

This proposition makes precise the sense in which a profinite group "is" the limit of its finite quotients. No external system of finite groups is needed: the group's own topology determines the system canonically.

The profinite completion now acquires a clean universal property.

Proposition 1.1.9: Universal Property of Profinite Completion

Let G be a group and $\iota\colon G \to \widehat{G}$ the natural map to its profinite completion. For any profinite group H and any group homomorphism $\varphi\colon G \to H$ whose image is dense, there exists a unique continuous surjection $\tilde{\varphi}\colon \widehat{G} \to H$ such that $\tilde{\varphi} \circ \iota = \varphi$. More generally, for any group homomorphism $\varphi\colon G \to H$ with H profinite, there exists a unique continuous homomorphism $\tilde{\varphi}\colon \widehat{G} \to H$ with $\tilde{\varphi} \circ \iota = \varphi$:

$$\begin{array}{ccc} G & \xrightarrow{\iota} & \widehat{G} \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \exists!\, \tilde{\varphi}} \\ & & H \end{array}$$

***Proof*:**
By proposition 1.1.8, write $H \cong \varprojlim_U H/U$ where U ranges over the open normal subgroups of H. For each such U, the composition $G \xrightarrow{\varphi} H \to H/U$ is a homomorphism to a finite group, so it factors through $G/\varphi^{-1}(U)$. Since U is open and has finite index in H, the preimage $\varphi^{-1}(U)$ is a finite-index normal subgroup of G, hence appears in the system defining $\widehat{G}$. The resulting maps $G/\varphi^{-1}(U) \to H/U$ are compatible as U varies, and the universal property of the limit $\widehat{G} = \varprojlim_N G/N$ produces a unique continuous homomorphism $\tilde{\varphi}\colon \widehat{G} \to H$. That $\tilde{\varphi} \circ \iota = \varphi$ is immediate from the construction.

The universal property says that the profinite completion is the best approximation of G by a profinite group: every homomorphism from G to a profinite group factors uniquely through $\widehat{G}$. In categorical language, the profinite completion functor is left adjoint to the forgetful functor from profinite groups to groups.

The stage is set for the main structural result of this section. The following theorem characterizes profinite groups intrinsically, without reference to any particular presentation as a limit.

Theorem 1.1.10: Characterizing Profinite Groups

A topological group G is profinite if and only if it is compact, Hausdorff, and totally disconnected.

***Proof*:**
Step 1: Profinite groups are compact, Hausdorff, and totally disconnected. Write $G = \varprojlim_{i \in I} G_i$ with each G_i finite and discrete. Each G_i is compact (finite), Hausdorff (discrete), and totally disconnected (discrete). The product $\prod G_i$ inherits all three properties: compactness by Ty-

chonoff's theorem, and the Hausdorff and total disconnectedness properties because they are preserved by arbitrary products and pass to subspaces. Since G is a closed subgroup of $\prod G_i$, it is compact, Hausdorff, and totally disconnected.

Step 2: Compact, Hausdorff, totally disconnected groups are profinite. Let G be a compact, Hausdorff, totally disconnected group.

Step 2a: An open normal subgroup inside a clopen neighbourhood of e. Let $C \subseteq G$ be a clopen (simultaneously open and closed) set with $e \in C$. The set $S = \{h \in G \mid hC \subseteq C\}$ is open. Let $h_0 \in S$. For each $c \in C$ the product $h_0 c$ lies in the open set C, so continuity of multiplication $G \times G \to G$ at (h_0, c) gives open sets $U_c \ni h_0$ and $V_c \ni c$ with $U_c V_c \subseteq C$. Since C is compact (closed in the compact space G), finitely many $V_{c_1}, \dots, V_{c_m}$ cover C, and $U = U_{c_1} \cap \cdots \cap U_{c_m}$ is an open neighbourhood of h_0 with $UC \subseteq U_{c_1} V_{c_1} \cup \cdots \cup U_{c_m} V_{c_m} \subseteq C$. So $U \subseteq S$, and S is open. An element h satisfies $hC = C$ if and only if $hC \subseteq C$ and $h^{-1}C \subseteq C$, so the stabilizer $\mathrm{Stab}_G(C) = \{h \in G \mid hC = C\}$ equals $S \cap S^{-1}$, which is open because inversion is a homeomorphism.

The left cosets of the open subgroup $\mathrm{Stab}_G(C)$ are open, pairwise disjoint, and cover the compact space G, so there are finitely many, $x_1\mathrm{Stab}_G(C), \dots, x_r\mathrm{Stab}_G(C)$. The left multiplication action of G on the finite set $X = G/\mathrm{Stab}_G(C)$ gives a homomorphism $\rho\colon G \to \mathrm{Sym}(X)$, whose kernel

$$N = \ker \rho = \bigcap_{i=1}^{r} x_i \mathrm{Stab}_G(C) x_i^{-1}$$

is an open normal subgroup, as a finite intersection of open subgroups. If $h \in N$, then $h\mathrm{Stab}_G(C) = \mathrm{Stab}_G(C)$, so $hC = C$, and since $e \in C$ this gives $h = he \in hC = C$. Thus $N \subseteq C$.

Step 2b: Completing the characterization. Let W be an open neighbourhood of e. The clopen subsets of the compact, Hausdorff, totally disconnected space G form a basis of its topology by *Topology* [17, Clopen Sets in Compact Hausdorff Spaces], so there is a clopen C with $e \in C \subseteq W$, and Step 2a produces an open normal subgroup $N \subseteq C \subseteq W$. The open normal subgroups therefore form a neighbourhood basis of e. For $g \neq e$, the set $G \setminus \{g\}$ is an open neighbourhood of e, as points are closed in the Hausdorff space G, so some open normal subgroup omits g, and the intersection of all open normal subgroups is trivial. Each open normal subgroup N has finite index, its cosets being finitely many by the compactness argument of Step 2a, and the open normal subgroups form a cofiltered system, since $N \cap N'$ is open and normal whenever N and N' are.

The natural homomorphism $\alpha\colon G \to \varprojlim_N G/N$, $g \mapsto (gN)_N$, is continuous, as each projection $G \to G/N$ to a discrete group has open fibres, the cosets of N. It is injective, as its kernel is the intersection of all open normal subgroups. It is surjective. Let $(g_N N)_N \in \varprojlim_N G/N$. Each coset $g_N N$ is nonempty and closed, its complement being the union of the other cosets of the open subgroup N. For finitely many open normal subgroups $N_1, \dots, N_k$, the subgroup $N = N_1 \cap \cdots \cap N_k$ is open normal, and the compatibility of the thread gives $g_N N_j = g_{N_j} N_j$, so $g_N \in g_{N_1} N_1 \cap \cdots \cap g_{N_k} N_k$. The closed sets $g_N N$ therefore have the finite intersection property, and compactness of G (*Topology* [17, FIP And Compactness]) gives an element $h \in \bigcap_N g_N N$. Then $hN = g_N N$ for every N, that is, $\alpha(h) = (g_N N)_N$. As a continuous bijection from a compact space to the Hausdorff space $\varprojlim_N G/N$, α is a homeomorphism. Since each G/N is

finite, G is profinite.

The characterization theorem reduces checking that a topological group is profinite to verifying three topological properties, none of which mentions limits or cofiltered systems. In practice, compactness and the Hausdorff property are often immediate, and the essential content is total disconnectedness (the absence of nontrivial connected subsets). For Galois groups, total disconnectedness is built into the Krull topology by construction, as will be seen in section 1.4.

Exercise 1.1.1

1. Let G be a group and $\iota\colon G \to \widehat{G}$ the natural map to its profinite completion. Show that $\iota(G)$ is dense in $\widehat{G}$.
2. Show that $\iota\colon G \to \widehat{G}$ is injective if and only if G is residually finite, and give an example of a group that is not residually finite.
3. Show that the profinite completion commutes with finite direct products: for any groups G and H, there is a canonical isomorphism $\widehat{G \times H} \cong \widehat{G} \times \widehat{H}$.
4. Let G be a profinite group and D a discrete group. Show that every continuous homomorphism $\varphi\colon G \to D$ has open kernel, and conclude that $\varphi(G)$ is finite.
5. Show that $\mathbb{Z}_p$ is profinite by verifying the three conditions of theorem 1.1.10 directly, without using the definition of $\mathbb{Z}_p$ as a limit.

1.2 Topological Properties of Profinite Groups

The characterization theorem (theorem 1.1.10) reduces the study of profinite groups to the study of compact, Hausdorff, totally disconnected groups. The next task is understanding the internal structure: which subgroups inherit the profinite property, how quotients behave, and what the profinite analogue of Sylow theory looks like. The key theme is that the topology and algebra of a profinite group are tightly coupled: for closed subgroups, being open and having finite index are the same condition.

The characterization theorem gives an immediate criterion for recognizing profinite subgroups.

Proposition 1.2.1: Closed Subgroups Are Profinite

Every closed subgroup of a profinite group is profinite.

***Proof*:**
Let H be a closed subgroup of a profinite group G. By theorem 1.1.10, G is compact, Hausdorff, and totally disconnected. As a closed subset of a compact space, H is compact. As a subspace of a Hausdorff space, H is Hausdorff. As a subspace of a totally disconnected space, H is totally disconnected. Applying theorem 1.1.10 again, H is profinite.

Non-closed subgroups, by contrast, need not be profinite: they may fail to be compact. For instance, $\mathbb{Z}$ embeds densely into $\widehat{\mathbb{Z}}$ as a non-closed subgroup, and $\mathbb{Z}$ with the subspace topology is not compact.

The relationship between the topological and algebraic notions of "smallness" for subgroups is governed by the following.

Proposition 1.2.2: Open and Finite-Index Subgroups

Let G be a profinite group and $H \leq G$. Then:

1. If H is open, then H is closed and has finite index in G.
2. If H is closed and has finite index, then H is open.
3. A closed subgroup is open if and only if it has finite index.

***Proof*:**
For (1), if H is open, its complement $G \setminus H = \bigcup_{g \notin H} gH$ is a union of open cosets (each gH is open since left multiplication is a homeomorphism), hence open. So H is closed. The cosets of H form an open cover of the compact space G, so finitely many suffice, giving $[G : H] < \infty$.

For (2), if H is closed with $[G : H] = n$, the complement $G \setminus H = g_2H \cup \cdots \cup g_nH$ is a finite union of closed sets (each g_iH is closed as a translate of the closed set H), hence closed. So H is open.

Part (3) combines (1) and (2).

Part (3) fails without the closure hypothesis: a dense subgroup of a profinite group can have finite index without being open (see exercise 1.2.1 (5)). The following result of Nikolov and Segal shows that for finitely generated profinite groups, this pathology does not arise.

Theorem 1.2.3: Nikolov-Segal Theorem

Let G be a topologically finitely generated profinite group. Then every subgroup of finite index in G is open.

The proof, which uses the classification of finite simple groups, is beyond the scope of this text. See [27]. The theorem implies that for finitely generated profinite groups, the algebraic structure (finite-index subgroups) completely determines the topology (open subgroups). In particular, the topology on such a group is the unique one making it profinite.

Quotients of profinite groups by closed normal subgroups remain profinite.

Proposition 1.2.4: Quotients of Profinite Groups

Let G be a profinite group and $N \trianglelefteq G$ a closed normal subgroup. Then G/N with the quotient topology is profinite, and the projection $\pi \colon G \to G/N$ is open.

***Proof*:**
The projection π is open: for any open subset $V \subseteq G$, the set $\pi^{-1}(\pi(V)) = VN = \bigcup_{n\in N} Vn$ is a union of open sets (each Vn is open), hence open, so $\pi(V)$ is open by the definition of the quotient topology. The quotient G/N is a topological group by *Topology* [17, Quotients by Normal Subgroups], compact as the continuous image of the compact space G, and Hausdorff because N is closed (*Topology* [17, Separation of the Quotient]).

For total disconnectedness, let $gN \neq hN$ in G/N, so that $x = g^{-1}h \notin N$. The set $x^{-1}(G \setminus N)$ is open, as N is closed, and contains e, so it contains an open normal subgroup U by proposition 1.1.8. Then $xU \cap N = \emptyset$. Since U is normal, $UN = NU$ is a subgroup, and $x \notin UN$: a factorization $x = nu$ with $n \in N$ and $u \in U$ would give $n = xu^{-1} \in xU \cap N$. The subgroup $UN/N = \pi(UN)$ of G/N is open, as π is open, so its cosets are open, and each of them is closed, being the complement of the union of the others. The coset gUN/N contains gN and not hN, since $h \in gUN$ would give $x \in UN$. If $Y \subseteq G/N$ contains gN and hN, then $Y \cap gUN/N$ is a clopen subset of Y containing gN and missing hN, so Y is not connected. No connected subset of G/N has two points, so G/N is totally disconnected, and by theorem 1.1.10 it is profinite.

A continuous homomorphism between profinite groups always has profinite image.

Corollary 1.2.5: Images of Continuous Homomorphisms

Let $f\colon G \to H$ be a continuous homomorphism of profinite groups. Then $f(G)$ is a closed subgroup of H, hence profinite.

***Proof*:**
The image $f(G)$ is compact (continuous image of the compact space G) and hence closed in the Hausdorff space H. By proposition 1.2.1, $f(G)$ is profinite.

The remainder of this section develops the profinite analogue of the Sylow theorems. The proofs illustrate a technique that recurs throughout the theory of profinite groups: reducing to the finite case and passing to the limit. The following observation makes this passage possible.

Lemma 1.2.6: Limits of Non-Empty Finite Sets

The limit of a cofiltered system of non-empty finite sets is non-empty.

***Proof*:**
Let $(X_i, f_{ji})_{i\in I}$ be such a system. Give each X_i the discrete topology, so that $\prod_{i\in I} X_i$ is compact by Tychonoff's theorem. For each finite subset $F \subseteq I$, define

$$Y_F = \left\{(x_i) \in \prod X_i \;\middle|\; f_{ji}(x_j) = x_i \text{ for all } i \leq j \text{ with } i, j \in F\right\}$$

. Each Y_F is closed (a finite intersection of closed conditions) and non-empty: by directedness, there exists $k \in I$ with $k \geq i$ for all $i \in F$, and any $x_k \in X_k$ determines compatible values $x_i = f_{ki}(x_k)$ for $i \in F$. Since $Y_{F_1\cup F_2} \subseteq Y_{F_1} \cap Y_{F_2}$, the family $\{Y_F\}$ has the finite intersection property. By compactness, $\bigcap_F Y_F = \varprojlim X_i$ is non-empty.

With this tool, the classical Sylow theorems extend to profinite groups.

Theorem 1.2.7: Sylow Theory for Profinite Groups

Let G be a profinite group and p a prime. Then:

1. G has a maximal pro-p-subgroup (a *Sylow pro-p-subgroup*).
2. Any two Sylow pro-p-subgroups of G are conjugate.
3. Every pro-p-subgroup of G is contained in a Sylow pro-p-subgroup.

***Proof*:**
Write $G \cong \varprojlim_U G/U$ where U ranges over the open normal subgroups, with projections $\pi_U \colon G \to G/U$ and transition maps $\pi_{VU} \colon G/U \to G/V$ for $U \subseteq V$. Each G/U is a finite group. The image of a Sylow p-subgroup of G/U under the surjection π_{VU} is a Sylow p-subgroup of G/V: if P is Sylow in G/U with $|P| = |G/U|_p$ and $K = \ker \pi_{VU} = V/U$, then $|P \cap K| = |K|_p$ (since $K \trianglelefteq G/U$ and a Sylow subgroup of G/U intersects any normal subgroup in a Sylow subgroup of that normal subgroup), giving $|\pi_{VU}(P)| = |P|/|P \cap K| = |G/U|_p/|K|_p = |G/V|_p$. So the assignment $P \mapsto \pi_{VU}(P)$ defines a map $\mathrm{Syl}_p(G/U) \to \mathrm{Syl}_p(G/V)$, and the sets $\mathrm{Syl}_p(G/U)$ form a cofiltered system of non-empty finite sets.

Step 1: Existence. By lemma 1.2.6, the limit $\varprojlim_U \mathrm{Syl}_p(G/U)$ is non-empty. A compatible family $(P_U)_U$ with $P_U \in \mathrm{Syl}_p(G/U)$ and $\pi_{VU}(P_U) = P_V$ for all $U \subseteq V$ defines a closed subgroup $S = \varprojlim_U P_U \subseteq G$. Each P_U is a p-group, so S is a pro-p-group. To see that S is maximal: if $S \subseteq T$ with T a pro-p-subgroup, then $\pi_U(T)$ is a p-subgroup of G/U containing the Sylow subgroup P_U, forcing $\pi_U(T) = P_U$ for all U. Since $T \subseteq \varprojlim_U \pi_U(T) = \varprojlim_U P_U = S$, maximality follows.

Step 2: Conjugacy. Let S_1, S_2 be Sylow pro-p-subgroups. For each U, the Sylow p-subgroups $\pi_U(S_1)$ and $\pi_U(S_2)$ of G/U are conjugate by finite Sylow theory. The set

$$C_U = \left\{ g \in G/U \;\middle|\; g\,\pi_U(S_1)\,g^{-1} = \pi_U(S_2) \right\}$$

is non-empty and finite (it is a coset of the normalizer of $\pi_U(S_1)$). The projection π_{VU} maps C_U into C_V, so $(C_U)_U$ is a cofiltered system of non-empty finite sets. By lemma 1.2.6, there exists $g = (g_U)_U \in \varprojlim_U C_U \subseteq G$ with $g\,S_1\,g^{-1} = S_2$.

Step 3: Containment. Let T be a pro-p-subgroup of G. For each U, the set

$$D_U = \left\{ P \in \mathrm{Syl}_p(G/U) \;\middle|\; \pi_U(T) \subseteq P \right\}$$

is non-empty (by finite Sylow theory). The transition maps send D_U into D_V: if $\pi_U(T) \subseteq P$, then $\pi_V(T) = \pi_{VU}(\pi_U(T)) \subseteq \pi_{VU}(P)$, and $\pi_{VU}(P) \in D_V$. By lemma 1.2.6, the limit of the system $(D_U)_U$ is non-empty, and a compatible family $(P_U)_U$ with $\pi_U(T) \subseteq P_U$ defines a Sylow pro-p-subgroup $S = \varprojlim_U P_U$ containing T.

The Sylow theorem for profinite groups recovers the finite Sylow theorems when G is finite (the only

open normal subgroup is the trivial group), and demonstrates the power of the limit-of-finite-sets technique: each part of the proof reduces to the corresponding finite statement, then passes to the limit using lemma 1.2.6. This technique will reappear in chapters 3 and 5.

Example 1.2.8: Sylow Structure

1. The profinite group $\widehat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p$ decomposes as a product of its Sylow pro-p-subgroups: the Sylow pro-p-subgroup is $\mathbb{Z}_p$, embedded as the factor corresponding to p. Since each Sylow subgroup is normal (in any abelian group), the decomposition $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ is the unique way to write $\widehat{\mathbb{Z}}$ as a product of its Sylow subgroups. This mirrors the decomposition $\mathbb{Z}/n\mathbb{Z} \cong \prod_{p|n} \mathbb{Z}/p^{v_p(n)}\mathbb{Z}$ from the Chinese Remainder Theorem.

2. (If the reader has read [11]) Let E/K be an elliptic curve over a number field and p a prime. The Galois action on the p-power torsion gives a continuous homomorphism $\rho_{E,p} \colon G_K \to \mathrm{GL}_2(\mathbb{Z}_p)$. The image $\rho_{E,p}(G_K)$ is a closed subgroup of $\mathrm{GL}_2(\mathbb{Z}_p)$ by corollary 1.2.5, hence profinite. The structure of this image (and in particular its Sylow subgroups) encodes arithmetic information about E.

Exercise 1.2.1

1. Let G be a profinite group and $H \leq G$ a subgroup (not necessarily closed). Show that:
$$\overline{H} = \bigcap_{U \trianglelefteq_{\text{open}} G} HU$$

2. Determine all open subgroups of $\mathbb{Z}_p$ and describe their inclusion relations.

3. Let $f \colon G \to H$ be a continuous surjective homomorphism of profinite groups. Show that f is an open map.

4. Let G be a profinite group and $N \trianglelefteq G$ a closed normal subgroup. Show that the open normal subgroups of G/N are exactly the images UN/N where U ranges over the open normal subgroups of G containing N.

5. Show that the closure hypothesis in proposition 1.2.2(2) is necessary: give an example of a profinite group G with a dense subgroup H of finite index that is not open. *Hint:* use the axiom of choice to construct a discontinuous $\mathbb{F}_2$-linear functional on $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$.

1.3 Pro-p-Groups and Supernatural Numbers

Among profinite groups, those built entirely from p-groups for a single prime p form the most tractable and best-understood class. These pro-p-groups are the objects on which the cohomological dimension theory of chapter 5 and the Golod-Shafarevich inequality of chapter 6 are best understood. To measure the "size" of a profinite group (which may have uncountably many elements), the classical notion of group order must be generalized. Supernatural numbers provide this generalization.

Definition 1.3.1: Pro-p-Group

Let p be a prime. A *pro-p-group* is a profinite group in which every finite quotient is a p-group.

Equivalently, a pro-p-group is a limit of a cofiltered system of finite p-groups. A pro-p-group G is the limit of its finite quotients G/U, U open normal, by proposition 1.1.8, and these are p-groups. Conversely, let $G = \varprojlim_i G_i$ with each G_i a finite p-group and projections π_i, and let U be an open normal subgroup. By proposition 1.1.8 some $\ker \pi_j$ lies in U, so G/U is a quotient of $G/\ker \pi_j$, which embeds in G_j, and G/U is a p-group.

Proposition 1.3.2: Characterizations of Pro-p-Groups

Let G be a profinite group and p a prime. The following are equivalent:

1. G is a pro-p-group.
2. Every open normal subgroup $U \trianglelefteq G$ has p-power index.
3. G is its own Sylow pro-p-subgroup: the only Sylow pro-p-subgroup of G is G itself.

***Proof*:**
The equivalence (1) $\Leftrightarrow$ (2) is immediate: the finite quotients of G are exactly the groups G/U for U open normal, and a finite group is a p-group if and only if its order is a power of p.

For (1) $\Rightarrow$ (3): if G is a pro-p-group, then G is a pro-p-subgroup of itself. Any Sylow pro-p-subgroup S contains G (by maximality and the fact that G is already pro-p), so $S = G$. For (3) $\Rightarrow$ (1): if G is its own Sylow pro-p-subgroup, then G is a pro-p-group by definition of Sylow.

Example 1.3.3: Pro-p-Groups

1. The p-adic integers $\mathbb{Z}_p = \varprojlim_k \mathbb{Z}/p^k\mathbb{Z}$ form a pro-p-group, with all finite quotients being cyclic p-groups.
2. For $n \geq 1$, the kernel of the reduction map $\mathrm{GL}_n(\mathbb{Z}_p) \to \mathrm{GL}_n(\mathbb{F}_p)$ is the subgroup $K_n = \{A \in \mathrm{GL}_n(\mathbb{Z}_p) \mid A \equiv I \pmod{p}\}$. Each finite quotient of K_n (obtained by reducing modulo p^k) is a p-group: the kernel of $\mathrm{GL}_n(\mathbb{Z}/p^k\mathbb{Z}) \to \mathrm{GL}_n(\mathbb{F}_p)$ has order $p^{(k-1)n^2}$. So K_n is a pro-p-group. The full group $\mathrm{GL}_n(\mathbb{Z}_p)$ is *not* a pro-p-group: the quotient $\mathrm{GL}_n(\mathbb{F}_p)$ has order $\prod_{i=0}^{n-1}(p^n - p^i)$, which is divisible by primes other than p for $n \geq 2$.
3. The profinite completion $\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell$ is not a pro-p-group for any prime p, since it has quotients of every finite order. Its Sylow pro-p-subgroup is $\mathbb{Z}_p$, embedded as the p-component of the product.

To express statements like "$G_{\mathbb{Q}_p}$ has pro-p part of infinite p-rank" or "the index $[G : H]$ is divisible by p^∞," a generalization of the natural numbers is needed.

Definition 1.3.4: Supernatural Numbers

A *supernatural number* is a formal product:

$$\alpha = \prod_p p^{n_p}$$

where p ranges over all primes and each exponent $n_p \in \{0, 1, 2, \ldots\} \cup \{\infty\}$. The supernatural numbers form a partially ordered set under divisibility: $\alpha \mid \beta$ if $n_p(\alpha) \leq n_p(\beta)$ for all p. The gcd and lcm of any family of supernatural numbers are defined by taking the componentwise infimum and supremum of exponents, respectively.

Every positive integer is a supernatural number (with all but finitely many exponents equal to zero, and none equal to ∞). The supernatural number $\prod_p p^\infty$ is the largest, divisible by every supernatural number. Unlike positive integers, supernatural numbers do not form a group under multiplication: there is no cancellation law.

The order and index of profinite groups take values in supernatural numbers.

Definition 1.3.5: Order and Index

Let G be a profinite group. The *order* of G is the supernatural number:

$$\#G = \operatorname{lcm}\{|G/U| \mid U \trianglelefteq_{\text{open}} G\}$$

If $H \leq G$ is a closed subgroup, the *index* of H in G is:

$$[G : H] = \operatorname{lcm}\{[G/U : \pi_U(H)] \mid U \trianglelefteq_{\text{open}} G\}$$

where $\pi_U \colon G \to G/U$ is the projection.

When G is finite, $\#G$ is the ordinary group order: the only open normal subgroup is $\{e\}$, and $|G/\{e\}| = |G|$. For a closed subgroup H of a finite group, $[G : H]$ is the ordinary index.

Proposition 1.3.6: Lagrange's Theorem for Profinite Groups

Let G be a profinite group and $H \leq G$ a closed subgroup. Then:

$$\#G = [G : H] \cdot \#H$$

as supernatural numbers.

***Proof*:**
The equality is verified prime by prime. Fix a prime p. For each open normal $U \trianglelefteq G$, the projection $\pi_U(H) = HU/U \cong H/(H \cap U)$ gives:

$$v_p(|G/U|) = v_p([G/U : \pi_U(H)]) + v_p(|H/(H \cap U)|)$$

Both terms on the right are non-decreasing as U shrinks (a finer quotient G/U' with $U' \subseteq U$ gives $[G/U' : \pi_{U'}(H)] \geq [G/U : \pi_U(H)]$ and $|H/(H \cap U')| \geq |H/(H \cap U)|$). Taking the supremum

over U: the supremum of a sum of non-decreasing functions equals the sum of the suprema, giving $v_p(\#G) = v_p([G:H]) + v_p(\#H)$.

Example 1.3.7: Supernatural Orders

1. $\#\widehat{\mathbb{Z}} = \prod_p p^\infty$, since $\widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$ for every $n \geq 1$, and $\operatorname{lcm}\{n : n \geq 1\} = \prod_p p^\infty$.
2. $\#\mathbb{Z}_p = p^\infty$ and $[\widehat{\mathbb{Z}} : \mathbb{Z}_p] = \prod_{\ell \neq p} \ell^\infty$, consistent with Lagrange: $\prod_\ell \ell^\infty = \prod_{\ell \neq p} \ell^\infty \cdot p^\infty$.
3. For a finite group G of order n, $\#G = n$. This recovers the classical Lagrange theorem from proposition 1.3.6.

Supernatural numbers give a clean characterization of pro-p-groups in terms of the order.

Corollary 1.3.8: Pro-p-Groups via Order

A profinite group G is a pro-p-group if and only if $\#G$ is a power of p (i.e., $v_\ell(\#G) = 0$ for all primes $\ell \neq p$).

***Proof*:**
By proposition 1.3.2, G is pro-p if and only if every open normal subgroup has p-power index. This holds if and only if $|G/U|$ is a power of p for every open normal U, which is equivalent to $v_\ell(\#G) = \sup_U v_\ell(|G/U|) = 0$ for all $\ell \neq p$.

The section closes with a structural result about pro-p-groups that will be essential in chapter 6: the Frattini subgroup controls the minimal number of topological generators.

Definition 1.3.9: Frattini Subgroup

Let G be a pro-p-group. The *Frattini subgroup* $\Phi(G)$ is the intersection of all maximal open subgroups of G. Equivalently, $\Phi(G)$ is the closure of the subgroup generated by all pth powers and commutators: $\Phi(G) = \overline{G^p[G,G]}$.

Each maximal open subgroup has index p in G (since maximal subgroups of a finite p-group have index p), so $\Phi(G)$ is a closed normal subgroup and $G/\Phi(G)$ is a quotient in which every element has order dividing p.

Proposition 1.3.10: Burnside Basis Theorem for Pro-p-Groups

Let G be a pro-p-group. Then:

1. The quotient $G/\Phi(G)$ is an elementary abelian p-group: there exists a cardinal d with $G/\Phi(G) \cong (\mathbb{Z}/p\mathbb{Z})^d$.
2. Elements $s_1, \ldots, s_n \in G$ topologically generate G (i.e., the closure $\overline{\langle s_1, \ldots, s_n \rangle} = G$) if and only if their images $\bar{s}_1, \ldots, \bar{s}_n$ generate $G/\Phi(G)$ as an $\mathbb{F}_p$-vector space.
3. The minimal number of topological generators of G is $d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$.

***Proof*:**
For (1): the quotient $G/\Phi(G)$ embeds into $\prod_M G/M$ where M ranges over maximal open subgroups. Each G/M has order p, so the product is an elementary abelian p-group, and a subgroup of an elementary abelian p-group is elementary abelian.

For (2): the forward direction is immediate. If $\{s_i\}$ generates G topologically, its image generates any quotient. For the converse, suppose $\{\bar{s}_i\}$ generates $G/\Phi(G)$, and let $H = \overline{\langle s_1, \ldots, s_n \rangle}$. Then $H\Phi(G) = G$ (since the images of the s_i generate $G/\Phi(G)$). For any open normal $U \trianglelefteq G$, the projection $\pi_U(H) \cdot \Phi(G/U) = G/U$, where $\Phi(G/U)$ is the Frattini subgroup of the finite p-group G/U (using $\pi_U(\Phi(G)) \subseteq \Phi(G/U)$, which follows from the correspondence between maximal subgroups of G containing U and maximal subgroups of G/U). The classical Burnside basis theorem for finite p-groups (see [9]) gives $\pi_U(H) = G/U$. Since this holds for all U, and H is closed, proposition 1.1.8 gives $H = G$.

Part (3) follows from (2): $d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$ because any generating set must span $G/\Phi(G)$, and any spanning set of $G/\Phi(G)$ lifts to a topological generating set.

The Burnside basis theorem reduces the question of topological generation to linear algebra over $\mathbb{F}_p$. For instance, $\mathbb{Z}_p$ has $\Phi(\mathbb{Z}_p) = p\mathbb{Z}_p$ and $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p$, so $d(\mathbb{Z}_p) = 1$: the group $\mathbb{Z}_p$ is topologically generated by any element not divisible by p.

Exercise 1.3.1

1. Compute the supernatural order $\#\mathrm{GL}_n(\mathbb{Z}_p)$ for $n \geq 1$.
2. Let G be an abelian profinite group. Show that G is isomorphic to the direct product of its Sylow pro-p-subgroups: $G \cong \prod_p G_p$, where G_p denotes the unique Sylow pro-p-subgroup of G.
3. Let G be a pro-p-group and $N \trianglelefteq G$ a closed normal subgroup. Show that both N and G/N are pro-p-groups.
4. Compute the Frattini subgroup and the minimal number of topological generators for: (a) $\mathbb{Z}_p$, (b) $\mathbb{Z}_p \times \mathbb{Z}_p$, (c) $\mathbb{Z}_p^n$ for general $n \geq 1$.
5. (If the reader has read [14]) Let K be a finite extension of $\mathbb{Q}_p$ with absolute Galois group G_K, inertia group I_K, and wild inertia group P_K. Verify that P_K is a pro-p-group and that I_K/P_K is a pro-p'-group (i.e., has order coprime to p).

1.4 The Absolute Galois Group

The preceding sections developed profinite groups as abstract topological objects. The reason this machinery matters for algebraic number theory is that Galois groups of infinite extensions are profinite. This section makes the connection precise, establishes the profinite structure of the absolute Galois group, and computes it for the fields that serve as running examples throughout the book.

An algebraic extension L/K, finite or infinite, is *Galois* if it is normal and separable (*Core Algebra* [9, Normal Extension, Separable Extension]), and $\mathrm{Gal}(L/K)$ denotes its group of K-automorphisms. For finite extensions this is the usual notion, by (1) $\Leftrightarrow$ (2) of *Core Algebra* [9, Fundamental Theorem

of Galois Theory I].

Theorem 1.4.1: Infinite Galois Theory

Let L/K be a Galois extension (possibly infinite), and let $\mathcal{F}$ denote the set of finite Galois extensions F/K contained in L, directed by inclusion. Then:

1. The restriction maps assemble into a topological isomorphism:

$$\mathrm{Gal}(L/K) \xrightarrow{\sim} \varprojlim_{F\in\mathcal{F}} \mathrm{Gal}(F/K)$$

where the right side carries the limit topology (each $\mathrm{Gal}(F/K)$ discrete). The induced topology on $\mathrm{Gal}(L/K)$ is called the *Krull topology*. In particular, $\mathrm{Gal}(L/K)$ is a profinite group, and the subgroups $\mathrm{Gal}(L/F)$ for $F \in \mathcal{F}$ are open normal subgroups forming a neighbourhood basis of the identity.

2. There is an inclusion-reversing bijection:

$$\{\text{closed subgroups of } \mathrm{Gal}(L/K)\} \longleftrightarrow \{\text{intermediate fields } K \subseteq M \subseteq L\}$$

given by $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$. Every open subgroup is closed (proposition 1.2.2), and under this bijection *open* subgroups correspond to *finite* extensions of K, with $[\mathrm{Gal}(L/K) : \mathrm{Gal}(L/M)] = [M : K]$ when M/K is finite, and open normal subgroups correspond to finite Galois extensions of K contained in L.

3. For every $F \in \mathcal{F}$ the restriction map $\mathrm{Gal}(L/K) \to \mathrm{Gal}(F/K)$ is surjective. More generally, if M is an intermediate field finite over K, every K-embedding $M \to L$ is the restriction of an element of $\mathrm{Gal}(L/K)$.

***Proof*:**

Step 1: The isomorphism. Each $x \in L$ lies in a member of $\mathcal{F}$: its minimal polynomial over K is separable and splits in L, because L/K is separable and normal, and its splitting field inside L is finite Galois over K by (1) $\Rightarrow$ (2) of *Core Algebra* [9, Fundamental Theorem of Galois Theory I]. The compositum of two members of $\mathcal{F}$ lies in $\mathcal{F}$ (*Core Algebra* [9, Galois Composite Fields II]), so $\mathcal{F}$ is directed and $L = \bigcup_{F\in\mathcal{F}} F$. For $\sigma \in \mathrm{Gal}(L/K)$ and $F \in \mathcal{F}$, item (5) of the same theorem gives $\sigma(F) = F$, so $\sigma|_F \in \mathrm{Gal}(F/K)$. Define $\varphi\colon \mathrm{Gal}(L/K) \to \varprojlim_F \mathrm{Gal}(F/K)$ by $\varphi(\sigma) = (\sigma|_F)_F$, a homomorphism. Injectivity: if $\sigma|_F = \mathrm{id}$ for all $F \in \mathcal{F}$, then σ fixes every element of $L = \bigcup_F F$, so $\sigma = \mathrm{id}$. Surjectivity: a compatible family $(\sigma_F)_F$ defines a map $\sigma\colon L \to L$ by $\sigma(x) = \sigma_F(x)$ for any $F \in \mathcal{F}$ containing x. Two members of $\mathcal{F}$ containing x lie in their compositum, so compatibility makes σ well defined, and any two elements of L lie in a common F, so σ is a homomorphism of fields fixing K. It is surjective because $\sigma(F) = \sigma_F(F) = F$ for every F, so $\sigma \in \mathrm{Gal}(L/K)$ and $\varphi(\sigma) = (\sigma_F)_F$. Give $\mathrm{Gal}(L/K)$ the topology making the group isomorphism φ a homeomorphism. Then $\mathrm{Gal}(L/K)$ is a topological group isomorphic to a limit of finite discrete groups, hence profinite.

Each $\mathrm{Gal}(F/K)$ is discrete, so a basic open neighbourhood of the identity in $\varprojlim_F \mathrm{Gal}(F/K) \subseteq \prod_F \mathrm{Gal}(F/K)$ is the set of threads $(\tau_F)_F$ with $\tau_{F_1} = \mathrm{id}, \ldots, \tau_{F_k} = \mathrm{id}$ for some $F_1, \ldots, F_k \in \mathcal{F}$. Its preimage under φ is the set of σ fixing $F_1 \cup \cdots \cup F_k$ pointwise, which is $\mathrm{Gal}(L/F_1 \cdots F_k)$,

and the compositum $F_1 \cdots F_k$ lies in $\mathcal{F}$. For a single $F \in \mathcal{F}$ this shows that $\mathrm{Gal}(L/F)$ is open, and it is normal as the kernel of $\sigma \mapsto \sigma|_F$. So the subgroups $\mathrm{Gal}(L/F)$ for $F \in \mathcal{F}$ are open normal subgroups forming a neighbourhood basis of the identity. This proves (1).

Step 2: Lifting. Let $F \in \mathcal{F}$ and $\tau_0 \in \mathrm{Gal}(F/K)$. The set $\mathcal{F}_F = \{E \in \mathcal{F} \mid F \subseteq E\}$ is directed, and for $E \in \mathcal{F}_F$ put $X_E = \{\tau \in \mathrm{Gal}(E/K) \mid \tau|_F = \tau_0\}$. Since F/K is Galois, restriction $\mathrm{Gal}(E/K) \to \mathrm{Gal}(F/K)$ is surjective (*Core Algebra* [9, Fundamental Theorem Of Galois Theory III]), so each X_E is a nonempty finite set, and for $E \subseteq E'$ in $\mathcal{F}_F$ restriction maps $X_{E'}$ into X_E, compatibly with composition. By lemma 1.2.6 there is a compatible family $(\tau_E)_{E \in \mathcal{F}_F}$ with $\tau_E \in X_E$. Every element of L lies in a member of $\mathcal{F}_F$, namely the compositum of F with a member of $\mathcal{F}$ containing it, so the surjectivity argument of Step 1, run over the directed set $\mathcal{F}_F$, produces $\sigma \in \mathrm{Gal}(L/K)$ with $\sigma|_E = \tau_E$ for every $E \in \mathcal{F}_F$. In particular $\sigma|_F = \tau_0$.

Now let M be an intermediate field finite over K and $\iota\colon M \to L$ a K-embedding. Write $M = K(a_1, \ldots, a_r)$ and choose $F_i \in \mathcal{F}$ containing a_i, so that $M \subseteq F = F_1 \cdots F_r \in \mathcal{F}$. For $a \in M$, the element $\iota(a)$ is a root of the minimal polynomial of a over K, which splits in the normal extension F, so $\iota(M) \subseteq F$. The field F is the splitting field over K of a separable polynomial, hence also its splitting field over M, so F/M is Galois. Item (3) of *Core Algebra* [9, Fundamental Theorem of Galois Theory I] gives $|\mathrm{Gal}(F/K)| = [F : K]$ and $|\mathrm{Gal}(F/M)| = [F : M]$, so by the tower law

$$[\mathrm{Gal}(F/K) : \mathrm{Gal}(F/M)] = [M : K]$$

The group $\mathrm{Gal}(F/K)$ acts on the finite set $\mathrm{Hom}_K(M, F)$ by composition, and the stabilizer of the inclusion $j\colon M \to F$ is $\mathrm{Gal}(F/M)$, so the orbit of j has $[M : K]$ elements. The set $\mathrm{Hom}_K(M, F)$ embeds in the set of K-embeddings of M into an algebraic closure of F, which has $[M : K]_s \leq [M : K]$ elements (*Core Algebra* [9, Finite Separable Extension Equivalence]). So the orbit of j is all of $\mathrm{Hom}_K(M, F)$, and $\iota = \tau_0 \circ j$ for some $\tau_0 \in \mathrm{Gal}(F/K)$. The first paragraph gives $\sigma \in \mathrm{Gal}(L/K)$ with $\sigma|_F = \tau_0$, so $\sigma|_M = \iota$. This proves (3).

Step 3: Intermediate fields. Let M be an intermediate field. For $x \in L$ the minimal polynomial of x over M divides that over K, which is separable and splits in L, so L/M is Galois. The subgroup $\mathrm{Gal}(L/M)$ of $\mathrm{Gal}(L/K)$ is closed. For $x \in M$ choose $F \in \mathcal{F}$ containing x. The stabilizer $\{\sigma \mid \sigma(x) = x\}$ contains the open subgroup $\mathrm{Gal}(L/F)$, so it is a union of cosets of $\mathrm{Gal}(L/F)$, hence an open subgroup, and its complement is a union of cosets of it, so it is closed. The intersection of these stabilizers over $x \in M$ is $\mathrm{Gal}(L/M)$.

The fixed field of $\mathrm{Gal}(L/M)$ is M. Every element of M is fixed. Let $x \in L \setminus M$. Step 1 applied to the Galois extension L/M gives a subfield $F' \subseteq L$ containing x and finite Galois over M, and item (4) of *Core Algebra* [9, Fundamental Theorem of Galois Theory I] gives $M = (F')^{\mathrm{Gal}(F'/M)}$, so some $\tau_0 \in \mathrm{Gal}(F'/M)$ moves x. Step 2 applied to L/M gives $\sigma \in \mathrm{Gal}(L/M)$ with $\sigma|_{F'} = \tau_0$, and $\sigma(x) \neq x$. So $L^{\mathrm{Gal}(L/M)} = M$.

Step 4: Closed subgroups. Let $H \leq \mathrm{Gal}(L/K)$ be closed. Then L^H is an intermediate field, and $H \subseteq \mathrm{Gal}(L/L^H)$. Let $\sigma \in \mathrm{Gal}(L/L^H)$ and $F \in \mathcal{F}$. The image H_F of H under restriction to F is a subgroup of $\mathrm{Gal}(F/K)$ with fixed field $F \cap L^H$, so $H_F = \mathrm{Gal}(F/F \cap L^H)$ by the bijectivity of the Galois correspondence for F/K (*Core Algebra* [9, Fundamental Theorem Of Galois Theory II]). The restriction $\sigma|_F$ fixes $F \cap L^H$, so $\sigma|_F = h|_F$ for some $h \in H$, that is, $h \in \sigma\mathrm{Gal}(L/F)$.

Left translation by σ is a homeomorphism, so by (1) the cosets $\sigma\mathrm{Gal}(L/F)$ for $F \in \mathcal{F}$ form a neighbourhood basis of σ, and each of them meets H. Hence $\sigma \in \overline{H} = H$, and $\mathrm{Gal}(L/L^H) = H$. With Step 3, the maps $H \mapsto L^H$ and $M \mapsto \mathrm{Gal}(L/M)$ are mutually inverse bijections between closed subgroups and intermediate fields, and both reverse inclusions.

Step 5: Open subgroups and finite extensions. Let M be an intermediate field and $H = \mathrm{Gal}(L/M)$. Suppose M/K is finite, and choose $F \in \mathcal{F}$ containing M as in Step 2. Then H contains the open subgroup $\mathrm{Gal}(L/F)$, so H is open. Restriction $r_F \colon \mathrm{Gal}(L/K) \to \mathrm{Gal}(F/K)$ is surjective by (3), and $r_F^{-1}(\mathrm{Gal}(F/M)) = H$, so

$$[\mathrm{Gal}(L/K) : H] = [\mathrm{Gal}(F/K) : \mathrm{Gal}(F/M)] = [M : K]$$

by the index computation of Step 2. Conversely, if H is open, then by (1) it contains $\mathrm{Gal}(L/F)$ for some $F \in \mathcal{F}$, so $M = L^H \subseteq L^{\mathrm{Gal}(L/F)} = F$ by Step 3, and M/K is finite.

Step 6: Normal subgroups and Galois extensions. For $\sigma \in \mathrm{Gal}(L/K)$ and an intermediate field M, $\sigma\mathrm{Gal}(L/M)\sigma^{-1} = \mathrm{Gal}(L/\sigma(M))$, since τ fixes $\sigma(M)$ pointwise if and only if $\sigma^{-1}\tau\sigma$ fixes M pointwise. Suppose $H = \mathrm{Gal}(L/M)$ is open and normal. Then M/K is finite by Step 5, and separable because L/K is. Let $a \in M$, let f be its minimal polynomial over K, and let $b \in L$ be a root of f. The K-embedding $K(a) \to L$ with $a \mapsto b$ is the restriction of some $\sigma \in \mathrm{Gal}(L/K)$ by (3), and then $\mathrm{Gal}(L/\sigma(M)) = \sigma H \sigma^{-1} = H = \mathrm{Gal}(L/M)$, so $\sigma(M) = M$ by Step 4 and $b = \sigma(a) \in M$. The polynomial f splits in L, so it splits in M, and M/K is normal. Hence $M \in \mathcal{F}$. Conversely, if $M \in \mathcal{F}$, item (5) of *Core Algebra* [9, Fundamental Theorem of Galois Theory I] gives $\sigma(M) = M$ for every $\sigma \in \mathrm{Gal}(L/K)$, so H is normal, and it is open by Step 5. This proves (2).

The infinite Galois theorem says that the Galois theory of L/K is *entirely* encoded in the profinite group $\mathrm{Gal}(L/K)$, but only through its *closed* subgroups. Non-closed subgroups exist and have no field-theoretic meaning. This distinction between closed and non-closed is absent in finite Galois theory (where all subgroups are closed) and is the fundamental reason why the profinite topology must be taken seriously.

Definition 1.4.2: Absolute Galois Group

Let K be a field. The *absolute Galois group* of K is the profinite group:

$$G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$$

where K^{sep} is a fixed separable closure of K. The field K is *separably closed* if it has no proper separable algebraic extension, that is, if $K^{\mathrm{sep}} = K$ (every separable algebraic extension of K embeds over K into an algebraic closure, by *Core Algebra* [9, Embedding Into Algebraic Extension], and its image consists of separable elements, so lies in K^{sep}).

The group G_K depends (up to inner automorphism) on the choice of separable closure, but all structural invariants of G_K (its closed subgroups, its quotients, and, as developed in chapter 2, its cohomology) are independent of this choice.

A *finite separable K-algebra* is a K-algebra isomorphic to a finite product $L_1 \times \cdots \times L_r$, with $r \geq 0$, of finite separable field extensions of K (the empty product is the zero algebra). An action of G_K on

a finite set is *continuous* if every stabilizer is open.

Theorem 1.4.3: Finite Separable Algebras and Finite Galois Sets

Let K be a field. For a finite separable K-algebra A, the set $\mathrm{F}(A) = \mathrm{Hom}_{K\text{-alg}}(A, K^{\mathrm{sep}})$ of K-algebra homomorphisms is finite, and G_K acts on it continuously by $\gamma \cdot j = \gamma \circ j$. The functor $A \mapsto \mathrm{F}(A)$, sending $f\colon A \to B$ to $\mathrm{F}(f)\colon j \mapsto j \circ f$, is an anti-equivalence from the category of finite separable K-algebras to the category of finite sets with a continuous G_K-action. A finite separable field extension L of K corresponds to a transitive G_K-set, isomorphic to $G_K/\mathrm{Gal}(K^{\mathrm{sep}}/\iota(L))$ for every K-embedding $\iota\colon L \to K^{\mathrm{sep}}$.

***Proof*:**

Step 1: The functor. Let $A = L_1 \times \cdots \times L_r$, with projections $p_i\colon A \to L_i$ and idempotents e_i, the unit of L_i placed in the i-th factor. Let E be a field and $j\colon A \to E$ a K-algebra homomorphism. Each $j(e_i)$ is an idempotent of E, so $j(e_i) \in \{0, 1\}$, and $\sum_i e_i = 1$ and $e_i e_{i'} = 0$ for $i \neq i'$ force exactly one $j(e_i)$ to be 1. For that i and every $x \in A$, $j(x) = \sum_{i'} j(x)j(e_{i'}) = j(xe_i)$, so j factors through p_i. Hence

$$\mathrm{Hom}_{K\text{-alg}}(A, E) = \bigsqcup_{i=1}^{r} \{g \circ p_i \mid g \in \mathrm{Hom}_K(L_i, E)\}$$

For $E = K^{\mathrm{sep}}$ this decomposes $\mathrm{F}(A)$ into the G_K-stable pieces $\mathrm{F}(L_i) \circ p_i$.

Let L be a finite separable field extension of K. A K-embedding of L into an algebraic closure of K containing K^{sep} sends each $a \in L$ to a root of the separable minimal polynomial of a, hence into K^{sep}. So $\mathrm{F}(L)$ has $[L : K]_s = [L : K]$ elements (*Core Algebra* [9, Finite Separable Extension Equivalence]), and in particular it is finite and nonempty. The stabilizer of $j \in \mathrm{F}(L)$ is $\mathrm{Gal}(K^{\mathrm{sep}}/j(L))$, open by theorem 1.4.1(2), as $j(L)$ is finite over K. For $j, j' \in \mathrm{F}(L)$, the K-embedding $j' \circ j^{-1}\colon j(L) \to K^{\mathrm{sep}}$ is the restriction of some $\gamma \in G_K$ by theorem 1.4.1(3), and then $\gamma \circ j = j'$. So $\mathrm{F}(L)$ is transitive, and $\gamma \mapsto \gamma \circ \iota$ identifies $G_K/\mathrm{Gal}(K^{\mathrm{sep}}/\iota(L))$ with $\mathrm{F}(L)$ for every $\iota \in \mathrm{F}(L)$. By the previous paragraph, $\mathrm{F}(A)$ is finite and the action on it is continuous. Each $\mathrm{F}(f)$ is G_K-equivariant, since $\gamma \circ (j \circ f) = (\gamma \circ j) \circ f$, and $\mathrm{F}(g \circ f) = \mathrm{F}(f) \circ \mathrm{F}(g)$.

Step 2: Essential surjectivity. Let X be a finite set with a continuous G_K-action, with orbits $O_1, \ldots, O_s$. Choose $x_t \in O_t$ and let U_t be its stabilizer, an open subgroup, so that $O_t \cong G_K/U_t$ as G_K-sets. By theorem 1.4.1(2) the fixed field $M_t = (K^{\mathrm{sep}})^{U_t}$ is finite over K with $\mathrm{Gal}(K^{\mathrm{sep}}/M_t) = U_t$, and it is separable over K as a subfield of K^{sep}. Step 1, applied to the inclusion $M_t \to K^{\mathrm{sep}}$, gives $\mathrm{F}(M_t) \cong G_K/U_t \cong O_t$, so the finite separable K-algebra $M_1 \times \cdots \times M_s$ satisfies $\mathrm{F}(M_1 \times \cdots \times M_s) \cong \bigsqcup_t \mathrm{F}(M_t) \cong X$.

Step 3: Full faithfulness. Let $A = L_1 \times \cdots \times L_r$ and $B = L'_1 \times \cdots \times L'_s$, with projections p_i and q_t. A K-algebra homomorphism $f\colon A \to B$ is the same as the family of its components $q_t \circ f\colon A \to L'_t$, and by the decomposition of Step 1 with $E = L'_t$ each component is $g_t \circ p_{i(t)}$ for a unique index $i(t)$ and a unique K-embedding $g_t\colon L_{i(t)} \to L'_t$. On the other side, $\mathrm{F}(B)$ is the disjoint union of the transitive pieces $\mathrm{F}(L'_t) \circ q_t$, and the image of a transitive G_K-set under a G_K-map lies in one orbit, hence in one piece $\mathrm{F}(L_i) \circ p_i$. So a G_K-map $\mathrm{F}(B) \to \mathrm{F}(A)$ is a family, indexed by t, of an index $i(t)$ and a G_K-map $\mathrm{F}(L'_t) \to \mathrm{F}(L_{i(t)})$. For f as above, $\mathrm{F}(f)$ sends

$j \circ q_t$ to $j \circ g_t \circ p_{i(t)}$, so it corresponds to the family $(i(t), \mathrm{F}(g_t))_t$. It therefore suffices to show that for finite separable field extensions L and L' of K the map $g \mapsto \mathrm{F}(g)$ from $\mathrm{Hom}_K(L, L')$ to $\mathrm{Hom}_{G_K}(\mathrm{F}(L'), \mathrm{F}(L))$ is bijective.

Fix $\iota' \in \mathrm{F}(L')$, with stabilizer $U' = \mathrm{Gal}(K^{\mathrm{sep}}/\iota'(L'))$. Since $\mathrm{F}(L')$ is transitive, a G_K-map $\psi\colon \mathrm{F}(L') \to \mathrm{F}(L)$ is determined by $\psi(\iota')$, which is fixed by U', and every U'-fixed $j \in \mathrm{F}(L)$ is $\psi(\iota')$ for the G_K-map $\gamma \circ \iota' \mapsto \gamma \circ j$, well defined because $\gamma \circ \iota' = \gamma' \circ \iota'$ forces $\gamma^{-1}\gamma' \in U'$. An embedding $j \in \mathrm{F}(L)$ is fixed by U' exactly when $j(L)$ lies in the fixed field of U', which is $\iota'(L')$ by theorem 1.4.1(2). These embeddings are exactly $\iota' \circ g$ for $g \in \mathrm{Hom}_K(L, L')$, with $g = (\iota')^{-1} \circ j$ determined by j. Since $\mathrm{F}(g)(\iota') = \iota' \circ g$, the map $g \mapsto \mathrm{F}(g)$ is bijective.

The functor F is therefore full, faithful and essentially surjective as a functor from the opposite of the category of finite separable K-algebras, hence an equivalence by *Category Theory* [7, Full, Faithful, Essentially Surjective Implies Equivalence]. The statement about field extensions is Step 1.

The theorem is Grothendieck's form of Galois theory for a field: finite separable field extensions of K correspond to the transitive G_K-sets, and a finite product of them to the disjoint union of the corresponding sets. It is the point case of the étale fundamental group (*Étale Cohomology* [12]).

The cyclotomic character, which records the action of Galois on roots of unity, is the first and most fundamental Galois representation.

Definition 1.4.4: Cyclotomic Character

Let K be a field, and for $n \geq 1$ let $\mu_n \subseteq K^{\mathrm{sep}}$ denote the group of nth roots of unity. The action of $\sigma \in G_K$ on μ_n is determined by a unique element $\chi_n(\sigma) \in (\mathbb{Z}/n\mathbb{Z})^\times$ satisfying $\sigma(\zeta) = \zeta^{\chi_n(\sigma)}$ for all $\zeta \in \mu_n$. The maps χ_n assemble into a continuous homomorphism:

$$\chi\colon G_K \to \widehat{\mathbb{Z}}^\times = \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^\times$$

called the *cyclotomic character* of K.

The cyclotomic character is surjective when $K = \mathbb{Q}$ (since $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ for all n), but not in general: if K contains all roots of unity, then χ is trivial. The kernel of χ is the absolute Galois group of $K(\mu_\infty) = K(\zeta_n : n \geq 1)$, the field obtained by adjoining all roots of unity to K.

Example 1.4.5: Running Examples: Absolute Galois Groups

The following Galois groups serve as running examples throughout the book. Each illustrates different features of the profinite theory, and their cohomology will be computed explicitly as the machinery is developed.

1. The only algebraic extension of $\mathbb{R}$ is $\mathbb{C}$, so $G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$, generated by complex conjugation. This is the simplest nontrivial profinite group: finite, cyclic of order 2. The cyclotomic character $\chi\colon G_{\mathbb{R}} \to \widehat{\mathbb{Z}}^\times$ sends complex conjugation to -1 (since $\overline{\zeta} = \zeta^{-1}$ for any root of unity ζ on the unit circle). The cohomology of $G_{\mathbb{R}}$ is the periodic cohomology of $\mathbb{Z}/2\mathbb{Z}$, which is nontrivial in every even degree (a reflection of the fact that passage from $\mathbb{C}$ to $\mathbb{R}$ involves genuine arithmetic obstructions).

2. The absolute Galois group $G_{\mathbb{Q}_p}$ has a rich internal structure governed by the ramification theory of [14]. The maximal unramified extension $\mathbb{Q}_p^{\mathrm{nr}} = \mathbb{Q}_p(\zeta_{p^n-1} : n \geq 1)$ gives a short exact sequence:
$$1 \to I_p \to G_{\mathbb{Q}_p} \xrightarrow{\mathrm{res}} \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \to 1$$
where $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is the *inertia group* and $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \cong G_{\mathbb{F}_p} \cong \widehat{\mathbb{Z}}$ is topologically generated by the *Frobenius* automorphism $\mathrm{Frob}_p \colon x \mapsto x^p$. The inertia group itself decomposes via the ramification filtration:
$$1 \to P_p \to I_p \to I_p/P_p \to 1$$
where P_p is the *wild inertia* (a pro-p-group by exercise 1.3.1 (5)) and $I_p/P_p \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ is the *tame inertia*. The supernatural order is $\#G_{\mathbb{Q}_p} = \prod_\ell \ell^\infty$: every prime appears with infinite multiplicity, reflecting the existence of extensions of $\mathbb{Q}_p$ of every degree.

3. The absolute Galois group $G_{\mathbb{Q}}$ is the most complex and least understood of the running examples. No explicit presentation by generators and relations is known. Its structure is constrained by the local Galois groups: for each prime p, a choice of embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$ determines a continuous injection $G_{\mathbb{Q}_p} \hookrightarrow G_{\mathbb{Q}}$ (a *decomposition group* at p, well-defined up to conjugacy), and the real embedding gives an inclusion $G_{\mathbb{R}} \hookrightarrow G_{\mathbb{Q}}$ (the decomposition group at the archimedean place). The collection of all these local subgroups constrains $G_{\mathbb{Q}}$ but does not determine it: the additional structure is captured by the global duality theorems developed in chapter 10.

4. (If the reader has read [11]) An elliptic curve $E/\mathbb{Q}$ gives a continuous Galois representation $\rho_{E,p} \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_p)$ via the action on the Tate module $T_p(E) = \varprojlim_n E[p^n]$. The image $\rho_{E,p}(G_{\mathbb{Q}})$ is a closed subgroup of $\mathrm{GL}_2(\mathbb{Z}_p)$ (by corollary 1.2.5), hence profinite. By Serre's open image theorem, this image is open in $\mathrm{GL}_2(\mathbb{Z}_p)$ for all but finitely many p when E has no complex multiplication. The cohomological study of such representations is a central application of the theory developed in this book.

The following criterion detects when the absolute Galois group is a pro-p-group.

Proposition 1.4.6: Pro-p Absolute Galois Groups

The absolute Galois group G_K is a pro-p-group if and only if K has no separable extensions of degree coprime to p (other than K itself).

***Proof*:**
By proposition 1.3.2, G_K is pro-p if and only if every finite quotient is a p-group. The finite quotients of G_K are the groups $\mathrm{Gal}(F/K)$ for finite Galois extensions F/K, by theorem 1.4.1. If G_K is pro-p, every finite Galois extension has p-power degree, and by taking normal closures, every finite separable extension has p-power degree. Conversely, if every finite separable extension has p-power degree, every finite Galois extension has p-power degree (since the degree of a Galois extension equals the order of its Galois group), and G_K is pro-p.

Exercise 1.4.1

1. Show that G_K is trivial if and only if K is separably closed.
2. Compute $G_{\mathbb{F}_q}$ for a finite field $\mathbb{F}_q$ and show it is isomorphic to $\widehat{\mathbb{Z}}$, topologically generated by the Frobenius automorphism $\mathrm{Frob}_q \colon x \mapsto x^q$.
3. Show that $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p^\times$, where $\mathbb{Q}(\zeta_{p^\infty}) = \bigcup_{n\geq 1} \mathbb{Q}(\zeta_{p^n})$. Decompose this as $\mathbb{Z}_p^\times \cong \mu_{p-1} \times \mathbb{Z}_p$ for odd p (where μ_{p-1} is the group of $(p-1)$th roots of unity in $\mathbb{Z}_p^\times$) and describe the corresponding decomposition of $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^\infty})/\mathbb{Q})$ into Galois groups of explicit sub-extensions.
4. Compute the supernatural order $\#G_{\mathbb{Q}_p}$ and verify that it equals $\prod_\ell \ell^\infty$.
5. (If the reader has read [6]) The Kronecker-Weber theorem states that every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_n)$. Use this to show that $G_{\mathbb{Q}}^{\mathrm{ab}} \cong \widehat{\mathbb{Z}}^\times$, where $G_{\mathbb{Q}}^{\mathrm{ab}} = G_{\mathbb{Q}}/\overline{[G_{\mathbb{Q}}, G_{\mathbb{Q}}]}$ is the Galois group of the maximal abelian extension of $\mathbb{Q}$. Identify the isomorphism with the cyclotomic character χ.

2

Cohomology of Discrete Groups

Given a group G acting on an abelian group M, the fixed-point subgroup M^G is natural invariant of the action. The functor $M \mapsto M^G$ is left-exact but not exact: a surjection $M \twoheadrightarrow N$ of G-modules need not induce a surjection $M^G \twoheadrightarrow N^G$. The right-derived functors of $(-)^G$, denoted $H^n(G, M)$, measure this failure. These are the *group cohomology* groups, and they encode arithmetic information ranging from the structure of unit groups (Hilbert's Theorem 90) to the classification of central simple algebras (the Brauer group).

This chapter develops group cohomology for discrete groups, first through derived functors (connecting to the general theory in [13]), then through explicit cocycle descriptions that make the low-degree groups computable and interpretable. The profinite refinement (where continuity of cochains becomes essential) is developed in chapter 3.

2.1 G-Modules and the Invariant Functor

To define cohomology of a group G, a category of coefficients is needed. For a topological space, cohomology takes coefficients in abelian groups (or sheaves). For a group, the natural coefficients are abelian groups equipped with a compatible G-action.

Definition 2.1.1: G-Module

Let G be a group. A *G-module* is an abelian group M equipped with a left G-action by group automorphisms: a map $G \times M \to M$, written $(g, m) \mapsto g \cdot m$, satisfying $g \cdot (m + m') = g \cdot m + g \cdot m'$, $1 \cdot m = m$, and $(gh) \cdot m = g \cdot (h \cdot m)$ for all $g, h \in G$ and $m, m' \in M$. Equivalently, a G-module is a left module over the group ring $\mathbb{Z}[G]$.

The equivalence between G-actions on abelian groups and $\mathbb{Z}[G]$-module structures is immediate: the

group ring $\mathbb{Z}[G] = \bigoplus_{g \in G} \mathbb{Z} \cdot g$ acts by $(\sum a_g g) \cdot m = \sum a_g (g \cdot m)$, and conversely any $\mathbb{Z}[G]$-module structure restricts to a G-action via the embedding $G \hookrightarrow \mathbb{Z}[G]^\times$. This perspective identifies the category of G-modules with the module category $\mathbb{Z}[G]$-Mod. Every k-linear representation of G (a $k[G]$-module as studied in [10]) is a G-module with the action restricted to the $\mathbb{Z}[G]$-structure, but the G-module framework is more general: it accommodates non-semisimple and torsion phenomena that are invisible in the representation-theoretic setting.

A morphism of G-modules is a group homomorphism $\varphi \colon M \to N$ that commutes with the G-action: $\varphi(g \cdot m) = g \cdot \varphi(m)$ for all $g \in G$ and $m \in M$. Equivalently, φ is a $\mathbb{Z}[G]$-module homomorphism. The set of all such morphisms is denoted $\mathrm{Hom}_G(M, N)$.

Proposition 2.1.2: The Category of G-Modules

The category G-Mod of G-modules is an abelian category with enough injectives and enough projectives.

***Proof*:**
The category G-Mod is isomorphic to $\mathbb{Z}[G]$-Mod, and the category of left modules over any ring, commutative or not, is abelian (*Homological Algebra* [13, Module and Sheaf Categories]). Every G-module M is a quotient of the free $\mathbb{Z}[G]$-module on the set M, and free modules are projective (*Core Algebra* [9, Equivalencies for Projective Modules]), so there are enough projectives.

For injectives, let M be a G-module. Its underlying abelian group embeds, by some $\iota \colon M \to D$, in an injective abelian group D (*Core Algebra* [9, $\mathbb{Z}$-module Submodules of Injective Modules]). Let $\mathrm{CoInd}^G(D) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], D)$ be the coinduced module of definition 2.2.4, with G acting by $(g \cdot f)(x) = f(xg)$. The map $u \colon M \to \mathrm{CoInd}^G(D)$, $u(m)(x) = \iota(x \cdot m)$ for $x \in \mathbb{Z}[G]$, is G-equivariant, since $u(g \cdot m)(x) = \iota(xg \cdot m) = u(m)(xg) = (g \cdot u(m))(x)$, and injective, since $u(m)(1) = \iota(m)$. The module $\mathrm{CoInd}^G(D)$ is injective. By proposition 2.2.5, whose proof does not use the present proposition, there is an isomorphism $\mathrm{Hom}_G(-, \mathrm{CoInd}^G(D)) \cong \mathrm{Hom}_{\mathbb{Z}}(-, D)$ natural in the G-module, and $\mathrm{Hom}_{\mathbb{Z}}(-, D)$ carries injective maps of G-modules to surjections, because D is injective. So every G-map from a submodule of a G-module N to $\mathrm{CoInd}^G(D)$ extends to N, and G-Mod has enough injectives.

The existence of enough injectives and projectives guarantees that derived functors can be computed, which is the foundation for defining group cohomology in section 2.2.

Example 2.1.3: G-Modules

1. For any abelian group A, the *trivial G-module* structure on A is defined by $g \cdot a = a$ for all $g \in G$ and $a \in A$. The most common trivial modules are $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$.
2. The group ring $\mathbb{Z}[G]$ itself is a G-module via left multiplication: $g \cdot (\sum a_h h) = \sum a_h (gh)$. This is the *free G-module of rank* 1: for any G-module M, the map $\mathrm{Hom}_G(\mathbb{Z}[G], M) \xrightarrow{\sim} M$ given by $\varphi \mapsto \varphi(1)$ is an isomorphism.
3. Let L/K be a Galois extension with $G = \mathrm{Gal}(L/K)$. The additive group L^+, the multiplicative group $L^\times$, and the group of units $\mathcal{O}_L^\times$ of the ring of integers are all G-modules via the natural Galois action. The fixed-point subgroups recover K-arithmetic: $(L^\times)^G = K^\times$ and $(\mathcal{O}_L^\times)^G = \mathcal{O}_K^\times$.

4. For a subgroup $H \leq G$, the *permutation module* $\mathbb{Z}[G/H]$ is the free abelian group on the left cosets G/H, with G acting by left multiplication on cosets. When $H = \{1\}$, this recovers $\mathbb{Z}[G]$.

5. (If the reader has read [11]) For an elliptic curve E over a field K, the group $E(K^{\text{sep}})$ of K^{sep}-rational points is a G_K-module, and $E(K^{\text{sep}})^{G_K} = E(K)$. The m-torsion subgroup $E[m] \subseteq E(K^{\text{sep}})$ is a finite G_K-submodule isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$ as an abelian group (for $\text{char}K \nmid m$), with a nontrivial G_K-action that encodes the arithmetic of E over K.

A fundamental piece of structure on $\mathbb{Z}[G]$ is the augmentation map, whose kernel plays a central role throughout the cohomological theory.

Definition 2.1.4: Augmentation Map and Augmentation Ideal

The *augmentation map* is the ring homomorphism $\epsilon\colon \mathbb{Z}[G] \to \mathbb{Z}$ defined by:

$$\epsilon\left(\sum_{g\in G} a_g\, g\right) = \sum_{g\in G} a_g$$

It is a G-module homomorphism when $\mathbb{Z}$ carries the trivial action. The *augmentation ideal* $I_G = \ker\epsilon$ is the kernel of this map, giving a short exact sequence of G-modules:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

As a $\mathbb{Z}$-module, I_G is free with basis $\{g - 1 : g \in G \setminus \{1\}\}$.

The augmentation ideal consists of formal $\mathbb{Z}$-linear combinations of group elements whose coefficients sum to zero. The generators $g - 1$ measure the deviation of each group element from the identity: on a G-module M, the element $g - 1$ acts as zero exactly when g acts trivially on M. The exact sequence $0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$ expresses $\mathbb{Z}$ (with trivial action) as the quotient of the free module $\mathbb{Z}[G]$ by the "non-trivial part" of the group action, and will appear repeatedly: in the standard resolution (section 2.3), in the computation of H_1 (section 2.5), and in many explicit cohomological calculations.

The two fundamental functors on G-modules extract, respectively, the elements fixed by the action and the elements modulo the action.

Definition 2.1.5: G-Invariants

Let M be a G-module. The subgroup of *G-invariants* (or *G-fixed points*) is:

$$M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\}$$

This is the largest submodule of M on which G acts trivially.

Corollary 2.1.6: Hom Representation of Invariants

$M^G \cong \text{Hom}_G(\mathbb{Z}, M)$, naturally in M, where $\mathbb{Z}$ carries the trivial G-action.

***Proof*:**
A G-module homomorphism $\varphi\colon \mathbb{Z} \to M$ is determined by $\varphi(1) \in M$, and the G-equivariance condition $\varphi(g \cdot 1) = g \cdot \varphi(1)$ becomes $\varphi(1) = g \cdot \varphi(1)$ for all g, i.e., $\varphi(1) \in M^G$. For a G-map $\alpha\colon M \to M'$ the composite $\alpha \circ \varphi$ corresponds to $\alpha(\varphi(1))$, which is naturality.

This identification is the reason group cohomology can be expressed as Ext: since $H^n(G, M) = R^n((-)^G)(M)$ and $(-)^G = \mathrm{Hom}_G(\mathbb{Z}, -)$, the derived functors give $H^n(G, M) = \mathrm{Ext}^n_G(\mathbb{Z}, M)$.

Definition 2.1.7: G-Coinvariants

Let M be a G-module. The *G-coinvariants* of M is the quotient:

$$M_G = \frac{M}{\langle g \cdot m - m \big| g \in G,\, m \in M \rangle}$$

This is the largest quotient of M on which G acts trivially.

Proposition 2.1.8: Tensor Representation of Coinvariants

$M_G \cong \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$, naturally in M, where $\mathbb{Z}$ is a right $\mathbb{Z}[G]$-module via the augmentation.

***Proof*:**
The tensor product $\mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ satisfies $1 \otimes (g \cdot m) = (1 \cdot g) \otimes m = 1 \otimes m$, since g acts trivially on $\mathbb{Z}$, so the relation $g \cdot m \sim m$ is imposed. The universal property of the tensor product identifies $\mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ with M_G, through $1 \otimes m \mapsto \bar{m}$, and this map commutes with every G-map $M \to M'$, which is naturality.

The invariant and coinvariant functors form an adjoint pair with the trivial-action functor, and this adjunction determines their exactness properties.

Proposition 2.1.9: Exactness of Invariants and Coinvariants

The functor $(-)^G\colon G\text{-Mod} \to \mathbf{Ab}$ is left-exact, and the functor $(-)_G\colon G\text{-Mod} \to \mathbf{Ab}$ is right-exact.

***Proof*:**
Let $T\colon \mathbf{Ab} \to G\text{-Mod}$ denote the trivial-action functor, which sends an abelian group A to A with trivial G-action. This functor is exact (a sequence in $\mathbf{Ab}$ is exact if and only if it is exact as a sequence of G-modules with trivial action). For any G-module M and abelian group A:

$$\mathrm{Hom}_G(T(A), M) \cong \mathrm{Hom}_{\mathbb{Z}}(A, M^G) \qquad \text{and} \qquad \mathrm{Hom}_G(M, T(A)) \cong \mathrm{Hom}_{\mathbb{Z}}(M_G, A)$$

The first adjunction shows $(-)^G$ is right adjoint to T, hence left-exact. The second shows $(-)_G$ is left adjoint to T, hence right-exact.

Neither functor is exact in general. This is the fundamental fact that motivates group cohomology. The following example shows the failure.

Example 2.1.10: Failure of Exactness

Let $G = C_2 = \langle \sigma \rangle$ and consider the augmentation sequence:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

Here $I_G = \mathbb{Z} \cdot (\sigma - 1)$ (free of rank 1 as a $\mathbb{Z}$-module). Taking $(-)^G$: the module $\mathbb{Z}[G]^G = \mathbb{Z} \cdot (1 + \sigma)$ consists of elements $a + a\sigma$ for $a \in \mathbb{Z}$. The augmentation restricted to this subgroup sends $a(1 + \sigma) \mapsto 2a$, so the image is $2\mathbb{Z} \subsetneq \mathbb{Z} = \mathbb{Z}^G$. The functor $(-)^G$ fails to preserve surjectivity.

The same sequence demonstrates that $(-)_G$ is not left-exact. Applying $(-)_G$: the coinvariants $\mathbb{Z}[G]_G = \mathbb{Z}[G]/(\sigma-1)\mathbb{Z}[G] \cong \mathbb{Z}$ (via ϵ), while $(I_G)_G = I_G/(\sigma-1)I_G$. Since $(\sigma-1)^2 = \sigma^2 - 2\sigma + 1 = -2(\sigma - 1)$ in $\mathbb{Z}[G]$, the coinvariants $(I_G)_G \cong \mathbb{Z}/2\mathbb{Z}$, and the induced map $(I_G)_G \to \mathbb{Z}[G]_G$ sends the generator to 0. The injection $I_G \hookrightarrow \mathbb{Z}[G]$ does not remain injective after taking coinvariants.

Exercise 2.1.1

1. Verify the two adjunctions in the proof of proposition 2.1.9 by constructing explicit natural bijections.
2. Let G be a finite group. Show that $\mathbb{Z}[G]^G$ is the free $\mathbb{Z}$-module of rank 1 generated by the *norm element* $N_G = \sum_{g \in G} g$, and that $\mathbb{Z}[G]_G \cong \mathbb{Z}$ via the augmentation.
3. Let $G = C_3 = \langle \sigma \rangle$ act on $M = \mathbb{Z}^2$ by $\sigma \cdot (a, b) = (-b, a - b)$. Compute M^G and M_G.
4. Show that the functor $(-)^G \colon G\text{-Mod} \to \mathbf{Ab}$ is exact if and only if $G = \{1\}$.
5. Let G be a group. Show that the map $G^{\mathrm{ab}} = G/[G, G] \to I_G/I_G^2$ defined by $g[G, G] \mapsto (g - 1) + I_G^2$ is a well-defined isomorphism of abelian groups.

2.2 Group Cohomology via Derived Functors

The failure of $(-)^G$ to preserve surjections, made concrete in example 2.1.10, induces the right-derived functors of $(-)^G$ measuring the obstruction to exactness in each degree:

Definition 2.2.1: Group Cohomology and Homology

Let G be a group and M a G-module. The *nth cohomology group* of G with coefficients in M is:

$$H^n(G, M) = R^n((-)^G)(M) = \mathrm{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

where $\mathbb{Z}$ carries the trivial G-action. The *nth homology group* of G with coefficients in M is:

$$H_n(G, M) = L_n((-)_G)(M) = \mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

The two expressions for cohomology agree because $M^G \cong \mathrm{Hom}_G(\mathbb{Z}, M)$ naturally in M (corollary 2.1.6). If $M \to Q^\bullet$ is an injective resolution, then $R^n((-)^G)(M)$ is the cohomology of $(Q^\bullet)^G \cong \mathrm{Hom}_G(\mathbb{Z}, Q^\bullet)$, and this is $\mathrm{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$, which a projective resolution of $\mathbb{Z}$ computes equally well, by *Homological Algebra* [13, Computation Of Tor and Ext] over the ring $\mathbb{Z}[G]$. In the

same way $M_G \cong \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ naturally in M (proposition 2.1.8), so for a projective resolution $P_\bullet \to M$ the group $L_n((-)_G)(M)$, the homology of $(P_\bullet)_G \cong \mathbb{Z} \otimes_{\mathbb{Z}[G]} P_\bullet$, is $\mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$ by the same theorem.

The Ext description is often more practical: rather than resolving each coefficient module M individually, one resolves $\mathbb{Z}$ once and applies $\mathrm{Hom}_G(-, M)$ to the resolution. The standard resolution of section 2.3 provides a canonical choice.

Proposition 2.2.2: Long Exact Sequence in Cohomology

A short exact sequence of G-modules $0 \to A \to B \to C \to 0$ induces a long exact sequence in cohomology:

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta^0} H^1(G, A) \to H^1(G, B) \to H^1(G, C) \xrightarrow{\delta^1} H^2(G, A) \to \cdots$$

and a long exact sequence in homology:

$$\cdots \to H_1(G, C) \xrightarrow{\partial_1} H_0(G, A) \to H_0(G, B) \to H_0(G, C) \to 0$$

The connecting homomorphisms δ^n and ∂_n are natural in the short exact sequence.

***Proof*:**
The category G-Mod is abelian with enough injectives and enough projectives (proposition 2.1.2). For cohomology, *Homological Algebra* [13, Long Exact Sequence From a Right-Derived Functor], applied to the additive functor $(-)^G$, gives a long exact sequence of the groups $H^n(G, -) = R^n((-)^G)$ beginning with $0 \to R^0((-)^G)(A)$, with connecting homomorphisms natural in the short exact sequence. The functor $(-)^G$ is left exact (proposition 2.1.9), so $R^0((-)^G) \cong (-)^G$ naturally (*Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]), which identifies the first three terms with $A^G \to B^G \to C^G$. For homology, *Homological Algebra* [13, Long Exact Sequence From Derived Functor], applied to $(-)_G$, gives the second sequence, ending with $H_0(G, C) \to 0$, and its naturality. In both cases the connecting homomorphisms arise from the Snake Lemma applied to the cochain (resp. chain) complexes obtained from injective (resp. projective) resolutions.

The long exact sequence is the primary computational tool in group cohomology: given information about two of the three modules in a short exact sequence, the connecting homomorphisms transfer it to the third. The concrete description of δ^0 will be made explicit through cocycles in section 2.4.

Proposition 2.2.3: Degree Zero

For any group G and G-module M: $H^0(G, M) = M^G$ and $H_0(G, M) = M_G$.

***Proof*:**
$H^0(G, M) = R^0((-)^G)(M) = M^G$ since the zeroth derived functor of a left-exact functor is the functor itself. Similarly, $H_0(G, M) = L_0((-)_G)(M) = M_G$.

The higher cohomology groups $H^n(G, M)$ for $n \geq 1$ therefore represent the "higher-order obstructions" beyond the fixed-point functor: H^1 measures the failure of surjectivity (as will be made precise

in section 2.4), H^2 classifies extensions, and so on. A module for which these obstructions all vanish is called acyclic.

To produce acyclic modules, the following construction is fundamental.

Definition 2.2.4: Induced and Coinduced Modules

Let G be a group and M an abelian group (with no G-action). The *coinduced module* is:

$$\mathrm{CoInd}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$$

with the G-action $(g \cdot f)(x) = f(xg)$ for $g \in G$, $x \in \mathbb{Z}[G]$, and $f \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$. The *induced module* is:

$$\mathrm{Ind}^G(M) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} M$$

with the G-action $g \cdot (x \otimes m) = (gx) \otimes m$.

Sending $g \otimes m$ to the $\mathbb{Z}$-linear map $f_{g,m} \colon \mathbb{Z}[G] \to M$ determined by its values on the basis G, $f_{g,m}(x) = m$ if $x = g^{-1}$ and $f_{g,m}(x) = 0$ for the other $x \in G$, defines a G-equivariant map $\alpha \colon \mathrm{Ind}^G(M) \to \mathrm{CoInd}^G(M)$ (the equivariance computation of exercise 2.2.1 (4) does not use the finiteness of G). Restriction to the basis G identifies $\mathrm{CoInd}^G(M)$ with the group of all functions $G \to M$, and α with the inclusion of the finitely supported ones. When G is finite, every function is finitely supported, and α is an isomorphism of G-modules. When G is infinite and $M \neq 0$, the map α is not surjective, and the two modules are not isomorphic at all: $\mathrm{CoInd}^G(M)^G \cong M \neq 0$ by exercise 2.2.1 (3), while $\mathrm{Ind}^G(M)^G = 0$. As $h \cdot \sum_g g \otimes m_g = \sum_g g \otimes m_{h^{-1}g}$, the finite set of g with $m_g \neq 0$ in an invariant element is stable under left multiplication by G, and is therefore empty, a nonempty such set being all of the infinite group G.

The coinduced module satisfies an adjunction that makes it the "right" notion for cohomological vanishing.

Proposition 2.2.5: Adjunction for Coinduced Modules

For any G-module N and abelian group M:

$$\mathrm{Hom}_G(N, \mathrm{CoInd}^G(M)) \cong \mathrm{Hom}_{\mathbb{Z}}(N, M)$$

The isomorphism is natural in both N and M, and sends $\psi \in \mathrm{Hom}_G(N, \mathrm{CoInd}^G(M))$ to the map $n \mapsto \psi(n)(1)$.

***Proof*:**

Given $\psi \colon N \to \mathrm{CoInd}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$, define $\Phi(\psi) \colon N \to M$ by $\Phi(\psi)(n) = \psi(n)(1)$. This is a $\mathbb{Z}$-module homomorphism. Conversely, given $\varphi \colon N \to M$, define $\Psi(\varphi) \colon N \to \mathrm{CoInd}^G(M)$ by $\Psi(\varphi)(n)(x) = \varphi(x \cdot n)$ for $x \in \mathbb{Z}[G]$. The G-equivariance of $\Psi(\varphi)$ follows from: $(g \cdot \Psi(\varphi)(n))(x) = \Psi(\varphi)(n)(xg) = \varphi(xg \cdot n)$ and $\Psi(\varphi)(g \cdot n)(x) = \varphi(x \cdot (g \cdot n)) = \varphi((xg) \cdot n)$, which agree. The maps Φ and Ψ are inverse to each other: $\Phi(\Psi(\varphi))(n) = \Psi(\varphi)(n)(1) = \varphi(1 \cdot n) = \varphi(n)$ and $\Psi(\Phi(\psi))(n)(x) = \Phi(\psi)(x \cdot n) = \psi(x \cdot n)(1) = (x \cdot \psi(n))(1) = \psi(n)(x)$ (using G-equivariance of ψ for the third equality).

Theorem 2.2.6: Coinduced Modules Are Acyclic

For any abelian group M: $H^n(G, \mathrm{CoInd}^G(M)) = 0$ for all $n \geq 1$.

***Proof*:**

Let $P_\bullet \to \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ in G-Mod. The cohomology $H^n(G, \mathrm{CoInd}^G(M))$ is the nth cohomology of the complex $\mathrm{Hom}_G(P_\bullet, \mathrm{CoInd}^G(M))$. By proposition 2.2.5, each term satisfies $\mathrm{Hom}_G(P_i, \mathrm{CoInd}^G(M)) \cong \mathrm{Hom}_{\mathbb{Z}}(P_i, M)$, and this isomorphism is compatible with the differentials. The complex $\mathrm{Hom}_{\mathbb{Z}}(P_\bullet, M)$ computes $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}, M)$: each P_i is projective over $\mathbb{Z}[G]$, and since $\mathbb{Z}[G]$ is free as a $\mathbb{Z}$-module, each P_i is projective (in fact free) as a $\mathbb{Z}$-module. So $P_\bullet \to \mathbb{Z}$ is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}$. Since $\mathbb{Z}$ is itself free over $\mathbb{Z}$, the resolution splits, giving $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}, M) = 0$ for $n \geq 1$.

Coinduced modules are the group-theoretic analogue of flasque sheaves in sheaf cohomology: they are "large enough" to absorb all cohomological obstructions. The corresponding statement for induced modules and homology holds by a dual argument: $H_n(G, \mathrm{Ind}^G(M)) = 0$ for $n \geq 1$. Rather than an adjunction, this follows from the associativity of the tensor product: the complex $P_\bullet \otimes_{\mathbb{Z}[G]} (\mathbb{Z}[G] \otimes_{\mathbb{Z}} M)$ is isomorphic to $P_\bullet \otimes_{\mathbb{Z}} M$, which computes $\mathrm{Tor}^{\mathbb{Z}}_n(\mathbb{Z}, M) = 0$.

Example 2.2.7: Acyclicity in Practice

1. For the trivial group $G = \{1\}$, every G-module M satisfies $M^G = M$, and the functor $(-)^G$ is exact. Thus $H^n(\{1\}, M) = 0$ for all $n \geq 1$ and all M.
2. Let G be a finite group. Since $\mathrm{Ind}^G(\mathbb{Z}) = \mathbb{Z}[G] \cong \mathrm{CoInd}^G(\mathbb{Z})$ (the isomorphism holding because G is finite), theorem 2.2.6 gives $H^n(G, \mathbb{Z}[G]) = 0$ for all $n \geq 1$. The group ring is cohomologically trivial. More generally, $H^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) = 0$ for any abelian group A and $n \geq 1$, since $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A \cong \mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$.
3. For any group G (possibly infinite), $\mathrm{CoInd}^G(\mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], \mathbb{Q}/\mathbb{Z})$ is acyclic and injective as a $\mathbb{Z}[G]$-module (since $\mathbb{Q}/\mathbb{Z}$ is injective over $\mathbb{Z}$ and $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], -)$ preserves injectivity).

Exercise 2.2.1

1. Show that $\{H^n(G, -)\}_{n \geq 0}$ is a universal δ-functor (as defined in [13]), using the fact that every G-module embeds into a coinduced module (hence an acyclic module).
2. Let $0 \to A \to B \to C \to 0$ and $0 \to A' \to B' \to C' \to 0$ be short exact sequences of G-modules, and let (f, g, h) be a morphism between them. Show that the connecting homomorphisms satisfy $\delta^n \circ h^* = f^* \circ \delta^n$ for all $n \geq 0$, where $h^* \colon H^n(G, C) \to H^n(G, C')$ and $f^* \colon H^{n+1}(G, A) \to H^{n+1}(G, A')$ are the induced maps.
3. Verify directly (without using derived functors) that $\mathrm{CoInd}^G(M)^G \cong M$ for any abelian group M, and describe the isomorphism explicitly.
4. Let G be a finite group. For $g \in G$ and $m \in M$, let $\alpha(g \otimes m)$ be the $\mathbb{Z}$-linear map $\mathbb{Z}[G] \to M$

with $\alpha(g \otimes m)(x) = \begin{cases} m & x = g^{-1} \\ 0 & x \neq g^{-1} \end{cases}$ for $x \in G$. Show that α extends additively to a G-module isomorphism $\alpha\colon \mathrm{Ind}^G(M) \to \mathrm{CoInd}^G(M)$.

5. Show that cohomology commutes with arbitrary direct products: for any family $\{M_i\}_{i \in I}$ of G-modules, the natural map $H^n(G, \prod_i M_i) \to \prod_i H^n(G, M_i)$ is an isomorphism for all $n \geq 0$.

2.3 The Standard Resolution and Explicit Cocycles

The derived functor definition of $H^n(G, M)$ is conceptually clean but not directly computable: it requires choosing a resolution and taking cohomology. This section provides a canonical projective resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$, the standard (or bar) resolution, and translates the abstract cohomology into explicit formulas for cocycles and coboundaries. These formulas make the low-degree cohomology groups computable by hand and are the foundation for the interpretations developed in section 2.4.

Definition 2.3.1: Standard Resolution

Let G be a group. The *standard resolution* (or *bar resolution*) of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules is the complex:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_n} \mathbb{Z}[G^n] \to \cdots \xrightarrow{d_1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\mathbb{Z}[G^{n+1}]$ is the free abelian group on $(n+1)$-tuples $(g_0, \ldots, g_n)$ of elements of G, with G acting diagonally: $g \cdot (g_0, \ldots, g_n) = (gg_0, \ldots, gg_n)$. The differentials are the alternating face maps:

$$d_n(g_0, \ldots, g_n) = \sum_{i=0}^{n} (-1)^i (g_0, \ldots, \widehat{g_i}, \ldots, g_n)$$

where $\widehat{g_i}$ indicates omission, and ϵ is the augmentation from definition 2.1.4.

Each $\mathbb{Z}[G^{n+1}]$ is a free $\mathbb{Z}[G]$-module: writing $(g_0, \ldots, g_n) = g_0 \cdot (1, g_0^{-1}g_1, \ldots, g_0^{-1}g_n)$, the module $\mathbb{Z}[G^{n+1}]$ is free on the set $\{(1, h_1, \ldots, h_n) : h_i \in G\}$, which is a copy of G^n.

Lemma 2.3.2: Exactness of the Standard Resolution

The standard resolution is exact.

***Proof*:**

Define $\mathbb{Z}$-module maps $s_n\colon \mathbb{Z}[G^{n+1}] \to \mathbb{Z}[G^{n+2}]$ by $s_n(g_0, \ldots, g_n) = (1, g_0, \ldots, g_n)$, and $s_{-1}\colon \mathbb{Z} \to \mathbb{Z}[G]$ by $s_{-1}(1) = (1)$. A direct computation gives $d_{n+1} \circ s_n + s_{n-1} \circ d_n = \mathrm{id}$ for all $n \geq 1$, $d_1 \circ s_0 + s_{-1} \circ \epsilon = \mathrm{id}_{\mathbb{Z}[G]}$, and $\epsilon \circ s_{-1} = \mathrm{id}_{\mathbb{Z}}$. This exhibits $\{s_n\}$ as a contracting homotopy for the augmented complex, proving exactness.

The identities are verified on generators. For $n \geq 1$,

$$d_{n+1}s_n(g_0, \ldots, g_n) = (g_0, \ldots, g_n) + \sum_{j=0}^{n} (-1)^{j+1} (1, g_0, \ldots, \widehat{g_j}, \ldots, g_n)$$

the first term being the $i = 0$ face of $(1, g_0, \dots, g_n)$, and

$$s_{n-1}d_n(g_0, \dots, g_n) = \sum_{j=0}^{n}(-1)^j(1, g_0, \dots, \widehat{g_j}, \dots, g_n)$$

so the sums cancel, leaving $(g_0, \dots, g_n)$. For $n = 0$, $d_1 s_0(g_0) = d_1(1, g_0) = (g_0) - (1)$ and $s_{-1}\epsilon(g_0) = s_{-1}(1) = (1)$, which sum to (g_0). Finally $\epsilon s_{-1}(1) = \epsilon((1)) = 1$.

For $G \neq \{1\}$ the homotopy s_n is *not* G-equivariant: it prepends the identity element 1, which $g \neq 1$ moves to g. No equivariant contracting homotopy exists. The additive functor $\mathrm{Hom}_G(-, M)$ would carry one to a contracting homotopy of the complex obtained by applying $\mathrm{Hom}_G(-, M)$ to the augmented resolution, forcing $H^n(G, M) = 0$ for all $n \geq 1$ and all M, and then the long exact sequence (proposition 2.2.2) would make $(-)^G$ exact, which fails for $G \neq \{1\}$ (exercise 2.1.1 (4)).

Applying $\mathrm{Hom}_G(-, M)$ to the standard resolution produces a cochain complex of functions on the powers of G, isomorphic to the complex of inhomogeneous cochains below. Its cohomology is $H^*(G, M)$ by proposition 2.3.4.

Definition 2.3.3: Inhomogeneous Cochains

Let G be a group and M a G-module. An *inhomogeneous n-cochain* is a function $f\colon G^n \to M$ (with no equivariance condition). The set of all n-cochains is $C^n(G, M) = \mathrm{Map}(G^n, M)$, an abelian group under pointwise addition. The *coboundary map* $d^n\colon C^n(G, M) \to C^{n+1}(G, M)$ is:

$$\begin{aligned}(d^n f)(g_1, \dots, g_{n+1}) = {} & g_1 \cdot f(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n}(-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \dots, g_n)\end{aligned}$$

The *n-cocycles* are $Z^n(G, M) = \ker d^n$ and the *n-coboundaries* are $B^n(G, M) = \mathrm{im} d^{n-1}$ (with $B^0 = 0$).

The inhomogeneous cochains arise from the isomorphism $\mathrm{Hom}_G(\mathbb{Z}[G^{n+1}], M) \cong \mathrm{Map}(G^n, M)$ that sends a G-equivariant map $\varphi\colon \mathbb{Z}[G^{n+1}] \to M$ to the function $\bar{\varphi}(g_1, \dots, g_n) = \varphi(1, g_1, g_1 g_2, \dots, g_1 \cdots g_n)$. The coboundary d^n is the transport of the dual differential $d^*_{n+1} = \mathrm{Hom}_G(d_{n+1}, M)$ through these isomorphisms: the face omitting g_0 gives the term $g_1 \cdot f(g_2, \dots, g_{n+1})$, the faces omitting the i-th coordinate for $1 \leq i \leq n$ merge g_i and g_{i+1}, and the last face gives $f(g_1, \dots, g_n)$, each with the sign $(-1)^i$. In particular $d^{n+1} \circ d^n = 0$, so $C^\bullet(G, M)$ is a cochain complex, isomorphic to $\mathrm{Hom}_G(\mathbb{Z}[G^{\bullet+1}], M)$.

Proposition 2.3.4: Cocycle Description of Cohomology

Let G be a group, and write $H^n(C^\bullet(G,M)) = Z^n(G,M)/B^n(G,M)$ for the cohomology of the complex of inhomogeneous cochains of definition 2.3.3.

1. For every short exact sequence $0 \to A \to B \to C \to 0$ of G-modules and every $n \geq 0$, the sequence
$$0 \to C^n(G,A) \to C^n(G,B) \to C^n(G,C) \to 0$$
given by composition with the maps of the short exact sequence is exact. With the connecting homomorphisms of the resulting long exact sequences of cohomology, the family $H^\bullet(C^\bullet(G,-)) = (H^n(C^\bullet(G,-)))_{n\geq 0}$ is a universal δ-functor from G-Mod to **Ab** in the sense of *Homological Algebra* [13, Universal δ-Functor].

2. There is exactly one isomorphism of δ-functors $H^\bullet(C^\bullet(G,-)) \xrightarrow{\sim} H^\bullet(G,-)$ whose degree-zero component is the natural isomorphism $H^0(C^\bullet(G,M)) = M^G \cong H^0(G,M)$. In particular, for every G-module M and every $n \geq 0$,
$$H^n(G,M) \cong \frac{Z^n(G,M)}{B^n(G,M)}$$
naturally in M, and these isomorphisms carry the connecting homomorphisms of the cochain long exact sequences of (1) to those of proposition 2.2.2.

***Proof*:**
Step 1: Exactness on cochains. Write $\alpha\colon A \to B$ and $\beta\colon B \to C$ for the maps of the sequence. Composition with α is injective because α is, and its image lies in the kernel of composition with β because $\beta \circ \alpha = 0$. If $f\colon G^n \to B$ satisfies $\beta \circ f = 0$, then every value of f lies in $\ker\beta = \alpha(A)$, so $f = \alpha \circ f'$ for the map $f'\colon G^n \to A$ sending each point to the unique preimage of its value. For surjectivity, choose for each $c \in C$ an element $\sigma(c) \in B$ with $\beta(\sigma(c)) = c$. For any $f\colon G^n \to C$ the cochain $\sigma \circ f$ satisfies $\beta \circ (\sigma \circ f) = f$.

Step 2: A δ-functor. Let $\varphi\colon M \to M'$ be a map of G-modules. Since φ is additive and $\varphi(g_1 \cdot m) = g_1 \cdot \varphi(m)$, the coboundary formula of definition 2.3.3 gives $d(\varphi \circ f) = \varphi \circ df$. So composition with φ is a map of cochain complexes $C^\bullet(G,M) \to C^\bullet(G,M')$, and since $(\varphi+\psi)\circ f = \varphi\circ f + \psi \circ f$ for a second map $\psi\colon M \to M'$, each $H^n(C^\bullet(G,-))$ is an additive functor G-Mod $\to$ **Ab**. By Step 1 a short exact sequence of G-modules induces a short exact sequence of cochain complexes, and the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) turns it into a long exact sequence
$$0 \to H^0(C^\bullet(G,A)) \to H^0(C^\bullet(G,B)) \to H^0(C^\bullet(G,C)) \xrightarrow{\delta} H^1(C^\bullet(G,A)) \to \cdots$$
whose unlabelled maps are induced by α and β. A morphism of short exact sequences of G-modules induces, by composition, a morphism of the corresponding short exact sequences of cochain complexes, and the naturality clause of the Snake Lemma makes the connecting homomorphisms commute with the induced maps. These are the two conditions of *Homological Algebra* [13, δ-Functor], so $H^\bullet(C^\bullet(G,-))$ is a δ-functor. In degree zero, $C^0(G,M) = M$ and $(d^0 m)(g) = g \cdot m - m$, so $H^0(C^\bullet(G,M)) = M^G$.

Step 3: Coinduced modules are acyclic for cochains. Let J be an abelian group. A $\mathbb{Z}$-linear map $\mathbb{Z}[G] \to J$ is determined by its values on the basis G of $\mathbb{Z}[G]$, and these values can be prescribed arbitrarily. Restriction to G therefore identifies the coinduced module $\mathrm{CoInd}^G(J)$ of definition 2.2.4 with the group $\mathrm{Map}(G, J)$ of all maps $G \to J$, on which G acts by $(g \cdot \varphi)(x) = \varphi(xg)$ for $g, x \in G$. Put $I = \mathrm{Map}(G, J)$, and for $n \geq 1$ define $s\colon C^n(G, I) \to C^{n-1}(G, I)$ by

$$(sf)(g_1, \dots, g_{n-1})(x) = f(x, g_1, \dots, g_{n-1})(1)$$

which for $n = 1$ is the element $x \mapsto f(x)(1)$ of $C^0(G, I) = I$. The map s satisfies $d(sf)+s(df) = f$ for every $f \in C^n(G, I)$ with $n \geq 1$. Since $(g \cdot \varphi)(x) = \varphi(xg)$ for $\varphi \in I$, evaluating the coboundary of f at $(x, g_1, \dots, g_n)$ and then at 1 gives

$$\begin{aligned}(s\,df)(g_1, \dots, g_n)(x) = {} & f(g_1, \dots, g_n)(x) - f(xg_1, g_2, \dots, g_n)(1) \\ & + \sum_{i=2}^{n} (-1)^i f(x, g_1, \dots, g_{i-1}g_i, \dots, g_n)(1) + (-1)^{n+1} f(x, g_1, \dots, g_{n-1})(1)\end{aligned}$$

the first term being $(x \cdot f(g_1, \dots, g_n))(1)$. Evaluating the coboundary of sf at $(g_1, \dots, g_n)$ and then at x gives

$$\begin{aligned}(d\,sf)(g_1, \dots, g_n)(x) = {} & f(xg_1, g_2, \dots, g_n)(1) \\ & + \sum_{i=1}^{n-1} (-1)^i f(x, g_1, \dots, g_ig_{i+1}, \dots, g_n)(1) + (-1)^n f(x, g_1, \dots, g_{n-1})(1)\end{aligned}$$

the first term being $(g_1 \cdot (sf)(g_2, \dots, g_n))(x)$. For $n = 1$ the sums are empty, $f(xg_1, g_2, \dots, g_n)$ stands for $f(xg_1)$, and $f(x, g_1, \dots, g_{n-1})$ stands for $f(x)$. In the sum of the two expressions the terms $\pm f(xg_1, g_2, \dots, g_n)(1)$ cancel, the term of index i in the second sum cancels the term of index $i + 1$ in the first, and the two last terms cancel, leaving $f(g_1, \dots, g_n)(x)$. So if f is a cocycle then $f = d(sf)$ is a coboundary, and $H^n(C^\bullet(G, I)) = 0$ for every $n \geq 1$.

Step 4: Universality. Every G-module M embeds in such a module. Define $u_M\colon M \to \mathrm{Map}(G, M)$, whose target uses only the abelian group underlying M, by $u_M(m)(x) = x \cdot m$. The map u_M is G-equivariant, since $u_M(g \cdot m)(x) = xg \cdot m = u_M(m)(xg) = (g \cdot u_M(m))(x)$, and injective, since evaluation at $x = 1$ recovers m. For $n \geq 1$ it induces a map $H^n(C^\bullet(G, M)) \to H^n(C^\bullet(G, \mathrm{Map}(G, M)))$ whose target is 0 by Step 3, so u_M effaces $H^n(C^\bullet(G, -))$ at M in the sense of *Homological Algebra* [13, Effaceable Functor]. The δ-functor of Step 2 is therefore effaceable in positive degrees, and it is universal by *Homological Algebra* [13, An Effaceable δ-Functor Is Universal]. This proves (1).

Step 5: Comparison. The category G-Mod is abelian with enough injectives (proposition 2.1.2), and $(-)^G$ is additive and left exact (proposition 2.1.9). By *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor], the derived functors $H^n(G, -)$ of definition 2.2.1, with the connecting homomorphisms of proposition 2.2.2, form a universal δ-functor, and there is a natural isomorphism $M^G \cong H^0(G, M)$. Composed with $H^0(C^\bullet(G, M)) = M^G$ from Step 2, this is a natural isomorphism $\varphi\colon H^0(C^\bullet(G, -)) \Rightarrow H^0(G, -)$. Universality of $H^\bullet(C^\bullet(G, -))$ extends φ to exactly one morphism of δ-functors $H^\bullet(C^\bullet(G, -)) \to H^\bullet(G, -)$, and as both δ-functors are universal, each of its components is an isomorphism by *Homological Algebra* [13, A Universal δ-Functor Is Determined by Its Degree-Zero Part]. This proves (2).

The cocycle description makes the low-degree coboundary maps explicit. In degree 0: $C^0(G, M) = M$ (functions from $G^0 = \{*\}$ to M), and $(d^0 m)(g) = g \cdot m - m$. So $Z^0(G, M) = M^G$ and $B^0 = 0$, recovering $H^0(G, M) = M^G$. The content of the cocycle description begins in degree 1.

Definition 2.3.5: Crossed Homomorphism

A *crossed homomorphism* (or 1-*cocycle*) from G to M is a function $f\colon G \to M$ satisfying:

$$f(gh) = g \cdot f(h) + f(g)$$

for all $g, h \in G$. The group of all crossed homomorphisms is $Z^1(G, M)$.

When G acts trivially on M, the cocycle condition reduces to $f(gh) = f(g) + f(h)$: ordinary group homomorphisms $G \to M$.

Definition 2.3.6: Principal Crossed Homomorphism

A *principal crossed homomorphism* (or 1-*coboundary*) is a crossed homomorphism of the form $f_m(g) = g \cdot m - m$ for some fixed $m \in M$. The group of all principal crossed homomorphisms is $B^1(G, M)$.

A principal crossed homomorphism is the "trivial" part of a cocycle: it measures how far an element m is from being G-fixed, and contributes nothing to $H^1(G, M)$.

In degree 2, the cocycle condition encodes the associativity constraint for group extensions.

Definition 2.3.7: 2-Cocycle

A 2-*cocycle* from G to M is a function $f\colon G \times G \to M$ satisfying the *cocycle condition*:

$$g_1 \cdot f(g_2, g_3) - f(g_1 g_2, g_3) + f(g_1, g_2 g_3) - f(g_1, g_2) = 0$$

for all $g_1, g_2, g_3 \in G$. The group of 2-cocycles is $Z^2(G, M)$, and the 2-coboundaries $B^2(G, M)$ consist of functions of the form $(g_1, g_2) \mapsto g_1 \cdot \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$ for a 1-cochain $\psi\colon G \to M$.

The cocycle condition for $f \in Z^2(G, M)$ is the precise analogue of the associativity axiom for a group multiplication twisted by f. The interpretation of $H^2(G, M)$ as classifying group extensions with this structure is developed in section 2.4.

The standard resolution is universal but not always efficient. For specific groups, smaller resolutions yield faster computations. The most important case is cyclic groups, where the relation $\sigma^n = 1$ produces a periodic resolution.

Example 2.3.8: Cohomology of Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n. The elements $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ (the norm) satisfy $N(\sigma - 1) = \sigma^n - 1 = 0$ and $(\sigma - 1)N = 0$ in $\mathbb{Z}[G]$. The sequence below is exact. The kernel of ϵ is $(\sigma - 1)\mathbb{Z}[G]$, since $\sum a_i \sigma^i$ with $\sum a_i = 0$ equals $\sum a_i(\sigma^i - 1)$ and each $\sigma^i - 1$ is a multiple of $\sigma - 1$. The kernel of $\sigma - 1$ is $\mathbb{Z}N$, since $\sigma f = f$ forces all coefficients of f to be equal. The kernel of N is $(\sigma - 1)\mathbb{Z}[G]$: identifying $\mathbb{Z}[G]$ with $\mathbb{Z}[x]/(x^n - 1)$, $Nf = f(1)N$, so $Nf = 0$ means $f(1) = 0$, that

is, f lies in the kernel of ϵ. So it is a periodic free resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

Applying $\mathrm{Hom}_G(-, M)$ and using $\mathrm{Hom}_G(\mathbb{Z}[G], M) \cong M$ gives the complex:

$$0 \to M \xrightarrow{\sigma-1} M \xrightarrow{N} M \xrightarrow{\sigma-1} M \xrightarrow{N} \cdots$$

where $(\sigma - 1)(m) = \sigma \cdot m - m$ and $N(m) = \sum_{i=0}^{n-1} \sigma^i \cdot m$. The cohomology is 2-periodic for $k \geq 1$:

$$H^k(C_n, M) = \begin{cases} M^G/NM & k \text{ even}, k \geq 2 \\ {}_NM/(\sigma - 1)M & k \text{ odd}, k \geq 1 \end{cases}$$

where ${}_NM = \ker(N\colon M \to M)$ is the *norm kernel*.

As a concrete application, take $M = \mathbb{Z}$ with trivial action. Then $\sigma - 1$ acts as 0 and N acts as multiplication by n. The cohomology is:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/n\mathbb{Z} & k \text{ even}, k \geq 2 \\ 0 & k \text{ odd}, k \geq 1 \end{cases}$$

This periodicity is special to cyclic groups and will be generalized by Tate cohomology in chapter 4.

The periodic description has a companion that the colimit computations of chapter 3 depend on: what the inflation maps do as the cyclic group grows. Surjecting a larger cyclic group onto a smaller one and inflating turns out to be multiplication by a power of the index, and the power grows with the degree, which is what makes cohomology in high degrees die in the limit.

Proposition 2.3.9: Inflation Between Cyclic Groups

Let $C = \langle \sigma \rangle$ be cyclic of order N, let $C' = \langle \sigma' \rangle$ be cyclic of order dN, and let $\pi\colon C' \to C$ be the surjection with $\pi(\sigma') = \sigma$. Regard a C-module M as a C'-module through π, so that $M^{C'} = M^C$ and the norm element of C' acts on M as $d\,N_C$. The *inflation map*

$$\mathrm{Inf}\colon H^i(C, M) \longrightarrow H^i(C', M)$$

is the map induced, through proposition 2.3.4, by the map of cochain complexes $C^\bullet(C, M) \to C^\bullet(C', M)$, $f \mapsto f \circ (\pi \times \cdots \times \pi)$ (the finite case of definition 3.3.1). Identify $H^i(C, M)$ with the cohomology of the periodic complex of example 2.3.8 through a chain map from the periodic resolution to the standard resolution, as fixed in Step 1 of the proof, and likewise for C'. Under these identifications, Inf is induced by multiplication by d^j on M, where $j = \lfloor i/2 \rfloor$.

***Proof*:**

Step 1: Inflation on resolutions. The map $f \mapsto f \circ (\pi \times \cdots \times \pi)$ commutes with the coboundaries of definition 2.3.3, because π is a homomorphism and C' acts on M through π. Write $B_\bullet$ and $B'_\bullet$ for the standard resolutions of $\mathbb{Z}$ over $\mathbb{Z}[C]$ and $\mathbb{Z}[C']$ (definition 2.3.1), and $P_\bullet$ and $P'_\bullet$ for the periodic resolutions of example 2.3.8. Let $\pi_*\colon B'_\bullet \to B_\bullet$ send $(g_0, \ldots, g_n)$ to $(\pi(g_0), \ldots, \pi(g_n))$.

It commutes with the face maps and the augmentations, and it is C'-linear when C' acts on $B_\bullet$ through π, so it is a map of complexes of C'-modules over $\mathrm{id}_\mathbb{Z}$. Under the isomorphisms $\mathrm{Hom}_C(B_n, M) \cong C^n(C, M)$ and $\mathrm{Hom}_{C'}(B'_n, M) \cong C^n(C', M)$ that follow definition 2.3.3, precomposition with π_* is the cochain map of the statement: for a C-map $\varphi\colon B_n \to M$ the cochain attached to $\varphi \circ \pi_*$ takes at $(g_1, \dots, g_n)$ the value

$$(\varphi \circ \pi_*)(1, g_1, g_1g_2, \dots, g_1 \cdots g_n) = \varphi(1, \pi(g_1), \pi(g_1)\pi(g_2), \dots, \pi(g_1) \cdots \pi(g_n))$$

which is the value of the cochain attached to φ at $(\pi(g_1), \dots, \pi(g_n))$. By *Homological Algebra* [13, Projective Resolution As Initial Objects], applied in C-modules to the free complex $P_\bullet$ and the resolution $B_\bullet$ (exact by lemma 2.3.2), and in the other direction, there are C-chain maps $u\colon P_\bullet \to B_\bullet$ and $v\colon B_\bullet \to P_\bullet$ over $\mathrm{id}_\mathbb{Z}$, and $v \circ u$ and $u \circ v$ are homotopic to the identities. The groups of example 2.3.8 are identified with $H^n(C, M)$ through u, with inverse induced by v, and this is the identification used here. Any two C-chain maps $P_\bullet \to B_\bullet$ over $\mathrm{id}_\mathbb{Z}$ are homotopic by the same lemma, so it does not depend on the choice of u. The same holds for C' with a chain map $u'\colon P'_\bullet \to B'_\bullet$ over $\mathrm{id}_\mathbb{Z}$. In terms of the periodic complexes, Inf is therefore induced by $\mathrm{Hom}(v \circ \pi_* \circ u', M)\colon \mathrm{Hom}_C(P_\bullet, M) \to \mathrm{Hom}_{C'}(P'_\bullet, M)$.

The composite $v \circ \pi_* \circ u'\colon P'_\bullet \to P_\bullet$ is a map of complexes of C'-modules over $\mathrm{id}_\mathbb{Z}$, where C' acts on $P_\bullet$ through π. The complex $P_\bullet$ is exact, hence a resolution of $\mathbb{Z}$ in C'-modules, and $P'_\bullet$ consists of free $\mathbb{Z}[C']$-modules. The same lemma makes every map of complexes of C'-modules $f_\bullet\colon P'_\bullet \to P_\bullet$ over $\mathrm{id}_\mathbb{Z}$ homotopic to $v \circ \pi_* \circ u'$. Homotopic maps remain homotopic after the additive functor $\mathrm{Hom}_{C'}(-, M)$, contravariant and so covariant on the opposite category (*Homological Algebra* [13, Homotopy Equivalence And Additive Functors]) and then induce the same map on cohomology (*Homological Algebra* [13, Chain-Homotopy and Cohomology]). So Inf is induced by $\mathrm{Hom}(f_\bullet, M)$ for any such $f_\bullet$, and it remains to construct one.

Step 2: An explicit lift. Write $D = \sigma - 1$ and N_C for the two elements of $\mathbb{Z}[C]$ appearing in the periodic resolution of example 2.3.8, and $D' = \sigma' - 1$, $N_{C'}$ for those of $\mathbb{Z}[C']$. Note $\pi(D') = D$ and

$$\pi(N_{C'}) = \sum_{t=0}^{dN-1} \sigma^t = d\, N_C$$

since π hits each power of σ exactly d times. A C'-map $f_k\colon \mathbb{Z}[C'] \to \mathbb{Z}[C]$ is determined by $a_k := f_k(1)$, namely $f_k(x) = \pi(x)\, a_k$. The conditions for $(f_k)_k$ to be a map of complexes over $\mathrm{id}_\mathbb{Z}$ read off one square at a time. Degree 0 needs $\epsilon(a_0) = 1$, satisfied by $a_0 = 1$. The square $f_0 D' = D f_1$ needs $D\, a_0 = D\, a_1$, satisfied by $a_1 = 1$. The square $f_1 N_{C'} = N_C f_2$ needs $d\, N_C\, a_1 = N_C\, a_2$, satisfied by $a_2 = d$. Inductively, if $a_{2j} = a_{2j+1} = d^j$, the next two squares need $d\, N_C\, a_{2j+1} = N_C\, a_{2j+2}$ and $D\, a_{2j+2} = D\, a_{2j+3}$, satisfied by $a_{2j+2} = a_{2j+3} = d^{j+1}$. So

$$a_{2j} = a_{2j+1} = d^j$$

defines a map of complexes $f_\bullet$ over $\mathrm{id}_\mathbb{Z}$.

Applying $\mathrm{Hom}_{C'}(-, M)$ and using $\mathrm{Hom}_{C'}(\mathbb{Z}[C'], M) \cong M$ and $\mathrm{Hom}_C(\mathbb{Z}[C], M) \cong M$, both given by evaluation at 1, the map induced by f_k in degree k sends φ to $\varphi \circ f_k$, with $(\varphi \circ f_k)(1) = \varphi(a_k) = a_k\, \varphi(1)$, so it is multiplication by a_k on M. Passing to cohomology identifies Inf in degree i with the map induced by multiplication by $d^{\lfloor i/2 \rfloor}$, as we sought to show.

Example 2.3.10: Cohomology of Free Groups
Let $F = F_S$ be the free group on a set S. The augmentation ideal I_F is a free $\mathbb{Z}[F]$-module with basis $\{s-1 : s \in S\}$ (this can be verified using the universal property of free groups, or by the Fox calculus). The augmentation sequence $0 \to I_F \to \mathbb{Z}[F] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ is therefore a free resolution of length 1, giving:

$$H^n(F_S, M) = 0 \quad \text{for all } n \geq 2$$

and $H^1(F_S, M) \cong \mathrm{Hom}_G(I_F, M) \cong \prod_{s \in S} M$ (choosing the image of each $s-1$ independently). Free groups have *cohomological dimension* 1 (a concept developed systematically in chapter 5).

2.3.11 Remark: Connection to Singular Cohomology (If the reader has read [23]) The standard resolution has a topological origin. The classifying space BG of a discrete group G satisfies $\pi_1(BG) = G$ and $\pi_n(BG) = 0$ for $n \geq 2$. Its singular chain complex, computed via the bar construction on $EG \to BG$, recovers the standard resolution. The resulting isomorphism:

$$H^n(G, M) \cong H^n_{\mathrm{Sing}}(BG, M)$$

holds for any G-module M (with local coefficients when the action is nontrivial). This identifies group cohomology with the singular cohomology of a topological space and explains the simplicial character of the cocycle formulas.

Exercise 2.3.1

1. Verify the identity $d_{n+1} \circ s_n + s_{n-1} \circ d_n = \mathrm{id}$ from the proof of lemma 2.3.2 by direct computation for $n = 1$.
2. Write out the coboundary maps d^0, d^1, d^2, and d^3 in inhomogeneous form explicitly, and verify that $d^1 \circ d^0 = 0$ and $d^2 \circ d^1 = 0$.
3. Compute $H^n(C_2, \mathbb{Z}/2\mathbb{Z})$ for all $n \geq 0$, where $C_2 = \langle \sigma \rangle$ acts trivially on $\mathbb{Z}/2\mathbb{Z}$.
4. Let $G = \mathbb{Z}$ (as a discrete group) and M a G-module, with the generator $1 \in \mathbb{Z}$ acting by an automorphism $\varphi \colon M \to M$. Using the resolution $0 \to \mathbb{Z}[\mathbb{Z}] \xrightarrow{t-1} \mathbb{Z}[\mathbb{Z}] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ (where t is the generator), show that $H^0(\mathbb{Z}, M) = M^{\mathbb{Z}} = \ker(\varphi - \mathrm{id})$, $H^1(\mathbb{Z}, M) = M/(\varphi - \mathrm{id})M$, and $H^n(\mathbb{Z}, M) = 0$ for $n \geq 2$.
5. (If the reader has read [13]) Show that for groups G_1, G_2 and a trivial $G_1 \times G_2$-module M, there is a Künneth isomorphism:

   $$H^n(G_1 \times G_2, M) \cong \bigoplus_{p+q=n} H^p(G_1, \mathbb{Z}) \otimes_{\mathbb{Z}} H^q(G_2, M) \oplus \bigoplus_{p+q=n+1} \mathrm{Tor}_1^{\mathbb{Z}}(H^p(G_1, \mathbb{Z}), H^q(G_2, M))$$

   Verify this for $G_1 = G_2 = C_2$ and $M = \mathbb{Z}$.

2.4 Low-Degree Cohomology: Interpretations

The abstract definition of $H^n(G, M)$ as a derived functor acquires concrete meaning in low degrees. The group H^0 is the invariant subgroup (already known). The group H^1 classifies crossed homomor-

phisms modulo principal ones, which amounts to measuring an obstruction to lifting fixed points. The group H^2 classifies group extensions. These interpretations connect the cohomological formalism to explicit algebraic problems.

The first interpretation makes the connecting homomorphism from the long exact sequence (proposition 2.2.2) concrete through the cocycle language of section 2.3.

Proposition 2.4.1: The Connecting Homomorphism via Cocycles

Let $0 \to A \xrightarrow{\iota} B \xrightarrow{\pi} C \to 0$ be a short exact sequence of G-modules, and let $c \in C^G$. Choose any lift $b \in B$ with $\pi(b) = c$. Then the function $\lambda\colon G \to A$ defined by $\iota(\lambda(g)) = g \cdot b - b$ is a crossed homomorphism (definition 2.3.5), and the connecting homomorphism of proposition 2.2.2 sends:

$$\delta^0(c) = [\lambda] \in H^1(G, A)$$

where $H^1(G, A)$ is identified with $Z^1(G, A)/B^1(G, A)$ by proposition 2.3.4. The class $[\lambda]$ is independent of the choice of lift b.

***Proof*:**

Since $c \in C^G$, the element $g \cdot b - b$ satisfies $\pi(g \cdot b - b) = g \cdot c - c = 0$, so $g \cdot b - b \in \ker \pi = \mathrm{im}\iota$. The function λ is well-defined because ι is injective. The crossed homomorphism condition follows from:

$$\begin{aligned}\iota(\lambda(gh)) &= gh \cdot b - b = g \cdot (h \cdot b - b) + (g \cdot b - b) \\ &= g \cdot \iota(\lambda(h)) + \iota(\lambda(g)) = \iota(g \cdot \lambda(h) + \lambda(g))\end{aligned}$$

so $\lambda(gh) = g \cdot \lambda(h) + \lambda(g)$. A different lift $b' = b + \iota(a)$ for some $a \in A$ changes the cocycle to $\lambda'(g) = \lambda(g) + g \cdot a - a$, which differs from λ by the principal crossed homomorphism $g \mapsto g \cdot a - a$ (definition 2.3.6). So $[\lambda'] = [\lambda]$ in $H^1(G, A)$.

By proposition 2.3.4(2), the isomorphisms $H^n(C^\bullet(G, -)) \cong H^n(G, -)$ carry the connecting homomorphism of the short exact sequence of cochain complexes $0 \to C^\bullet(G, A) \to C^\bullet(G, B) \to C^\bullet(G, C) \to 0$ of proposition 2.3.4(1) to δ^0, and in degree zero they are the identification $H^0(C^\bullet(G, C)) = C^G = H^0(G, C)$. The construction of the connecting homomorphism in the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) computes it on the 0-cocycle $c \in C^0(G, C) = C$: lift c to $b \in C^0(G, B) = B$, apply d^0 to obtain the 1-cochain $g \mapsto g \cdot b - b$, whose values lie in $\iota(A)$, and take the class of the 1-cochain with values in A that ι carries to it. That 1-cochain is λ, so $\delta^0(c) = [\lambda]$.

The connecting homomorphism δ^0 measures a precise obstruction: the element $c \in C^G$ lifts to a G-fixed element of B if and only if $\delta^0(c) = 0$. The lift exists precisely when the crossed homomorphism λ is principal (when b can be adjusted by an element of A to become G-fixed).

Example 2.4.2: The Connecting Homomorphism in Action

Let $G = C_2 = \langle \sigma \rangle$ act on $\mathbb{Z}$ by negation ($\sigma \cdot m = -m$), and consider the exact sequence $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$, where the first map is multiplication by 2. All three modules carry the negation action. Since $-1 \equiv 1 \pmod 2$, the action on $\mathbb{Z}/2\mathbb{Z}$ is trivial, giving $(\mathbb{Z}/2\mathbb{Z})^{C_2} = \mathbb{Z}/2\mathbb{Z}$.

Meanwhile $\mathbb{Z}^{C_2} = 0$ (the only integer fixed by negation is 0).

The connecting homomorphism $\delta^0 \colon \mathbb{Z}/2\mathbb{Z} \to H^1(C_2, \mathbb{Z})$ sends $\bar{1} \in \mathbb{Z}/2\mathbb{Z}$ to the cocycle class computed as follows: lift $\bar{1}$ to $b = 1 \in \mathbb{Z}$. The cocycle is $\lambda(\sigma) = \frac{1}{2}(\sigma \cdot 1 - 1) = \frac{1}{2}(-1 - 1) = -1$ (using ι^{-1}, i.e., dividing by 2). Using the periodic resolution, $H^1(C_2, \mathbb{Z}) = {}_N\mathbb{Z}/(\sigma - 1)\mathbb{Z} = \mathbb{Z}/2\mathbb{Z}$ (with the negation action: $(\sigma - 1)m = -2m$ and $N(m) = m + (-m) = 0$). The cocycle $\lambda(\sigma) = -1$ represents the nontrivial class, so δ^0 is an isomorphism $\mathbb{Z}/2\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/2\mathbb{Z}$.

Proposition 2.4.3: H^1 with Trivial Action

If G acts trivially on M, then $H^1(G, M) \cong \mathrm{Hom}(G, M)$.

***Proof*:**
With trivial action, a crossed homomorphism $f \colon G \to M$ satisfies $f(gh) = f(g) + f(h)$ (by definition 2.3.5 with $g \cdot f(h) = f(h)$), so $Z^1(G, M) = \mathrm{Hom}(G, M)$. A principal crossed homomorphism $f_m(g) = g \cdot m - m = 0$ (trivial action), so $B^1(G, M) = 0$. The quotient gives $H^1(G, M) = \mathrm{Hom}(G, M)$.

For finite G, this gives $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z}) = 0$ (no nontrivial homomorphisms from a finite group to a torsion-free group). In contrast, $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ is the Pontryagin dual of G^{ab} and is typically nonzero.

For the profinite group G_K the same computation on continuous cochains gives $H^1(G_K, \mathbb{Z}/n\mathbb{Z}) = \mathrm{Hom}_{\mathrm{cts}}(G_K, \mathbb{Z}/n\mathbb{Z})$ (proposition 3.1.7). A continuous homomorphism $\chi \colon G_K \to \mathbb{Z}/n\mathbb{Z}$ has image the subgroup of some order $d \mid n$, and its kernel is an open normal subgroup whose fixed field is a cyclic extension of K of degree d (theorem 1.4.1). Every cyclic extension of K inside K^{sep} of degree dividing n arises in this way, and two homomorphisms give the same extension exactly when they have the same kernel, that is, when they differ by an automorphism of their common image $\mathbb{Z}/d\mathbb{Z}$. So $H^1(G_K, \mathbb{Z}/n\mathbb{Z})$ classifies the pairs consisting of a cyclic extension L/K inside K^{sep} of some degree $d \mid n$ and an isomorphism of $\mathrm{Gal}(L/K)$ onto the subgroup of order d of $\mathbb{Z}/n\mathbb{Z}$, and each such L occurs in $\varphi(d)$ pairs, φ being Euler's function. The non-trivial action case, where H^1 captures genuine arithmetic obstructions, is exemplified by Hilbert's Theorem 90 ($H^1(\mathrm{Gal}(L/K), L^\times) = 0$), which will be proved in chapter 8.

The second cohomology group classifies group extensions. This is the analogue, for the group G, of the classification of principal bundles in topology or of torsors in algebraic geometry.

Definition 2.4.4: Group Extension

Let G be a group and M a G-module. A *group extension* of G by M (with the given action) is a short exact sequence of groups:

$$1 \to M \xrightarrow{\iota} E \xrightarrow{\pi} G \to 1$$

such that the conjugation action of E on $\iota(M)$ agrees with the G-action on M: for all $e \in E$ and $m \in M$, $e\,\iota(m)\,e^{-1} = \iota(\pi(e) \cdot m)$. Two extensions E and E' are *equivalent* if there exists a group isomorphism $\varphi\colon E \xrightarrow{\sim} E'$ making the diagram commute:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M & \xrightarrow{\iota} & E & \xrightarrow{\pi} & G & \longrightarrow & 1 \\ & & \| & & \downarrow{\scriptstyle \sim}\varphi & & \| & & \\ 1 & \longrightarrow & M & \xrightarrow{\iota'} & E' & \xrightarrow{\pi'} & G & \longrightarrow & 1 \end{array}$$

The set of equivalence classes of extensions is denoted $\mathrm{Ext}(G, M)$.

The split extension $M \rtimes G$ (semidirect product with the given action) is always an extension, serving as the "trivial" class.

Theorem 2.4.5: Classification of Extensions by H^2

Let G be a group and M a G-module. There is a bijection:

$$H^2(G, M) \xrightarrow{\sim} \mathrm{Ext}(G, M)$$

Under this bijection, the zero element of $H^2(G, M)$ corresponds to the split extension $M \rtimes G$.

***Proof*:**

Step 1: From extensions to cocycles. Given an extension $1 \to M \xrightarrow{\iota} E \xrightarrow{\pi} G \to 1$, choose a set-theoretic section $s\colon G \to E$ with $\pi \circ s = \mathrm{id}_G$ and $s(1) = 1$. For $g_1, g_2 \in G$, both $s(g_1)s(g_2)$ and $s(g_1g_2)$ map to g_1g_2 under π, so their ratio lies in $\ker \pi = \iota(M)$. Define $f\colon G \times G \to M$ by:

$$\iota(f(g_1, g_2)) = s(g_1)\,s(g_2)\,s(g_1g_2)^{-1}$$

The function f is a 2-cocycle: the associativity $(s(g_1)s(g_2))s(g_3) = s(g_1)(s(g_2)s(g_3))$ in E, combined with the relation $s(g)^{-1}\iota(m)s(g) = \iota(g \cdot m)$, yields the cocycle condition of definition 2.3.7.

Step 2: Independence of section. A different section $s'(g) = s(g)\,\iota(\psi(g))$ for some function $\psi\colon G \to M$ produces a new cocycle:

$$f'(g_1, g_2) = f(g_1, g_2) + g_1 \cdot \psi(g_2) - \psi(g_1g_2) + \psi(g_1)$$

so $f' - f$ is a 2-coboundary and $[f'] = [f]$ in $H^2(G, M)$.

Step 3: From cocycles to extensions. Given a 2-cocycle $f\colon G \times G \to M$, define a group $E_f = M \times G$ (as a set) with multiplication:

$$(m_1, g_1) \cdot (m_2, g_2) = (m_1 + g_1 \cdot m_2 + f(g_1, g_2),\ g_1g_2)$$

The cocycle condition ensures associativity (the computation is a direct expansion of both sides of $(a\cdot b)\cdot c = a\cdot(b\cdot c)$, using the cocycle identity to match the M-components). The normalization $f(1,g) = f(g,1) = 0$ (which can always be arranged by modifying f by a coboundary) ensures $(0,1)$ is the identity. The maps $\iota(m) = (m,1)$ and $\pi(m,g) = g$ give an extension.

Step 4: Bijection. Steps 1-3 define inverse maps between $H^2(G,M)$ and $\mathrm{Ext}(G,M)$. Equivalent extensions produce cohomologous cocycles (Step 2), and cohomologous cocycles produce equivalent extensions (the isomorphism $E_f \to E_{f'}$ is $(m,g) \mapsto (m+\psi(g),g)$ where $f' - f = d^1\psi$). The split extension $M \rtimes G$ corresponds to $f = 0$.

The classification shows that $H^2(G,M)$ is the complete obstruction to splitting an extension: a group E mapping onto G with kernel M splits (as a semidirect product) if and only if the corresponding cohomology class vanishes.

Example 2.4.6: Extensions Classified by H^2

1. By example 2.3.8, $H^2(C_n,\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (with trivial action). The n equivalence classes of extensions $0 \to \mathbb{Z} \to E \to C_n \to 1$ are represented by $E_k = \langle a,t \,|\, tat^{-1} = a,\ t^n = a^k\rangle$ for $0 \le k < n$, where a generates $\mathbb{Z}$ and t maps to a generator of C_n. The class $k = 0$ is the split extension $\mathbb{Z} \times C_n$. For $n = 2$ and $k = 1$: $E_1 = \langle a,t \,|\, t^2 = a\rangle \cong \mathbb{Z}$ (generated by t, with $a = t^2$), giving the non-split extension $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$.
2. The Brauer group of a field K is $\mathrm{Br}(K) \cong H^2(G_K,(K^{\mathrm{sep}})^\times)$, where the 2-cocycle $f\colon G_K \times G_K \to (K^{\mathrm{sep}})^\times$ encodes the multiplication table of a central simple K-algebra via the crossed product construction. This identification is developed in chapter 7.
3. (If the reader has read [23]) For a connected space X with $\pi_1(X) = G$, the group $H^2(G,A)$ (trivial action, A abelian) classifies central extensions $1 \to A \to E \to G \to 1$, which correspond to fibrations over BG with fibre $K(A,1) = BA$ (principal A-bundles are classified by $H^1(BG;A)$, one degree lower).

In higher degrees, the cohomological interpretations become progressively more complex. The group $H^3(G,M)$ classifies certain "crossed modules" (equivalently, 2-group extensions), and appears as the obstruction to lifting a homomorphism $G \to \mathrm{Out}(E)$ to a group extension with kernel E. These higher-degree interpretations, while structurally significant, are rarely needed in the number-theoretic applications of this book.

Exercise 2.4.1

1. Classify (up to equivalence) all group extensions $1 \to \mathbb{Z}/2\mathbb{Z} \to E \to \mathbb{Z}/2\mathbb{Z} \to 1$ where $\mathbb{Z}/2\mathbb{Z}$ acts trivially on $\mathbb{Z}/2\mathbb{Z}$. Identify each extension explicitly as a group of order 4.
2. Let $0 \to M \to E \to G \to 1$ be a group extension. Show that the extension splits (i.e., the corresponding class in $H^2(G,M)$ is trivial) if and only if π admits a group homomorphism section $s\colon G \to E$ with $\pi \circ s = \mathrm{id}_G$.
3. Let $0 \to A \to B \xrightarrow{\pi} C \to 0$ be a short exact sequence of G-modules. Show that if $H^1(G,A) = 0$, then $\pi^G\colon B^G \to C^G$ is surjective. Is the converse true? Give an example where π^G is surjective but $H^1(G,A) \neq 0$.

4. Let G be a finite group. The *Schur multiplier* of G is $H^2(G, \mathbb{C}^\times)$ (with trivial action). Show that $H^2(G, \mathbb{C}^\times) \cong H^2(G, \mathbb{Q}/\mathbb{Z}) \cong H_2(G, \mathbb{Z})^*$ where $(-)^*$ denotes the Pontryagin dual. *Hint:* use the exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ and the universal coefficient theorem.

5. Compute $H^2(C_2, \mathbb{Z}/2\mathbb{Z})$ (trivial action) using the periodic resolution, and describe the corresponding extensions explicitly.

2.5 Homology and Coinvariants

Cohomology derives the invariant functor $(-)^G$. Homology derives the coinvariant functor $(-)_G$. The two theories extract different information from the same action: where cohomology measures obstructions to lifting fixed points and classifies extensions, homology captures the abelianization and relations among relations. The reason cohomology takes precedence in this book (and in the Galois-theoretic literature more broadly) is threefold: the invariant functor $(-)^G$ is arithmetically natural (fixed fields, invariant elements), cup products endow $H^*(G, -)$ with a graded ring structure that homology lacks, and the passage to profinite groups in chapter 3 is cleaner for cohomology with discrete coefficients. Homology is nonetheless indispensable: the results of this section are used in the structure theory of pro-p-groups (chapter 6) and in the Tate cohomology of chapter 4, which unifies homology and cohomology into a single theory.

Group homology was defined in definition 2.2.1 as $H_n(G, M) = \operatorname{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M)$. The long exact sequence in homology (proposition 2.2.2) and the identification $H_0(G, M) = M_G$ (proposition 2.2.3) are already in hand. The first task is to compute homology explicitly for the groups whose cohomology was computed in section 2.3.

Example 2.5.1: Homology of Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$. The periodic resolution from example 2.3.8, applied to homology via tensoring with $\mathbb{Z}$ over $\mathbb{Z}[G]$, gives the complex:

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$$

where $(\sigma - 1)$ acts as 0 on $\mathbb{Z}$ (trivial action) and the norm N acts as multiplication by n. The homology is:

$$H_k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}/n\mathbb{Z} & k \text{ odd} \\ 0 & k \text{ even},\ k \geq 2 \end{cases}$$

Comparing with the cohomology $H^k(C_n, \mathbb{Z})$ from example 2.3.8: the nonzero groups appear in complementary degrees (odd for homology, even for cohomology). This complementarity reflects the periodicity that Tate cohomology (chapter 4) makes explicit.

More generally, for a C_n-module M with arbitrary action, the periodic resolution gives $H_k(C_n, M) = M^G/NM$ for even $k \geq 2$ and $H_k(C_n, M) = {}_N M/(\sigma - 1)M$ for odd $k \geq 1$, with $H_0(C_n, M) = M_G = M/(\sigma - 1)M$. Comparing with the cohomology (example 2.3.8): the formulas for H_k and H^k coincide in degrees $k \geq 1$. The difference lies entirely in degree 0: $H_0 = M_G$ (coinvariants) while $H^0 = M^G$ (invariants). This coincidence in positive degrees is specific to cyclic groups and reflects the periodicity of the resolution.

For general groups, homology is computed by tensoring a projective resolution of $\mathbb{Z}$ with M over $\mathbb{Z}[G]$ (rather than applying $\mathrm{Hom}_G(-, M)$ as for cohomology). The standard resolution of definition 2.3.1 serves this purpose: tensoring $\mathbb{Z}[G^{n+1}]$ with M over $\mathbb{Z}[G]$ gives M^{n+1} (via $\mathbb{Z}[G^{n+1}] \otimes_{\mathbb{Z}[G]} M \cong M \otimes_{\mathbb{Z}} \mathbb{Z}[G^n] \cong M^{|G^n|}$ when G is finite, or more generally, the free M-module on G^n). The resulting chain complex produces the homology via the same face maps as in definition 2.3.1, now acting on the M-component through the G-action.

The most important structural result about H_1 is that it recovers the abelianization.

Theorem 2.5.2: H_1 and the Abelianization

For any group G: $H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}} = G/[G, G]$.

***Proof*:**

Apply the long exact sequence in homology to the augmentation sequence $0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$ (definition 2.1.4):

$$H_1(G, \mathbb{Z}[G]) \to H_1(G, \mathbb{Z}) \xrightarrow{\partial} (I_G)_G \xrightarrow{\bar{\iota}} (\mathbb{Z}[G])_G \xrightarrow{\bar{\epsilon}} \mathbb{Z}_G \to 0$$

Step 1: Vanishing of $H_1(G, \mathbb{Z}[G])$. Since $\mathbb{Z}[G] \cong \mathrm{Ind}^G(\mathbb{Z})$, the associativity of the tensor product gives $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}[G] \cong P_\bullet$ for any projective resolution $P_\bullet \to \mathbb{Z}$, so $H_n(G, \mathbb{Z}[G]) = \mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, \mathbb{Z}[G]) = 0$ for $n \geq 1$.

Step 2: Computing $(I_G)_G$. The coinvariants of I_G are $I_G/\langle g \cdot x - x : g \in G, x \in I_G \rangle$. For $x \in I_G$, the element $g \cdot x - x = (g-1) \cdot x$ lies in $I_G \cdot I_G = I_G^2$ (since $g - 1 \in I_G$). Conversely, every generator $(g_1 - 1)(g_2 - 1)$ of I_G^2 has the form $g_1 \cdot (g_2 - 1) - (g_2 - 1)$, a coinvariant relation. So $(I_G)_G = I_G/I_G^2$.

Step 3: The map $\bar{\iota}$ vanishes. The inclusion $\iota \colon I_G \hookrightarrow \mathbb{Z}[G]$ on coinvariants gives $\bar{\iota} \colon I_G/I_G^2 \to \mathbb{Z}[G]/I_G$. Since $\iota(I_G) = I_G$ maps to 0 in $\mathbb{Z}[G]/I_G$, the map $\bar{\iota}$ is the zero map.

Step 4: Conclusion. The vanishing of $H_1(G, \mathbb{Z}[G])$ and $\bar{\iota}$ reduces the long exact sequence to $0 \to H_1(G, \mathbb{Z}) \xrightarrow{\sim} I_G/I_G^2$. The isomorphism $I_G/I_G^2 \cong G^{\mathrm{ab}}$ from exercise 2.1.1 (5) (sending $\overline{g - 1} \mapsto g[G, G]$) completes the proof.

The abelianization G^{ab} is the first and coarsest invariant of G: it captures the "linear" part of the group, discarding all higher-order commutator structure. The theorem says that $H_1(G, \mathbb{Z})$ detects precisely this information. For instance, $H_1(S_n, \mathbb{Z}) = S_n^{\mathrm{ab}} = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$ (the sign homomorphism), and $H_1(F_S, \mathbb{Z}) = F_S^{\mathrm{ab}} = \mathbb{Z}^S$ for a free group on the set S.

The universal coefficient theorem relates homology and cohomology, allowing the computation of one from the other when the coefficient module is a trivial G-module.

Theorem 2.5.3: Universal Coefficient Theorem

Let G be a group and A an abelian group with trivial G-action. For each $n \geq 0$, there is a natural short exact sequence:

$$0 \to \operatorname{Ext}^1_{\mathbb{Z}}(H_{n-1}(G,\mathbb{Z}), A) \to H^n(G,A) \to \operatorname{Hom}_{\mathbb{Z}}(H_n(G,\mathbb{Z}), A) \to 0$$

This sequence splits (non-canonically).

***Proof*:**
This is a special case of the universal coefficient theorem for the cohomology of a chain complex (see [13]), applied to the chain complex $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}$ (which computes $H_*(G,\mathbb{Z})$) and the functor $\operatorname{Hom}_{\mathbb{Z}}(-, A)$ (which, applied to $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}$, computes $H^*(G,A)$ for trivial A).

Example 2.5.4: The UCT in Action
For $G = C_m$ and $A = \mathbb{Z}/k\mathbb{Z}$ (trivial action), the UCT gives:

$$0 \to \operatorname{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_m,\mathbb{Z}), \mathbb{Z}/k\mathbb{Z}) \to H^n(C_m, \mathbb{Z}/k\mathbb{Z}) \to \operatorname{Hom}_{\mathbb{Z}}(H_n(C_m,\mathbb{Z}), \mathbb{Z}/k\mathbb{Z}) \to 0$$

Using $H_n(C_m,\mathbb{Z})$ from example 2.5.1: for odd n, $H_n(C_m,\mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$ and $H_{n-1}(C_m,\mathbb{Z}) = 0$, giving $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \operatorname{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$. For even $n \geq 2$, $H_n(C_m,\mathbb{Z}) = 0$ and $H_{n-1}(C_m,\mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$, giving $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \operatorname{Ext}^1(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$. In both cases, $H^n(C_m,\mathbb{Z}/k\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,k)\mathbb{Z}$ for $n \geq 1$.

The second homology group admits an interpretation in terms of generators and relations.

Theorem 2.5.5: Hopf Formula

Let G be a group with a free presentation $1 \to R \to F \xrightarrow{\pi} G \to 1$ (i.e., F is a free group and $R = \ker \pi$). Then:

$$H_2(G,\mathbb{Z}) \cong \frac{R \cap [F,F]}{[R,F]}$$

***Proof*:**
The low-degree exact sequence in homology of the group extension $1 \to R \to F \to G \to 1$, which comes from the Lyndon-Hochschild-Serre spectral sequence in homology [33, p. 6.8.7] and is not derived in this book, gives:

$$H_2(F,\mathbb{Z}) \to H_2(G,\mathbb{Z}) \to R/[R,F] \to F^{\mathrm{ab}} \to G^{\mathrm{ab}} \to 0$$

Since F is free, $H_2(F,\mathbb{Z}) = 0$ (free groups have cohomological dimension 1 by example 2.3.10, and the same holds for homology). The map $R/[R,F] \to F^{\mathrm{ab}} = F/[F,F]$ sends $r[R,F] \mapsto r[F,F]$, with kernel $(R \cap [F,F])/[R,F]$. Exactness at $H_2(G,\mathbb{Z})$ gives the isomorphism.

The numerator $R \cap [F,F]$ consists of relators that are also commutators in the free group: "relations that hold for trivial reasons in the abelianization." The denominator $[R,F]$ consists of consequences of

the relations that are automatically trivial. The quotient $H_2(G,\mathbb{Z})$ therefore measures the "relations among relations" that are not accounted for by formal consequences. The group $H_2(G,\mathbb{Z})$ is also called the *Schur multiplier* of G (see exercise 2.4.1 (4)).

For finite groups, the Tate cohomology of chapter 4 merges homology and cohomology into a single sequence $\hat{H}^n(G,M)$ indexed by all integers $n \in \mathbb{Z}$, with $\hat{H}^n = H^n$ for $n \geq 1$ and $\hat{H}^{-n} = H_{n-1}$ for $n \geq 2$ (up to modification in degrees 0 and -1). The periodicity visible in the cyclic group computations then extends to a single 2-periodic sequence.

Exercise 2.5.1

1. Compute $H_n(C_m, M)$ for all $n \geq 0$ and an arbitrary C_m-module M, using the periodic resolution. Express the answer in terms of the norm map N and the action of $\sigma - 1$.
2. Compute $H_1(S_3,\mathbb{Z})$ using theorem 2.5.2, and verify the result independently by computing H_1 from a free presentation of S_3.
3. Verify the universal coefficient theorem for $G = C_2$ and $A = \mathbb{Z}/2\mathbb{Z}$ in degrees $n = 1$ and $n = 2$: check that both sides of the UCT short exact sequence agree with the direct computation from exercise 2.3.1 (3).
4. Show that $H_n(G,\mathbb{Z}[G]) = 0$ for all $n \geq 1$, using the isomorphism $P_\bullet \otimes_{\mathbb{Z}[G]} \mathbb{Z}[G] \cong P_\bullet$ for a projective resolution $P_\bullet \to \mathbb{Z}$.
5. Use the Hopf formula to compute $H_2(C_2 \times C_2, \mathbb{Z})$, starting from the presentation $\langle a, b \,|\, a^2, b^2, [a,b]\rangle$. Verify the result using the Künneth formula.

3

Profinite Group Cohomology

The cohomology theory developed in chapter 2 applies to any group G acting on an abelian group M: one forms the derived functors of $M \mapsto M^G$ and obtains a sequence of abelian groups $H^n(G, M)$ that measure the failure of the invariant functor to be exact. Nothing in that construction uses the topology of G, and when G is profinite, this is a serious defect. The group $G = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, for instance, acts on $\overline{\mathbb{Q}}^\times$ through a continuous action, and the cohomology that captures arithmetic information must respect this continuity. Ignoring the topology produces groups that are too large, too poorly behaved, and disconnected from the Galois-theoretic applications that motivate the entire subject.

This chapter rebuilds group cohomology in the profinite setting. The essential modification is to restrict attention to *discrete* G-modules (those on which G acts continuously when the module carries the discrete topology) and to define cohomology as the derived functors of the continuous invariant functor on this restricted category. The payoff is immediate: the resulting cohomology groups are colimits of the finite-group cohomology from chapter 2, making them computable by reduction to finite quotients. With this foundation in place, the standard functorial apparatus (inflation, restriction, corestriction, the Lyndon-Hochschild-Serre spectral sequence, and Shapiro's lemma) transfers to the profinite setting with only minor modifications.

3.1 Continuous Cochains and Discrete Modules

The first task is to identify the correct category of coefficient modules for a profinite group G. The naive approach (taking all $\mathbb{Z}[G]$-modules and computing derived functors) ignores the topology of G entirely. The correct approach requires the G-action to be continuous, which forces a topological constraint on the module.

Throughout this section, G denotes a profinite group with its natural topology (compact, Hausdorff,

and totally disconnected, as established in chapter 1).

Definition 3.1.1: Discrete G-Module

Let G be a profinite group. A *discrete G-module* is an abelian group M equipped with a left G-action $G \times M \to M$ such that the stabilizer $\mathrm{Stab}_G(m) = \{g \in G \mid g \cdot m = m\}$ is open in G for every $m \in M$. Equivalently, M is a discrete G-module if and only if

$$M = \bigcup_{U \trianglelefteq_o G} M^U,$$

where the union runs over all open normal subgroups U of G, and $M^U = \{m \in M \mid u \cdot m = m \text{ for all } u \in U\}$ is the submodule of U-invariants.

The equivalence in the definition is immediate: if every stabilizer is open, then for each $m \in M$ there exists an open normal subgroup $U \subseteq \mathrm{Stab}_G(m)$ (since every open neighbourhood of the identity in a profinite group contains an open normal subgroup, by proposition 1.1.8), so $m \in M^U$. Conversely, if $m \in M^U$ for some open normal U, then $U \subseteq \mathrm{Stab}_G(m)$, and a subgroup containing an open subgroup is itself open.

The terminology "discrete" refers to the topology on M: when M carries the discrete topology, the action $G \times M \to M$ is continuous if and only if every stabilizer is open. With this topology, $G \times M$ is the disjoint union of the open sets $G \times \{m\}$, so the action is continuous if and only if each orbit map $g \mapsto g \cdot m$ is continuous into the discrete space M, that is, has open fibres. The fibre over m is $\mathrm{Stab}_G(m)$, and the fibre over $h \cdot m$ is the coset $h\mathrm{Stab}_G(m)$, so all fibres are open exactly when the stabilizers are (the condition in the definition).

Proposition 3.1.2: The Category of Discrete G-Modules

Let G be a profinite group. The category $\mathrm{Mod}_G^{\mathrm{disc}}$ of discrete G-modules, with G-equivariant group homomorphisms as morphisms, is an abelian category, whose kernels and cokernels are those of Mod_G.

***Proof*:**

The category $\mathrm{Mod}_G^{\mathrm{disc}}$ is a full subcategory of the category Mod_G of all $\mathbb{Z}[G]$-modules. It is closed under taking submodules, quotients, and arbitrary direct sums. For submodules: if $N \subseteq M$ is a G-stable subgroup and M is discrete, then for each $n \in N$ the stabilizer $\mathrm{Stab}_G(n)$ is the same whether n is viewed in N or in M, so it is open. For quotients: if $\pi : M \to M/N$ is the projection and $\overline{m} = \pi(m)$, then $\mathrm{Stab}_G(\overline{m}) \geqslant \mathrm{Stab}_G(m)$, which is open. For direct sums: if $(M_i)_{i \in I}$ is a family of discrete G-modules, each element of $\bigoplus_i M_i$ has finite support, so its stabilizer is a finite intersection of open subgroups, hence open. The kernel of a morphism of discrete G-modules is a submodule of its source and the cokernel a quotient of its target, so $\mathrm{Mod}_G^{\mathrm{disc}}$ is a full subcategory of the abelian category Mod_G containing 0 and closed under finite direct sums, kernels and cokernels. Such a subcategory is additive, and the kernels and cokernels computed in Mod_G satisfy their universal properties in it, as it is full. If f is a monomorphism of $\mathrm{Mod}_G^{\mathrm{disc}}$, the inclusion k of its kernel in Mod_G is a morphism of $\mathrm{Mod}_G^{\mathrm{disc}}$ with $f \circ k = 0 = f \circ 0$, so $k = 0$ and the kernel is 0. So f is a monomorphism of Mod_G, and therefore the kernel in Mod_G of its cokernel in Mod_G. That cokernel lies in $\mathrm{Mod}_G^{\mathrm{disc}}$, so f is the kernel of its cokernel

there. Dually, every epimorphism of $\mathrm{Mod}_G^{\mathrm{disc}}$ is the cokernel of its kernel. So $\mathrm{Mod}_G^{\mathrm{disc}}$ is abelian, with the kernels and cokernels of Mod_G.

Derived functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ require enough injectives. In Mod_G the injectives of proposition 2.1.2 are the coinduced modules $\mathrm{CoInd}^G(D)$, which consist of all functions $G \to D$, and restricting to continuous functions makes them discrete.

Proposition 3.1.3: Enough Injectives

The category $\mathrm{Mod}_G^{\mathrm{disc}}$ has enough injectives. Specifically, for every discrete G-module M, there exists an injective object I in $\mathrm{Mod}_G^{\mathrm{disc}}$ and a G-equivariant injection $M \hookrightarrow I$.

***Proof*:**
Let M be a discrete G-module. Its underlying abelian group embeds, by some $\iota\colon M \to J$, in an injective abelian group J (*Core Algebra* [9, $\mathbb{Z}$-module Submodules of Injective Modules]). Give J the discrete topology, and let $I = \mathrm{Map}^{\mathrm{cts}}(G, J)$ be the group of continuous maps $G \to J$ under pointwise addition. The group G acts on I by $(g \cdot f)(x) = f(xg)$ for $g, x \in G$, and $g \cdot f$ is continuous when f is, because right multiplication by g is continuous.

Step 1: I is a discrete G-module. Fix $f \in I$. For each $x \in G$ the fibre $f^{-1}(f(x))$ is open, so continuity of multiplication at $(x, 1)$ gives open neighbourhoods W_x of x and V_x of 1 with $W_x V_x \subseteq f^{-1}(f(x))$. Finitely many $W_{x_1}, \dots, W_{x_r}$ cover the compact space G, and $V = V_{x_1} \cap \cdots \cap V_{x_r}$ is an open neighbourhood of 1. If $x \in W_{x_i}$ and $v \in V$, then both $x = x \cdot 1$ and xv lie in $W_{x_i} V_{x_i}$, so $f(xv) = f(x_i) = f(x)$. Hence $\mathrm{Stab}_G(f) \geqslant V$, and a subgroup containing an open neighbourhood of 1 is open, being the union of the translates hV of that neighbourhood by its elements h.

Step 2: The embedding $M \hookrightarrow I$. Define $\varphi\colon M \to I$ by $\varphi(m)(g) = \iota(g \cdot m)$. The map $\varphi(m)$ is continuous because m has open stabilizer, so $g \mapsto g \cdot m$ is constant on the cosets $g\mathrm{Stab}_G(m)$. The map φ is G-equivariant, since $\varphi(h \cdot m)(g) = \iota(gh \cdot m) = \varphi(m)(gh) = (h \cdot \varphi(m))(g)$. It is injective: if $\varphi(m) = 0$, then taking $g = 1$ gives $\iota(m) = 0$, hence $m = 0$.

Step 3: I is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$. The module I is the coinduced module $\mathrm{CoInd}_{\{1\}}^G(J)$ of definition 3.5.1, for the closed subgroup $\{1\}$ and the abelian group J. By proposition 3.5.4, whose proof does not use the present proposition, there is an isomorphism $\mathrm{Hom}_G(B, I) \cong \mathrm{Hom}_{\mathbb{Z}}(B, J)$ natural in the discrete G-module B. Let $A \hookrightarrow B$ be an injection of discrete G-modules. Restriction $\mathrm{Hom}_{\mathbb{Z}}(B, J) \to \mathrm{Hom}_{\mathbb{Z}}(A, J)$ is surjective because J is injective, so by naturality restriction $\mathrm{Hom}_G(B, I) \to \mathrm{Hom}_G(A, I)$ is surjective, and every G-map $A \to I$ extends to B.

With the abelian category structure and enough injectives established, the derived functor definition of cohomology proceeds exactly as in the discrete case.

Definition 3.1.4: Continuous Cohomology

Let G be a profinite group and M a discrete G-module. The *continuous cohomology groups* $H^n(G, M)$ for $n \geq 0$ are defined as the right derived functors of the invariant functor

$$(-)^G : \mathrm{Mod}_G^{\mathrm{disc}} \to \mathbf{Ab}, \qquad M \mapsto M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\},$$

applied to M. That is, choosing an injective resolution $0 \to M \to I^0 \to I^1 \to \cdots$ in $\mathrm{Mod}_G^{\mathrm{disc}}$, one sets

$$H^n(G, M) = H^n\big((I^0)^G \to (I^1)^G \to (I^2)^G \to \cdots\big).$$

By the general theory of derived functors ([13, Unique Projection Resolution, Long Exact Sequence From a Right-Derived Functor]), the groups $H^n(G, M)$ are independent of the choice of injective resolution, and the short exact sequence $0 \to M' \to M \to M'' \to 0$ of discrete G-modules induces a long exact sequence

$$0 \to (M')^G \to M^G \to (M'')^G \xrightarrow{\delta} H^1(G, M') \to H^1(G, M) \to \cdots$$

The connecting homomorphisms δ are natural in the short exact sequence.

As in section 2.3, one can compute these derived functors using cochains rather than injective resolutions. The profinite analogue of the standard cochain complex requires a continuity constraint.

Definition 3.1.5: Continuous Cochain Complex

Let G be a profinite group and M a discrete G-module. The group of *continuous n-cochains* is

$$C^n_{\mathrm{cts}}(G, M) = \{f : G^n \to M \mid f \text{ is continuous}\},$$

where G^n carries the product topology and M the discrete topology. Since M is discrete, a map $f : G^n \to M$ is continuous if and only if it is *locally constant*: every point of G^n has an open neighbourhood on which f is constant. The coboundary maps $d^n : C^n_{\mathrm{cts}}(G, M) \to C^{n+1}_{\mathrm{cts}}(G, M)$ are defined by the same formula as in definition 2.3.3:

$$\begin{aligned}(d^n f)(g_1, \ldots, g_{n+1}) = {} & g_1 \cdot f(g_2, \ldots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \ldots, g_n).\end{aligned}$$

These maps preserve continuity (the action of g_1 on a locally constant function remains locally constant since the action map is continuous), so $C^\bullet_{\mathrm{cts}}(G, M)$ is a subcomplex of the complex of all inhomogeneous cochains of definition 2.3.3, and $d^{n+1} \circ d^n = 0$. Set $C^{-1}_{\mathrm{cts}}(G, M) = 0$ and $d^{-1} = 0$. The *continuous cocycles*, *continuous coboundaries*, and *continuous cochain cohomology* groups are

$$Z^n_{\mathrm{cts}}(G, M) = \ker d^n, \qquad B^n_{\mathrm{cts}}(G, M) = \mathrm{im} d^{n-1}, \qquad \tilde{H}^n(G, M) = Z^n_{\mathrm{cts}}/B^n_{\mathrm{cts}}.$$

Proposition 3.1.6: Equivalence of Definitions

Let G be a profinite group.

1. For every short exact sequence $0 \to A \to B \to C \to 0$ of discrete G-modules and every $n \geq 0$, the sequence

$$0 \to C^n_{\mathrm{cts}}(G, A) \to C^n_{\mathrm{cts}}(G, B) \to C^n_{\mathrm{cts}}(G, C) \to 0$$

   given by composition with the maps of the short exact sequence is exact. With the connecting homomorphisms of the resulting long exact sequences of cohomology, the family $\tilde{H}^\bullet(G, -) = (\tilde{H}^n(G, -))_{n \geq 0}$ is a universal δ-functor from $\mathrm{Mod}_G^{\mathrm{disc}}$ to $\mathbf{Ab}$ in the sense of *Homological Algebra* [13, Universal δ-Functor].

2. There is exactly one isomorphism of δ-functors $\tilde{H}^\bullet(G, -) \xrightarrow{\sim} H^\bullet(G, -)$ whose degree-zero component is the natural isomorphism $\tilde{H}^0(G, M) = M^G \cong H^0(G, M)$. In particular $\tilde{H}^n(G, M) \cong H^n(G, M)$ for every discrete G-module M and every $n \geq 0$, naturally in M, and these isomorphisms carry the connecting homomorphisms of the cochain long exact sequences of (1) to those of the derived functor long exact sequences.

***Proof*:**

Step 1: Exactness on cochains. The category $\mathrm{Mod}_G^{\mathrm{disc}}$ is abelian, with the kernels and cokernels of Mod_G (proposition 3.1.2), so a short exact sequence of discrete G-modules is exact as a sequence of abelian groups. Write $\alpha\colon A \to B$ and $\beta\colon B \to C$ for the maps of the sequence. A continuous map $f\colon G^n \to A$ is locally constant (definition 3.1.5), so $\alpha \circ f$ is locally constant, and composition with α and with β carries continuous cochains to continuous cochains. Since α is injective and $\beta \circ \alpha = 0$, composition with α is injective and its image lies in the kernel of composition with β. If $f\colon G^n \to B$ is continuous with $\beta \circ f = 0$, then every value of f lies in $\ker \beta = \alpha(A)$, so $f = \alpha \circ f'$ for a unique map $f'\colon G^n \to A$, and f' is locally constant because f is. For surjectivity, let $f\colon G^n \to C$ be continuous. Its fibres $f^{-1}(c)$ are open, pairwise disjoint, and cover the compact space G^n, so only finitely many of them are nonempty, and f takes finitely many values $c_1, \ldots, c_r$. Choose $b_i \in B$ with $\beta(b_i) = c_i$, and let $F\colon G^n \to B$ take the value b_i on $f^{-1}(c_i)$. Then F is constant on each of the open sets $f^{-1}(c_i)$, hence continuous, and $\beta \circ F = f$.

Step 2: A δ-functor. Composition with a map of discrete G-modules preserves local constancy, so it carries continuous cochains to continuous cochains. With this and Step 1, the argument of Step 2 of the proof of proposition 2.3.4 applies word for word to continuous cochains. Each $\tilde{H}^n(G, -)$ is therefore an additive functor $\mathrm{Mod}_G^{\mathrm{disc}} \to \mathbf{Ab}$, and $\tilde{H}^\bullet(G, -)$ is a δ-functor whose connecting homomorphisms are those of the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) applied to the short exact sequences of continuous cochain complexes of Step 1, with $\tilde{H}^0(G, M) = M^G$. The opening segment $0 \to A^G \to B^G \to C^G$ of its long exact sequences shows that $(-)^G$ is left exact on $\mathrm{Mod}_G^{\mathrm{disc}}$.

Step 3: Coinduced modules. For an abelian group J with the discrete topology, let $\mathrm{Map}^{\mathrm{cts}}(G, J)$ be the group of continuous maps $G \to J$. It is a G-submodule of the module $\mathrm{Map}(G, J)$ of Step 3 of the proof of proposition 2.3.4, on which G, regarded as an abstract group, acts by $(g \cdot f)(x) = f(xg)$: if f is continuous then so is $g \cdot f$, because right multiplication by g is

continuous. It is the module $\mathrm{CoInd}_{\{1\}}^{G}(J)$ of definition 3.5.1, and it is discrete by Step 1 of the proof of proposition 3.1.3.

Every discrete G-module M embeds in such a module, as in Step 2 of the proof of proposition 3.1.3. The G-equivariant injection $u_M\colon M \to \mathrm{Map}(G,M)$, $u_M(m)(x) = x\cdot m$, of Step 4 of the proof of proposition 2.3.4 takes values in $\mathrm{Map}^{\mathrm{cts}}(G,M)$, because $x \mapsto x\cdot m$ is constant on the cosets of the open subgroup $\mathrm{Stab}_G(m)$, hence continuous.

Step 4: Coinduced modules are acyclic for cochains. Fix J and put $I = \mathrm{Map}^{\mathrm{cts}}(G,J)$. Since I is a G-submodule of $\mathrm{Map}(G,J)$ by Step 3, the group $C^n_{\mathrm{cts}}(G,I)$ is a subgroup of $C^n(G,\mathrm{Map}(G,J))$, and its coboundary is the restriction of the coboundary of definition 2.3.3. For $n \geq 1$ let s be the map of Step 3 of the proof of proposition 2.3.4, given by

$$(sf)(g_1,\dots,g_{n-1})(x) = f(x,g_1,\dots,g_{n-1})(1)$$

which for $n = 1$ is the element $x \mapsto f(x)(1)$ of $C^0(G,\mathrm{Map}(G,J)) = \mathrm{Map}(G,J)$. Let $f \in C^n_{\mathrm{cts}}(G,I)$. The map $\Phi\colon G^{n-1}\times G \to J$, $(y,x)\mapsto (sf)(y)(x)$, is the composite of a rearrangement of coordinates $G^{n-1}\times G \to G^n$, the continuous map f, and evaluation at 1, which is continuous because I and J are discrete. So Φ is locally constant, and each $(sf)(y) = \Phi(y,-)$ lies in I. To see that $y\mapsto (sf)(y)$ is locally constant, fix y. Each point (y,x) has a neighbourhood $V_x \times W_x$, with V_x open in G^{n-1} and W_x open in G, on which Φ is constant, finitely many $W_{x_1},\dots,W_{x_r}$ cover the compact space G, and for y' in the open set $V = V_{x_1}\cap\cdots\cap V_{x_r}$ and $x \in W_{x_i}$ one has $\Phi(y',x) = \Phi(y,x_i) = \Phi(y,x)$. So $(sf)(y') = (sf)(y)$ for $y' \in V$, and sf is a continuous cochain.

The identity $d(sf) + s(df) = f$, proved in Step 3 of the proof of proposition 2.3.4 for every $f \in C^n(G,\mathrm{Map}(G,J))$ with $n \geq 1$, holds in particular for $f \in C^n_{\mathrm{cts}}(G,I)$, and there sf is a continuous cochain. So if f is a continuous cocycle then $f = d(sf)$ is the coboundary of a continuous cochain, and $\tilde{H}^n(G,I) = 0$ for every $n \geq 1$.

Step 5: Universality. Let M be a discrete G-module and $n \geq 1$. The monomorphism u_M of Step 3 induces a map $\tilde{H}^n(G,M) \to \tilde{H}^n(G,\mathrm{Map}^{\mathrm{cts}}(G,M))$ whose target is 0 by Step 4, so u_M effaces $\tilde{H}^n(G,-)$ at M in the sense of *Homological Algebra* [13, Effaceable Functor]. The δ-functor of Step 2 is therefore effaceable in positive degrees, and it is universal by *Homological Algebra* [13, An Effaceable δ-Functor Is Universal]. This proves (1).

Step 6: Comparison. The category $\mathrm{Mod}_G^{\mathrm{disc}}$ is abelian (proposition 3.1.2) and has enough injectives (proposition 3.1.3), and $(-)^G$ is additive and, by Step 2, left exact. By *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor], the derived functors $H^n(G,-)$ of definition 3.1.4, with the connecting homomorphisms of their long exact sequences, form a universal δ-functor, and the augmentation of an injective resolution induces a natural isomorphism $M^G \cong H^0(G,M)$. Composed with $\tilde{H}^0(G,M) = M^G$, this is a natural isomorphism $\varphi\colon \tilde{H}^0(G,-) \Rightarrow H^0(G,-)$. Universality of $\tilde{H}^\bullet(G,-)$ extends φ to exactly one morphism of δ-functors $\tilde{H}^\bullet(G,-) \to H^\bullet(G,-)$, and as both δ-functors are universal, each of its components is an isomorphism by *Homological Algebra* [13, A Universal δ-Functor Is Determined by Its Degree-Zero Part]. This proves (2).

Henceforth, $H^n(G, M)$ refers to either the derived functor definition or the continuous cochain definition interchangeably, and by proposition 3.1.6(2) the same holds for the connecting homomorphisms of its long exact sequences.

The low-degree continuous cohomology groups admit the same concrete interpretations as in section 2.4. In degree zero, $H^0(G, M) = M^G$ by definition. In degree one, $H^1(G, M)$ classifies continuous crossed homomorphisms modulo principal ones: a continuous 1-cocycle is a continuous map $f : G \to M$ satisfying $f(gh) = g \cdot f(h) + f(g)$, and two such maps are cohomologous if they differ by $g \mapsto g \cdot m - m$ for some fixed $m \in M$. In degree two, $H^2(G, M)$ classifies topological group extensions of G by M (continuous sections, compatible with the profinite topology on G). These interpretations carry over verbatim from the discrete theory because the continuity constraint is built into the cochain complex.

Proposition 3.1.7: H^1 of a Profinite Group with Trivial Action

Let G be a profinite group and M a discrete G-module on which G acts trivially. Then $H^1(G, M) \cong \mathrm{Hom}_{\mathrm{cts}}(G, M)$, the group of continuous homomorphisms $G \to M$, naturally in M. A homomorphism $\chi\colon G \to M$ is continuous if and only if its kernel is open.

***Proof*:**

By proposition 3.1.6(2), $H^1(G, M) \cong \tilde{H}^1(G, M) = Z^1_{\mathrm{cts}}(G, M)/B^1_{\mathrm{cts}}(G, M)$, naturally in M. With the trivial action, the coboundary of definition 3.1.5 on $f \in C^1_{\mathrm{cts}}(G, M)$ is $(d^1 f)(g, h) = f(h) - f(gh) + f(g)$, so the continuous 1-cocycles are exactly the continuous homomorphisms $G \to M$. The coboundary of $m \in C^0_{\mathrm{cts}}(G, M) = M$ is $g \mapsto g \cdot m - m = 0$, so $B^1_{\mathrm{cts}}(G, M) = 0$. A map of modules acts on both sides by composition, which gives the naturality. The fibres of a homomorphism χ are the cosets of $\ker \chi$, and M is discrete, so χ is continuous exactly when $\ker \chi$ is open.

Example 3.1.8: Continuous Cohomology of Running Examples

1. Let $G = \hat{\mathbb{Z}}$ act trivially on $M = \mathbb{Z}$. An element $m \in M$ is G-invariant if and only if $g \cdot m = m$ for all $g \in G$, which is automatic since the action is trivial. Hence $H^0(\hat{\mathbb{Z}}, \mathbb{Z}) = \mathbb{Z}$. A continuous 1-cocycle $f : \hat{\mathbb{Z}} \to \mathbb{Z}$ is a continuous group homomorphism (since the action is trivial, the cocycle condition reduces to $f(g + h) = f(g) + f(h)$). But $\hat{\mathbb{Z}}$ is compact and $\mathbb{Z}$ is discrete, so the image of f is a compact subgroup of $\mathbb{Z}$, hence $\{0\}$. Therefore $H^1(\hat{\mathbb{Z}}, \mathbb{Z}) = 0$.

2. Let $G = G_{\mathbb{Q}_p} = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ act on $M = \overline{\mathbb{Q}}_p^{\times}$ by Galois automorphisms. The invariants are

$$H^0(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^{\times}) = (\overline{\mathbb{Q}}_p^{\times})^{G_{\mathbb{Q}_p}} = \mathbb{Q}_p^{\times},$$

   by the fundamental theorem of Galois theory (theorem 1.4.1). The group $H^1(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^{\times})$ will be computed later: Hilbert's Theorem 90 (theorem 8.1.1(2)) asserts that it vanishes. The group $H^2(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^{\times})$ will be identified with $\mathbb{Q}/\mathbb{Z}$ via the invariant map of the Brauer group (chapter 7).

3. Let $G = G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ act on $\mathbb{C}^\times$ by complex conjugation. Since G is finite (hence discrete and profinite simultaneously), continuous cohomology coincides with the discrete group cohomology of chapter 2. The invariants are $H^0(G_{\mathbb{R}}, \mathbb{C}^\times) = \mathbb{R}^\times$. By example 2.3.8, the cohomology of a cyclic group of order 2 acting on a module M is 2-periodic in positive degrees, with

$$H^{2k+1}(G_{\mathbb{R}}, \mathbb{C}^\times) \cong \ker(\mathrm{Nm})/\mathrm{im}(g-1), \qquad H^{2k}(G_{\mathbb{R}}, \mathbb{C}^\times) \cong M^G/\mathrm{Nm}(M),$$

where $\mathrm{Nm} = 1 + \sigma$ is the norm element (σ being complex conjugation) and $g - 1 = \sigma - 1$. Since $\mathrm{Nm}(z) = z \cdot \overline{z} = |z|^2$ and $(\sigma - 1)(z) = \overline{z}/z$, the norm map $\mathrm{Nm} : \mathbb{C}^\times \to \mathbb{R}^\times$ is surjective onto $\mathbb{R}^\times_{>0}$, so $H^2(G_{\mathbb{R}}, \mathbb{C}^\times) \cong \mathbb{R}^\times/\mathbb{R}^\times_{>0} \cong \mathbb{Z}/2\mathbb{Z}$. The kernel of the norm consists of elements with $|z|^2 = 1$, i.e., the unit circle S^1, while $\mathrm{im}(\sigma - 1) = \{\overline{z}/z \mid z \in \mathbb{C}^\times\}$ also equals S^1 (since $|\overline{z}/z| = 1$ and every element of S^1 has the form $e^{i\theta} = \overline{e^{i\theta/2}}/e^{i\theta/2}$). Hence $H^1(G_{\mathbb{R}}, \mathbb{C}^\times) = 0$.

The computation for $G_{\mathbb{R}}$ is a preview of a general phenomenon: Hilbert's Theorem 90 asserts $H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$ for *any* field K, and $H^2(G_K, (K^{\mathrm{sep}})^\times)$ is the Brauer group $\mathrm{Br}(K)$. These identifications are central to the arithmetic applications in chapters 8 and 10.

Proposition 3.1.9: Continuity of Cochains Is Essential

Let $G = \mathbb{Z}_p$, viewed as a profinite group under addition, act trivially on the discrete module $M = \mathbb{Q}$. Then the continuous cohomology $H^1(\mathbb{Z}_p, \mathbb{Q})$ is 0, while the group of all, not necessarily continuous, 1-cocycles modulo coboundaries is $\mathrm{Hom}_{\mathrm{grp}}(\mathbb{Z}_p, \mathbb{Q})$, which is nonzero.

***Proof*:**
The action is trivial, so a 1-cocycle f satisfies $f(g+g') = f(g)+f(g')$, that is, it is a homomorphism, and every coboundary $g \mapsto g \cdot m - m$ is 0. For the continuous group, let $\varphi\colon \mathbb{Z}_p \to \mathbb{Q}$ be a continuous homomorphism. Its kernel $\varphi^{-1}(0)$ is open, because $\mathbb{Q}$ is discrete, and an open subgroup of the compact group $\mathbb{Z}_p$ has finite index. So $\varphi(\mathbb{Z}_p)$ is a finite subgroup of $\mathbb{Q}$. The only finite subgroup of the torsion-free group $\mathbb{Q}$ is 0, so $\varphi = 0$ and $H^1(\mathbb{Z}_p, \mathbb{Q}) = 0$.

For the abstract group, $\mathbb{Q}$ is divisible, hence an injective $\mathbb{Z}$-module by part (2) of [9, Baer's Criterion]. The homomorphism $\mathbb{Z} \to \mathbb{Q}$ sending 1 to 1 therefore extends along the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$ to a homomorphism $\mathbb{Z}_p \to \mathbb{Q}$, which is nonzero because it sends 1 to 1, as claimed.

Abstract cocycles depend only on the algebraic structure of $\mathbb{Z}_p$, a torsion-free group of uncountable rank. Continuous cocycles depend on its topology, which forces every continuous homomorphism to a discrete torsion-free group to vanish.

Exercise 3.1.1

1. Let G be a profinite group and M an abelian group with a G-action. Prove that the following are equivalent:

   (a) M is a discrete G-module.

   (b) The action map $G \times M \to M$ is continuous when M carries the discrete topology.

   (c) For every $m \in M$, the orbit map $G \to M$ defined by $g \mapsto g \cdot m$ is continuous.

(d) $M = \varinjlim_{U \trianglelefteq_o G} M^U$, where the colimit is taken in the category of abelian groups over the directed system of open normal subgroups ordered by reverse inclusion.

2. Let G be a profinite group and let M, N be discrete G-modules. Show that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ (with the conjugation action $(g \cdot f)(m) = g \cdot f(g^{-1} \cdot m)$) need not be a discrete G-module in general, but that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is discrete whenever M is finitely generated as a $\mathbb{Z}$-module.

3. Let $G = \hat{\mathbb{Z}}$ and let $M = \mathbb{Z}$ with the trivial G-action. Use the continuous cochain definition to verify directly that $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$. (*Hint:* A continuous 2-cocycle $f : \hat{\mathbb{Z}} \times \hat{\mathbb{Z}} \to \mathbb{Z}$ factors through some $(\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$. Use the result from example 2.3.8 for cyclic groups and pass to the colimit.)

4. Let G be a profinite group and M a (not necessarily discrete) $\mathbb{Z}[G]$-module. Define $M^{\mathrm{disc}} = \bigcup_{U \trianglelefteq_o G} M^U$. Show that M^{disc} is the largest discrete G-submodule of M, and that the functor $M \mapsto M^{\mathrm{disc}}$ from Mod_G to $\mathrm{Mod}_G^{\mathrm{disc}}$ is right adjoint to the inclusion functor $\mathrm{Mod}_G^{\mathrm{disc}} \hookrightarrow \mathrm{Mod}_G$.

5. Let p be an odd prime, and take as given the form of local class field theory in which the local Artin map (theorem 8.4.4) induces an isomorphism of profinite groups from the profinite completion of $\mathbb{Q}_p^\times$ onto $G_{\mathbb{Q}_p}^{\mathrm{ab}}$. Using proposition 3.1.7, compute $H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z})$ for each $n \geq 1$, with $G_{\mathbb{Q}_p}$ acting trivially on $\mathbb{Z}/n\mathbb{Z}$.

3.2 Cohomology as a Colimit

The definition of continuous cohomology in section 3.1 is conceptually clean but not yet directly computable: it requires constructing injective resolutions or working with continuous cochains on a profinite group, which is an uncountable topological space. The theorem of this section eliminates both difficulties at once. It shows that the continuous cohomology of a profinite group G is the colimit of the ordinary (discrete) cohomology of its finite quotients (a reduction that makes every computation in this book possible).

The mechanism is elementary: a continuous cochain on G is locally constant, so it factors through a finite quotient. Making this precise requires showing that the entire cochain complex decomposes as a colimit.

Lemma 3.2.1: The Continuous Cochain Complex as a Colimit

Let G be a profinite group and M a discrete G-module. For each open normal subgroup U of G, the finite group G/U acts on M^U, and the ordinary cochain complex $C^\bullet(G/U, M^U)$ from definition 2.3.3 embeds into $C^\bullet_{\mathrm{cts}}(G, M)$ via inflation: a cochain $f : (G/U)^n \to M^U$ is sent to the composition $G^n \twoheadrightarrow (G/U)^n \xrightarrow{f} M^U \hookrightarrow M$. This embedding is compatible with the coboundary maps, giving an inclusion of cochain complexes $C^\bullet(G/U, M^U) \hookrightarrow C^\bullet_{\mathrm{cts}}(G, M)$. The resulting colimit map

$$\varinjlim_{U \trianglelefteq_o G} C^n(G/U, M^U) \xrightarrow{\sim} C^n_{\mathrm{cts}}(G, M)$$

is an isomorphism for each $n \geq 0$, where the colimit is taken over open normal subgroups of G ordered by reverse inclusion (so that the transition maps are inflation).

***Proof*:**
Step 1: Surjectivity. Let $f : G^n \to M$ be a continuous n-cochain. Since M is discrete, f is locally constant: each $x \in G^n$ has an open neighbourhood on which f is constant. The open normal subgroups of G form a neighbourhood basis of the identity (Step 2 of the proof of theorem 1.1.10), so the sets $x \cdot V^n$, for V open normal in G, form a neighbourhood basis of x in G^n, and there is an open normal subgroup U_x with f constant on $x \cdot U_x^n$. These sets cover G^n, and by compactness finitely many of them, say $x_1 \cdot U_{x_1}^n, \ldots, x_r \cdot U_{x_r}^n$, already do. Put $U = U_{x_1} \cap \cdots \cap U_{x_r}$, an open normal subgroup of G. Every $y \in G^n$ lies in some $x_i \cdot U_{x_i}^n$, and then $y \cdot U^n \subseteq x_i \cdot U_{x_i}^n U^n = x_i \cdot U_{x_i}^n$, on which f is constant. So f is constant on the cosets of U^n in G^n, which means that f factors through the projection $G^n \twoheadrightarrow (G/U)^n$.

It remains to check that f takes values in M^U, possibly after shrinking U. Fix a value $m = f(g_1, \ldots, g_n)$ in the image of f. Since M is a discrete G-module, the stabilizer $\mathrm{Stab}_G(m)$ is open, so it contains some open normal subgroup V_m. The image of f is finite (as f is locally constant on the compact space G^n), so the intersection $V = \bigcap_{m \in \mathrm{im}(f)} V_m$ is a finite intersection of open subgroups and hence open. Replacing U by $U \cap V$ (which is still open and normal), the cochain f factors through $(G/(U \cap V))^n$ and every value of f is fixed by $U \cap V$. Hence f defines an element of $C^n(G/(U \cap V), M^{U \cap V})$, as required.

Step 2: Injectivity. The colimit is a filtered colimit of abelian groups, so an element of the colimit is zero if and only if it becomes zero in some term of the system. An element of $C^n(G/U, M^U)$ that maps to zero in $C^n_{\mathrm{cts}}(G, M)$ is a cochain $f : (G/U)^n \to M^U$ such that the inflated cochain $G^n \to M$ is zero. Since the projection $G^n \to (G/U)^n$ is surjective, f itself must be zero.

Step 3: Compatibility with differentials. The inflation map $C^n(G/U, M^U) \to C^n_{\mathrm{cts}}(G, M)$ commutes with the coboundary operator d^n because the coboundary formula involves only the group operation and the G-action, both of which are compatible with the projection $G \twoheadrightarrow G/U$.

The colimit formula for cohomology is now an immediate consequence.

Theorem 3.2.2: Cohomology as a Colimit

Let G be a profinite group and M a discrete G-module. For every $n \geq 0$, there is a canonical isomorphism

$$H^n(G, M) \cong \varinjlim_{U \trianglelefteq_o G} H^n(G/U, M^U),$$

where the colimit runs over the directed system of open normal subgroups of G (ordered by reverse inclusion), and the transition maps are the inflation homomorphisms $\mathrm{Inf} : H^n(G/U, M^U) \to H^n(G/V, M^V)$ for $V \subseteq U$.

***Proof*:**
By proposition 3.1.6, $H^n(G, M)$ is the n-th cohomology of the continuous cochain complex $C^\bullet_{\mathrm{cts}}(G, M)$. By lemma 3.2.1, this complex is the colimit of the cochain complexes $C^\bullet(G/U, M^U)$:

$$H^n(G, M) = H^n(C^\bullet_{\mathrm{cts}}(G, M)) = H^n\left(\varinjlim_U C^\bullet(G/U, M^U)\right).$$

The open normal subgroups of G form a directed set under reverse inclusion (for any $U_1, U_2 \trianglelefteq_{\mathrm{o}} G$, the intersection $U_1 \cap U_2$ is again open and normal, and is contained in both). The colimit of the cochain complexes is therefore a filtered colimit. Since filtered colimits in the category of abelian groups are exact (they commute with finite limits, and in particular with kernels and cokernels), taking cohomology commutes with the colimit:

$$H^n\left(\varinjlim_U C^\bullet(G/U, M^U)\right) \cong \varinjlim_U H^n\big(C^\bullet(G/U, M^U)\big) = \varinjlim_U H^n(G/U, M^U).$$

The last equality uses the cocycle description of finite-group cohomology from proposition 2.3.4.

This theorem is the computational engine of profinite group cohomology. It converts every cohomological question about a profinite group into a question about the cohomology of its finite quotients, where the tools of chapter 2 apply directly. For Galois cohomology, it says that the cohomology of an absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ is computed by taking the colimit over all finite Galois extensions L/K:

$$H^n(G_K, M) = \varinjlim_{L/K} H^n(\mathrm{Gal}(L/K), M^{G_L}),$$

where L ranges over finite Galois extensions of K contained in a fixed separable closure K^{sep}, and $G_L = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ is the corresponding open normal subgroup. Every element of $H^n(G_K, M)$ "lives at a finite level": it is represented by a cocycle of some finite Galois group.

The first payoff is the complete cohomology of the simplest infinite profinite group. It is worth having in closed form, because $\widehat{\mathbb{Z}}$ is the absolute Galois group of every finite field, and the answer is what makes arithmetic over finite fields tractable: one invariant module in degree zero, one coinvariant module in degree one, and nothing above.

Theorem 3.2.3: The Cohomology of $\widehat{\mathbb{Z}}$

Let $\Gamma = \widehat{\mathbb{Z}}$ with topological generator φ, and let M be a *torsion* discrete Γ-module. Then

$$H^0(\Gamma, M) = M^\Gamma, \qquad H^1(\Gamma, M) = M_\Gamma, \qquad H^i(\Gamma, M) = 0 \quad (i \geq 2)$$

where $M^\Gamma = \ker(\varphi - 1)$ and $M_\Gamma = M/(\varphi - 1)M$ are the invariants and coinvariants of definition 2.1.7.

***Proof*:**

First, the descriptions of M^Γ and M_Γ hold for every discrete Γ-module M. The stabilizer of $x \in M$ is open, hence of some finite index r (proposition 1.2.2), and it contains $r\widehat{\mathbb{Z}}$, as Γ is abelian. The image of φ generates the finite quotient $\Gamma/r\widehat{\mathbb{Z}} \cong \mathbb{Z}/r\mathbb{Z}$, since $\langle\varphi\rangle$ is dense. So every $g \in \Gamma$ acts on x as φ^k for some $0 \leq k < r$. Hence x is invariant if and only if $\varphi x = x$, and $gx - x = (\varphi^k - 1)x = (\varphi - 1)(1 + \varphi + \cdots + \varphi^{k-1})x$ lies in $(\varphi - 1)M$. So the subgroup of definition 2.1.7 generated by the elements $gx - x$ is $(\varphi - 1)M$.

Next, reduce to finite M. A torsion discrete Γ-module is the filtered union of its finite Γ-submodules: each element has finite orbit, since its stabilizer is open, and finite order, so it generates a finite submodule. Cohomology commutes with that union: a continuous cochain $\Gamma^n \to M$ takes finitely many values, as Γ^n is compact, so it lands in a finite submodule. Hence

$C^\bullet_{\mathrm{cts}}(\Gamma, M)$ is the union of the subcomplexes $C^\bullet_{\mathrm{cts}}(\Gamma, M_i)$ over the finite submodules M_i, and filtered colimits of abelian groups are exact. Invariants and coinvariants commute with the union too. For finite M the isomorphisms constructed below are natural in M: they are induced by the identity of M on the periodic complexes of example 2.3.8, and these are identified with cohomology through chain maps between resolutions of $\mathbb{Z}$ that do not depend on M (proposition 2.3.9). So they are compatible with the inclusions of finite submodules and pass to the colimit, and it suffices to treat M finite. Let e be an exponent for M.

The open subgroups of $\widehat{\mathbb{Z}}$ are the $m\widehat{\mathbb{Z}}$ with quotient $\mathbb{Z}/m\mathbb{Z}$, cyclic of order m generated by the image of φ. Fix m with $M^{m\widehat{\mathbb{Z}}} = M$, which is possible because the action is continuous and M finite. Then for every multiple the module of invariants is all of M, and theorem 3.2.2 gives

$$H^i(\Gamma, M) = \varinjlim_{c} H^i\big(\mathbb{Z}/cm\mathbb{Z},\ M\big)$$

the colimit over $c \geq 1$ with inflation as transition maps.

Each term is computed by example 2.3.8, and by proposition 2.3.9, applied with $d = c'/c$, the transition from level cm to level $c'm$, for $c \mid c'$, is induced by multiplication by $(c'/c)^{\lfloor i/2 \rfloor}$ on M.

In degree zero every term is $M^{\mathbb{Z}/cm\mathbb{Z}} = M^\Gamma$, every transition is the identity, and the colimit is M^Γ. In degree one the exponent $\lfloor i/2 \rfloor$ is zero, so the transition from level cm to level $c'm$ is the map

$$\ker(N_{cm})/(\varphi - 1)M \longrightarrow \ker(N_{c'm})/(\varphi - 1)M$$

induced by the identity of M, where N_{cm} denotes the norm element of $\mathbb{Z}/cm\mathbb{Z}$. By proposition 2.3.9, applied to $\mathbb{Z}/cm\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$, the element N_{cm} acts on M as $c\,N_m$. When $e \mid c$ this is zero, so $\ker(N_{cm}) = M$ and the term at level cm is $M/(\varphi - 1)M$, with identity transitions among such levels. The levels cm with $e \mid c$ are cofinal, so the colimit is $M/(\varphi - 1)M = M_\Gamma$.

For $i \geq 2$ the exponent is at least one. Every group in the system is a subquotient of M, hence killed by e, and the transition from level cm to level ecm is multiplication by $e^{\lfloor i/2 \rfloor}$, which is zero on such a group. Every element of the colimit therefore dies at a finite stage, so the colimit vanishes, completing the proof.

The vanishing above degree one is exactly the statement that $\widehat{\mathbb{Z}}$ has cohomological dimension one, recovered independently in example 5.1.5, and the degree-one answer is the one used whenever a Frobenius acts: invariants below, coinvariants above, and the two have the same cardinality when M is finite, since the kernel and the cokernel of the endomorphism $\varphi - 1$ of a finite abelian group have the same order.

Corollary 3.2.4: Torsion Property

Let G be a profinite group and M a discrete G-module. For every $n \geq 1$, the group $H^n(G, M)$ is torsion: every element has finite order.

***Proof*:**

By theorem 3.2.2, every element $\alpha \in H^n(G, M)$ lies in the image of $H^n(G/U, M^U)$ for some open normal subgroup U. The group $Q = G/U$ is finite, and it suffices to show that $|Q| \cdot H^n(Q, A) = 0$ for every Q-module A and every $n \geq 1$. By proposition 2.3.4, $H^n(Q, A)$ is computed by the

inhomogeneous cochains of definition 2.3.3. For $f \in C^n(Q, A)$ define $F \in C^{n-1}(Q, A)$ by

$$F(g_1, \dots, g_{n-1}) = \sum_{g \in Q} f(g_1, \dots, g_{n-1}, g)$$

Sum the coboundary formula for $(df)(g_1, \dots, g_n, g)$ over $g \in Q$. The first term gives $g_1 \cdot F(g_2, \dots, g_n)$. For $1 \le i \le n-1$ the term with $g_i g_{i+1}$ gives $(-1)^i F(g_1, \dots, g_i g_{i+1}, \dots, g_n)$. The term with $g_n g$ gives $(-1)^n F(g_1, \dots, g_{n-1})$, after substituting $g_n^{-1} g$ for g. The last term gives $(-1)^{n+1}|Q|\, f(g_1, \dots, g_n)$. Comparing with the coboundary formula for F,

$$\sum_{g \in Q} (df)(g_1, \dots, g_n, g) = (dF)(g_1, \dots, g_n) + (-1)^{n+1}|Q|\, f(g_1, \dots, g_n)$$

If f is a cocycle the left side vanishes, so $|Q| \cdot f = (-1)^n\, dF$ is a coboundary. Hence $|Q| \cdot H^n(Q, A) = 0$, and in particular $|G/U| \cdot \alpha = 0$.

The torsion property has an immediate and important structural consequence: $H^n(G, M)$ decomposes as a direct sum of its p-primary components $H^n(G, M)\{p\}$, the group of $x \in H^n(G, M)$ with $p^k x = 0$ for some $k \ge 1$, over all primes p. Many arguments in Galois cohomology proceed by reducing to a single prime at a time, and the torsion property guarantees that no information is lost in doing so.

A word of caution: the torsion property does *not* imply that $H^n(G, M)$ has bounded exponent. The orders of the finite quotients G/U can grow without bound, and different elements of the colimit may be killed by different orders. For example, $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$ is torsion but not of bounded exponent: every positive integer appears as the order of some element.

Another consequence of the colimit formula is that continuous cohomology commutes with colimits in the coefficient module.

Corollary 3.2.5: Commutation with Coefficient Colimits

Let G be a profinite group and $(M_i)_{i \in I}$ a directed system of discrete G-modules. Then for each $n \ge 0$,

$$H^n\left(G,\ \varinjlim_i M_i\right) \cong \varinjlim_i H^n(G, M_i).$$

***Proof*:**
Put $M = \varinjlim_i M_i$, with canonical maps $\lambda_i \colon M_i \to M$. It is a discrete G-module, as each element is $\lambda_i(x)$ for some i and x, and its stabilizer contains that of x. Composition with the λ_i gives a map of complexes $\Lambda \colon \varinjlim_i C^\bullet_{\mathrm{cts}}(G, M_i) \to C^\bullet_{\mathrm{cts}}(G, M)$, and Λ is an isomorphism in each degree n. A continuous cochain $G^n \to M$, or $G^n \to M_i$, takes finitely many values, its fibres being open, disjoint and covering the compact space G^n. For surjectivity, let $f \colon G^n \to M$ be continuous, with values $c_1, \dots, c_r$. Each c_l lies in the image of some λ_{i_l}, and at an upper bound i of $i_1, \dots, i_r$ there are $x_l \in M_i$ with $\lambda_i(x_l) = c_l$. The cochain equal to x_l on $f^{-1}(c_l)$ is continuous, and λ_i carries it to f. For injectivity, let $f_i \colon G^n \to M_i$ be continuous with $\lambda_i \circ f_i = 0$. Each of the finitely many values of f_i maps to 0 in M, hence to 0 in some M_j with $j \ge i$, as the colimit is filtered, and at an upper bound k of these indices the image of f_i in $C^n_{\mathrm{cts}}(G, M_k)$ is 0. So f_i represents 0 in the colimit. Taking cohomology, and using the exactness of filtered colimits of

abelian groups and proposition 3.1.6(2), gives the result.

This commutation property is used constantly in Galois cohomology, often without explicit mention. For instance, a discrete G_K-module M that is a torsion abelian group can be written as $M = \varinjlim M_i$ where each M_i is a finitely generated (hence finite) G_K-submodule, and the cohomology of M reduces to the cohomology of its finite submodules.

Example 3.2.6: Cohomology of $\hat{\mathbb{Z}}$ via the Colimit

1. The open normal subgroups of $\hat{\mathbb{Z}}$ are $n\hat{\mathbb{Z}}$ for $n \geq 1$, with quotients $\hat{\mathbb{Z}}/n\hat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$. By theorem 3.2.2, for any discrete $\hat{\mathbb{Z}}$-module M with trivial action,

$$H^q(\hat{\mathbb{Z}}, M) = \varinjlim_n H^q(\mathbb{Z}/n\mathbb{Z}, M),$$

where the colimit is over positive integers ordered by divisibility. The cohomology of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ with trivial coefficients in M is periodic (example 2.3.8):

$$H^q(\mathbb{Z}/n\mathbb{Z}, M) \cong \begin{cases} M & q = 0, \\ M[n] & q \text{ odd}, q \geq 1, \\ M/nM & q \text{ even}, q \geq 2, \end{cases}$$

where $M[n] = \ker(n : M \to M)$ is the n-torsion submodule and M/nM is the quotient by the image of multiplication by n. By proposition 2.3.9, for $n \mid n'$ the transition map from level n to level n' in degree q is induced by multiplication by $(n'/n)^{\lfloor q/2 \rfloor}$ on M.

For $M = \mathbb{Z}$: the n-torsion $\mathbb{Z}[n] = 0$, so $H^q(\hat{\mathbb{Z}}, \mathbb{Z}) = 0$ for all odd $q \geq 1$. In degree 2 the transition $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n'\mathbb{Z}$ is multiplication by n'/n, which is the canonical embedding $k/n \mapsto k/n$ in $\mathbb{Q}/\mathbb{Z}$ (as computed in exercise 3.1.1 (3)), giving

$$H^2(\hat{\mathbb{Z}}, \mathbb{Z}) = \varinjlim_n \mathbb{Z}/n\mathbb{Z} = \mathbb{Q}/\mathbb{Z}$$

In degree $2j$ with $j \geq 2$ the transition from level n to level n^2 is the map $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n^2\mathbb{Z}$ induced by multiplication by n^j, which is zero because n^2 divides n^j. Every element of the colimit dies at a finite stage, and $H^q(\hat{\mathbb{Z}}, \mathbb{Z}) = 0$ for every $q \geq 3$.

2. For $M = \mathbb{Z}/m\mathbb{Z}$ (trivial action): the n-torsion is $(\mathbb{Z}/m\mathbb{Z})[n] = \mathbb{Z}/\gcd(m, n)\mathbb{Z}$, and the quotient is $(\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}$. Once $m \mid n$, both are $\mathbb{Z}/m\mathbb{Z}$. In degree 1 the transitions are induced by the identity of M, so $H^1(\hat{\mathbb{Z}}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z}$. In degree $q \geq 2$ the transition from level n to level mn is multiplication by $m^{\lfloor q/2 \rfloor}$, which is zero on $\mathbb{Z}/m\mathbb{Z}$, so

$$H^q(\hat{\mathbb{Z}}, \mathbb{Z}/m\mathbb{Z}) = 0 \qquad \text{for all } q \geq 2$$

as theorem 3.2.3 predicts for the torsion module $\mathbb{Z}/m\mathbb{Z}$.

The cohomology of $\hat{\mathbb{Z}}$ is not periodic, although that of each finite cyclic quotient is. In degree q the transition maps are multiplication by the powers $(n'/n)^{\lfloor q/2 \rfloor}$ of the index, and they kill every class in the colimit when $q \geq 3$, and already when $q \geq 2$ if the coefficients are torsion. The cohomological dimension of a profinite group (the largest n for which $H^n(G, M)$ can be nonzero) will be a central invariant in chapter 5.

Example 3.2.7: Galois Cohomology of the Multiplicative Group

1. Let K be a field and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ its absolute Galois group. The multiplicative group $(K^{\mathrm{sep}})^\times$ is a discrete G_K-module: for any $\alpha \in (K^{\mathrm{sep}})^\times$, the stabilizer $\mathrm{Stab}_{G_K}(\alpha)$ is the open subgroup $G_{K(\alpha)} = \mathrm{Gal}(K^{\mathrm{sep}}/K(\alpha))$. The colimit formula gives

$$H^n(G_K, (K^{\mathrm{sep}})^\times) = \varinjlim_{L/K} H^n(\mathrm{Gal}(L/K), L^\times),$$

   where L ranges over finite Galois extensions of K inside K^{sep}.

2. For $K = \mathbb{Q}_p$, the colimit formula applied in degree 2 gives

$$H^2(G_{\mathbb{Q}_p}, \overline{\mathbb{Q}}_p^\times) = \varinjlim_{L/\mathbb{Q}_p} H^2(\mathrm{Gal}(L/\mathbb{Q}_p), L^\times).$$

   Each group $H^2(\mathrm{Gal}(L/\mathbb{Q}_p), L^\times)$ classifies central simple algebras over $\mathbb{Q}_p$ split by L (chapter 7), and the colimit assembles into the Brauer group $\mathrm{Br}(\mathbb{Q}_p)$. The identification $\mathrm{Br}(\mathbb{Q}_p) \cong \mathbb{Q}/\mathbb{Z}$ via the invariant map is one of the central results of local class field theory (chapter 8).

3. For $K = \mathbb{R}$: the only nontrivial finite Galois extension is $\mathbb{C}/\mathbb{R}$, so the colimit stabilizes immediately and $H^n(G_{\mathbb{R}}, \mathbb{C}^\times) = H^n(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$ for all n. By the computation in example 3.1.8, $H^1(G_{\mathbb{R}}, \mathbb{C}^\times) = 0$ and $H^2(G_{\mathbb{R}}, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$. The latter is $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$, reflecting the fact that the only non-split central simple algebra over $\mathbb{R}$ is the quaternion algebra $\mathbb{H}$.

The colimit formula also computes connecting homomorphisms. Let $0 \to M' \to M \xrightarrow{\beta} M'' \to 0$ be a short exact sequence of discrete G-modules. For an open normal subgroup U the sequence $0 \to (M')^U \to M^U \to (M'')^U$ is exact, but $M^U \to (M'')^U$ need not be surjective, so the long exact sequence of the finite group G/U need not apply to these three modules. By proposition 3.1.6(2) the long exact sequence

$$\cdots \to H^n(G, M') \to H^n(G, M) \to H^n(G, M'') \xrightarrow{\delta} H^{n+1}(G, M') \to \cdots \tag{3.1}$$

of the derived functors is the long exact sequence of the short exact sequence of continuous cochain complexes in proposition 3.1.6(1), and its connecting homomorphism is computed at a finite level. Let α be a continuous n-cocycle with values in M''. Lifting each of its finitely many values to M, as in Step 1 of the proof of proposition 3.1.6, gives a continuous cochain F with $\beta \circ F = \alpha$. By lemma 3.2.1 there is an open normal subgroup V such that F is inflated from a cochain of G/V with values in M^V. Then dF takes values in M', because $\beta \circ dF = d\alpha = 0$. It is inflated from the

coboundary of that cochain of G/V (Step 3 of the proof of lemma 3.2.1), a cocycle of the finite group G/V with values in $M^V \cap M' = (M')^V$, and $\delta[\alpha] = [dF]$ by the construction of the connecting homomorphism in the Snake Lemma (*Homological Algebra* [13, Snake Lemma]). Every connecting homomorphism in profinite cohomology is therefore a diagram chase in the cochain complex of a finite quotient.

Exercise 3.2.1

1. Let G be a profinite group and $0 \to \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$ the short exact sequence of discrete G-modules (with trivial G-action on all three terms). Write out the long exact sequence in cohomology and use the colimit formula to compute the connecting homomorphism $\delta : H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z})$. Identify δ as the map $\mathrm{Hom}_{\mathrm{cts}}(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z})$ that sends a continuous homomorphism $\chi : G \to \mathbb{Z}/p\mathbb{Z}$ to the class of the extension $0 \to \mathbb{Z} \to E_\chi \to G/\ker\chi \to 0$.

2. Let $G = \mathbb{Z}_p$ (the additive group of p-adic integers, viewed as a pro-p-group). Compute $H^n(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z})$ for all $n \geq 0$ using the colimit formula and the cohomology of $\mathbb{Z}/p^k\mathbb{Z}$ with coefficients in $\mathbb{Z}/p\mathbb{Z}$ (trivial action). (*Hint:* The open subgroups of $\mathbb{Z}_p$ are $p^k\mathbb{Z}_p$ with quotient $\mathbb{Z}/p^k\mathbb{Z}$. Compute the transition maps with proposition 2.3.9.)

3. Give an example of a short exact sequence $0 \to M' \to M \to M'' \to 0$ of discrete G-modules and an open normal subgroup U for which $M^U \to (M'')^U$ is not surjective. Then give a direct proof, not using the derived functor formalism, that the long exact sequence (3.1) of continuous cochain cohomology is exact, by passing to the colimit over U in the sequences of cochain complexes $C^\bullet(G/U, (M')^U) \to C^\bullet(G/U, M^U) \to C^\bullet(G/U, (M'')^U)$.

4. Show that the torsion property (corollary 3.2.4) fails in degree zero: give an example of a profinite group G and a discrete G-module M such that $H^0(G, M) = M^G$ contains elements of infinite order.

5. Let G be a profinite group and M a *finite* discrete G-module (i.e., $|M| < \infty$). Show that $|M| \cdot H^n(G, M) = 0$ for every $n \geq 0$, so that $H^n(G, M)$ is a torsion group of exponent dividing $|M|$. (*Hint:* Use dimension shifting: embed M into a coinduced module and apply induction on n. The cokernel of the embedding need not be finite, so run the induction over all discrete G-modules killed by $|M|$.)

3.3 Inflation, Restriction, and Corestriction

The colimit formula of theorem 3.2.2 expresses $H^n(G, M)$ in terms of the cohomology of finite quotients, with the transition maps being a specific instance of a general construction: *inflation*. This section develops three functorial operations (inflation, restriction, and corestriction) that relate the cohomology of a profinite group to the cohomology of its subgroups and quotients. These maps are the basic tools for computing cohomology in practice, and their interactions (particularly the inflation-restriction exact sequence and the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$) underlie most of the structural results in Parts II and III.

Throughout this section, G denotes a profinite group and M a discrete G-module.

Definition 3.3.1: Inflation

Let N be a closed normal subgroup of G. The quotient G/N is a profinite group (by proposition 1.2.4), and M^N is a discrete G/N-module. The *inflation map*

$$\mathrm{Inf} = \mathrm{Inf}^G_{G/N} : H^n(G/N, M^N) \to H^n(G, M)$$

is defined at the cochain level by precomposition with the projection $\pi : G \twoheadrightarrow G/N$. Explicitly, if $f : (G/N)^n \to M^N$ is a continuous n-cocycle, then

$$\mathrm{Inf}(f)(g_1, \ldots, g_n) = f(\pi(g_1), \ldots, \pi(g_n)) \in M^N \subseteq M.$$

Since π is a continuous group homomorphism and $M^N \hookrightarrow M$ is G-equivariant (with G acting on M^N through G/N), the inflated cochain is continuous and the coboundary of an inflated cochain is the inflation of the coboundary. Hence Inf is a well-defined homomorphism on cohomology.

Inflation is the map that "extends a cocycle from a quotient to the full group by making it constant on N-cosets." The transition maps in the colimit formula of theorem 3.2.2 are precisely inflation maps: for open normal subgroups $V \subseteq U$, the map $H^n(G/U, M^U) \to H^n(G/V, M^V)$ is the inflation along $G/V \twoheadrightarrow G/U$.

Definition 3.3.2: Restriction

Let H be a closed subgroup of G. Every discrete G-module M is also a discrete H-module (if $\mathrm{Stab}_G(m)$ is open in G, then $\mathrm{Stab}_H(m) = H \cap \mathrm{Stab}_G(m)$ is open in H). The *restriction map*

$$\mathrm{Res} = \mathrm{Res}^H_G : H^n(G, M) \to H^n(H, M)$$

is defined at the cochain level by restricting the domain: if $f : G^n \to M$ is a continuous n-cocycle, then $\mathrm{Res}(f) = f|_{H^n} : H^n \to M$ is a continuous n-cocycle for H. The coboundary formula is unchanged, so restriction is a chain map and induces a well-defined map on cohomology.

In degree zero, restriction is the inclusion $M^G \hookrightarrow M^H$: if m is fixed by all of G, it is certainly fixed by the subgroup H. In degree one, restriction sends a continuous crossed homomorphism $f : G \to M$ to its restriction $f|_H : H \to M$, which is again a continuous crossed homomorphism.

Both inflation and restriction are natural in M, since precomposition with π and restriction of the domain commute with composition by a module map. Restriction also commutes with the connecting homomorphisms of the long exact sequence of a short exact sequence $0 \to M' \to M \to M'' \to 0$ of discrete G-modules. Inflation commutes with them for short exact sequences of discrete G/N-modules, regarded as G-modules through π. For a general sequence of G-modules the N-invariants need not form a short exact sequence. Both statements are part of lemma 3.3.6.

The first fundamental relation between inflation and restriction constrains how cohomology classes can "come from a quotient."

Proposition 3.3.3: Inflation-Restriction Exact Sequence

Let N be a closed normal subgroup of G and M a discrete G-module. The sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\text{Inf}} H^1(G, M) \xrightarrow{\text{Res}} H^1(N, M)$$

is exact. That is: inflation is injective on H^1, and a class in $H^1(G, M)$ restricts to zero in $H^1(N, M)$ if and only if it is inflated from $H^1(G/N, M^N)$.

***Proof*:**

Step 1: Inf *is injective.* Let $f : G/N \to M^N$ be a continuous 1-cocycle such that $\text{Inf}(f) = f \circ \pi$ is a coboundary: $f(\pi(g)) = g \cdot m - m$ for some $m \in M$ and all $g \in G$. For $n \in N$, this gives $f(\pi(n)) = f(\bar{1}) = 0 = n \cdot m - m$, so $m \in M^N$. Then $f(\bar{g}) = g \cdot m - m$ for all $\bar{g} \in G/N$ (where g is any lift of $\bar{g}$, and the right side depends only on $\bar{g}$ because $m \in M^N$). Hence f is a coboundary in $C^1(G/N, M^N)$.

Step 2: $\text{im}(\text{Inf}) \subseteq \ker(\text{Res})$. If $f : G/N \to M^N$ is a 1-cocycle, then $\text{Res}(\text{Inf}(f))(n) = f(\pi(n)) = f(\bar{1}) = 0$ for all $n \in N$. So $\text{Res} \circ \text{Inf} = 0$.

Step 3: $\ker(\text{Res}) \subseteq \text{im}(\text{Inf})$. Let $f\colon G \to M$ be a continuous 1-cocycle whose class restricts to 0 in $H^1(N, M)$, so there is $m_0 \in M$ with $f(n) = n \cdot m_0 - m_0$ for all $n \in N$. Replacing f by the cohomologous cocycle $f - d^0(m_0)$, we may assume $f|_N = 0$. For $g \in G$ and $n \in N$ the cocycle condition gives $f(gn) = g \cdot f(n) + f(g) = f(g)$, so f is constant on the cosets gN and factors through G/N. It also gives $f(ng) = n \cdot f(g) + f(n) = n \cdot f(g)$. Since N is normal, $ng = g(g^{-1}ng)$ lies in gN, so $f(ng) = f(g)$. Hence $n \cdot f(g) = f(g)$ for all $n \in N$, that is, $f(g) \in M^N$. The induced map $\bar{f}\colon G/N \to M^N$ is continuous, as G/N carries the quotient topology, and it is a 1-cocycle, since $\bar{f}(\bar{g}\bar{g}') = f(gg') = g \cdot f(g') + f(g) = \bar{g} \cdot \bar{f}(\bar{g}') + \bar{f}(\bar{g})$ with G/N acting on M^N through G. So $f = \text{Inf}(\bar{f})$, as we sought to show.

The exact sequence extends to a five-term sequence involving H^2. The extended version requires the *transgression map*, which arises from the Lyndon-Hochschild-Serre spectral sequence of section 3.4, and its proof is proposition 3.4.9 there.

Theorem 3.3.4: Five-Term Exact Sequence

Let N be a closed normal subgroup of G and M a discrete G-module. The sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\text{Inf}} H^1(G, M) \xrightarrow{\text{Res}} H^1(N, M)^{G/N} \xrightarrow{\text{tg}} H^2(G/N, M^N) \xrightarrow{\text{Inf}} H^2(G, M)$$

is exact, where $H^1(N, M)^{G/N}$ denotes the G/N-invariants of $H^1(N, M)$ under the conjugation action, and tg is the transgression map.

The conjugation action of G/N on $H^1(N, M)$ is the case $q = 1$ of definition 3.4.1: a lift $g \in G$ of $\bar{g} \in G/N$ sends a continuous 1-cocycle $f\colon N \to M$ to the cocycle $g_\# f$, $(g_\# f)(n) = g \cdot f(g^{-1}ng)$. The cocycle $g_\# f$ depends on the lift. For $G = N = \mathbb{Z}/2\mathbb{Z} = \{1, \sigma\}$ acting on $M = \mathbb{Z}$ by $\sigma \cdot m = -m$, the cocycle f with $f(1) = 0$ and $f(\sigma) = 1$ has $(\sigma_\# f)(\sigma) = -1 \neq f(\sigma)$, although σ and 1 both lift the trivial coset. The class of $g_\# f$ does not depend on the lift, by lemma 3.4.2(1) and (2). So the class

of f is G/N-invariant if and only if $g_\# f$ is cohomologous to f for every $g \in G$.

The third fundamental operation moves cohomology classes in the opposite direction to restriction: from a subgroup back to the full group. It is defined only for subgroups of finite index.

Definition 3.3.5: Corestriction

Let H be an open (hence finite-index) subgroup of G, and let $\{s_1, \dots, s_m\}$ be a set of left coset representatives for H in G (so $G = \bigsqcup_{i=1}^m s_i H$ and $m = [G : H]$). For each $g \in G$ and each index i, there is a unique index $\sigma_g(i)$ and a unique element $h_g^{(i)} \in H$ such that $g s_i = s_{\sigma_g(i)} h_g^{(i)}$. The *corestriction* (or *transfer*) map

$$\mathrm{Cor} = \mathrm{Cor}_H^G : H^n(H, M) \to H^n(G, M)$$

is induced by the following map on continuous cochains. For a continuous n-cochain $f : H^n \to M$, set

$$\mathrm{Cor}(f)(g_1, \dots, g_n) = \sum_{j=1}^{m} s_{\sigma_{g_1 \cdots g_n}(j)} \cdot f\Big(h_{g_1}^{(\sigma_{g_2 \cdots g_n}(j))},\ h_{g_2}^{(\sigma_{g_3 \cdots g_n}(j))},\ \dots,\ h_{g_n}^{(j)} \Big)$$

whose r-th argument is $h_{g_r}^{(\sigma_{g_{r+1} \cdots g_n}(j))}$, with σ of the empty product equal to the identity.

The assignment $g \mapsto \sigma_g$ is an action of G on the indices: the identity

$$g g' s_i = g s_{\sigma_{g'}(i)} h_{g'}^{(i)} = s_{\sigma_g \sigma_{g'}(i)} h_g^{(\sigma_{g'}(i))} h_{g'}^{(i)}$$

gives $\sigma_{gg'} = \sigma_g \sigma_{g'}$ and $h_{gg'}^{(i)} = h_g^{(\sigma_{g'}(i))} h_{g'}^{(i)}$. Every $x \in G$ is $x = \rho(x) s_k^{-1}$ for exactly one index k, the one with $x^{-1} \in s_k H$, and one $\rho(x) \in H$. The map $\rho \colon G \to H$ is continuous, as $\rho(x) = x s_k$ on each of the finitely many open sets $H s_k^{-1}$, which cover G, and $\rho(hx) = h\rho(x)$ for $h \in H$. As in the discussion before proposition 2.3.4, the assignment $f \mapsto \varphi$, $\varphi(x_0, \dots, x_n) = x_0 \cdot f(x_0^{-1} x_1, \dots, x_{n-1}^{-1} x_n)$, is an isomorphism of complexes from continuous inhomogeneous cochains onto continuous *homogeneous* cochains, the continuous maps φ with $\varphi(g x_0, \dots, g x_n) = g \cdot \varphi(x_0, \dots, x_n)$, whose coboundary is the alternating sum of the functions obtained by omitting one variable. On homogeneous cochains of H put

$$\mathrm{Cor}(\varphi)(x_0, \dots, x_n) = \sum_{i=1}^{m} s_i \cdot \varphi\big(\rho(s_i^{-1} x_0), \dots, \rho(s_i^{-1} x_n)\big)$$

This is a homogeneous cochain of G. For $g \in G$ write $g^{-1} s_i = s_k h$ with $h \in H$, so that $s_i = g s_k h$ and $\rho(s_i^{-1} g x) = \rho(h^{-1} s_k^{-1} x) = h^{-1} \rho(s_k^{-1} x)$. The term of index i at $(g x_0, \dots, g x_n)$ is then $g s_k h \cdot h^{-1} \cdot \varphi(\rho(s_k^{-1} x_0), \dots, \rho(s_k^{-1} x_n))$, which is g times the term of index k at $(x_0, \dots, x_n)$, and $i \mapsto k$ is a bijection. The construction commutes with omitting a variable, hence with the coboundary, and it commutes with composition by maps of modules. At $(1, g_1, g_1 g_2, \dots, g_1 \cdots g_n)$ it gives the formula of the definition: $\rho(s_i^{-1}) = 1$, and $y_r = \rho(s_i^{-1} g_1 \cdots g_r) = s_i^{-1} g_1 \cdots g_r s_{k_r}$ satisfies $y_{r-1}^{-1} y_r = s_{k_{r-1}}^{-1} g_r s_{k_r} = h_{g_r}^{(k_r)}$ with $k_{r-1} = \sigma_{g_r}(k_r)$ and $k_0 = i$. Indexing the sum by $j = k_n$ gives $k_r = \sigma_{g_{r+1} \cdots g_n}(j)$ and $i = \sigma_{g_1 \cdots g_n}(j)$.

So the formula defines a map of complexes $C^\bullet_{\mathrm{cts}}(H, M) \to C^\bullet_{\mathrm{cts}}(G, M)$, natural in the discrete

G-module M. In degree 0 the corestriction $\mathrm{Cor} : M^H \to M^G$ is the *trace* (or *norm*) map:

$$\mathrm{Cor}(m) = \sum_{i=1}^{m} s_i \cdot m$$

This is independent of the coset representatives and lands in M^G (since $g \cdot \mathrm{Cor}(m) = \sum_i g s_i \cdot m = \sum_i s_{\sigma_g(i)} h_g^{(i)} \cdot m = \sum_i s_{\sigma_g(i)} \cdot m = \mathrm{Cor}(m)$, because $h_g^{(i)} \cdot m = m$ for $m \in M^H$ and σ_g is a permutation).

Lemma 3.3.6: Inflation, Restriction, and Corestriction Are Morphisms of δ-Functors

Let G be a profinite group.

1. Let H be a closed subgroup of G. Composed with the exact restriction functor $\mathrm{Mod}_G^{\mathrm{disc}} \to \mathrm{Mod}_H^{\mathrm{disc}}$, the family $H^\bullet(H, -)$ is a universal δ-functor on $\mathrm{Mod}_G^{\mathrm{disc}}$. The maps $\mathrm{Res}\colon H^\bullet(G, -) \to H^\bullet(H, -)$ form a morphism of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$, in the sense of *Homological Algebra* [13, Morphism of δ-Functors], whose degree-zero component is the inclusion $M^G \subseteq M^H$, and it is the only such morphism.

2. Let H be an open subgroup of G. The maps $\mathrm{Cor}\colon H^\bullet(H, -) \to H^\bullet(G, -)$ form a morphism of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ whose degree-zero component is the trace, and it is the only such morphism. So the maps Cor induces on cohomology do not depend on the coset representatives, although the cochain formula does.

3. Let N be a closed normal subgroup of G, with projection $\pi\colon G \to G/N$. For a discrete G/N-module B write π^*B for the discrete G-module with the same underlying group and the action $g \cdot b = \pi(g) \cdot b$, so that $(\pi^*B)^N = B$. The functor $\pi^*\colon \mathrm{Mod}_{G/N}^{\mathrm{disc}} \to \mathrm{Mod}_G^{\mathrm{disc}}$ is exact, and the inflation maps $\mathrm{Inf}\colon H^\bullet(G/N, B) \to H^\bullet(G, \pi^*B)$ of definition 3.3.1 form a morphism of δ-functors on $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$ whose degree-zero component is the identity of $B^{G/N} = (\pi^*B)^G$, and it is the only such morphism.

***Proof*:**

Step 1: Compatibility with connecting homomorphisms. The G-module π^*B is discrete, since the stabilizer of b in G is the preimage under the continuous map π of its stabilizer in G/N. The functor π^* changes neither underlying groups nor maps, so it is exact, and so is the restriction functor of definition 3.3.2. Restriction of the domain is a map of complexes $C^\bullet_{\mathrm{cts}}(G, M) \to C^\bullet_{\mathrm{cts}}(H, M)$ for every closed subgroup H (definition 3.3.2), the corestriction formula is one for H open (the discussion after definition 3.3.5), and precomposition with π is one from $C^\bullet_{\mathrm{cts}}(G/N, B)$ to $C^\bullet_{\mathrm{cts}}(G, \pi^*B)$ (definition 3.3.1). All three are natural in the module. For a short exact sequence of discrete G-modules, or of discrete G/N-modules in the case of inflation, each of them is therefore a morphism between the short exact sequences of cochain complexes of proposition 3.1.6(1) for the two groups involved, and the naturality clause of the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) makes it commute with the connecting homomorphisms. Since restriction and π^* are exact, the families $H^\bullet(H, -)$ and $H^\bullet(G, \pi^*(-))$ are δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ and on $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$. Transporting along the isomorphisms of δ-functors of proposition 3.1.6(2) for the groups G, H and G/N gives the three morphisms of δ-functors. In degree 0 the cochain groups are the modules themselves, so restriction is the inclusion $M^G \subseteq M^H$, corestriction is the trace, and inflation is the identity of $B^{G/N} = (\pi^*B)^G$.

Step 2: Universality of $H^\bullet(H,-)$. Let H be closed. By [30, Ch. I, §1.2, Prop. 1] the projection $G \to G/H$ onto the space of left cosets has a continuous section s. The map $G/H \times H \to G$, $(c,h) \mapsto s(c)h$, is a continuous bijection, with continuous inverse $x \mapsto (xH, s(xH)^{-1}x)$, and right multiplication by $h' \in H$ on G corresponds to $(c,h) \mapsto (c, hh')$. Let J be an abelian group with the discrete topology. For a continuous $f\colon G \to J$ and $h \in H$ let $F_f(h)\colon G/H \to J$ be $c \mapsto f(s(c)h)$, a locally constant map. The map $h \mapsto F_f(h)$ is locally constant: fix h, choose for each $c \in G/H$ open sets $V_c \ni c$ and $W_c \ni h$ with $(c',h') \mapsto f(s(c')h')$ constant on $V_c \times W_c$, cover the compact space G/H by finitely many $V_{c_1}, \dots, V_{c_r}$, and put $W = W_{c_1} \cap \cdots \cap W_{c_r}$. For $h' \in W$ and $c \in V_{c_i}$, $F_f(h')(c) = f(s(c_i)h) = F_f(h)(c)$. Conversely, a locally constant map $F\colon H \to \mathrm{Map}^{\mathrm{cts}}(G/H, J)$ gives the locally constant function $s(c)h \mapsto F(h)(c)$ on G. So $f \mapsto F_f$ is an isomorphism from $\mathrm{Map}^{\mathrm{cts}}(G,J)$ onto $\mathrm{Map}^{\mathrm{cts}}(H,A)$, where $A = \mathrm{Map}^{\mathrm{cts}}(G/H, J)$ is regarded as a discrete abelian group, and as $(h' \cdot f)(s(c)h) = f(s(c)hh')$ it carries the action of H on $\mathrm{Map}^{\mathrm{cts}}(G,J)$ to right translation on $\mathrm{Map}^{\mathrm{cts}}(H,A)$. The group H is profinite (proposition 1.2.1), and Step 4 of the proof of proposition 3.1.6, applied to H, with proposition 3.1.6(2), gives $H^q(H, \mathrm{Map}^{\mathrm{cts}}(H,A)) = 0$ for $q \geq 1$. Hence $H^q(H, \mathrm{Map}^{\mathrm{cts}}(G,J)) = 0$ for $q \geq 1$. A discrete G-module M embeds in $\mathrm{Map}^{\mathrm{cts}}(G,M)$ by the map u_M of Step 3 of the same proof, so u_M effaces $H^q(H,-)$ at M for $q \geq 1$, and the δ-functor $H^\bullet(H,-)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$ is universal by *Homological Algebra* [13, An Effaceable δ-Functor Is Universal].

Step 3: Uniqueness. The derived functors $H^\bullet(G,-)$ and $H^\bullet(G/N,-)$ are universal δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ and on $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$ by *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor], the group G/N being profinite (proposition 1.2.4) and its category of discrete modules abelian with enough injectives (propositions 3.1.2 and 3.1.3). The δ-functor $H^\bullet(H,-)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$ is universal by Step 2. A morphism of δ-functors out of a universal δ-functor is determined by its degree-zero component (*Homological Algebra* [13, Universal δ-Functor]), which gives the three uniqueness statements. The trace does not depend on the coset representatives, and therefore neither does Cor on cohomology.

In degree 1, if $f : H \to M$ is a continuous 1-cocycle, then

$$\mathrm{Cor}(f)(g) = \sum_{j=1}^{m} s_{\sigma_g(j)} \cdot f(h_g^{(j)}) = \sum_{j=1}^{m} s_{\sigma_g(j)} \cdot f(s_{\sigma_g(j)}^{-1} g s_j) \tag{3.2}$$

The central algebraic identity governing the interaction of restriction and corestriction is the following.

Proposition 3.3.7: The Formula $\mathrm{Cor} \circ \mathrm{Res} = [G:H]$

Let H be an open subgroup of G with $[G:H] = m$. For every $n \geq 0$ and every $\alpha \in H^n(G,M)$,

$$\mathrm{Cor}(\mathrm{Res}(\alpha)) = m \cdot \alpha.$$

***Proof*:**

Restriction and corestriction are morphisms of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ (lemma 3.3.6(1) and (2)), so $\mathrm{Cor} \circ \mathrm{Res}$ is an endomorphism of the δ-functor $H^\bullet(G,-)$. Its degree-zero component sends $a \in M^G$ to $\sum_{i=1}^m s_i \cdot a = ma$. Multiplication by m is a second endomorphism of $H^\bullet(G,-)$ with the same degree-zero component: it is a natural transformation, and it commutes with the connecting homomorphisms because these are additive. The category $\mathrm{Mod}_G^{\mathrm{disc}}$ is abelian

with enough injectives (propositions 3.1.2 and 3.1.3), so the δ-functor $H^\bullet(G,-)$ is universal (*Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]), and the two endomorphisms coincide, as claimed.

This formula has several immediate and important consequences.

Corollary 3.3.8: Consequences of $\mathrm{Cor}\circ\mathrm{Res} = [G:H]$

Let H be an open subgroup of G with $[G:H] = m$.

1. *Restriction detects m-torsion-free elements:* if $\alpha \in H^n(G,M)$ satisfies $\mathrm{Res}(\alpha) = 0$, then $m \cdot \alpha = 0$. In particular, if $H^n(G,M)$ has no m-torsion, then Res is injective.
2. *Corestriction surjects onto m-divisible elements:* the image of Cor contains $m \cdot H^n(G,M)$. In particular, if m is invertible in $H^n(G,M)$, then Cor is surjective.
3. *p-primary decomposition:* if p is a prime not dividing $m = [G:H]$, then Res is injective on the p-primary component $H^n(G,M)\{p\}$, and the composition $\frac{1}{m}\,\mathrm{Cor} : H^n(H,M)\{p\} \to H^n(G,M)\{p\}$ is a left inverse of Res.
4. *Annihilation by the order:* if G is finite, then $|G| \cdot H^n(G,M) = 0$ for $n \geq 1$. (Apply the formula with $H = \{1\}$ and note that $H^n(\{1\},M) = 0$ for $n \geq 1$.)

Part (3) is the primary tool for reducing Galois cohomology computations to one prime at a time. If H is a Sylow p-subgroup of a finite group G (or, in the profinite case, a pro-p-Sylow subgroup), then $[G:H]$ is prime to p, and restriction gives an injection on p-primary parts. This reduces many questions about $H^n(G,M)\{p\}$ to questions about the pro-p-group H, where the theory of chapter 6 applies.

Example 3.3.9: Inflation and Restriction for $G_{\mathbb{Q}_p}$

Let $K = \mathbb{Q}_p$ and consider the absolute Galois group $G_{\mathbb{Q}_p}$. The inertia subgroup $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is a closed normal subgroup with quotient $G_{\mathbb{Q}_p}/I_p \cong \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \cong \hat{\mathbb{Z}}$ (generated topologically by the Frobenius automorphism Frob_p).

1. For any discrete $G_{\mathbb{Q}_p}$-module M, the inflation-restriction exact sequence gives

$$0 \to H^1(\hat{\mathbb{Z}}, M^{I_p}) \xrightarrow{\mathrm{Inf}} H^1(G_{\mathbb{Q}_p}, M) \xrightarrow{\mathrm{Res}} H^1(I_p, M)^{\hat{\mathbb{Z}}}.$$

   When M is an unramified module (meaning I_p acts trivially, so $M^{I_p} = M$), the restriction map lands in $\mathrm{Hom}_{\mathrm{cts}}(I_p, M)^{\hat{\mathbb{Z}}}$, which captures the "ramified part" of $H^1(G_{\mathbb{Q}_p}, M)$. The inflation from $H^1(\hat{\mathbb{Z}}, M)$ captures the "unramified part."

2. Take $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action. Since I_p acts trivially, $M^{I_p} = \mathbb{Z}/n\mathbb{Z}$, and the inflation-restriction sequence reads

$$0 \to H^1(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\mathrm{Inf}} H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\mathrm{Res}} H^1(I_p, \mathbb{Z}/n\mathbb{Z})^{\hat{\mathbb{Z}}}.$$

> The group $H^1(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) = \mathrm{Hom}_{\mathrm{cts}}(\hat{\mathbb{Z}}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (generated by the homomorphism sending Frob_p to 1). This is the subgroup of $H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/n\mathbb{Z})$ classifying unramified $\mathbb{Z}/n\mathbb{Z}$-extensions of $\mathbb{Q}_p$ (cyclic extensions contained in $\mathbb{Q}_p^{\mathrm{nr}}$). The restriction map detects the ramification: $\mathrm{Res}(\chi) = 0$ if and only if the extension classified by χ is unramified.

This example illustrates a pattern that pervades the local theory of chapter 8: the inflation-restriction sequence for the pair $(G_{\mathbb{Q}_p}, I_p)$ separates the "unramified" and "ramified" contributions to Galois cohomology, and the structure of I_p (a pro-p-group times a tame quotient) provides a further decomposition.

Exercise 3.3.1

1. Let N be a closed normal subgroup of G and $0 \to B' \to B \to B'' \to 0$ a short exact sequence of discrete G/N-modules, regarded as G-modules through $G \to G/N$. Show directly on cochains that inflation commutes with the connecting homomorphisms: the diagram
$$\begin{array}{ccc} H^n(G/N, B'') & \xrightarrow{\delta} & H^{n+1}(G/N, B') \\ \downarrow{\scriptstyle \mathrm{Inf}} & & \downarrow{\scriptstyle \mathrm{Inf}} \\ H^n(G, B'') & \xrightarrow{\delta} & H^{n+1}(G, B') \end{array}$$
commutes. State and prove the analogous result for restriction. Then give a short exact sequence $0 \to M' \to M \to M'' \to 0$ of discrete G-modules whose sequence of N-invariants is not short exact, so that the top row of such a diagram is not available for arbitrary G-modules.

2. Let $G = \mathbb{Z}/6\mathbb{Z}$ and $H = 2\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z}$ (the unique subgroup of index 2). Let $M = \mathbb{Z}$ with trivial G-action. Compute $\mathrm{Cor} : H^2(H, \mathbb{Z}) \to H^2(G, \mathbb{Z})$ explicitly using the cochain formula and coset representatives $\{0, 1\}$. Verify that $\mathrm{Cor} \circ \mathrm{Res} = 2$ on $H^2(G, \mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}$.

3. Let G be a profinite group and H an open subgroup. Show that $\mathrm{Res} \circ \mathrm{Cor} : H^n(H, M) \to H^n(H, M)$ is given by
$$\mathrm{Res}(\mathrm{Cor}(\alpha)) = \sum_{s \in H \backslash G / H} \mathrm{Cor}^H_{H \cap sHs^{-1}} \left(\mathrm{Res}^{H \cap sHs^{-1}}_H (s_* \alpha) \right),$$
where the sum runs over double coset representatives and s_* denotes the conjugation action. This is the *Mackey formula* (or double coset formula) for group cohomology.

4. Let G be a profinite group and M a discrete G-module. Use proposition 3.3.7 to give a short proof that $H^n(G, M)\{p\} = 0$ for $n \geq 1$ whenever G is a pro-p'-group (i.e., every open normal subgroup U satisfies $p \nmid [G : U]$).

5. Let $G = \mathbb{Z}/4\mathbb{Z}$, $N = 2\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, and $M = \mathbb{Z}$ with trivial action. Write out the five-term exact sequence (theorem 3.3.4) and compute all five terms explicitly. Identify the transgression map $\mathrm{tg} : H^1(N, \mathbb{Z})^{G/N} \to H^2(G/N, \mathbb{Z})$ and verify exactness.

3.4 The Lyndon-Hochschild-Serre Spectral Sequence

The inflation-restriction exact sequence of proposition 3.3.3 captures only the beginning of the relationship between $H^\bullet(G, M)$, $H^\bullet(N, M)$, and $H^\bullet(G/N, -)$. The five-term exact sequence (theorem 3.3.4) extends this one step further, but a complete description requires a spectral sequence. The

Lyndon-Hochschild-Serre (LHS) spectral sequence, also called the Hochschild-Serre spectral sequence, is the fundamental tool for computing cohomology inductively: it assembles $H^n(G, M)$ from the cohomology of the quotient G/N with coefficients in the cohomology of the normal subgroup N.

The general framework is the Grothendieck spectral sequence for a composition of derived functors, treated in *Homological Algebra* [13, The Grothendieck Spectral Sequence].

The coefficients on the E_2 page are the groups $H^q(N, M)$ with an action of G/N. Let N be a closed normal subgroup of G and M a discrete G-module.

Definition 3.4.1: Conjugation Action

For $g \in G$, define the *conjugation map* $c_g : N \to N$ by $c_g(n) = g^{-1}ng$ and the *twist map* $\tau_g : M \to M$ by $\tau_g(m) = g \cdot m$. They are compatible, $\tau_g(c_g(n) \cdot m) = n \cdot \tau_g(m)$, and $f \mapsto \tau_g \circ f \circ c_g$ is a map on continuous cochains

$$g_\# : C^q_{\text{cts}}(N, M) \to C^q_{\text{cts}}(N, M), \qquad (g_\# f)(n_1, \dots, n_q) = g \cdot f(g^{-1}n_1g, \dots, g^{-1}n_qg).$$

This map commutes with the coboundary operator, hence induces a map $g_\# : H^q(N, M) \to H^q(N, M)$, and $(gh)_\# = g_\# h_\#$ for all $g, h \in G$ (lemma 3.4.2(1)). For every $n \in N$ the map $n_\#$ is the identity on $H^q(N, M)$ (lemma 3.4.2(2)), so $(gn)_\# = g_\# n_\# = g_\#$ on cohomology, and $g_\#$ depends only on the coset $\bar{g} = gN \in G/N$. This defines the *conjugation action* of G/N on $H^q(N, M)$.

Lemma 3.4.2: Inner Conjugation Acts Trivially on Cohomology

Let N be a closed normal subgroup of a profinite group G and M a discrete G-module.

1. For every $g \in G$ the map $g_\#$ of definition 3.4.1 carries continuous cochains to continuous cochains and commutes with the coboundary, and $(gh)_\# = g_\# \circ h_\#$ for all $g, h \in G$.
2. For every $n \in N$ and every $q \geq 0$, the map $n_\#$ is the identity on $H^q(N, M)$.

In particular, taking $N = G$, the map $g_\# \colon H^q(G, M) \to H^q(G, M)$ is the identity for every $g \in G$.

***Proof*:**

Part (1). Fix $g \in G$ and recall $c_g(n) = g^{-1}ng$ from definition 3.4.1. Normality of N makes c_g a map $N \to N$, and it is a continuous group homomorphism. For $f \in C^q_{\text{cts}}(N, M)$ the cochain $g_\# f$ is the composite of the continuous map $c_g \times \cdots \times c_g$, the map f, and the action of g on the discrete group M, so it is continuous. Let $n_1, \dots, n_{q+1} \in N$. The first term of $(d(g_\# f))(n_1, \dots, n_{q+1})$ is $n_1 \cdot (g \cdot f(c_g(n_2), \dots, c_g(n_{q+1})))$, and the first term of $(g_\#(df))(n_1, \dots, n_{q+1}) = g \cdot (df)(c_g(n_1), \dots, c_g(n_{q+1}))$ is $g\, c_g(n_1) \cdot f(c_g(n_2), \dots, c_g(n_{q+1}))$. These agree because $g\, c_g(n_1) = n_1 g$. The remaining terms agree because $c_g(n_i n_{i+1}) = c_g(n_i)c_g(n_{i+1})$ and the action of g on M is additive. So $d \circ g_\# = g_\# \circ d$. For $h \in G$,

$$(g_\#(h_\# f))(n_1, \dots, n_q) = gh \cdot f(h^{-1}g^{-1}n_1gh, \dots, h^{-1}g^{-1}n_qgh) = ((gh)_\# f)(n_1, \dots, n_q)$$

Part (2). Fix $n \in N$. The group N is profinite (proposition 1.2.1), and the restriction of M

to N is a discrete N-module, because $\mathrm{Stab}_N(m) = \mathrm{Stab}_G(m) \cap N$ is open in N. The formula $(n_\# f)(n_1, \dots, n_q) = n \cdot f(n^{-1}n_1 n, \dots, n^{-1}n_q n)$ uses only the action of N, so it defines $n_\#$ on $C^\bullet_{\mathrm{cts}}(N, M')$ for every discrete N-module M', and Part (1), applied to the closed normal subgroup N of the profinite group N, makes it a map of cochain complexes. For a map $\varphi\colon M' \to M''$ of discrete N-modules, $\varphi \circ (n_\# f) = n_\#(\varphi \circ f)$ because $\varphi(n \cdot m) = n \cdot \varphi(m)$. So $n_\#$ is a natural transformation $\tilde{H}^q(N, -) \Rightarrow \tilde{H}^q(N, -)$ for each q. For a short exact sequence of discrete N-modules, the maps $n_\#$ therefore form an endomorphism of the short exact sequence of cochain complexes of proposition 3.1.6(1), and the naturality clause of the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) shows that $n_\#$ commutes with the connecting homomorphisms. The family $(n_\#)_{q \geq 0}$ is thus an endomorphism of the δ-functor $\tilde{H}^\bullet(N, -)$ in the sense of *Homological Algebra* [13, Morphism of δ-Functors]. In degree zero, $C^0_{\mathrm{cts}}(N, M') = M'$ and $n_\#$ is $m \mapsto n \cdot m$, which is the identity on $\tilde{H}^0(N, M') = (M')^N$. The identity family is a second endomorphism of $\tilde{H}^\bullet(N, -)$ with the same degree-zero component, and $\tilde{H}^\bullet(N, -)$ is universal by proposition 3.1.6(1) applied to N, so the two coincide. Hence $n_\#$ is the identity on $\tilde{H}^q(N, M')$ for every $q \geq 0$. Taking M' to be the restriction of M, where $n_\#$ is the map of definition 3.4.1, and transporting along the isomorphism of proposition 3.1.6(2) gives the identity on $H^q(N, M)$. The final sentence of the statement is the case $N = G$.

See also [26, Ch. I, §6, (1.6.3)], where part (2) is proved by dimension shifting from degree 0.

Proposition 3.4.3: $H^q(N, M)$ as a Discrete G/N-Module

The conjugation action makes $H^q(N, M)$ into a discrete G/N-module.

***Proof*:**
Let $x \in H^q(N, M)$ be the class of a continuous cocycle $f\colon N^q \to M$ (proposition 3.1.6). The group N is profinite (proposition 1.2.1), so by lemma 3.2.1 for N there is an open normal subgroup W of N such that f is constant on the cosets of W^q in N^q, and f takes finitely many values. The topology of N is the subspace topology, and the open normal subgroups of G form a neighbourhood basis of the identity (Step 2 of the proof of theorem 1.1.10), so there is an open normal subgroup U of G with $U \cap N \subseteq W$. Shrinking U, we may assume that U also fixes the finitely many values of f, whose stabilizers are open. Let $u \in U$ and $n \in N$. Then $u^{-1}nu = n \cdot n^{-1}u^{-1}nu$, and $n^{-1}u^{-1}nu$ lies in U, as U is normal, and in N, as N is normal. So $u^{-1}n_i u \in n_i(U \cap N) \subseteq n_i W$ for $n_1, \dots, n_q \in N$, and

$$(u_\# f)(n_1, \dots, n_q) = u \cdot f(u^{-1}n_1 u, \dots, u^{-1}n_q u) = u \cdot f(n_1, \dots, n_q) = f(n_1, \dots, n_q)$$

So $u_\# x = x$ for every $u \in U$, and the stabilizer of x in G/N contains the image UN/N of U, which is open because the projection $G \to G/N$ is open (proposition 1.2.4). Hence $H^q(N, M)$ is a discrete G/N-module (definition 3.1.1).

The Grothendieck spectral sequence of the functors $(-)^N$ and $(-)^{G/N}$, which compose to $(-)^G$, requires that $(-)^N$ send injective discrete G-modules to $(-)^{G/N}$-acyclic discrete G/N-modules. It sends them to injective ones.

Lemma 3.4.4: N-Invariants Preserve Injectives

Let N be a closed normal subgroup of G. If I is an injective object in $\mathrm{Mod}_G^{\mathrm{disc}}$, then I^N is an injective object in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$. In particular, I^N is acyclic for the functor $(-)^{G/N}$.

***Proof*:**
Let $0 \to A \to B$ be an injection of discrete G/N-modules, and let $\varphi : A \to I^N$ be a G/N-equivariant map. View A and B as discrete G-modules via the projection $G \twoheadrightarrow G/N$ (the G-action factors through G/N, so every element has open stabilizer). The composite $A \xrightarrow{\varphi} I^N \hookrightarrow I$ is G-equivariant, and since I is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$, this extends to a G-equivariant map $\psi : B \to I$.

The image of ψ lies in I^N: for any $b \in B$ and $n \in N$, the element $\psi(b)$ satisfies $n \cdot \psi(b) = \psi(n \cdot b) = \psi(b)$, since N acts trivially on B (as B is a G/N-module). Hence ψ factors through I^N, giving a G/N-equivariant extension $\bar{\psi} : B \to I^N$ of φ. An injective object is acyclic for every additive functor (*Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]), which gives the last assertion.

Theorem 3.4.5: The Lyndon-Hochschild-Serre Spectral Sequence

Let G be a profinite group, N a closed normal subgroup, and M a discrete G-module. There is a first-quadrant cohomological spectral sequence, natural in M,

$$E_2^{p,q} = H^p(G/N,\ H^q(N, M)) \implies H^{p+q}(G, M)$$

where $H^q(N, M)$ carries the conjugation action of G/N from definition 3.4.1. It is the Grothendieck spectral sequence of the factorization $(-)^G = (-)^{G/N} \circ (-)^N$. Its second page $R^p(-)^{G/N}(R^q(-)^N(M))$ is identified with the displayed one through isomorphisms of discrete G/N-modules $\alpha_M \colon R^q(-)^N(M) \to H^q(N, M)$, which, as M varies, form an isomorphism of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ whose degree-zero component is the identity of M^N.

***Proof*:**
Step 1: The Grothendieck spectral sequence. For a discrete G-module M, the group M^N is a discrete G/N-module: the stabilizer in G/N of $m \in M^N$ is the image of $\mathrm{Stab}_G(m)$, which is open because the projection $G \to G/N$ is open (proposition 1.2.4). So the invariant functor $(-)^G \colon \mathrm{Mod}_G^{\mathrm{disc}} \to \mathbf{Ab}$ factors as the composition

$$\mathrm{Mod}_G^{\mathrm{disc}} \xrightarrow{(-)^N} \mathrm{Mod}_{G/N}^{\mathrm{disc}} \xrightarrow{(-)^{G/N}} \mathbf{Ab}$$

Both functors are additive and left exact, by proposition 2.1.9, as the kernels in both categories are those of abelian groups (proposition 3.1.2). The group G/N is profinite (proposition 1.2.4), so both categories are abelian with enough injectives (propositions 3.1.2 and 3.1.3). By lemma 3.4.4, the functor $(-)^N$ sends injective objects to $(-)^{G/N}$-acyclic ones. The Grothendieck spectral sequence (*Homological Algebra* [13, The Grothendieck Spectral Sequence]) gives a first-quadrant spectral sequence, natural in M,

$$E_2^{p,q} = R^p(-)^{G/N}\big(R^q(-)^N(M)\big) \implies R^{p+q}(-)^G(M) = H^{p+q}(G, M)$$

where $R^q(-)^N(M) = H^q((I^\bullet)^N)$ for an injective resolution $M \to I^\bullet$ in $\mathrm{Mod}_G^{\mathrm{disc}}$, and $R^p(-)^{G/N} = H^p(G/N, -)$.

Step 2: The underlying groups. Let $U\colon \mathrm{Mod}_{G/N}^{\mathrm{disc}} \to \mathbf{Ab}$ be the forgetful functor, which is exact. The derived functors $R^\bullet(-)^N$ form a δ-functor that vanishes on injective objects in positive degrees (*Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]), so $U \circ R^\bullet(-)^N$ is a δ-functor on $\mathrm{Mod}_G^{\mathrm{disc}}$ with the same property, and it is universal by *Homological Algebra* [13, Universality and Effaceability Agree When There Are Enough Injectives]. The δ-functor $H^\bullet(N, -)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$ is universal by lemma 3.3.6(1). In degree zero, $R^0(-)^N(M) \cong M^N = H^0(N, M)$ naturally in M. By *Homological Algebra* [13, A Universal δ-Functor Is Determined by Its Degree-Zero Part] there is an isomorphism of δ-functors $\alpha\colon U \circ R^\bullet(-)^N \to H^\bullet(N, -)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$ whose degree-zero component is this identification, and by universality it is the only morphism of δ-functors with that degree-zero component.

Step 3: The actions of G/N. Fix $g \in G$. The action of g on the discrete G/N-modules $R^q(-)^N(M)$ gives maps $\rho_g\colon UR^q(-)^N(M) \to UR^q(-)^N(M)$, natural in M because the maps $R^q(-)^N(\varphi)$ are morphisms of G/N-modules, and commuting with the connecting homomorphisms because these are morphisms of G/N-modules too. So ρ_g is an endomorphism of the δ-functor $U \circ R^\bullet(-)^N$, and its degree-zero component is $m \mapsto g \cdot m$ on M^N. The maps $g_\#$ of definition 3.4.1 form an endomorphism of the δ-functor $H^\bullet(N, -)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$. On continuous cochains $g_\#$ is a map of complexes (lemma 3.4.2(1)), natural in the discrete G-module M, as a G-map φ satisfies $\varphi(g \cdot m) = g \cdot \varphi(m)$. For a short exact sequence of discrete G-modules it is therefore an endomorphism of the short exact sequence of cochain complexes of proposition 3.1.6(1) for N, and the naturality clause of the Snake Lemma (*Homological Algebra* [13, Snake Lemma]) makes it commute with the connecting homomorphisms, which proposition 3.1.6(2) for N carries to those of $H^\bullet(N, -)$. In degree zero $g_\#$ is $m \mapsto g \cdot m$ on M^N. So $\alpha\rho_g\alpha^{-1}$ and $g_\#$ are endomorphisms of the universal δ-functor $H^\bullet(N, -)$ with the same degree-zero component, and they coincide (*Homological Algebra* [13, Universal δ-Functor]). Hence $\alpha_M(\bar{g} \cdot x) = g_\# \, \alpha_M(x)$ for $x \in R^q(-)^N(M)$: the map α_M is an isomorphism of G/N-modules from $R^q(-)^N(M)$ onto $H^q(N, M)$ with the conjugation action. In particular the conjugation action makes $H^q(N, M)$ a discrete G/N-module, as proposition 3.4.3 also shows.

Step 4: The second page. Applying the functor $H^p(G/N, -)$ to α_M gives isomorphisms $E_2^{p,q} = H^p(G/N, R^q(-)^N(M)) \cong H^p(G/N, H^q(N, M))$, natural in M because α is natural, completing the proof.

The spectral sequence is a first-quadrant spectral sequence, meaning $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$. This has the consequence that only finitely many differentials can affect any given term: the differentials $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$ eventually land outside the first quadrant, so $E_r^{p,q}$ stabilizes to a limiting value $E_\infty^{p,q}$ for r sufficiently large.

The convergence $E_2^{p,q} \Rightarrow H^{p+q}(G, M)$ means that for each $n \geq 0$, the group $H^n(G, M)$ admits a finite descending filtration

$$H^n(G, M) = F^0H^n \geqslant F^1H^n \geqslant \cdots \geqslant F^nH^n \geqslant F^{n+1}H^n = 0 \tag{3.3}$$

with graded pieces $F^pH^{p+q}/F^{p+1}H^{p+q} \cong E_\infty^{p,q}$. The group $H^n(G, M)$ is thus assembled from the subquotients $E_\infty^{p,q}$ with $p + q = n$, but the extensions relating these subquotients are not determined

by the spectral sequence alone.

Lemma 3.4.6: Inflation Through Resolutions

Let N be a closed normal subgroup of G, let B be a discrete G/N-module, and write π^*B for B regarded as a discrete G-module through $\pi\colon G \to G/N$, as in lemma 3.3.6(3). Let $B \to K^\bullet$ be an injective resolution in $\mathrm{Mod}^{\mathrm{disc}}_{G/N}$ and $\pi^*B \to L^\bullet$ one in $\mathrm{Mod}^{\mathrm{disc}}_{G}$. There is a chain map $\chi\colon \pi^*K^\bullet \to L^\bullet$ inducing the identity of π^*B on H^0, unique up to homotopy, and the map it induces,

$$H^n(G/N, B) = H^n\big((K^\bullet)^{G/N}\big) = H^n\big((\pi^*K^\bullet)^G\big) \longrightarrow H^n\big((L^\bullet)^G\big) = H^n(G, \pi^*B)$$

is the inflation map of definition 3.3.1.

***Proof*:**

Step 1: The map. The functor π^* is exact (lemma 3.3.6(3)), so $\pi^*K^\bullet$ is a resolution of π^*B, and $L^\bullet$ is a complex of injective objects concentrated in non-negative degrees. *Homological Algebra* [13, Projective Resolution As Initial Objects], in its dual form for complexes of injective objects, therefore gives χ, and any two chain maps $\pi^*K^\bullet \to L^\bullet$ inducing the same map on H^0 are homotopic. Since N acts trivially on π^*K^q, the group $(\pi^*K^q)^G$ is $(K^q)^{G/N}$. The additive functor $(-)^G$ preserves homotopies, so the map $\Theta_B = H^n(\chi^G)$ does not depend on χ. It is natural in B. Let $b\colon B \to C$ be a map of discrete G/N-modules, with injective resolutions $C \to K^\bullet_C$ and $\pi^*C \to L^\bullet_C$ and a chain map $\chi_C\colon \pi^*K^\bullet_C \to L^\bullet_C$ as above. Lifts $\tilde{b}\colon K^\bullet \to K^\bullet_C$ of b and $\hat{b}\colon L^\bullet \to L^\bullet_C$ of π^*b give chain maps $\chi_C \circ \pi^*\tilde{b}$ and $\hat{b} \circ \chi$ from $\pi^*K^\bullet$ to $L^\bullet_C$, which both induce π^*b on H^0 and are therefore homotopic. Taking G-invariants and cohomology gives $\Theta_C \circ H^n(G/N, b) = H^n(G, \pi^*b) \circ \Theta_B$, as the derived functors take the values $H^n(\tilde{b}^{G/N})$ and $H^n(\hat{b}^G)$ on b and π^*b. The case $b = \mathrm{id}_B$ shows that Θ_B is compatible with the comparison isomorphisms between different choices of resolutions. In degree 0, χ^0 restricts to the identity of π^*B through the augmentations, so Θ_B is the identity of $B^{G/N} = (\pi^*B)^G$.

Step 2: Connecting homomorphisms. Let $0 \to B' \to B \to B'' \to 0$ be a short exact sequence of discrete G/N-modules. By *Homological Algebra* [13, Horseshoe Lemma], in its injective form, it lifts to a degreewise split short exact sequence $0 \to K'^\bullet \to K^\bullet \to K''^\bullet \to 0$ of injective resolutions, and the connecting homomorphisms of $H^\bullet(G/N, -)$ are those of the Snake Lemma for $0 \to (K'^\bullet)^{G/N} \to (K^\bullet)^{G/N} \to (K''^\bullet)^{G/N} \to 0$, as in the proof of *Homological Algebra* [13, Long Exact Sequence From a Right-Derived Functor]. In the same way a degreewise split short exact sequence $0 \to L'^\bullet \to L^\bullet \xrightarrow{p} L''^\bullet \to 0$ of injective resolutions of $0 \to \pi^*B' \to \pi^*B \to \pi^*B'' \to 0$ computes the connecting homomorphisms of $H^\bullet(G, \pi^*(-))$. Applying π^* to the first sequence gives a degreewise split short exact sequence $0 \to \pi^*K'^\bullet \to \pi^*K^\bullet \xrightarrow{p_K} \pi^*K''^\bullet \to 0$. Choose $\chi\colon \pi^*K^\bullet \to L^\bullet$ and $\chi''\colon \pi^*K''^\bullet \to L''^\bullet$ as in Step 1, and a degreewise section s of p. The chain maps $p \circ \chi$ and $\chi'' \circ p_K$ from $\pi^*K^\bullet$ to $L''^\bullet$ induce the same map $\pi^*B \to \pi^*B''$ on H^0, so by the uniqueness clause used in Step 1, now with target $L''^\bullet$, they differ by $dh + hd$ for maps $h\colon \pi^*K^q \to L''^{q-1}$. Then $\chi_1 = \chi - (d\,sh + sh\,d)$ is a chain map homotopic to χ, and $p \circ \chi_1 = p \circ \chi - (dh + hd) = \chi'' \circ p_K$, because p is a chain map and $ps = \mathrm{id}$. So χ_1 carries $\pi^*K'^\bullet = \ker(p_K)$ into $\ker(p) = L'^\bullet$, and its restriction $\chi'\colon \pi^*K'^\bullet \to L'^\bullet$ is a chain map inducing the identity of π^*B' on H^0. Then (χ', χ_1, χ'') is a morphism of short exact sequences of complexes. Applying $(-)^G$, the naturality clause of the Snake Lemma (*Homological Algebra* [13, Snake

Lemma]) gives $\delta \circ \Theta_{B''} = \Theta_{B'} \circ \delta$, the maps Θ being computed with χ', χ_1 and χ'' by Step 1.

Step 3: Conclusion. By Steps 1 and 2, the maps Θ_B form a morphism of δ-functors from $H^\bullet(G/N, -)$ to $H^\bullet(G, \pi^*(-))$ on $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$ whose degree-zero component is the identity of $B^{G/N} = (\pi^* B)^G$. By the uniqueness in lemma 3.3.6(3) it is inflation.

Proposition 3.4.7: Edge Maps

Identify the second page of the spectral sequence of theorem 3.4.5 with $H^p(G/N, H^q(N, M))$ through the isomorphisms α_M of that theorem. Its two edge morphisms (*Homological Algebra* [13, Edge Morphisms]) are inflation and restriction:

1. *Inflation edge:* For each $n \geq 0$, the composition

$$H^n(G/N, M^N) = E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^n H^n / F^{n+1} H^n \hookrightarrow H^n(G, M)$$

   (where the first map is the quotient by the images of all differentials landing on the $(n, 0)$ spot, and the second map is the inclusion of the step $F^n H^n$ of the filtration, the last before $F^{n+1} H^n = 0$) coincides with the inflation map $\mathrm{Inf} : H^n(G/N, M^N) \to H^n(G, M)$ of definition 3.3.1.

2. *Restriction edge:* For each $n \geq 0$, the composition

$$H^n(G, M) = F^0 H^n \twoheadrightarrow F^0 H^n / F^1 H^n = E_\infty^{0,n} \hookrightarrow E_2^{0,n} = H^n(N, M)^{G/N}$$

   (where the first map is the quotient, and the second map is the inclusion of E_∞ into E_2) coincides with the restriction map $\mathrm{Res} : H^n(G, M) \to H^n(N, M)$ of definition 3.3.2, which therefore takes values in $H^n(N, M)^{G/N}$.

***Proof*:**

Step 1: The two edge morphisms. Let $M \to I^\bullet$ be the injective resolution in $\mathrm{Mod}_G^{\mathrm{disc}}$ from which the spectral sequence is built, and put $X = (I^\bullet)^N$. It is a complex in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$, concentrated in non-negative degrees, whose terms are injective (lemma 3.4.4), with $H^0(X) = M^N$ by left exactness of $(-)^N$ and $H^q(X) = R^q(-)^N(M)$. By *Homological Algebra* [13, Edge Morphisms of the Grothendieck Spectral Sequence], the first edge morphism is $\mathbb{R}^n(-)^{G/N}(\iota)$, for the inclusion $\iota : M^N \to X$ of M^N as a complex concentrated in degree 0, composed with the isomorphisms $\mathbb{R}^n(-)^{G/N}(M^N) \cong H^n(G/N, M^N)$ and $\mathbb{R}^n(-)^{G/N}(X) \cong H^n(X^{G/N}) = H^n((I^\bullet)^G) = H^n(G, M)$ of *Homological Algebra* [13, Hypercohomology Recovers the Ordinary Derived Functors]. The second edge morphism is the map $\kappa_X : H^n(X^{G/N}) \to H^n(X)^{G/N}$ induced by the projection from the cocycles of X onto $H^n(X)$.

Step 2: The first edge morphism through a resolution. Let $M^N \to K^\bullet$ be an injective resolution in $\mathrm{Mod}_{G/N}^{\mathrm{disc}}$. By *Homological Algebra* [13, Projective Resolution As Initial Objects], in its dual form for complexes of injective objects, there is a chain map $\varphi : K^\bullet \to X$ inducing the identity of M^N on H^0, unique up to homotopy. Compute the hypercohomology of the complex M^N, concentrated in degree 0, with the Cartan-Eilenberg resolution whose column 0 is $K^\bullet$ and whose other columns vanish, which is allowed because hypercohomology does not depend on

the resolution (*Homological Algebra* [13, Hypercohomology is Well Defined]). The conditions of *Homological Algebra* [13, Cartan-Eilenberg Resolution] hold, since the complex M^N has zero differential, so that its cycles and cohomology in degree 0 are M^N and its coboundaries vanish. The total complex is $K^\bullet$, and the isomorphism $\mathbb{R}^n(-)^{G/N}(M^N) \cong H^n(G/N, M^N)$, the collapse of the second hypercohomology spectral sequence onto its row $q = 0$, is induced by the identity on representatives, as in Step 3 of the proof of *Homological Algebra* [13, Edge Morphisms of the Grothendieck Spectral Sequence]. So it is the identity of $H^n((K^\bullet)^{G/N})$. Let $J^{\bullet,\bullet}$ be the Cartan-Eilenberg resolution of X, with augmentation $\varepsilon\colon X \to J^{\bullet,0}$, and let $T = \mathrm{Tot}(J^{\bullet,\bullet})$, a complex of injective objects concentrated in non-negative degrees. The augmentation is a chain map $\varepsilon\colon X \to T$, because each column of $J^{\bullet,\bullet}$ is a resolution, and by Step 3 of the same proof the isomorphism $H^n(X^{G/N}) \cong H^n(T^{G/N})$ is induced by $\varepsilon^{G/N}$. By *Homological Algebra* [13, Comparison Theorem for Cartan-Eilenberg Resolutions], ι lifts to a morphism of double complexes Λ from $K^\bullet$, placed in column 0, to $J^{\bullet,\bullet}$, compatible with the augmentations, and $\mathbb{R}^n(-)^{G/N}(\iota)$ is the map induced by $\mathrm{Tot}(\Lambda)^{G/N}$. The chain maps $\mathrm{Tot}(\Lambda)$ and $\varepsilon \circ \varphi$ from $K^\bullet$ to T induce the same map $b \mapsto [\varepsilon(\iota(b))]$ from M^N to $H^0(T)$: the first by the compatibility of Λ with the augmentations, the second because φ induces the identity of M^N. By the dual form of *Homological Algebra* [13, Projective Resolution As Initial Objects] they are homotopic, so they induce the same map on the cohomology of G/N-invariants. The first edge morphism is therefore

$$H^n(\varepsilon^{G/N})^{-1} \circ H^n\big(\mathrm{Tot}(\Lambda)^{G/N}\big) = H^n(\varepsilon^{G/N})^{-1} \circ H^n(\varepsilon^{G/N}) \circ H^n(\varphi^{G/N}) = H^n(\varphi^{G/N})$$

Step 3: The first edge morphism is inflation. Write $\pi^*(M^N)$ for M^N regarded as a G-module through $\pi\colon G \to G/N$, and $j\colon \pi^*(M^N) \to M$ for the inclusion, a map of discrete G-modules. Since N acts trivially on π^*K^q, a G-map $\pi^*K^q \to I^q$ is the same as a G/N-map $K^q \to (I^q)^N$, so φ is a chain map $\tilde{\varphi}\colon \pi^*K^\bullet \to I^\bullet$, and it induces j on H^0. Let $\pi^*(M^N) \to L^\bullet$ be an injective resolution in $\mathrm{Mod}_G^{\mathrm{disc}}$, let $\chi\colon \pi^*K^\bullet \to L^\bullet$ be as in lemma 3.4.6, and let $\hat{\jmath}\colon L^\bullet \to I^\bullet$ be a chain map inducing j on H^0, which exists by the dual form of *Homological Algebra* [13, Projective Resolution As Initial Objects]. The chain maps $\hat{\jmath} \circ \chi$ and $\tilde{\varphi}$ from the resolution $\pi^*K^\bullet$ of $\pi^*(M^N)$ to the complex of injective objects $I^\bullet$ both induce j on H^0, so they are homotopic by the same lemma. On G-invariants $\tilde{\varphi}^G = \varphi^{G/N}$, as $(I^q)^G = ((I^q)^N)^{G/N}$, so

$$H^n(\varphi^{G/N}) = H^n(\hat{\jmath}^G) \circ H^n(\chi^G)$$

By lemma 3.4.6, $H^n(\chi^G)$ is the inflation map $H^n(G/N, M^N) \to H^n(G, \pi^*(M^N))$, and $H^n(\hat{\jmath}^G) = H^n(G, j)$ by the definition of derived functors on morphisms. On continuous cochains, inflation into $H^n(G, \pi^*(M^N))$ followed by composition with j is the inflation map of definition 3.3.1, and the isomorphisms of proposition 3.1.6(2) are natural in the module. With Step 2 this proves (1).

Step 4: The second edge morphism is restriction. The cocycles of $X^{G/N} = (I^\bullet)^G$ are the G/N-invariant cocycles of X, so the composite of κ_X with the inclusion $H^n(X)^{G/N} \subseteq H^n(X)$ is the map $\lambda_M\colon H^n((I^\bullet)^G) \to H^n((I^\bullet)^N)$ induced by the inclusion of complexes $(I^\bullet)^G \subseteq (I^\bullet)^N$. Let $U\colon \mathrm{Mod}_{G/N}^{\mathrm{disc}} \to \mathbf{Ab}$ be the forgetful functor. As M varies, λ is a morphism of δ-functors from $H^\bullet(G, -)$ to $U \circ R^\bullet(-)^N$. It is natural in M, because a lift of a morphism of modules to injective resolutions commutes with the two inclusions. For a short exact sequence of discrete G-modules, one degreewise split short exact sequence $0 \to I'^\bullet \to I^\bullet \to I''^\bullet \to 0$ of injective resolutions lifting it (*Homological Algebra* [13, Horseshoe Lemma]) computes the connecting

homomorphisms of both δ-functors, by the Snake Lemma applied to its G-invariants and to its N-invariants. The inclusions of G-invariants into N-invariants form a morphism between these two short exact sequences of complexes, and the naturality clause of the Snake Lemma gives the compatibility with the connecting homomorphisms. In degree zero λ_M is the inclusion $M^G \subseteq M^N$. By theorem 3.4.5, $\alpha \circ \lambda \colon H^\bullet(G,-) \to H^\bullet(N,-)$ is a morphism of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ whose degree-zero component is the inclusion $M^G \subseteq M^N$, so it is Res by the uniqueness in lemma 3.3.6(1). Since $E_2^{0,n} = H^n(X)^{G/N}$ is identified with $H^n(N,M)^{G/N}$ through α_M, this proves (2).

Definition 3.4.8: Transgression

The *transgression* is the differential

$$\mathrm{tg} = d_2^{0,q} : E_2^{0,q} \to E_2^{2,q-1}$$

of the spectral sequence of theorem 3.4.5, which maps $E_2^{0,q} = H^q(N,M)^{G/N}$ to $E_2^{2,q-1} = H^2(G/N, H^{q-1}(N,M))$.

The transgression in degree $q = 1$ is the map

$$\mathrm{tg} : H^1(N,M)^{G/N} \to H^2(G/N, M^N)$$

that appeared in the five-term exact sequence of theorem 3.3.4. Its kernel is the image of restriction and its image is the kernel of inflation, by proposition 3.4.9 below. Up to sign it agrees with the map described on cochains in [26, Ch. I, §6, (1.6.6)], recalled below. Kernel and image, the only properties used below, do not depend on the sign. Let $x \colon N \to M$ be a continuous 1-cocycle whose class is G/N-invariant. There is a continuous 1-cochain $y \colon G \to M$ with $y|_N = x$ such that dy takes values in M^N and depends only on the cosets of its arguments modulo N, so that dy is the inflation of a 2-cocycle c of G/N with values in M^N. The class of c does not depend on the choice of y, and [26, Ch. II, §4, (2.4.3)] identifies $[x] \mapsto [c]$ with the differential $d_2^{0,1}$ of the Hochschild-Serre spectral sequence as constructed there.

The five-term exact sequence stated in theorem 3.3.4 is the low-degree exact sequence of the LHS spectral sequence.

Proposition 3.4.9: Five-Term Exact Sequence from the Spectral Sequence

The spectral sequence of theorem 3.4.5 yields the exact sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^1(G,M) \xrightarrow{\mathrm{Res}} H^1(N,M)^{G/N} \xrightarrow{\mathrm{tg}} H^2(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^2(G,M).$$

***Proof*:**
Examine the E_2 page in the region relevant to total degree $n = 1$ and $n = 2$:

$q=2$	$H^0(G/N, H^2(N,M))$	$\cdots$	
$q=1$	$H^0(G/N, H^1(N,M))$	$H^1(G/N, H^1(N,M))$	$\cdots$
$q=0$	$H^0(G/N, M^N)$	$H^1(G/N, M^N)$	$H^2(G/N, M^N)$
	$p=0$	$p=1$	$p=2$

The differential $d_2^{p,q} : E_2^{p,q} \to E_2^{p+2,q-1}$ maps two steps to the right and one step down. The

relevant differentials are:

- $d_2^{0,1} : E_2^{0,1} = H^1(N, M)^{G/N} \to E_2^{2,0} = H^2(G/N, M^N)$. This is the transgression.
- $d_2^{1,0} : E_2^{1,0} = H^1(G/N, M^N) \to E_2^{3,-1} = 0$. This is the zero map.

Step 1: $E_\infty^{1,0} = E_2^{1,0}$. The term $E_2^{1,0}$ can only be affected by differentials from $E_2^{-1,1} = 0$ (incoming) and $d_2^{1,0}$ to $E_2^{3,-1} = 0$ (outgoing). Hence $E_3^{1,0} = E_2^{1,0}$. All subsequent differentials $d_r^{1,0} : E_r^{1,0} \to E_r^{1+r,1-r}$ land in the region $q < 0$ for $r \geq 2$, and differentials $d_r^{1-r,r-1} : E_r^{1-r,r-1} \to E_r^{1,0}$ originate in the region $p < 0$ for $r \geq 2$. So $E_\infty^{1,0} = E_2^{1,0} = H^1(G/N, M^N)$.

Step 2: $E_\infty^{0,1} = \ker(\mathrm{tg})$. The term $E_2^{0,1}$ can only receive differentials from $E_2^{-2,2} = 0$ and emits $d_2^{0,1} = \mathrm{tg}$. Hence $E_3^{0,1} = \ker(\mathrm{tg})$. For $r \geq 3$, all subsequent differentials to and from $E_r^{0,1}$ involve terms with $p < 0$, so $E_\infty^{0,1} = \ker(\mathrm{tg})$.

Step 3: The filtration on $H^1(G, M)$. The convergence gives a filtration $F^1H^1 \subseteq H^1(G, M)$ with $H^1(G, M)/F^1H^1 \cong E_\infty^{0,1} = \ker(\mathrm{tg})$ and $F^1H^1 \cong E_\infty^{1,0} = H^1(G/N, M^N)$. By the edge map identifications (proposition 3.4.7), $F^1H^1 = \mathrm{im}(\mathrm{Inf})$ and the quotient map $H^1(G, M) \twoheadrightarrow E_\infty^{0,1}$ is Res. This gives the exact sequence

$$0 \to H^1(G/N, M^N) \xrightarrow{\mathrm{Inf}} H^1(G, M) \xrightarrow{\mathrm{Res}} H^1(N, M)^{G/N} \xrightarrow{\mathrm{tg}} H^2(G/N, M^N)$$

Step 4: Exactness at $H^2(G/N, M^N)$. The filtration on $H^2(G, M)$ gives $E_\infty^{2,0} \cong F^2H^2 \subseteq H^2(G, M)$, and $E_\infty^{2,0} = E_2^{2,0}/\mathrm{im}(\mathrm{tg}) = H^2(G/N, M^N)/\mathrm{im}(\mathrm{tg})$ (since the only differential landing on $E_2^{2,0}$ at the E_2 level is $d_2^{0,1} = \mathrm{tg}$, and subsequent differentials from $E_r^{2-r,r-1}$ with $r \geq 3$ originate in $p < 0$). The edge map identification shows that $F^2H^2 \hookrightarrow H^2(G, M)$ is the inflation map. Hence $\ker(\mathrm{Inf} : H^2(G/N, M^N) \to H^2(G, M)) = \mathrm{im}(\mathrm{tg})$, completing the five-term exact sequence.

Beyond the five-term exact sequence, the d_2 differentials

$$d_2^{p,q} : H^p(G/N, H^q(N, M)) \to H^{p+2}(G/N, H^{q-1}(N, M))$$

are the first differentials on the E_2 page. Two cases recur in this book:

- $d_2^{0,q}$: the transgression of definition 3.4.8, whose kernel and image for $q = 1$ are fixed by the five-term exact sequence.
- $d_2^{p,0}$: always zero, since the source is $H^p(G/N, M^N)$ and the target is $H^{p+2}(G/N, H^{-1}(N, M)) = 0$.

When all the differentials on a page vanish, the spectral sequence *degenerates* at that page, and the filtration on $H^n(G, M)$ is determined by the E_2 terms alone.

Example 3.4.10: A Central Extension

Let p be a prime and consider the central extension

$$1 \to N \to G \to Q \to 1$$

with $N = \mathbb{Z}/p\mathbb{Z}$, $G = \mathbb{Z}/p^2\mathbb{Z}$, and $Q = G/N \cong \mathbb{Z}/p\mathbb{Z}$. Take $M = \mathbb{Z}/p\mathbb{Z}$ with trivial G-action. Since the extension is central ($N \subseteq Z(G)$), the conjugation action of Q on $H^q(N, M)$ is trivial, which simplifies the E_2 page considerably.

1. The E_2 page. Since $N = \mathbb{Z}/p\mathbb{Z}$ and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action, the cohomology $H^q(N, M)$ is computed by the periodic cohomology of cyclic groups (example 2.3.8):

   $$H^q(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z} \quad \text{for all } q \geq 0.$$

   Since the Q-action is trivial, the E_2 page is

   $$E_2^{i,j} = H^i(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z} \quad \text{for all } i, j \geq 0$$

   Every entry in the first quadrant is a copy of $\mathbb{Z}/p\mathbb{Z}$:

$j = 3$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$j = 2$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$j = 1$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
$j = 0$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p\mathbb{Z}$	$\cdots$
	$i = 0$	$i = 1$	$i = 2$	$i = 3$	

2. The differentials. The spectral sequence converges to $H^n(G, M) = H^n(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$. By the periodic cohomology of cyclic groups, $H^n(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ for all $n \geq 0$, since the norm acts as multiplication by p^2, which is zero on $\mathbb{Z}/p\mathbb{Z}$. So the abutment is $\mathbb{Z}/p\mathbb{Z}$ in every total degree.

   On the E_2 page, the total degree n contributes $n+1$ copies of $\mathbb{Z}/p\mathbb{Z}$, from $E_2^{0,n}$, $E_2^{1,n-1}$, $\ldots$, $E_2^{n,0}$. Since the abutment is a single $\mathbb{Z}/p\mathbb{Z}$ in each degree, the differentials must remove all but one copy on each diagonal. Each differential $d_2^{i,j} : E_2^{i,j} \to E_2^{i+2,j-1}$ is a map $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, hence either zero or an isomorphism.

3. The transgression. The differential $d_2^{0,1} : H^1(N, M)^Q \to H^2(Q, M^N)$ is a map $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. By the five-term exact sequence, $\ker(\mathrm{tg}) = E_\infty^{0,1}$ contributes to the filtration on $H^1(G, M) \cong \mathbb{Z}/p\mathbb{Z}$, and $E_\infty^{1,0} = H^1(Q, M) \cong \mathbb{Z}/p\mathbb{Z}$ also contributes. Since $H^1(G, M) \cong \mathbb{Z}/p\mathbb{Z}$ has order p and must be an extension of $E_\infty^{0,1}$ by $E_\infty^{1,0}$, one of these must be zero. The inflation map $\mathrm{Inf} : H^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \to H^1(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z})$ sends a homomorphism $\mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ to its inflation along $\mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, which is the composition $\mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. This is nonzero (the inflation of the identity map is the reduction-mod-p map, which is a nonzero element of $\mathrm{Hom}(\mathbb{Z}/p^2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$). Hence $E_\infty^{1,0} = \mathbb{Z}/p\mathbb{Z}$, forcing $E_\infty^{0,1} = 0$, so $\mathrm{tg} : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ is an isomorphism.

4. Conclusion. The differential $d_2^{0,1}$ is an isomorphism. Counting does not force the remaining differentials, since a diagonal can lose its excess copies in several patterns. They are determined by the multiplicative structure of the spectral sequence, which this book does not develop.

When the E_2 page is "too large" for the known abutment, the differentials must kill the excess, and independent knowledge of the abutment and of the edge maps can force individual differentials, as it forced the transgression in the central extension example.

Example 3.4.11: The LHS Spectral Sequence for $G_{\mathbb{Q}_p}$

Let p be a prime, $K = \mathbb{Q}_p$, and consider the short exact sequence of profinite groups

$$1 \to I_p \to G_{\mathbb{Q}_p} \to \hat{\mathbb{Z}} \to 1,$$

where $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p^{\mathrm{nr}})$ is the inertia group and $\hat{\mathbb{Z}} = \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p)$ is generated topologically by the Frobenius Frob_p. Let ℓ be a prime with $\ell \neq p$, and take $M = \mathbb{Z}/\ell\mathbb{Z}$ with trivial $G_{\mathbb{Q}_p}$-action.

1. The E_2 page. The cohomology of the inertia group with trivial $\mathbb{Z}/\ell\mathbb{Z}$-coefficients is:

$$H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathbb{Z}/\ell\mathbb{Z},$$
$$H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathrm{Hom}_{\mathrm{cts}}(I_p, \mathbb{Z}/\ell\mathbb{Z}).$$

   The group I_p has tame quotient $I_p^{\mathrm{t}} = I_p/P_p \cong \prod_{\ell' \neq p} \mathbb{Z}_{\ell'}$ (where P_p is the wild inertia, a pro-p-group). Since $\ell \neq p$, every continuous homomorphism $I_p \to \mathbb{Z}/\ell\mathbb{Z}$ factors through the tame quotient, and $\mathrm{Hom}_{\mathrm{cts}}(I_p^{\mathrm{t}}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ (the ℓ-component of the tame quotient contributes one copy). So $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$.

   The conjugation action of $\hat{\mathbb{Z}}$ on $H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathbb{Z}/\ell\mathbb{Z}$ is trivial. Its action on $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z})$, which is $\mathrm{Hom}_{\mathrm{cts}}(I_p, \mathbb{Z}/\ell\mathbb{Z})$, is determined by the conjugation action of a lift of Frob_p to $G_{\mathbb{Q}_p}$ on I_p, and does not depend on the lift (lemma 3.4.2(1) and (2)). The Frobenius acts on the tame inertia by $\mathrm{Frob}_p \sigma \mathrm{Frob}_p^{-1} = \sigma^p$, where p is a unit of $\prod_{\ell' \neq p} \mathbb{Z}_{\ell'}$, so $\mathrm{Frob}_p^{-1} \sigma \mathrm{Frob}_p = \sigma^{p^{-1}}$. By definition 3.4.1, with the trivial action on $\mathbb{Z}/\ell\mathbb{Z}$, the Frobenius sends a homomorphism f to the homomorphism $\sigma \mapsto f(\mathrm{Frob}_p^{-1} \sigma \mathrm{Frob}_p)$, whose value at σ is $f(\sigma^{p^{-1}})$, that is, $p^{-1} f(\sigma)$. Hence $\hat{\mathbb{Z}}$ acts on $H^1(I_p, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ through $\mathrm{Frob}_p \mapsto p^{-1}$, where p^{-1} is the inverse of p modulo ℓ. Write A for $\mathbb{Z}/\ell\mathbb{Z}$ with this action.

   The rows $j \geq 2$ vanish. The wild inertia P_p is pro-p, so $\mathrm{cd}_\ell(P_p) = 0$ by proposition 5.1.4, and the tame quotient is isomorphic to the closed subgroup $\prod_{\ell' \neq p} \mathbb{Z}_{\ell'}$ of $\hat{\mathbb{Z}} \cong \prod_{\ell'} \mathbb{Z}_{\ell'}$, so $\mathrm{cd}_\ell(I_p^{\mathrm{t}}) \leq 1$ by example 5.1.5 and proposition 5.1.3(1). By proposition 5.1.3(2), $\mathrm{cd}_\ell(I_p) \leq 1$, and $H^j(I_p, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for $j \geq 2$.

2. The row $j = 0$. Since the action on $H^0(I_p, \mathbb{Z}/\ell\mathbb{Z}) = \mathbb{Z}/\ell\mathbb{Z}$ is trivial, the bottom row is $E_2^{i,0} = H^i(\hat{\mathbb{Z}}, \mathbb{Z}/\ell\mathbb{Z})$. By theorem 3.2.3 (or example 3.2.6), this is $\mathbb{Z}/\ell\mathbb{Z}$ for $i = 0, 1$ and 0 for $i \geq 2$.

3. The row $j = 1$. The row is $E_2^{i,1} = H^i(\hat{\mathbb{Z}}, A)$. By theorem 3.2.3, $H^0(\hat{\mathbb{Z}}, A) = \ker(p^{-1} - 1)$, $H^1(\hat{\mathbb{Z}}, A) = A/(p^{-1} - 1)A$, and $H^i(\hat{\mathbb{Z}}, A) = 0$ for $i \geq 2$.
   - If $p \equiv 1 \pmod{\ell}$, the action is trivial, and $E_2^{0,1} \cong E_2^{1,1} \cong \mathbb{Z}/\ell\mathbb{Z}$.
   - If $p \not\equiv 1 \pmod{\ell}$, then $p^{-1} - 1$ is invertible on $\mathbb{Z}/\ell\mathbb{Z}$, so $E_2^{0,1} = E_2^{1,1} = 0$.

4. Consequences. The second page is concentrated in the columns $i = 0$ and $i = 1$, and every differential d_r with $r \geq 2$ changes i by r, so all differentials vanish and the spectral sequence degenerates at E_2. When $p \not\equiv 1 \pmod{\ell}$, the only nonzero terms are $E_2^{0,0}$ and $E_2^{1,0}$, so $H^0(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) \cong H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$ and $H^n(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for $n \geq 2$.

   When $p \equiv 1 \pmod{\ell}$, the four terms $E_2^{0,0}$, $E_2^{1,0}$, $E_2^{0,1}$ and $E_2^{1,1}$ are copies of $\mathbb{Z}/\ell\mathbb{Z}$. So $H^1(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z})$ has order ℓ^2, $H^2(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) \cong \mathbb{Z}/\ell\mathbb{Z}$, and $H^n(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for $n \geq 3$. This case is precisely when $\mathbb{Q}_p$ contains the ℓ-th roots of unity, and the extra classes reflect the Kummer theory of chapter 8.

The two examples exhibit the two basic behaviours of the LHS spectral sequence: degeneration at E_2 (when the differentials are forced to vanish for dimensional reasons) and nontrivial differentials (when the E_2 page is too large for the abutment). In the local field example an arithmetic condition, whether $\mathbb{Q}_p$ contains the ℓ-th roots of unity, decides which terms of the second page are nonzero.

Exercise 3.4.1

1. Let N be a closed normal subgroup of G and M a discrete G-module. Verify that the conjugation action of G/N on $H^1(N, M)$ is well defined: show that if $f : N \to M$ is a continuous 1-cocycle and $n_0 \in N$, then the cocycles f and ${}_{n_0\#}f$ (defined by $({}_{n_0\#}f)(n) = n_0 \cdot f(n_0^{-1} n n_0)$) differ by the coboundary of the element $f(n_0) \in M$. (*Hint:* Use the cocycle identity $f(n_0 \cdot n_0^{-1} n n_0) = n_0 \cdot f(n_0^{-1} n n_0) + f(n_0)$.)

2. Let $G = D_4$ (the dihedral group of order 8), $N = \langle r^2 \rangle \cong \mathbb{Z}/2\mathbb{Z}$ (the center), and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. Compute the E_2 page of the LHS spectral sequence. Then use the five-term exact sequence and $H^1(D_4, \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^2$ to show that the transgression $d_2^{0,1}$ is injective, so that the spectral sequence does not degenerate at E_2.

3. Let $1 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 1$ be the unique nontrivial central extension with $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action). Use the five-term exact sequence to show that the transgression $\mathrm{tg} : H^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}/2\mathbb{Z}} \to H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ is an isomorphism, and confirm with proposition 2.3.9 that inflation $H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \to H^2(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ is zero. Compare with exercise 3.3.1 (5), where $M = \mathbb{Z}$.

4. Let G be a profinite group and N a closed normal subgroup such that G/N has p-cohomological dimension ≤ 1 (i.e., $H^n(G/N, A) = 0$ for all $n \geq 2$ and all p-torsion discrete G/N-modules A). Show that for every p-torsion discrete G-module M, the restriction map $\mathrm{Res} : H^n(G, M)\{p\} \to H^n(N, M)\{p\}$ is surjective onto the G/N-invariants for every $n \geq 0$. (*Hint:* The $E_2^{p,q}$ terms with $p \geq 2$ vanish, so the spectral sequence degenerates.)

5. Use the LHS spectral sequence to prove the following: if N is a closed normal subgroup of G with $\mathrm{cd}_p(N) \leq m$ and $\mathrm{cd}_p(G/N) \leq n$ (where cd_p denotes p-cohomological dimension, to be

defined formally in chapter 5), then $\mathrm{cd}_p(G) \leq m+n$. (*Hint:* The $E_2^{p,q}$ terms vanish for $p > n$ or $q > m$.)

3.5 Shapiro's Lemma and Induced Modules

The coinduced modules introduced in definition 2.2.4 provided acyclic objects for computing cohomology via derived functors in the discrete setting. This section develops their profinite analogues and establishes Shapiro's lemma, which asserts that the cohomology of a profinite group G with coefficients in a coinduced module reduces to the cohomology of the subgroup. This reduction is one of the most frequently used tools in Galois cohomology: it allows cohomological questions about an absolute Galois group G_K to be transferred to questions about G_L for a field extension L/K, simply by choosing the right coefficient module.

Throughout this section, G denotes a profinite group and H a closed subgroup of G.

Definition 3.5.1: Coinduced Module

Let H be a closed subgroup of a profinite group G, and let M be a discrete H-module. The *(continuously) coinduced module* is

$$\begin{aligned}\mathrm{CoInd}_H^G(M) &= \mathrm{Map}_H^{\mathrm{cts}}(G, M) \\ &= \{f : G \to M \mid f \text{ is continuous and } f(hg) = h \cdot f(g) \text{ for all } h \in H,\, g \in G\},\end{aligned}$$

with G-action given by right translation: $(g' \cdot f)(g) = f(gg')$ for $g' \in G$.

The continuity condition means that $f : G \to M$ is locally constant (since M is discrete). The H-equivariance condition $f(hg) = h \cdot f(g)$ determines f on each right coset Hg from its value at a single point: if $f(g)$ is known, then $f(hg) = h \cdot f(g)$ for all $h \in H$.

The coinduced module $\mathrm{CoInd}_H^G(M)$ is a discrete G-module. To see this, fix $f \in \mathrm{CoInd}_H^G(M)$. Since f is locally constant and G is compact, f factors through G/U for some open normal subgroup U of G. For any $u \in U$, the translate $(u \cdot f)(g) = f(gu) = f(g)$ (since f is constant on right U-cosets), so $U \subseteq \mathrm{Stab}_G(f)$, and the stabilizer is open.

Definition 3.5.2: Induced Module

Let H be an *open* subgroup of G (so that $[G:H]$ is finite), and let M be a discrete H-module. The *(continuously) induced module* is

$$\mathrm{Ind}_H^G(M) = \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} M,$$

where G acts on the left tensor factor by left multiplication. Since H has finite index, this is a finite direct sum: choosing left coset representatives $s_1, \dots, s_m$ for H in G,

$$\mathrm{Ind}_H^G(M) = \bigoplus_{i=1}^{m} s_i \otimes_H M \cong \bigoplus_{i=1}^{m} M$$

as abelian groups, with G-action permuting the summands according to the coset action and acting on the module entry via H.

For H open, the induced module is a discrete G-module: each element is a finite sum with entries in M (a discrete H-module), and the stabilizer is a finite intersection of open subgroups, hence open.

The induced and coinduced modules are related by an isomorphism when H is open.

Proposition 3.5.3: Induced Equals Coinduced for Open Subgroups

Let H be an open subgroup of G and M a discrete H-module. There is a canonical isomorphism of discrete G-modules

$$\mathrm{Ind}_H^G(M) \xrightarrow{\sim} \mathrm{CoInd}_H^G(M).$$

***Proof*:**

For $g \in G$ and $m \in M$ let $f_{g,m} \colon G \to M$ be the function with $f_{g,m}(x) = xg \cdot m$ if $xg \in H$ and $f_{g,m}(x) = 0$ otherwise. It lies in $\mathrm{CoInd}_H^G(M)$. For $h \in H$, $hxg \in H$ exactly when $xg \in H$, and then $f_{g,m}(hx) = h \cdot (xg \cdot m) = h \cdot f_{g,m}(x)$. The set $\{x \in G \mid xg \in H\} = Hg^{-1}$ is open, its complement is a union of right cosets of the open subgroup H, hence open, and on Hg^{-1} the map $x \mapsto xg \cdot m$ is locally constant because $\mathrm{Stab}_H(m)$ is open in H. So $f_{g,m}$ is continuous. It is additive in m, and $f_{gh,m} = f_{g,h\cdot m}$ for $h \in H$, since $xgh \in H$ exactly when $xg \in H$ and then both functions take the value $xgh \cdot m$. So there is a homomorphism

$$\varphi \colon \mathrm{Ind}_H^G(M) = \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} M \to \mathrm{CoInd}_H^G(M), \qquad g \otimes m \mapsto f_{g,m}$$

It is G-equivariant: for $g' \in G$, the value $(g' \cdot f_{g,m})(x) = f_{g,m}(xg')$ is $xg'g \cdot m$ if $xg'g \in H$ and 0 otherwise, so $g' \cdot f_{g,m} = f_{g'g,m} = \varphi(g' \cdot (g \otimes m))$.

Let $s_1, \dots, s_r$ be left coset representatives, $G = \bigsqcup_i s_i H$, and define $\psi \colon \mathrm{CoInd}_H^G(M) \to \mathrm{Ind}_H^G(M)$ by $\psi(f) = \sum_i s_i \otimes f(s_i^{-1})$. For $x \in G$ exactly one index i satisfies $xs_i \in H$, namely the one with $s_i H = x^{-1} H$. For that i, the H-equivariance of f gives

$$\varphi(\psi(f))(x) = \sum_k f_{s_k, f(s_k^{-1})}(x) = xs_i \cdot f(s_i^{-1}) = f(xs_i s_i^{-1}) = f(x)$$

Similarly, for $g \in G$ exactly one index i satisfies $s_i^{-1} g \in H$, and then $g = s_i h$ with $h = s_i^{-1} g$, so $\psi(\varphi(g \otimes m)) = s_i \otimes h \cdot m = s_i h \otimes m = g \otimes m$. The elements $g \otimes m$ generate $\mathrm{Ind}_H^G(M)$, so ψ is a

two-sided inverse of φ, and φ is an isomorphism of discrete G-modules. The map φ involves no choices, so $\psi = \varphi^{-1}$ does not depend on the coset representatives.

When H is closed but not open (so $[G : H]$ is infinite), the induced and coinduced modules differ: $\mathrm{Ind}_H^G(M)$ involves compactly supported functions (finitely many nonzero cosets), while $\mathrm{CoInd}_H^G(M)$ allows all continuous H-equivariant functions. For Shapiro's lemma (the main result of this section), the coinduced module is the correct object.

The key property of the coinduced module is an adjunction: it is the right adjoint of restriction.

Proposition 3.5.4: Adjunction for Coinduced Modules

For any discrete G-module A and any discrete H-module M, there is a natural isomorphism

$$\mathrm{Hom}_G(A, \mathrm{CoInd}_H^G(M)) \cong \mathrm{Hom}_H(A, M),$$

where the right side views A as an H-module by restriction.

***Proof*:**

Given a G-equivariant map $\Phi : A \to \mathrm{CoInd}_H^G(M)$, define $\varphi : A \to M$ by $\varphi(a) = \Phi(a)(1)$ (evaluation at the identity). This is H-equivariant: for $h \in H$, $\varphi(h \cdot a) = \Phi(h \cdot a)(1) = (h \cdot \Phi(a))(1) = \Phi(a)(1 \cdot h) = \Phi(a)(h) = h \cdot \Phi(a)(1) = h \cdot \varphi(a)$, using the G-equivariance of Φ and the H-equivariance of elements of $\mathrm{CoInd}_H^G(M)$.

Conversely, given an H-equivariant map $\varphi\colon A \to M$, define $\Phi\colon A \to \mathrm{CoInd}_H^G(M)$ by $\Phi(a)(g) = \varphi(g \cdot a)$. The map $\Phi(a)$ lies in $\mathrm{CoInd}_H^G(M)$: it satisfies $\Phi(a)(hg) = \varphi(h \cdot (g \cdot a)) = h \cdot \varphi(g \cdot a) = h \cdot \Phi(a)(g)$ by the H-equivariance of φ, and it is continuous because a has an open stabilizer in G, so $g \mapsto g \cdot a$ is locally constant. The map Φ is G-equivariant, since $\Phi(g' \cdot a)(g) = \varphi(gg' \cdot a) = \Phi(a)(gg') = (g' \cdot \Phi(a))(g)$.

The two constructions are mutually inverse. Starting from φ, evaluation at 1 gives $\Phi(a)(1) = \varphi(a)$. Starting from Φ, the map built from $\varphi = \Phi(-)(1)$ sends a to the function $g \mapsto \Phi(g \cdot a)(1) = (g \cdot \Phi(a))(1) = \Phi(a)(g)$, that is, to $\Phi(a)$. Both constructions are natural in A and in M, because precomposing with a G-map $A' \to A$, or composing with an H-map $M \to M'$, commutes with evaluation at 1, as claimed.

The adjunction of proposition 3.5.4 is the key input to Shapiro's lemma.

Theorem 3.5.5: Shapiro's Lemma

Let H be a closed subgroup of a profinite group G.

1. The functor $\mathrm{CoInd}_H^G\colon \mathrm{Mod}_H^{\mathrm{disc}} \to \mathrm{Mod}_G^{\mathrm{disc}}$ is exact and carries injective objects to injective objects.

2. With the connecting homomorphisms of the long exact sequences of the short exact sequences $0 \to \mathrm{CoInd}_H^G(M') \to \mathrm{CoInd}_H^G(M) \to \mathrm{CoInd}_H^G(M'') \to 0$, which are exact by (1), the family $H^\bullet(G, \mathrm{CoInd}_H^G(-))$ is a δ-functor from $\mathrm{Mod}_H^{\mathrm{disc}}$ to **Ab**, and there is exactly one isomorphism of δ-functors

$$\mathrm{Sh}\colon H^\bullet(G, \mathrm{CoInd}_H^G(-)) \xrightarrow{\sim} H^\bullet(H, -)$$

whose degree-zero component, under the natural identifications $H^0(G,-) \cong (-)^G$ and $H^0(H,-) \cong (-)^H$, is the isomorphism $\mathrm{CoInd}_H^G(M)^G \to M^H$, $f \mapsto f(1)$. In particular, for every discrete H-module M and every $n \geq 0$ there is a canonical isomorphism

$$H^n(G, \mathrm{CoInd}_H^G(M)) \cong H^n(H, M)$$

natural in M, and these isomorphisms carry the connecting homomorphisms of the long exact sequence of $0 \to \mathrm{CoInd}_H^G(M') \to \mathrm{CoInd}_H^G(M) \to \mathrm{CoInd}_H^G(M'') \to 0$ to those of $0 \to M' \to M \to M'' \to 0$.

When H is open, composing with the isomorphism of proposition 3.5.3 gives $H^n(G, \mathrm{Ind}_H^G(M)) \cong H^n(H, M)$.

***Proof*:**

Step 1: The functor CoInd_H^G *preserves injectives.* Since CoInd_H^G is right adjoint to the exact functor Res_G^H (proposition 3.5.4), it preserves injectives: if I is injective in $\mathrm{Mod}_H^{\mathrm{disc}}$, then $\mathrm{Hom}_G(-, \mathrm{CoInd}_H^G(I)) \cong \mathrm{Hom}_H(-, I)$ is exact (since Res_G^H is exact and I is injective), so $\mathrm{CoInd}_H^G(I)$ is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$.

Step 2: The functor CoInd_H^G *is exact.* The projection $G \to H\backslash G$ onto the space of cosets Hg has a continuous section t: for a continuous section s of $G \to G/H$, which exists by [30, Ch. I, §1.2, Prop. 1], set $t(Hx) = s(x^{-1}H)^{-1}$, which lies in Hx. Every $g \in G$ is then $g = h\,t(Hg)$ with $h = g\,t(Hg)^{-1} \in H$ depending continuously on g. So $f \mapsto f \circ t$ is an isomorphism of abelian groups, natural in M, from $\mathrm{CoInd}_H^G(M)$ onto the group of locally constant maps $H\backslash G \to M$, with inverse $\psi \mapsto \big(g \mapsto g\,t(Hg)^{-1} \cdot \psi(Hg)\big)$. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of discrete H-modules. The induced sequence of groups of locally constant maps on $H\backslash G$ is exact at its first two places pointwise. For surjectivity, a locally constant $\psi''\colon H\backslash G \to M''$ takes finitely many values, as $H\backslash G$ is compact, each on a clopen set, and these sets partition $H\backslash G$. Choosing a preimage in M of each value gives a locally constant lift of ψ''. So CoInd_H^G is exact, which with Step 1 proves (1).

Step 3: A universal δ*-functor.* A map $\varphi\colon M \to M'$ of discrete H-modules acts on $\mathrm{CoInd}_H^G(M)$ by composition, and $(\varphi + \psi) \circ f = \varphi \circ f + \psi \circ f$ for a second map $\psi\colon M \to M'$, so CoInd_H^G is an additive functor and each $T^n = H^n(G, \mathrm{CoInd}_H^G(-))$ is an additive functor $\mathrm{Mod}_H^{\mathrm{disc}} \to \mathbf{Ab}$.

By Step 2, CoInd_H^G carries a short exact sequence ξ of discrete H-modules to a short exact sequence $\mathrm{CoInd}_H^G(\xi)$ of discrete G-modules, and a morphism of short exact sequences $\xi \to \eta$ to a morphism $\mathrm{CoInd}_H^G(\xi) \to \mathrm{CoInd}_H^G(\eta)$. The derived functors $H^\bullet(G,-)$ form a δ-functor on $\mathrm{Mod}_G^{\mathrm{disc}}$ by *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor], so the long exact sequence of $\mathrm{CoInd}_H^G(\xi)$ and the naturality of its connecting homomorphisms are the two conditions of *Homological Algebra* [13, δ-Functor] for $T^\bullet$. If I is injective in $\mathrm{Mod}_H^{\mathrm{disc}}$, then $\mathrm{CoInd}_H^G(I)$ is injective in $\mathrm{Mod}_G^{\mathrm{disc}}$ by Step 1, so $T^n(I) = 0$ for $n > 0$ by the same corollary of *Homological Algebra*. The group H is profinite (proposition 1.2.1), so $\mathrm{Mod}_H^{\mathrm{disc}}$ is abelian with enough injectives (propositions 3.1.2 and 3.1.3), and $T^\bullet$ is universal by *Homological Algebra* [13, Universality and Effaceability Agree When There Are Enough Injectives].

Step 4: Comparison. The functor $(-)^H$ on $\mathrm{Mod}_H^{\mathrm{disc}}$ is additive and left exact, by proposition 2.1.9, because the kernels of $\mathrm{Mod}_H^{\mathrm{disc}}$ are those of Mod_H (proposition 3.1.2). So its derived functors $H^\bullet(H,-)$ form a universal δ-functor, and $H^0(H,M) \cong M^H$ naturally in M, by *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]. The same corollary gives $H^0(G,N) \cong N^G$ naturally in the discrete G-module N. The adjunction of proposition 3.5.4, with $A = \mathbb{Z}$ carrying the trivial action, gives

$$\mathrm{CoInd}_H^G(M)^G = \mathrm{Hom}_G(\mathbb{Z}, \mathrm{CoInd}_H^G(M)) \cong \mathrm{Hom}_H(\mathbb{Z}, M) = M^H$$

where the outer identifications send a map to the image of 1 (corollary 2.1.6). This isomorphism sends f to $f(1)$, and it is natural in M because the adjunction is natural in its second variable. Composing it with the two isomorphisms of degree zero gives a natural isomorphism $\varphi\colon T^0 \Rightarrow H^0(H,-)$. Universality of $T^\bullet$ (Step 3) extends φ to exactly one morphism of δ-functors $\mathrm{Sh}\colon T^\bullet \to H^\bullet(H,-)$, and as both δ-functors are universal, each component of Sh is an isomorphism by *Homological Algebra* [13, A Universal δ-Functor Is Determined by Its Degree-Zero Part]. An isomorphism of δ-functors with degree-zero component φ is a morphism extending φ, so it equals Sh. This proves (2), and the statement for $\mathrm{Ind}_H^G(M)$ follows from proposition 3.5.3.

Shapiro's lemma says that "cohomology does not see the coinduction process": computing $H^n(G,-)$ with coinduced coefficients is the same as computing $H^n(H,-)$ with the original coefficients, long exact sequences included.

A special case recovers the acyclicity result from theorem 2.2.6.

Corollary 3.5.6: Acyclicity of Coinduced Modules

Let H be a closed subgroup of G and M a discrete H-module.

1. If M is injective in $\mathrm{Mod}_H^{\mathrm{disc}}$, then $\mathrm{CoInd}_H^G(M)$ is injective (hence acyclic) in $\mathrm{Mod}_G^{\mathrm{disc}}$.
2. Taking $H = \{1\}$: for any abelian group J, the module $\mathrm{CoInd}_{\{1\}}^G(J) = \mathrm{Map}^{\mathrm{cts}}(G,J)$ (continuous functions $G \to J$ with the right-translation action) satisfies $H^n(G, \mathrm{CoInd}_{\{1\}}^G(J)) = H^n(\{1\}, J) = 0$ for all $n \geq 1$.

***Proof*:**
In Part (1), the module $\mathrm{CoInd}_H^G(M)$ is injective by theorem 3.5.5(1), hence acyclic by *Homological Algebra* [13, The Right-Derived Functors Form a Universal δ-Functor]. Part (2) is the special case $H = \{1\}$ of Shapiro's lemma: $H^n(G, \mathrm{CoInd}_{\{1\}}^G(J)) \cong H^n(\{1\}, J) = 0$ for $n \geq 1$, since the trivial group has no higher cohomology.

The Shapiro isomorphism interacts with the restriction and corestriction maps in a precise way. These compatibilities are essential for the computations in chapters 8 and 10.

Proposition 3.5.7: Compatibility of Shapiro with Restriction and Corestriction

Let H be a closed subgroup of G, and let Sh be the isomorphism of theorem 3.5.5(2). For a discrete G-module A, write $\mathrm{CoInd}_H^G(A)$ for the coinduced module of the restriction of A to H.

1. For every discrete H-module M, the evaluation $\mathrm{ev}\colon \mathrm{CoInd}_H^G(M) \to M$, $f \mapsto f(1)$, is H-equivariant, and
$$\mathrm{Sh} = \mathrm{ev}_* \circ \mathrm{Res}_G^H \colon H^n(G, \mathrm{CoInd}_H^G(M)) \to H^n(H, M)$$
2. For every discrete G-module A, the map $j_A\colon A \to \mathrm{CoInd}_H^G(A)$, $j_A(a)(g) = g \cdot a$, is G-equivariant, and
$$\mathrm{Res}_G^H = \mathrm{Sh} \circ (j_A)_* \colon H^n(G, A) \to H^n(H, A)$$
3. Let H be open, with left coset representatives $s_1, \ldots, s_r$. For every discrete G-module A, the map $\pi_A\colon \mathrm{CoInd}_H^G(A) \to A$, $\pi_A(f) = \sum_{i=1}^r s_i \cdot f(s_i^{-1})$, is G-equivariant, surjective, and independent of the representatives, and
$$\mathrm{Cor}_H^G = (\pi_A)_* \circ \mathrm{Sh}^{-1} \colon H^n(H, A) \to H^n(G, A)$$

***Proof*:**
Step 1: The three maps. For $h \in H$, $\mathrm{ev}(h \cdot f) = f(h) = h \cdot f(1)$, so ev is H-equivariant. The function $j_A(a)$ is continuous because a has an open stabilizer, and $j_A(a)(hg) = h \cdot j_A(a)(g)$, so $j_A(a) \in \mathrm{CoInd}_H^G(A)$. The map j_A is G-equivariant, since $j_A(g' \cdot a)(g) = gg' \cdot a = j_A(a)(gg') = (g' \cdot j_A(a))(g)$. Replacing s_i by $s_i h$ with $h \in H$ replaces $s_i \cdot f(s_i^{-1})$ by $s_i h \cdot f(h^{-1} s_i^{-1}) = s_i h h^{-1} \cdot f(s_i^{-1})$, so π_A does not depend on the representatives. For $g \in G$, write $g^{-1} s_i = s_k h$ with $h \in H$. Then $s_i = g s_k h$ and $s_i^{-1} g = h^{-1} s_k^{-1}$, so

$$s_i \cdot (g \cdot f)(s_i^{-1}) = s_i \cdot f(s_i^{-1} g) = g s_k h \cdot h^{-1} \cdot f(s_k^{-1}) = g \cdot \left(s_k \cdot f(s_k^{-1})\right)$$

and as $i \mapsto k$ is a bijection, $\pi_A(g \cdot f) = g \cdot \pi_A(f)$. For surjectivity, take $s_1 = 1$ and, for $a \in A$, let $f(x) = x \cdot a$ for $x \in H$ and $f(x) = 0$ for $x \notin H$. Then $f \in \mathrm{CoInd}_H^G(A)$, as H is open and closed, and $s_i^{-1} \in H$ only for $i = 1$, so $\pi_A(f) = f(1) = a$.

Step 2: Part (1). Write $T^\bullet = H^\bullet(G, \mathrm{CoInd}_H^G(-))$, a universal δ-functor on $\mathrm{Mod}_H^{\mathrm{disc}}$ by Step 3 of the proof of theorem 3.5.5. The restriction Res is a morphism of δ-functors on $\mathrm{Mod}_G^{\mathrm{disc}}$ (lemma 3.3.6(1)), and applying it to the short exact sequences $\mathrm{CoInd}_H^G(\xi)$ makes it a morphism of δ-functors from $T^\bullet$ to $H^\bullet(H, \mathrm{CoInd}_H^G(-))$. The maps ev form a natural transformation from

$\mathrm{CoInd}_H^G(-)$, viewed in $\mathrm{Mod}_H^{\mathrm{disc}}$, to the identity functor, so for each short exact sequence ξ of discrete H-modules they form a morphism of short exact sequences $\mathrm{CoInd}_H^G(\xi) \to \xi$. The naturality of the connecting homomorphisms of $H^\bullet(H,-)$ therefore makes ev_* a morphism of δ-functors. So $\mathrm{ev}_* \circ \mathrm{Res}$ and Sh are morphisms of δ-functors $T^\bullet \to H^\bullet(H,-)$. In degree zero both send $f \in \mathrm{CoInd}_H^G(M)^G$ to $f(1)$, the restriction being the inclusion of invariants, so they coincide by universality of $T^\bullet$.

Step 3: Part (2). Restriction is natural in the module, so it commutes with the maps induced by the G-equivariant map j_A. By Part (1) applied to $M = A$,

$$\mathrm{Sh} \circ (j_A)_* = \mathrm{ev}_* \circ \mathrm{Res} \circ (j_A)_* = \mathrm{ev}_* \circ (j_A)_* \circ \mathrm{Res} = (\mathrm{ev} \circ j_A)_* \circ \mathrm{Res} = \mathrm{Res}$$

because $\mathrm{ev}(j_A(a)) = 1 \cdot a = a$.

Step 4: Part (3). Write $S^\bullet$ for $H^\bullet(H,-)$ on $\mathrm{Mod}_G^{\mathrm{disc}}$. By lemma 3.3.6(1) and (2), $S^\bullet$ is universal and $\mathrm{Cor}\colon S^\bullet \to H^\bullet(G,-)$ is a morphism of δ-functors whose degree-zero component is the trace $a \mapsto \sum_i s_i \cdot a$ on A^H. The isomorphism Sh^{-1} is a morphism of δ-functors $H^\bullet(H,-) \to T^\bullet$ on $\mathrm{Mod}_H^{\mathrm{disc}}$, and restriction to H is an exact functor $\mathrm{Mod}_G^{\mathrm{disc}} \to \mathrm{Mod}_H^{\mathrm{disc}}$, so the components of Sh^{-1} at restrictions of discrete G-modules form a morphism of δ-functors from $S^\bullet$ to $H^\bullet(G, \mathrm{CoInd}_H^G(-))$ on $\mathrm{Mod}_G^{\mathrm{disc}}$. The maps π_A are natural in A, so as in Step 2 they induce a morphism of δ-functors from $H^\bullet(G, \mathrm{CoInd}_H^G(-))$ to $H^\bullet(G,-)$. In degree zero, a G-invariant f satisfies $f(x) = (x \cdot f)(1) = f(1)$, so Sh^{-1} sends $a \in A^H$ to the constant function c_a with value a, and $\pi_A(c_a) = \sum_i s_i \cdot a$ is the trace of a. So $(\pi_A)_* \circ \mathrm{Sh}^{-1}$ and Cor are morphisms of δ-functors out of $S^\bullet$ with the same degree-zero component, and they coincide by universality of $S^\bullet$.

Example 3.5.8: Shapiro's Lemma for Galois Extensions

1. Let L/K be a finite Galois extension with $\Gamma = \mathrm{Gal}(L/K)$. The absolute Galois groups are related by $G_L \trianglelefteq_{\mathrm{o}} G_K$ with $G_K/G_L \cong \Gamma$. For any discrete G_L-module M, Shapiro's lemma gives
$$H^n(G_K, \mathrm{CoInd}_{G_L}^{G_K}(M)) \cong H^n(G_L, M).$$
By proposition 3.5.3 (since G_L is open in G_K), the coinduced module equals the induced module:
$$\mathrm{CoInd}_{G_L}^{G_K}(M) \cong \mathrm{Ind}_{G_L}^{G_K}(M) = \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}} M \cong \bigoplus_{\sigma \in \Gamma} M,$$
with G_K acting by permuting the copies of M according to the Γ-action and acting on each copy via G_L. The Shapiro isomorphism thus provides a "base change" formula: the G_K-cohomology of a module induced from G_L equals the G_L-cohomology of the original module.

2. A concrete instance: let $K = \mathbb{Q}_p$, $L = \mathbb{Q}_p(\zeta_p)$ (a cyclic extension of degree $p-1$ for p odd), and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial G_L-action. Then
$$H^n(G_{\mathbb{Q}_p}, \mathrm{Ind}_{G_L}^{G_{\mathbb{Q}_p}}(\mathbb{Z}/p\mathbb{Z})) \cong H^n(G_L, \mathbb{Z}/p\mathbb{Z}).$$

The left side involves the $G_{\mathbb{Q}_p}$-cohomology of the module $\mathbb{Z}/p\mathbb{Z}[\mathrm{Gal}(L/\mathbb{Q}_p)] \cong \mathbb{Z}/p\mathbb{Z}^{p-1}$ (a $(p-1)$-dimensional $\mathbb{Z}/p\mathbb{Z}$-representation of $G_{\mathbb{Q}_p}$), and the right side is the cohomology of the simpler group G_L with constant coefficients. This kind of reduction is the mechanism behind Shapiro-based proofs of local duality (chapter 8).

3. The isomorphism $H^1(G_K, \mathrm{Ind}_{G_L}^{G_K}((K^{\mathrm{sep}})^\times)) \cong H^1(G_L, (K^{\mathrm{sep}})^\times)$ relates twisted G_K-cohomology to untwisted G_L-cohomology. Since $H^1(G_L, (K^{\mathrm{sep}})^\times) = 0$ by Hilbert's Theorem 90 (theorem 8.1.1(2)), this gives $H^1(G_K, \mathrm{Ind}_{G_L}^{G_K}((K^{\mathrm{sep}})^\times)) = 0$: the induced module of the multiplicative group is cohomologically trivial in degree 1. This vanishing is a key input to the computation of the Brauer group via the exact sequence of chapter 7.

Exercise 3.5.1

1. Verify the adjunction of proposition 3.5.4 directly: given an H-equivariant map $\varphi : A \to M$ (where A is a discrete G-module and M is a discrete H-module), construct the corresponding G-equivariant map $\Phi : A \to \mathrm{CoInd}_H^G(M)$ and check that $\Phi(a) \in \mathrm{CoInd}_H^G(M)$ (i.e., that the function $g \mapsto \varphi(g \cdot a)$ is continuous and H-equivariant in the correct sense).

2. Let H be an open subgroup of G. Use Shapiro's lemma to give a short proof of the formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ from proposition 3.3.7. (*Hint:* For a discrete G-module M, consider the G-equivariant maps $M \xrightarrow{j_M} \mathrm{CoInd}_H^G(M) \xrightarrow{\pi_M} M$ of proposition 3.5.7. Show that $\pi_M \circ j_M = [G : H] \cdot \mathrm{id}_M$, and use parts (2) and (3) of that proposition.)

3. Let H be an open subgroup of G and M a discrete H-module. Write out the Shapiro isomorphism $H^1(G, \mathrm{CoInd}_H^G(M)) \cong H^1(H, M)$ explicitly at the cocycle level: given a continuous 1-cocycle $F : G \to \mathrm{CoInd}_H^G(M)$, describe the corresponding 1-cocycle $f : H \to M$, and vice versa.

4. Let $H = N$ be a closed normal subgroup of G and M a discrete N-module. The Shapiro isomorphism gives $H^n(G, \mathrm{CoInd}_N^G(M)) \cong H^n(N, M)$. Apply the LHS spectral sequence (theorem 3.4.5) to the left side (with the group extension $1 \to N \to G \to G/N \to 1$ and the module $\mathrm{CoInd}_N^G(M)$). Show that the spectral sequence degenerates, and explain how this is consistent with the Shapiro isomorphism.

5. Let $G = \mathbb{Z}_p$ and $H = \{0\}$ (the trivial subgroup, which is closed but not open). Let $M = \mathbb{Z}/p\mathbb{Z}$ with trivial H-action. Describe $\mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z})$ as an abelian group with $\mathbb{Z}_p$-action, and show that it is not finitely generated as a $\mathbb{Z}$-module. Compute $H^0(\mathbb{Z}_p, \mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z}))$ and verify that Shapiro's lemma gives $H^0(\mathbb{Z}_p, \mathrm{CoInd}_{\{0\}}^{\mathbb{Z}_p}(\mathbb{Z}/p\mathbb{Z})) \cong H^0(\{0\}, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$.

4

Tate Cohomology and Cup Products

The cohomology groups $H^n(G, M)$ constructed in chapters 2 and 3 are indexed by non-negative integers $n \geq 0$. This indexing is natural from the derived functor perspective (the right derived functors of $(-)^G$ start in degree zero), but for finite groups it conceals a symmetry. The group homology $H_n(G, M)$, defined as the left derived functors of the coinvariant functor $M \mapsto M_G$, carries complementary information in non-negative degrees $n \geq 0$, and the two theories are linked by the norm map $N_G : M_G \to M^G$. Tate's insight was to splice cohomology ($n \geq 1$), homology ($n \leq -2$), and the norm map ($n = 0, -1$) into a single cohomological functor $\hat{H}^n(G, M)$ indexed by *all* integers $n \in \mathbb{Z}$. The resulting theory, *Tate cohomology* or *complete cohomology*, admits unrestricted dimension shifting, yields the Herbrand quotient as a multiplicative invariant, and is the natural domain for the cup products and duality pairings of the remaining sections of this chapter.

4.1 Complete Cohomology for Finite Groups

Throughout this section, G denotes a *finite* group and M a $\mathbb{Z}[G]$-module (equivalently, a G-module in the sense of definition 2.1.1). There is no topology to consider: every G-module is automatically discrete.

The invariant functor $(-)^G$ and the coinvariant functor $(-)_G$ are the two natural ways to extract abelian-group-valued information from a G-module. The invariants M^G, the elements of M fixed by every $g \in G$, form the largest submodule on which G acts trivially. The coinvariants $M_G = M/I_G M$ (where I_G is the augmentation ideal of $\mathbb{Z}[G]$, as in definition 2.1.4) form the largest quotient on which G acts trivially. Neither functor is exact, and their derived functors give group cohomology and group homology respectively. The norm map connects them.

Definition 4.1.1: The Norm Map

The *norm map* (or *trace map*) is the $\mathbb{Z}[G]$-module homomorphism

$$N_G : M \to M, \qquad N_G(m) = \sum_{g \in G} g \cdot m.$$

Since $N_G(g' \cdot m) = \sum_g gg' \cdot m = N_G(m)$ for all $g' \in G$ (as $g \mapsto gg'$ permutes G), the image of N_G lies in M^G. Since $N_G((\sigma - 1) \cdot m) = \sum_g g\sigma \cdot m - \sum_g g \cdot m = 0$ for any $\sigma \in G$, the kernel of N_G contains $I_G M$. Hence N_G descends to a map

$$\bar{N}_G : M_G \to M^G.$$

The norm map measures the obstruction to passing between invariants and coinvariants. When $|G|$ is invertible in M (for instance, M is a $\mathbb{Q}$-vector space), the map $m \mapsto \frac{1}{|G|} N_G(m)$ provides a G-equivariant splitting of the inclusion $M^G \hookrightarrow M$, and $\bar{N}_G$ is an isomorphism. In general, both the kernel and cokernel of $\bar{N}_G$ carry significant information.

Definition 4.1.2: Modified Cohomology in Degrees 0 and -1

Let G be a finite group and M a G-module. The *modified zeroth Tate cohomology* is

$$\hat{H}^0(G, M) = M^G / N_G(M) = \operatorname{coker}(\bar{N}_G),$$

and the *modified minus-first Tate cohomology* is

$$\hat{H}^{-1}(G, M) = \ker(N_G) / I_G M = \ker(\bar{N}_G).$$

These differ from ordinary cohomology and homology: $H^0(G, M) = M^G$ (the full invariants, not the quotient by the norm image), and $H_0(G, M) = M_G$ (the full coinvariants, not the kernel of the norm). The modification "quotients out" the contribution of the norm in degree zero and "restricts to" its kernel in degree -1, producing groups that are better behaved for the dimension-shifting arguments of §3.

The full Tate cohomology extends $\hat{H}^0$ and $\hat{H}^{-1}$ to all degrees using a doubly infinite resolution.

Definition 4.1.3: The Complete Standard Resolution

Let G be a finite group, and let $P_n = \mathbb{Z}[G^{n+1}]$ ($n \geq 0$) be the terms of the standard resolution $\cdots \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ of definition 2.3.1. For a G-module P write $P^* = \operatorname{Hom}_{\mathbb{Z}}(P, \mathbb{Z})$, with G acting by $(g \cdot \varphi)(x) = \varphi(g^{-1}x)$, and write $\alpha^* \colon \varphi \mapsto \varphi \circ \alpha$ for the dual of a map α. The *complete standard resolution* of $\mathbb{Z}$ over $\mathbb{Z}[G]$ is the doubly infinite complex

$$\hat{P}_\bullet : \quad \cdots \to \hat{P}_1 \xrightarrow{d_1} \hat{P}_0 \xrightarrow{d_0} \hat{P}_{-1} \xrightarrow{d_{-1}} \hat{P}_{-2} \to \cdots$$

with $\hat{P}_n = P_n$ for $n \geq 0$ and the standard differentials d_n for $n \geq 1$, with $\hat{P}_{-n-1} = P_n^*$ and $d_{-n-1} = d_{n+1}^* \colon P_n^* \to P_{n+1}^*$ for $n \geq 0$, and with $d_0 = \varepsilon^* \circ \varepsilon \colon P_0 \to \mathbb{Z} \to P_0^*$, where $\mathbb{Z}^* = \mathbb{Z}$.

The complex $\hat{P}_\bullet$ is exact. At $\hat{P}_n$ with $n \geq 1$ this is lemma 2.3.2. The contracting homotopy

of the augmented standard resolution in the proof of lemma 2.3.2 consists of homomorphisms of abelian groups, and their duals contract the dual sequence $0 \to \mathbb{Z} \xrightarrow{\varepsilon^*} P_0^* \xrightarrow{d_1^*} P_1^* \to \cdots$, which is therefore exact. This gives exactness at $\hat{P}_n$ for $n \leq -2$. At the splice, ε^* is injective, so $\ker(d_0) = \ker(\varepsilon) = \operatorname{im}(d_1)$, and ε is surjective, so $\operatorname{im}(d_0) = \operatorname{im}(\varepsilon^*) = \ker(d_{-1})$. Each $\hat{P}_n$ is a finitely generated free $\mathbb{Z}[G]$-module. For $n \geq 0$ this is the remark after definition 2.3.1. For P_k^*, the functionals δ_y dual to the basis G^{k+1} of P_k form a basis of P_k^*, because this basis is finite, and they satisfy $g \cdot \delta_y = \delta_{gy}$, and G acts freely on G^{k+1}, so P_k^* is free on the δ_y with y running through representatives of the G-orbits. In particular $P_0^* \cong \mathbb{Z}[G]$ by $\delta_{(g)} \mapsto g$, and this isomorphism sends $\varepsilon = \sum_g \delta_{(g)}$ to the norm element $N_G = \sum_{g \in G} g$. Under it, d_0 is $x \mapsto \varepsilon(x) N_G = N_G x$, multiplication by the norm element on $\mathbb{Z}[G]$, and $\operatorname{im}(d_0) = \mathbb{Z} \cdot N_G$. The norm element thus connects the positive and negative halves.

Definition 4.1.4: Tate Cohomology

Let G be a finite group and M a G-module. The *Tate cohomology groups* are defined for all $n \in \mathbb{Z}$ by

$$\hat{H}^n(G, M) = H^n(\operatorname{Hom}_G(\hat{P}_\bullet, M)),$$

where $\hat{P}_\bullet$ is the complete standard resolution of definition 4.1.3.

The relationship between Tate cohomology, ordinary cohomology, and group homology is summarized in the following proposition.

Proposition 4.1.5: Identification of Tate Cohomology

Let G be a finite group and M a G-module. Then:

1. $\hat{H}^n(G, M) \cong H^n(G, M)$ for all $n \geq 1$ (ordinary group cohomology).
2. $\hat{H}^n(G, M) \cong H_{-n-1}(G, M)$ for all $n \leq -2$ (group homology).
3. $\hat{H}^0(G, M) \cong M^G / N_G(M)$ and $\hat{H}^{-1}(G, M) \cong \ker(N_G)/I_G M$.

***Proof*:**

Step 1: Positive degrees. The complex $\operatorname{Hom}_G(\hat{P}_\bullet, M)$ in degrees ≥ 0, with the differentials between those degrees, coincides with $\operatorname{Hom}_G(P_\bullet, M)$, where $P_\bullet$ is the standard resolution, and for $n \geq 1$ the group in degree n depends only on degrees $n-1$, n and $n+1$. The standard resolution is a resolution of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules (definition 2.3.1, lemma 2.3.2), so the cohomology of $\operatorname{Hom}_G(P_\bullet, M)$ is $\operatorname{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M) = H^n(G, M)$ (definition 2.2.1). Hence $\hat{H}^n(G, M) = H^n(G, M)$ for $n \geq 1$.

Step 2: Negative degrees. For a finitely generated free $\mathbb{Z}[G]$-module P, form $P \otimes_G M$ with the right action $x \cdot g = g^{-1}x$ on P, and define $\theta_P \colon P \otimes_G M \to \operatorname{Hom}_G(P^*, M)$ by

$$\theta_P(x \otimes m)(\varphi) = \sum_{g \in G} \varphi(gx)\, g \cdot m$$

The map is balanced: substituting $g' = gh^{-1}$ turns $\theta_P(h^{-1}x \otimes m)(\varphi) = \sum_g \varphi(gh^{-1}x)\, g \cdot m$ into $\sum_{g'} \varphi(g'x)\, g'h \cdot m = \theta_P(x \otimes h \cdot m)(\varphi)$. Its values are G-equivariant: substituting $g = g'k$

gives $\theta_P(x\otimes m)(g'\cdot\varphi)=\sum_g \varphi(g'^{-1}gx)\,g\cdot m = g'\cdot\theta_P(x\otimes m)(\varphi)$. For a G-map $\alpha\colon P\to Q$, $\theta_Q(\alpha(x)\otimes m)(\varphi)=\theta_P(x\otimes m)(\varphi\circ\alpha)$, so θ carries $\alpha\otimes 1$ to $\mathrm{Hom}_G(\alpha^*,M)$. For $P=\mathbb{Z}[G]$, the module $\mathbb{Z}[G]^*$ is free on δ_1 (the discussion after definition 4.1.3), and $\theta_{\mathbb{Z}[G]}(1\otimes m)(\delta_1)=m$, so under $\mathbb{Z}[G]\otimes_G M\cong M$ and evaluation at δ_1, the map $\theta_{\mathbb{Z}[G]}$ is the identity of M. Both sides are additive in P, so θ_P is an isomorphism for every finitely generated free P. For $n\le -1$, the term of $\mathrm{Hom}_G(\hat{P}_\bullet,M)$ in degree n is $\mathrm{Hom}_G(P^*_{-n-1},M)$, and the coboundary from degree $n-1$ to degree n is $\mathrm{Hom}_G(d^*_{-n},M)$. The isomorphisms θ_{P_k} therefore identify the part of $\mathrm{Hom}_G(\hat{P}_\bullet,M)$ in degrees ≤ -1 with the complex $P_\bullet\otimes_G M$, the degree n corresponding to $P_{-n-1}\otimes_G M$. For $n\le -2$ the group $\hat{H}^n(G,M)$ involves only the degrees $n-1$, n and $n+1\le -1$, so $\hat{H}^n(G,M)\cong H_{-n-1}(P_\bullet\otimes_G M)=\mathrm{Tor}^{\mathbb{Z}[G]}_{-n-1}(\mathbb{Z},M)=H_{-n-1}(G,M)$ (definition 2.2.1).

Step 3: Degrees 0 *and* -1. Identify $\hat{P}_0=\mathbb{Z}[G]$ and $\hat{P}_{-1}=P_0^*\cong\mathbb{Z}[G]$ as in the discussion after definition 4.1.3, so that d_0 is multiplication by N_G, and identify $\mathrm{Hom}_G(\mathbb{Z}[G],M)$ with M by evaluation at 1. With the indexing of definition 4.1.3, the coboundary of $\mathrm{Hom}_G(\hat{P}_\bullet,M)$ from degree k to degree $k+1$ is $d^\vee_{k+1}=\mathrm{Hom}_G(d_{k+1},M)$, precomposition with $d_{k+1}\colon \hat{P}_{k+1}\to\hat{P}_k$. So

$$\hat{H}^0(G,M)=\ker(d_1^\vee)/\mathrm{im}(d_0^\vee),\qquad \hat{H}^{-1}(G,M)=\ker(d_0^\vee)/\mathrm{im}(d_{-1}^\vee)$$

A map $\varphi\colon\hat{P}_0\to M$ with $\varphi(1)=m$ has $\varphi(d_1(g_0,g_1))=g_1\cdot m-g_0\cdot m$, so $\ker(d_1^\vee)=M^G$. A map $\psi\colon\hat{P}_{-1}\to M$ with $\psi(1)=m$ has $(\psi\circ d_0)(x)=N_Gx\cdot m=\varepsilon(x)N_G\cdot m$, so $\mathrm{im}(d_0^\vee)=N_G(M)$ and $\ker(d_0^\vee)=\ker(N_G)$. This gives $\hat{H}^0(G,M)=M^G/N_G(M)$. Finally, $\hat{P}_{-2}=P_1^*$ is free on the functionals $\delta_{(1,h)}$ with $h\in G$, so a G-map $\chi\colon P_1^*\to M$ is given by arbitrary values $m_h=\chi(\delta_{(1,h)})$, and $\chi(\delta_{(a,b)})=a\cdot m_{a^{-1}b}$. The functional $d_{-1}(\delta_{(1)})=\delta_{(1)}\circ d_1$ takes the value $[g_1=1]-[g_0=1]$ at (g_0,g_1), so it equals $\sum_g\delta_{(g,1)}-\sum_g\delta_{(1,g)}$, and

$$(\chi\circ d_{-1})(\delta_{(1)})=\sum_{g\in G} g\cdot m_{g^{-1}}-\sum_{g\in G}m_g=\sum_{g\in G}(g-1)\cdot m_{g^{-1}}$$

As the m_h range over M independently, these elements fill I_GM, because the elements $g-1$ generate I_G as an abelian group. So $\mathrm{im}(d_{-1}^\vee)=I_GM$ and $\hat{H}^{-1}(G,M)=\ker(N_G)/I_GM$.

The central advantage of Tate cohomology is the long exact sequence extending in both directions.

Proposition 4.1.6: Long Exact Sequence for Tate Cohomology

A short exact sequence $0\to M'\to M\to M''\to 0$ of G-modules induces an exact sequence

$$\cdots\to\hat{H}^{n-1}(G,M'')\xrightarrow{\delta}\hat{H}^n(G,M')\to\hat{H}^n(G,M)\to\hat{H}^n(G,M'')\xrightarrow{\delta}\hat{H}^{n+1}(G,M')\to\cdots$$

for all $n\in\mathbb{Z}$, with connecting homomorphisms δ that are natural in the short exact sequence.

***Proof*:**

The complex $\hat{P}_\bullet$ consists of free (hence flat and projective) $\mathbb{Z}[G]$-modules in every degree. Applying $\mathrm{Hom}_G(\hat{P}_\bullet,-)$ to the short exact sequence gives a short exact sequence of cochain complexes (since each $\hat{P}_n$ is projective, the $\mathrm{Hom}_G(\hat{P}_n,-)$ functor is exact). The associated long exact sequence in cohomology is the stated sequence.

A key feature distinguishing Tate cohomology from ordinary cohomology is the behaviour of induced modules.

Proposition 4.1.7: Induced Modules Are Tate-Acyclic

Let G be a finite group. For every abelian group A, the induced module $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A$ satisfies $\hat{H}^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) = 0$ for all $n \in \mathbb{Z}$.

***Proof*:**

Since G is finite, $\mathbb{Z}[G] \otimes_{\mathbb{Z}} A = \mathrm{Ind}^G(A)$ is isomorphic to $\mathrm{CoInd}^G(A)$ as a G-module (the discussion after definition 2.2.4, and exercise 2.2.1 (4)). With proposition 2.2.5 this gives isomorphisms $\mathrm{Hom}_G(\hat{P}_n, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) \cong \mathrm{Hom}_{\mathbb{Z}}(\hat{P}_n, A)$, natural in $\hat{P}_n$ and hence compatible with the differentials. So $\hat{H}^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A)$ is the n-th cohomology of the complex $\mathrm{Hom}_{\mathbb{Z}}(\hat{P}_\bullet, A)$.

The complex $\hat{P}_\bullet$ is exact and consists of finitely generated free abelian groups (the discussion after definition 4.1.3). Put $B_n = \mathrm{im}(d_{n+1}) = \ker(d_n) \subseteq \hat{P}_n$. A subgroup of a finitely generated free abelian group is free, so each exact sequence $0 \to B_n \to \hat{P}_n \xrightarrow{d_n} B_{n-1} \to 0$ splits. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, A)$ therefore gives exact sequences

$$0 \to \mathrm{Hom}_{\mathbb{Z}}(B_{n-1}, A) \to \mathrm{Hom}_{\mathbb{Z}}(\hat{P}_n, A) \to \mathrm{Hom}_{\mathbb{Z}}(B_n, A) \to 0$$

The coboundary from degree $n-1$ to degree n is the composite

$$\mathrm{Hom}_{\mathbb{Z}}(\hat{P}_{n-1}, A) \to \mathrm{Hom}_{\mathbb{Z}}(B_{n-1}, A) \to \mathrm{Hom}_{\mathbb{Z}}(\hat{P}_n, A)$$

of a surjection, as B_{n-1} is a direct summand of $\hat{P}_{n-1}$, and an injection, as d_n maps $\hat{P}_n$ onto B_{n-1}. By the exact sequence above, its image is the kernel of $\mathrm{Hom}_{\mathbb{Z}}(\hat{P}_n, A) \to \mathrm{Hom}_{\mathbb{Z}}(B_n, A)$. That kernel is also the kernel of the next coboundary, which is this map followed by the injection $\mathrm{Hom}_{\mathbb{Z}}(B_n, A) \to \mathrm{Hom}_{\mathbb{Z}}(\hat{P}_{n+1}, A)$. Hence $\mathrm{Hom}_{\mathbb{Z}}(\hat{P}_\bullet, A)$ is exact, and $\hat{H}^n(G, \mathbb{Z}[G] \otimes_{\mathbb{Z}} A) = 0$ for every $n \in \mathbb{Z}$.

The vanishing $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for *all* $n \in \mathbb{Z}$ is strictly stronger than the vanishing $H^n(G, \mathbb{Z}[G]) = 0$ for $n \geq 1$ from ordinary cohomology (theorem 2.2.6). It reflects the fact that $\mathbb{Z}[G]$ is both induced and coinduced, and it is the property that makes dimension shifting work in both directions (see §3).

The Tate cohomology of a cyclic group has a particularly clean form, which motivates the Herbrand quotient.

Example 4.1.8: Tate Cohomology of Cyclic Groups

1. Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n and $M = \mathbb{Z}$ with trivial action. The norm map is $N_G(m) = nm$, so $\hat{H}^0(C_n, \mathbb{Z}) = \mathbb{Z}^G / N_G(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$. The kernel of N_G is zero (since $nm = 0$ implies $m = 0$ in $\mathbb{Z}$), so $\hat{H}^{-1}(C_n, \mathbb{Z}) = 0$. By the periodicity that will be established in §3, $\hat{H}^q(C_n, \mathbb{Z}) \cong \hat{H}^{q+2}(C_n, \mathbb{Z})$ for all q, giving

$$\hat{H}^q(C_n, \mathbb{Z}) \cong \begin{cases} \mathbb{Z}/n\mathbb{Z} & q \text{ even}, \\ 0 & q \text{ odd}, \end{cases}$$

for all $q \in \mathbb{Z}$. This extends the computation of example 2.3.8 to negative degrees.

2. Let $G = C_2 = \{1, \sigma\}$ and $M = \mathbb{Z}$ with the sign action: $\sigma \cdot m = -m$. Then $M^G = 0$, $N_G(m) = m + \sigma \cdot m = m - m = 0$, and $\ker(N_G) = \mathbb{Z}$. The augmentation ideal $I_G = (\sigma - 1)\mathbb{Z}[G]$ acts on M by $(\sigma - 1) \cdot m = -2m$, so $I_G M = 2\mathbb{Z}$. Hence $\hat{H}^0(C_2, \mathbb{Z}^-) = 0$ and $\hat{H}^{-1}(C_2, \mathbb{Z}^-) = \mathbb{Z}/2\mathbb{Z}$. By periodicity, $\hat{H}^q(C_2, \mathbb{Z}^-) \cong \mathbb{Z}/2\mathbb{Z}$ for q odd and 0 for q even (the opposite parity from the trivial action).

3. (If the reader has read [23]) The group $\hat{H}^n(C_2, \mathbb{Z})$ with the trivial action computes the cohomology of the infinite real projective space: $H^n(\mathbb{RP}^\infty, \mathbb{Z}) \cong \hat{H}^n(C_2, \mathbb{Z})$ for $n \geq 1$, since $\mathbb{RP}^\infty$ is the classifying space BC_2. Tate cohomology extends this to negative degrees, which corresponds to the homology of $\mathbb{RP}^\infty$ via the identification $\hat{H}^{-n-1}(C_2, \mathbb{Z}) \cong H_n(\mathbb{RP}^\infty, \mathbb{Z})$. The 2-periodic pattern in Tate cohomology is the algebraic form of the $\mathbb{Z}/2\mathbb{Z}$-graded structure of real topological K-theory.

Definition 4.1.9: The Herbrand Quotient

Let G be a cyclic group and M a G-module such that $\hat{H}^0(G, M)$ and $\hat{H}^{-1}(G, M)$ are both finite. The *Herbrand quotient* of M is

$$h(G, M) = \frac{|\hat{H}^0(G, M)|}{|\hat{H}^{-1}(G, M)|} \in \mathbb{Q}_{>0}.$$

The Herbrand quotient is defined only for cyclic groups, where the 2-periodicity of Tate cohomology means that $h(G, M) = |\hat{H}^{2k}(G, M)|/|\hat{H}^{2k-1}(G, M)|$ for any k. Its power lies in its multiplicativity and its invariance under passage to finite-index submodules.

Theorem 4.1.10: Multiplicativity of the Herbrand Quotient

Let G be a cyclic group and $0 \to M' \to M \to M'' \to 0$ a short exact sequence of G-modules. If any two of the three Herbrand quotients $h(G, M')$, $h(G, M)$, $h(G, M'')$ are defined, then so is the third, and

$$h(G, M) = h(G, M') \cdot h(G, M'').$$

***Proof*:**
The long exact sequence of proposition 4.1.6, combined with the 2-periodicity for cyclic groups, gives an exact hexagon:

$$\begin{aligned}
\hat{H}^0(G, M') &\to \hat{H}^0(G, M) \\
&\to \hat{H}^0(G, M'') \xrightarrow{\delta} \hat{H}^{-1}(G, M') \\
&\to \hat{H}^{-1}(G, M) \\
&\to \hat{H}^{-1}(G, M'') \xrightarrow{\delta} \hat{H}^0(G, M') \\
&\to \cdots
\end{aligned}$$

(The sequence is 2-periodic, so after six terms it repeats.) If any two of the three Herbrand quotients are defined, then all six groups in the hexagon are finite (since the maps between them have finite kernels and cokernels by the exact sequence). The alternating product of the orders in an exact sequence of finite groups equals 1:

$$\frac{|\hat{H}^0(G,M')|\cdot|\hat{H}^0(G,M'')|\cdot|\hat{H}^{-1}(G,M)|}{|\hat{H}^0(G,M)|\cdot|\hat{H}^{-1}(G,M')|\cdot|\hat{H}^{-1}(G,M'')|}=1,$$

which rearranges to $h(G,M)=h(G,M')\cdot h(G,M'')$.

The multiplicativity of the Herbrand quotient makes it a powerful computational tool: to compute $h(G,M)$ for a complicated module M, one can filter M by submodules and multiply the Herbrand quotients of the graded pieces.

Proposition 4.1.11: Herbrand Quotient of Finite Modules

Let G be a cyclic group and M a finite G-module. Then $h(G,M)=1$.

***Proof*:**
Step 1: The case $M=\mathbb{Z}/m\mathbb{Z}$ with trivial action. By the computation of example 4.1.8 (replacing $\mathbb{Z}$ by $\mathbb{Z}/m\mathbb{Z}$), both $\hat{H}^0(G,\mathbb{Z}/m\mathbb{Z})$ and $\hat{H}^{-1}(G,\mathbb{Z}/m\mathbb{Z})$ have order $\gcd(|G|,m)$, so $h(G,\mathbb{Z}/m\mathbb{Z})=1$.

Step 2: General M. Filter M by the ascending chain $0=M_0\subseteq M_1\subseteq\cdots\subseteq M_k=M$ of G-stable submodules with each quotient M_i/M_{i-1} a simple $\mathbb{Z}[G]$-module. Each simple module is finite, and the Herbrand quotient of a simple module is 1 by a direct computation (or by a generalization of Step 1 using the structure of simple $\mathbb{Z}[G]$-modules for G cyclic). By multiplicativity (theorem 4.1.10), $h(G,M)=\prod_i h(G,M_i/M_{i-1})=1$.

The proposition says that the Herbrand quotient detects the "infinite part" of a module: for a finitely generated G-module M, $h(G,M)$ depends only on $M\otimes_{\mathbb{Z}}\mathbb{Q}$ (since M fits in an exact sequence $0\to M_{\text{tors}}\to M\to M/M_{\text{tors}}\to 0$ with M_{tors} finite, and $h(G,M_{\text{tors}})=1$).

Example 4.1.12: Herbrand Quotient Computations

1. Let $G=C_p$ (cyclic of prime order p) act trivially on $M=\mathbb{Z}$. Then $\hat{H}^0(C_p,\mathbb{Z})=\mathbb{Z}/p\mathbb{Z}$ and $\hat{H}^{-1}(C_p,\mathbb{Z})=0$, giving $h(C_p,\mathbb{Z})=p$.

2. Let $G=C_p$ act on $M=\mathbb{Z}[G]\cong\mathbb{Z}^p$ (the regular representation). By proposition 4.1.7, $\hat{H}^n(G,\mathbb{Z}[G])=0$ for all n, so $h(C_p,\mathbb{Z}[G])=1$. The short exact sequence $0\to I_G\to\mathbb{Z}[G]\xrightarrow{\varepsilon}\mathbb{Z}\to 0$ then gives $h(C_p,I_G)=h(C_p,\mathbb{Z}[G])/h(C_p,\mathbb{Z})=1/p$ by multiplicativity. Since I_G is torsion-free of rank $p-1$, this means $|\hat{H}^{-1}(C_p,I_G)|=p\cdot|\hat{H}^0(C_p,I_G)|$, so the "homological" side is p times larger than the "cohomological" side.

3. Let $G = C_n$ act on $M = \mathbb{Z}^r$ by a permutation action (permuting a set of r elements). Filter M by the G-orbits: each orbit of size d (with $d|n$) contributes a summand $\mathbb{Z}[G/H]$ where $H \cong C_{n/d}$ is the stabilizer. The Herbrand quotient of $\mathbb{Z}[G/H]$ is $h(C_n, \mathbb{Z}[G/H])$. The short exact sequence $0 \to I_{G/H} \to \mathbb{Z}[G/H] \to \mathbb{Z} \to 0$ gives $h(C_n, \mathbb{Z}[G/H]) = h(C_n, \mathbb{Z})/h(C_n, I_{G/H})$. For a transitive action ($r = d$, a single orbit), $\mathbb{Z}[G/H] \cong \mathrm{Ind}_H^G(\mathbb{Z})$, and by Shapiro's lemma (theorem 3.5.5) combined with the Tate-acyclicity of induced modules, $h(C_n, \mathbb{Z}[G/H]) = 1$ when $H = \{1\}$ (i.e., the free orbit, $d = n$). For a fixed point ($d = 1$, stabilizer $H = G$), the summand is $\mathbb{Z}$ with trivial action, contributing $h(C_n, \mathbb{Z}) = n$.

Exercise 4.1.1

1. Verify by hand that the complete standard resolution $\hat{P}_\bullet$ of definition 4.1.3 is exact at the splice point. Specifically, identify $\hat{P}_0$ and $\hat{P}_{-1}$ with $\mathbb{Z}[G]$, show that d_0 becomes multiplication by N_G, and show that $\ker(d_0) = I_G = \mathrm{im}(d_1)$ and $\mathrm{im}(d_0) = \mathbb{Z} \cdot N_G = \ker(d_{-1})$.

2. Let $G = C_4 = \langle \sigma \rangle$ and $M = \mathbb{Z}$ with trivial action. Compute $\hat{H}^{-2}(C_4, \mathbb{Z})$ using the identification $\hat{H}^{-2} \cong H_1$ and the formula $H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$. Verify that the result matches the periodicity prediction $\hat{H}^{-2}(C_4, \mathbb{Z}) \cong \hat{H}^0(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.

3. Let $G = C_p$ (cyclic of prime order) and consider the G-module $M = \mathbb{Z}[G]/(N_G)$, where (N_G) is the ideal generated by the norm element. Show that M is torsion-free of rank $p - 1$ and compute $h(C_p, M)$.

4. Give a direct proof that $h(G, M) = 1$ for any finite G-module M (with G cyclic) using only the long exact sequence: show that the six-term exact hexagon has all groups finite and apply the alternating-product formula for exact sequences of finite abelian groups.

5. Let $G = C_3 = \langle \sigma \rangle$ and let $M = \mathbb{Z}^2$ with the action $\sigma \cdot (a, b) = (b, -a - b)$ (the standard representation of C_3 on the root lattice of type A_2). Compute $\hat{H}^0(C_3, M)$, $\hat{H}^{-1}(C_3, M)$, and the Herbrand quotient $h(C_3, M)$. (*Hint:* Compute N_G and $I_G M$ explicitly using the matrix of σ.)

4.2 Cup Products

The cohomology groups $H^n(G, M)$ are abelian groups, but they carry additional multiplicative structure: a bilinear pairing

$$H^p(G, M) \times H^q(G, N) \to H^{p+q}(G, M \otimes_{\mathbb{Z}} N)$$

called the *cup product*. This pairing turns the graded group $H^\bullet(G, M)$ (with $N = M$) into a graded ring, and it extends to Tate cohomology for finite groups. The cup product is the mechanism behind the periodicity isomorphism of §3, the duality pairings of §4, and the proof of the local reciprocity law in chapter 8.

Definition 4.2.1: Cup Product on Cochains

Let G be a group, M and N two G-modules. For inhomogeneous cochains $f \in C^p(G, M)$ and $g \in C^q(G, N)$, the *cup product* $f \smile g \in C^{p+q}(G, M \otimes_{\mathbb{Z}} N)$ is defined by

$$(f \smile g)(g_1, \dots, g_{p+q}) = f(g_1, \dots, g_p) \otimes (g_1 \cdots g_p) \cdot g(g_{p+1}, \dots, g_{p+q}). \tag{4.1}$$

The tensor product $m \otimes n$ lies in $M \otimes_{\mathbb{Z}} N$, which is a G-module with diagonal action $g \cdot (m \otimes n) = (g \cdot m) \otimes (g \cdot n)$.

The formula records the "first p variables go to f, the last q variables go to g" splitting of the $(p+q)$-tuple, with the G-action on g's output twisted by the product of the first p group elements. This twist ensures compatibility with the coboundary operator.

Proposition 4.2.2: The Leibniz Rule

The cup product satisfies the *graded Leibniz rule*: for $f \in C^p(G, M)$ and $g \in C^q(G, N)$,

$$d^{p+q}(f \smile g) = (d^p f) \smile g + (-1)^p f \smile (d^q g).$$

In particular, the cup product of two cocycles is a cocycle, the cup product of a cocycle with a coboundary is a coboundary, and the cup product descends to a well-defined bilinear pairing

$$\smile : H^p(G, M) \times H^q(G, N) \to H^{p+q}(G, M \otimes_{\mathbb{Z}} N).$$

***Proof*:**
The verification is a direct computation from the coboundary formula (definition 2.3.3) and the cup product formula (4.1). One expands $d(f \smile g)(g_1, \dots, g_{p+q+1})$ using the coboundary formula, which has $p+q+2$ terms (one for the action of g_1, $p+q$ terms for the "face" insertions, and one for the last variable). Separating these into terms where the insertion occurs in the first $p+1$ positions (which contribute to $(df) \smile g$) and terms where it occurs in the last $q+1$ positions (which contribute to $(-1)^p f \smile (dg)$, with the sign coming from the offset) yields the stated identity.

The consequences for cohomology: if $df = 0$ and $dg = 0$, then $d(f \smile g) = 0$, so the cup product of cocycles is a cocycle. If $f = dh$ is a coboundary and $dg = 0$, then $f \smile g = (dh) \smile g = d(h \smile g) - (-1)^{p-1} h \smile (dg) = d(h \smile g)$, so the cup product is a coboundary. The case $df = 0$ and $g = dk$ is analogous with a sign.

For finite groups, the cup product extends to Tate cohomology: a pairing $\hat{H}^p(G, M) \times \hat{H}^q(G, N) \to \hat{H}^{p+q}(G, M \otimes_{\mathbb{Z}} N)$ defined for all $p, q \in \mathbb{Z}$.

Definition 4.2.3: Cup Product on Tate Cohomology

Let G be a finite group, M and N two G-modules. A *complete diagonal approximation* on the complete standard resolution $\hat{P}_\bullet$ (definition 4.1.3) is a family of G-maps $\Delta_{p,q}\colon \hat{P}_{p+q} \to \hat{P}_p \otimes_{\mathbb{Z}} \hat{P}_q$, one for every pair of integers (p, q), with G acting diagonally on the targets, such that

$$\Delta_{p,q} \circ d = (d \otimes 1)\Delta_{p+1,q} + (-1)^p (1 \otimes d)\Delta_{p,q+1} \qquad \text{and} \qquad (\varepsilon \otimes \varepsilon)\Delta_{0,0} = \varepsilon$$

The *cup product on Tate cohomology*

$$\smile\colon \hat{H}^p(G, M) \times \hat{H}^q(G, N) \to \hat{H}^{p+q}(G, M \otimes_{\mathbb{Z}} N)$$

sends the classes of cocycles $f \in \mathrm{Hom}_G(\hat{P}_p, M)$ and $g \in \mathrm{Hom}_G(\hat{P}_q, N)$ to the class of $f \smile g = (f \otimes g) \circ \Delta_{p,q} \in \mathrm{Hom}_G(\hat{P}_{p+q}, M \otimes_{\mathbb{Z}} N)$.

The first condition says that Δ is a chain map from $\hat{P}_\bullet$ to the *completed* tensor product, whose term in degree n is the product $\prod_{p+q=n} \hat{P}_p \otimes_{\mathbb{Z}} \hat{P}_q$. In each degree infinitely many components occur, which is why the target is a product. The second condition, preservation of the augmentation, excludes chain maps such as $\Delta = 0$. Complete diagonal approximations exist [26, Ch. I, §4, (1.4.7)]. The first condition makes $f \smile g$ a cocycle whose class depends only on the classes of f and g, and the resulting product does not depend on the choice of Δ: it is the unique family of products that is natural in the modules, is induced by $M^G \times N^G \to (M \otimes_{\mathbb{Z}} N)^G$ in degrees $(0, 0)$, and is compatible with connecting homomorphisms [26, Ch. I, §4, (1.4.7)].

For $p, q \geq 0$ the components $\Delta_{p,q}$ can be taken to be the Alexander-Whitney maps $(g_0, \ldots, g_{p+q}) \mapsto (g_0, \ldots, g_p) \otimes (g_p, \ldots, g_{p+q})$, and in these degrees the cup product on Tate cohomology is the cochain-level cup product of definition 4.2.1. The Alexander-Whitney map says nothing about the components with $p < 0$ or $q < 0$, which are given by separate explicit formulas built from the duality between P_n and $\hat{P}_{-n-1}$ [26, Ch. I, §4, (1.4.7)].

Theorem 4.2.4: Properties of the Cup Product

The cup product on $H^\bullet(G,-)$ (and on $\hat{H}^\bullet(G,-)$ for G finite) satisfies:

1. *Bilinearity.* For fixed $\beta \in H^q(G,N)$, the map $\alpha \mapsto \alpha \smile \beta$ is a group homomorphism, and similarly in the second variable.
2. *Associativity.* For $\alpha \in H^p(G,L)$, $\beta \in H^q(G,M)$, $\gamma \in H^r(G,N)$:
$$(\alpha \smile \beta) \smile \gamma = \alpha \smile (\beta \smile \gamma) \in H^{p+q+r}(G, L \otimes M \otimes N).$$
3. *Graded-commutativity.* For $\alpha \in H^p(G,M)$ and $\beta \in H^q(G,N)$:
$$\alpha \smile \beta = (-1)^{pq}\, \tau_*(\beta \smile \alpha) \in H^{p+q}(G, M \otimes N),$$
where $\tau : N \otimes M \to M \otimes N$ is the twist isomorphism $n \otimes m \mapsto m \otimes n$.
4. *Unit.* The element $1 \in H^0(G,\mathbb{Z}) = \mathbb{Z}^G = \mathbb{Z}$ (in Tate cohomology, its image $\bar{1} \in \hat{H}^0(G,\mathbb{Z})$) acts as a two-sided identity: $1 \smile \alpha = \alpha = \alpha \smile 1$ under the identifications $\mathbb{Z} \otimes M \cong M \cong M \otimes \mathbb{Z}$.

***Proof*:**
Step 1: Bilinearity. The cup product formula (4.1) is $\mathbb{Z}$-bilinear in f and g at the cochain level, so it is bilinear on cohomology.

Step 2: Associativity. At the cochain level, $((f \smile g) \smile h)(g_1,\dots,g_{p+q+r})$ and $(f \smile (g \smile h))(g_1,\dots,g_{p+q+r})$ both equal $f(g_1,\dots,g_p) \otimes (g_1 \cdots g_p) \cdot g(g_{p+1},\dots,g_{p+q}) \otimes (g_1 \cdots g_{p+q}) \cdot h(g_{p+q+1},\dots,g_{p+q+r})$, using the associativity of the tensor product $L \otimes M \otimes N$.

Step 3: Graded-commutativity. This is more involved. One constructs, for each pair (p,q), an explicit chain homotopy between the Alexander-Whitney diagonal approximation and its twist by the transposition of tensor factors. The homotopy introduces the sign $(-1)^{pq}$. The classical argument uses the acyclic models theorem or an explicit cup-i-product construction. See [13] for the general framework.

Step 4: Unit. The element $1 \in H^0(G,\mathbb{Z})$ is represented by the cocycle $f_1 \in C^0(G,\mathbb{Z})$ with $f_1 = 1$ (the constant function). The cup product $f_1 \smile g = g$ for any $g \in C^q(G,M)$, since $(f_1 \smile g)(g_1,\dots,g_q) = 1 \otimes g(g_1,\dots,g_q) = g(g_1,\dots,g_q)$ under $\mathbb{Z} \otimes M \cong M$.

Step 5: Tate cohomology. For G finite, the cup product of definition 4.2.3 is the product of [31, Ch. VIII, §3, Prop. 5]. In degrees $p = q = 0$ it is induced by $M^G \otimes N^G \to (M \otimes N)^G$, so $\bar{1} \smile \alpha = \alpha$ for $\alpha \in \hat{H}^0(G,M)$, and the four properties in all degrees follow from the degree-zero case by dimension shifting, which this book does not carry out [31, Ch. VIII, §3].

The cup product interacts with the three functorial maps from section 3.3 in the following way.

Proposition 4.2.5: Compatibility with Restriction, Corestriction, and Inflation

Let G be a profinite group, M and N discrete G-modules.

1. *Restriction.* For $H \leq G$ closed and $\alpha \in H^p(G, M)$, $\beta \in H^q(G, N)$:
$$\mathrm{Res}^H_G(\alpha \smile \beta) = \mathrm{Res}^H_G(\alpha) \smile \mathrm{Res}^H_G(\beta).$$
2. *Projection formula.* For $H \leq G$ open, $\alpha \in H^p(G, M)$, and $\beta \in H^q(H, N)$:
$$\mathrm{Cor}^G_H(\mathrm{Res}^H_G(\alpha) \smile \beta) = \alpha \smile \mathrm{Cor}^G_H(\beta).$$
3. *Inflation.* For $N \trianglelefteq G$ closed normal and $\alpha \in H^p(G/N, M^N)$, $\beta \in H^q(G/N, (N')^N)$:
$$\mathrm{Inf}^G_{G/N}(\alpha \smile \beta) = \mathrm{Inf}^G_{G/N}(\alpha) \smile \mathrm{Inf}^G_{G/N}(\beta).$$

***Proof*:**
Part (1) is immediate from the cochain-level formula: restriction acts by restricting the domain of cochains from G^n to H^n, and the cup product formula involves only the group operation and action, which are unchanged.

Part (2), the projection formula, is the most substantial. At the cochain level, $\mathrm{Cor}(\mathrm{Res}(\alpha) \smile \beta)(g_1, \ldots, g_{p+q})$ involves a sum over coset representatives of the cup product of $\alpha|_H$ with β, and regrouping the sum using the cocycle property of α on G produces $\alpha \smile \mathrm{Cor}(\beta)$.

Part (3) follows from the cochain formula for inflation: $\mathrm{Inf}(f)(g_1, \ldots, g_n) = f(\bar{g}_1, \ldots, \bar{g}_n)$, and the cup product formula applied to inflated cochains produces the inflation of the cup product.

The projection formula in part (2) is particularly important: it says that corestriction is a module map over the cup product ring $H^\bullet(G, -)$, viewing $H^\bullet(H, -)$ as an $H^\bullet(G, -)$-module via restriction.

The cup product also interacts with the connecting homomorphisms of the long exact sequence.

Proposition 4.2.6: Cup Product and Connecting Homomorphisms

Let $0 \to M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \to 0$ be a short exact sequence of G-modules, and let N be a G-module such that the two induced sequences below are exact (for instance, N free as an abelian group, or the given sequence split as a sequence of abelian groups).

1. For $\alpha \in H^p(G, M'')$ and $\beta \in H^q(G, N)$:

$$\delta(\alpha) \smile \beta = \delta(\alpha \smile \beta) \in H^{p+q+1}(G, M' \otimes N),$$

   where δ on the left is the connecting homomorphism from the sequence $0 \to M' \to M \to M'' \to 0$, and on the right from the induced sequence $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$.

2. For $\alpha \in H^p(G, N)$ and $\beta \in H^q(G, M'')$:

$$\alpha \smile \delta(\beta) = (-1)^p \delta(\alpha \smile \beta) \in H^{p+q+1}(G, N \otimes M'),$$

   with the analogous connecting homomorphism from $0 \to N \otimes M' \to N \otimes M \to N \otimes M'' \to 0$.

The same identities hold for Tate cohomology when G is finite.

***Proof*:**

Both identities are verified by a diagram chase at the cochain level. For part (1): represent α by a cochain $\tilde{f} \in C^p(G, M)$ lifting $f \in Z^p(G, M'')$ (so $\pi \circ \tilde{f} = f$ and $d\tilde{f}$ takes values in M'). Then $\delta(\alpha)$ is represented by $d\tilde{f} \in Z^{p+1}(G, M')$. The Leibniz rule gives $d(\tilde{f} \smile g) = (d\tilde{f}) \smile g + (-1)^p \tilde{f} \smile dg = (d\tilde{f}) \smile g$ (since $dg = 0$). Hence $\delta(\alpha) \smile \beta$ is the class of $d(\tilde{f} \smile g)$, which is δ applied to the class of $f \smile g$ in $H^{p+q}(G, M'' \otimes N)$.

Part (2) uses the same argument with the roles of the two factors interchanged, producing the sign $(-1)^p$ from the Leibniz rule.

For Tate cohomology of a finite group the two identities are properties iii) and iv) of the cup product of [31, Ch. VIII, §3, Prop. 5], which is the product of definition 4.2.3 defined on a complete resolution, under the same exactness hypothesis. This book does not prove them in negative degrees.

This compatibility is the algebraic engine of dimension shifting: given a short exact sequence $0 \to M \to I \to Q \to 0$ with I acyclic, the connecting homomorphism $\delta : H^n(G, Q) \xrightarrow{\sim} H^{n+1}(G, M)$ is an isomorphism for $n \geq 1$. The cup product identity $\delta(\alpha \smile \beta) = \delta(\alpha) \smile \beta$ shows that this isomorphism is compatible with the cup product: shifting the degree of one factor by 1 via δ shifts the degree of the cup product by 1.

Example 4.2.7: Cup Products in Low Degrees and for Cyclic Groups

1. In Tate cohomology, the cup product $\hat{H}^0(G, M) \times \hat{H}^0(G, N) \to \hat{H}^0(G, M \otimes N)$ is the map induced by $m \otimes n \mapsto m \otimes n$ on $M^G/N_G(M) \times N^G/N_G(N) \to (M \otimes N)^G/N_G(M \otimes N)$. This is well defined because $N_G(m) \otimes n = N_G(m \otimes n)$ for $n \in N^G$ (as $g \cdot (m \otimes n) = (g \cdot m) \otimes n$ when n is G-invariant).

2. Let $G = C_n = \langle \sigma \rangle$ and consider the periodicity generator $u \in \hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (corresponding to 1 mod n, and constructed in section 4.3). The cup product $- \smile u : \hat{H}^q(G, M) \to \hat{H}^{q+2}(G, M \otimes \mathbb{Z}) \cong \hat{H}^{q+2}(G, M)$ is an isomorphism for all $q \in \mathbb{Z}$. This is the content of the periodicity theorem, theorem 4.3.4, and the cup product is the mechanism that produces it.

3. Let $G = C_2 = \{1, \sigma\}$ and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. The group $H^1(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is generated by the unique nontrivial homomorphism $\chi : C_2 \to \mathbb{Z}/2\mathbb{Z}$. The cup product $\chi \smile \chi \in H^2(C_2, \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}) \cong H^2(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is computed via the cochain formula: $(\chi \smile \chi)(\sigma^a, \sigma^b) = \chi(\sigma^a) \cdot \sigma^a \cdot \chi(\sigma^b) = a \cdot b \pmod 2$ (where $a, b \in \{0, 1\}$ and the action is trivial). This is the cocycle $c(a, b) = ab \bmod 2$, which represents the nonzero element of $H^2(C_2, \mathbb{Z}/2\mathbb{Z})$. By induction, $\chi^n \neq 0$ for all $n \geq 1$, and the cup product ring is

$$H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi], \qquad |\chi| = 1.$$

This is a polynomial ring in one variable of degree 1.

4. (If the reader has read [23]) The identification $H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi]$ is the algebraic counterpart of the cohomology ring $H^\bullet(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[w_1]$, where w_1 is the first Stiefel-Whitney class of the tautological line bundle. The classifying space $BC_2 = \mathbb{R}\mathrm{P}^\infty$ realizes group cohomology as singular cohomology, and the cup product on group cohomology corresponds to the topological cup product. This correspondence extends to all finite groups via the classifying space construction.

Exercise 4.2.1

1. Verify the associativity of the cup product at the cochain level: for $f \in C^p(G, L)$, $g \in C^q(G, M)$, $h \in C^r(G, N)$, show that $((f \smile g) \smile h)(g_1, \ldots, g_{p+q+r}) = (f \smile (g \smile h))(g_1, \ldots, g_{p+q+r})$ by expanding both sides using the formula (4.1).

2. Let $G = C_4 = \langle \sigma \rangle$, $H = C_2 = \langle \sigma^2 \rangle$, and $M = N = \mathbb{Z}$ with trivial action. Compute the cup product $\mathrm{Res}(\alpha) \smile \beta$ for α a generator of $H^2(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$ and β a generator of $H^2(C_2, \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, and verify the projection formula $\mathrm{Cor}(\mathrm{Res}(\alpha) \smile \beta) = \alpha \smile \mathrm{Cor}(\beta)$ by computing both sides independently.

3. Let $G = C_p$ (cyclic of prime order p) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action. Compute the cup product ring $H^\bullet(C_p, \mathbb{Z}/p\mathbb{Z})$.

   (a) Show that $H^\bullet(C_p, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}[\chi, u]/(\chi^2)$ where $|\chi| = 1$ and $|u| = 2$, for p odd. (*Hint:* $\chi^2 = 0$ by graded-commutativity since p is odd, and $\chi \smile u \neq 0$ generates H^3.)

   (b) Show that $H^\bullet(C_2, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}[\chi]$ where $|\chi| = 1$ (the polynomial ring from example 4.2.7). Why does $\chi^2 \neq 0$ in this case?

4. Let G be a finite group and $0 \to \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ the short exact sequence with trivial

G-action. Use the compatibility of the cup product with connecting homomorphisms (proposition 4.2.6) to express the Bockstein homomorphism $\beta : H^q(G, \mathbb{Z}/n\mathbb{Z}) \to H^{q+1}(G, \mathbb{Z}/n\mathbb{Z})$ (defined as the composition of $\delta : H^q(G, \mathbb{Z}/n\mathbb{Z}) \to H^{q+1}(G, \mathbb{Z})$ with reduction mod n) in terms of the cup product with the class $\chi \in H^1(G, \mathbb{Z}/n\mathbb{Z})$ given by $\chi(\sigma) = 1$, when $G = \langle\sigma\rangle$ is cyclic of order n.

5. Let G be a profinite group of p-cohomological dimension $\leq d$ (foreshadowing chapter 5: $H^n(G, M) = 0$ for all $n > d$ and all p-torsion discrete G-modules M). Show that the cup product $H^p(G, M) \times H^q(G, N) \to H^{p+q}(G, M \otimes N)$ is zero whenever $p + q > d$ and $M \otimes N$ is p-torsion.

4.3 Dimension Shifting and Periodicity

Dimension shifting is the technique of computing cohomology in one degree by embedding the coefficient module into an acyclic module and using the connecting homomorphism to shift the degree. In ordinary cohomology, this works only in the positive direction (from degree n to degree $n + 1$), because the connecting homomorphism $\delta : H^n(G, M'') \to H^{n+1}(G, M')$ requires $n \geq 0$. In Tate cohomology, the long exact sequence extends to all $n \in \mathbb{Z}$ (proposition 4.1.6), so dimension shifting works in both directions without restriction. This unrestricted shifting is the key to the periodicity theorem for cyclic groups, which asserts that cup product with a canonical class $u \in \hat{H}^2(G, \mathbb{Z})$ gives an isomorphism $\hat{H}^q(G, M) \cong \hat{H}^{q+2}(G, M)$ for all $q \in \mathbb{Z}$.

Proposition 4.3.1: Dimension Shifting

Let G be a finite group and M a G-module.

1. *Upward shift.* Let $I = \mathrm{CoInd}_{\{1\}}^G(M) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], M)$ be the coinduced module, with the natural inclusion $\iota : M \hookrightarrow I$ (sending m to the map $g \mapsto g \cdot m$). Let $Q = I/\iota(M)$. Since I is Tate-acyclic (proposition 4.1.7), the connecting homomorphism in the long exact sequence for $0 \to M \to I \to Q \to 0$ gives isomorphisms
$$\delta : \hat{H}^n(G, Q) \xrightarrow{\sim} \hat{H}^{n+1}(G, M) \qquad \text{for all } n \in \mathbb{Z}.$$

2. *Downward shift.* Let $J = \mathbb{Z}[G] \otimes_{\mathbb{Z}} M$ be the induced module, with the natural surjection $\varepsilon : J \twoheadrightarrow M$ (sending $g \otimes m$ to $g \cdot m$). Let $K = \ker(\varepsilon)$. Since J is Tate-acyclic, the connecting homomorphism for $0 \to K \to J \to M \to 0$ gives isomorphisms
$$\delta : \hat{H}^{n-1}(G, M) \xrightarrow{\sim} \hat{H}^n(G, K) \qquad \text{for all } n \in \mathbb{Z}.$$

***Proof*:**
Both statements follow from the long exact sequence of proposition 4.1.6 applied to the respective short exact sequences, using $\hat{H}^n(G, I) = \hat{H}^n(G, J) = 0$ for all n.

Dimension shifting replaces the study of $\hat{H}^{n+1}(G, M)$ with the study of $\hat{H}^n(G, Q)$, where Q depends on M but is often more tractable. Iterating: $\hat{H}^{n+k}(G, M) \cong \hat{H}^n(G, Q_k)$ for a module Q_k obtained by k successive shifts. The downward version similarly gives $\hat{H}^{n-k}(G, M) \cong \hat{H}^n(G, K_k)$. The interplay

between upward and downward shifting is possible only in Tate cohomology, where the long exact sequence has no lower bound on the degree.

For cyclic groups, dimension shifting has a dramatic consequence: the Tate cohomology is 2-periodic, and this periodicity is realized by the cup product with a canonical class. The first step is to exhibit the periodic structure of the free resolution.

Proposition 4.3.2: The Periodic Resolution for Cyclic Groups

Let $G = C_n = \langle \sigma \rangle$ be cyclic of order n. Define the elements $D = \sigma - 1$ and $N = N_G = 1 + \sigma + \cdots + \sigma^{n-1}$ in $\mathbb{Z}[G]$. The sequence

$$\cdots \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{D} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{D} \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \to 0 \tag{4.2}$$

is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$, called the *periodic resolution*. The differentials alternate between multiplication by $D = \sigma - 1$ and multiplication by $N = N_G$.

***Proof*:**
The identities $D \cdot N = (\sigma - 1)(1 + \sigma + \cdots + \sigma^{n-1}) = \sigma^n - 1 = 0$ and $N \cdot D = 0$ (by the same computation) show that consecutive maps compose to zero. To verify exactness:

Step 1: At the augmentation. The kernel of $\varepsilon : \mathbb{Z}[G] \to \mathbb{Z}$ is the augmentation ideal $I_G = (D) = (\sigma - 1) \cdot \mathbb{Z}[G]$, which is the image of $D : \mathbb{Z}[G] \to \mathbb{Z}[G]$.

Step 2: At D-to-N transitions. The kernel of $D : \mathbb{Z}[G] \to \mathbb{Z}[G]$ consists of elements $f \in \mathbb{Z}[G]$ with $(\sigma - 1)f = 0$. Writing $f = \sum_{i=0}^{n-1} a_i \sigma^i$, the condition $\sigma f = f$ means $a_0 = a_1 = \cdots = a_{n-1}$, so $f = a \cdot N$ for some $a \in \mathbb{Z}$. Hence $\ker(D) = \mathbb{Z} \cdot N = \mathrm{im}(N)$.

Step 3: At N-to-D transitions. The kernel of $N : \mathbb{Z}[G] \to \mathbb{Z}[G]$ consists of elements f with $N \cdot f = 0$. Since $\mathbb{Z}[G] \cong \mathbb{Z}[x]/(x^n - 1)$ and $N = (x^n - 1)/(x - 1)$, the condition $N \cdot f = 0$ in $\mathbb{Z}[G]$ means $(x - 1) \mid f$ in $\mathbb{Z}[x]/(x^n - 1)$, i.e., $f = (x - 1)g$ for some g. Hence $\ker(N) = (\sigma - 1) \cdot \mathbb{Z}[G] = \mathrm{im}(D)$.

The periodic resolution has period 2: the pattern of differentials $(D, N, D, N, \dots)$ repeats every two steps. Splicing the periodic resolution with its dual, as the standard resolution is spliced in definition 4.1.3, gives a doubly infinite exact complex of period 2 for a cyclic group. This periodicity at the level of resolutions translates into periodicity in Tate cohomology via the cup product with the following class.

Definition 4.3.3: The Periodicity Generator

Let $G = C_n$ be cyclic of order n. The *periodicity generator* $u_G \in \hat{H}^2(G, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ is the class corresponding to 1 mod n under the canonical identification. Concretely, u_G is the extension class of the short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \xrightarrow{\sigma - 1} I_G \to 0, \tag{4.3}$$

where the first map sends $1 \mapsto N_G$ and the second is multiplication by $\sigma - 1$. (The sequence is exact by the proof of proposition 4.3.2: $\ker(\sigma - 1) = \mathbb{Z} \cdot N_G = \mathrm{im}(N_G)$, and $\mathrm{im}(\sigma - 1) = I_G = \ker(\varepsilon)$.)

The extension (4.3) splices two consecutive steps of the periodic resolution, and its class in $H^2(G, \mathbb{Z}) = \mathrm{Ext}^1_{\mathbb{Z}[G]}(I_G, \mathbb{Z})$ measures the "twisting" between the D- and N-steps. The identification $H^2(C_n, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ sends this class to 1 mod n. This can be verified using the explicit cocycle representing the extension (the "carry cocycle" from example 2.3.8).

Theorem 4.3.4: Periodicity for Cyclic Groups

Let $G = C_n$ be cyclic of order n, M a G-module, and $u_G \in \hat{H}^2(G, \mathbb{Z})$ the periodicity generator. The cup product map

$$- \smile u_G : \hat{H}^q(G, M) \xrightarrow{\sim} \hat{H}^{q+2}(G, M)$$

(using the identification $M \otimes_{\mathbb{Z}} \mathbb{Z} \cong M$) is an isomorphism for every $q \in \mathbb{Z}$. It is natural in M: for every homomorphism of G-modules $f : M \to M'$ and every $\alpha \in \hat{H}^q(G, M)$,

$$f_*(\alpha \smile u_G) = f_*(\alpha) \smile u_G$$

***Proof*:**

Fix a generator σ of G, and write $\mu_M : M \otimes_{\mathbb{Z}} \mathbb{Z} \to M$, $m \otimes k \mapsto km$, for the isomorphism of G-modules used in the statement. All tensor products below are over $\mathbb{Z}$, with G acting diagonally.

Step 1: Two short exact sequences. Consider the sequences of G-modules

$$E_1 : \ 0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \to 0 \qquad E_2 : \ 0 \to \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \xrightarrow{\sigma - 1} I_G \to 0$$

The first is exact by the definition of I_G (definition 2.1.4), and the second is the exact sequence (4.3) of definition 4.3.3. Every term is a free abelian group: $\mathbb{Z}[G]$ has the basis G, and I_G has the basis $\{g - 1 : g \neq 1\}$, because an element $\sum_g a_g g$ with $\sum_g a_g = 0$ equals $\sum_{g \neq 1} a_g (g - 1)$. A surjection onto a free abelian group splits, so E_1 and E_2 split as sequences of abelian groups, and for every G-module M the sequences $M \otimes E_1$ and $M \otimes E_2$ are exact. Their maps are G-homomorphisms because those of E_1 and E_2 are: multiplication by $\sigma - 1$ commutes with the action of G since G is abelian, and $gN_G = N_G$ for every $g \in G$.

Step 2: The connecting isomorphisms. Let M_0 be the abelian group underlying M, and let G act on $\mathbb{Z}[G] \otimes M_0$ through the left factor only. The map $\mathbb{Z}[G] \otimes M_0 \to M \otimes \mathbb{Z}[G]$, $g \otimes m \mapsto gm \otimes g$, is bijective, with inverse $m \otimes g \mapsto g \otimes g^{-1}m$, and G-equivariant, since $h \cdot (g \otimes m) = hg \otimes m$ goes

to $hgm \otimes hg = h \cdot (gm \otimes g)$. By proposition 4.1.7, $\hat{H}^k(G, M \otimes \mathbb{Z}[G]) = 0$ for every $k \in \mathbb{Z}$. In the long exact sequences of proposition 4.1.6 for $M \otimes E_1$ and $M \otimes E_2$ the connecting homomorphisms

$$\delta_1^M \colon \hat{H}^q(G, M \otimes \mathbb{Z}) \to \hat{H}^{q+1}(G, M \otimes I_G) \qquad \delta_2^M \colon \hat{H}^{q+1}(G, M \otimes I_G) \to \hat{H}^{q+2}(G, M \otimes \mathbb{Z})$$

are therefore isomorphisms for every $q \in \mathbb{Z}$, and so is $\theta_M = \delta_2^M \circ \delta_1^M \colon \hat{H}^q(G, M \otimes \mathbb{Z}) \to \hat{H}^{q+2}(G, M \otimes \mathbb{Z})$.

Step 3: Naturality of θ. A homomorphism of G-modules $f \colon M \to M'$ induces morphisms of short exact sequences $f \otimes 1 \colon M \otimes E_i \to M' \otimes E_i$ for $i = 1, 2$. The connecting homomorphisms of proposition 4.1.6 are natural in the short exact sequence, so $(f \otimes 1)_* \circ \delta_i^M = \delta_i^{M'} \circ (f \otimes 1)_*$, and hence $(f \otimes 1)_* \circ \theta_M = \theta_{M'} \circ (f \otimes 1)_*$.

Step 4: θ is a cup product. Let $\bar{1} \in \hat{H}^0(G, \mathbb{Z})$ be the class of 1, let $\delta_i^{\mathbb{Z}}$ be the connecting homomorphism of E_i, and put $u = \delta_2^{\mathbb{Z}}(\delta_1^{\mathbb{Z}}(\bar{1})) \in \hat{H}^2(G, \mathbb{Z})$. For $\alpha \in \hat{H}^q(G, M)$, part (2) of proposition 4.2.6, applied with $N = M$ first to E_2 and then to E_1 (the tensored sequences being exact by Step 1), gives

$$\alpha \smile u = (-1)^q \, \delta_2^M\big(\alpha \smile \delta_1^{\mathbb{Z}}(\bar{1})\big) = (-1)^q (-1)^q \, \delta_2^M \delta_1^M (\alpha \smile \bar{1}) = \theta_M(\alpha \smile \bar{1})$$

By the unit property (theorem 4.2.4(4)), $\alpha \smile \bar{1} = (\mu_M)_*^{-1}(\alpha)$. Hence

$$(\mu_M)_*(\alpha \smile u) = (\mu_M)_* \, \theta_M \, (\mu_M)_*^{-1}(\alpha)$$

so the map $\alpha \mapsto (\mu_M)_*(\alpha \smile u)$ is an isomorphism $\hat{H}^q(G, M) \to \hat{H}^{q+2}(G, M)$ by Step 2.

Step 5: Passage to u_G. By proposition 4.1.5(3), $\hat{H}^0(G, \mathbb{Z}) \cong \mathbb{Z}^G / N_G(\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$, since N_G acts on the trivial module $\mathbb{Z}$ as multiplication by n. This group is generated by $\bar{1}$. The connecting homomorphisms $\delta_1^{\mathbb{Z}}$ and $\delta_2^{\mathbb{Z}}$ are isomorphisms, by Step 2 for $M = \mathbb{Z}$, with $\mathbb{Z} \otimes E_i$ identified with E_i termwise through the isomorphisms $\mathbb{Z} \otimes A \to A$, $k \otimes x \mapsto kx$, which commute with the maps of E_i. So the group $\hat{H}^2(G, \mathbb{Z})$ is cyclic of order n with generator $u = \delta_2^{\mathbb{Z}} \delta_1^{\mathbb{Z}}(\bar{1})$. The periodicity generator u_G generates this group (definition 4.3.3), so $u_G = ku$ for an integer k prime to n, and $\alpha \smile u_G = k(\alpha \smile u)$ by bilinearity (theorem 4.2.4(1)). As $nu = 0$, and every element of $\hat{H}^{q+2}(G, M)$ has the form $(\mu_M)_*(\alpha \smile u)$ by Step 4, the group $\hat{H}^{q+2}(G, M)$ is killed by n. Choosing k' with $kk' \equiv 1 \pmod{n}$, multiplication by k on this group is an automorphism, with inverse multiplication by k'. Hence $- \smile u_G$ is the composite of the isomorphism of Step 4 with an automorphism, and is an isomorphism.

Step 6: Naturality. Let $f \colon M \to M'$ be a homomorphism of G-modules. Then $f \circ \mu_M = \mu_{M'} \circ (f \otimes 1)$, so $f_*(\mu_M)_* = (\mu_{M'})_*(f \otimes 1)_*$ and $(f \otimes 1)_*(\mu_M)_*^{-1} = (\mu_{M'})_*^{-1} f_*$. With Steps 3 and 4,

$$f_*(\mu_M)_*(\alpha \smile u) = (\mu_{M'})_*(f \otimes 1)_* \theta_M (\mu_M)_*^{-1}(\alpha) = (\mu_{M'})_* \theta_{M'} (\mu_{M'})_*^{-1}(f_*\alpha)$$

and the right side is $(\mu_{M'})_*(f_*\alpha \smile u)$ by Step 4 for M'. Multiplying by k gives $f_*(\alpha \smile u_G) = f_*(\alpha) \smile u_G$, completing the proof.

The periodicity theorem says that all of the Tate cohomology of a cyclic group is encoded in the two

groups $\hat{H}^0(G,M) = M^G/N_G(M)$ and $\hat{H}^{-1}(G,M) = \ker(N_G)/I_G M$, repeating with period 2. The periodicity generator u_G is the algebraic mechanism behind this repetition: cupping with u_G shifts the degree by 2, and the fact that this is an isomorphism reflects the 2-periodic structure of the free resolution for cyclic groups.

Periodicity is special to cyclic groups. For a non-cyclic finite group G, the Tate cohomology $\hat{H}^n(G,M)$ can exhibit growth in n (for instance, $|\hat{H}^n(C_p \times C_p, \mathbb{Z}/p\mathbb{Z})| = p^{n+1}$ for $n \geq 0$, which is unbounded). The cohomological dimension and structure theory of profinite groups developed in chapter 5 provide the correct framework for understanding the behaviour of $\hat{H}^n$ as n grows for non-cyclic groups.

The Herbrand quotient of definition 4.1.9 is a direct consequence of periodicity: since $\hat{H}^q(G,M)$ depends only on the parity of q (for G cyclic), the ratio $h(G,M) = |\hat{H}^0|/|\hat{H}^{-1}|$ captures all the information about the "size" of the cohomology, and its multiplicativity (theorem 4.1.10) follows from the long exact sequence combined with the 2-periodic collapse.

The periodicity theorem has a powerful converse: Tate's criterion characterizes modules with vanishing Tate cohomology.

Theorem 4.3.5: Tate's Theorem on Cohomological Triviality

Let G be a finite group and M a G-module. The following are equivalent:

1. $\hat{H}^n(H,M) = 0$ for all $n \in \mathbb{Z}$ and all subgroups $H \leq G$.
2. $\hat{H}^{-1}(H,M) = \hat{H}^0(H,M) = 0$ for all subgroups $H \leq G$.
3. For every prime p there is a Sylow p-subgroup G_p of G with $\hat{H}^{-1}(G_p,M) = \hat{H}^0(G_p,M) = 0$.

A module satisfying these equivalent conditions is called *cohomologically trivial*.

***Proof Key ideas*:**
The implications (1) ⇒ (2) ⇒ (3) are immediate. The implication (3) ⇒ (1) is the theorem of Tate and Nakayama, which this book does not prove. It is [31, Ch. IX, §5, Thm. 8], and also [2, Ch. IV, §9, Thm. 9]. The proof treats p-groups first. For a p-group P and a P-module A with $pA = 0$, the vanishing of a single group $\hat{H}^q(P,A)$ forces A to be a free $\mathbb{F}_p[P]$-module, by Nakayama's lemma for the ring $\mathbb{F}_p[P]$ [31, Ch. IX, §4, Thm. 5]. For a P-module A without p-torsion, vanishing in two consecutive degrees passes to A/pA through the sequence $0 \to A \xrightarrow{p} A \to A/pA \to 0$, so A/pA is free and A is cohomologically trivial [31, Ch. IX, §4, Thm. 6]. For a general finite group, one writes M as a quotient $0 \to R \to L \to M \to 0$ of a free $\mathbb{Z}[G]$-module L. The hypothesis on the Sylow subgroups shows that the $\mathbb{Z}$-free module R is $\mathbb{Z}[G]$-projective [31, Ch. IX, §5, Thm. 7], and then M is cohomologically trivial because R and L are, by the long exact sequence of proposition 4.1.6, completing the sketch.

Tate's theorem is a reduction principle: to verify cohomological triviality it suffices to check the two degrees 0 and -1 on one Sylow subgroup for each prime. A single degree does not suffice: exercise 4.1.1 (3) provides a module with $\hat{H}^0 = 0$ but $\hat{H}^{-1} \neq 0$. Cyclic subgroups do not suffice in place of Sylow subgroups. Let $G = \langle a, b \rangle \cong C_2 \times C_2$, let $\lambda \in \mathbb{F}_4 \setminus \mathbb{F}_2$, and let G act on $M = \mathbb{F}_4^2$ through $a \mapsto 1 + E$ and $b \mapsto 1 + \lambda E$, where E is the matrix with $Ee_1 = e_2$ and $Ee_2 = 0$, so that $E^2 = 0$. The three subgroups of order 2 are generated by elements acting as $1 + \mu E$ with $\mu \in \{1, \lambda, 1+\lambda\}$, all nonzero. For such a subgroup H the norm N_H acts as μE, the characteristic

being 2, and $\ker(\mu E) = \mathrm{im}(\mu E) = \mathbb{F}_4 e_2$, so $\hat{H}^0(H, M) = \hat{H}^{-1}(H, M) = 0$ (definition 4.1.2). But $N_G = 1 + a + b + ab$ acts as $4 + (1 + \lambda + 1 + \lambda)E = 0$, so $\hat{H}^0(G, M) = M^G = \mathbb{F}_4 e_2 \neq 0$.

Example 4.3.6: Cohomologically Trivial Modules

1. The module $M = \mathbb{Z}[G]$ is cohomologically trivial by proposition 4.1.7: $\hat{H}^n(G, \mathbb{Z}[G]) = 0$ for all n. This can also be verified via Tate's theorem: $\hat{H}^0(H, \mathbb{Z}[G]) = (\mathbb{Z}[G])^H / N_H(\mathbb{Z}[G])$. The invariants $(\mathbb{Z}[G])^H \cong \mathbb{Z}[G/H]$ (the formal sums over cosets), and $N_H(\mathbb{Z}[G]) = N_H \cdot \mathbb{Z}[G]$. One checks that $N_H \cdot \mathbb{Z}[G] = (\mathbb{Z}[G])^H$ by observing that $N_H \cdot g = \sum_{h \in H} hg$, which is the sum over the right H-coset Hg of g. Hence $\hat{H}^0(H, \mathbb{Z}[G]) = 0$. A similar computation gives $\hat{H}^{-1}(H, \mathbb{Z}[G]) = 0$.

2. The module $M = \mathbb{Z}/n\mathbb{Z}$ with trivial C_n-action is *not* cohomologically trivial: $\hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})/n(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z} \neq 0$. The periodicity theorem gives $\hat{H}^q(C_n, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ for all $q \in \mathbb{Z}$.

3. For $G = C_n$ and $M = \mathbb{Z}[G]/(N_G)$: the sequence $0 \to \mathbb{Z} \xrightarrow{N_G} \mathbb{Z}[G] \to M \to 0$ is exact, and $\mathbb{Z}[G]$ is Tate-acyclic (proposition 4.1.7), so the long exact sequence of proposition 4.1.6 gives $\hat{H}^q(C_n, M) \cong \hat{H}^{q+1}(C_n, \mathbb{Z})$ for every q. Hence $\hat{H}^0(C_n, M) \cong H^1(C_n, \mathbb{Z}) = \mathrm{Hom}(C_n, \mathbb{Z}) = 0$ and $\hat{H}^{-1}(C_n, M) \cong \hat{H}^0(C_n, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z} \neq 0$ (proposition 4.1.5), so M is not cohomologically trivial. The criterion fails in degree -1 only, so a single vanishing degree does not suffice.

4. Let G be a finite group and M a finitely generated $\mathbb{Z}[G]$-module that is $\mathbb{Z}$-free. Then M is cohomologically trivial if and only if M is a projective $\mathbb{Z}[G]$-module. If M is cohomologically trivial, then so is its restriction to each Sylow subgroup of G, and M is projective by [31, Ch. IX, §5, Thm. 7]. Conversely, a projective module is a direct summand of a free $\mathbb{Z}[G]$-module, whose restriction to any subgroup H is a free $\mathbb{Z}[H]$-module and so Tate-acyclic by proposition 4.1.7, and a direct summand of a Tate-acyclic module is Tate-acyclic.

Exercise 4.3.1

1. Let $G = C_6$ and $M = \mathbb{Z}$ with trivial action. Use dimension shifting with the augmentation sequence $0 \to I_G \to \mathbb{Z}[C_6] \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ to compute $\hat{H}^{-2}(C_6, \mathbb{Z})$. Verify that the result matches the periodicity prediction $\hat{H}^{-2}(C_6, \mathbb{Z}) \cong \hat{H}^0(C_6, \mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z}$.

2. For $G = C_4 = \langle \sigma \rangle$, write down the periodic resolution (4.2) explicitly (with $D = \sigma - 1$ and $N = 1 + \sigma + \sigma^2 + \sigma^3$). Construct the extension (4.3) and verify that its class in $H^2(C_4, \mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$ corresponds to 1 mod 4 by computing the associated 2-cocycle.

3. Let G be a finite group and $H \leq G$ a subgroup with $H \neq \{1\}$. Show that the permutation module $\mathbb{Z}[G/H]$ is not cohomologically trivial by computing $\hat{H}^0(G, \mathbb{Z}[G/H])$. (*Hint:* The invariants are the multiples of the sum of all cosets.)

4. Show that the condition in theorem 4.3.5 cannot be weakened to a single degree: give an example of a cyclic group G and a G-module M with $\hat{H}^0(G,M) = 0$ but $\hat{H}^{-1}(G,M) \neq 0$. (*Hint:* The module $\mathbb{Z}[G]/(N_G)$ from exercise 4.1.1 (3) works.)

5. Let $G = C_p$ (p prime) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action. Let $u \in \hat{H}^2(C_p,\mathbb{Z})$ be the periodicity generator and $\bar{u}$ its image under the reduction map $\hat{H}^2(C_p,\mathbb{Z}) \to \hat{H}^2(C_p,\mathbb{Z}/p\mathbb{Z})$. Show that $\bar{u}$ equals the class u from the cup product ring $H^\bullet(C_p,\mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}[\chi,u]/(\chi^2)$ of exercise 4.2.1 (3) (for p odd). Explain how the periodicity isomorphism $- \smile u_G$ on $\hat{H}^\bullet(C_p,\mathbb{Z})$ descends to the periodicity $- \smile \bar{u}$ on $\hat{H}^\bullet(C_p,\mathbb{Z}/p\mathbb{Z})$.

4.4 Duality for Finite Groups

The periodicity theorem of theorem 4.3.4 shows that for a cyclic group G, the Tate cohomology $\hat{H}^p(G,M)$ in positive degrees is mirrored by $\hat{H}^{-p-1}(G,M)$ in negative degrees (with a shift). This mirroring becomes a precise duality when the cup product is used to pair $\hat{H}^p(G,M)$ with $\hat{H}^{-p-1}(G,M^*)$, where M^* is the Pontryagin dual. The resulting pairing is perfect for cyclic groups acting on finite modules (a purely group-theoretic duality that will be the model for the arithmetic duality theorems of chapters 8 and 10).

The section also establishes Nakayama's criterion, which connects the vanishing of Tate cohomology to the module-theoretic property of projectivity. This criterion is the tool that converts cohomological computations ("$\hat{H}^0$ and $\hat{H}^{-1}$ vanish") into structural conclusions ("the module is projective"), and it is used throughout Part III.

Definition 4.4.1: Pontryagin Dual with G-Action

Let G be a finite group and M a G-module. The *Pontryagin dual* of M is

$$M^* = \mathrm{Hom}_{\mathbb{Z}}(M,\mathbb{Q}/\mathbb{Z})$$

equipped with the *contragredient* G-action: $(g \cdot f)(m) = f(g^{-1} \cdot m)$ for $g \in G$, $f \in M^*$, $m \in M$. When M is finite, M^* is also finite with $|M^*| = |M|$.

The contragredient action ensures that M^* transforms "dually" to M: if G acts on M by a representation $\rho : G \to \mathrm{Aut}(M)$, then G acts on M^* by the contragredient representation $\rho^*(g) = \rho(g^{-1})^T$ (the transpose-inverse). For M finite, the canonical isomorphism $M \xrightarrow{\sim} M^{**}$ is G-equivariant, so the double dual recovers M with its original action.

The evaluation map $\mathrm{ev} : M \otimes_{\mathbb{Z}} M^* \to \mathbb{Q}/\mathbb{Z}$, defined by $\mathrm{ev}(m \otimes f) = f(m)$, is G-equivariant: $\mathrm{ev}(g\cdot m \otimes g\cdot f) = (g\cdot f)(g\cdot m) = f(g^{-1}\cdot g\cdot m) = f(m) = \mathrm{ev}(m\otimes f)$. When M is finite, the evaluation is a perfect pairing of abelian groups: the induced map $M \to (M^*)^*$ is an isomorphism. The map ev induces a homomorphism $\mathrm{ev}_* : \hat{H}^n(G, M\otimes M^*) \to \hat{H}^n(G,\mathbb{Q}/\mathbb{Z})$ for all n, and combining this with the cup product produces the duality pairing.

Definition 4.4.2: The Tate Duality Pairing

Let G be a finite group, M a G-module. For $p \in \mathbb{Z}$, the *Tate duality pairing*

$$\langle -, - \rangle : \hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$$

is the composition

$$\hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \xrightarrow{\smile} \hat{H}^{-1}(G, M \otimes M^*) \xrightarrow{\mathrm{ev}_*} \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\mathrm{inv}} \mathbb{Q}/\mathbb{Z}$$

where the first map is the cup product (definition 4.2.3), the second is induced by the evaluation, and inv is defined as follows. For G cyclic of order n acting trivially on $\mathbb{Q}/\mathbb{Z}$: the norm map N_G acts on $\mathbb{Q}/\mathbb{Z}$ by multiplication by n, and $\hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) = \ker(N_G : \mathbb{Q}/\mathbb{Z} \to \mathbb{Q}/\mathbb{Z})/I_G(\mathbb{Q}/\mathbb{Z}) = \frac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$ (since $\ker(n) = \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ and $I_G(\mathbb{Q}/\mathbb{Z}) = 0$ for the trivial action). The *invariant map* $\mathrm{inv} : \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$ is the natural inclusion $\frac{1}{n}\mathbb{Z}/\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$.

The target $\mathbb{Q}/\mathbb{Z}$ is an injective abelian group, so the pairing $\langle -, - \rangle$ induces a homomorphism $\hat{H}^p(G, M) \to \mathrm{Hom}(\hat{H}^{-p-1}(G, M^*), \mathbb{Q}/\mathbb{Z}) = \hat{H}^{-p-1}(G, M^*)^*$ (the Pontryagin dual of the Tate cohomology group). The pairing is $\mathbb{Z}$-bilinear in both variables (by the bilinearity of the cup product, theorem 4.2.4) and natural in M: a G-equivariant map $\varphi : M \to M'$ induces maps $\hat{H}^p(G, M) \to \hat{H}^p(G, M')$ and $\hat{H}^{-p-1}(G, (M')^*) \to \hat{H}^{-p-1}(G, M^*)$ (via φ^*), and the pairings for M and M' are compatible. The duality theorem asserts that the induced homomorphism is an isomorphism for cyclic groups.

Theorem 4.4.3: Tate Duality for Cyclic Groups

Let G be a cyclic group of order n and M a finite G-module. The Tate duality pairing

$$\langle -, - \rangle : \hat{H}^p(G, M) \times \hat{H}^{-p-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$$

is a perfect pairing of finite abelian groups for every $p \in \mathbb{Z}$. That is, the induced maps

$$\hat{H}^p(G, M) \to \hat{H}^{-p-1}(G, M^*)^* \quad \text{and} \quad \hat{H}^{-p-1}(G, M^*) \to \hat{H}^p(G, M)^*$$

are isomorphisms.

***Proof Key ideas*:**
This book does not prove the theorem, which holds for every finite group G. For a finite group G and any G-module A, with $A^* = \mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z})$, the cup product pairing $\hat{H}^i(G, A^*) \times \hat{H}^{-i-1}(G, A) \to \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) = \frac{1}{|G|}\mathbb{Z}/\mathbb{Z}$ induces an isomorphism $\hat{H}^i(G, A^*) \to \hat{H}^{-i-1}(G, A)^*$ [26, Ch. III, §1, (3.1.1)]. With $A = M$ and $i = -p-1$ this is the second map of the statement. With $A = M^*$ and $i = p$, and $M^{**} = M$ for finite M, it is the first. The pairing there evaluates $\mathrm{Hom}(A, \mathbb{Q}/\mathbb{Z}) \otimes A \to \mathbb{Q}/\mathbb{Z}$, the factors in the order opposite to the evaluation of definition 4.4.2, which changes the pairing by a sign (theorem 4.2.4(3)) and so does not affect perfectness. The proof there treats $i = 0$ directly: a G-homomorphism $f : A \to \mathbb{Q}/\mathbb{Z}$ is the same as a homomorphism $A/I_G A \to \mathbb{Q}/\mathbb{Z}$, and f is a norm $N_G h$ exactly when it vanishes on the norm-kernel ${}_{N_G}A$, which gives $\hat{H}^0(G, A^*) \cong ({}_{N_G}A/I_G A)^* = \hat{H}^{-1}(G, A)^*$ by proposition 4.1.5(3). The other degrees follow by dimension shifting, using the compatibility of cup products with

connecting homomorphisms (proposition 4.2.6), completing the sketch.

Cyclicity plays no role in the duality itself: [26, Ch. III, §1, (3.1.1)] proves it for every finite group G, with $\hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) = \frac{1}{|G|}\mathbb{Z}/\mathbb{Z}$ as in definition 4.4.2. The local and global duality theorems of chapters 8 and 10 provide the arithmetic analogues, where the role of the cyclic group is played by the decomposition group of a prime and the role of M^* is played by an arithmetically defined dual (the Cartier dual for finite flat group schemes, or the Pontryagin dual with a Tate twist for Galois modules).

Tate duality relates the Herbrand quotients of M and M^*: the perfect pairing gives $|\hat{H}^0(G, M)| = |\hat{H}^{-1}(G, M^*)|$ and $|\hat{H}^{-1}(G, M)| = |\hat{H}^0(G, M^*)|$, so $h(G, M^*) = h(G, M)^{-1}$ for finite M (exercise 4.4.1 (3)), consistent with $h(G, M) = h(G, M^*) = 1$ (proposition 4.1.11).

The duality pairing interacts with the Nakayama lemma for group rings to produce a criterion for cohomological triviality in terms of module-theoretic projectivity. This connection is the algebraic foundation for the cohomological approach to local class field theory.

Theorem 4.4.4: Nakayama's Criterion

Let G be a p-group and M a finitely generated $\mathbb{Z}_p[G]$-module that is $\mathbb{Z}_p$-free. The following are equivalent:

1. M is a projective $\mathbb{Z}_p[G]$-module.
2. M is cohomologically trivial: $\hat{H}^n(H, M) = 0$ for all $n \in \mathbb{Z}$ and all subgroups $H \leq G$.
3. $\hat{H}^{-1}(G, M) = \hat{H}^0(G, M) = 0$.

***Proof*:**
The implication (1) $\Rightarrow$ (2) is immediate: projective $\mathbb{Z}_p[G]$-modules are direct summands of free modules $\mathbb{Z}_p[G]^r$, and $\hat{H}^n(H, \mathbb{Z}_p[G]) = 0$ for all n and H (since $\mathbb{Z}_p[G]$ is induced from the trivial subgroup). The implication (2) $\Rightarrow$ (3) is trivial.

(3) $\Rightarrow$ (1). The Sylow p-subgroup of the p-group G is G itself, and for a prime $\ell \neq p$ the Sylow ℓ-subgroup is trivial, so (3) is condition (3) of theorem 4.3.5, and M is cohomologically trivial. Since M is $\mathbb{Z}_p$-free, multiplication by p is injective on M, and the long exact sequence of proposition 4.1.6 for $0 \to M \xrightarrow{p} M \to M/pM \to 0$ shows that $\hat{H}^n(H, M/pM) = 0$ for all n and all $H \leq G$. A module over the p-group G annihilated by p with this property is a free $\mathbb{F}_p[G]$-module [31, Ch. IX, §4, Thm. 5], a consequence of Nakayama's lemma for $\mathbb{F}_p[G]$ that this book does not prove.

Choose $m_1, \dots, m_r \in M$ whose images form an $\mathbb{F}_p[G]$-basis of M/pM, and let $\varphi\colon \mathbb{Z}_p[G]^r \to M$ send the i-th basis vector to m_i. The image M' of φ satisfies $M' + pM = M$. As M is a finitely generated $\mathbb{Z}_p$-module, Nakayama's lemma over the local ring $\mathbb{Z}_p$ gives $M' = M$, so φ is surjective. Both $\mathbb{Z}_p[G]^r$ and M are free $\mathbb{Z}_p$-modules of rank $r|G| = \dim_{\mathbb{F}_p} M/pM$, and a surjective homomorphism between free $\mathbb{Z}_p$-modules of the same finite rank is an isomorphism, its kernel being a direct summand of rank zero. Hence $M \cong \mathbb{Z}_p[G]^r$ is free, in particular projective, as claimed.

Nakayama's criterion is the bridge between the cohomological triviality studied in theorem 4.3.5 and the module-theoretic notion of projectivity. For a general finite group G (not necessarily a p-group), the criterion generalizes as follows: a finitely generated $\mathbb{Z}[G]$-module M that is $\mathbb{Z}$-free is projective if and only if it is cohomologically trivial. The proof reduces to p-Sylow subgroups via the restriction-corestriction formula: M is projective over $\mathbb{Z}[G]$ if and only if M is projective over $\mathbb{Z}_p[G_p]$ for every prime p and every p-Sylow subgroup $G_p \leq G$, and the criterion above applies to each $\mathbb{Z}_p[G_p]$.

The criterion is checked by computing cohomology: two vanishing conditions at each Sylow subgroup suffice. No projective resolution of M, which can be difficult to construct, is involved. In the applications of chapter 8, the module in question is the unit group of a local field extension, and the Tate cohomology is computed using the filtration by higher unit groups (a computation that is tractable because the graded pieces are modules for the residue field).

Example 4.4.5: The Duality Pairing in Practice

1. Let $G = C_n$ and $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action. The Pontryagin dual $M^* = \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (also with trivial action, since the contragredient of the trivial action is trivial). The Tate cohomology is $\hat{H}^q(C_n, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ for all q (by example 4.3.6(2)). The duality pairing in degree $p = 0$:
$$\langle -, - \rangle : \hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) \times \hat{H}^{-1}(C_n, \mathbb{Z}/n\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}.$$
Here $\hat{H}^0(C_n, \mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})/n(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$ (since $n \cdot \bar{a} = 0$ for all $\bar{a}$, the norm image is zero) and $\hat{H}^{-1}(C_n, \mathbb{Z}/n\mathbb{Z}) = \ker(n : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z})/0 = \mathbb{Z}/n\mathbb{Z}$. The cup product sends $(\bar{a}, \bar{b}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ to $\bar{a} \cdot \bar{b} \in \hat{H}^{-1}(C_n, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, composed with the inclusion $\mathbb{Z}/n\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$ sending $\bar{1} \mapsto 1/n$. Hence $\langle \bar{a}, \bar{b} \rangle = ab/n \in \mathbb{Q}/\mathbb{Z}$. This is the standard perfect pairing $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$.

2. Let $G = C_p$ (p odd prime) and $M = (\mathbb{Z}/p\mathbb{Z})^2$ with the action of a generator σ given by the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (a unipotent action). The invariants are $M^G = \ker(A - I) = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \langle e_1 \rangle \cong \mathbb{Z}/p\mathbb{Z}$ (where e_1, e_2 is the standard basis). The norm $N_G = I + A + \cdots + A^{p-1}$. Since $A = I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$, $A^k = I + kN$, so $N_G = pI + \frac{p(p-1)}{2}N = \frac{p(p-1)}{2}N$ (since $pI = 0$ in $(\mathbb{Z}/p\mathbb{Z})^2$). For p odd, $\frac{p-1}{2}$ is an integer and $\frac{p(p-1)}{2} \equiv 0 \pmod{p}$, so $N_G = 0$. Hence $\hat{H}^0(G, M) = M^G/N_G(M) = \mathbb{Z}/p\mathbb{Z}/0 = \mathbb{Z}/p\mathbb{Z}$.
The augmentation ideal acts by $(\sigma - 1) \cdot (a, b) = N(a, b) = (b, 0)$, so $I_G M = \langle e_1 \rangle$. Since $\ker(N_G) = M$ (as $N_G = 0$), $\hat{H}^{-1}(G, M) = M/I_G M = (\mathbb{Z}/p\mathbb{Z})^2/\langle e_1 \rangle \cong \mathbb{Z}/p\mathbb{Z}$.
The Pontryagin dual $M^* \cong (\mathbb{Z}/p\mathbb{Z})^2$ with the contragredient action $\sigma \cdot f = f \circ \sigma^{-1}$. Since $A^{-1} = I - N = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, the matrix of σ on M^* (in the dual basis) is $A^{-T} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$.
The invariants $(M^*)^G = \ker(A^{-T} - I) = \ker \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = \langle e_2^* \rangle \cong \mathbb{Z}/p\mathbb{Z}$, and by the same computation $\hat{H}^{-1}(G, M^*) \cong \mathbb{Z}/p\mathbb{Z}$. The duality pairing $\hat{H}^0(G, M) \times \hat{H}^{-1}(G, M^*) \to \mathbb{Q}/\mathbb{Z}$ is a pairing $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$, which is perfect by theorem 4.4.3.

The duality pairing is functorial in the group: for $H \leq G$ a subgroup, the restriction map Res_G^H on the first factor and the corestriction map Cor_H^G on the second factor are adjoint with respect to the pairing. That is, $\langle \mathrm{Res}(\alpha), \beta \rangle_H = \langle \alpha, \mathrm{Cor}(\beta) \rangle_G$ for $\alpha \in \hat{H}^p(G, M)$ and $\beta \in \hat{H}^{-p-1}(H, M^*)$. This adjunction is a consequence of the projection formula (proposition 4.2.5(2)) and the compatibility of the invariant map with restriction. The functoriality ensures that the duality pairings for different subgroups are compatible, which is essential for the inductive arguments in Part III.

Exercise 4.4.1

1. Let $G = C_4 = \langle \sigma \rangle$ and $M = \mathbb{Z}/4\mathbb{Z}$ with the action $\sigma \cdot \bar{a} = -\bar{a}$. Compute $\hat{H}^0(G, M)$, $\hat{H}^{-1}(G, M)$, and the Pontryagin dual M^* with its G-action. Verify that $|\hat{H}^0(G, M)| = |\hat{H}^{-1}(G, M^*)|$ and $|\hat{H}^{-1}(G, M)| = |\hat{H}^0(G, M^*)|$, as predicted by theorem 4.4.3.

2. Let $G = C_p$ and $M = \mathbb{Z}_p^2$ with the action $\sigma \cdot (a, b) = (a + pb, b)$ (a $\mathbb{Z}_p$-linear action). Verify Nakayama's criterion (theorem 4.4.4) by:

   (a) Computing $\hat{H}^0(G, M)$ and $\hat{H}^{-1}(G, M)$.

   (b) Determining whether M is a projective $\mathbb{Z}_p[C_p]$-module.

   (*Hint:* The norm N_G and augmentation ideal I_G can be computed using the matrix $A = I + pN$ where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.)

3. Use theorem 4.4.3 to show that $h(G, M^*) = h(G, M)^{-1}$ for G cyclic and M finite, and check that this agrees with proposition 4.1.11.

4. Let $G = C_n$ and M a finite G-module. Show that the duality pairing in degree p is related to the pairing in degree $p + 2$ by the periodicity generator: $\langle \alpha \smile u_G, \beta \rangle = \langle \alpha, u_G \smile \beta \rangle$ for $\alpha \in \hat{H}^p(G, M)$ and $\beta \in \hat{H}^{-p-3}(G, M^*)$. Deduce that $\langle \alpha, \gamma \rangle = \langle \alpha \smile u_G, u_G^{-1} \smile \gamma \rangle$ for $\gamma \in \hat{H}^{-p-1}(G, M^*)$, where $u_G^{-1} \in \hat{H}^{-2}(G, \mathbb{Z})$ is the class with $u_G^{-1} \smile u_G = \bar{1}$. (Both are formal consequences of the properties of the cup product.)

5. A G-module M is *self-dual* if $M \cong M^*$ as G-modules. Let $G = C_2$ and $M = \mathbb{Z}/2\mathbb{Z}$ with trivial action. Show that M is self-dual and that the duality pairing $\hat{H}^0(G, M) \times \hat{H}^{-1}(G, M) \to \mathbb{Q}/\mathbb{Z}$ becomes a *symmetric* pairing $\hat{H}^0 \times \hat{H}^{-1} \to \mathbb{Q}/\mathbb{Z}$. Compute this pairing explicitly.

Part II

Structure Theory

The abstract cohomology of Part I applies to any profinite group and any discrete module. Part II narrows the focus to the structural invariants that distinguish the profinite groups and modules arising in arithmetic: cohomological dimension, pro-p-groups, and the Brauer group.

Chapter 5 introduces cohomological dimension $\mathrm{cd}(G)$, the supremum of degrees in which $H^n(G, M) \neq 0$ for torsion modules M, and computes it for the profinite groups that appear in field arithmetic: $\mathrm{cd}(\hat{\mathbb{Z}}) = 1$, $\mathrm{cd}(G_K) = 2$ for p-adic fields K, and $\mathrm{cd}(G_K) \leq 2$ for number fields (under suitable hypotheses). Chapter 6 studies pro-p-groups: their generators and relations, the Golod-Shafarevich inequality (which resolved the class field tower problem), and Demushkin groups (the pro-p-completions of local Galois groups when $\mu_p \subseteq K$). Chapter 7 develops the Brauer group $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$: central simple algebras, the crossed product construction, and the period-index problem. The Brauer group is the bridge between the abstract cohomology of Part I and the field arithmetic of Part III: the invariant map $\mathrm{Br}(K) \to \mathbb{Q}/\mathbb{Z}$ for local fields and the Albert-Brauer-Hasse-Noether exact sequence for global fields are the inputs that drive the duality theorems and reciprocity laws.

5

Cohomological Dimension

The Tate cohomology and cup product machinery of chapter 4 provides the tools. The question now is how to measure the *complexity* of a profinite group's cohomology. For many groups of arithmetic interest (absolute Galois groups of fields, decomposition groups at primes, pro-p-completions of fundamental groups), the cohomology groups $H^n(G, M)$ vanish beyond a certain degree n, at least when the coefficients are restricted to p-torsion modules for a fixed prime p. This vanishing threshold is the *p-cohomological dimension* of G, and it is the single most important numerical invariant of a profinite group from the cohomological perspective. In dimension 1 the pro-p-groups are exactly the free ones (chapter 6), and in dimension 2 the Demushkin groups satisfy a Poincaré-type duality. The passage from dimension 2 to higher dimensions marks the boundary of what the methods of class field theory can reach.

This chapter develops cohomological dimension as an abstract invariant of profinite groups (section 5.1), computes it for the absolute Galois groups of familiar fields (section 5.2), and introduces the duality groups that satisfy a cohomological analogue of Poincaré duality (section 5.3).

5.1 p-Cohomological Dimension

Let G be a profinite group and p a prime. The colimit formula of theorem 3.2.2 shows that $H^n(G, M)$ is a torsion group for $n \geq 1$, so it decomposes into p-primary components. The p-cohomological dimension measures how far in degree the p-primary part of the cohomology can reach.

Definition 5.1.1: p-Cohomological Dimension

Let G be a profinite group and p a prime. The *p-cohomological dimension* of G is

$$\mathrm{cd}_p(G) = \sup\{n \geq 0 \mid H^n(G, M) \neq 0 \text{ for some } p\text{-torsion discrete } G\text{-module } M\}$$

with the convention $\sup \emptyset = 0$. The *cohomological dimension* is $\mathrm{cd}(G) = \sup_p \mathrm{cd}_p(G)$.

A module M is *p-torsion* if every element of M is annihilated by a power of p (equivalently, $M = \bigcup_k M[p^k]$). The restriction to p-torsion coefficients is essential: without it, the group $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z}$ (computed in example 3.2.6) would force $\mathrm{cd}(\hat{\mathbb{Z}}) \geq 2$, even though $\hat{\mathbb{Z}}$ has cohomological dimension 1 when measured with p-torsion coefficients at each prime p.

The first major result reduces the computation of $\mathrm{cd}_p(G)$ from checking all p-torsion modules to checking a single module.

Theorem 5.1.2: Serre's Criterion

Let G be a profinite group, p a prime, and $n \geq 0$. The following are equivalent:

1. $\mathrm{cd}_p(G) \leq n$.
2. $H^{n+1}(G, M) = 0$ for every p-torsion discrete G-module M.
3. $H^{n+1}(G, M) = 0$ for every simple discrete G-module M with $pM = 0$.

If G is a pro-p-group, they are also equivalent to:

4. $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z}) = 0$, where $\mathbb{Z}/p\mathbb{Z}$ carries the trivial G-action.

***Proof*:**
The implication (1) $\Rightarrow$ (2) is the definition of cd_p, and (2) $\Rightarrow$ (3) is a special case.

Step 1: Finitely generated modules are finite. Let M be a p-torsion discrete G-module. The stabilizer of $m \in M$ is open, hence of finite index (proposition 1.2.2), so the orbit $G \cdot m$ is finite. The submodule generated by finitely many elements is the subgroup generated by their finitely many orbits, a finitely generated abelian group whose elements are killed by powers of p. It is therefore a finite group of p-power order.

Step 2: (3) $\Rightarrow$ (2). Let M be a finite discrete G-module with $p^k M = 0$. We show $H^{n+1}(G, M) = 0$ by induction on $|M|$, the case $M = 0$ being clear. If $M \neq 0$, choose a nonzero submodule S of smallest order. It is simple, and pS is a submodule of S different from S, as otherwise $S = p^k S = 0$, so $pS = 0$. The quotient M/S is a discrete G-module (proposition 3.1.2) of smaller order killed by p^k, and the long exact sequence of $0 \to S \to M \to M/S \to 0$ contains the exact segment $H^{n+1}(G, S) \to H^{n+1}(G, M) \to H^{n+1}(G, M/S)$. Its outer terms vanish, by (3) and by induction, so $H^{n+1}(G, M) = 0$. A general p-torsion discrete G-module M is the union of the directed family of its finitely generated submodules M_i, which are finite and killed by powers of p by Step 1, so $H^{n+1}(G, M) = \varinjlim_i H^{n+1}(G, M_i) = 0$ by corollary 3.2.5.

Step 3: (2) ⇒ (1). We show by induction on $q \geq n+1$ that $H^q(G,M) = 0$ for every p-torsion discrete G-module M, the case $q = n+1$ being (2). The map u_M of Step 3 of the proof of proposition 3.1.6 embeds M in $I = \mathrm{Map}^{\mathrm{cts}}(G,M)$. A continuous map $G \to M$ takes finitely many values, as G is compact, so it is killed by a power of p. Hence I and $Q = I/u_M(M)$ are p-torsion. Since $H^q(G,I) = 0$ for $q \geq 1$ (corollary 3.5.6(2)), the long exact sequence of $0 \to M \to I \to Q \to 0$ makes $H^q(G,Q) \to H^{q+1}(G,M)$ surjective, and its source vanishes by induction. Hence $H^q(G,M) = 0$ for all $q > n$ and all p-torsion M, which is (1).

Step 4: Groups acting through a p-group. Let Γ be a profinite group acting on a finite nonzero $\mathbb{F}_p$-vector space V through a finite p-group Γ/U. The orbits of Γ/U on V have p-power sizes and $|V|$ is divisible by p, so the number of fixed points is divisible by p. As 0 is fixed, $V^\Gamma \neq 0$. If V is simple, the nonzero submodule V^Γ equals V, so Γ acts trivially and $V \cong \mathbb{Z}/p\mathbb{Z}$. Choosing $0 \neq v \in V^\Gamma$ and applying the same argument to $V/\mathbb{F}_p v$ gives, by induction on $\dim V$, a filtration of V by submodules whose successive quotients are $\mathbb{Z}/p\mathbb{Z}$ with the trivial action.

Step 5: (3) ⇔ (4) for pro-p-groups. Since $\mathbb{Z}/p\mathbb{Z}$ is simple and killed by p, (3) implies (4). Conversely, let G be a pro-p-group and M a simple discrete G-module with $pM = 0$. Being simple, M is generated by any of its nonzero elements, so M is finite by Step 1, so the intersection U of the stabilizers of its elements is an open subgroup, normal because $\mathrm{Stab}_G(g\cdot m) = g\mathrm{Stab}_G(m)g^{-1}$. The finite quotient G/U is a p-group (definition 1.3.1), so $M \cong \mathbb{Z}/p\mathbb{Z}$ by Step 4, and (4) gives (3).

For a pro-p-group, Serre's criterion reduces the bound $\mathrm{cd}_p(G) \leq n$ to the vanishing of the single group $H^{n+1}(G, \mathbb{Z}/p\mathbb{Z})$. For a general profinite group the simple modules killed by p need not be trivial, and condition (4) does not suffice. For $G = S_3$ and $p = 3$, the group $H^1(S_3, \mathbb{Z}/3\mathbb{Z}) = \mathrm{Hom}(S_3, \mathbb{Z}/3\mathbb{Z})$ vanishes (proposition 3.1.7), as the abelianization of S_3 has order 2, while $\mathrm{cd}_3(S_3) = \infty$, as S_3 contains the cyclic group $\mathbb{Z}/3\mathbb{Z}$ of infinite 3-cohomological dimension (proposition 5.1.3(1), example 5.2.11). The next results establish the basic functorial properties of cd_p.

Proposition 5.1.3: Monotonicity

Let G be a profinite group and p a prime.

1. If H is a closed subgroup of G, then $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(G)$.
2. If N is a closed normal subgroup of G, then $\mathrm{cd}_p(G) \leq \mathrm{cd}_p(N) + \mathrm{cd}_p(G/N)$.

***Proof*:**

Part (1): Let M be a p-torsion discrete H-module. An element f of $\mathrm{CoInd}_H^G(M)$ is a continuous map from the compact group G to the discrete group M, so it takes finitely many values and is killed by a power of p. So $\mathrm{CoInd}_H^G(M)$ is a p-torsion discrete G-module, and Shapiro's lemma (theorem 3.5.5) gives $H^q(H,M) \cong H^q(G, \mathrm{CoInd}_H^G(M))$ for every $q \geq 0$. If $H^q(H,M) \neq 0$, then $q \leq \mathrm{cd}_p(G)$ by the definition of $\mathrm{cd}_p(G)$. Hence $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(G)$.

Part (2): Set $m = \mathrm{cd}_p(N)$ and $n = \mathrm{cd}_p(G/N)$. If either is infinite there is nothing to prove. Let M be a p-torsion discrete G-module and consider the LHS spectral sequence (theorem 3.4.5) $E_2^{p,q} = H^p(G/N, H^q(N,M)) \Rightarrow H^{p+q}(G,M)$. The coefficient module $H^q(N,M)$ is a subquotient of the group of continuous cochains $N^q \to M$ (proposition 3.1.6), and such a cochain takes finitely

many values on the compact space N^q, each killed by a power of p, so $H^q(N,M)$ is p-torsion. The conjugation action makes it a discrete G/N-module (proposition 3.4.3), so $E_2^{p,q}=0$ for $p>n$. And $E_2^{p,q}=0$ for $q>m$, since already $H^q(N,M)=0$. For total degree $k>m+n$, every term $E_2^{p,q}$ on the diagonal $p+q=k$ has $p>n$ or $q>m$, so the whole diagonal vanishes. Each $E_\infty^{p,q}$ is a subquotient of $E_2^{p,q}$, so the filtration of $H^k(G,M)$ has all graded pieces zero, and $H^k(G,M)=0$. Hence $\mathrm{cd}_p(G)\leq m+n$.

The quotient G/N obeys no bound in terms of G. The group $\mathbb{Z}_p$ has $\mathrm{cd}_p(\mathbb{Z}_p)=1$ (example 5.1.5), while its finite quotient $\mathbb{Z}/p\mathbb{Z}$ has $\mathrm{cd}_p(\mathbb{Z}/p\mathbb{Z})=\infty$ (example 5.2.11).

Proposition 5.1.4: Characterization of $\mathrm{cd}_p(G)=0$

Let G be a profinite group. Then $\mathrm{cd}_p(G)=0$ if and only if p does not divide the order $\#G$ of definition 1.3.5, that is, $p\nmid[G:U]$ for every open normal subgroup U of G.

***Proof*:**
Suppose $p\nmid[G:U]$ for every open normal U, and let M be a p-torsion discrete G-module, $n\geq 1$ and $\alpha\in H^n(G,M)$. By theorem 3.2.2, α is inflated from $H^n(G/U,M^U)$ for some open normal U, a group killed by $|G/U|$ (corollary 3.3.8(4)), so $|G/U|\cdot\alpha=0$. The module M is the union of its finitely generated submodules M_i, which are finite of p-power order (Step 1 of the proof of theorem 5.1.2). Multiplication by $|M_i|$ is zero on M_i, hence on $H^n(G,M_i)$ by additivity, and $H^n(G,M)=\varinjlim_i H^n(G,M_i)$ (corollary 3.2.5), so α is also killed by a power of p. As $|G/U|$ is prime to p, $\alpha=0$. Hence $\mathrm{cd}_p(G)=0$.

Conversely, suppose p divides $[G:U]$ for some open normal U. Let $\bar{P}$ be a Sylow p-subgroup of the finite group G/U (*Core Algebra* [9, Sylow's Theorems](1)), which is nontrivial, and let H be its preimage in G, an open subgroup. The nontrivial finite p-group $\bar{P}$ is solvable (*Core Algebra* [9, Finite P-Groups]), so its derived series reaches 1. As $\bar{P}\neq 1$, its commutator subgroup $\bar{P}'$ is a proper subgroup, and $A=\bar{P}/\bar{P}'$ is a nontrivial finite abelian p-group. Then $A/pA\neq 0$, since $A=pA$ would give $A=p^kA=0$ for large k, and the $\mathbb{F}_p$-vector space A/pA maps onto $\mathbb{F}_p$. This gives a surjective homomorphism $\chi\colon H\to H/U=\bar{P}\to\mathbb{Z}/p\mathbb{Z}$, continuous because U is open. With the trivial action, χ is a continuous 1-cocycle that is not a coboundary, since all coboundaries $h\mapsto h\cdot m-m$ vanish. So $H^1(H,\mathbb{Z}/p\mathbb{Z})\neq 0$, $\mathrm{cd}_p(H)\geq 1$, and $\mathrm{cd}_p(G)\geq 1$ by proposition 5.1.3(1).

The condition is stronger than the absence of elements of order p: the group $\mathbb{Z}_p$ has no elements of finite order other than 0, while $\mathrm{cd}_p(\mathbb{Z}_p)=1$ (example 5.1.5).

The next example computes cd_p for the procyclic groups that have appeared throughout Parts I and II.

Example 5.1.5: Cohomological Dimension of Procyclic Groups

1. The group $\hat{\mathbb{Z}}=\varprojlim_n\mathbb{Z}/n\mathbb{Z}$ satisfies $\mathrm{cd}_p(\hat{\mathbb{Z}})=1$ for every prime p, and therefore $\mathrm{cd}(\hat{\mathbb{Z}})=1$. This is immediate from theorem 3.2.3: every p-torsion module is torsion, so $H^i(\hat{\mathbb{Z}},M)=0$ for $i\geq 2$ and every p-torsion M, which is the bound $\mathrm{cd}_p(\hat{\mathbb{Z}})\leq 1$. The lower bound is $H^1(\hat{\mathbb{Z}},\mathbb{Z}/p\mathbb{Z})=(\mathbb{Z}/p\mathbb{Z})_{\hat{\mathbb{Z}}}=\mathbb{Z}/p\mathbb{Z}\neq 0$, the action being trivial.

The vanishing of $H^2(\hat{\mathbb{Z}}, -)$ on p-torsion coefficients contrasts with $H^2(\hat{\mathbb{Z}}, \mathbb{Z}) \cong \mathbb{Q}/\mathbb{Z} \neq 0$. The module $\mathbb{Z}$ is not p-torsion, so this nonvanishing does not affect cd_p.

2. The pro-p-group $\mathbb{Z}_p$ satisfies $\mathrm{cd}_p(\mathbb{Z}_p) = 1$. It is a closed subgroup of $\hat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell$, so $\mathrm{cd}_p(\mathbb{Z}_p) \leq 1$ by proposition 5.1.3(1). The reduction map $\mathbb{Z}_p \to \mathbb{Z}/p\mathbb{Z}$ is a continuous 1-cocycle for the trivial action that is not a coboundary, since all coboundaries $g \mapsto g \cdot m - m$ vanish, so $H^1(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z}) \neq 0$. For $\ell \neq p$, $\mathrm{cd}_\ell(\mathbb{Z}_p) = 0$ by proposition 5.1.4, since every finite quotient $\mathbb{Z}/p^k\mathbb{Z}$ of $\mathbb{Z}_p$ has order prime to ℓ.

3. More generally, $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$ for any $d \geq 1$. This follows from the Künneth formula: $H^n(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \cong \bigoplus_{n_1+\cdots+n_d=n} \bigotimes_{i=1}^d H^{n_i}(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z})$. Since $H^0(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ and $H^1(\mathbb{Z}_p, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ (and $H^n = 0$ for $n \geq 2$), the nonzero contributions have each $n_i \in \{0, 1\}$, so $H^n(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \neq 0$ exactly when $0 \leq n \leq d$. The top nonvanishing degree is $n = d$ (with $H^d(\mathbb{Z}_p^d, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$, the "top class"), giving $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$.

4. (If the reader has read [23]) The equality $\mathrm{cd}_p(\mathbb{Z}_p^d) = d$ is the profinite analogue of the fact that $H^n(T^d; \mathbb{Z}/p\mathbb{Z}) \neq 0$ for $0 \leq n \leq d$ and $= 0$ for $n > d$, where $T^d = (\mathbb{R}/\mathbb{Z})^d$ is the d-torus. The classifying space $B\mathbb{Z}_p^d$ is the p-completion of T^d, and profinite cohomological dimension equals the topological cohomological dimension of the classifying space.

The characterization of $\mathrm{cd}_p = 1$ is especially important because of its connection to free pro-p-groups.

Proposition 5.1.6: Free Pro-p-Groups Have $\mathrm{cd}_p = 1$

Let F be a free pro-p-group of rank ≥ 1. Then $\mathrm{cd}_p(F) = 1$.

***Proof*:**

A free pro-p-group F on a set S of generators satisfies $H^1(F, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})^{|S|}$ (continuous homomorphisms $F \to \mathbb{Z}/p\mathbb{Z}$ correspond to functions $S \to \mathbb{Z}/p\mathbb{Z}$), so $\mathrm{cd}_p(F) \geq 1$ when $|S| \geq 1$. For the upper bound, the key fact is that free pro-p-groups have a "tree-like" structure that forces $H^2(F, M) = 0$ for all p-torsion M. This follows from the characterization of free pro-p-groups as those pro-p-groups for which every embedding problem with p-group kernel has a solution: any extension $0 \to M \to E \to F \to 1$ with M a p-torsion F-module splits (because F maps onto any finite p-quotient of E), so the obstruction class in $H^2(F, M)$ vanishes. By Serre's criterion, $\mathrm{cd}_p(F) \leq 1$.

The converse (that a pro-p-group with $\mathrm{cd}_p = 1$ is free) is also true and will be proved in chapter 6. Together, the two directions give a cohomological characterization of free pro-p-groups: a pro-p-group G with $G \neq 1$ is free if and only if $\mathrm{cd}_p(G) = 1$.

The following proposition shows how cd_p behaves under products, complementing the spectral

sequence bound of proposition 5.1.3(2).

Proposition 5.1.7: Cohomological Dimension of Products

Let G_1 and G_2 be profinite groups. Then $\mathrm{cd}_p(G_1 \times G_2) = \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$.

***Proof*:**
The inequality $\mathrm{cd}_p(G_1 \times G_2) \leq \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$ follows from the LHS spectral sequence applied to the normal subgroup $N = G_2$ of $G = G_1 \times G_2$ with quotient $G/N = G_1$ (proposition 5.1.3(2)).

For the reverse inequality, the Künneth formula in group cohomology gives a natural map

$$\bigoplus_{i+j=n} H^i(G_1, \mathbb{Z}/p\mathbb{Z}) \otimes H^j(G_2, \mathbb{Z}/p\mathbb{Z}) \to H^n(G_1 \times G_2, \mathbb{Z}/p\mathbb{Z})$$

When G_1 and G_2 are pro-p-groups, this map is an isomorphism (since the cohomology groups are $\mathbb{Z}/p\mathbb{Z}$-vector spaces and the Künneth formula holds over a field with no Tor correction term). In particular, $H^{\mathrm{cd}_p(G_1)}(G_1, \mathbb{Z}/p\mathbb{Z}) \neq 0$ and $H^{\mathrm{cd}_p(G_2)}(G_2, \mathbb{Z}/p\mathbb{Z}) \neq 0$, so $H^{\mathrm{cd}_p(G_1)+\mathrm{cd}_p(G_2)}(G_1 \times G_2, \mathbb{Z}/p\mathbb{Z}) \neq 0$.

For general profinite groups, the argument reduces to pro-p-Sylow subgroups via restriction-corestriction: $\mathrm{cd}_p(G) = \mathrm{cd}_p(G_p)$ where G_p is a pro-p-Sylow subgroup (proposition 5.1.3(1) gives $\leq$, and restriction to the Sylow gives $\geq$ since Res is injective on p-primary parts by $\mathrm{Cor} \circ \mathrm{Res} = [G : G_p]$ with $p \nmid [G : G_p]$). Then $\mathrm{cd}_p(G_1 \times G_2) = \mathrm{cd}_p((G_1)_p \times (G_2)_p) = \mathrm{cd}_p((G_1)_p) + \mathrm{cd}_p((G_2)_p) = \mathrm{cd}_p(G_1) + \mathrm{cd}_p(G_2)$.

Exercise 5.1.1

1. Let G be a profinite group with $\mathrm{cd}_p(G) \leq n$. Use Serre's criterion and the long exact sequence for $0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p^2\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$ to show that $H^{n+1}(G, \mathbb{Z}/p^k\mathbb{Z}) = 0$ for all $k \geq 1$.
2. Let $G = N \rtimes Q$ be a semidirect product of profinite groups. Show that $\mathrm{cd}_p(G) \leq \mathrm{cd}_p(N) + \mathrm{cd}_p(Q)$. Give an example showing this inequality can be strict. (*Hint:* For strictness, let G be the free pro-p-group on two generators x, y (definition 6.1.1), let Q be the free pro-p-group on one generator x, and let N be the kernel of the map $G \to Q$ with $x \mapsto x$ and $y \mapsto 1$.)
3. Let G be a profinite group, p a prime, and H an open subgroup. Show that $\mathrm{cd}_p(H) = \mathrm{cd}_p(G)$ if $p \nmid [G : H]$ or if $\mathrm{cd}_p(G) < \infty$, and give an example where $\mathrm{cd}_p(H) < \mathrm{cd}_p(G)$. (*Hint:* In the first case restriction is injective on p-primary parts. In the second, show that corestriction is surjective in degree $\mathrm{cd}_p(G)$, using proposition 3.5.7(3).)
4. Let G be a profinite group and G_p a pro-p-Sylow subgroup. Prove that $\mathrm{cd}_p(G) = \mathrm{cd}_p(G_p)$.
5. Let G be a profinite group. Show that $\mathrm{cd}_p(G) \leq \sup_U \mathrm{cd}_p(G/U)$, where U ranges over the open normal subgroups of G. Show that the right side is either 0 or ∞, and deduce that the inequality is strict for $G = \mathbb{Z}_p$. (*Hint:* Use the colimit formula of theorem 3.2.2 for the first part, and example 5.2.11 for the second.)

5.2 Cohomological Dimension of Fields

The p-cohomological dimension of a profinite group becomes an arithmetic invariant when applied to the absolute Galois group of a field. For a field K with separable closure K^{sep} and absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$, the number $\mathrm{cd}_p(K) := \mathrm{cd}_p(G_K)$ encodes fundamental properties of the arithmetic and geometry of K. Separably closed fields have $\mathrm{cd} = 0$; finite fields and function fields of curves over algebraically closed fields have $\mathrm{cd} = 1$; local fields have $\mathrm{cd} = 2$. This section computes $\mathrm{cd}_p(K)$ for the fields that have appeared as running examples throughout the book, introduces the C_r-property as a geometric condition that controls cohomological dimension, and bounds cd_p under algebraic and transcendental extensions.

Definition 5.2.1: Cohomological Dimension of a Field

Let K be a field. The *p-cohomological dimension of K* is $\mathrm{cd}_p(K) = \mathrm{cd}_p(G_K)$, and the *cohomological dimension of K* is $\mathrm{cd}(K) = \mathrm{cd}(G_K)$.

The definition depends only on the separable closure K^{sep} (or equivalently, on the isomorphism class of G_K as a profinite group), not on any embedding of K into an algebraically closed field. A separable closure of K is unique up to K-isomorphism, being the set of separable elements of an algebraic closure, which is unique up to K-isomorphism (*Core Algebra* [9, Uniqueness Of Algebraic Closure]), so G_K is determined by K up to isomorphism (and up to inner automorphism once the separable closure is fixed up to K-automorphism), and $\mathrm{cd}_p(K)$ is a well-defined invariant of the field.

Proposition 5.2.2: $\mathrm{cd}(K) = 0$ Characterizes Separably Closed Fields

A field K has $\mathrm{cd}(K) = 0$ if and only if K is separably closed.

***Proof*:**
If K is separably closed (definition 1.4.2), then $K^{\mathrm{sep}} = K$ and $G_K = 1$. The trivial group has $\mathrm{cd}_p = 0$ for every prime p by proposition 5.1.4, as its order is 1, so $\mathrm{cd}(K) = 0$. Conversely, suppose K is not separably closed, and take $x \in K^{\mathrm{sep}} \setminus K$. The Galois closure F of $K(x)$ in K^{sep} is a finite Galois extension of K with $F \neq K$, so $U = \mathrm{Gal}(K^{\mathrm{sep}}/F)$ is an open normal subgroup of G_K of index $[F : K] > 1$ by theorem 1.4.1(2). A prime p dividing $[G_K : U]$ gives $\mathrm{cd}_p(G_K) \geq 1$ by proposition 5.1.4. Hence $\mathrm{cd}(K) \geq 1$.

The next case, $\mathrm{cd}(K) \leq 1$, has a cohomological characterization involving the Brauer group, which will be constructed in chapter 7. The following proposition uses the identification $H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$ (the Brauer group of K), which is proved in section 7.2.

Proposition 5.2.3: Characterization of $\mathrm{cd}_p(K) \leq 1$

Let K be a field and p a prime different from the characteristic of K. The following are equivalent:

1. $\mathrm{cd}_p(K) \leq 1$.
2. $H^2(G_L, (L^{\mathrm{sep}})^\times)\{p\} = 0$ for every finite separable extension L/K.
3. $\mathrm{Br}(L)\{p\} = 0$ for every finite separable extension L/K.

***Proof*:**
The equivalence (2) $\Leftrightarrow$ (3) is the identification $H^2(G_L, (L^{\mathrm{sep}})^\times) = \mathrm{Br}(L)$ of theorem 7.2.8, and (1) $\Leftrightarrow$ (3) is corollary 7.2.10. Both are proved in chapter 7, without using this proposition. The proof of corollary 7.2.10 uses theorem 5.1.2, proposition 5.1.3, theorem 3.2.2, corollary 3.3.8, theorem 1.4.1, and Hilbert's Theorem 90 (theorem 8.1.1, proved in chapter 8).

The hypothesis on the characteristic cannot be dropped. Every field K of characteristic p has $\mathrm{cd}_p(K) \leq 1$ [30, Ch. II, §2.2, Prop. 3], while the local field $\mathbb{F}_p((t))$ has $\mathrm{Br}(\mathbb{F}_p((t))) \cong \mathbb{Q}/\mathbb{Z}$ (theorem 7.3.1), whose p-primary part is nonzero.

Example 5.2.4: Fields of Cohomological Dimension ≤ 1
The condition $\mathrm{cd}_p(K) \leq 1$ says that the Galois group of K is "one-dimensional" from the perspective of p-torsion cohomology: all higher obstructions vanish. For p different from the characteristic, the equivalence with the vanishing of the Brauer group shows this: there are no nontrivial central simple algebras of p-power degree over K or its finite separable extensions. The following fields satisfy this condition.

1. *Finite fields.* Let $K = \mathbb{F}_q$ be a finite field with q elements. The absolute Galois group is $G_{\mathbb{F}_q} \cong \hat{\mathbb{Z}}$ (example 1.4.5), and $\mathrm{cd}_p(\hat{\mathbb{Z}}) = 1$ for all primes p (example 5.1.5). Hence $\mathrm{cd}(\mathbb{F}_q) = 1$.

2. *Function fields of curves over algebraically closed fields.* Let k be algebraically closed and $K = k(C)$ the function field of a smooth projective curve C over k. The field K has transcendence degree 1 over k, so it is C_1 by proposition 5.2.7(3), and theorem 5.2.8(2) gives $\mathrm{cd}(K) \leq 1$. The field K is not separably closed: for $t \in K$ transcendental over k, the extension $K/k(t)$ is finite, and for a prime $\ell > [K : k(t)]$ different from the characteristic, a root α of the separable polynomial $X^\ell - t$ satisfies $[k(t, \alpha) : k(t)] = \ell$ (Eisenstein at t), so $[K(\alpha) : K] \geq \ell/[K : k(t)] > 1$. Hence $\mathrm{cd}(K) = 1$ by proposition 5.2.2.

3. *p-adic fields and completions (preview).* Let $K = \mathbb{Q}_p$. The absolute Galois group $G_{\mathbb{Q}_p}$ has $\mathrm{cd}(\mathbb{Q}_p) = 2$. It is corollary 8.3.3, whose proof is cited. The group $H^2(G_{\mathbb{Q}_p}, \mu_p) \cong \mathrm{Br}(\mathbb{Q}_p)[p]$ is nonzero (so $\mathrm{cd}_p \geq 2$), while $H^3(G_{\mathbb{Q}_p}, M) = 0$ for all p-torsion M (so $\mathrm{cd}_p \leq 2$).

4. *Number fields (preview).* Let K be a number field. Then $\mathrm{cd}_p(K) = 2$ for p odd, and also for $p = 2$ when K is totally imaginary, while $\mathrm{cd}_2(K) = \infty$ when K has a real embedding, since $\mathrm{cd}_2(\mathbb{R}) = \infty$ (see below). The references are given after the table.

The fields of $\mathrm{cd} \leq 1$ include finite fields and function fields of curves over algebraically closed fields. These examples share a geometric property: every homogeneous polynomial equation of "low" degree in "many" variables has a nontrivial solution. This is the C_r-property, introduced by Lang and studied by Tsen.

Definition 5.2.5: C_r-Fields

A field K is a *C_r-field* (for $r \geq 0$) if every homogeneous polynomial $f \in K[x_1, \ldots, x_n]$ of degree $d \geq 1$ with $n > d^r$ has a nontrivial zero in K^n.

A C_0-field is algebraically closed. For a monic $g \in K[x]$ of degree $d \geq 1$, the binary form $y^d g(x/y)$ has a nontrivial zero (a, b), and $b \neq 0$, since the form takes the value a^d at $(a, 0)$ and $(a, b) \neq (0, 0)$, so $g(a/b) = 0$. Since $d^{r+1} \geq d^r$ for $d \geq 1$, every C_r-field is a C_{r+1}-field.

Lemma 5.2.6: Forms in More Variables than Equations

Let k be an algebraically closed field and $P_1, \ldots, P_s \in k[x_1, \ldots, x_m]$ homogeneous polynomials of positive degree with $s < m$. Then $P_1, \ldots, P_s$ have a common zero in $k^m \setminus \{0\}$.

***Proof*:**
Each P_l vanishes at 0, being homogeneous of positive degree. Set $X_0 = \mathbb{A}_k^m$, an irreducible closed set of dimension m containing 0. Suppose an irreducible closed subset $X_{l-1} \subseteq \mathbb{A}_k^m$ has been found that contains 0, has dimension at least $m - l + 1$, and on which $P_1, \ldots, P_{l-1}$ vanish. If P_l vanishes on X_{l-1}, set $X_l = X_{l-1}$. Otherwise the restriction of P_l to X_{l-1} is a nonzero element of the coordinate ring of X_{l-1} vanishing at the point 0, hence not a unit, and by *Classical Algebraic Geometry* [18, Hypersurface Sections Drop Dimension by One] every irreducible component of $\{x \in X_{l-1} \mid P_l(x) = 0\}$ has dimension $\dim X_{l-1} - 1 \geq m - l$. Let X_l be such a component containing 0. After s steps, X_s is an irreducible closed set containing 0, of dimension at least $m - s \geq 1$, on which every P_l vanishes. A single point has dimension 0, so X_s contains a point other than 0.

Proposition 5.2.7: Tsen-Lang Theorem

1. Algebraically closed fields are C_0.
2. Finite fields are C_1 (the Chevalley-Warning theorem).
3. If k is algebraically closed, every extension of k of transcendence degree 1 is a C_1-field. In particular $k(t)$ is C_1 (Tsen's theorem).
4. Every algebraic extension of a C_1-field is a C_1-field.

***Proof*:**
Part (1). Let $f \in K[x_1, \dots, x_n]$ be homogeneous of degree $d \geq 1$ with $n \geq 2$, over an algebraically closed field K, and let c be the coefficient of x_1^d in f. If $c = 0$, then $f(1, 0, \dots, 0) = c = 0$. If $c \neq 0$, the polynomial $f(x, 1, 0, \dots, 0) \in K[x]$ has degree d and so a root $a \in K$, and $(a, 1, 0, \dots, 0)$ is a nontrivial zero of f.

Part (2). Let $K = \mathbb{F}_q$, with q a power of the prime p, and let $f \in K[x_1, \dots, x_n]$ be homogeneous of degree $d \geq 1$ with $n > d$. Write N for the number of zeros of f in K^n. Since $\beta^{q-1} = 1$ for $\beta \in K^\times$, the element $1 - f(\alpha)^{q-1}$ of K is 1 when $f(\alpha) = 0$ and 0 otherwise, so in K

$$N \cdot 1 = \sum_{\alpha \in K^n} \big(1 - f(\alpha)^{q-1}\big) = q^n \cdot 1 - \sum_{\alpha \in K^n} f(\alpha)^{q-1} = - \sum_{\alpha \in K^n} f(\alpha)^{q-1}$$

The polynomial f^{q-1} has degree $d(q-1) < n(q-1)$, so every monomial $x_1^{r_1} \cdots x_n^{r_n}$ occurring in it has some exponent $r_i < q-1$. For $0 \leq r < q-1$ the sum $\sum_{\beta \in K} \beta^r$ vanishes, with the convention $0^0 = 1$: for $r = 0$ it is $q \cdot 1 = 0$, and for $r \geq 1$ it is $\sum_{j=0}^{q-2} \omega^{jr} = (\omega^{r(q-1)} - 1)/(\omega^r - 1) = 0$, where ω generates the cyclic group $K^\times$, so that $\omega^r \neq 1$. Hence $\sum_{\alpha \in K^n} \alpha_1^{r_1} \cdots \alpha_n^{r_n} = \prod_i \sum_{\beta \in K} \beta^{r_i} = 0$ for every monomial of f^{q-1}, so $N \cdot 1 = 0$ in K and $p \big| N$. The zero $0 \in K^n$ gives $N \geq 1$, so $N \geq p \geq 2$ and f has a nontrivial zero.

Part (4). Let K be a C_1-field, L/K an algebraic extension, and $F \in L[x_1, \dots, x_n]$ homogeneous of degree $d \geq 1$ with $n > d$. The coefficients of F generate a finite extension of K inside L, and a nontrivial zero there is one in L^n, so assume $m = [L : K]$ is finite. For $z \in L$ write $N_{L/K}(z)$ for the determinant of the K-linear map $L \to L$, $w \mapsto zw$. It is nonzero when $z \neq 0$, as this map is then invertible. Choose a K-basis $v_1, \dots, v_m$ of L, let $M_k \in M_m(K)$ be the matrix of multiplication by v_k, and substitute $x_i = \sum_j y_{ij} v_j$ with new variables y_{ij}. Expanding F and writing each coefficient in the basis gives $F(x) = \sum_k G_k(y) v_k$ with each $G_k \in K[y]$ homogeneous of degree d or zero, so

$$N_{L/K}\big(F(x)\big) = \det\Big(\sum_k G_k(y) M_k\Big)$$

is a polynomial in the mn variables y_{ij} with coefficients in K, homogeneous of degree md or zero. Since $mn > md$ and K is C_1, it has a nontrivial zero $(b_{ij}) \in K^{mn}$. The vector $\beta = (\sum_j b_{ij} v_j)_i \in L^n$ is nonzero and $N_{L/K}(F(\beta)) = 0$, so $F(\beta) = 0$. This is the argument of [30, Ch. II, §3.2, Prop. 8(a)].

Part (3). An extension of k of transcendence degree 1 is algebraic over $k(t)$ for any element t of it transcendental over k, so by part (4) it suffices to treat $K = k(t)$. Let $f \in K[x_1, \dots, x_n]$ be homogeneous of degree $d \geq 1$ with $n > d$. Clearing denominators, assume the coefficients of f lie in $k[t]$ and have degree at most e in t. Put $N = e + 1$ and substitute $x_i = \sum_{j=0}^{N} a_{ij} t^j$ with new variables a_{ij}. Since f is homogeneous of degree d and each x_i is linear in the a_{ij},

$$f(x_1, \dots, x_n) = \sum_{l=0}^{dN+e} P_l(a)\, t^l$$

with each $P_l \in k[a_{ij}]$ homogeneous of degree d or zero. These are $dN + e + 1$ polynomials in $n(N+1)$ variables, and

$$n(N+1) - (dN + e + 1) = (n-d)N + n - e - 1 \geq N + n - e - 1 = n > 0$$

By lemma 5.2.6, the nonzero P_l have a common zero $(a_{ij}) \neq 0$ in $k^{n(N+1)}$. The resulting $x_i \in k[t]$ are not all zero, because the powers of t are linearly independent over k, and $f(x_1, \ldots, x_n) = 0$.

Lang and Nagata extended parts (3) and (4) to every r: an algebraic extension of a C_r-field is C_r, and an extension of transcendence degree n of a C_r-field is C_{r+n} [30, Ch. II, §4.5]. In particular an extension of transcendence degree r of an algebraically closed field is C_r. For $r \geq 2$ the norm argument of part (4) breaks down, since $mn > (md)^r$ does not follow from $n > d^r$.

Theorem 5.2.8: C_1-Fields Have $\mathrm{cd} \leq 1$

Let K be a C_1-field. Then:

1. $\mathrm{Br}(K) = 0$.
2. $\mathrm{cd}(K) \leq 1$.

***Proof*:**

Part (1). By theorem 7.1.2 and proposition 7.1.7, it suffices to show that a finite-dimensional division algebra D with center K equals K. By proposition 7.2.9, $\dim_K D = d^2$ for some $d \geq 1$, and for a K-basis $e_1, \ldots, e_{d^2}$ of D there is a homogeneous polynomial $N_D \in K[X_1, \ldots, X_{d^2}]$ of degree d with $N_D(x) \neq 0$ whenever $\sum_i x_i e_i$ is invertible. Every nonzero element of D is invertible, so N_D has no nontrivial zero in K^{d^2}. If $d \geq 2$, then $d^2 > d$, and the C_1-property gives a nontrivial zero. Hence $d = 1$ and $D = K$.

Part (2). For p equal to the characteristic of K, every field of characteristic p has $\mathrm{cd}_p \leq 1$ [30, Ch. II, §2.2, Prop. 3]. For a prime p different from the characteristic, corollary 7.2.10 reduces $\mathrm{cd}_p(K) \leq 1$ to the vanishing of $\mathrm{Br}(L)\{p\}$ for every finite separable extension L/K. Such an L is a C_1-field by proposition 5.2.7(4), so $\mathrm{Br}(L) = 0$ by part (1). Hence $\mathrm{cd}_p(K) \leq 1$ for every prime p, and $\mathrm{cd}(K) \leq 1$.

The converse is false: Ax constructed a field of characteristic 0 with $\mathrm{cd} \leq 1$ that is not C_1 [30, Ch. II, §3.2, Exercise]. The C_r-property is a *geometric* condition on the field (it constrains the solution set of polynomial equations), while cohomological dimension is a *cohomological* condition on the Galois group. The implication $C_1 \Rightarrow \mathrm{cd} \leq 1$ shows that geometry constrains cohomology, but the two invariants are not equivalent.

Serre expects every C_r-field to have $\mathrm{cd} \leq r$, and the case $r = 2$ follows from a theorem of Merkurjev and Suslin [30, Ch. II, §4.5]. The case $r = 2$ is connected to the period-index problem for the Brauer group, discussed in section 7.4.

Serre's expectation holds for the C_r-fields that the theorem of Lang and Nagata produces from an algebraically closed field: an extension of transcendence degree r of an algebraically closed field has $\mathrm{cd} \leq r$ by theorem 5.2.10 below. The absolute Galois groups of a field and of a subfield are compared by restricting automorphisms to the elements separable over the subfield.

Proposition 5.2.9: Restriction of Absolute Galois Groups

Let L/K be a field extension, Ω a separable closure of L, and $K^{\text{sep}} \subseteq \Omega$ the set of elements of Ω separable over K. Write $G_L = \text{Gal}(\Omega/L)$ and $G_K = \text{Gal}(K^{\text{sep}}/K)$.

1. If K has characteristic $p > 0$, every element x of an extension of K that is algebraic over K has a power x^{p^m}, for some $m \geq 0$, separable over K.
2. The field K^{sep} is a separable closure of K, every $\sigma \in G_L$ satisfies $\sigma(K^{\text{sep}}) = K^{\text{sep}}$, and the restriction $\rho\colon G_L \to G_K$, $\sigma \mapsto \sigma|_{K^{\text{sep}}}$, is a continuous homomorphism.
3. If L/K is algebraic, then ρ is injective and an isomorphism of profinite groups from G_L onto the closed subgroup $\rho(G_L)$ of G_K. In particular $\text{cd}_p(L) \leq \text{cd}_p(K)$ for every prime p.
4. If K has characteristic $p > 0$ and every element of L has a p^m-th power in K for some $m \geq 0$, then ρ is an isomorphism of profinite groups, and $\text{cd}_\ell(L) = \text{cd}_\ell(K)$ for every prime ℓ.

***Proof*:**

1. Let $f \in K[X]$ be the minimal polynomial of x, and choose $m \geq 0$ as large as possible with $f(X) = g(X^{p^m})$ for some $g \in K[X]$. Such an m exists, as $m = 0$ qualifies and $p^m \leq \deg f$. A factorization of g into nonconstant factors would give one of f, so g is irreducible. If $g' = 0$, the coefficient of X^i in g vanishes whenever $p \nmid i$, so $g(X) = h(X^p)$ for some $h \in K[X]$ and $f(X) = h(X^{p^{m+1}})$, contradicting the choice of m. So $g' \neq 0$, and the irreducible polynomial g is separable by *Core Algebra* [9, Separability and irreducible polynomial]. Since $g(x^{p^m}) = f(x) = 0$, the minimal polynomial of x^{p^m} divides g, so it is separable.
2. The field Ω is the set of elements separable over L of an algebraic closure of L, so every separable polynomial over L splits in Ω, and Ω/L is Galois. For $x, y \in K^{\text{sep}}$ the finite extension $K(x, y)/K$ is generated by separable elements, so it is separable by *Core Algebra* [9, Finite Separable Extension Equivalence], and $x \pm y$, xy and, for $x \neq 0$, x^{-1} lie in K^{sep}. So K^{sep} is a subfield of Ω, algebraic and separable over K. A separable polynomial $g \in K[X]$ is separable as a polynomial over L, so its roots lie in Ω, and they are separable over K, so they lie in K^{sep}. Let C be an algebraic closure of K^{sep}, which is also an algebraic closure of K by *Core Algebra* [9, Composing Algebraic extensions]. An element of C separable over K is a root of its separable minimal polynomial g over K. The $\deg g$ distinct roots of g lie in K^{sep}, so g splits over K^{sep} and every root of g in C is one of them, and the element lies in K^{sep}. Hence K^{sep} is the set of elements of C separable over K, a separable closure of K.

   For $\sigma \in G_L$ and $x \in K^{\text{sep}}$, the element $\sigma(x)$ is a root of the minimal polynomial of x over K, since σ fixes $K \subseteq L$, so $\sigma(x) \in K^{\text{sep}}$. The same holds for σ^{-1}, so $\sigma(K^{\text{sep}}) = K^{\text{sep}}$, and ρ is a well-defined homomorphism. For continuity, let $E \subseteq K^{\text{sep}}$ be a finite Galois extension of K, say $E = K(\alpha_1, \ldots, \alpha_r)$. An element $\sigma \in G_L$ satisfies $\rho(\sigma) \in \text{Gal}(K^{\text{sep}}/E)$

exactly when it fixes $\alpha_1, \dots, \alpha_r$, so

$$\rho^{-1}\big(\mathrm{Gal}(K^{\mathrm{sep}}/E)\big) = \mathrm{Gal}\big(\Omega/L(\alpha_1, \dots, \alpha_r)\big)$$

which is open by theorem 1.4.1(2), as $L(\alpha_1, \dots, \alpha_r)/L$ is finite. By theorem 1.4.1(1), the cosets $\tau\mathrm{Gal}(K^{\mathrm{sep}}/E)$ form a neighbourhood basis of each $\tau \in G_K$, and ρ maps the open neighbourhood $\sigma\rho^{-1}(\mathrm{Gal}(K^{\mathrm{sep}}/E))$ of $\sigma \in G_L$ into $\rho(\sigma)\mathrm{Gal}(K^{\mathrm{sep}}/E)$. So ρ is continuous.

3. Let $\sigma \in G_L$ with $\rho(\sigma) = 1$, and let $x \in \Omega$. As x is algebraic over L and L/K is algebraic, x is algebraic over K by *Core Algebra* [9, Composing Algebraic extensions]. If K has characteristic 0, then x is separable over K by *Core Algebra* [9, Separability Over Characteristic 0], so $x \in K^{\mathrm{sep}}$ and $\sigma(x) = x$. If K has characteristic $p > 0$, part (1) gives $q = p^m$ with $x^q \in K^{\mathrm{sep}}$, and then $(\sigma(x) - x)^q = \sigma(x^q) - x^q = 0$, so again $\sigma(x) = x$. Hence $\sigma = 1$, and ρ is injective. The group G_L is compact and G_K is Hausdorff, so the image $\rho(G_L)$ is compact, hence closed, and the continuous bijection $\rho\colon G_L \to \rho(G_L)$ is a homeomorphism. By proposition 5.1.3(1), $\mathrm{cd}_p(L) = \mathrm{cd}_p(\rho(G_L)) \le \mathrm{cd}_p(G_K) = \mathrm{cd}_p(K)$, where $\mathrm{cd}_p(K)$ does not depend on the separable closure used to define it, as noted after definition 5.2.1.

4. The hypothesis makes L/K algebraic, so by part (3) it suffices to show that $\rho(G_L) = G_K$. Let $\tau \in G_K$ and let $E \subseteq K^{\mathrm{sep}}$ be a finite Galois extension of K. By *Core Algebra* [9, Primitive Element Theorem], $E = K(\alpha)$ for some α, and the minimal polynomial g of α over K splits in E as $g = \prod_i (X - \alpha_i)$ with distinct α_i. By unique factorization in $\Omega[X]$, a monic factor g_1 of g in $L[X]$ is the product of some of the $X - \alpha_i$, so its coefficients lie in $L \cap K^{\mathrm{sep}}$. An element $c \in L \cap K^{\mathrm{sep}}$ has $c^q \in K$ for some $q = p^m$, so its minimal polynomial over K divides $X^q - c^q = (X - c)^q$, and being separable it equals $X - c$, so $c \in K$. Hence $g_1 \in K[X]$. Division with remainder by g_1 gives the same quotient in $K[X]$ as in $L[X]$, so g_1 divides g in $K[X]$, and $g_1 \in \{1, g\}$ because g is irreducible over K. So g is irreducible over L, and since $\tau(\alpha)$ is a root of g in Ω, the assignment $\alpha \mapsto \tau(\alpha)$ defines an L-embedding $L(\alpha) \cong L[X]/(g) \to \Omega$. By theorem 1.4.1(3), applied to the Galois extension Ω/L and its intermediate field $L(\alpha)$, finite over L, this embedding is the restriction of some $\sigma \in G_L$. Then $\rho(\sigma)$ and τ agree on K and on α, hence on E, so $\rho(\sigma) \in \tau\mathrm{Gal}(K^{\mathrm{sep}}/E)$. These cosets form a neighbourhood basis of τ (theorem 1.4.1(1)), so τ lies in the closure of $\rho(G_L)$, which is $\rho(G_L)$ by part (3). Hence ρ is surjective, and by part (3) it is an isomorphism of profinite groups, so $\mathrm{cd}_\ell(L) = \mathrm{cd}_\ell(K)$ for every prime ℓ, completing the proof.

For a field k and an element t transcendental over k, restriction maps $G_{k(t)}$ to G_k with kernel the absolute Galois group of $k^{\mathrm{sep}}(t)$. Tsen's theorem (proposition 5.2.7(3)) and theorem 5.2.8 bound the cohomological dimension of $\overline{k}(t)$ by 1. When k is imperfect of characteristic p, the field $\overline{k}(t)$ is a proper extension of $k^{\mathrm{sep}}(t)$ in which every element has a p^m-th power in $k^{\mathrm{sep}}(t)$, and it has the same absolute Galois group by proposition 5.2.9(4).

Theorem 5.2.10: cd_p Grows by at Most the Transcendence Degree

Let K/k be a field extension of finite transcendence degree $n = \mathrm{trdeg}_k K$. Then for every prime p,

$$\mathrm{cd}_p(K) \le \mathrm{cd}_p(k) + n$$

with the convention $\infty + n = \infty$.

***Proof*:**
Step 1: Reduction to one variable. Let $\{t_1, \dots, t_n\}$ be a transcendence basis of K/k (*Core Algebra* [9, Transcendence Basis, Transcendence Degree]), so that $t_1, \dots, t_n$ are algebraically independent over k and K is algebraic over $k(t_1, \dots, t_n)$. Put $k_0 = k$ and $k_i = k_{i-1}(t_i)$ for $1 \leq i \leq n$. Each t_i is transcendental over k_{i-1}, since clearing denominators in a nonzero polynomial over k_{i-1} vanishing at t_i gives a nonzero polynomial over k vanishing at $(t_1, \dots, t_i)$. As K/k_n is algebraic, $\mathrm{cd}_p(K) \leq \mathrm{cd}_p(k_n)$ by proposition 5.2.9(3). It therefore suffices to show that

$$\mathrm{cd}_p(F(t)) \leq \mathrm{cd}_p(F) + 1$$

for every field F and every t transcendental over F. Applied to $F = k_{i-1}$ and $t = t_i$ for $i = 1, \dots, n$, this gives $\mathrm{cd}_p(k_n) \leq \mathrm{cd}_p(k) + n$.

Step 2: Restriction to the constants. Let F be a field, t an element transcendental over F, and C an algebraic closure of $F(t)$. Let $\Omega \subseteq C$ be the set of elements separable over $F(t)$, a separable closure of $F(t)$, so that $G_{F(t)} = \mathrm{Gal}(\Omega/F(t))$, and let $F^{\mathrm{sep}} \subseteq \Omega$ be the set of elements of Ω separable over F. By proposition 5.2.9(2), applied to the extension $F(t)/F$, the field F^{sep} is a separable closure of F and restriction is a continuous homomorphism $\rho\colon G_{F(t)} \to G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$. Let $M = F^{\mathrm{sep}}(t) \subseteq \Omega$. An element of $G_{F(t)}$ fixes t, so it fixes F^{sep} pointwise if and only if it fixes M pointwise, and

$$N := \ker \rho = \mathrm{Gal}(\Omega/M)$$

This is a closed subgroup of $G_{F(t)}$ by theorem 1.4.1(2), and normal as a kernel. The induced homomorphism $\bar{\rho}\colon G_{F(t)}/N \to G_F$ is injective and continuous for the quotient topology. Its source is compact and G_F is Hausdorff, so its image $\rho(G_{F(t)})$ is closed and $\bar{\rho}$ is a homeomorphism onto it. By proposition 5.1.3(1),

$$\mathrm{cd}_p(G_{F(t)}/N) = \mathrm{cd}_p\big(\rho(G_{F(t)})\big) \leq \mathrm{cd}_p(F)$$

Step 3: The kernel has cohomological dimension at most 1. Let $\overline{F} \subseteq C$ be the set of elements algebraic over F, and $\overline{M} = \overline{F}(t) \subseteq C$. For $x, y \in \overline{F}$ the extension $F(x, y)/F$ is finite, hence algebraic, so $\overline{F}$ is a subfield of C. A root in C of a nonconstant polynomial over $\overline{F}$ is algebraic over F by *Core Algebra* [9, Composing Algebraic extensions], so $\overline{F}$ is algebraically closed. The same result shows that t, being transcendental over F, is transcendental over $\overline{F}$. So $\overline{M}$ is an extension of transcendence degree 1 of the algebraically closed field $\overline{F}$, hence a C_1-field by proposition 5.2.7(3), and $\mathrm{cd}(\overline{M}) \leq 1$ by theorem 5.2.8(2).

It remains to identify N with the absolute Galois group of $\overline{M}$. The field C is algebraically closed and algebraic over $F(t)$, so it is an algebraic closure of M and of $\overline{M}$. The set Ω consists of the elements of C separable over M. An element of Ω is separable over M, as its minimal polynomial over M divides the one over $F(t)$. Conversely, let $x \in C$ be separable over M. The extension $M(x)/M$ is separable by *Core Algebra* [9, Finite Separable Extension Equivalence], and so is $M/F(t)$: each element of M lies in $F(t)(a_1, \dots, a_r)$ for finitely many $a_i \in F^{\mathrm{sep}}$, which are separable over $F(t)$, so that $F(t)(a_1, \dots, a_r)/F(t)$ is separable by the same result. So $M(x)/F(t)$ is separable by *Core Algebra* [9, Tower Law For Separable Extensions], and $x \in \Omega$. Let $\Omega' \subseteq C$ be the set of elements separable over $\overline{M}$, a separable closure of $\overline{M}$. Every element of C separable over M is separable over $\overline{M}$, so Ω is also the set of elements of Ω' separable over M.

An element $a \in \overline{F}$ separable over F is separable over $F(t)$, so it lies in Ω, and hence in F^{sep}. If F has characteristic 0, every element of $\overline{F}$ is separable over F by *Core Algebra* [9, Separability Over Characteristic 0], so $\overline{F} = F^{\text{sep}}$ and $\overline{M} = M$. Then $\Omega' = \Omega$, and $N = \text{Gal}(\Omega/M)$ is the absolute Galois group of $\overline{M}$. If F has characteristic $p > 0$, each $a \in \overline{F}$ has a power a^{p^m} separable over F by proposition 5.2.9(1), which lies in F^{sep}. An element $h = P/Q$ of $\overline{M}$, with $P, Q \in \overline{F}[t]$, then has a power in M: choose $q = p^m$ such that the q-th powers of the finitely many coefficients of P and Q lie in F^{sep}. Since $\left(\sum_i a_i t^i\right)^q = \sum_i a_i^q t^{iq}$ in characteristic p, the element $h^q = P^q/Q^q$ lies in M. By proposition 5.2.9(4), applied to the extension $\overline{M}/M$ with the separable closure Ω' of $\overline{M}$, in which the elements separable over M form Ω, restriction is an isomorphism of profinite groups

$$G_{\overline{M}} = \text{Gal}(\Omega'/\overline{M}) \longrightarrow \text{Gal}(\Omega/M) = N$$

In both cases $\text{cd}_p(N) = \text{cd}_p(\overline{M}) \leq 1$.

Step 4: Conclusion. By proposition 5.1.3(2), applied to the closed normal subgroup N of $G_{F(t)}$, and by Steps 2 and 3,

$$\text{cd}_p(F(t)) \leq \text{cd}_p(N) + \text{cd}_p(G_{F(t)}/N) \leq 1 + \text{cd}_p(F)$$

which by Step 1 completes the proof.

A field K of transcendence degree n over a separably closed field k therefore has $\text{cd}(K) \leq n$, since $\text{cd}(k) = 0$ (proposition 5.2.2). For $n = 1$ and k algebraically closed the bound is attained by the function field of a curve, which has $\text{cd} = 1$ (example 5.2.4). For p equal to the characteristic of k and $n \geq 1$, the bound $\text{cd}_p(k) + n$ is at least 1, while $\text{cd}_p(K) \leq 1$ holds for every field of characteristic p [30, Ch. II, §2.2, Prop. 3], a bound the proof uses through theorem 5.2.8(2).

The real numbers provide an instructive example of how orderings affect cohomological dimension.

Example 5.2.11: The Real Numbers and Orderings

The absolute Galois group $G_{\mathbb{R}} = \text{Gal}(\mathbb{C}/\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ is a finite group of order 2. For any finite group G and any prime p dividing $|G|$, the periodic cohomology of cyclic subgroups (theorem 4.3.4) gives $H^n(G, \mathbb{Z}/p\mathbb{Z}) \neq 0$ for infinitely many n, so $\text{cd}_p(G) = \infty$. In particular:

1. $\text{cd}_2(\mathbb{R}) = \infty$, since $H^n(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ for all $n \geq 0$.
2. $\text{cd}_p(\mathbb{R}) = 0$ for p odd, since the order 2 of $G_{\mathbb{R}} = \mathbb{Z}/2\mathbb{Z}$ is prime to p (proposition 5.1.4).

The infinite 2-cohomological dimension is a consequence of the ordering on $\mathbb{R}$: the nontrivial element of $G_{\mathbb{R}}$ is complex conjugation, which preserves the ordering, and the resulting periodicity in cohomology prevents the cohomology from vanishing in high degrees. This phenomenon extends to any formally real field K (one admitting an ordering): if K has a real closure K^{rc}, then $\text{Gal}(K^{\text{rc}}/K)$ contains $\mathbb{Z}/2\mathbb{Z}$, and $\text{cd}_2(K) = \infty$.

For number fields: if K is a totally imaginary number field (no real embeddings), then the obstruction disappears, and $\text{cd}_2(K) = 2$.

The dichotomy between $\text{cd}_2 = \infty$ (formally real fields) and $\text{cd}_2 < \infty$ (fields with no orderings) motivates the notion of *virtual cohomological dimension*.

Definition 5.2.12: Virtual Cohomological Dimension

Let G be a profinite group. The *virtual p-cohomological dimension* of G is

$$\operatorname{vcd}_p(G) = \inf\{\operatorname{cd}_p(H) \mid H \leq G \text{ open, torsion-free}\}$$

when a torsion-free open subgroup exists. For a field K, $\operatorname{vcd}_p(K) = \operatorname{vcd}_p(G_K)$.

When G has no torsion elements, $\operatorname{vcd}_p(G) = \operatorname{cd}_p(G)$. For fields: a number field K with real places has $\operatorname{cd}_2(K) = \infty$, but $\operatorname{vcd}_2(K) = 2$ (the open subgroup $G_{K(\sqrt{-1})}$ corresponds to the totally imaginary field $K(\sqrt{-1})$, which has $\operatorname{cd}_2 = 2$). The virtual cohomological dimension captures the "true" complexity of the Galois group, discounting the contribution of finite torsion subgroups.

The following table summarizes the cohomological dimensions computed or stated in this section, for reference throughout the rest of the book.

Field K	$\operatorname{cd}_p(K)$ (p odd)	$\operatorname{cd}_2(K)$
Separably closed	0	0
$\mathbb{F}_q$ (finite)	1	1
$k(C)$, $k = \overline{k}$	1	1
$\operatorname{trdeg}_k K = n$, $k = k^{\mathrm{sep}}$	$\leq n$	$\leq n$
$\mathbb{Q}_p$, p odd	2	2
K number field, totally imaginary	2	2
K number field, real places	2	∞ ($\operatorname{vcd}_2 = 2$)
$\mathbb{R}$	0	∞

The entry for $\mathbb{Q}_p$ is corollary 8.3.3, with its proof cited. For number fields $\operatorname{cd}_p \leq 2$ when p is odd or K is totally imaginary [30, Ch. II, §4.4, Prop. 13], with equality because $H^2(G_K, \mu_p) \cong \operatorname{Br}(K)[p]$ is nonzero (corollary 10.1.3). The others have been established above. The pattern $\operatorname{cd} = 0, 1, 2$ for separably closed, finite, and local fields reflects the increasing arithmetic complexity of these fields, and $\operatorname{cd} = 2$ is the threshold at which class field theory operates.

Exercise 5.2.1

1. Let L/K be an algebraic extension of fields, so that $\operatorname{cd}_p(L) \leq \operatorname{cd}_p(K)$ by proposition 5.2.9(3). When is equality guaranteed? (*Hint:* Consider separately the cases where $[L : K]$ is finite or infinite, and whether p divides $[L : K]$.)

2. Let K be a field of characteristic 0 with $\operatorname{trdeg}_{\mathbb{Q}} K = n$ finite. Show that $\operatorname{cd}_p(K) \leq n + 2$ for every odd prime p, that the absolute Galois group of $K(\sqrt{-1})$ is torsion-free, and that $\operatorname{vcd}_2(K) \leq n+2$. (*Hint:* Apply theorem 5.2.10 over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{-1})$, with the number-field entries of the table, and use example 5.2.11 for the torsion.)

3. Let K be a formally real field (i.e., -1 is not a sum of squares in K). Show that $\operatorname{cd}_2(K) = \infty$. (*Hint:* K has a real closure K^{rc} with $\operatorname{Gal}(K^{\mathrm{rc}}/K) \cong \mathbb{Z}/2\mathbb{Z}$. Use the periodic cohomology of $\mathbb{Z}/2\mathbb{Z}$.)

4. (If the reader has read [6]) Let $K = \mathbb{Q}(\mu_{p^\infty})$ be the field obtained by adjoining all p-power roots of unity to $\mathbb{Q}$ (p an odd prime). Show that $\operatorname{cd}_p(K) \leq 1$. (*Hint:* By corollary 7.2.10 it suffices to show $\operatorname{Br}(L)\{p\} = 0$ for every finite extension L/K. Bring a class of $\operatorname{Br}(L)$ down to

a number field, and combine the Hasse principle (theorem 7.3.4) with the fact that the local degrees of $\mathbb{Q}(\mu_{p^m})$ over $\mathbb{Q}$ are divisible by arbitrarily large powers of p as m grows.)

5. Assuming the result $\mathrm{cd}(\mathbb{Q}_p) = 2$ (corollary 8.3.3), compute $\mathrm{cd}_p(L)$ and $\mathrm{cd}_\ell(L)$ (for $\ell \neq p$) for a finite extension $L/\mathbb{Q}_p$. Use exercise 5.1.1 (3) and the structure of $G_{\mathbb{Q}_p}$.

5.3 Duality Groups and Poincaré Duality Groups

The finite-group Tate duality of theorem 4.4.3 paired $\hat{H}^p(G, M)$ with $\hat{H}^{-p-1}(G, M^*)$ for cyclic groups G acting on finite modules. For profinite groups of finite cohomological dimension, an analogous duality can hold: the cohomology in degree i is paired with the cohomology in the complementary degree $d - i$, where $d = \mathrm{cd}_p(G)$. Profinite groups admitting such a pairing are called *duality groups*, and those where the pairing takes the simplest possible form are *Poincaré duality groups*. These notions are the algebraic analogues of the topological Poincaré duality for closed manifolds, and they organize the profinite groups of arithmetic interest (absolute Galois groups of local fields, fundamental groups of arithmetic surfaces) into a structural hierarchy.

Definition 5.3.1: Duality Group

Let G be a profinite group, p a prime, and $d = \mathrm{cd}_p(G) < \infty$. The group G is a *duality group of dimension d at p* if there exists a discrete G-module D (the *dualizing module*) such that for every finite p-torsion discrete G-module M and every $0 \leq i \leq d$, the cup product pairing

$$H^i(G, M) \times H^{d-i}(G, \mathrm{Hom}(M, D)) \to H^d(G, D) \xrightarrow{\theta} \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing of finite groups, where $\theta : H^d(G, D) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is a fixed isomorphism (the *trace map* or *fundamental class*).

The dualizing module D and the trace map θ are part of the structure: a duality group is a triple (G, D, θ). The pairing sends $(\alpha, \beta) \in H^i(G, M) \times H^{d-i}(G, \mathrm{Hom}(M, D))$ to $\theta(\alpha \smile \mathrm{ev}_*(\beta))$, where ev_* is induced by the evaluation $M \otimes \mathrm{Hom}(M, D) \to D$. Perfectness means that the induced maps $H^i(G, M) \to H^{d-i}(G, \mathrm{Hom}(M, D))^*$ and $H^{d-i}(G, \mathrm{Hom}(M, D)) \to H^i(G, M)^*$ are both isomorphisms (where $(-)^* = \mathrm{Hom}(-, \mathbb{Q}_p/\mathbb{Z}_p)$ is the Pontryagin dual).

The dualizing module is unique when it exists: if (D, θ) and (D', θ') both make G a duality group of the same dimension, then $D \cong D'$ as G-modules (the isomorphism is determined by $\theta' \circ \theta^{-1}$). This follows from the Yoneda lemma applied to the functor $M \mapsto H^d(G, \mathrm{Hom}(M, D))^*$, which is representable by the dualizing module.

The duality pairing has a transparent interpretation at the extremes of the degree range. At $i = 0$: the pairing $H^0(G, M) \times H^d(G, \mathrm{Hom}(M, D)) \to \mathbb{Q}_p/\mathbb{Z}_p$ says that the G-invariants of M are dual to the top cohomology with dual coefficients. At $i = d$: the pairing $H^d(G, M) \times H^0(G, \mathrm{Hom}(M, D)) \to \mathbb{Q}_p/\mathbb{Z}_p$ says that the top cohomology of M is dual to the G-invariants of $\mathrm{Hom}(M, D)$. These two extremes are the "easy" cases. The content of the duality theorem is that the pairing is perfect in all intermediate degrees as well.

A consequence of duality: for a duality group G of dimension d and a finite p-torsion module M, the groups $H^i(G, M)$ and $H^{d-i}(G, \mathrm{Hom}(M, D))$ have the same order. The alternating product

$\prod_{i=0}^{d} |H^i(G,M)|^{(-1)^i}$, the *Euler-Poincaré characteristic* $\chi(G,M)$, therefore satisfies $\chi(G,M) = \chi(G,\mathrm{Hom}(M,D))^{(-1)^d}$, a symmetry that constrains the possible sizes of cohomology groups. The Euler-Poincaré characteristic will be computed for local and global fields in chapters 8 and 10.

The simplest case of a duality group is when the dualizing module is as small as possible.

Definition 5.3.2: Poincaré Duality Group

A profinite group G is a *Poincaré duality group of dimension d at p* (or a *PD_d-group at p*) if it is a duality group of dimension d at p with dualizing module $D \cong \mathbb{Q}_p/\mathbb{Z}_p$ (equipped with some continuous G-action). Equivalently, G is a PD_d-group at p if $\mathrm{cd}_p(G) = d$ and the functor $M \mapsto H^d(G,M)^*$ is naturally isomorphic to $H^0(G,\mathrm{Hom}(M,\mathbb{Q}_p/\mathbb{Z}_p))$ on finite p-torsion G-modules.

The G-action on $D = \mathbb{Q}_p/\mathbb{Z}_p$ is called the *orientation character*: it is a continuous homomorphism $\omega : G \to \mathbb{Z}_p^\times$, with g acting on $\mathbb{Q}_p/\mathbb{Z}_p$ by multiplication by $\omega(g)$. When ω is trivial (G acts trivially on D), the group G is called an *orientable* Poincaré duality group. The terminology comes from topology: for the pro-p-completion of the fundamental group of a closed manifold, the orientation character is trivial if and only if the manifold is orientable. In the arithmetic setting, the orientation character of G_K (for a local field K) is the cyclotomic character $\chi_p : G_K \to \mathbb{Z}_p^\times$, which is always nontrivial (since K does not contain all p-power roots of unity). Hence absolute Galois groups of local fields are *non-orientable* duality groups.

Example 5.3.3: Poincaré Duality for $\mathbb{Z}_p^d$

1. The group $G = \mathbb{Z}_p$ is a PD_1-group at p. The cohomological dimension $\mathrm{cd}_p(\mathbb{Z}_p) = 1$ was established in example 5.1.5. The dualizing module is $D = \mathbb{Q}_p/\mathbb{Z}_p$ with trivial action, and the trace map $\theta : H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is the isomorphism $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$ (a continuous homomorphism $\mathbb{Z}_p \to \mathbb{Q}_p/\mathbb{Z}_p$ is determined by the image of 1). The duality pairing in the only nontrivial case ($i = 0$, $d - i = 1$):

$$H^0(\mathbb{Z}_p, M) \times H^1(\mathbb{Z}_p, \mathrm{Hom}(M, \mathbb{Q}_p/\mathbb{Z}_p)) \to H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$$

   pairs $M^{\mathbb{Z}_p}$ with $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, M^*) \cong M^*$ (using $H^1 = \mathrm{Hom}_{\mathrm{cts}}$ for trivial action and the structure of M^*). Since M is finite, $M^{\mathbb{Z}_p} = M$ (because $\mathbb{Z}_p$ is pro-p and M is p-torsion, the action factors through a finite quotient $\mathbb{Z}/p^k\mathbb{Z}$; for the trivial action, $M^G = M$). The pairing is the natural evaluation $M \times M^* \to \mathbb{Q}_p/\mathbb{Z}_p$, which is perfect.

   For a nontrivial action: if $\mathbb{Z}_p$ acts on $M = \mathbb{Z}/p\mathbb{Z}$ through the quotient $\mathbb{Z}/p\mathbb{Z}$ by a nontrivial character, then $M^{\mathbb{Z}_p} = 0$ and $H^1(\mathbb{Z}_p, M^*) = \mathrm{Hom}_{\mathrm{cts}}(\mathbb{Z}_p, M^*)/(\text{conjugation})$. Both sides of the pairing are zero, so the pairing is vacuously perfect.

2. More generally, $G = \mathbb{Z}_p^d$ is a PD_d-group at p (orientable, since the action on D is trivial). The proof uses the Künneth formula: $H^d(\mathbb{Z}_p^d, \mathbb{Q}_p/\mathbb{Z}_p) \cong \bigotimes_{i=1}^{d} H^1(\mathbb{Z}_p, \mathbb{Q}_p/\mathbb{Z}_p) \cong (\mathbb{Q}_p/\mathbb{Z}_p)^{\otimes d}$. Since $\mathbb{Q}_p/\mathbb{Z}_p \otimes \mathbb{Q}_p/\mathbb{Z}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$ (as the tensor product of injective $\mathbb{Z}_p$-modules), this gives $H^d(\mathbb{Z}_p^d, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathbb{Q}_p/\mathbb{Z}_p$, providing the trace map. The cup product pairing reduces to the exterior algebra structure: $H^i(\mathbb{Z}_p^d, M)$ pairs with $H^{d-i}(\mathbb{Z}_p^d, M^*)$ via the wedge product on the exterior algebra $\bigwedge^\bullet (\mathbb{Z}_p^d)^*$, and this pairing is perfect.

3. (If the reader has read [23]) The PD_d-property of $\mathbb{Z}_p^d$ is the profinite analogue of Poincaré duality for the d-torus $T^d = (\mathbb{R}/\mathbb{Z})^d$: the classifying space $B\mathbb{Z}_p^d$ is the p-completion of T^d, and Poincaré duality for T^d (a closed orientable manifold of dimension d) translates into the algebraic duality for $\mathbb{Z}_p^d$. The connection extends: if X is a closed orientable aspherical manifold of dimension d (meaning X is a $K(\pi, 1)$ with $\pi = \pi_1(X)$), then the pro-p-completion of π is a PD_d-group at p.

Poincaré duality groups can be characterized without explicit reference to the dualizing module, using only the cohomology of the group ring.

Proposition 5.3.4: Characterization of PD_d-Groups

Let G be a pro-p-group with $\mathrm{cd}_p(G) = d < \infty$. Then G is a PD_d-group at p if and only if

$$H^i(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \cong \begin{cases} \mathbb{Z}/p\mathbb{Z} & i = d, \\ 0 & i \neq d, \end{cases}$$

where $\mathbb{Z}/p\mathbb{Z}[[G]] = \varprojlim_U \mathbb{Z}/p\mathbb{Z}[G/U]$ is the completed group ring.

***Proof Sketch*:**
The completed group ring $\mathbb{Z}/p\mathbb{Z}[[G]]$ plays the role of the "chains on the universal cover" in the topological setting. For a closed manifold X with fundamental group π, Poincaré duality is equivalent to the statement that the cellular chain complex of the universal cover $\tilde{X}$ has homology concentrated in degree d (this is the "fundamental class"). The algebraic analogue: the condition $H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \cong \mathbb{Z}/p\mathbb{Z}$ says that the top cohomology of G with "universal" coefficients is one-dimensional, and the vanishing $H^i(G, \mathbb{Z}/p\mathbb{Z}[[G]]) = 0$ for $i \neq d$ says that the universal coefficients contribute nothing in other degrees.

The dualizing module is recovered as $D = H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]])^* \otimes_{\mathbb{Z}/p\mathbb{Z}} \mathbb{Q}_p/\mathbb{Z}_p$ with the induced G-action. The G-action on $H^d(G, \mathbb{Z}/p\mathbb{Z}[[G]])$ (which is $\mathbb{Z}/p\mathbb{Z}$ as an abelian group but carries a G-action through the coefficients) determines the orientation character. The cup product pairing is constructed from the cap product on $\mathbb{Z}/p\mathbb{Z}[[G]]$: the cap product $H^i(G, M) \otimes H_d(G, \mathbb{Z}/p\mathbb{Z}[[G]]) \to H_{d-i}(G, M)$ composed with the identification $H_{d-i}(G, M) \cong H^{d-i}(G, \mathrm{Hom}(M, D))^*$ (via the universal coefficient theorem) produces the duality pairing.

The most important class of Poincaré duality groups in arithmetic are the *Demushkin groups*: pro-p-groups of $\mathrm{cd}_p = 2$ satisfying PD_2-duality. These groups are the pro-p-completions of the absolute Galois groups of p-adic fields, and their classification is one of the main results of chapter 6.

Definition 5.3.5: Demushkin Group

A *Demushkin group* is a pro-p-group G satisfying:

1. $\mathrm{cd}_p(G) = 2$.
2. $H^1(G, \mathbb{Z}/p\mathbb{Z})$ is finite (i.e., G is topologically finitely generated).
3. $H^2(G, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$.

Equivalently (by proposition 5.3.4), a Demushkin group is a topologically finitely generated pro-p-PD_2-group.

The conditions are stringent: condition (3) says that the "second Betti number" of G is exactly 1, which constrains the number of relations in any presentation of G as a pro-p-group. If $\dim_{\mathbb{Z}/p\mathbb{Z}} H^1(G, \mathbb{Z}/p\mathbb{Z}) = d$ (so G requires d generators), then the cup product pairing $H^1(G, \mathbb{Z}/p\mathbb{Z}) \times H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ is a nondegenerate bilinear form on $(\mathbb{Z}/p\mathbb{Z})^d$. For p odd, graded-commutativity forces this form to be alternating ($\alpha \smile \alpha = 0$), so nondegeneracy requires d to be even. For $p = 2$, the form need not be alternating (the cup product $\alpha \smile \alpha$ can be nonzero mod 2), allowing odd d.

Demushkin groups were classified by Demushkin (for p odd) and by Serre and Labute (for $p = 2$). The classification, which will be developed in section 6.4, shows that a Demushkin group is determined up to isomorphism by the pair (d, q), where $d = \dim H^1(G, \mathbb{Z}/p\mathbb{Z})$ is the number of generators and q is a power of p (or $q = 0$) encoding the structure of the cup product pairing. The group is presented by d generators $x_1, \ldots, x_d$ subject to a single relation r whose form depends on (d, q).

Example 5.3.6: Demushkin Groups in Arithmetic

1. Let K be a finite extension of $\mathbb{Q}_p$ with p odd. The maximal pro-p-quotient $G_K(p)$ of the absolute Galois group G_K is a Demushkin group. The number of generators is $d = \dim H^1(G_K(p), \mathbb{Z}/p\mathbb{Z}) = [K : \mathbb{Q}_p] + 1 + \delta$, where $\delta = 1$ if K contains a primitive p-th root of unity and $\delta = 0$ otherwise. Since $[K : \mathbb{Q}_p] \geq 1$ and p is odd, the parity constraint forces d to be even. For example: $K = \mathbb{Q}_p$ gives $d = 2$ (when $p \geq 3$ and $\zeta_p \notin \mathbb{Q}_p$), and $G_{\mathbb{Q}_p}(p) \cong \mathbb{Z}_p^2$ (the unique Demushkin group on 2 generators with $q = p$). The cup product pairing on H^1 is related to the Hilbert symbol of local class field theory, and the classification of Demushkin groups determines the pro-p-Galois structure of p-adic fields. The full classification is in section 6.4.

2. The dualizing module for $G_{\mathbb{Q}_p}$ (as a duality group, not just its pro-p-quotient) is $D = \mu_{p^\infty} = \varinjlim_k \mu_{p^k}$, the p-power roots of unity in $\overline{\mathbb{Q}}_p^\times$. The $G_{\mathbb{Q}_p}$-action on μ_{p^∞} is through the p-adic cyclotomic character $\chi_p : G_{\mathbb{Q}_p} \to \mathbb{Z}_p^\times$, and the trace map $\theta : H^2(G_{\mathbb{Q}_p}, \mu_{p^\infty}) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ is the invariant map of local class field theory. The proof that $G_{\mathbb{Q}_p}$ is a duality group of dimension 2 with this dualizing module is the content of local Tate duality (chapter 8).

3. Not every pro-p-group of $\mathrm{cd}_p = 2$ is Demushkin: the Demushkin condition $\dim H^2 = 1$

is restrictive. The group $\mathbb{Z}_p^2$ has $\mathrm{cd}_p = 2$ (example 5.1.5) and $H^2(\mathbb{Z}_p^2, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ (the exterior square of the 2-dimensional H^1), so $\mathbb{Z}_p^2$ is Demushkin with $d = 2$. On the other hand, if G is the pro-p-group with three generators x_1, x_2, x_3 and two independent relations, then $\dim H^2(G, \mathbb{Z}/p\mathbb{Z}) = 2 \neq 1$, so G is not Demushkin despite having $\mathrm{cd}_p = 2$ (assuming the relations are chosen so that G is infinite). Such groups arise as Galois groups of maximal p-extensions of global fields, where $\dim H^2$ can be larger than 1.

The hierarchy of profinite groups studied in this chapter mirrors the hierarchy of fields in section 5.2. Its levels are the groups of $\mathrm{cd}_p = 0$ (pro-p'-groups), $\mathrm{cd}_p = 1$ (free pro-p-groups), $\mathrm{cd}_p = 2$ (Demushkin and more general PD_2-groups), and higher. The abstract characterizations ($\mathrm{cd}_p = 1$ iff free, PD_2 iff Demushkin with one relation) transform cohomological computations into structural conclusions about the group. The structure theory of pro-p-groups in chapter 6 exploits this hierarchy systematically, and the connection between the abstract duality theory of this section and the arithmetic duality theorems of chapters 8 and 10 is a central theme of Part III.

Exercise 5.3.1

1. Let G be a duality group of dimension d at p with dualizing modules (D, θ) and (D', θ'). Show that $D \cong D'$ as G-modules. (*Hint:* Consider the functor $M \mapsto H^d(G, M)^*$ on finite p-torsion G-modules and apply the representability criterion: this functor is represented by D via the cup product pairing.)

2. For $G = \mathbb{Z}_p^2$, write out the cup product pairing $H^1(G, \mathbb{Z}/p\mathbb{Z}) \times H^1(G, \mathbb{Z}/p\mathbb{Z}) \to H^2(G, \mathbb{Z}/p\mathbb{Z})$ explicitly. Show that $H^1 \cong (\mathbb{Z}/p\mathbb{Z})^2$ (with generators χ_1, χ_2 dual to the standard generators of $\mathbb{Z}_p^2$), $H^2 \cong \mathbb{Z}/p\mathbb{Z}$ (generated by $\chi_1 \smile \chi_2$), and that the pairing is the standard symplectic form $(\chi_i, \chi_j) \mapsto \chi_i \smile \chi_j$, with $\chi_1 \smile \chi_2 = -\chi_2 \smile \chi_1$ being the unique generator.

3. Let G be a Demushkin group with p odd and $d = \dim H^1(G, \mathbb{Z}/p\mathbb{Z})$. Show that d is even. (*Hint:* The cup product pairing $H^1 \times H^1 \to H^2 \cong \mathbb{Z}/p\mathbb{Z}$ is a nondegenerate alternating form for p odd, by graded-commutativity. An alternating form on a finite-dimensional vector space over a field of odd characteristic is nondegenerate only if the dimension is even.)

4. Let G be a duality group of dimension d at p with dualizing module D, and let $H \leq G$ be an open subgroup. Show that H is also a duality group of dimension d at p, with the same dualizing module D (restricted to H). (*Hint:* Use the Shapiro isomorphism and the compatibility of the cup product with restriction.)

5. (If the reader has read [13]) Let G be a pro-p-group with $\mathrm{cd}_p(G) = 2$ and $H^1(G, \mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/p\mathbb{Z})^d$ for some finite d. Show that G is Demushkin if and only if G has a single defining relation in its minimal pro-p-presentation (i.e., $G \cong F/\overline{\langle r \rangle}$ where F is a free pro-p-group on d generators and $r \in F$). (*Hint:* The number of relations equals $\dim H^2(G, \mathbb{Z}/p\mathbb{Z})$.)

6

Pro-p-Groups and the Golod-Shafarevich Inequality

Chapter 5 introduced cohomological dimension as the fundamental invariant of a profinite group's cohomology. The groups of p-cohomological dimension 0 are those whose order is prime to p (proposition 5.1.4), and among pro-p-groups those of cohomological dimension 1 turn out to be precisely the free ones. This chapter develops the structure theory of pro-p-groups through the lens of cohomological dimension: the minimal number of generators and relations of a pro-p-group are read off from H^1 and H^2, and the Golod-Shafarevich inequality provides a lower bound on the relation count that resolved the class field tower problem. The chapter closes with the classification of Demushkin groups (the pro-p-groups of cohomological dimension 2 satisfying Poincaré duality), which arise in both topology and arithmetic.

6.1 Free Profinite Groups

Among all pro-p-groups, the free ones are the simplest: they admit no nontrivial relations. The cohomological characterization is sharp: a pro-p-group is free if and only if its p-cohomological dimension is at most 1 (theorem 6.1.4 below). This is the pro-p analogue of the Stallings-Swan theorem for discrete groups, and it reduces a group-theoretic property (freeness) to a cohomological vanishing condition.

Definition 6.1.1: Free Profinite Group and Free Pro-p-Group

Let S be a set. The *free profinite group* on S is a profinite group $\hat{F}(S)$ together with a map $\iota\colon S \to \hat{F}(S)$ satisfying the following universal property: for every profinite group H and every map $\varphi\colon S \to H$, there exists a unique continuous homomorphism $\tilde{\varphi}\colon \hat{F}(S) \to H$ with $\tilde{\varphi} \circ \iota = \varphi$:

$$\begin{array}{ccc} S & \xrightarrow{\iota} & \hat{F}(S) \\ & \searrow^{\varphi} & \downarrow{\scriptstyle \exists!\, \tilde{\varphi}} \\ & & H \end{array}$$

The *free pro-p-group* on S, denoted $F_p(S)$ (or $F(d)$ when $|S| = d$ is finite), is defined by the same universal property with H restricted to pro-p-groups. Equivalently, $F_p(S)$ is the maximal pro-p quotient of $\hat{F}(S)$.

The free profinite group $\hat{F}(S)$ is the profinite completion of the discrete free group $F(S)$ on S (by proposition 1.1.9: the profinite completion satisfies the same universal property). The free pro-p-group $F_p(S)$ is the pro-p-completion: the limit of all finite p-group quotients of $F(S)$.

Proposition 6.1.2: Basic Properties of Free Pro-p-Groups

Let $F = F_p(S)$ be the free pro-p-group on a set S. Then:

1. The image $\iota(S)$ topologically generates F: the closure $\overline{\langle \iota(S) \rangle} = F$.
2. The Frattini quotient satisfies $F/\Phi(F) \cong \mathbb{F}_p^{|S|}$ (where $\Phi(F)$ is the Frattini subgroup from definition 1.3.9). In particular, $d(F) = |S|$ (the generator rank from proposition 1.3.10).
3. Every surjection $G \twoheadrightarrow F$ of pro-p-groups splits: there exists a continuous homomorphism $s\colon F \to G$ with $\pi \circ s = \mathrm{id}_F$.

***Proof*:**

For (1): by the universal property, any open normal subgroup $U \trianglelefteq F$ satisfies $F/U = \overline{\langle \iota(S) \rangle} U/U$, since the finite p-group F/U is generated by the images of $\iota(S)$. Taking the limit over all U gives $F = \overline{\langle \iota(S) \rangle}$ (by proposition 1.1.8).

For (2): the Burnside basis theorem (proposition 1.3.10) gives $d(F) = \dim_{\mathbb{F}_p} F/\Phi(F)$, and the generators $\iota(S)$ are a minimal topological generating set (any proper subset fails to generate: removing $s \in S$ and applying the universal property to the quotient $\mathbb{F}_p$ that kills s but not the others shows $\overline{\langle \iota(S \setminus \{s\}) \rangle} \neq F$).

For (3): given $\pi\colon G \twoheadrightarrow F$, the universal property provides a section: the inclusion $\iota\colon S \to F$ lifts to a map $S \to G$ (choosing arbitrary preimages), which extends to $s\colon F \to G$ by the universal property. Then $\pi \circ s$ and id_F agree on the generating set $\iota(S)$, hence are equal.

Property (3) says that free pro-p-groups are projective objects in the category of pro-p-groups: every

surjection onto F splits. The converse (that projective pro-p-groups are free) is also true, and combined with the cohomological characterization of proposition 5.1.6, gives the main theorem of this section.

The cohomology of free pro-p-groups is concentrated in degrees 0 and 1. This was established in proposition 5.1.6: $\mathrm{cd}_p(F_p(S)) \leq 1$ for any set S. The first cohomology group has a clean description.

Proposition 6.1.3: Cohomology of Free Pro-p-Groups

Let $F = F_p(S)$ be the free pro-p-group on a set S, and let M be a discrete p-torsion F-module. Then:

1. $H^1(F, \mathbb{F}_p) \cong \mathrm{Map}(S, \mathbb{F}_p)$ (continuous maps from S with the discrete topology to $\mathbb{F}_p$). When S is finite, $H^1(F, \mathbb{F}_p) \cong \mathbb{F}_p^{|S|}$.
2. $H^n(F, M) = 0$ for all $n \geq 2$.

***Proof*:**
Part (2) is proposition 5.1.6. For part (1), $\mathbb{F}_p$ carries the trivial action, so $H^1(F, \mathbb{F}_p) \cong \mathrm{Hom}_{\mathrm{cts}}(F, \mathbb{F}_p)$ by proposition 3.1.7. The finite group $\mathbb{F}_p$ is a pro-p-group, so by the universal property of F (definition 6.1.1) the map $\chi \mapsto \chi \circ \iota$ is a bijection $\mathrm{Hom}_{\mathrm{cts}}(F, \mathbb{F}_p) \to \mathrm{Map}(S, \mathbb{F}_p)$. It is additive, as both groups are added pointwise. So $H^1(F, \mathbb{F}_p) \cong \mathrm{Map}(S, \mathbb{F}_p)$.

The main theorem characterizes free pro-p-groups by the vanishing of H^2.

Theorem 6.1.4: Cohomological Characterization of Free Pro-p-Groups

A pro-p-group G is free if and only if $\mathrm{cd}_p(G) \leq 1$.

***Proof*:**
Step 1: Free implies $\mathrm{cd}_p \leq 1$. This is proposition 5.1.6.

Step 2: $\mathrm{cd}_p \leq 1$ *implies projective.* Suppose $\mathrm{cd}_p(G) \leq 1$, and let $\pi \colon H \twoheadrightarrow G$ be a surjection of pro-p-groups with kernel $N = \ker \pi$. The goal is to construct a continuous section $s \colon G \to H$. Since H is pro-p, it suffices to construct compatible sections $s_U \colon G/\pi(U) \to H/U$ for each open normal subgroup $U \trianglelefteq H$ contained in N (as the limit of these sections gives s).

Fix $U \trianglelefteq H$ open with $U \leq N$. The extension $1 \to N/U \to H/U \to G \to 1$ is classified by a class $\alpha \in H^2(G, N/U)$ (by theorem 2.4.5, applied to N/U as a G-module via conjugation). Since N/U is a finite p-group (as a quotient of the pro-p-group N by an open subgroup) and $\mathrm{cd}_p(G) \leq 1$, it follows that $H^2(G, N/U) = 0$ (by the definition of cd_p, since N/U is p-torsion). So $\alpha = 0$, meaning the extension splits, and a section s_U exists.

The set of sections $s_U \colon G \to H/U$ forms a cofiltered system of non-empty finite sets (for each $U' \leq U$, the projection $H/U' \to H/U$ sends a section for U' to a section for U). By lemma 1.2.6, the limit is non-empty, producing a continuous section $s \colon G \to H$.

Step 3: Projective implies free. Let G be a projective pro-p-group. Choose a surjection $\pi\colon F(d(G)) \twoheadrightarrow G$ from the free pro-p-group on $d(G)$ generators (such a surjection exists by proposition 1.3.10). Since G is projective, π splits: there exists $s\colon G \to F(d(G))$ with $\pi \circ s = \mathrm{id}$. Then s is injective and identifies G with a retract of $F(d(G))$. A retract of a free pro-p-group is free: let $R = \ker \pi$, so that $F(d(G)) = s(G) \ltimes R$ (semidirect product). Freeness of $G \cong s(G)$ follows from the Nielsen-Schreier property for pro-p-groups (see [29]), which states that subgroups of free pro-p-groups are free.

The theorem reduces a structural property (freeness) to a single cohomological condition ($H^2(G, M) = 0$ for all p-torsion M, which by Serre's criterion (theorem 5.1.2) reduces further to $H^2(G, \mathbb{F}_p) = 0$). In practice, the characterization is used in both directions: to prove a group is free by checking a vanishing condition, and to prove cohomological vanishing by establishing freeness through other means.

Example 6.1.5: Free Pro-p-Groups

1. The group $\mathbb{Z}_p$ is the free pro-p-group of rank 1: $F(1) \cong \mathbb{Z}_p$. The universal property holds because every element of a pro-p-group generates a closed subgroup isomorphic to a quotient of $\mathbb{Z}_p$.
2. The absolute Galois group $G_{\mathbb{F}_q} \cong \widehat{\mathbb{Z}}$ (exercise 1.4.1 (2)) is a free profinite group of rank 1. Its maximal pro-p-quotient is $\mathbb{Z}_p$, a free pro-p-group.
3. The absolute Galois group of the function field $\overline{\mathbb{F}}_q(t)$, over the algebraic closure $\overline{\mathbb{F}}_q$, is a free profinite group. This is a nontrivial theorem of Harbater and of Pop. It does not follow from $\mathrm{cd}(\overline{\mathbb{F}}_q(t)) = 1$ (example 5.2.4) alone.
4. (If the reader has read [23]) Let F_d be the discrete free group on d generators. Its pro-p-completion $\widehat{F_d}^{(p)}$ is the free pro-p-group $F(d)$: the pro-p-completion functor preserves the universal property with respect to pro-p-group targets.

Subgroups of free pro-p-groups inherit freeness, with a rank formula generalizing the classical Schreier index formula for discrete free groups.

Theorem 6.1.6: Schreier Index Formula for Pro-p-Groups

Let $F = F(d)$ be the free pro-p-group of finite rank $d \geq 1$, and let $H \leq F$ be an open subgroup of index $[F : H] = n$. Then H is a free pro-p-group of rank:

$$d(H) = n(d - 1) + 1$$

***Proof*:**
Since H is an open subgroup of F, it is a closed subgroup of a profinite group, hence profinite (by proposition 1.2.1). As a subgroup of the pro-p-group F, it is pro-p. The Euler characteristic gives the rank: the cohomological dimension of H is at most 1 (since $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(F) = 1$ by proposition 5.1.3), so H is free by theorem 6.1.4. For the rank formula, the Euler characteristic $\chi(G) = 1 - d(G)$ is multiplicative under finite-index inclusions: $\chi(H) = [F : H] \cdot \chi(F)$ (this is a consequence of the multiplicativity of the Herbrand quotient, or equivalently of the formula

> $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ from proposition 3.3.7, applied to the Euler characteristic computed from H^0 and H^1 with $\mathbb{F}_p$-coefficients). Substituting: $1 - d(H) = n(1-d)$, giving $d(H) = n(d-1)+1$.

The Schreier formula has a satisfying interpretation: a free pro-p-group of rank d has "$d-1$ degrees of freedom" (subtracting the single relation imposed by the group identity), and passing to a subgroup of index n multiplies the degrees of freedom by n, then adds back the group identity.

More generally, closed subgroups of free pro-p-groups are free (not just open ones). This stronger result uses a limit argument.

Proposition 6.1.7: Closed Subgroups of Free Pro-p-Groups Are Free

Every closed subgroup of a free pro-p-group is free.

> ***Proof*:**
> Let H be a closed subgroup of $F = F_p(S)$. Then $\mathrm{cd}_p(H) \leq \mathrm{cd}_p(F) \leq 1$ by proposition 5.1.3, and H is free by theorem 6.1.4.

The brevity of this proof (a single application of monotonicity and the characterization theorem) illustrates the power of the cohomological approach. The corresponding result for discrete free groups (the Nielsen-Schreier theorem) requires significantly more combinatorial work.

Exercise 6.1.1

1. Verify the universal property of $F(1) = \mathbb{Z}_p$ directly: show that for every pro-p-group H and every $h \in H$, there exists a unique continuous homomorphism $\varphi\colon \mathbb{Z}_p \to H$ with $\varphi(1) = h$.
2. Show that the profinite completion of the discrete free group F_d (on d generators) is the free profinite group $\hat{F}(d)$. Conclude that the pro-p-completion of F_d is $F(d)$.
3. Let $F = F(2)$ be the free pro-p-group of rank 2, and let $H \leq F$ be an open normal subgroup of index p. Compute $d(H)$ using the Schreier formula, and verify this independently by computing $\dim_{\mathbb{F}_p} H^1(H, \mathbb{F}_p)$ using the corestriction map.
4. Show that the free pro-p-group on a countably infinite set S is not topologically finitely generated, and compute its p-cohomological dimension.
5. Let F be a free pro-p-group and $\varphi\colon F \to F$ a continuous surjective endomorphism. Show that $\ker \varphi$ is a free pro-p-group. Is every normal closed subgroup of F necessarily free?

6.2 Generators and Relations for Pro-p-Groups

Every pro-p-group can be expressed as a quotient of a free pro-p-group. The number of generators and relations in such a presentation are not arbitrary: they are cohomological invariants, determined by H^1 and H^2 with $\mathbb{F}_p$-coefficients. This correspondence between presentations and cohomology is the algebraic engine behind the Golod-Shafarevich inequality of section 6.3.

Definition 6.2.1: Pro-p-Presentation

A *pro-p-presentation* of a pro-p-group G is a short exact sequence of pro-p-groups:

$$1 \to R \to F \xrightarrow{\pi} G \to 1$$

where F is a free pro-p-group (definition 6.1.1) and $R = \ker \pi$ is the *relation subgroup*. The presentation is *minimal* if $F = F(d(G))$, the free pro-p-group on $d(G)$ generators (i.e., π maps a free basis of F to a minimal topological generating set of G).

Every pro-p-group admits a minimal presentation: the Burnside basis theorem (proposition 1.3.10) provides a minimal generating set $\{g_1, \dots, g_d\}$ of G (with $d = d(G) = \dim_{\mathbb{F}_p} G/\Phi(G)$), and the universal property of $F(d)$ gives a surjection $F(d) \twoheadrightarrow G$ mapping the free generators to the g_i.

The generator rank $d(G)$ was already identified in proposition 1.3.10 as the dimension of the Frattini quotient. The following reformulation through H^1 makes the cohomological nature of $d(G)$ explicit.

Theorem 6.2.2: Generator Rank

Let G be a finitely generated pro-p-group. Then:

$$d(G) = \dim_{\mathbb{F}_p} H^1(G, \mathbb{F}_p)$$

***Proof*:**
Since G acts trivially on $\mathbb{F}_p$, proposition 3.1.7 gives $H^1(G, \mathbb{F}_p) \cong \mathrm{Hom}_{\mathrm{cts}}(G, \mathbb{F}_p)$, and Hom below denotes continuous group homomorphisms. A continuous homomorphism $G \to \mathbb{F}_p$ factors through $G/\Phi(G)$ (since $\Phi(G)$ is the intersection of all maximal open subgroups, each of index p, and $\mathbb{F}_p$ has order p). Conversely, every $\mathbb{F}_p$-linear functional on $G/\Phi(G) \cong \mathbb{F}_p^{d(G)}$ lifts to a homomorphism $G \to \mathbb{F}_p$. So $\mathrm{Hom}(G, \mathbb{F}_p) \cong (G/\Phi(G))^* \cong \mathbb{F}_p^{d(G)}$, and $\dim_{\mathbb{F}_p} H^1(G, \mathbb{F}_p) = d(G)$.

The generator rank is thus the $\mathbb{F}_p$-dimension of the first cohomology group with the simplest possible coefficients. The relation rank admits an analogous description through H^2.

Theorem 6.2.3: Relation Rank

Let G be a finitely generated pro-p-group with minimal presentation $1 \to R \to F(d) \to G \to 1$. Then:

$$r(G) = \dim_{\mathbb{F}_p} H^2(G, \mathbb{F}_p)$$

The number $r(G)$ is independent of the choice of minimal presentation and is called the *relation rank* of G. It equals $d(R)$, the minimal number of topological generators of R as a normal subgroup of $F(d)$ (i.e., the minimal number of elements whose normal closure in $F(d)$ is R).

***Proof*:**
Apply the five-term exact sequence (theorem 3.3.4) to the extension $1 \to R \to F \to G \to 1$ with

coefficients in $\mathbb{F}_p$:

$$0 \to H^1(G, \mathbb{F}_p) \xrightarrow{\text{Inf}} H^1(F, \mathbb{F}_p) \xrightarrow{\text{Res}} H^1(R, \mathbb{F}_p)^G \xrightarrow{\text{tg}} H^2(G, \mathbb{F}_p) \to H^2(F, \mathbb{F}_p)$$

Since F is free, $H^2(F, \mathbb{F}_p) = 0$ (by proposition 6.1.3). The transgression is therefore surjective:

$$\text{tg}\colon H^1(R, \mathbb{F}_p)^G \twoheadrightarrow H^2(G, \mathbb{F}_p)$$

The inflation map is an isomorphism onto its image: $\text{Inf}\colon H^1(G, \mathbb{F}_p) \xrightarrow{\sim} \ker(\text{Res})$. The presentation is minimal ($F = F(d(G))$), so $\dim H^1(F, \mathbb{F}_p) = d(G) = \dim H^1(G, \mathbb{F}_p)$ (by proposition 6.1.3 and theorem 6.2.2). The inflation map between spaces of the same dimension with trivial kernel is an isomorphism $H^1(G, \mathbb{F}_p) \cong H^1(F, \mathbb{F}_p)$, forcing $\text{Res} = 0$. The five-term sequence collapses to:

$$H^1(R, \mathbb{F}_p)^G \cong H^2(G, \mathbb{F}_p)$$

The left side is $\text{Hom}(R, \mathbb{F}_p)^G$: the G-invariant continuous homomorphisms from R to $\mathbb{F}_p$. This is identified with $\text{Hom}(R/\overline{[R,F]}\Phi(R), \mathbb{F}_p)$, where $\overline{[R,F]}$ is the closure of the commutator subgroup of R with F (the G-invariance eliminates the F-conjugation) and $\Phi(R)$ is the Frattini subgroup (the $\mathbb{F}_p$ target forces factoring through the Frattini quotient). The dimension $\dim_{\mathbb{F}_p} \text{Hom}(R/\overline{[R,F]}\Phi(R), \mathbb{F}_p)$ is the minimal number of topological generators of R as a closed normal subgroup of F, which is $r(G)$.

Independence of the presentation: $H^2(G, \mathbb{F}_p)$ depends only on G, not on the choice of free resolution.

The generator and relation ranks together determine the "complexity" of a pro-p-group's presentation. A free pro-p-group has $r(G) = 0$ (since $H^2(G, \mathbb{F}_p) = 0$ by proposition 6.1.3), recovering the characterization of theorem 6.1.4. A group with $r(G) = 0$ is free, and $r(G) > 0$ measures the number of independent relations needed to define G.

Example 6.2.4: Generator and Relation Ranks

1. $G = \mathbb{Z}_p$: $d(G) = 1$ (topologically generated by 1), and $r(G) = \dim_{\mathbb{F}_p} H^2(\mathbb{Z}_p, \mathbb{F}_p) = 0$ (since $\text{cd}_p(\mathbb{Z}_p) = 1$). The minimal presentation is $F(1) \xrightarrow{\sim} \mathbb{Z}_p$ with trivial relation subgroup.

2. $G = \mathbb{Z}/p\mathbb{Z}$: $d(G) = 1$ and $r(G) = \dim_{\mathbb{F}_p} H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$ (by example 2.3.8, $H^2(C_p, \mathbb{F}_p) = \mathbb{F}_p$). The minimal presentation is $1 \to R \to \mathbb{Z}_p \to \mathbb{Z}/p\mathbb{Z} \to 1$, where $R = p\mathbb{Z}_p$ is the single relation $x^p = 1$ (more precisely, R is the closed normal subgroup generated by g^p, where g is the free generator).

3. $G = \mathbb{Z}_p^n$ (the free abelian pro-p-group of rank n): $d(G) = n$ (by proposition 1.3.10, since $G/\Phi(G) = \mathbb{F}_p^n$). The relation rank is $r(G) = \binom{n}{2}$: the relations are the commutators $[x_i, x_j]$ for $1 \le i < j \le n$. This can be verified directly from $H^2(\mathbb{Z}_p^n, \mathbb{F}_p)$: the Künneth formula (or the LHS spectral sequence applied iteratively) gives $H^2(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^2 H^1(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^2 \mathbb{F}_p^n$, which has dimension $\binom{n}{2}$.

4. $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$: $d(G) = 2$ and $r(G) = \dim H^2(G, \mathbb{F}_p)$. By the Künneth formula, $H^2((\mathbb{Z}/p\mathbb{Z})^2, \mathbb{F}_p) \cong (H^1 \otimes H^1) \oplus H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) \oplus H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p)$. Using $\dim H^1(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$

and $\dim H^2(\mathbb{Z}/p\mathbb{Z}, \mathbb{F}_p) = 1$: $r(G) = 1 + 1 + 1 = 3$. The three relations are x_1^p, x_2^p, and $[x_1, x_2]$.

The generator and relation ranks combine into a single invariant that measures the "efficiency" of a presentation.

Definition 6.2.5: Euler-Poincaré Characteristic

Let G be a finitely generated pro-p-group with $d(G)$ and $r(G)$ both finite. The *Euler-Poincaré characteristic* of G (at p) is:

$$\chi_p(G) = 1 - d(G) + r(G) = \sum_{i=0}^{\infty} (-1)^i \dim_{\mathbb{F}_p} H^i(G, \mathbb{F}_p)$$

when the sum converges (in particular, when $\mathrm{cd}_p(G) < \infty$).

The alternating sum formula extends to higher degrees: $\dim H^0(G, \mathbb{F}_p) = 1$ (the trivial module $\mathbb{F}_p$ has one-dimensional invariants), $\dim H^1(G, \mathbb{F}_p) = d(G)$, and $\dim H^2(G, \mathbb{F}_p) = r(G)$. For groups of cohomological dimension ≤ 2, the characteristic is $\chi_p(G) = 1 - d(G) + r(G)$, and all higher terms vanish.

Proposition 6.2.6: Multiplicativity of the Euler-Poincaré Characteristic

Let G be a finitely generated pro-p-group with $\mathrm{cd}_p(G) < \infty$, and let $H \leq G$ be an open subgroup of index $[G : H] = n$. Then:

$$\chi_p(H) = n \cdot \chi_p(G)$$

***Proof*:**
The formula $\mathrm{Cor} \circ \mathrm{Res} = [G : H]$ (proposition 3.3.7) implies that the restriction map on each $H^i(G, \mathbb{F}_p)$ has image satisfying $[G : H] \cdot H^i(G, \mathbb{F}_p) \subseteq \mathrm{Cor}(H^i(H, \mathbb{F}_p))$. Since $[G : H]$ is a power of p and the cohomology groups are $\mathbb{F}_p$-vector spaces, a direct argument using the LHS spectral sequence (theorem 3.4.5) for $H \trianglelefteq G$ (or a transfer argument for H not necessarily normal) gives the multiplicativity. The argument is identical to the one used for the Schreier formula (theorem 6.1.6): the Euler characteristic $\chi_p(G) = \sum (-1)^i \dim H^i(G, \mathbb{F}_p)$ transforms under restriction to an open subgroup of index n by multiplication by n.

Example 6.2.7: Euler-Poincaré Characteristics

1. $G = F(d)$ (free pro-p of rank d): $\chi_p(F(d)) = 1 - d + 0 = 1 - d$. The Schreier formula (theorem 6.1.6) for an open subgroup H of index n gives $\chi_p(H) = 1 - (n(d-1) + 1) = n(1 - d) = n \cdot \chi_p(G)$, confirming multiplicativity.

2. $G = \mathbb{Z}_p^n$: the cohomology $H^i(\mathbb{Z}_p^n, \mathbb{F}_p) \cong \bigwedge^i \mathbb{F}_p^n$ has dimension $\binom{n}{i}$, so $\chi_p(\mathbb{Z}_p^n) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} = (1-1)^n = 0$ for $n \geq 1$.

3. $G = \mathbb{Z}/p\mathbb{Z}$: $\chi_p(\mathbb{Z}/p\mathbb{Z}) = 1 - 1 + 1 - 1 + \cdots$ does not converge in the usual sense. The finite group $\mathbb{Z}/p\mathbb{Z}$ has infinite cohomological dimension (its cohomology is periodic and nontrivial

in every degree), so the Euler-Poincaré characteristic is not defined. This is consistent with the definition requiring $\mathrm{cd}_p(G) < \infty$.

Exercise 6.2.1

1. Verify the relation rank $r(\mathbb{Z}_p^3) = 3$ by writing an explicit minimal presentation $1 \to R \to F(3) \to \mathbb{Z}_p^3 \to 1$ and identifying three generators of R as a normal subgroup of $F(3)$.
2. Show that for a nontrivial finite p-group G, both $d(G) \geq 1$ and $r(G) \geq 1$. Conclude that every nontrivial finite p-group requires at least one relation.
3. Derive the relation rank formula $r(G) = \dim H^2(G, \mathbb{F}_p)$ for a not-necessarily-minimal presentation $1 \to R \to F(m) \to G \to 1$ (with $m \geq d(G)$). Show that the formula still holds: $\dim H^2(G, \mathbb{F}_p)$ is independent of the presentation.
4. Let $G = \mathbb{Z}_p^2$ and $H = p\mathbb{Z}_p \times \mathbb{Z}_p$ (an open subgroup of index p). Compute $\chi_p(H)$ directly and verify the multiplicativity $\chi_p(H) = [G : H] \cdot \chi_p(G)$.
5. (If the reader has read [23]) Let Σ_g be a closed orientable surface of genus $g \geq 1$. The fundamental group $\pi_1(\Sigma_g)$ has the standard presentation $\langle a_1, b_1, \ldots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle$. Compute d and r for the pro-p-completion $G = \widehat{\pi_1(\Sigma_g)}^{(p)}$, and verify that $\chi_p(G) = 2 - 2g$.

6.3 The Golod-Shafarevich Inequality and the Class Field Tower

Given a finitely presented pro-p-group G with $d = d(G)$ generators and $r = r(G)$ relations, how are d and r constrained? The Golod-Shafarevich inequality provides a fundamental answer: if G is a nontrivial finite p-group, then $r > d^2/4$. The contrapositive (if $r \leq d^2/4$, then G is infinite) resolved the class field tower problem, one of the central open questions in algebraic number theory in the first half of the 20th century.

The proof uses the Poincaré series (Hilbert series) of the associated graded algebra. The key idea is that the growth of the graded pieces of a finitely presented algebra is constrained by the number of generators and relations, and a finite group forces the growth to terminate, which is incompatible with having too few relations.

Definition 6.3.1: Augmentation Filtration and Poincaré Series

Let G be a pro-p-group and $\mathbb{F}_p[[G]]$ its completed group algebra. The *augmentation ideal* I is the kernel of the augmentation $\mathbb{F}_p[[G]] \to \mathbb{F}_p$. The *augmentation filtration* is $I \supseteq I^2 \supseteq I^3 \supseteq \cdots$. The *Poincaré series* of G is the formal power series:

$$H_G(t) = \sum_{n=0}^{\infty} a_n t^n, \qquad a_n = \dim_{\mathbb{F}_p} I^n / I^{n+1}$$

with the convention $I^0 = \mathbb{F}_p[[G]]$, so $a_0 = 1$.

For a finite p-group G, the augmentation ideal I is nilpotent ($I^N = 0$ for N sufficiently large, since $|G|$ is a power of p and $\mathbb{F}_p[[G]] = \mathbb{F}_p[G]$ has dimension $|G|$). In this case, $H_G(t)$ is a polynomial of degree $N-1$, and $\sum_n a_n = \dim_{\mathbb{F}_p} \mathbb{F}_p[G] = |G|$.

For the free pro-p-group $F(d)$, the completed group algebra $\mathbb{F}_p[[F(d)]]$ is isomorphic to the ring of formal power series $\mathbb{F}_p\langle\langle x_1, \ldots, x_d\rangle\rangle$ in d non-commuting variables. The augmentation ideal J is generated by $x_1, \ldots, x_d$, and $a_n = d^n$ (the monomials of degree n in d non-commuting variables form a basis for J^n/J^{n+1}). The Poincaré series is $H_{F(d)}(t) = 1/(1-dt)$.

A presentation $1 \to R \to F(d) \to G \to 1$ induces a surjection $\mathbb{F}_p[[F(d)]] \twoheadrightarrow \mathbb{F}_p[[G]]$, and the relations constrain the growth of the Poincaré series. The following inequality makes this constraint precise.

Lemma 6.3.2: The Fundamental Inequality for Poincaré Series

Let G be a finitely generated pro-p-group with minimal presentation $1 \to R \to F(d) \to G \to 1$, where $d = d(G)$ and $r = r(G)$. Let $a_n = \dim_{\mathbb{F}_p} I^n/I^{n+1}$ be the coefficients of the Poincaré series of G. Then for all $n \geq 2$:

$$a_n \geq d \cdot a_{n-1} - r \cdot a_{n-2}$$

with $a_0 = 1$ and $a_1 = d$.

***Proof*:**
The surjection $\mathbb{F}_p[[F(d)]] \twoheadrightarrow \mathbb{F}_p[[G]]$ sends J^n onto I^n. The space I^n/I^{n+1} is generated by products $\bar{x}_i \cdot m$ where $\bar{x}_1, \ldots, \bar{x}_d$ are the images of the generators in I/I^2 and $m \in I^{n-1}/I^n$. These $d \cdot a_{n-1}$ products span I^n/I^{n+1}, but some may be linearly dependent due to the relations.

The relation subgroup R is contained in the Frattini subgroup $\Phi(F(d))$ of the free group (since the presentation is minimal). In terms of the completed group algebra, $\Phi(F(d))$ corresponds to $1 + J^2$, so each relation $\rho_j \in R$ has the form $\rho_j = 1 + f_j$ with $f_j \in J^2$. The image of f_j in J^2/J^3 contributes a relation of weight 2 in the associated graded algebra. Each such relation, when multiplied by elements of J^{n-2}/J^{n-1}, can kill at most a_{n-2} independent elements in J^n/J^{n+1}. With r independent relations, the total number of dependencies is at most $r \cdot a_{n-2}$, giving $a_n \geq d \cdot a_{n-1} - r \cdot a_{n-2}$.

The inequality can be reformulated in terms of power series. Define $c_n = a_n - d \cdot a_{n-1} + r \cdot a_{n-2}$ for $n \geq 2$, with $c_0 = 1$ and $c_1 = 0$. The lemma gives $c_n \geq 0$ for all n. In power series form:

$$H_G(t) \cdot (1 - dt + rt^2) = C(t) = \sum_{n=0}^{\infty} c_n t^n$$

where $C(t)$ has non-negative coefficients and constant term $c_0 = 1$. This reformulation is the key to the proof.

Theorem 6.3.3: Golod-Shafarevich Inequality

Let G be a nontrivial finite p-group with $d = d(G)$ generators and $r = r(G)$ relations. Then $r > d^2/4$.

***Proof*:**
Step 1: Setup. Since G is finite, $H_G(t)$ is a polynomial with non-negative coefficients (all $a_n \geq 0$) and $H_G(0) = 1$. For any $t > 0$, the value $H_G(t) \geq 1$ (since $a_0 = 1$ and all $a_n \geq 0$). The power series identity $H_G(t)(1 - dt + rt^2) = C(t)$ holds with $C(t)$ having non-negative coefficients and $C(0) = 1$, so $C(t) \geq 1$ for all $t > 0$.

Step 2: Contradiction from a positive root. Suppose $r \leq d^2/4$. The quadratic $q(t) = 1 - dt + rt^2$ has discriminant $\Delta = d^2 - 4r \geq 0$, so it has real roots. The smaller root is:

$$t_0 = \frac{d - \sqrt{d^2 - 4r}}{2r} > 0$$

(taking $r > 0$; the case $r = 0$ gives $q(t) = 1 - dt$ with root $1/d > 0$). Since $q(0) = 1 > 0$ and $q(t_0) = 0$, the quadratic is positive on $(0, t_0)$.

For $0 < t < t_0$: the identity $H_G(t) = C(t)/q(t)$ holds with both $C(t) \geq 1$ and $q(t) > 0$, giving $H_G(t) \geq 1/q(t)$. As $t \to t_0^-$, the denominator $q(t) \to 0^+$, so $H_G(t) \to +\infty$.

Step 3: Finite groups have polynomial Poincaré series. Since G is finite, $H_G(t)$ is a polynomial, and in particular $H_G(t_0)$ is a finite real number. This contradicts $H_G(t) \to +\infty$ as $t \to t_0^-$. The assumption $r \leq d^2/4$ is false, so $r > d^2/4$.

The inequality extends to infinite pro-p-groups in contrapositive form.

Theorem 6.3.4: Golod-Shafarevich Criterion for Infinite Groups

Let G be a finitely presented pro-p-group with $d = d(G)$ and $r = r(G)$. If $r \leq d^2/4$, then G is infinite.

***Proof*:**
If G were finite, theorem 6.3.3 would give $r > d^2/4$, contradicting the hypothesis.

The bound $d^2/4$ is essentially sharp: for each d, there exist finite p-groups with r slightly larger than $d^2/4$.

The original application of the Golod-Shafarevich inequality was to the class field tower problem, which asks whether the tower of successive Hilbert class fields of a number field must terminate.

Definition 6.3.5: The p-Class Field Tower

Let K be a number field and p a prime. The *p-class field tower* of K is the sequence of fields:

$$K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

where K_{i+1} is the maximal unramified abelian p-extension of K_i. The tower *terminates* if $K_n = K_{n+1}$ for some n (equivalently, if $\mathrm{Cl}(K_n)$ has trivial p-part). The union $K_\infty = \bigcup K_i$ is a Galois extension of K, and $G = \mathrm{Gal}(K_\infty/K)$ is a pro-p-group.

The tower terminates if and only if G is finite. The generator and relation ranks of G are determined by the arithmetic of K. By class field theory [25, Ch. VI, (6.9)], the first layer K_1/K has $\mathrm{Gal}(K_1/K) \cong \mathrm{Cl}(K)\{p\}$ (the p-primary part of the class group), where for $p = 2$ "unramified" includes the infinite places, a condition automatic for odd p. So the abelianization $G^{\mathrm{ab}} \cong \mathrm{Cl}(K)\{p\}$ and:

$$d(G) = \dim_{\mathbb{F}_p} \mathrm{Cl}(K)/p\,\mathrm{Cl}(K) = \text{the } p\text{-rank of } \mathrm{Cl}(K)$$

The relation rank $r(G)$ is bounded above by the number of primes of K lying above p together with contributions from roots of unity and the signature of K (see [26] for the precise formula).

Theorem 6.3.6: Existence of Infinite Class Field Towers

Let K be a number field, p a prime, and $G = \mathrm{Gal}(K_\infty/K)$ the Galois group of the p-class field tower. If $d(G)^2 > 4r(G)$, then G is infinite and the p-class field tower does not terminate.

***Proof*:**
Direct application of theorem 6.3.4: the condition $r(G) < d(G)^2/4$ (equivalently $d(G)^2 > 4r(G)$) forces G to be infinite.

In practice, $d(G) = \mathrm{rk}_p(\mathrm{Cl}(K))$ is computable from the class group, while $r(G)$ is bounded above by arithmetic data of K. The precise bound (see [26]) takes the form $r(G) \leq d(G) + \rho$, where ρ depends on the number of primes above p, the unit rank, and the roots of unity in K. Substituting into the criterion: the tower is infinite whenever $d(G)^2 - 4d(G) - 4\rho > 0$, i.e., $d(G) > 2 + 2\sqrt{1+\rho}$.

Example 6.3.7: The First Infinite Class Field Tower

1. The first explicit example was given by Golod and Shafarevich in 1964: let $K = \mathbb{Q}(\sqrt{-2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13}) = \mathbb{Q}(\sqrt{-30030})$. The discriminant of K is divisible by $t = 6$ finite primes. By genus theory (see [6]), the 2-rank of $\mathrm{Cl}(K)$ satisfies $d = d(G) \geq t - 1 = 5$. The relation rank is bounded by the arithmetic of K: the bound from [26] gives $r(G) \leq d(G) + \rho$ where ρ depends on the primes above 2 and the signature. For this specific field, the computation yields $d(G)^2 > 4r(G)$, so the 2-class field tower is infinite by theorem 6.3.6.

2. More generally, for any prime p, there exist number fields with infinite p-class field towers. For $p = 2$: the field $K = \mathbb{Q}(\sqrt{-\ell_1 \cdots \ell_t})$ where $\ell_1, \ldots, \ell_t$ are distinct odd primes satisfying $\ell_i \equiv 1 \pmod 4$ has 2-rank of $\mathrm{Cl}(K)$ at least $t - 1$. Choosing t large enough (the criterion $d^2 > 4(d + \rho)$ is satisfied for $t \geq 7$ with standard bounds on ρ) guarantees an infinite 2-class field tower.

The existence of infinite class field towers answered a question that had been open since Furtwängler's work in the 1930s: the Hilbert class field tower does not always terminate, so there is no general procedure for "unwinding" the class group to reach a field with trivial class number. This was one of the first applications of pro-p-group theory to a problem in algebraic number theory, and it demonstrated the power of cohomological methods in a setting where direct computation is infeasible.

6.3.8 Remark: Golod-Shafarevich Beyond Number Theory The Golod-Shafarevich inequality has applications far beyond class field towers. Golod used it to construct the first examples of infinite finitely generated p-groups (answering the *general Burnside problem* in the negative for p-groups): if $d \geq 2$ and $r \leq d^2/4$, the quotient $F(d)/\overline{\langle \rho_1, \ldots, \rho_r \rangle}$ is an infinite pro-p-group, and its discrete quotients provide infinite finitely generated torsion groups. In ring theory, the same technique produces infinite-dimensional nil algebras over $\mathbb{F}_p$, disproving the Kurosh conjecture.

Exercise 6.3.1

1. Verify the Golod-Shafarevich inequality directly for the cyclic group C_{p^2}: compute $d(C_{p^2})$ and $r(C_{p^2})$ from a minimal pro-p-presentation, and check that $r > d^2/4$.
2. Show that the free pro-p-group $F(d)$ with $d \geq 2$ does not contradict the Golod-Shafarevich inequality (since $F(d)$ is infinite, the inequality does not apply). Compute $H_{F(d)}(t)$ and verify that $H_{F(d)}(t)(1 - dt) = 1$.
3. Let $G = \mathbb{Z}/p\mathbb{Z}$. Compute the Poincaré series $H_G(t)$ directly by determining $\dim_{\mathbb{F}_p} I^n/I^{n+1}$ for the augmentation ideal I of $\mathbb{F}_p[G]$. Verify that $H_G(t)(1 - t + t^2) = C(t)$ for a specific polynomial $C(t)$ with non-negative coefficients.
4. (If the reader has read [6]) Let $p = 2$ and $K = \mathbb{Q}(\sqrt{-d})$ where d is a positive squarefree integer divisible by t distinct odd primes, all congruent to 1 (mod 4). Show that the 2-rank of $\mathrm{Cl}(K)$ is at least $t - 1$ (using genus theory), and determine the smallest t for which the Golod-Shafarevich criterion guarantees an infinite 2-class field tower.
5. Formulate and prove a graded version of the Golod-Shafarevich inequality: if the r relations in a minimal presentation have weights $w_1 \leq w_2 \leq \cdots \leq w_r$ (where w_j is the largest k such that $\rho_j \in I^k$), show that the Poincaré series satisfies $H_G(t)(1 - dt + \sum_{j=1}^{r} t^{w_j}) = C(t)$ with $C(t)$ having non-negative coefficients.

6.4 Demushkin Groups

At the opposite extreme from free pro-p-groups ($\mathrm{cd}_p = 1$, no relations) sit the Demushkin groups: pro-p-groups of cohomological dimension 2 whose cohomology satisfies Poincaré duality. These groups were introduced in definition 5.3.5 and previewed in example 5.3.6. This section develops their classification through the cup product pairing on H^1, which determines the group up to a single discrete invariant.

Recall the definition: a *Demushkin group* is a pro-p-group G with $\mathrm{cd}_p(G) = 2$, $H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^d$ for some finite $d \geq 2$, and $H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$. By theorem 6.2.2 and theorem 6.2.3, these conditions translate to: $d(G) = d$ generators and $r(G) = 1$ relation. A Demushkin group is a one-relator pro-p-group of cohomological dimension 2.

The cup product from theorem 4.2.4 endows the cohomology of any profinite group with a graded ring structure. For a Demushkin group, the cup product in degree 1 produces the key structural invariant.

Proposition 6.4.1: The Cup Product Pairing

Let G be a Demushkin group with $d(G) = d$. The cup product:

$$\smile : H^1(G, \mathbb{F}_p) \times H^1(G, \mathbb{F}_p) \to H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$$

is a non-degenerate bilinear pairing.

***Proof*:**
By proposition 5.3.4, a Demushkin group is a Poincaré duality group of dimension 2 at p. The Poincaré duality pairing in dimensions i and $2 - i$ (with $\mathbb{F}_p$-coefficients) is a perfect pairing $H^i(G, \mathbb{F}_p) \times H^{2-i}(G, \mathbb{F}_p) \to H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$, realized by the cup product. Taking $i = 1$ gives the stated non-degenerate pairing.

The pairing $\smile$ is a bilinear form on the $\mathbb{F}_p$-vector space $V = H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^d$. Its symmetry type is determined by the prime p.

Proposition 6.4.2: Symmetry of the Cup Product Pairing

Let G be a Demushkin group. The cup product pairing on $H^1(G, \mathbb{F}_p)$ is:

1. alternating (i.e., $\alpha \smile \alpha = 0$ for all $\alpha \in H^1(G, \mathbb{F}_p)$) when p is odd;
2. symmetric when $p = 2$.

In particular, for p odd, $d(G)$ is necessarily even.

***Proof*:**
The graded-commutativity of the cup product (theorem 4.2.4) gives $\alpha \smile \beta = (-1)^{pq}\beta \smile \alpha$ for $\alpha \in H^p$, $\beta \in H^q$. With $p = q = 1$: $\alpha \smile \beta = -\beta \smile \alpha$. Taking $\beta = \alpha$: $\alpha \smile \alpha = -\alpha \smile \alpha$, so $2(\alpha \smile \alpha) = 0$.

For p odd, 2 is invertible in $\mathbb{F}_p$, so $\alpha \smile \alpha = 0$: the pairing is alternating. A non-degenerate alternating form on $\mathbb{F}_p^d$ exists only when d is even (the standard classification of alternating forms over fields of odd characteristic).

For $p = 2$, the condition $2(\alpha \smile \alpha) = 0$ is automatic (since $2 = 0$ in $\mathbb{F}_2$), so no constraint beyond symmetry is imposed.

The classification of Demushkin groups proceeds from the classification of the cup product pairing. For p odd, a non-degenerate alternating form on $\mathbb{F}_p^d$ (with d even) has a unique isomorphism class (the standard symplectic form), so the pairing alone does not distinguish Demushkin groups. The additional invariant comes from the action on p-power roots of unity.

Definition 6.4.3: Type of a Demushkin Group

Let G be a Demushkin group. The *type* of G is the supernatural number $q \in \{p, p^2, p^3, \dots\} \cup \{0\}$ defined as follows. Consider the maximal abelian quotient $G^{\mathrm{ab}} = G/[G,G]$ and the p-adic cyclotomic character $\chi\colon G^{\mathrm{ab}} \to \mathbb{Z}_p^\times$ classifying the action of G on the Tate module $\mathbb{Z}_p(1) = \varprojlim_n \mu_{p^n}$. The type q is the order of $\mathrm{im}(\chi)$ in $\mathbb{Z}_p^\times$ (with $q = 0$ when χ is trivial, i.e., G acts trivially on all p-power roots of unity).

For the reader unfamiliar with the cyclotomic character in this abstract setting: the type q can equivalently be defined through the relation in a minimal presentation. Given a minimal presentation $1 \to R \to F(d) \to G \to 1$ with $R = \overline{\langle \rho \rangle}^{F(d)}$ (the normal closure of a single element $\rho \in F(d)$), the element ρ determines q through its expression in terms of commutators and p-powers of the free generators.

Theorem 6.4.4: Classification of Demushkin Groups

Let p be an odd prime. A Demushkin group G with $d(G) = d$ (necessarily even) is determined up to isomorphism by the pair (d, q), where q is the type of G. For each even $d \geq 2$ and each $q \in \{p, p^2, p^3, \dots\} \cup \{0\}$, there exists a unique Demushkin group $G(d,q)$, presented as:

$$G(d,q) = \langle x_1, \dots, x_d \,\big|\, x_1^q [x_1, x_2][x_3, x_4] \cdots [x_{d-1}, x_d] = 1 \rangle$$

where $x_1^0 = 1$ is interpreted as the absence of the power term (i.e., the relation is $[x_1, x_2][x_3, x_4] \cdots [x_{d-1}, x_d] = 1$ when $q = 0$).

The proof of the classification uses the structure theory of non-degenerate alternating forms over $\mathbb{Z}_p$ and a careful analysis of the relation in the completed group algebra. See Labute [21] and Serre [30] for complete proofs.

The classification for $p = 2$ is more involved: the cup product pairing is symmetric rather than alternating, and the classification of symmetric bilinear forms over $\mathbb{F}_2$ depends on additional invariants (the Arf invariant and the rank modulo 2). The result is that Demushkin 2-groups are classified by a triple (d, q, ϵ) where $d \geq 2$, $q \in \{2, 4, 8, \dots\} \cup \{0\}$, and $\epsilon \in \{0, 1\}$ is the Arf invariant of the quadratic form associated to the cup product pairing (see [21] for the precise statement).

Example 6.4.5: Demushkin Groups

1. The simplest Demushkin group has $d = 2$ and relation $x_1^q [x_1, x_2] = 1$ (for p odd). When $q = 0$: the group is $G(2,0) = \langle x_1, x_2 \,|\, [x_1, x_2] = 1 \rangle \cong \mathbb{Z}_p^2$ (the free abelian pro-p-group of rank 2). For $q = p$: $G(2,p) = \langle x_1, x_2 \,|\, x_1^p [x_1, x_2] = 1 \rangle$, a non-abelian group in which x_1 has finite order modulo the commutator subgroup.

2. A Demushkin group of type $q = 0$ has the property that all generators have infinite order and the single relation is purely a product of commutators. The group $G(d,0) = \langle x_1, \dots, x_d \,|\, [x_1, x_2] \cdots [x_{d-1}, x_d] = 1 \rangle$ is torsion-free: no nonzero element has finite order.

3. (If the reader has read [23]) Let Σ_g be a closed orientable surface of genus $g \geq 1$. The fundamental group $\pi_1(\Sigma_g)$ has presentation $\langle a_1, b_1, \dots, a_g, b_g \,|\, [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle$. The pro-$p$-completion $\widehat{\pi_1(\Sigma_g)}^{(p)}$ is a Demushkin group with $d = 2g$ and type $q = 0$: the

presentation is exactly the Demushkin presentation $G(2g, 0)$. This connects the classification of Demushkin groups to the topology of surfaces: the Demushkin groups of type 0 are precisely the pro-p surface groups. The Euler-Poincaré characteristic is $\chi_p = 1 - 2g + 1 = 2 - 2g$ (exercise 6.2.1 (5)), matching the topological Euler characteristic of Σ_g.

4. The Golod-Shafarevich inequality (theorem 6.3.3) is automatically satisfied for Demushkin groups: $r(G) = 1$ and $d(G) \geq 2$, so $d^2/4 \geq 1 = r$ (with equality when $d = 2$). This is consistent with Demushkin groups being infinite.

The Demushkin classification has consequences for the structure of the cohomology ring:

Proposition 6.4.6: Cohomology Ring of a Demushkin Group

Let G be a Demushkin group with $d(G) = d$ and p odd. The cohomology ring $H^*(G, \mathbb{F}_p)$ is an exterior algebra on d generators in degree 1 modulo the ideal generated by the kernel of the cup product pairing. Explicitly:

1. $H^0(G, \mathbb{F}_p) = \mathbb{F}_p$.
2. $H^1(G, \mathbb{F}_p) = \mathbb{F}_p^d$.
3. $H^2(G, \mathbb{F}_p) = \mathbb{F}_p$ (generated by any nonzero cup product $\alpha_i \smile \alpha_j$ with α_i, α_j forming part of a symplectic basis).
4. $H^n(G, \mathbb{F}_p) = 0$ for $n \geq 3$ (since $\mathrm{cd}_p(G) = 2$).

***Proof*:**
Parts (1), (2), and (4) follow from the definition. For (3): $H^2(G, \mathbb{F}_p) \cong \mathbb{F}_p$ by definition, and the cup product $H^1 \times H^1 \to H^2$ is surjective (since it is non-degenerate by proposition 6.4.1 and $H^2 \cong \mathbb{F}_p$). Choose a symplectic basis $\alpha_1, \beta_1, \ldots, \alpha_{d/2}, \beta_{d/2}$ for $H^1(G, \mathbb{F}_p)$ such that $\alpha_i \smile \beta_i$ is a generator of H^2 for each i, and all other pairings vanish. Then H^2 is generated by any single $\alpha_i \smile \beta_i$.

The maximal pro-p quotients of absolute Galois groups of local fields are Demushkin groups (for p odd). The identification and computation of their invariants from local field data is carried out in chapter 8 using local Tate duality.

Exercise 6.4.1

1. Let G be a Demushkin group with p odd and $d(G) = 4$. Write out the cup product pairing on $H^1(G, \mathbb{F}_p) \cong \mathbb{F}_p^4$ in matrix form with respect to a symplectic basis, and verify it is non-degenerate and alternating.
2. For p odd, describe all Demushkin groups with $d = 2$ by listing the possible types q and writing the explicit presentation for each. Show that $G(2, 0) \cong \mathbb{Z}_p^2$.
3. Show that a Demushkin group of type $q = 0$ is torsion-free. *Hint:* the relation $[x_1, x_2] \cdots [x_{d-1}, x_d] = 1$ does not force any generator to have finite order. Use the fact that the abelianization $G^{\mathrm{ab}} \cong \mathbb{Z}_p^d$ is torsion-free.

4. Verify that the Golod-Shafarevich inequality $r > d^2/4$ fails for a Demushkin group G with $d(G) = 2$ (i.e., $r = 1 \leq 1 = d^2/4$). Explain why this does not contradict theorem 6.3.3.

5. (If the reader has read [23]) Let $G = \widehat{\pi_1(\Sigma_g)}^{(p)}$ be the pro-p-completion of the fundamental group of a genus-g surface, with p odd. Compute the full cohomology ring $H^*(G, \mathbb{F}_p)$ and compare it with the singular cohomology ring $H^*(\Sigma_g, \mathbb{F}_p)$.

7

The Brauer Group

The cohomology group $H^2(G, M)$ classifies extensions of G by M (theorem 2.4.5). When $G = G_K$ is the absolute Galois group and $M = (K^{\text{sep}})^\times$, this classification acquires an independent algebraic identity: the Brauer group $\text{Br}(K)$, which parametrizes central simple algebras over K up to Morita equivalence. This chapter develops the Brauer group from both the algebraic and cohomological perspectives, establishes the identification $\text{Br}(K) \cong H^2(G_K, (K^{\text{sep}})^\times)$ via crossed product algebras, and computes $\text{Br}(K)$ for local and global fields. The Brauer group is the object connecting the abstract cohomological machinery of Parts I and II to the arithmetic of Part III.

7.1 Central Simple Algebras and the Brauer Group

The Brauer group arises from a classification problem in algebra: given a field K, describe all division algebras with center K, up to the identification that treats a division algebra and its matrix rings as equivalent. The resulting group $\text{Br}(K)$ is an invariant of the field that turns out to be computable by cohomological methods.

Definition 7.1.1: Central Simple Algebra

A *central simple algebra* (CSA) over a field K is a finite-dimensional K-algebra A satisfying:

1. the center of A is K (centrality), and
2. A has no two-sided ideals other than 0 and A (simplicity).

The two conditions interact: centrality prevents A from decomposing as a product of algebras over a proper extension of K, while simplicity prevents A from having a nontrivial quotient. Together, they force A to be a matrix algebra over a division ring.

Theorem 7.1.2: Wedderburn's Structure Theorem

Every central simple algebra A over K is isomorphic to $M_n(D)$ for a unique (up to isomorphism) division algebra D with center K and a unique positive integer n. The division algebra D is the *Wedderburn component* of A, and $\dim_K A = n^2 \dim_K D$.

***Proof*:**
Every module over the finite-dimensional simple K-algebra A is semisimple, in particular A as a left module over itself, so A is a semisimple ring, and the Artin-Wedderburn theorem writes it as a finite product of matrix algebras over division algebras, finite-dimensional over K, with the factors unique up to isomorphism (*Core Representation Theory* [10, Representation of Simple Algebra], [10, Artin-Wedderburn] and [10, Artin-Wedderburn For k-Algebras]). Each factor is a two-sided ideal of the product, so the simplicity of A leaves a single factor $M_n(D)$. The center of $M_n(D)$ is the center of D (scalar matrices), so the centrality of A forces D to have center K. Counting dimensions, $\dim_K A = n^2 \dim_K D$.

The cited uniqueness is up to ring isomorphism, and D is in fact unique up to K-algebra isomorphism. Let $A \cong M_n(D)$ as K-algebras, and let $S = D^{\oplus n}$ be the module of column vectors, a simple left A-module. Every simple left A-module is isomorphic to S (*Core Representation Theory* [10, Structure of Simple $M_n(D)$-Module]), so the ring $\mathrm{End}_A(S)$ depends, up to isomorphism, only on A. It is a K-algebra through the action of the central subfield K of A, and conjugation by an isomorphism of A-modules preserves this structure, because the isomorphism commutes with the action of A, hence of K. The map $d \mapsto (x \mapsto xd)$ is a ring isomorphism $D^{\mathrm{op}} \to \mathrm{End}_A(S)$ (*Core Representation Theory* [10, Endomorphism Of Simple $M_n(D)$-Module]), and it is K-linear, since $\lambda \in K$ acts on S through the scalar matrix λI, which sends x to $x\lambda$. So the K-algebra D^{op}, and hence D, is determined by A up to isomorphism, and then n is determined by $\dim_K A = n^2 \dim_K D$.

Proposition 7.1.3: The Dimension of a Central Simple Algebra Is a Square

Let A be a central simple algebra over K. There are an integer $r \geq 1$ and a finite extension L/K with $A \otimes_K L \cong M_r(L)$ as L-algebras, and $\dim_K A = r^2$.

***Proof*:**
The splitting $A \otimes_K L \cong M_r(L)$ is [19, Thm. 2.2.1], proved there by base change to an algebraic closure of K, without maximal subfields. Extending scalars preserves dimension, so $\dim_K A = \dim_L(A \otimes_K L) = \dim_L M_r(L) = r^2$ [19, Cor. 2.2.3].

A division algebra D with center K, finite-dimensional over K, is a central simple algebra over K, as a division ring has no two-sided ideals other than 0 and D. By proposition 7.1.3, $\dim_K D = m^2$ for an integer $m \geq 1$, the *degree* of D, and if $A \cong M_n(D)$, then $\dim_K A = (nm)^2$.

Example 7.1.4: Central Simple Algebras

1. The matrix algebra $M_n(K)$ is a CSA over K with Wedderburn component $D = K$. These are the *split* CSAs.

2. The real quaternion algebra $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ (with $i^2 = j^2 = -1$, $ij = k = -ji$) is a CSA over $\mathbb{R}$ of dimension 4. Since $\mathbb{H}$ is a division algebra ($\mathbb{H}$ has no zero divisors), its Wedderburn component is $D = \mathbb{H}$ with $n = 1$.

3. For any $a, b \in K^\times$, the *quaternion algebra* $(a, b)_K$ is the 4-dimensional K-algebra generated by elements i, j subject to $i^2 = a$, $j^2 = b$, $ij = -ji$. This is a CSA over K. The algebra $(a, b)_K$ is either isomorphic to $M_2(K)$ (split) or is a division algebra (non-split), and these are the only two possibilities since $\dim_K (a, b)_K = 4 = 2^2$.

4. (If the reader has read [10]) The endomorphism algebra $\mathrm{End}_D(V)$ of a finite-dimensional right D-module V is a CSA over the center of D. When $D = K$ and $V = K^n$, this recovers $M_n(K)$.

The quaternion algebra $(a, b)_K$ is the most important family of CSAs for arithmetic applications. The condition for splitting, $(a, b)_K \cong M_2(K)$, can be tested by checking whether the equation $ax^2 + by^2 = 1$ has a solution in K (the norm form of the quaternion algebra).

Two CSAs are identified if they have the same Wedderburn component.

Definition 7.1.5: Brauer Equivalence

Two central simple algebras A and B over K are *Brauer equivalent*, written $A \sim B$, if their Wedderburn components are isomorphic: $A \cong M_n(D)$ and $B \cong M_m(D)$ for the same division algebra D.

Equivalently, $A \sim B$ if and only if $A \otimes_K M_r(K) \cong B \otimes_K M_s(K)$ for some $r, s \geq 1$. The tensor product of two CSAs over K is again a CSA over K, and Brauer equivalence classes form an abelian group under tensor product (*Core Representation Theory* [10, Tensor Product of Two CSA's, Brauer Group]).

Definition 7.1.6: The Brauer Group

The *Brauer group* of K, denoted $\mathrm{Br}(K)$, is the set of Brauer equivalence classes of central simple algebras over K, with group operation $[A] \cdot [B] = [A \otimes_K B]$.

The identity element is $[K] = [M_n(K)]$ (the class of all split algebras). The inverse of $[A]$ is $[A^{\mathrm{op}}]$ (the opposite algebra, with multiplication reversed), since $A \otimes_K A^{\mathrm{op}} \cong M_{n^2}(K)$ where $n^2 = \dim_K A$ (the isomorphism sends $a \otimes b$ to the endomorphism $x \mapsto axb$).

Proposition 7.1.7: Vanishing of the Brauer Group

$\mathrm{Br}(K) = 0$ if and only if every central simple algebra over K is split, i.e., every finite-dimensional division algebra with center K is K itself.

***Proof*:**
The Brauer group is trivial if and only if every CSA is Brauer equivalent to K, i.e., every CSA is isomorphic to some $M_n(K)$. By theorem 7.1.2, this holds if and only if the only division algebra with center K is K itself.

Example 7.1.8: Brauer Groups of Familiar Fields

1. $\mathrm{Br}(\mathbb{C}) = 0$: the field $\mathbb{C}$ is algebraically closed, so every polynomial has a root. A division algebra D with center $\mathbb{C}$ would have $D^\times$ containing an element α with $\mathbb{C}(\alpha)/\mathbb{C}$ a nontrivial extension, but no such extensions exist.
2. $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$: the only division algebras with center $\mathbb{R}$ are $\mathbb{R}$ itself (the trivial class) and $\mathbb{H}$ (the quaternion class). The isomorphism sends $[\mathbb{H}] \mapsto 1 \in \mathbb{Z}/2\mathbb{Z}$. The class $[\mathbb{H}]$ has order 2: $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$ (as verified in exercise 7.1.1 (1)).
3. $\mathrm{Br}(\mathbb{F}_q) = 0$ for any finite field $\mathbb{F}_q$: by Wedderburn's theorem (every finite division ring is a field, [9, Wedderburn's Little Theorem]), every finite-dimensional division algebra over $\mathbb{F}_q$ with center $\mathbb{F}_q$ is $\mathbb{F}_q$ itself.
4. $\mathrm{Br}(K) = 0$ whenever K is separably closed: if D is a division algebra with center K, any maximal subfield $L \subseteq D$ is a finite extension of K, hence equals K (since K is separably closed and L/K is separable). So $D = K$, and every CSA is split.

Field extensions induce homomorphisms between Brauer groups.

Definition 7.1.9: Base Change and the Relative Brauer Group

For a field extension L/K, the *base change* (or *scalar extension*) map is the group homomorphism:

$$\mathrm{res}_{L/K} \colon \mathrm{Br}(K) \to \mathrm{Br}(L), \qquad [A] \mapsto [A \otimes_K L]$$

The kernel $\mathrm{Br}(L/K) = \ker(\mathrm{res}_{L/K})$ is the *relative Brauer group*: the set of classes split by L.

An element $[A] \in \mathrm{Br}(L/K)$ is a CSA over K that becomes a matrix algebra after extending scalars to L. The relative Brauer groups exhaust $\mathrm{Br}(K)$: every CSA is split by some finite extension (in fact, by a finite Galois extension, a splitting field for the Wedderburn component).

Proposition 7.1.10: Splitting by Finite Galois Extensions

$\mathrm{Br}(K) = \bigcup_{L/K} \mathrm{Br}(L/K)$, where the union ranges over finite Galois extensions L/K inside K^{sep}.

***Proof*:**
Let $[D] \in \mathrm{Br}(K)$ with D a division algebra with center K. By proposition 7.1.3, $\dim_K D = m^2$ for an integer $m \geq 1$, the degree of D. The algebra D contains a subfield L, separable over K, with $[L : K] = m$ [19, Prop. 4.5.4], the theorem of Noether and Köthe. Such an L splits D. Regard D as a right L-vector space, of dimension $\dim_K D/[L : K] = m$. Letting D act on itself by left and L by right multiplication gives a homomorphism of L-algebras $D \otimes_K L \to \mathrm{End}_L(D) \cong M_m(L)$, $d \otimes l \mapsto (x \mapsto dxl)$, as L is commutative. Since L/K is finite, $D \otimes_K L$ is a central simple algebra over L [19, Lemma 2.2.2], so this nonzero homomorphism is injective, and hence an isomorphism, both sides having dimension $m^2 = \dim_K D$ over L. Finally, L/K is finite separable, so L is K-isomorphic to a subfield of K^{sep}, and the Galois closure L' of that subfield over K is a finite Galois extension inside K^{sep} with $D \otimes_K L' \cong (D \otimes_K L) \otimes_L L' \cong M_m(L')$. So $[D] \in \mathrm{Br}(L'/K)$.

The period and index of a Brauer class measure its complexity from two perspectives.

Definition 7.1.11: Period and Index

Let $[A] \in \mathrm{Br}(K)$ with Wedderburn component D. The *period* of $[A]$ is:

$$\mathrm{per}(A) = \mathrm{ord}\left([A]\right) \text{ in } \mathrm{Br}(K)$$

The *index* of $[A]$ is:

$$\mathrm{ind}(A) = \deg(D) = \sqrt{\dim_K D}$$

Proposition 7.1.12: Period Divides Index

For any $[A] \in \mathrm{Br}(K)$: $\mathrm{per}(A) \big| \mathrm{ind}(A)$, and $\mathrm{per}(A)$ and $\mathrm{ind}(A)$ have the same prime factors.

***Proof*:**
Let D be the Wedderburn component of A, with $\deg(D) = m$. By the proof of proposition 7.1.10, D contains a subfield L, separable over K with $[L : K] = m$, such that $D \otimes_K L \cong M_m(L)$, so $[A] = [D]$ lies in $\mathrm{Br}(L/K)$. The divisibility $\mathrm{per}(A) \big| \mathrm{ind}(A)$ follows because $[L : K] \cdot [A] = 0$ for every finite separable splitting field L of A, a consequence of the corestriction formula established through the cohomological identification in section 7.2.

The statement about prime factors follows from the primary decomposition of $\mathrm{Br}(K)$: if $[A] = \sum_p [A_p]$ with $[A_p] \in \mathrm{Br}(K)\{p\}$, then $\mathrm{ind}(A) = \prod_p \mathrm{ind}(A_p)$ and $\mathrm{per}(A) = \mathrm{lcm}_p \, \mathrm{per}(A_p)$. If $p \big| \mathrm{ind}(A)$, then $\mathrm{ind}(A_p) > 1$, forcing $\mathrm{per}(A_p) \geq p$ (since the trivial class is the only class of index 1), so $p \big| \mathrm{per}(A)$.

The question of when $\mathrm{per}(A) = \mathrm{ind}(A)$ (and, more generally, what bounds exist on ind in terms of per) is the period-index problem, addressed in section 7.4.

Exercise 7.1.1

1. Show that $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$. *Hint:* exhibit an explicit isomorphism using the left and right multiplication actions of $\mathbb{H}$ on itself.
2. Let K be a field with $\mathrm{char} K \neq 2$ and $a, b \in K^\times$. Show that the quaternion algebra $(a, b)_K$ splits (is isomorphic to $M_2(K)$) if and only if the equation $ax^2 + by^2 = 1$ has a solution $(x, y) \in K^2$.
3. Show that if $L_1 \subseteq L_2$ are Galois extensions of K, then $\mathrm{Br}(L_1/K) \subseteq \mathrm{Br}(L_2/K)$. Conclude that the union in proposition 7.1.10 is a filtered colimit.
4. Let $[A] \in \mathrm{Br}(K)$ with $\mathrm{per}(A) = e$ and $\mathrm{ind}(A) = m$. Show that if p is a prime dividing m, then $p \big| e$. *Hint:* consider the p-primary component of $[A]$ in $\mathrm{Br}(K)$.
5. Show that for any fields K_1 and K_2, the Brauer group of their product (as a ring, if defined) satisfies $\mathrm{Br}(K_1 \times K_2) \cong \mathrm{Br}(K_1) \times \mathrm{Br}(K_2)$. What goes wrong if K_1 and K_2 are not fields?

7.2 Crossed Product Algebras and H^2

The Brauer group was defined in section 7.1 as an algebraic object classifying central simple algebras. The central result of this chapter is that $\mathrm{Br}(K)$ is a cohomology group: the explicit construction of a CSA from a 2-cocycle (the crossed product) provides an isomorphism $H^2(\mathrm{Gal}(L/K), L^\times) \cong \mathrm{Br}(L/K)$ for any finite Galois extension L/K, and passing to the colimit gives $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$.

Definition 7.2.1: Crossed Product Algebra

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$ and let $f \in Z^2(G, L^\times)$ be a 2-cocycle (definition 2.3.7). The *crossed product algebra* $A_f = (L/K, f)$ is the left L-vector space with basis $\{u_\sigma\}_{\sigma \in G}$:

$$A_f = \bigoplus_{\sigma \in G} L \cdot u_\sigma$$

equipped with the multiplication rule:

$$(a \cdot u_\sigma)(b \cdot u_\tau) = a \cdot \sigma(b) \cdot f(\sigma, \tau) \cdot u_{\sigma\tau}$$

for $a, b \in L$ and $\sigma, \tau \in G$, extended K-linearly.

The multiplication rule encodes two pieces of structure: the Galois action ($\sigma(b)$ appears when commuting u_σ past $b \in L$) and the failure of the formal symbols u_σ to form a group ($u_\sigma u_\tau = f(\sigma, \tau) u_{\sigma\tau}$, not $u_{\sigma\tau}$). The 2-cocycle f measures this failure.

Proposition 7.2.2: Crossed Products Are Central Simple

The crossed product $A_f = (L/K, f)$ is a central simple algebra over K of dimension $[L : K]^2$. Its identity element is $f(1,1)^{-1} u_1$, and $b \mapsto b f(1,1)^{-1} u_1$ embeds L as a subfield of A_f. With L embedded in this way, left multiplication by $b \in L$ is the given L-vector space structure, $u_\sigma b = \sigma(b) u_\sigma$ for $b \in L$ and $\sigma \in G$, each u_σ is invertible, and $A_f = \bigoplus_\sigma u_\sigma L$.

***Proof*:**

Step 1: Associativity. The product of three elements $(a \cdot u_\sigma)(b \cdot u_\tau)(c \cdot u_\rho)$ can be computed in two ways. On one hand: $((a \cdot u_\sigma)(b \cdot u_\tau))(c \cdot u_\rho) = a\sigma(b) f(\sigma, \tau) \cdot \sigma\tau(c) \cdot f(\sigma\tau, \rho) \cdot u_{\sigma\tau\rho}$. On the other: $(a \cdot u_\sigma)((b \cdot u_\tau)(c \cdot u_\rho)) = a\sigma(b\tau(c) f(\tau, \rho)) \cdot f(\sigma, \tau\rho) \cdot u_{\sigma\tau\rho}$. Equality of these expressions reduces to the 2-cocycle condition: $\sigma(f(\tau, \rho)) \cdot f(\sigma, \tau\rho) = f(\sigma, \tau) \cdot f(\sigma\tau, \rho)$, which holds by definition 2.3.7.

Step 2: Unit and the subfield L. The cocycle condition at $(1, 1, \rho)$ reads $f(1, \rho)^2 = f(1,1) f(1, \rho)$, and at $(\sigma, 1, 1)$ it reads $\sigma(f(1,1)) f(\sigma, 1) = f(\sigma, 1)^2$. Hence $f(1, \rho) = f(1, 1)$ and $f(\sigma, 1) = \sigma(f(1,1))$ for all $\rho, \sigma \in G$. Put $e = f(1,1)^{-1} u_1$. For $c, b \in L$ and $\tau \in G$,

$$(ce)(bu_\tau) = cf(1,1)^{-1} b f(1, \tau) u_\tau = cbu_\tau, \qquad (bu_\tau)(ce) = b\tau(c)\tau(f(1,1))^{-1} f(\tau, 1) u_\tau = b\tau(c) u_\tau$$

With $c = 1$ these show that e is the identity. With $\tau = 1$ they show that $c \mapsto ce$ is an injective ring homomorphism $L \to A_f$, and the first identity says that left multiplication by ce is scalar multiplication by c. The second, with $b = 1$, reads $u_\tau c = \tau(c) u_\tau$. Next, $u_\sigma u_{\sigma^{-1}} = f(\sigma, \sigma^{-1}) u_1$ and $u_{\sigma^{-1}} u_\sigma = f(\sigma^{-1}, \sigma) u_1$ are nonzero elements of L, as $u_1 = f(1,1) e$, hence invertible, so u_σ

has a right and a left inverse and is invertible. Finally $bu_\sigma = u_\sigma \sigma^{-1}(b)$, so $Lu_\sigma = u_\sigma L$ and $A_f = \bigoplus_\sigma u_\sigma L$.

Step 3: Dimension. As an L-vector space, A_f has dimension $|G| = [L : K]$, so $\dim_K A_f = [L : K]^2$.

Step 4: Center. An element $x = \sum_\sigma a_\sigma u_\sigma$ lies in the center if and only if $xb = bx$ for all $b \in L$ and $xu_\tau = u_\tau x$ for all $\tau \in G$. By Step 2, the condition $xb = bx$ forces $a_\sigma \sigma(b) = a_\sigma b$ for all σ and b, which gives $a_\sigma = 0$ whenever $\sigma \neq \mathrm{id}$ (since $\sigma \neq \mathrm{id}$ means $\sigma(b) \neq b$ for some b). So $x = au_1$ with $a \in L$. The condition $xu_\tau = u_\tau x$ then reads $af(1,\tau)u_\tau = \tau(a)f(\tau,1)u_\tau$, that is, $af(1,1) = \tau(af(1,1))$ for all τ by Step 2. So $af(1,1) \in L^G = K$, and $x = (af(1,1))e$ lies in Ke.

Step 5: Simplicity. Let $I \subseteq A_f$ be a nonzero two-sided ideal. Choose a nonzero $x = \sum_\sigma a_\sigma u_\sigma \in I$ with the minimal number of nonzero coefficients, and some σ with $a_\sigma \neq 0$. By Step 2, $u_\sigma^{-1} = cu_{\sigma^{-1}}$ for some $c \in L^\times$, and $a_\tau u_\tau c u_{\sigma^{-1}} = a_\tau \tau(c) f(\tau, \sigma^{-1}) u_{\tau\sigma^{-1}}$. So $xu_\sigma^{-1} \in I$ has as many nonzero coefficients as x, among them the coefficient of u_1. Multiplying on the left by the inverse of that coefficient, assume $a_1 = 1$. For any $b \in L$, the commutator $bx - xb = \sum_{\sigma \neq \mathrm{id}} (b - \sigma(b)) a_\sigma u_\sigma$ lies in I and has fewer nonzero terms (the u_1-coefficient vanishes). By minimality, $bx = xb$ for all b, forcing $a_\sigma = 0$ for $\sigma \neq \mathrm{id}$ (as in Step 4). So $x = u_1 = f(1,1)e$, a unit of A_f, and $I = A_f$.

Different cocycles can produce the same Brauer class: cohomologous cocycles give Brauer-equivalent algebras.

Proposition 7.2.3: Cohomologous Cocycles Give Equivalent Algebras

Let $f, f' \in Z^2(G, L^\times)$ with $f' = f \cdot d^1\psi$ for some $\psi \colon G \to L^\times$ (i.e., $f'(\sigma,\tau) = f(\sigma,\tau) \cdot \psi(\sigma) \cdot \sigma(\psi(\tau)) \cdot \psi(\sigma\tau)^{-1}$). Then $A_{f'} \cong A_f$ as K-algebras. In particular, $[A_{f'}] = [A_f]$ in $\mathrm{Br}(K)$.

***Proof*:**
Define $\varphi \colon A_{f'} \to A_f$ by $\varphi(a \cdot u'_\sigma) = a \cdot \psi(\sigma) \cdot u_\sigma$. A direct computation shows φ is a K-algebra homomorphism: $\varphi((a \cdot u'_\sigma)(b \cdot u'_\tau)) = a\sigma(b) f'(\sigma,\tau)\psi(\sigma\tau) u_{\sigma\tau}$ and $\varphi(a \cdot u'_\sigma)\varphi(b \cdot u'_\tau) = a\psi(\sigma)\sigma(b\psi(\tau)) f(\sigma,\tau) u_{\sigma\tau}$. These are equal by the relation $f'(\sigma,\tau) = f(\sigma,\tau)\psi(\sigma)\sigma(\psi(\tau))\psi(\sigma\tau)^{-1}$. Since φ is a bijection (it maps a basis to a basis), it is an isomorphism.

A central simple algebra split by L acts on an L-vector space, and the Galois group then acts on that space by semilinear maps commuting with the algebra. These maps compose like the elements of G only up to scalars in $L^\times$, and the scalars form a 2-cocycle. This is the matrix form of the Galois descent of [19, §§2.3, 4.4].

Definition 7.2.4: Semilinear Splitting

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$, and let A be a finite-dimensional K-algebra. A *semilinear splitting* of A over L is a triple (V, ρ, F) with the following properties.

1. V is a nonzero finite-dimensional L-vector space.
2. $\rho\colon A \to \mathrm{End}_L(V)$ is a homomorphism of K-algebras with $\rho(1) = \mathrm{id}_V$ whose L-linear extension $A \otimes_K L \to \mathrm{End}_L(V)$, $a \otimes c \mapsto c\rho(a)$, is bijective.
3. $F = (F_\sigma)_{\sigma \in G}$ is a family of additive bijections $F_\sigma\colon V \to V$ with $F_\sigma(cv) = \sigma(c)F_\sigma(v)$ and $F_\sigma \rho(a) = \rho(a)F_\sigma$ for all $c \in L$, $v \in V$, $a \in A$.

Lemma 7.2.5: The Cocycle of a Semilinear Splitting

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$, and let (V, ρ, F) be a semilinear splitting of a finite-dimensional K-algebra A over L.

1. For $\sigma, \tau \in G$ there is a unique $f(\sigma, \tau) \in L^\times$ with $F_\sigma F_\tau = f(\sigma, \tau)F_{\sigma\tau}$, and $f \in Z^2(G, L^\times)$. This f is the *cocycle* of (V, ρ, F).
2. $A_f \otimes_K A \cong \mathrm{End}_K(V)$. In particular, if A is a central simple algebra over K, then $[A_f] = [A]^{-1}$ in $\mathrm{Br}(K)$.

***Proof*:**

Part (1). The map $T = F_\sigma F_\tau F_{\sigma\tau}^{-1}$ is an additive bijection, and $T(cv) = \sigma\tau(\sigma\tau)^{-1}(c)\,T(v) = cT(v)$, so T is L-linear. It commutes with every $\rho(a)$ and with every scalar, hence with the L-span of $\rho(A)$, which is $\mathrm{End}_L(V)$ by the bijectivity in definition 7.2.4. The center of $\mathrm{End}_L(V) \cong M_d(L)$, $d = \dim_L V$, consists of the scalars, so $T = f(\sigma, \tau)\,\mathrm{id}_V$ with $f(\sigma, \tau) \in L$, nonzero because T is bijective and $V \neq 0$. It is unique because $F_{\sigma\tau}$ maps onto $V \neq 0$. For $\sigma, \tau, \rho \in G$,

$$(F_\sigma F_\tau)F_\rho = f(\sigma, \tau)f(\sigma\tau, \rho)F_{\sigma\tau\rho}, \qquad F_\sigma(F_\tau F_\rho) = \sigma(f(\tau, \rho))F_\sigma F_{\tau\rho} = \sigma(f(\tau, \rho))f(\sigma, \tau\rho)F_{\sigma\tau\rho}$$

using the σ-semilinearity of F_σ in the second computation. Comparing gives $f(\sigma, \tau)f(\sigma\tau, \rho) = \sigma(f(\tau, \rho))f(\sigma, \tau\rho)$, the cocycle condition of definition 2.3.7 written multiplicatively.

Part (2). Let $n = [L : K]$ and $d = \dim_L V$. Define a K-linear map $\Psi_1\colon A_f \to \mathrm{End}_K(V)$ by $\Psi_1(\sum_\sigma a_\sigma u_\sigma) = \sum_\sigma a_\sigma F_\sigma$, where $a_\sigma \in L$ acts by scalar multiplication. Since $(aF_\sigma)(bF_\tau) = a\sigma(b)F_\sigma F_\tau = a\sigma(b)f(\sigma, \tau)F_{\sigma\tau}$, which matches the multiplication of definition 7.2.1, Ψ_1 is multiplicative. Applying F_1^{-1} to $F_1F_1 = f(1, 1)F_1$ gives $F_1 = f(1, 1)\,\mathrm{id}_V$, so Ψ_1 sends the identity $f(1, 1)^{-1}u_1$ of A_f (proposition 7.2.2) to id_V. Each $\rho(a)$ is L-linear, so it commutes with the scalars, and it commutes with each F_σ by definition 7.2.4. Hence $x \otimes a \mapsto \Psi_1(x)\rho(a)$ defines a homomorphism of K-algebras $\Psi\colon A_f \otimes_K A \to \mathrm{End}_K(V)$.

The map Ψ is surjective. Its image contains the maps $c\rho(a)$ for $c \in L$ and $a \in A$, whose sums form $\mathrm{End}_L(V)$, and the F_σ. Choose an L-basis of V, identifying $V = L^d$, and let $s_\sigma\colon V \to V$ apply σ to each coordinate. Then $F_\sigma s_\sigma^{-1}$ is an invertible L-linear map, so $\mathrm{End}_L(V)s_\sigma = \mathrm{End}_L(V)F_\sigma$ lies in the image of Ψ. A K-linear map $V \to V$ is a $d \times d$ matrix (φ_{ij}) of K-linear maps $L \to L$.

By Dedekind's lemma (*Core Algebra* [9, linear independence of Automorphisms (Dedekind's Lemma)]), the n maps $\sigma \in G$ are linearly independent over L, so $\sum_\sigma L\sigma \subseteq \mathrm{End}_K(L)$ has K-dimension $n^2 = \dim_K \mathrm{End}_K(L)$ and equals $\mathrm{End}_K(L)$. Writing $\varphi_{ij} = \sum_\sigma c_{ij\sigma}\sigma$ with $c_{ij\sigma} \in L$ gives $(\varphi_{ij}) = \sum_\sigma (c_{ij\sigma})_{i,j}\, s_\sigma$, which lies in the image.

By definition 7.2.4, $\dim_K A = \dim_L(A \otimes_K L) = \dim_L \mathrm{End}_L(V) = d^2$, so $\dim_K(A_f \otimes_K A) = n^2d^2 = (\dim_K V)^2 = \dim_K \mathrm{End}_K(V)$, using proposition 7.2.2. Hence Ψ is bijective. As $\mathrm{End}_K(V) \cong M_{nd}(K)$, when A is central simple the product $[A_f][A] = [A_f \otimes_K A]$ is the identity of $\mathrm{Br}(K)$, and $[A_f] = [A]^{-1}$.

Lemma 7.2.6: Examples of Semilinear Splittings

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$.

1. *(Existence.)* A central simple algebra A over K with $[A] \in \mathrm{Br}(L/K)$ has a semilinear splitting over L.
2. *(Crossed products.)* Let $f \in Z^2(G, L^\times)$. Let L act on $V = A_f$ by right multiplication, let $\rho(a)$ be left multiplication by $a \in A_f$, and let $F_\sigma(v) = vu_\sigma^{-1}$. Then (V, ρ, F) is a semilinear splitting of A_f over L with cocycle f^{-1}.
3. *(Tensor products.)* If (V, ρ, F) and (W, ρ', F') are semilinear splittings over L of A and of B, with cocycles f and f', then $V \otimes_L W$ with $a \otimes b \mapsto \rho(a) \otimes \rho'(b)$ and $(F_\sigma \otimes F'_\sigma)_{\sigma \in G}$ is a semilinear splitting of $A \otimes_K B$ over L with cocycle ff'.
4. *(Inflation.)* Let $L' \supseteq L$ be a finite Galois extension of K, with $G' = \mathrm{Gal}(L'/K)$ and restriction $\pi\colon G' \to G$. If (V, ρ, F) is a semilinear splitting of A over L with cocycle f, then $V \otimes_L L'$ with $a \mapsto \rho(a) \otimes \mathrm{id}_{L'}$ and $(F_{\pi(\sigma')} \otimes \sigma')_{\sigma' \in G'}$ is a semilinear splitting of A over L' whose cocycle is the inflation $(\sigma', \tau') \mapsto f(\pi(\sigma'), \pi(\tau'))$ of f (definition 3.3.1).
5. *(Restriction.)* Let $K \subseteq M \subseteq L$ be an intermediate field, with $H = \mathrm{Gal}(L/M)$. If (V, ρ, F) is a semilinear splitting of A over L with cocycle f, then V with $a \otimes c \mapsto c\rho(a)$ and $(F_h)_{h \in H}$ is a semilinear splitting of the M-algebra $A \otimes_K M$ over L whose cocycle is the restriction of f to $H \times H$.

***Proof*:**

Part (1). The class of $A \otimes_K L$ in $\mathrm{Br}(L)$ is trivial, so by definition 7.1.5 and theorem 7.1.2 there is an isomorphism of L-algebras $\Phi\colon A \otimes_K L \to \mathrm{End}_L(V)$ with $V = L^d$ for some $d \geq 1$. Put $\rho(a) = \Phi(a \otimes 1)$. The L-linear extension of ρ is Φ, which is bijective. For $\sigma \in G$, the map $\sigma_A = \mathrm{id}_A \otimes \sigma$ is a ring automorphism of $A \otimes_K L$ fixing $A \otimes 1$, with $\sigma_A(cx) = \sigma(c)\sigma_A(x)$ for $c \in L$. Let $s_\sigma\colon V \to V$ apply σ to each coordinate, so $s_\sigma c = \sigma(c)s_\sigma$. The map $\beta_\sigma(\varphi) = s_\sigma^{-1}\, \Phi\sigma_A\Phi^{-1}(\varphi)\, s_\sigma$ is a ring automorphism of $\mathrm{End}_L(V)$, and it is L-linear: $\beta_\sigma(c\varphi) = s_\sigma^{-1}\sigma(c)\, \Phi\sigma_A\Phi^{-1}(\varphi)\, s_\sigma = c\, \beta_\sigma(\varphi)$. Every L-algebra automorphism of the matrix algebra $\mathrm{End}_L(V) \cong M_d(L)$ is inner [19, Lemma 2.4.1], so $\beta_\sigma(\varphi) = h_\sigma \varphi h_\sigma^{-1}$ for some invertible $h_\sigma \in \mathrm{End}_L(V)$. Put $F_\sigma = s_\sigma h_\sigma$, an additive bijection with $F_\sigma(cv) = \sigma(c)F_\sigma(v)$. For $a \in A$,

$$F_\sigma \rho(a) F_\sigma^{-1} = s_\sigma \beta_\sigma(\rho(a)) s_\sigma^{-1} = \Phi\big(\sigma_A(a \otimes 1)\big) = \Phi(a \otimes 1) = \rho(a)$$

Part (2). By proposition 7.2.2, $V = \bigoplus_\sigma u_\sigma L$ has L-dimension $n = [L : K]$, each $\rho(a)$ is L-linear by associativity, and $\rho(1) = \mathrm{id}_V$. The L-linear extension $A_f \otimes_K L \to \mathrm{End}_L(V)$, $a \otimes c \mapsto (v \mapsto avc)$, is a nonzero homomorphism of L-algebras. Its source is a central simple algebra over L [19, Lemma 2.2.2], since A_f is central simple over K (proposition 7.2.2) and L/K is finite, so it is injective, and both sides have L-dimension n^2, so it is bijective. The map F_σ is an additive bijection with inverse $v \mapsto vu_\sigma$ that commutes with left multiplications. Multiplying $u_\sigma c = \sigma(c)u_\sigma$ by u_σ^{-1} on both sides gives $cu_\sigma^{-1} = u_\sigma^{-1}\sigma(c)$, so $F_\sigma(vc) = F_\sigma(v)\sigma(c)$. Finally

$$F_\sigma F_\tau(v) = vu_\tau^{-1}u_\sigma^{-1} = v(u_\sigma u_\tau)^{-1} = v\big(f(\sigma,\tau)u_{\sigma\tau}\big)^{-1} = vu_{\sigma\tau}^{-1}f(\sigma,\tau)^{-1}$$

which is $f(\sigma,\tau)^{-1}$ times $F_{\sigma\tau}(v)$ in the right L-structure.

Part (3). The L-vector space $V \otimes_L W$ is nonzero and $\rho(a) \otimes \rho'(b)$ is L-linear, so $a \otimes b \mapsto \rho(a) \otimes \rho'(b)$ is a unital homomorphism of K-algebras. Its L-linear extension is the composite of the isomorphisms $(A \otimes_K B) \otimes_K L \cong (A \otimes_K L) \otimes_L (B \otimes_K L)$, of the tensor product of the two bijections of definition 7.2.4, and of $\mathrm{End}_L(V) \otimes_L \mathrm{End}_L(W) \cong \mathrm{End}_L(V \otimes_L W)$, so it is bijective. The additive map $(v, w) \mapsto F_\sigma(v) \otimes F'_\sigma(w)$ satisfies $F_\sigma(cv) \otimes F'_\sigma(w) = F_\sigma(v) \otimes F'_\sigma(cw)$, as both equal $\sigma(c)F_\sigma(v) \otimes F'_\sigma(w)$. So it induces an additive bijection $F_\sigma \otimes F'_\sigma$ of $V \otimes_L W$, with inverse $F_\sigma^{-1} \otimes F_\sigma'^{-1}$, which is σ-semilinear and commutes with every $\rho(a) \otimes \rho'(b)$. Its cocycle is ff', because $(F_\sigma \otimes F'_\sigma)(F_\tau \otimes F'_\tau) = f(\sigma,\tau)F_{\sigma\tau} \otimes f'(\sigma,\tau)F'_{\sigma\tau} = f(\sigma,\tau)f'(\sigma,\tau)(F_{\sigma\tau} \otimes F'_{\sigma\tau})$.

Part (4). The L'-linear extension of $a \mapsto \rho(a) \otimes \mathrm{id}_{L'}$ is the composite of $A \otimes_K L' \cong (A \otimes_K L) \otimes_L L'$, of the base change of the bijection of definition 7.2.4, and of $\mathrm{End}_L(V) \otimes_L L' \cong \mathrm{End}_{L'}(V \otimes_L L')$, so it is bijective. For $\sigma' \in G'$ with $\sigma = \pi(\sigma')$, the restriction of σ' to L is σ, so $F_\sigma(cv) \otimes \sigma'(c') = F_\sigma(v) \otimes \sigma'(cc')$ for $c \in L$ and $c' \in L'$, and $F_\sigma \otimes \sigma'$ is a well-defined additive bijection of $V \otimes_L L'$. It is σ'-semilinear and commutes with each $\rho(a) \otimes \mathrm{id}_{L'}$. For $\sigma', \tau' \in G'$ with images σ, τ,

$$(F_\sigma \otimes \sigma')(F_\tau \otimes \tau') = F_\sigma F_\tau \otimes \sigma'\tau' = f(\sigma,\tau)\,(F_{\sigma\tau} \otimes \sigma'\tau')$$

and $\pi(\sigma'\tau') = \sigma\tau$.

Part (5). The extension L/M is Galois with group H. The map $A \otimes_K M \to \mathrm{End}_L(V)$, $a \otimes c \mapsto c\rho(a)$, is a unital homomorphism of M-algebras, and its L-linear extension is the bijection of definition 7.2.4 composed with $(A \otimes_K M) \otimes_M L \cong A \otimes_K L$. For $h \in H$ and $c \in M$, $F_h(cv) = h(c)F_h(v) = cF_h(v)$, so F_h commutes with every $c\rho(a)$. The relation $F_hF_{h'} = f(h,h')F_{hh'}$ for $h, h' \in H$ gives the cocycle.

Theorem 7.2.7: Cohomological Identification of the Relative Brauer Group

Let L/K be a finite Galois extension with $G = \mathrm{Gal}(L/K)$. The map $f \mapsto [A_f]$ induces an isomorphism of abelian groups:

$$H^2(G, L^\times) \xrightarrow{\sim} \mathrm{Br}(L/K)$$

***Proof*:**
Step 1: Well-definedness. By proposition 7.2.2, A_f is a central simple algebra over K, and by proposition 7.2.3 its class depends only on $[f] \in H^2(G, L^\times)$. By lemma 7.2.6(2), $A_f \otimes_K L \cong \mathrm{End}_L(A_f) \cong M_n(L)$ with $n = [L : K]$, so $[A_f] \in \mathrm{Br}(L/K)$. Write $\theta([f]) = [A_f]$.

Step 2: Homomorphism. Let $f, g \in Z^2(G, L^\times)$. By lemma 7.2.6(2) and (3), the central simple algebra $A_f \otimes_K A_g$ has a semilinear splitting over L with cocycle $f^{-1}g^{-1}$, and A_{fg} has one with cocycle $(fg)^{-1} = f^{-1}g^{-1}$. By lemma 7.2.5(2), $[A_{f^{-1}g^{-1}}]$ equals both $[A_f \otimes_K A_g]^{-1}$ and $[A_{fg}]^{-1}$. Hence $\theta([fg]) = [A_{fg}] = [A_f \otimes_K A_g] = \theta([f])\,\theta([g])$.

Step 3: Surjectivity. Let $[A] \in \mathrm{Br}(L/K)$. By lemma 7.2.6(1), A has a semilinear splitting over L, and by lemma 7.2.5(2) its cocycle f satisfies $\theta([f]) = [A]^{-1}$. As θ is a homomorphism, $\theta([f]^{-1}) = [A]$.

Step 4: Injectivity. Suppose $\theta([f]) = 0$, so that $A_f \cong M_n(K)$, the size n being forced by $\dim_K A_f = n^2$, and $M_n(K) = \mathrm{End}_K(K^n)$. Through this isomorphism $S = K^n$ is a left A_f-module of K-dimension n, and through the subfield $L \subseteq A_f$ of proposition 7.2.2 an L-vector space of dimension 1, so $S = Lv$ for any nonzero $v \in S$. For $\sigma \in G$, $u_\sigma v = c_\sigma v$ with $c_\sigma \in L$, and $c_\sigma \neq 0$ because u_σ is invertible. Using $u_\sigma c = \sigma(c)u_\sigma$,

$$f(\sigma,\tau)c_{\sigma\tau}v = f(\sigma,\tau)u_{\sigma\tau}v = u_\sigma u_\tau v = u_\sigma c_\tau v = \sigma(c_\tau)c_\sigma v$$

so $f(\sigma,\tau) = c_\sigma \sigma(c_\tau) c_{\sigma\tau}^{-1}$, a 2-coboundary (definition 2.3.7, written multiplicatively). So θ has trivial kernel.

Theorem 7.2.8: The Brauer Group as H^2

Let K be a field and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$.

1. For finite Galois extensions $L \subseteq L'$ of K inside K^{sep} and $f \in Z^2(\mathrm{Gal}(L/K), L^\times)$, the inflation $\mathrm{Inf} f \in Z^2(\mathrm{Gal}(L'/K), L'^\times)$ satisfies $[A_{\mathrm{Inf} f}] = [A_f]$ in $\mathrm{Br}(K)$.
2. There is an isomorphism of abelian groups $\theta_K \colon H^2(G_K, (K^{\mathrm{sep}})^\times) \to \mathrm{Br}(K)$ sending the inflation of the class of $f \in Z^2(\mathrm{Gal}(L/K), L^\times)$ to $[A_f]$, for every finite Galois extension L/K inside K^{sep}.
3. For a finite separable extension E/K inside K^{sep}, so that $E^{\mathrm{sep}} = K^{\mathrm{sep}}$, the restriction $\mathrm{Res} \colon H^2(G_K, (K^{\mathrm{sep}})^\times) \to H^2(G_E, (K^{\mathrm{sep}})^\times)$ satisfies $\theta_E \circ \mathrm{Res} = \mathrm{res}_{E/K} \circ \theta_K$.

In particular, $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$.

***Proof*:**
Part (1). By lemma 7.2.6(2) and (4), A_f has a semilinear splitting over L' with cocycle $\mathrm{Inf}(f^{-1}) = (\mathrm{Inf} f)^{-1}$. By lemma 7.2.5(2), $[A_{(\mathrm{Inf} f)^{-1}}] = [A_f]^{-1}$, and by theorem 7.2.7 for L'/K, $[A_{(\mathrm{Inf} f)^{-1}}] = [A_{\mathrm{Inf} f}]^{-1}$. Hence $[A_{\mathrm{Inf} f}] = [A_f]$.

Part (2). The open normal subgroups of G_K are the $G_L = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ for finite Galois

extensions L/K inside K^{sep}, with $G_K/G_L \cong \mathrm{Gal}(L/K)$ (restriction is onto with kernel G_L, theorem 1.4.1(3)) and $((K^{\mathrm{sep}})^\times)^{G_L} = L^\times$ (theorem 1.4.1(2)). So theorem 3.2.2 identifies $H^2(G_K, (K^{\mathrm{sep}})^\times)$ with the colimit of the groups $H^2(\mathrm{Gal}(L/K), L^\times)$ along the inflation maps. Let $\theta_L \colon H^2(\mathrm{Gal}(L/K), L^\times) \to \mathrm{Br}(L/K)$ be the isomorphism of theorem 7.2.7. By part (1), $\theta_{L'} \circ \mathrm{Inf} = \theta_L$ for $L \subseteq L'$. An element c of the colimit is the image of some $c_L \in H^2(\mathrm{Gal}(L/K), L^\times)$, and $\theta_K(c) = \theta_L(c_L)$ does not depend on the choice, since two such representatives have equal inflations to a common finite Galois extension L'. The map θ_K is a homomorphism, as any two elements are represented at a common L. It is surjective because $\mathrm{Br}(K)$ is the union of the $\mathrm{Br}(L/K)$ (proposition 7.1.10) and each θ_L maps onto $\mathrm{Br}(L/K)$. It is injective, since $\theta_K(c) = 0$ forces $\theta_L(c_L) = 0$, so $c_L = 0$ and $c = 0$.

Part (3). Let $c \in H^2(G_K, (K^{\mathrm{sep}})^\times)$, inflated from the class of $f \in Z^2(\mathrm{Gal}(L/K), L^\times)$ with L/K finite Galois. Replacing f by its inflation to a finite Galois extension of K containing L and E, assume $E \subseteq L$, and put $H = \mathrm{Gal}(L/E)$. Inflation is precomposition with the projection and restriction is restriction of the domain, so $\mathrm{Res}(c)$ is inflated from the class of $f|_{H\times H}$, and $\theta_E(\mathrm{Res}c) = [A_{f|_{H\times H}}]$, a crossed product over E. By lemma 7.2.6(2) and (5), the E-algebra $A_f \otimes_K E$, which is central simple over E [19, Lemma 2.2.2], has a semilinear splitting over L with cocycle $(f|_{H\times H})^{-1}$. By lemma 7.2.5(2) over E and theorem 7.2.7 for L/E, $[A_{f|_{H\times H}}]^{-1} = [A_f \otimes_K E]^{-1}$ in $\mathrm{Br}(E)$. Hence $\theta_E(\mathrm{Res}c) = [A_f \otimes_K E] = \mathrm{res}_{E/K}(\theta_K(c))$.

Proposition 7.2.9: The Reduced Norm

Let A be a central simple algebra over K. Then $\dim_K A = d^2$ for an integer $d \geq 1$, and for every K-basis $e_1, \ldots, e_{d^2}$ of A there is a homogeneous polynomial $N_A \in K[X_1, \ldots, X_{d^2}]$ of degree d with $N_A(x_1, \ldots, x_{d^2}) \neq 0$ whenever $\sum_i x_i e_i$ is invertible in A.

***Proof*:**
By proposition 7.1.10, $[A] \in \mathrm{Br}(L/K)$ for a finite Galois extension L/K, with group G, and by lemma 7.2.6(1) A has a semilinear splitting (V, ρ, F) over L. With $d = \dim_L V$, definition 7.2.4 gives $\dim_K A = \dim_L \mathrm{End}_L(V) = d^2$. Fix an L-basis of V, identifying $\mathrm{End}_L(V) = M_d(L)$, and for a matrix M over L or $L[X_1, \ldots, X_{d^2}]$ write $\sigma(M)$ for σ applied to the coefficients of its entries. With s_σ applying σ to each coordinate, $s_\sigma M s_\sigma^{-1} = \sigma(M)$ for $M \in M_d(L)$, and $g_\sigma = F_\sigma s_\sigma^{-1}$ is an invertible L-linear map. Since F_σ commutes with $\rho(a)$,

$$\rho(a) = F_\sigma \rho(a) F_\sigma^{-1} = g_\sigma \, \sigma(\rho(a)) \, g_\sigma^{-1} \qquad (a \in A)$$

Let $M(X) = \sum_i X_i \rho(e_i)$, a $d \times d$ matrix of linear forms in $L[X_1, \ldots, X_{d^2}]$, and $P = \det M(X)$. Since the determinant is a polynomial with integer coefficients in the entries,

$$\sigma(P) = \det\Big(\sum_i X_i \, \sigma(\rho(e_i))\Big) = \det\big(g_\sigma^{-1} M(X) g_\sigma\big) = P$$

for every $\sigma \in G$, so the coefficients of P lie in $L^G = K$. Being the determinant of a $d \times d$ matrix of linear forms, P is homogeneous of degree d, and it is nonzero because its value at the coordinates of 1_A is $\det \rho(1_A) = 1$. If $a = \sum_i x_i e_i$ is invertible, then $\rho(a)\rho(a^{-1}) = \mathrm{id}_V$, so $P(x_1, \ldots, x_{d^2}) = \det \rho(a) \neq 0$. Take $N_A = P$, the reduced norm of [19, Construction 2.6.1].

The identification converts the algebraic classification of CSAs into a cohomological computation, making the full machinery of Parts I and II applicable. For a finite separable extension L/K of degree n, theorem 7.2.8(3) and the corestriction-restriction formula (proposition 3.3.7) show that $\mathrm{res}_{L/K}$ followed by the corestriction, transported to Br through θ_L and θ_K, is multiplication by n on $\mathrm{Br}(K)$. So $n[A] = 0$ for every class $[A]$ split by L, and $\mathrm{per}(A) \big| [L : K]$ for every finite separable splitting field L of A, completing the argument deferred in proposition 7.1.12.

Corollary 7.2.10: $\mathrm{cd}_p \leq 1$ and the Brauer Group

For a field K and a prime p different from the characteristic of K: $\mathrm{cd}_p(K) \leq 1$ if and only if $\mathrm{Br}(L)\{p\} = 0$ for every finite separable extension L/K.

***Proof*:**
Let L/K be finite separable, so that $L^{\mathrm{sep}} = K^{\mathrm{sep}}$ and G_L is an open subgroup of G_K. Since p is not the characteristic, the polynomial $X^p - a$ is separable for $a \neq 0$, so $x \mapsto x^p$ is onto $(K^{\mathrm{sep}})^\times$, with kernel the group μ_p of order p. The group $H^1(G_L, (K^{\mathrm{sep}})^\times)$ vanishes by Hilbert's Theorem 90 (theorem 8.1.1(2) for the field L). So the long exact sequence of $1 \to \mu_p \to (K^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^p} (K^{\mathrm{sep}})^\times \to 1$ identifies $H^2(G_L, \mu_p)$ with the kernel of multiplication by p on $H^2(G_L, (K^{\mathrm{sep}})^\times)$, and the group isomorphism of theorem 7.2.8 gives $H^2(G_L, \mu_p) \cong \mathrm{Br}(L)[p]$. A p-primary torsion group with no element of order p is zero, so $\mathrm{Br}(L)\{p\} = 0$ if and only if $H^2(G_L, \mu_p) = 0$.

If $\mathrm{cd}_p(K) \leq 1$, then $\mathrm{cd}_p(L) \leq 1$ by proposition 5.1.3(1), so $H^2(G_L, \mu_p) = 0$ and $\mathrm{Br}(L)\{p\} = 0$.

Conversely, assume $\mathrm{Br}(L)\{p\} = 0$ for every finite separable L/K. By theorem 5.1.2 it suffices to show $H^2(G_K, M) = 0$ for every simple discrete G_K-module M with $pM = 0$. Such an M is finite (Step 1 of the proof of theorem 5.1.2), so the intersection of the stabilizers of its elements is an open normal subgroup, equal to G_E for a finite Galois extension E/K (theorem 1.4.1). Replacing E by $E(\mu_p)$, assume $\mu_p \subseteq E$. Let P be a Sylow p-subgroup of $\mathrm{Gal}(E/K)$ and $K' = E^P$. Then $[K' : K] = [\mathrm{Gal}(E/K) : P]$ is prime to p. Multiplication by p is zero on M, hence on $H^2(G_K, M)$, so $\mathrm{Res}\colon H^2(G_K, M) \to H^2(G_{K'}, M)$ is injective by corollary 3.3.8(3). The degree $[K'(\mu_p) : K']$ divides $p - 1$ and also $[E : K'] = |P|$, so $\mu_p \subseteq K'$, and $\mu_p \cong \mathbb{Z}/p\mathbb{Z}$ as $G_{K'}$-modules. The group $G_{K'}$ acts on M through the p-group $\mathrm{Gal}(E/K') = P$, so by Step 4 of the proof of theorem 5.1.2, M has a filtration by $G_{K'}$-submodules whose successive quotients are $\mathbb{Z}/p\mathbb{Z} \cong \mu_p$. Since $H^2(G_{K'}, \mu_p) = 0$ by the first paragraph, the long exact sequences along this filtration give $H^2(G_{K'}, M) = 0$, hence $H^2(G_K, M) = 0$.

The proof of proposition 5.2.3 rests on this corollary.

Example 7.2.11: Crossed Products in Practice

1. The trivial cocycle $f(\sigma, \tau) = 1$ for all σ, τ gives the split crossed product: $A_1 = \bigoplus_\sigma L \cdot u_\sigma$ with $u_\sigma u_\tau = u_{\sigma\tau}$ and identity u_1. Letting $a u_\sigma$ act on L by $v \mapsto a\sigma(v)$ defines a homomorphism of K-algebras $A_1 \to \mathrm{End}_K(L) \cong M_n(K)$, $n = [L : K]$, which is injective because A_1 is simple (proposition 7.2.2) and bijective because both sides have dimension n^2. So $A_1 \cong M_n(K)$ (exercise 7.2.1 (2)).

2. Let char$K \neq 2$, let $a \in K^\times$ be a non-square, and let $L = K(\sqrt{a})$, with $G = \{1, \sigma\}$ and $\sigma(\sqrt{a}) = -\sqrt{a}$. Every $f \in Z^2(G, L^\times)$ is cohomologous to a *normalized* cocycle, one with $f(1,1) = 1$: for $\psi\colon G \to L^\times$ with $\psi(1) = f(1,1)^{-1}$, the cocycle $f \cdot d^1\psi$ of proposition 7.2.3 takes the value $f(1,1)\psi(1) = 1$ at $(1,1)$. A normalized cocycle has $f(1,\sigma) = f(\sigma,1) = 1$ (Step 2 of the proof of proposition 7.2.2), and the cocycle condition at (σ,σ,σ) gives $\sigma(f(\sigma,\sigma)) = f(\sigma,\sigma)$, so f is determined by $b = f(\sigma,\sigma) \in K^\times$. Conversely, every $b \in K^\times$ defines a normalized cocycle in this way. For normalized f the identity of A_f is u_1, and A_f has basis $\{1, \sqrt{a}, u_\sigma, \sqrt{a} \cdot u_\sigma\}$ over K, with $u_\sigma^2 = b$ and $u_\sigma\sqrt{a} = -\sqrt{a} \cdot u_\sigma$. Setting $i = \sqrt{a}$ and $j = u_\sigma$: $i^2 = a$, $j^2 = b$, $ij = -ji$. This is the quaternion algebra $(a,b)_K$, so every class of $H^2(G, L^\times)$ maps to the class of a quaternion algebra $(a,b)_K$ under theorem 7.2.7.

3. (If the reader has read [11]) Let n be prime to the characteristic of K. The Weil pairing on an elliptic curve E/K gives a map $E[n] \times E[n] \to \mu_n$, which induces a cup product pairing $H^1(G_K, E[n]) \times H^1(G_K, E[n]) \to H^2(G_K, \mu_n)$. The argument of the first paragraph of the proof of corollary 7.2.10, with n in place of p, identifies $H^2(G_K, \mu_n)$ with $\mathrm{Br}(K)[n]$, so the pairing takes values in Brauer classes of period dividing n.

Exercise 7.2.1

1. Carry out the full associativity verification of proposition 7.2.2 in detail: expand both bracketings of $(au_\sigma)(bu_\tau)(cu_\rho)$ and show they agree using the cocycle condition.

2. Let L/K be a finite Galois extension and $n = [L : K]$. Show that the crossed product A_1 with the trivial cocycle ($f = 1$) is isomorphic to $M_n(K)$ by constructing an explicit isomorphism. *Hint:* let A_1 act on L, a K-vector space of dimension n, by $(au_\sigma)(v) = a\sigma(v)$.

3. Compute $H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times)$ directly from the periodic resolution of C_2 and identify the nontrivial class with the quaternion algebra $\mathbb{H}$.

4. Using the identification $\mathrm{Br}(K) = H^2(G_K, (K^{\mathrm{sep}})^\times)$ and the corestriction formula, show that every element of $\mathrm{Br}(K)$ has finite order. Conclude that $\mathrm{Br}(K)$ is a torsion group.

5. (If the reader has read [10]) State and prove the Skolem-Noether theorem: any two homomorphisms of K-algebras from a finite-dimensional simple K-algebra B into a central simple algebra A over K differ by conjugation by an element of $A^\times$. Deduce that every automorphism of A is inner, and recover the case of matrix algebras used in the proof of lemma 7.2.6(1).

7.3 The Brauer Group of Local and Global Fields

The identification $\mathrm{Br}(K) \cong H^2(G_K, (K^{\mathrm{sep}})^\times)$ from theorem 7.2.8 converts the classification of central simple algebras into a cohomological computation. For local and global fields, this computation yields a complete and explicit description: $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$ for a local field K, and the Brauer group of a number field is determined by local data via an exact sequence. The local result is constructed in chapter 8 and the global sequence is corollary 10.1.3 in chapter 10, both resting on inputs cited there. This section states the results and develops their consequences through explicit computations.

The Brauer group of a local field is as simple as possible: cyclic algebras suffice.

Theorem 7.3.1: The Invariant Map for Local Fields

Let K be a local field (a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$). There is a canonical isomorphism:

$$\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$$

called the *invariant map* (or *Hasse invariant*). It has the following properties:

1. For the maximal unramified extension K^{nr}/K of degree n (i.e., $[K_n^{\mathrm{nr}} : K] = n$ where K_n^{nr} is the unique unramified extension of degree n), the restriction $\mathrm{inv}_K|_{\mathrm{Br}(K_n^{\mathrm{nr}}/K)}$ is an isomorphism $\mathrm{Br}(K_n^{\mathrm{nr}}/K) \xrightarrow{\sim} \frac{1}{n}\mathbb{Z}/\mathbb{Z}$.
2. Every element of $\mathrm{Br}(K)$ is split by an unramified extension. That is, $\mathrm{Br}(K) = \bigcup_n \mathrm{Br}(K_n^{\mathrm{nr}}/K)$.
3. For a finite extension L/K of local fields, the diagram

$$\begin{array}{ccc} \mathrm{Br}(K) & \xrightarrow{\mathrm{inv}_K} & \mathbb{Q}/\mathbb{Z} \\ {\scriptstyle \mathrm{res}_{L/K}}\downarrow & & \downarrow{\scriptstyle \times[L:K]} \\ \mathrm{Br}(L) & \xrightarrow[\mathrm{inv}_L]{} & \mathbb{Q}/\mathbb{Z} \end{array}$$

commutes: $\mathrm{inv}_L(\mathrm{res}_{L/K}([A])) = [L : K] \cdot \mathrm{inv}_K([A])$.

Part (1) is proved in chapter 8 as corollary 8.1.4, and part (2) is theorem 8.1.5, whose proof is cited from [31, Ch. XII, §1, Thm. 1]. The invariant map is constructed from the identification $\mathrm{Br}(K^{\mathrm{nr}}/K) = H^2(\mathrm{Gal}(K^{\mathrm{nr}}/K), (K^{\mathrm{nr}})^\times)$ and the valuation $v \colon (K^{\mathrm{nr}})^\times \to \mathbb{Z}$. Since $\mathrm{Gal}(K^{\mathrm{nr}}/K) \cong \widehat{\mathbb{Z}}$ (topologically generated by the Frobenius), the cohomology is computable by the methods of example 2.3.8. Part (3) expresses the compatibility of the invariant map with field extensions and is the local ingredient in the global theory. This book does not prove it. Its proof treats unramified and totally ramified extensions separately: in the first case the Frobenius of L is the $[L : K]$-th power of that of K, and in the second the valuation of L restricts to $[L : K]$ times that of K. It is [31, Ch. XIII, §3, Prop. 7].

Example 7.3.2: Computing Local Invariants

1. For $K = \mathbb{Q}_p$ and the quaternion algebra $(a, b)_{\mathbb{Q}_p}$: the invariant $\mathrm{inv}_{\mathbb{Q}_p}((a, b)_{\mathbb{Q}_p}) \in \{0, 1/2\} \subseteq \mathbb{Q}/\mathbb{Z}$ (since quaternion algebras have period dividing 2). The value is 0 iff $(a, b)_{\mathbb{Q}_p}$ splits, i.e., iff $ax^2 + by^2 = 1$ is solvable in $\mathbb{Q}_p$ (exercise 7.1.1 (2)).
2. The quaternion algebra $(-1, -1)_{\mathbb{Q}_2}$ does not split over $\mathbb{Q}_2$: the equation $-x^2 - y^2 = 1$ has no solution in $\mathbb{Q}_2$ (since -1 is not a sum of two squares in $\mathbb{Q}_2$, verifiable by checking modulo 8). So $\mathrm{inv}_{\mathbb{Q}_2}((-1, -1)_{\mathbb{Q}_2}) = 1/2$.
3. For p odd: $(-1, -1)_{\mathbb{Q}_p}$ splits if and only if -1 is a sum of two squares in $\mathbb{Q}_p$. By Hensel's lemma, this reduces to checking modulo p: $(-1, -1)_{\mathbb{Q}_p}$ splits for all $p \equiv 1 \pmod 4$ (since -1 is a square mod p) and for $p = 2$ the previous item gives non-splitting. For $p \equiv 3 \pmod 4$: the equation $-x^2 - y^2 \equiv 1 \pmod p$ is solvable (counting solutions via character sums gives $p - 1 > 0$ solutions), so $(-1, -1)_{\mathbb{Q}_p}$ splits.
4. An element $[A] \in \mathrm{Br}(\mathbb{Q}_p)$ with $\mathrm{inv}_{\mathbb{Q}_p}([A]) = 1/n$ is represented by a cyclic algebra: take

the unramified extension $L/\mathbb{Q}_p$ of degree n and a uniformizer π of $\mathbb{Q}_p$, and form the cyclic algebra $(L/\mathbb{Q}_p, \mathrm{Frob}, \pi) = \bigoplus_{i=0}^{n-1} L \cdot u^i$ with $u^n = \pi$ and $u \cdot \ell = \mathrm{Frob}(\ell) \cdot u$ for $\ell \in L$. This is a division algebra of degree n with invariant $1/n$.

For the archimedean places, the Brauer group is already known.

Proposition 7.3.3: Archimedean Brauer Groups

$\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$ (with $\mathrm{inv}_{\mathbb{R}}([\mathbb{H}]) = 1/2$) and $\mathrm{Br}(\mathbb{C}) = 0$. The invariant map for $\mathbb{R}$ is $\mathrm{inv}_{\mathbb{R}} \colon \mathrm{Br}(\mathbb{R}) \xrightarrow{\sim} \frac{1}{2}\mathbb{Z}/\mathbb{Z} \subseteq \mathbb{Q}/\mathbb{Z}$.

***Proof*:**
$\mathrm{Br}(\mathbb{C}) = 0$ since $\mathbb{C}$ is algebraically closed (example 7.1.8). For $\mathrm{Br}(\mathbb{R})$: $\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}/\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{C}^\times) = H^2(C_2, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$ (exercise 7.2.1 (3)). The nontrivial class is $[\mathbb{H}]$, with $\mathrm{inv}_{\mathbb{R}}([\mathbb{H}]) = 1/2$.

The global theory assembles the local invariants into a single exact sequence.

Theorem 7.3.4: Hasse Principle for Central Simple Algebras

Let K be a number field. A central simple algebra A over K is split ($[A] = 0$ in $\mathrm{Br}(K)$) if and only if $A \otimes_K K_v$ is split for every completion K_v of K (at all finite and archimedean places v).

The ABHN theorem is a local-global principle for central simple algebras: the Brauer group $\mathrm{Br}(K)$ embeds into the product of local Brauer groups. The theorem says that the only obstruction to splitting a CSA over a number field is a collection of local obstructions, one at each place. This is the Brauer-group analogue of the Hasse-Minkowski theorem for quadratic forms.

The ABHN theorem is sharpened by the following exact sequence, which describes the image of the localization map.

Theorem 7.3.5: Brauer Group Exact Sequence

For a number field K, there is an exact sequence of abelian groups:

$$0 \to \mathrm{Br}(K) \xrightarrow{\bigoplus_v \mathrm{res}_v} \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

where the first sum is over all places v of K (finite and archimedean), and the second map sends a tuple of local Brauer classes to the sum of their invariants.

Both theorems come from chapter 10: theorem 7.3.4 is theorem 10.1.2(1) applied to a finite Galois extension of K that splits A (proposition 7.1.10), and the sequence is corollary 10.1.3. The injectivity on the left is theorem 7.3.4. Exactness in the middle has two halves. The sum of the local invariants of a global Brauer class is zero, a cohomological form of the global reciprocity law. Conversely, for any collection of local invariants $\alpha_v \in \mathbb{Q}/\mathbb{Z}$ (almost all zero, with $\alpha_v \in \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ at the real places and $\alpha_v = 0$ at the complex places) summing to zero, there exists a CSA over K with exactly those local

invariants. Surjectivity on the right is the easy part: a single inv_v at a finite place is already onto $\mathbb{Q}/\mathbb{Z}$.

Example 7.3.6: The Brauer Group Exact Sequence in Practice

1. Consider the quaternion algebra $(-1,-1)_{\mathbb{Q}}$ over $\mathbb{Q}$. At each place v, the local invariant is $\mathrm{inv}_v((-1,-1)_{\mathbb{Q}_v}) \in \{0, 1/2\}$. From example 7.3.2: the algebra is non-split at $v = 2$ (invariant $1/2$) and at $v = \infty$ (invariant $1/2$, since $(-1,-1)_{\mathbb{R}} = \mathbb{H}$). At all odd primes p: the algebra splits (invariant 0). The sum of invariants is $1/2 + 1/2 = 0$ in $\mathbb{Q}/\mathbb{Z}$, confirming the reciprocity constraint. Since the invariant is nonzero at $v = 2$, the algebra $(-1,-1)_{\mathbb{Q}}$ does not split over $\mathbb{Q}$: it is a division algebra ($(-1,-1)_{\mathbb{Q}} \cong \mathbb{H}$, viewed as a $\mathbb{Q}$-algebra... but $\mathbb{H}$ has center $\mathbb{R}$, not $\mathbb{Q}$). More precisely, $(-1,-1)_{\mathbb{Q}}$ is a 4-dimensional division algebra over $\mathbb{Q}$ that becomes $\mathbb{H}$ after extending scalars to $\mathbb{R}$.

2. The quaternion algebra $(2,5)_{\mathbb{Q}}$: compute the local invariants. At $v = \infty$: $(2,5)_{\mathbb{R}}$ splits (the equation $2x^2 + 5y^2 = 1$ has the real solution $x = 1/\sqrt{2}$, $y = 0$). So $\mathrm{inv}_\infty = 0$. At $v = p$ for $p \neq 2, 5$: the algebra splits by Hensel's lemma (since 2 and 5 are units in $\mathbb{Z}_p$ for $p \neq 2, 5$, and the norm form represents 1 modulo p by a counting argument). At $v = 5$: $2x^2 + 5y^2 = 1$ in $\mathbb{Q}_5$ requires $2x^2 \equiv 1 \pmod 5$, i.e., $x^2 \equiv 3 \pmod 5$. Since 3 is not a square modulo 5 ($1^2 = 1$, $2^2 = 4$, $3^2 = 4$, $4^2 = 1$), the algebra does not split at $v = 5$: $\mathrm{inv}_5 = 1/2$. At $v = 2$: a direct check modulo 8 shows $2x^2 + 5y^2 \equiv 1 \pmod 8$ is solvable (take $x = 1, y = 1$: $2 + 5 = 7 \equiv -1 \pmod 8$; take $x = 0, y = 1$: $5 \not\equiv 1$; take $x = 1, y = 0$: $2 \not\equiv 1$). A more careful Hensel-lifting analysis gives $\mathrm{inv}_2 = 1/2$. The reciprocity check: $\mathrm{inv}_2 + \mathrm{inv}_5 = 1/2 + 1/2 = 0$ in $\mathbb{Q}/\mathbb{Z}$. So $(2,5)_{\mathbb{Q}}$ is a non-split quaternion algebra over $\mathbb{Q}$, ramified at 2 and 5.

3. The exact sequence implies $\mathrm{Br}(\mathbb{Q}) \hookrightarrow \mathrm{Br}(\mathbb{R}) \oplus \bigoplus_p \mathrm{Br}(\mathbb{Q}_p) \cong \frac{1}{2}\mathbb{Z}/\mathbb{Z} \oplus \bigoplus_p \mathbb{Q}/\mathbb{Z}$. An element of $\mathrm{Br}(\mathbb{Q})$ is determined by a finite set of places where the invariant is nonzero, subject to the constraint that the invariants sum to zero. For quaternion algebras (period 2): the ramification set must have even cardinality (since each invariant is $1/2$ and the sum must be 0).

Corollary 7.3.7: Period Equals Index for Local and Global Fields

For a local field K: $\mathrm{per}(A) = \mathrm{ind}(A)$ for every $[A] \in \mathrm{Br}(K)$. For a number field K: $\mathrm{per}(A) = \mathrm{ind}(A)$ for every $[A] \in \mathrm{Br}(K)$.

***Proof*:**
For a local field: by theorem 7.3.1, $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$. An element of order n in $\mathbb{Q}/\mathbb{Z}$ is a/n with $\gcd(a,n) = 1$. The corresponding CSA is a cyclic algebra of degree n (constructed from the unramified extension of degree n), which is a division algebra (its invariant a/n has denominator n, so it is not split by any proper subextension). So $\mathrm{ind} = n = \mathrm{per}$.

For a number field: the ABHN theorem reduces the problem to the local case. An element $[A] \in \mathrm{Br}(K)$ with $\mathrm{per}(A) = n$ has local invariants $\mathrm{inv}_v([A]) \in \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ at each place v. By the Grunwald-Wang theorem (see [26]), there exists a cyclic extension L/K of degree n that splits A at every place. The CSA split by L has index dividing $[L : K] = n$, and since $\mathrm{per} \mid \mathrm{ind}$ (proposition 7.1.12), equality follows.

Exercise 7.3.1

1. Compute $\mathrm{inv}_{\mathbb{Q}_p}((p,p)_{\mathbb{Q}_p})$ for an odd prime p. *Hint:* determine whether $px^2 + py^2 = 1$ is solvable in $\mathbb{Q}_p$.
2. Show that a quaternion algebra $(a,b)_{\mathbb{Q}}$ (with $a,b \in \mathbb{Q}^\times$) is non-split if and only if the set of places where it has invariant $1/2$ is nonempty. Determine this set for $(a,b) = (-1,3)$.
3. Using the exact sequence of theorem 7.3.5, show that the natural map $\mathrm{Br}(\mathbb{Q}) \to \mathrm{Br}(\mathbb{Q}(\sqrt{5}))$ has kernel isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and identify the nontrivial element.
4. Let K be a totally imaginary number field (no real embeddings). Show that the archimedean contribution to the Brauer group exact sequence is trivial (all complex places give $\mathrm{Br}(K_v) = 0$), and the exact sequence becomes $0 \to \mathrm{Br}(K) \to \bigoplus_{\mathfrak{p}} \mathrm{Br}(K_{\mathfrak{p}}) \to \mathbb{Q}/\mathbb{Z} \to 0$.
5. Using the exact sequence, show that every element of $\mathrm{Br}(K)$ (for K a number field) has finite order, and that the exponent of $\mathrm{Br}(K)[n]$ (the n-torsion) is n.

7.4 The Period-Index Problem

The period and index of a Brauer class (definition 7.1.11) measure its complexity from two perspectives: the period is the order in the Brauer group (an algebraic invariant), and the index is the degree of the Wedderburn component (a geometric invariant, related to the dimension of the underlying division algebra). The divisibility $\mathrm{per}(A) \mid \mathrm{ind}(A)$ (proposition 7.1.12) and the equality per = ind for local and number fields (corollary 7.3.7) are the starting point. The period-index problem asks: for a given field K, what is the precise relationship between these two invariants?

Definition 7.4.1: The Period-Index Problem

For a field K, the *period-index problem* asks for the optimal bound $\mathrm{ind}(A) \mid \mathrm{per}(A)^{e(K)}$ that holds for all $[A] \in \mathrm{Br}(K)$, where $e(K) \geq 0$ is the smallest exponent with this property. When $e(K) = 0$, the field satisfies per = ind.

The problem is trivial for C_1-fields ($\mathrm{Br}(K) = 0$ by theorem 5.2.8(1), so there is nothing to check) and is settled for local and number fields ($e(K) = 0$ by corollary 7.3.7). The first nontrivial cases arise for function fields of curves over local fields and of surfaces over finite fields.

A guiding heuristic relates the exponent $e(K)$ to cohomological dimension: for a field K of cohomological dimension δ, one may hope that $\mathrm{ind}(A) \mid \mathrm{per}(A)^{\delta-1}$ for every $[A] \in \mathrm{Br}(K)$ [1, §0.1]. It is not a theorem, and in this generality it is false.

1. Merkurjev constructed fields of cohomological dimension 2 carrying division algebras of period 2 and index 2^m for arbitrarily large m [19, Rem. 8.4.6].
2. At the characteristic it fails prime by prime. Every field K of characteristic p has $\mathrm{cd}_p(K) \leq 1$ [30, Ch. II, §2.2, Prop. 3], so the prime-by-prime form of the heuristic, $\mathrm{ind}(A) \mid \mathrm{per}(A)^{\mathrm{cd}_p(K)-1}$ for classes of p-power period, would force $\mathrm{Br}(K)\{p\} = 0$. But the class of $\mathrm{Br}(\mathbb{F}_p((t))) \cong \mathbb{Q}/\mathbb{Z}$ with invariant $1/p$ (theorem 7.3.1) has period and index p (corollary 7.3.7). So the heuristic can hold at most for periods prime to the characteristic, and only those are considered below.

At primes ℓ different from the characteristic, the case $\delta \leq 1$ holds: if $\mathrm{cd}_\ell(K) \leq 1$, then $\mathrm{Br}(K)\{\ell\} = 0$ by corollary 7.2.10, so every class of ℓ-power period is trivial. A local field has $\mathrm{cd}_\ell = 2$ for ℓ different from its characteristic (proof of corollary 8.3.3), a number field has $\mathrm{cd}_\ell = 2$ for odd ℓ (the table of section 5.2), and on both per = ind for every class (corollary 7.3.7), as the heuristic predicts. Two theorems on function fields of surfaces fit the heuristic as well.

Theorem 7.4.2: de Jong, Lieblich

Let S be a connected smooth projective surface over a field k, and let $\alpha \in \mathrm{Br}(k(S))$.

1. (de Jong) If k is algebraically closed, then $\mathrm{ind}(\alpha) = \mathrm{per}(\alpha)$.
2. (Lieblich) If k is finite, then $\mathrm{ind}(\alpha) \,\big|\, \mathrm{per}(\alpha)^2$.

Both statements are quoted from [1, §0.1], and their proofs are beyond this book. For a prime ℓ different from the characteristic of k, the field $k(S)$ has $\mathrm{cd}_\ell(k(S)) = 2 + \mathrm{cd}_\ell(k)$ [30, Ch. II, §4.2, Prop. 11], which is 2 over an algebraically closed field and 3 over a finite field, so the exponents are those of the heuristic. The case of $\mathbb{C}(x, y, z)$, of cohomological dimension 3, is wide open [1, §0.1].

The divisibility $\mathrm{ind} \,\big|\, \mathrm{per}^e$ can be checked prime by prime: it suffices to prove $\mathrm{ind}(A_p) \,\big|\, \mathrm{per}(A_p)^e$ for each p-primary component A_p of $[A]$ (exercise 7.4.1 (3)).

Example 7.4.3: Period and Index for Specific Fields

1. $\mathrm{cd}(K) = 1$ (e.g., $K = \mathbb{F}_q$ or $K = \overline{\mathbb{F}}_q(t)$): both fields are C_1 (proposition 5.2.7), so $\mathrm{Br}(K) = 0$ by theorem 5.2.8(1), as the heuristic predicts.
2. $\mathrm{cd}(K) = 2$: for a local field K, per = ind by corollary 7.3.7, and for the function field of a surface over an algebraically closed field by theorem 7.4.2(1). Merkurjev's fields show that $\mathrm{cd}(K) = 2$ does not force this equality.
3. $\mathrm{cd}(K) = 3$: for the function field of a surface over a finite field, $\mathrm{ind} \,\big|\, \mathrm{per}^2$ by theorem 7.4.2(2). For the function field of a curve over $\mathbb{Q}_p$, Saltman bounds $\mathrm{ind}(A)$ in terms of $\mathrm{per}(A)$ when $\mathrm{per}(A)$ is prime to p [19, Rem. 4.5.15].
4. Number fields: $\mathrm{cd}_\ell(K) = 2$ for odd primes ℓ, and per = ind holds for every class (corollary 7.3.7).

The period-index problem also has a local-to-global aspect. For a number field K, the equality per = ind is a consequence of the local-global principle (ABHN theorem, theorem 7.3.4): the global index is the lcm of local indices, and at each local place $\mathrm{per}_v = \mathrm{ind}_v$. For function fields $K(C)$ of curves C over other base fields, the analogous local-global statement (the "local-global principle for the index") is itself a nontrivial conjecture.

The period-index problem connects to several other areas of mathematics.

7.4.4 Remark: Connections to Algebraic Geometry and K-Theory (If the reader has read [4] or [5]) The period-index problem has geometric and K-theoretic reformulations. A Brauer class $[A] \in \mathrm{Br}(K)$ of period n determines an element $\alpha \in H^2_{\text{ét}}(\operatorname{Spec} K, \mu_n)$, and the index of $[A]$ is the minimum degree of a finite field extension splitting α. In the language of twisted sheaves on an algebraic variety X with function field K: the period of α is the order of the associated gerbe $\mathcal{G}_\alpha$ over X, while the index measures the minimum rank of a twisted sheaf on $\mathcal{G}_\alpha$. For $K = k(X)$, with X a variety of dimension d over an algebraically closed field k, the field K has $\mathrm{cd}_\ell(K) = d$ for primes ℓ different from the characteristic [30, Ch. II, §4.2, Prop. 11], and the heuristic $\mathrm{ind} \,\big|\, \mathrm{per}^{d-1}$ becomes a statement about the geometry of twisted sheaves on X. From the K-theoretic perspective, the period-index problem is related to the structure of the Grothendieck group K_0 of the category of A-modules (for a CSA A) and the behavior of Chern classes in étale cohomology.

The period-index problem remains one of the central open questions in the theory of the Brauer group. For the purposes of this book, the equality per = ind for local and number fields (corollary 7.3.7) is the essential result: it ensures that the Brauer group of these fields is completely determined by the invariant map, and that division algebras over these fields are controlled by their cohomological data.

Exercise 7.4.1

1. Show that every quaternion algebra $(a, b)_K$ satisfies $\mathrm{per}(A) = \mathrm{ind}(A)$ over any field K with $\mathrm{char} K \neq 2$. *Hint:* per and ind are both either 1 or 2.
2. Show that if K is a C_1-field (definition 5.2.5), then $\mathrm{Br}(K) = 0$, and conclude that the period-index problem is vacuous for such fields.
3. Show that the period-index problem reduces to the case $\mathrm{per}(A) = p^k$ (a prime power): if $\mathrm{ind}(A) \,\big|\, \mathrm{per}(A)^e$ holds for all p-primary classes and all primes p, then it holds for all classes.
4. Let $K = \overline{\mathbb{F}}_q(t)$ (the rational function field over the algebraic closure of a finite field). Show that $\mathrm{Br}(K) = 0$, using that K is a C_1-field by Tsen's theorem (proposition 5.2.7(3)). What happens for $K = \mathbb{F}_q((t))$?
5. (If the reader has read [5]) A *symbol algebra* $(a, b)_{K,n}$ is the K-algebra generated by x, y with $x^n = a$, $y^n = b$, $yx = \zeta xy$ (where $\zeta \in K$ is a primitive nth root of unity). Show that $(a, b)_{K,n}$ is a CSA of degree n and period dividing n, and that the symbol algebras generate $\mathrm{Br}(K)[n]$ when K contains the nth roots of unity (this is the Merkurjev-Suslin theorem).

Part III

Galois Cohomology of Fields

Parts I and II built the cohomological machine and identified its structural invariants. Part III applies the machine to the Galois groups of local and global fields, proving the central theorems of arithmetic duality and class field theory in their cohomological formulation.

Chapter 8 computes the Galois cohomology of local fields: Hilbert 90 (both multiplicative and additive), the filtered cohomology of the unit group, local Tate duality, and the local reciprocity law via class formations and the Tate-Nakayama theorem. Chapter 9 develops the adèle ring and idèle group, providing the global Galois modules needed for chapter 10. Chapter 10 states the Albert-Brauer-Hasse-Noether theorem and the global fundamental class, whose proofs rest on the inequalities of global class field theory cited from [2, Ch. VII], presents the Poitou-Tate exact sequence and the global Euler-Poincaré characteristic, states the global reciprocity law, and connects the theory to Selmer groups and the Shafarevich-Tate group. Chapter 11 moves beyond the abelian setting: Kummer theory, Galois descent, and non-abelian H^1 and H^2.

8

Local Fields

The abstract cohomological machine of Parts I and II now meets the arithmetic of local fields. A local field K, a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$, has absolute Galois group G_K whose structure was described in example 1.4.5: a filtration by inertia and wild inertia, with the unramified quotient topologically generated by the Frobenius. This chapter computes $H^n(G_K, M)$ for the coefficient modules that arise in number theory, records local Tate duality and the local Euler-Poincaré characteristic with their proofs cited, and constructs the local reciprocity map from the local class formation. The results fulfill the promises of chapter 7: the invariant map, the structure of $\mathrm{Br}(K)$, and the period-index equality rest on the computations developed here and on the results cited with them.

8.1 Cohomology of Local Fields: Unramified and Tamely Ramified Extensions

The first step in understanding the Galois cohomology of local fields is the vanishing theorem of Hilbert, which holds over every field: the first cohomology of the multiplicative group is trivial. This result, together with the explicit structure of unramified extensions, determines the Brauer group of a local field and constructs the invariant map.

Theorem 8.1.1: Hilbert's Theorem 90 (Multiplicative Form)

Let K be any field.

1. For any finite Galois extension L/K: $H^1(\mathrm{Gal}(L/K), L^\times) = 0$.
2. $H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$, where $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$.

***Proof*:**
Part (1). Write $G = \mathrm{Gal}(L/K)$. A 1-cocycle $a\colon G \to L^\times$ satisfies $a_{\sigma\tau} = a_\sigma \cdot \sigma(a_\tau)$ for all $\sigma, \tau \in G$ (definition 2.3.5). By Dedekind's lemma (*Core Algebra* [9, linear independence of Automorphisms (Dedekind's Lemma)]), the distinct automorphisms $\sigma \in G$, viewed as functions $L \to L$, are linearly independent over L. The coefficients a_σ are nonzero, so the function $c \mapsto \sum_{\sigma\in G} a_\sigma \cdot \sigma(c)$ is not identically zero, and there is $c \in L$ with $b = \sum_{\sigma\in G} a_\sigma \cdot \sigma(c) \neq 0$. For such c, the cocycle condition gives:

$$\tau(b) = \sum_\sigma \tau(a_\sigma) \cdot \tau\sigma(c) = \sum_\sigma a_\tau^{-1} a_{\tau\sigma} \cdot \tau\sigma(c) = a_\tau^{-1} \sum_\rho a_\rho \cdot \rho(c) = a_\tau^{-1} \cdot b$$

where the second equality uses the cocycle condition in the form $\tau(a_\sigma) = a_\tau^{-1} a_{\tau\sigma}$, and the third substitutes $\rho = \tau\sigma$. Rearranging: $a_\tau = b/\tau(b)$ for all $\tau \in G$. This is a 1-coboundary (definition 2.3.6), so $[a] = 0$ in $H^1(G, L^\times)$.

Part (2). The open normal subgroups of G_K are the subgroups $G_L = \mathrm{Gal}(K^{\mathrm{sep}}/L)$ for finite Galois extensions L/K inside K^{sep}, with $G_K/G_L \cong \mathrm{Gal}(L/K)$ and $((K^{\mathrm{sep}})^\times)^{G_L} = L^\times$ (theorem 1.4.1). By theorem 3.2.2, $H^1(G_K, (K^{\mathrm{sep}})^\times)$ is the colimit of the groups $H^1(\mathrm{Gal}(L/K), L^\times)$, which vanish by part (1).

For L/K cyclic, with σ a generator of $G = \mathrm{Gal}(L/K)$, Hilbert's Theorem 90 says that the elements of norm 1 in $L^\times$ are exactly the quotients $b/\sigma(b)$ with $b \in L^\times$. By theorem 4.3.4, $\hat{H}^{-1}(G, L^\times) \cong \hat{H}^1(G, L^\times)$, which is $H^1(G, L^\times) = 0$ by proposition 4.1.5(1) and theorem 8.1.1(1). Written multiplicatively, proposition 4.1.5(3) identifies $\hat{H}^{-1}(G, L^\times)$ with the kernel of $N_{L/K}$ modulo the subgroup generated by the quotients $\tau(m)/m$ with $\tau \in G$ and $m \in L^\times$. As $\sigma^k(m)/m = \prod_{i=0}^{k-1} \sigma(\sigma^i(m))/\sigma^i(m)$, that subgroup is $\{\sigma(m)/m \mid m \in L^\times\}$, the set of quotients $b/\sigma(b)$ with $b = m^{-1}$.

Theorem 8.1.2: Hilbert's Theorem 90 (Additive Form)

Let K be any field.

1. For any finite Galois extension L/K: $H^n(\mathrm{Gal}(L/K), L) = 0$ for all $n \geq 1$.
2. $H^n(G_K, K^{\mathrm{sep}}) = 0$ for all $n \geq 1$, where $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$.

***Proof*:**
Part (1). Write $G = \mathrm{Gal}(L/K)$. The normal basis theorem [9, Normal Basis Theorem] states that $L \cong K[G]$ as G-modules: there exists $\theta \in L$ such that $\{\sigma(\theta)\}_{\sigma\in G}$ is a K-basis for L. Thus L is a free $K[G]$-module of rank 1, so $L \cong K[G] = \mathbb{Z}[G] \otimes_{\mathbb{Z}} K = \mathrm{Ind}^G(K)$ as G-modules. Since G is finite, $\mathrm{Ind}^G(K) \cong \mathrm{CoInd}^G(K)$ (the discussion after definition 2.2.4, and exercise 2.2.1 (4)), and coinduced modules are acyclic (theorem 2.2.6), giving $H^n(G, L) = 0$ for all $n \geq 1$.

Part (2). As in the proof of theorem 8.1.1(2), with $(K^{\mathrm{sep}})^{G_L} = L$, theorem 3.2.2 identifies $H^n(G_K, K^{\mathrm{sep}})$ with the colimit of the groups $H^n(\mathrm{Gal}(L/K), L)$, which vanish by part (1).

From here to the end of this section, K denotes a local field with ring of integers $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K$, residue field $\mathbb{F}_K = \mathcal{O}_K/\mathfrak{m}_K$ of cardinality q, and uniformizer π_K. For a finite Galois extension

L/K, write $G = \mathrm{Gal}(L/K)$ and use the analogous notation $\mathcal{O}_L, \mathfrak{m}_L, \mathbb{F}_L, \pi_L$ for L.

With Hilbert 90 established, the cohomology of unramified extensions can be computed explicitly. The key input is the decomposition of $L^\times$ by the valuation.

Let L/K be an unramified extension of degree n, so that $G = \mathrm{Gal}(L/K)$ is cyclic of order n, generated by the Frobenius Frob (example 1.4.5). The valuation gives a short exact sequence of G-modules:

$$1 \to \mathcal{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbb{Z} \to 0$$

where G acts trivially on $\mathbb{Z}$ (since $v_L(\sigma(x)) = v_L(x)$ for all $\sigma \in G$ in an unramified extension).

Theorem 8.1.3: Cohomology of Unramified Extensions

Let L/K be an unramified extension of local fields with $G = \mathrm{Gal}(L/K)$ cyclic of order n. Then:

1. $H^1(G, \mathcal{O}_L^\times) = 0$.
2. $H^2(G, L^\times) \cong \mathbb{Z}/n\mathbb{Z}$.
3. $\hat{H}^0(G, L^\times) = K^\times / N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$.
4. $\hat{H}^0(G, \mathcal{O}_L^\times) = 0$, that is, $N_{L/K}(\mathcal{O}_L^\times) = \mathcal{O}_K^\times$.

***Proof*:**

Let π be a uniformizer of K and $q = |\mathbb{F}_K|$. Since L/K is unramified, $[L : K] = [\mathbb{F}_L : \mathbb{F}_K]$ [14, Unramified Extension], so the ramification index is 1 [14, The Fundamental Identity For Local Fields]: π is a uniformizer of L, and v_L restricts to v_K on K. Reduction modulo π maps G isomorphically onto $\mathrm{Gal}(\mathbb{F}_L/\mathbb{F}_K)$ [14, Unramified Extension And Residue Field Category Equivalence].

Norms of units. Write $N = N_{L/K}$, so $N(x) = \prod_{g \in G} g(x)$. Each $g \in G$ preserves $\mathcal{O}_L$ and $\pi\mathcal{O}_L$, so modulo π the norm of a unit reduces to the norm $\mathbb{F}_L^\times \to \mathbb{F}_K^\times$ of the extension $\mathbb{F}_L/\mathbb{F}_K$ of degree n. That norm is $y \mapsto y^{(q^n-1)/(q-1)}$ on the cyclic group $\mathbb{F}_L^\times$ of order $q^n - 1$, and its image is the subgroup of order $q - 1$, which is $\mathbb{F}_K^\times$. Next let $i \geq 1$ and $a \in \mathcal{O}_L$. Expanding the product,

$$N(1 + \pi^i a) = \prod_{g \in G} \big(1 + \pi^i g(a)\big) \equiv 1 + \pi^i \,\mathrm{Tr}_{L/K}(a) \pmod{\pi^{i+1}}$$

and $\mathrm{Tr}_{L/K}$ reduces modulo π to the trace of $\mathbb{F}_L/\mathbb{F}_K$, which is $\mathbb{F}_K$-linear and nonzero, being the sum of the distinct automorphisms of $\mathbb{F}_L$ over $\mathbb{F}_K$, which are linearly independent (*Core Algebra* [9, linear independence of Automorphisms (Dedekind's Lemma)]), hence surjective onto $\mathbb{F}_K$. So for every $b \in \mathcal{O}_K$ there is $a \in \mathcal{O}_L$ with $N(1 + \pi^i a) \in (1 + \pi^i b)(1 + \pi^{i+1}\mathcal{O}_K)$. Now let $u \in \mathcal{O}_K^\times$. Choose $x_0 \in \mathcal{O}_L^\times$ with $u_1 = u\,N(x_0)^{-1} \in 1 + \pi\mathcal{O}_K$, and inductively $x_i \in 1 + \pi^i\mathcal{O}_L$ with $u_{i+1} = u_i\,N(x_i)^{-1} \in 1 + \pi^{i+1}\mathcal{O}_K$. The partial products $y_m = x_0 x_1 \cdots x_m$ satisfy $y_{m+1} - y_m \in \pi^{m+1}\mathcal{O}_L$, so they converge to some $y \in \mathcal{O}_L^\times$, the field L being complete [14, Extension of DVR's, Completion]. As $N(y_m) = u\,u_{m+1}^{-1}$ tends to u and N is continuous, $N(y) = u$. Hence $N(\mathcal{O}_L^\times) = \mathcal{O}_K^\times$.

1. The valuation sequence $1 \to \mathcal{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbb{Z} \to 0$, with G acting trivially on $\mathbb{Z}$, gives the

exact sequence

$$K^\times \xrightarrow{v_K} \mathbb{Z} \xrightarrow{\delta} H^1(G, \mathcal{O}_L^\times) \to H^1(G, L^\times)$$

in which the first map is v_K because v_L restricts to v_K on K. Since v_K is surjective, $\delta = 0$, so $H^1(G, \mathcal{O}_L^\times)$ injects into $H^1(G, L^\times)$, which is 0 by theorem 8.1.1.

2. By theorem 4.3.4 and proposition 4.1.5(1), $H^2(G, L^\times) = \hat{H}^2(G, L^\times) \cong \hat{H}^0(G, L^\times)$, and the exact sequence displayed in (3) shows that this group is $\mathbb{Z}/n\mathbb{Z}$.

3. The long exact sequence of proposition 4.1.6 for the valuation sequence contains

$$\hat{H}^0(G, \mathcal{O}_L^\times) \to \hat{H}^0(G, L^\times) \to \hat{H}^0(G, \mathbb{Z}) \to \hat{H}^1(G, \mathcal{O}_L^\times)$$

The first term is $\mathcal{O}_K^\times / N(\mathcal{O}_L^\times) = 0$ by the surjectivity of the norm on units shown above, and the last vanishes by (1). The norm acts on the trivial module $\mathbb{Z}$ as multiplication by n, so $\hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$ (proposition 4.1.5(3)). Hence $\hat{H}^0(G, L^\times) \cong \mathbb{Z}/n\mathbb{Z}$, and $\hat{H}^0(G, L^\times) = K^\times / N_{L/K}(L^\times)$ by definition 4.1.2.

4. The invariants of $\mathcal{O}_L^\times$ are $\mathcal{O}_K^\times$, so $\hat{H}^0(G, \mathcal{O}_L^\times) = \mathcal{O}_K^\times / N(\mathcal{O}_L^\times)$ (definition 4.1.2), which is trivial by the surjectivity of the norm on units, completing the proof.

Part (3) identifies $K^\times / N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$ for an unramified extension of degree n. The norm map $N_{L/K}$ surjects onto $\mathcal{O}_K^\times$ (part (4)), and the cokernel of $N_{L/K}$ on $K^\times$ is generated by the class of any uniformizer π_K (since $v_K(\pi_K) = 1$ generates $\mathbb{Z}/n\mathbb{Z}$ under the isomorphism $K^\times / N_{L/K}(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$).

The construction of the invariant map follows.

Corollary 8.1.4: Construction of the Invariant Map

Let K be a local field with residue field $\mathbb{F}_q$. For $n \geq 1$ let K_n^{nr}/K be the unramified extension of degree n, with group G_n, and let $\mathrm{Frob}_n \in G_n$ be the Frobenius, the element acting as $y \mapsto y^q$ on the residue field. Let inv_n be the composite

$$H^2(G_n, (K_n^{\mathrm{nr}})^\times) \xrightarrow{v_*} H^2(G_n, \mathbb{Z}) \xleftarrow{\delta} H^1(G_n, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G_n, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\chi \mapsto \chi(\mathrm{Frob}_n)} \mathbb{Q}/\mathbb{Z}$$

where v is the valuation of K_n^{nr} and δ is the connecting homomorphism of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$, with G_n acting trivially. Then v_* and δ are isomorphisms, inv_n is an isomorphism onto $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, and $\mathrm{inv}_n \circ \mathrm{Inf} = \mathrm{inv}_m$ on $H^2(G_m, (K_m^{\mathrm{nr}})^\times)$ for $m \mid n$. Inflation into $H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$ (theorem 7.2.8) maps $H^2(G_n, (K_n^{\mathrm{nr}})^\times)$ isomorphically onto $\mathrm{Br}(K_n^{\mathrm{nr}}/K)$, and through these maps the inv_n define an isomorphism $\mathrm{inv}_K \colon \mathrm{Br}(K^{\mathrm{nr}}/K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$.

***Proof*:**

Step 1: The three maps. Write $L = K_n^{\mathrm{nr}}$. The valuation sequence $1 \to \mathcal{O}_L^\times \to L^\times \to \mathbb{Z} \to 0$ gives an exact sequence $H^2(G_n, \mathcal{O}_L^\times) \to H^2(G_n, L^\times) \xrightarrow{v_*} H^2(G_n, \mathbb{Z}) \to H^3(G_n, \mathcal{O}_L^\times)$. By theorem 4.3.4 the outer groups are isomorphic to $\hat{H}^0(G_n, \mathcal{O}_L^\times)$ and $\hat{H}^1(G_n, \mathcal{O}_L^\times)$ (proposition 4.1.5(1)), which vanish by theorem 8.1.3(4) and (1), so v_* is an isomorphism. For $q \geq 1$ the group $H^q(G_n, \mathbb{Q})$ is killed by n (corollary 3.3.8(4)), which acts invertibly on it, so it vanishes and δ is an isomorphism. Reduction maps G_n isomorphically onto $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ [14, Unramified Extension And Residue

Field Category Equivalence], which is cyclic of order n generated by $y \mapsto y^q$, so Frob_n generates G_n and evaluation at Frob_n maps $\mathrm{Hom}(G_n, \mathbb{Q}/\mathbb{Z})$ isomorphically onto $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$.

Step 2: Inflation. Let $m \mid n$, so that $K_m^{\mathrm{nr}} \subseteq K_n^{\mathrm{nr}}$. The valuation of K_n^{nr} restricts to that of K_m^{nr}, both extensions being unramified, so inflation commutes with v_*. Inflation commutes with δ, being a morphism of cohomological functors [31, Ch. VII, §5]. On $\mathrm{Hom}(G_m, \mathbb{Q}/\mathbb{Z})$ inflation is composition with the projection $G_n \to G_m$, which maps Frob_n to Frob_m, as both act as $y \mapsto y^q$ on residue fields. So $\mathrm{inv}_n(\mathrm{Inf}x) = \mathrm{inv}_m(x)$.

Step 3: The union. Inflation $H^2(G_n, (K_n^{\mathrm{nr}})^\times) \to \mathrm{Br}(K)$ is injective with image the classes split by K_n^{nr}, that is $\mathrm{Br}(K_n^{\mathrm{nr}}/K)$ [31, Ch. X, §4, Prop. 6 and Cor.]. If classes x and y over K_m^{nr} and K_n^{nr} have the same image in $\mathrm{Br}(K)$, their inflations to G_N, for N a common multiple of m and n, coincide, inflation being transitive and injective, so $\mathrm{inv}_m(x) = \mathrm{inv}_N(\mathrm{Inf}x) = \mathrm{inv}_N(\mathrm{Inf}y) = \mathrm{inv}_n(y)$ by Step 2. So the inv_n define a homomorphism on $\mathrm{Br}(K^{\mathrm{nr}}/K) = \bigcup_n \mathrm{Br}(K_n^{\mathrm{nr}}/K)$, injective because each inv_n is, with image $\bigcup_n \frac{1}{n}\mathbb{Z}/\mathbb{Z} = \mathbb{Q}/\mathbb{Z}$, completing the proof.

The remaining input to theorem 7.3.1 is that every element of $\mathrm{Br}(K)$ is split by an unramified extension.

Theorem 8.1.5: Every CSA Is Split by an Unramified Extension

$\mathrm{Br}(K) = \mathrm{Br}(K^{\mathrm{nr}}/K)$. In particular, $\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$.

***Proof Key ideas*:**
This book does not prove the first assertion. It is [31, Ch. XII, §1, Thm. 1], whose proof uses $\mathrm{Br}(K^{\mathrm{nr}}) = 0$: a class of $\mathrm{Br}(K)$ dies in $\mathrm{Br}(K^{\mathrm{nr}})$, and as K^{nr} is the union of the finite unramified extensions K' of K, it already dies in some $\mathrm{Br}(K')$. A second proof, which finds an unramified maximal subfield in every central division algebra over K, is [31, Ch. XII, §2, Prop. 2]. With $\mathrm{Br}(K) = \mathrm{Br}(K^{\mathrm{nr}}/K)$, the second assertion is corollary 8.1.4, completing the sketch.

Example 8.1.6: Explicit Computations for $\mathbb{Q}_p$

1. For $K = \mathbb{Q}_p$ and $L = \mathbb{Q}_{p^2}$ (the unramified quadratic extension): $G = \langle \mathrm{Frob} \rangle \cong C_2$. The norm $N_{L/K} \colon L^\times \to K^\times$ satisfies $K^\times / N(L^\times) \cong \mathbb{Z}/2\mathbb{Z}$, generated by the class of p (any uniformizer). The units $\mathbb{Z}_p^\times$ are norms (since the norm surjects onto $\mathcal{O}_K^\times$ by theorem 8.1.3(4)). The Brauer class with invariant $1/2$ is represented by the crossed product $(L/K, \mathrm{Frob}, p)$: the cyclic algebra $L \oplus L \cdot u$ with $u^2 = p$ and $u \cdot \ell = \mathrm{Frob}(\ell) \cdot u$ for $\ell \in L$.
2. The fundamental class $u_{\mathbb{Q}_{p^n}/\mathbb{Q}_p} \in H^2(C_n, \mathbb{Q}_{p^n}^\times)$ is the cocycle $f(\mathrm{Frob}^i, \mathrm{Frob}^j) = \begin{cases} 1 & i+j < n \\ p & i+j \geq n \end{cases}$ (for $0 \leq i, j < n$). Its invariant is $1/n \in \mathbb{Q}/\mathbb{Z}$. The crossed product with this cocycle is the unique division algebra over $\mathbb{Q}_p$ of degree n.
3. (If the reader has read [14]) The Hilbert symbol $(a,b)_p = \mathrm{inv}_{\mathbb{Q}_p}((a,b)_{\mathbb{Q}_p}) \in \{0, 1/2\}$ for $a, b \in \mathbb{Q}_p^\times$ computes the invariant of the quaternion algebra. The values computed in example 7.3.2 are recovered: $(a,b)_p = 0$ iff $ax^2 + by^2 = 1$ is solvable in $\mathbb{Q}_p$.

Exercise 8.1.1

1. Let $L = K(\sqrt{a})$ be a quadratic extension of a local field K (with $\mathrm{char} K \neq 2$). Verify Hilbert 90 directly: show that if $x \in L^\times$ satisfies $N_{L/K}(x) = 1$, then $x = b/\sigma(b)$ for some $b \in L^\times$, where σ is the nontrivial automorphism.
2. Let L/K be the unramified extension of degree n. Show that the norm map $N_{L/K} \colon \mathcal{O}_L^\times \to \mathcal{O}_K^\times$ is surjective by reducing to the surjectivity of the norm $N_{\mathbb{F}_L/\mathbb{F}_K} \colon \mathbb{F}_L^\times \to \mathbb{F}_K^\times$ on residue fields and applying Hensel's lemma.
3. Compute the invariant of the cyclic algebra $(\mathbb{Q}_{p^3}/\mathbb{Q}_p, \mathrm{Frob}, p^2)$ (the crossed product with $u^3 = p^2$). Show that $\mathrm{inv}_{\mathbb{Q}_p} = 2/3 \in \mathbb{Q}/\mathbb{Z}$.
4. Let L/K be a totally ramified cyclic extension of degree p (with $\mathrm{char} K = 0$). Show that $H^1(\mathrm{Gal}(L/K), \mathcal{O}_L^\times) \neq 0$ in general. *Hint:* use the exact sequence $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$ and the fact that G acts trivially on $\mathbb{F}_L^\times$ (since L/K is totally ramified).
5. Give a direct proof that the quaternion algebra $(-1, p)_{\mathbb{Q}_p}$ (for p odd) is split by the unramified quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$ by exhibiting an explicit embedding $\mathbb{Q}_{p^2} \hookrightarrow M_2(\mathbb{Q}_p)$ that commutes with the quaternion algebra structure.

8.2 Filtered Cohomology: Higher Unit Groups

For unramified extensions, the cohomology of local units vanishes (theorem 8.1.3(1)). For ramified extensions, the situation is more complex: the unit group $\mathcal{O}_L^\times$ has a natural filtration by the higher unit groups $U_L^{(i)}$ that reflects the ramification filtration of the Galois group, and the cohomology of this filtration connects the abstract theory to the arithmetic of ramification.

Definition 8.2.1: Higher Unit Groups

Let L be a local field with ring of integers $\mathcal{O}_L$ and maximal ideal $\mathfrak{m}_L$. The *higher unit groups* are:

$$U_L^{(0)} = \mathcal{O}_L^\times, \qquad U_L^{(i)} = 1 + \mathfrak{m}_L^i \quad \text{for } i \geq 1$$

These form a descending filtration $\mathcal{O}_L^\times = U_L^{(0)} \supset U_L^{(1)} \supset U_L^{(2)} \supset \cdots$ with $\bigcap_i U_L^{(i)} = \{1\}$.

The graded pieces of this filtration have a clean algebraic structure.

Proposition 8.2.2: Graded Pieces of the Unit Filtration

There are isomorphisms of abelian groups:

1. $U_L^{(0)}/U_L^{(1)} \cong \mathbb{F}_L^\times$ (the residue field multiplicative group), via $u \mapsto u \bmod \mathfrak{m}_L$.
2. $U_L^{(i)}/U_L^{(i+1)} \cong \mathbb{F}_L^+$ (the residue field additive group) for $i \geq 1$, via $1 + a\pi_L^i \mapsto a \bmod \mathfrak{m}_L$, where π_L is a uniformizer.

If L/K is Galois with $G = \mathrm{Gal}(L/K)$, each $U_L^{(i)}$ is a G-submodule of $\mathcal{O}_L^\times$, and the isomorphisms are G-equivariant (where G acts on $\mathbb{F}_L$ via the natural action on the residue field).

***Proof*:**
For (1): the map $\mathcal{O}_L^\times \to \mathbb{F}_L^\times$ sending $u \mapsto \bar{u}$ is a surjective group homomorphism with kernel $U_L^{(1)} = 1 + \mathfrak{m}_L$. For (2): the map $U_L^{(i)} \to \mathbb{F}_L^+$ sending $1 + a\pi_L^i \mapsto \bar{a}$ is a surjective group homomorphism (using $(1 + a\pi_L^i)(1 + b\pi_L^i) = 1 + (a+b)\pi_L^i + ab\pi_L^{2i} \equiv 1 + (a+b)\pi_L^i \pmod{\mathfrak{m}_L^{i+1}}$) with kernel $U_L^{(i+1)}$. The G-equivariance follows from $\sigma(1 + a\pi_L^i) = 1 + \sigma(a)\sigma(\pi_L)^i$ and the fact that $\sigma(\pi_L)/\pi_L \in U_L^{(0)}$ (so the leading term modulo $\mathfrak{m}_L^{i+1}$ is $\sigma(a) \cdot (\sigma(\pi_L)/\pi_L)^i$, which reduces to $\bar{\sigma}(\bar{a})$ in $\mathbb{F}_L$).

The filtration interacts with the ramification filtration of the Galois group in a precise way.

Proposition 8.2.3: Ramification Groups and Unit Groups

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. The ramification subgroups G_i (in lower numbering, see [14, Ramification Groups in the Lower Numbering]) satisfy:

1. G_0 (the inertia group) acts trivially on $\mathbb{F}_L^\times$: $\sigma(\bar{u}) = \bar{u}$ for all $\sigma \in G_0$ and $u \in \mathcal{O}_L^\times$.
2. G_i acts trivially on $U_L^{(0)}/U_L^{(i)}$ for each $i \geq 1$.
3. G_1 (the wild inertia) acts trivially on $U_L^{(0)}/U_L^{(1)} \cong \mathbb{F}_L^\times$.

***Proof*:**
For (1): $G_0 = \mathrm{Gal}(L/L \cap K^{\mathrm{nr}})$ fixes the maximal unramified subextension, hence acts trivially on the residue field $\mathbb{F}_L$ (which is determined by the unramified part). For (2): by definition, $\sigma \in G_i$ satisfies $v_L(\sigma(\pi_L) - \pi_L) \geq i + 1$ (see [14, Ramification Groups in the Lower Numbering]), so $\sigma(1 + a\pi_L^j) = 1 + \sigma(a)\sigma(\pi_L)^j \equiv 1 + a\pi_L^j \pmod{\mathfrak{m}_L^{j+i}}$ for $j < i$, giving trivial action on $U_L^{(j)}/U_L^{(j+i)}$. In particular, G_i acts trivially on each quotient $U_L^{(j)}/U_L^{(j+1)}$ for $j < i$, hence on $U_L^{(0)}/U_L^{(i)}$. Part (3) is the special case $i = 1$.

The interaction between ramification and the unit filtration governs the cohomology. The tamely ramified case (where the wild inertia G_1 is trivial) is significantly simpler than the wildly ramified case.

Theorem 8.2.4: Cohomology of Tamely Ramified Extensions

Let L/K be a finite tamely ramified Galois extension (i.e., $G_1 = \{1\}$, so the ramification is concentrated in the tame inertia $G_0/G_1 = G_0$, which has order coprime to the residue characteristic p). Then:

1. $H^1(G, U_L^{(1)}) = 0$.
2. The long exact sequence for $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$ gives $H^1(G, \mathcal{O}_L^\times) \cong \ker(H^1(G_0, \mathbb{F}_L^\times) \to H^2(G, U_L^{(1)}))$.

***Proof*:**

Step 1: Vanishing of $H^1(G, U_L^{(1)})$. Each graded piece $U_L^{(i)}/U_L^{(i+1)} \cong \mathbb{F}_L^+$ (for $i \geq 1$) is a finite G-module. The additive form of Hilbert 90 (theorem 8.1.2) applied to the residue field extension $\mathbb{F}_L/\mathbb{F}_K$ gives $H^1(G_0/G_1, \mathbb{F}_L^+) = 0$ (since $G_0 = G_0/G_1$ acts on $\mathbb{F}_L$ and the normal basis theorem applies to $\mathbb{F}_L/\mathbb{F}_K$). The vanishing propagates up the filtration: the long exact sequence for $1 \to U_L^{(i+1)} \to U_L^{(1)} \to U_L^{(1)}/U_L^{(i+1)} \to 1$ and induction on i give $H^1(G, U_L^{(1)}/U_L^{(i)}) = 0$ for all i. Passing to the limit (using $U_L^{(1)} = \varprojlim_i U_L^{(1)}/U_L^{(i)}$ and the compatibility of cohomology with limits of discrete modules) gives $H^1(G, U_L^{(1)}) = 0$.

Part (2) follows from the long exact sequence applied to the short exact sequence $1 \to U_L^{(1)} \to \mathcal{O}_L^\times \to \mathbb{F}_L^\times \to 1$: the vanishing $H^1(G, U_L^{(1)}) = 0$ gives $H^1(G, \mathcal{O}_L^\times) \hookrightarrow H^1(G, \mathbb{F}_L^\times)$, and the connecting map $\mathbb{F}_K^\times \to H^1(G, U_L^{(1)}) = 0$ shows the inflation $H^1(G/G_0, (\mathbb{F}_L^\times)^{G_0}) \to H^1(G, \mathbb{F}_L^\times)$ captures the full H^1.

For wildly ramified extensions ($G_1 \neq \{1\}$), the cohomology of $U_L^{(1)}$ is nontrivial. The wild part of the cohomology is a p-group and is controlled by the higher ramification groups.

Proposition 8.2.5: Wild Cohomology

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$ and $G_1 \neq \{1\}$ (wild ramification). Then $H^1(G_1, U_L^{(1)})$ is a p-group, and its order is related to the discriminant of L/K via the formula:

$$|H^1(G_1, U_L^{(1)})| = |G_1|/e_{\text{wild}}$$

where $e_{\text{wild}} = |G_1|$ is the wild part of the ramification index (the precise relationship involves the higher ramification breaks; see [14] and [30]).

The detailed structure of $H^1(G_1, U_L^{(i)})$ for each layer of the filtration depends on the ramification breaks of L/K. A full treatment requires the Hasse-Arf theorem and the upper numbering of ramification groups. The essential point for the applications in this chapter is the Herbrand quotient computation, which holds uniformly regardless of ramification type.

Theorem 8.2.6: Herbrand Quotient for Local Fields

Let L/K be a cyclic extension of local fields of degree n with $G = \mathrm{Gal}(L/K) \cong C_n$. Then:

1. $h(G, L^\times) = n$.
2. $h(G, \mathcal{O}_L^\times) = 1$.
3. $|\hat{H}^0(G, L^\times)| = |K^\times / N_{L/K}(L^\times)| = n$.

***Proof*:**

1. The valuation sequence $1 \to \mathcal{O}_L^\times \to L^\times \xrightarrow{v_L} \mathbb{Z} \to 0$ is a sequence of G-modules with G acting trivially on $\mathbb{Z}$: for $\sigma \in G$ the map $x \mapsto |\sigma(x)|$ is an absolute value on L extending that of K, so it equals $|x|$ by uniqueness [14, Extension of DVR's, Completion]. The Herbrand quotient of $\mathbb{Z}$ is $h(G, \mathbb{Z}) = |\hat{H}^0(G, \mathbb{Z})|/|\hat{H}^{-1}(G, \mathbb{Z})| = n/1 = n$ (example 4.1.8), and $h(G, \mathcal{O}_L^\times) = 1$ by (2). Hence $h(G, L^\times) = h(G, \mathcal{O}_L^\times)\, h(G, \mathbb{Z}) = n$ by theorem 4.1.10.
2. The group $\mathcal{O}_L^\times$ contains an open G-stable subgroup V with $\hat{H}^q(G, V) = 0$ for all $q \in \mathbb{Z}$. This book does not prove it. It is [2, Ch. VI, §1.4, Prop. 3], where $V = 1 + \pi_K^i A$ for a large i and the lattice $A = \bigoplus_{\sigma \in G} \mathcal{O}_K\, \sigma(\alpha)$ spanned by a normal basis of L/K, and the filtration $V \supseteq 1 + \pi_K^{i+1} A \supseteq \cdots$ has graded pieces isomorphic to the free $(\mathcal{O}_K/\pi_K\mathcal{O}_K)[G]$-module $A/\pi_K A$. Granting V, the quotient $\mathcal{O}_L^\times/V$ is finite, because $\mathcal{O}_L^\times$ is compact, being closed in the compact ring $\mathcal{O}_L$ [14, Compactness Of Local Fields], and V is open. Hence $h(G, \mathcal{O}_L^\times) = h(G, V)\, h(G, \mathcal{O}_L^\times/V) = 1$ by theorem 4.1.10 and proposition 4.1.11.
3. By theorem 4.3.4, $\hat{H}^{-1}(G, L^\times) \cong \hat{H}^1(G, L^\times)$, which is $H^1(G, L^\times) = 0$ by proposition 4.1.5(1) and theorem 8.1.1. So $|\hat{H}^0(G, L^\times)| = h(G, L^\times) = n$ by (1), and $\hat{H}^0(G, L^\times) = K^\times / N_{L/K}(L^\times)$ by definition 4.1.2, completing the proof.

Part (3) determines the norm index: for a cyclic extension of local fields of degree n, the group $K^\times/N_{L/K}(L^\times)$ has order exactly n. This is the "local norm index equality," and it is the local input for the reciprocity law of section 8.4.

Example 8.2.7: The Unit Filtration for $\mathbb{Q}_p$

1. For $K = \mathbb{Q}_p$ with p odd: $\mathcal{O}_K^\times = \mathbb{Z}_p^\times$ has the filtration $\mathbb{Z}_p^\times \supset 1 + p\mathbb{Z}_p \supset 1 + p^2\mathbb{Z}_p \supset \cdots$ with graded pieces $\mathbb{F}_p^\times$ (of order $p-1$) and $\mathbb{F}_p^+$ (of order p) in each subsequent layer. The logarithm map $\log\colon 1 + p\mathbb{Z}_p \xrightarrow{\sim} p\mathbb{Z}_p$ (converging for p odd) gives an isomorphism $U_{\mathbb{Q}_p}^{(1)} \cong \mathbb{Z}_p$ of topological groups.
2. For a totally ramified cyclic extension $L/\mathbb{Q}_p$ of degree p (e.g., $L = \mathbb{Q}_p(\zeta_p)$ when p is odd): $G = C_p$ acts nontrivially on $U_L^{(1)}$, and $H^1(G, U_L^{(1)}) \neq 0$ (by exercise 8.1.1 (4)). The Herbrand quotient $h(G, \mathcal{O}_L^\times) = 1$ means that $|\hat{H}^0(G, \mathcal{O}_L^\times)| = |H^1(G, \mathcal{O}_L^\times)|$: the "cohomological" and "homological" sides have the same order, even though neither vanishes.
3. For $K = \mathbb{Q}_p$ and the unramified extension $L = \mathbb{Q}_{p^n}$: the unit filtration is preserved by $G = \langle \mathrm{Frob} \rangle$, and Frob acts on each graded piece $\mathbb{F}_{p^n}^+$ by the Frobenius $x \mapsto x^p$. The

vanishing $H^1(G, U_L^{(1)}) = 0$ (theorem 8.2.4, since unramified implies tamely ramified) confirms theorem 8.1.3(1) through the filtration.

Exercise 8.2.1

1. Let L/K be an unramified extension of degree n. Compute $H^1(G, \mathbb{F}_L^\times)$ directly (where G acts on $\mathbb{F}_L^\times$ via the Frobenius), and verify that it vanishes.
2. Let L/K be a totally ramified cyclic extension of degree p (with $\mathrm{char} K = 0$). Verify $h(G, L^\times) = p$ by computing $|\hat{H}^0(G, L^\times)| = |K^\times / N(L^\times)|$ and $|\hat{H}^{-1}(G, L^\times)| = |H^1(G, L^\times)| = 1$ (Hilbert 90) independently.
3. Let $K = \mathbb{Q}_p$ (with p odd) and $L = K(\sqrt{u})$ where $u \in \mathbb{Z}_p^\times$ is a non-square. This is an unramified quadratic extension. Determine $K^\times / N_{L/K}(L^\times)$ explicitly: show that a uniformizer p is not a norm, while every unit is a norm.
4. Let $L = \mathbb{Q}_p(\zeta_p)$ for p an odd prime. This is a totally ramified cyclic extension of $\mathbb{Q}_p$ of degree $p-1$. Compute $h(G, \mathcal{O}_L^\times)$ and determine $|H^1(G, \mathcal{O}_L^\times)|$. *Hint:* first compute $|\hat{H}^0(G, \mathcal{O}_L^\times)|$ using the exact sequence $1 \to \mathcal{O}_L^\times \to L^\times \to \mathbb{Z} \to 0$.
5. For $K = \mathbb{Q}_2$ and $L = \mathbb{Q}_2(\sqrt{-1})$ (a totally ramified quadratic extension): write out the unit filtration of $\mathcal{O}_L^\times$, identify the G-action on each graded piece, and compute $H^1(G, U_L^{(i)}/U_L^{(i+1)})$ for $i = 0, 1, 2$.

8.3 Local Tate Duality

The cohomology of a local field satisfies a duality that pairs H^r with H^{2-r}: a local analogue of Poincaré duality for manifolds. This duality, the local Tate duality theorem, is the most important structural result about the cohomology of local Galois groups. It identifies the maximal pro-p quotients as Demushkin groups and enters the computation of the Euler-Poincaré characteristic that controls the sizes of the cohomology groups.

Definition 8.3.1: Cartier Dual

Let K be a local field and M a finite discrete G_K-module. The *Cartier dual* of M is the finite G_K-module:

$$M^* = \mathrm{Hom}_{\mathbb{Z}}(M, (K^{\mathrm{sep}})^\times)$$

with G_K acting by $(g \cdot \varphi)(m) = g(\varphi(g^{-1}m))$ for $g \in G_K$, $\varphi \in M^*$, $m \in M$.

When the order of M is prime to the characteristic of K, the Cartier dual is a finite G_K-module of the same order as M. For $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action: $M^* = \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, (K^{\mathrm{sep}})^\times) \cong \mu_n$ (the nth roots of unity, with the natural Galois action). For $M = \mu_n$: $M^* \cong \mathbb{Z}/n\mathbb{Z}$ (with trivial action when $\mu_n \subseteq K$). The double dual satisfies $M^{**} \cong M$ canonically.

Theorem 8.3.2: Local Tate Duality

Let K be a local field and M a finite discrete G_K-module whose order is prime to the characteristic of K. For $r = 0, 1, 2$, the cup product followed by the invariant map:

$$H^r(G_K, M) \times H^{2-r}(G_K, M^*) \xrightarrow{\smile} H^2(G_K, (K^{\text{sep}})^\times) \xrightarrow{\text{inv}_K} \mathbb{Q}/\mathbb{Z}$$

induces a perfect pairing of finite groups. That is, the induced map $H^r(G_K, M) \to \text{Hom}(H^{2-r}(G_K, M^*), \mathbb{Q}/\mathbb{Z})$ is an isomorphism.

***Proof Key ideas*:**
This is Tate's local duality theorem, which this book does not prove. The module M^* is the dual $M^D = \text{Hom}(M, (K^{\text{sep}})^\times)$ of [22], and the statement is [22, Ch. I, Cor. 2.3]: cup product gives isomorphisms $H^r(G_K, M^*) \to H^{2-r}(G_K, M)^*$ for $r = 1, 2$ and an isomorphism $H^0(G_K, M^*)^\wedge \to H^2(G_K, M)^*$, in which the completion changes nothing because $H^0(G_K, M^*)$ is finite. The finiteness of the groups $H^r(G_K, M)$ is part of [22, Ch. I, Thm. 2.1], whose proof derives it from the Kummer sequence $1 \to \mu_m \to (K^{\text{sep}})^\times \to (K^{\text{sep}})^\times \to 1$ and the Hochschild-Serre spectral sequence for a finite Galois extension over which M becomes a sum of modules μ_m.

Milne deduces the duality from the duality theorem for class formations [22, Ch. I, Thm. 1.8], applied to $(G_K, (K^{\text{sep}})^\times)$ with the invariant map of theorem 7.3.1 and the reciprocity isomorphism of local class field theory, including the existence theorem [31, Ch. XIII, §4 and Ch. XIV, §6], together with the vanishing of $\text{Ext}^r_{\mathbb{Z}}(M, (K^{\text{sep}})^\times)$ for $r > 0$, the group $(K^{\text{sep}})^\times$ being divisible by every integer prime to the characteristic of K. The argument uses neither the cohomological dimension of G_K nor the Euler-Poincaré characteristic, both treated after this theorem, completing the sketch.

The cohomological dimension of a local field is computed independently of duality, from the maximal unramified extension.

Corollary 8.3.3: Cohomological Dimension of Local Fields

For a local field K: $\text{cd}(K) = 2$.

***Proof Key ideas*:**
The residue field of K is finite, so its absolute Galois group is $\widehat{\mathbb{Z}}$, which has $\text{cd}_p = 1$ for every prime p (example 5.1.5). For a field complete under a discrete valuation with residue field k, $\text{cd}_p(G_K) \leq 1 + \text{cd}_p(G_k)$ for every prime p, with equality when $\text{cd}_p(G_k)$ is finite and p is different from the characteristic of K [30, Ch. II, §4.3, Prop. 12], which this book does not prove. The proof there uses the maximal unramified extension K^{nr}: the group $\text{Gal}(K^{\text{sep}}/K^{\text{nr}})$ has cohomological dimension at most 1, and the Hochschild-Serre spectral sequence (theorem 3.4.5) for $\text{Gal}(K^{\text{sep}}/K^{\text{nr}}) \subseteq G_K$ bounds $\text{cd}_p(G_K)$ by the sum. Hence $\text{cd}_p(K) \leq 2$ for every p, with equality for p different from the characteristic of K, and $\text{cd}(K) = 2$, completing the sketch.

The local Euler-Poincaré characteristic controls the sizes of the three cohomology groups, and its

proof uses local duality.

Theorem 8.3.4: Local Euler-Poincaré Characteristic

Let K be a finite extension of $\mathbb{Q}_p$ and M a finite G_K-module of order m. Then:

$$\frac{|H^0(G_K, M)| \cdot |H^2(G_K, M)|}{|H^1(G_K, M)|} = p^{-[K:\mathbb{Q}_p]\, v_p(m)}$$

where $v_p(m)$ is the exponent of p in m. In particular the quotient is $|M|^{-[K:\mathbb{Q}_p]}$ when m is a power of p, and 1 when m is prime to p.

***Proof Key ideas*:**
This is Tate's local Euler-Poincaré formula, which this book does not prove. It is [22, Ch. I, Thm. 2.8], whose right side $(R : mR)^{-1}$, for the ring of integers R of K, equals $p^{-[K:\mathbb{Q}_p]\, v_p(m)}$. Both sides are multiplicative in short exact sequences of finite modules: on the left because $H^r(G_K, -)$ vanishes on finite modules for $r \geq 3$ (corollary 8.3.3) and the alternating product of the orders in a finite exact sequence is 1. When m is prime to p, the quotient is 1 [22, Ch. I, Lemma 2.9], by the Hochschild-Serre spectral sequences for the inertia group and its Sylow p-subgroup. For M annihilated by p, both sides define homomorphisms on the Grothendieck group of $\mathbb{F}_p[G]$-modules for a finite quotient G of G_K through which G_K acts on M. That group is generated up to torsion by modules induced from cyclic subgroups of order prime to p [22, Ch. I, Lemma 2.10], and Shapiro's lemma (theorem 3.5.5) reduces the formula to a finite extension L of K and a module on which G_L acts through a cyclic group of order prime to p. That case is [22, Ch. I, Lemma 2.11], whose proof computes the groups $H^r(G_L, \mathbb{Z}/p\mathbb{Z})$ from the order of $L^\times/L^{\times p}$ and the roots of unity in L by local duality (theorem 8.3.2), completing the sketch.

The identification of local Galois groups as Demushkin groups, foreshadowed in definition 5.3.5 and example 5.3.6, is now a consequence of Tate duality.

Theorem 8.3.5: Structure of the Maximal Pro-p Quotient

Let K be a finite extension of $\mathbb{Q}_p$ with p odd, and let $G_K^{(p)}$ denote the maximal pro-p quotient of G_K. Then:

1. If $\mu_p \not\subseteq K$: the group $G_K^{(p)}$ is a free pro-p-group (theorem 6.1.4) of rank $d(G_K^{(p)}) = [K : \mathbb{Q}_p] + 1$.
2. If $\mu_p \subseteq K$: the group $G_K^{(p)}$ is a Demushkin group (definition 5.3.5) with $d(G_K^{(p)}) = [K : \mathbb{Q}_p] + 2$, $H^2(G_K^{(p)}, \mathbb{F}_p) \cong \mathbb{F}_p$, and type $q = |\mu_{p^\infty}(K)|$ (definition 6.4.3).

***Proof*:**
Step 1: Generator rank. By theorem 6.2.2, $d(G_K^{(p)}) = \dim_{\mathbb{F}_p} H^1(G_K, \mathbb{F}_p)$ (since H^1 with $\mathbb{F}_p$-coefficients factors through the maximal pro-p quotient). The Euler-Poincaré characteristic

(theorem 8.3.4) with $M = \mathbb{F}_p$ (trivial action) gives:

$$\frac{p \cdot |H^2(G_K, \mathbb{F}_p)|}{|H^1(G_K, \mathbb{F}_p)|} = p^{-[K:\mathbb{Q}_p]}$$

By Tate duality (theorem 8.3.2), $|H^2(G_K, \mathbb{F}_p)| = |H^0(G_K, \mu_p)| = |\mu_p(K)|$. If $\mu_p \not\subseteq K$: $|\mu_p(K)| = 1$, giving $|H^1| = p^{[K:\mathbb{Q}_p]+1}$, so $d = [K : \mathbb{Q}_p] + 1$. If $\mu_p \subseteq K$: $|\mu_p(K)| = p$, giving $|H^1| = p^{[K:\mathbb{Q}_p]+2}$, so $d = [K : \mathbb{Q}_p] + 2$.

Step 2: The case $\mu_p \not\subseteq K$. Tate duality gives $|H^2(G_K, \mathbb{F}_p)| = |\mu_p(K)| = 1$, so $H^2(G_K^{(p)}, \mathbb{F}_p) = 0$. By Serre's criterion (theorem 5.1.2), $\mathrm{cd}_p(G_K^{(p)}) \leq 1$. Since $G_K^{(p)}$ is nontrivial (it has nontrivial H^1), the characterization of theorem 6.1.4 gives $G_K^{(p)} \cong F([K : \mathbb{Q}_p] + 1)$, a free pro-$p$-group.

Step 3: The case $\mu_p \subseteq K$. Tate duality gives $|H^2(G_K, \mathbb{F}_p)| = |\mu_p(K)| = p$, so $H^2(G_K^{(p)}, \mathbb{F}_p) \cong \mathbb{F}_p$. Serre's criterion gives $\mathrm{cd}_p(G_K^{(p)}) = 2$ (since $H^2 \neq 0$ and $H^3 = 0$ by corollary 8.3.3). The conditions for a Demushkin group (definition 5.3.5) are satisfied: $\mathrm{cd}_p = 2$, $H^1 \cong \mathbb{F}_p^d$ with d finite, and $H^2 \cong \mathbb{F}_p$.

Step 4: The type. When $\mu_p \subseteq K$, the type q is determined by the action of $G_K^{(p)}$ on the p-power roots of unity: $q = |\mu_{p^\infty}(K)| = p^k$ where k is the largest integer with $\mu_{p^k} \subseteq K$. This identifies $G_K^{(p)}$ with the Demushkin group $G(d, q)$ from theorem 6.4.4.

The dichotomy in the theorem reflects a fundamental arithmetic distinction: when K contains the pth roots of unity, Kummer theory provides a nontrivial H^2 (the Brauer group has p-torsion classes that are not split by p-extensions), and the resulting constraint on the pro-p Galois group forces the Demushkin structure. When $\mu_p \not\subseteq K$, no such constraint exists, and the group is as free as its generator count allows.

Example 8.3.6: Tate Duality in Practice

1. For $M = \mu_p$ with p odd and $K \supseteq \mu_p$: the Cartier dual $M^* = \mathrm{Hom}(\mu_p, (K^{\mathrm{sep}})^\times) \cong \mathbb{Z}/p\mathbb{Z}$ (with trivial action, since K contains μ_p). Tate duality gives perfect pairings: $H^0(G_K, \mu_p) \times H^2(G_K, \mathbb{Z}/p\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$ (pairing $\mu_p(K) \cong \mathbb{F}_p$ with $H^2(G_K, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{F}_p$) and $H^1(G_K, \mu_p) \times H^1(G_K, \mathbb{Z}/p\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$. The group $H^1(G_K, \mu_p) \cong K^\times/(K^\times)^p$ (by Kummer theory) has dimension $[K : \mathbb{Q}_p]+2$ over $\mathbb{F}_p$, matching $|H^1(G_K, \mathbb{Z}/p\mathbb{Z})| = p^{[K:\mathbb{Q}_p]+2}$. The pairing identifies cyclic p-extensions of K (classified by $H^1(G_K, \mathbb{Z}/p\mathbb{Z}) = \mathrm{Hom}(G_K, \mathbb{Z}/p\mathbb{Z})$) with elements of $K^\times/(K^\times)^p$ (a precursor to the local reciprocity map of section 8.4).

2. For $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action and $\gcd(n, \mathrm{char}\mathbb{F}_K) = 1$: the dual is $M^* = \mu_n$, and Tate duality gives $H^1(G_K, \mathbb{Z}/n\mathbb{Z}) \times H^1(G_K, \mu_n) \to \frac{1}{n}\mathbb{Z}/\mathbb{Z}$. This pairing sends a character $\chi\colon G_K \to \mathbb{Z}/n\mathbb{Z}$ (classifying a cyclic extension L/K of degree dividing n) and a Kummer class $a \in K^\times/(K^\times)^n$ to $\mathrm{inv}_K((a, L/K))$, where $(a, L/K)$ is the norm residue symbol.

3. (If the reader has read [11]) For an elliptic curve E/K and $M = E[p]$ (the p-torsion): the Weil pairing gives $E[p]^* \cong E[p]$ (self-duality). Tate duality gives a perfect pairing $H^1(G_K, E[p]) \times H^1(G_K, E[p]) \to \mathbb{Q}/\mathbb{Z}$, and the image of the local Kummer map $E(K)/pE(K) \hookrightarrow H^1(G_K, E[p])$ is its own orthogonal complement under this pairing. This

is the local condition defining the Selmer group (developed further in section 10.5).

Exercise 8.3.1

1. Let $K = \mathbb{Q}_p$ with p odd and $M = \mathbb{Z}/p\mathbb{Z}$ (trivial action). Compute $|H^r(G_{\mathbb{Q}_p}, M)|$ for $r = 0, 1, 2$ using the Euler-Poincaré characteristic and Tate duality. Determine which case ($\mu_p \subseteq \mathbb{Q}_p$ or not) applies.
2. A finite G_K-module M is *self-dual* if $M^* \cong M$ as G_K-modules. Show that Tate duality gives a perfect pairing $H^1(G_K, M) \times H^1(G_K, M) \to \mathbb{Q}/\mathbb{Z}$ for self-dual M.
3. Verify $\mathrm{cd}(\mathbb{Q}_p) = 2$ directly (for p odd) by showing $H^2(G_{\mathbb{Q}_p}, \mathbb{Z}/2\mathbb{Z}) \neq 0$ (using Tate duality and $\mu_2(\mathbb{Q}_p) = \{\pm 1\}$) and $H^3(G_{\mathbb{Q}_p}, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for all primes ℓ (using the LHS spectral sequence for $1 \to I_p \to G_{\mathbb{Q}_p} \to \widehat{\mathbb{Z}} \to 1$).
4. Determine the structure of $G_K^{(p)}$ (free or Demushkin, with invariants d and q) for: (a) $p \geq 5$ and $K = \mathbb{Q}_p$; (b) $p \geq 3$ and $K = \mathbb{Q}_p(\zeta_p)$.
5. Using the Euler-Poincaré characteristic, compute $|H^1(G_K, \mu_n)|$ for $K = \mathbb{Q}_p$ and $n = p$. Verify the result agrees with the Kummer theory computation $H^1(G_K, \mu_p) \cong K^\times/(K^\times)^p$.

8.4 The Cohomological Reciprocity Isomorphism

The invariant map $\mathrm{inv}_K \colon \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ (corollary 8.1.4, theorem 8.1.5) and the cup product pairing combine to produce the local reciprocity map: a canonical homomorphism $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$ that describes the abelian extensions of K in terms of the multiplicative group. Its construction is cohomological: it inverts the Tate-Nakayama isomorphism of theorem 8.4.3, which rests on Tate's theorem on cup products with a fundamental class [31, Ch. IX, §8, Thm. 14]. The abstract structure underlying the construction is that of a *class formation*: an axiomatic framework isolating the properties of $(G_K, (K^{\mathrm{sep}})^\times)$ that make reciprocity work.

What this section constructs is the reciprocity map as an isomorphism *of the cohomological formalism*: rec_K is the degree -2 image of the fundamental class under Tate-Nakayama. The inputs that the construction cites, Tate's cup-product theorem, the compatibility of the reciprocity maps in towers and the local existence theorem, come from [31]. Its *development* into class field theory belongs to *Class Field Theory* [24]: the identification of rec_K with the norm-residue symbol, the norm-group and conductor description of the abelian extensions of K, the local existence theorem, and the explicit reciprocity laws. That book takes the isomorphism constructed here as its starting point.

Definition 8.4.1: Class Formation

A *class formation* is a pair (G, C) consisting of a profinite group G and a discrete G-module C, together with an isomorphism

$$\mathrm{inv}_{H'/H} \colon H^2(H'/H, C^H) \xrightarrow{\sim} \frac{1}{[H' : H]}\mathbb{Z}/\mathbb{Z}$$

for every pair of open subgroups $H \trianglelefteq H'$ of G, satisfying three axioms:

1. *(Hilbert 90)* For every open subgroup $H \leq G$: $H^1(H, C) = 0$.
2. *(Compatibility with inflation)* For open subgroups $H \subseteq H' \subseteq H''$ of G with H and H' normal in H'', the diagram

$$\begin{array}{ccc} H^2(H''/H', C^{H'}) & \xrightarrow{\mathrm{inv}} & \frac{1}{[H'':H']}\mathbb{Z}/\mathbb{Z} \\ \Big\downarrow{\scriptstyle \mathrm{Inf}} & & \Big\downarrow \\ H^2(H''/H, C^H) & \xrightarrow[\mathrm{inv}]{} & \frac{1}{[H'':H]}\mathbb{Z}/\mathbb{Z} \end{array}$$

   commutes, the right vertical map being the inclusion.

3. *(Compatibility with restriction)* For open subgroups $H \subseteq H' \subseteq H''$ of G with H normal in H'', and every $x \in H^2(H''/H, C^H)$,

$$\mathrm{inv}_{H'/H}(\mathrm{Res} x) = [H'' : H'] \cdot \mathrm{inv}_{H''/H}(x)$$

   where Res is restriction to the subgroup H'/H of H''/H.

Axiom (1) abstracts Hilbert's Theorem 90. The invariant maps, part of the data, say that H^2 of the formation module, for any finite quotient of G, is cyclic of the expected order. Axioms (2) and (3) abstract the behaviour of the invariant map on the Brauer group: these identifications are compatible across the tower of quotients and under passage to subgroups, the second being the counterpart of theorem 7.3.1(3). These are the axioms of [31, Ch. XI, §2], written for pairs of open subgroups. Axiom (3) is what makes the restriction of the fundamental class of H''/H to the subgroup H'/H again a fundamental class.

Theorem 8.4.2: The Local Class Formation

For a local field K, the pair $(G_K, (K^{\mathrm{sep}})^\times)$ is a class formation.

***Proof*:**
For open subgroups $H \trianglelefteq H'$ of G_K with fixed fields $L \supseteq K'$, the extension L/K' is finite Galois with group H'/H, and $C^H = L^\times$. Let $\mathrm{inv}_{H'/H}$ be the composite

$$H^2(H'/H, L^\times) \xrightarrow{\mathrm{Inf}} H^2(H', (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K') \xrightarrow{\mathrm{inv}_{K'}} \mathbb{Q}/\mathbb{Z}$$

the middle identification being theorem 7.2.8 for K', whose absolute Galois group is H'.

The invariant maps. The inflation map is injective with image the kernel of the restriction $\mathrm{Br}(K') \to \mathrm{Br}(L)$ [31, Ch. X, §4, Prop. 6 and Cor.]. By theorem 7.3.1(3), $\mathrm{inv}_L \circ \mathrm{Res} = [L : K'] \cdot \mathrm{inv}_{K'}$, and inv_L is injective (theorem 8.1.5), so this kernel is the set of classes whose invariant is killed by $[L : K'] = [H' : H]$. As $\mathrm{inv}_{K'}$ is an isomorphism $\mathrm{Br}(K') \to \mathbb{Q}/\mathbb{Z}$ (theorem 8.1.5), $\mathrm{inv}_{H'/H}$ is an isomorphism onto $\frac{1}{[H':H]}\mathbb{Z}/\mathbb{Z}$.

Axiom (1): for any open subgroup $H = \mathrm{Gal}(K^{\mathrm{sep}}/L)$,

$$H^1(H, (K^{\mathrm{sep}})^\times) = \varinjlim_{L'/L} H^1(\mathrm{Gal}(L'/L), (L')^\times) = 0$$

by theorem 3.2.2 and Hilbert 90 (theorem 8.1.1).

Axiom (2): let K'' be the fixed field of H''. Inflation from H''/H' to H''/H followed by inflation to H'' is inflation from H''/H' to H'', the maps on cohomology attached to compatible pairs of group homomorphisms and module maps composing [31, Ch. VII, §5]. So both invariant maps in the diagram are $\mathrm{inv}_{K''}$ composed with inflation into $H^2(H'', (K^{\mathrm{sep}})^\times)$, and the diagram commutes.

Axiom (3): with K'' and K' the fixed fields of H'' and H', and $x \in H^2(H''/H, L^\times)$, restriction to H'/H followed by inflation to H' equals inflation to H'' followed by restriction to H', for the same reason. So $\mathrm{inv}_{H'/H}(\mathrm{Res}x) = \mathrm{inv}_{K'}(\mathrm{ResInf}x) = [K' : K''] \cdot \mathrm{inv}_{K''}(\mathrm{Inf}x) = [H'' : H'] \cdot \mathrm{inv}_{H''/H}(x)$ by theorem 7.3.1(3), which is [31, Ch. XIII, §3, Prop. 7]. The verification of all three axioms is also [31, Ch. XIII, §4, Thm. 1], completing the proof.

The class formation axioms are the input for the Tate-Nakayama theorem, which produces the reciprocity map. The argument proceeds through two key identifications from earlier chapters. The first, from the Tate cohomology of finite groups (proposition 4.1.5, theorem 2.5.2): the group $\hat{H}^{-2}(G, \mathbb{Z}) \cong H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$ for any finite group G, where $G^{\mathrm{ab}} = G/[G, G]$ is the abelianization. The second, from the periodicity theorem (theorem 4.3.4): for a cyclic group G, the cup product with the periodicity generator $u_G \in \hat{H}^2(G, \mathbb{Z})$ of definition 4.3.3 gives an isomorphism $\hat{H}^r(G, M) \xrightarrow{\sim} \hat{H}^{r+2}(G, M)$ for all $r \in \mathbb{Z}$ and all G-modules M. The Tate-Nakayama theorem generalizes this: the cup product with the fundamental class $u_{L/K} \in H^2(\mathrm{Gal}(L/K), L^\times)$ of a class formation gives the same periodicity, even when $\mathrm{Gal}(L/K)$ is not cyclic.

Theorem 8.4.3: Tate-Nakayama for the Local Class Formation

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. The *fundamental class* $u_{L/K} \in H^2(G, L^\times)$ is the unique element with $\mathrm{inv}_K(u_{L/K}) = 1/[L : K] \in \mathbb{Q}/\mathbb{Z}$. The cup product with $u_{L/K}$ gives an isomorphism:

$$\hat{H}^r(G, \mathbb{Z}) \xrightarrow{- \smile u_{L/K}} \hat{H}^{r+2}(G, L^\times)$$

for all $r \in \mathbb{Z}$.

***Proof Key ideas*:**
The fundamental class exists and is unique: $H^2(G, L^\times) \cong \mathrm{Br}(L/K)$ by theorem 7.2.7, and inv_K maps $\mathrm{Br}(L/K)$ isomorphically onto $\frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$, as part of the data of the local class formation (definition 8.4.1, theorem 8.4.2).

The isomorphism is Tate's theorem on cup products with a class whose restrictions to the Sylow subgroups generate their second cohomology, which this book does not prove. Let G be a finite group, A a G-module and $a \in H^2(G, A)$, and suppose that for every prime p and a Sylow p-subgroup G_p of G, $H^1(G_p, A) = 0$ and $H^2(G_p, A)$ has order $|G_p|$ and is generated by $\mathrm{Res}(a)$. Then cup product with a is an isomorphism $\hat{H}^r(G, \mathbb{Z}) \to \hat{H}^{r+2}(G, A)$ for every $r \in \mathbb{Z}$ [31, Ch. IX, §8, Thm. 14 and Cor.]. Its proof compares $\mathbb{Z}$ and A through a mapping cylinder and applies the criterion for cohomological triviality of theorem 4.3.5 [31, Ch. IX, §7, Thm. 12].

The hypotheses hold for $A = L^\times$ and $a = u_{L/K}$. Fix a prime p, a Sylow p-subgroup G_p of G and its fixed field K_p, so that $G_p = \mathrm{Gal}(L/K_p)$. First, $H^1(G_p, L^\times) = 0$ by theorem 8.1.1. Second, the invariant map of the local class formation (definition 8.4.1, theorem 8.4.2) identifies $H^2(G_p, L^\times)$ with the cyclic group $\frac{1}{[L:K_p]}\mathbb{Z}/\mathbb{Z}$ of order $|G_p|$. Third, by axiom (3), the invariant of $\mathrm{Res}(u_{L/K})$ is $[K_p : K] \cdot \frac{1}{[L:K]} = \frac{1}{[L:K_p]}$, so $\mathrm{Res}(u_{L/K})$ generates $H^2(G_p, L^\times)$. Tate's theorem then gives the isomorphisms, as in [31, Ch. XIII, §4, Prop. 8], completing the sketch.

The local reciprocity law is the case $r = -2$ of the Tate-Nakayama isomorphism.

Theorem 8.4.4: Local Reciprocity

Let L/K be a finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. There is a canonical isomorphism:

$$\mathrm{rec}_{L/K} \colon K^\times / N_{L/K}(L^\times) \xrightarrow{\sim} \mathrm{Gal}(L/K)^{\mathrm{ab}}$$

Passing to the limit over all finite Galois extensions L/K inside K^{sep} gives the *local Artin map*:

$$\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$$

a continuous homomorphism with dense image, whose kernel is the intersection of all norm groups $\bigcap_L N_{L/K}(L^\times) = \{1\}$.

***Proof*:**
Step 1: The isomorphism for one extension. The Tate-Nakayama isomorphism (theorem 8.4.3) at $r = -2$ gives:

$$\hat{H}^{-2}(G, \mathbb{Z}) \xrightarrow{\sim} \hat{H}^0(G, L^\times)$$

The left side is $\hat{H}^{-2}(G, \mathbb{Z}) \cong H_1(G, \mathbb{Z}) \cong G^{\mathrm{ab}}$ (proposition 4.1.5, theorem 2.5.2). The right side is $\hat{H}^0(G, L^\times) = (L^\times)^G / N_G(L^\times) = K^\times / N_{L/K}(L^\times)$ (definition 4.1.2). The isomorphism $G^{\mathrm{ab}} \xrightarrow{\sim} K^\times / N_{L/K}(L^\times)$ is inverted to give $\mathrm{rec}_{L/K} \colon K^\times / N_{L/K}(L^\times) \xrightarrow{\sim} G^{\mathrm{ab}}$.

Step 2: Compatibility in towers. For finite Galois extensions $K \subseteq L \subseteq L'$ inside K^{sep} and $x \in K^\times$, the projection $\mathrm{Gal}(L'/K)^{\mathrm{ab}} \to \mathrm{Gal}(L/K)^{\mathrm{ab}}$ sends $\mathrm{rec}_{L'/K}(x)$ to $\mathrm{rec}_{L/K}(x)$. This step is not proved in this book. It is not a formal consequence of Step 1, because inflation does not commute

with the Tate-Nakayama isomorphisms: it carries $u_{L/K}$ to $[L' : L]\, u_{L'/K}$. The comparison runs through the characterization of $\mathrm{rec}_{L/K}(x)$ by the values that the characters of $\mathrm{Gal}(L/K)$ take on it, a cup-product computation pairing $\hat{H}^{-2}(G, \mathbb{Z})$ with $H^2(G, \mathbb{Z}) \cong \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ that this book does not develop. The statement is [31, Ch. XIII, §4, Prop. 12], where $\mathrm{rec}_{L/K}(x)$ is written $(x, L/K)$ and is defined, as in Step 1, by inverting the cup product with the fundamental class in degree -2.

Step 3: The local Artin map. For finite Galois L/K the group $\mathrm{Gal}(L/K)^{\mathrm{ab}}$ is the Galois group of the largest abelian subextension of L/K, the fixed field of the commutator subgroup, so the limit may be taken over the finite abelian extensions L/K, for which $\mathrm{Gal}(L/K)^{\mathrm{ab}} = \mathrm{Gal}(L/K)$. By theorem 1.4.1(1) applied to the maximal abelian extension K^{ab}, $\varprojlim_L \mathrm{Gal}(L/K) = \mathrm{Gal}(K^{\mathrm{ab}}/K) = G_K^{\mathrm{ab}}$. By Step 2 the maps $x \mapsto \mathrm{rec}_{L/K}(x)$ for finite abelian L/K are compatible, so they define a homomorphism $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$.

The map rec_K is continuous. For a finite abelian L/K, the composite of rec_K with the projection to the discrete group $\mathrm{Gal}(L/K)$ has kernel $N_{L/K}(L^\times)$ by Step 1, so it suffices that this subgroup is open. The absolute value of L is $|y| = |N_{L/K}(y)|^{1/[L:K]}$ [14, Extension of DVR's, Completion], so $y \in L^\times$ is a unit exactly when $N_{L/K}(y)$ is one, and $N_{L/K}(L^\times) \cap \mathcal{O}_K^\times = N_{L/K}(\mathcal{O}_L^\times)$. This group is compact, as the continuous image of $\mathcal{O}_L^\times$, which is closed in the compact ring $\mathcal{O}_L$ [14, Compactness Of Local Fields], and it has finite index in $\mathcal{O}_K^\times$, since $\mathcal{O}_K^\times / N_{L/K}(\mathcal{O}_L^\times)$ injects into the finite group $K^\times / N_{L/K}(L^\times)$. A closed subgroup of finite index is open, its complement being a finite union of closed cosets. So $N_{L/K}(\mathcal{O}_L^\times)$ is open in $\mathcal{O}_K^\times = \{x \in K \mid |x| = 1\}$, hence in $K^\times$, and so is the subgroup $N_{L/K}(L^\times)$ containing it. The image of rec_K is dense: its composite with each projection $G_K^{\mathrm{ab}} \to \mathrm{Gal}(L/K)$ is onto by Step 1, and the fibres of these projections form a basis of the topology of G_K^{ab}. By Step 1 again, the kernel of rec_K is the intersection of the groups $N_{L/K}(L^\times)$ over the finite abelian L/K, and this intersection contains the intersection over all finite Galois L/K.

Step 4: The kernel is trivial. The intersection of the norm groups $N_{L/K}(L^\times)$ over all finite abelian extensions L/K is $\{1\}$ [31, Ch. XIV, §6, Cor. 2(i)]. This is the first intersection of Step 3, which contains the second, so both equal $\{1\}$, and so does the kernel of rec_K. This step is not proved in this book. It rests on the local existence theorem, that every closed subgroup of finite index in $K^\times$ is the norm group $N_{L/K}(L^\times)$ of a finite abelian extension L/K [31, Ch. XIV, §6, Thm. 1], applied to the subgroups generated by π_K^m and the higher unit group $U_K^{(m)}$, whose intersection over all $m \geq 1$ is $\{1\}$. The proof of the existence theorem uses the theory of universal norms in a class formation, which is not developed here. This completes the proof.

The local Artin map has explicit descriptions on specific elements of $K^\times$.

Proposition 8.4.5: Properties of the Local Artin Map

Let K be a local field with residue field $\mathbb{F}_q$. The local Artin map $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$ satisfies:

1. A uniformizer π_K maps to a lift of the arithmetic Frobenius $\mathrm{Frob}_K \in \mathrm{Gal}(K^{\mathrm{nr}}/K)$ (the automorphism acting as $x \mapsto x^q$ on the residue field of K^{nr}).
2. The unit group $\mathcal{O}_K^\times$ maps onto the inertia subgroup $I_K^{\mathrm{ab}} \subseteq G_K^{\mathrm{ab}}$: $\mathrm{rec}_K(\mathcal{O}_K^\times) = I_K^{\mathrm{ab}}$.
3. For a finite abelian extension L/K: the kernel of rec_K composed with $G_K^{\mathrm{ab}} \twoheadrightarrow \mathrm{Gal}(L/K)$ is $N_{L/K}(L^\times)$.

***Proof*:**
For (1), let L/K be a finite unramified extension. As rec_K is the limit of the maps $\mathrm{rec}_{L/K}$ (theorem 8.4.4), the image of $\mathrm{rec}_K(\pi_K)$ in $\mathrm{Gal}(L/K)$ is $\mathrm{rec}_{L/K}(\pi_K)$, and $\mathrm{rec}_{L/K}(x) = \mathrm{Frob}_{L/K}^{v_K(x)}$ for $x \in K^\times$, where $\mathrm{Frob}_{L/K}$ acts as $y \mapsto y^q$ on the residue field of L [31, Ch. XIII, §4, Prop. 13]. Serre defines that reciprocity map, as in theorem 8.4.4, by inverting the cup product with the fundamental class, and his invariant map on unramified extensions is the map of corollary 8.1.4. This book does not prove the formula: its proof evaluates characters χ of $\mathrm{Gal}(L/K)$ on the reciprocity map through $\chi(\mathrm{rec}_{L/K}(x)) = \mathrm{inv}_K(\bar{x} \smile \delta\chi)$ [31, Ch. XI, §3, Prop. 2]. As K^{nr} is the union of the finite unramified L, the element $\mathrm{rec}_K(\pi_K)$ acts on K^{nr} as Frob_K.

For (2), write $I_K^{\mathrm{ab}} = \mathrm{Gal}(K^{\mathrm{ab}}/K^{\mathrm{nr}})$. Let $u \in \mathcal{O}_K^\times$. Every finite unramified extension L/K is cyclic, u is a norm from L by theorem 8.1.3(4), and so $\mathrm{rec}_{L/K}(u) = 1$ by theorem 8.4.4. By the construction of rec_K, the image of $\mathrm{rec}_K(u)$ in $\mathrm{Gal}(L/K)$ is $\mathrm{rec}_{L/K}(u)$. As K^{nr} is the union of these L, the element $\mathrm{rec}_K(u)$ acts trivially on K^{nr}, that is, $\mathrm{rec}_K(\mathcal{O}_K^\times) \subseteq I_K^{\mathrm{ab}}$.

For the reverse inclusion, let L/K be a finite abelian extension. The image of I_K^{ab} in $\mathrm{Gal}(L/K)$ is the inertia subgroup $\mathrm{Gal}(L/L \cap K^{\mathrm{nr}})$, and the reciprocity map sends $\mathcal{O}_K^\times$ onto this subgroup [31, Ch. XIII, §4, Cor. to Prop. 13]. This book does not prove it: the proof writes an element of the inertia subgroup as $\mathrm{rec}_{L/K}(a)$, uses the explicit form of the reciprocity map on the unramified extension $(L \cap K^{\mathrm{nr}})/K$ to see that the residue degree of L/K divides $v_K(a)$, and divides a by a norm from L of the same valuation. So the set $S = \mathrm{rec}_K(\mathcal{O}_K^\times)$ and the subgroup I_K^{ab} have the same image in every quotient $\mathrm{Gal}(L/K) = G_K^{\mathrm{ab}}/U$ with $U = \mathrm{Gal}(K^{\mathrm{ab}}/L)$, that is, $SU = I_K^{\mathrm{ab}}U$. Both are closed in G_K^{ab}: S is compact, as the continuous image of the compact group $\mathcal{O}_K^\times$ (theorem 8.4.4), and I_K^{ab} is closed by theorem 1.4.1. A closed subset T of G_K^{ab} is the intersection of the sets TU: if $x \notin T$, some coset xU is disjoint from T, since these cosets form a basis of neighbourhoods of x (theorem 1.4.1), and then $x \notin TU$. Hence $S = I_K^{\mathrm{ab}}$.

Part (3) is the definition of the reciprocity map: $\mathrm{rec}_{L/K}$ is an isomorphism $K^\times/N_{L/K}(L^\times) \xrightarrow{\sim} \mathrm{Gal}(L/K)$.

Example 8.4.6: The Local Artin Map

1. For $K = \mathbb{Q}_p$ (with p odd): $K^\times \cong p^{\mathbb{Z}} \times \mathbb{Z}_p^\times \cong \mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$. The Artin map sends p

to an element acting on $\mathbb{Q}_p^{\mathrm{nr}}$ as the arithmetic Frobenius Frob_p, the automorphism $y \mapsto y^p$ of the residue field, and maps $\mathbb{Z}_p^\times$ onto the inertia subgroup I_p^{ab} (proposition 8.4.5(1) and (2)).

2. For $K = \mathbb{R}$: $K^\times = \mathbb{R}^\times$, $G_K = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \cong C_2$, and $N_{\mathbb{C}/\mathbb{R}}(z) = |z|^2$. So $\mathbb{R}^\times/N(\mathbb{C}^\times) = \mathbb{R}^\times/\mathbb{R}_{>0} \cong \{+1, -1\} \cong C_2$. The Artin map sends $-1 \mapsto \sigma$ (complex conjugation) and $\mathbb{R}_{>0} \mapsto 1$.

3. For $K = \mathbb{Q}_p$ and the unramified extension $L = \mathbb{Q}_{p^n}$: the norm group $N(L^\times)$ contains all units $\mathbb{Z}_p^\times$ (by theorem 8.1.3(4)) and the powers of $N(p) = p^n$, and it has index n (theorem 8.1.3(3)), so it is $p^{n\mathbb{Z}} \times \mathbb{Z}_p^\times$ and $\mathbb{Q}_p^\times/N(L^\times) \cong \mathbb{Z}/n\mathbb{Z}$, generated by the class of p. By proposition 8.4.5(1), $\mathrm{rec}_{L/K}(p) = \mathrm{Frob}_p$, the arithmetic Frobenius of L/K, which generates $\mathrm{Gal}(L/K)$.

The local reciprocity law constructed in this section is used in two directions. The book [24] cites it as the cohomological construction of the local Artin map, and chapter 10 uses the local reciprocity maps as ingredients in the global theory.

Exercise 8.4.1

1. For $\mathbb{Q}_{p^2}/\mathbb{Q}_p$, show that $H^2(\mathrm{Gal}(\mathbb{Q}_{p^2}/\mathbb{Q}_p), \mathbb{Q}_{p^2}^\times) \cong \mathbb{Z}/2\mathbb{Z}$ and identify the invariant map $\mathrm{inv}_{H'/H}$ of definition 8.4.1 for the local class formation (theorem 8.4.2).

2. Let $K = \mathbb{Q}_p$ with p odd and $L = K(\sqrt{p})$, a totally ramified extension of degree 2 ($x^2 - p$ being an Eisenstein polynomial). Determine the norm group $N_{L/K}(L^\times) \subseteq \mathbb{Q}_p^\times$ and describe $\mathrm{rec}_{L/K}$ explicitly.

3. Let L/K be a finite, totally ramified abelian extension of local fields. Show that the map $\mathcal{O}_K^\times \to K^\times/N_{L/K}(L^\times)$ is onto, and deduce that $\mathrm{rec}_{L/K}(\mathcal{O}_K^\times) = \mathrm{Gal}(L/K)$.

4. Let $K \subseteq L \subseteq M$ be a tower of finite Galois extensions of local fields. Show that $\mathrm{Inf}(u_{L/K}) = [M : L] \cdot u_{M/K}$ in $H^2(\mathrm{Gal}(M/K), M^\times)$, and explain why this does not formally give Step 2 of theorem 8.4.4.

5. (If the reader has read [24]) The *existence theorem* of local class field theory states that every open subgroup of finite index in $K^\times$ is the norm group $N_{L/K}(L^\times)$ for a unique finite abelian extension L/K. Using this, show that the local Artin map $\mathrm{rec}_K \colon K^\times \to G_K^{\mathrm{ab}}$ induces a bijection between open subgroups of finite index in $K^\times$ and open subgroups of G_K^{ab}.

9

Adèles and Idèles

The global duality theorems of chapter 10 require Galois modules that assemble local information at every place of a number field simultaneously. The adèle ring and idèle group provide the correct framework: they are restricted products that combine the completions K_v across all places v of a number field K, with a topology that remembers the integral structure at almost all places. This chapter develops the definitions and essential properties, assuming the theory of places and completions from [14] and the classical finiteness theorems (finite class number, Dirichlet unit theorem) from [6]. The treatment is independent of [24]. Readers of that book will find a parallel development there with different emphasis.

9.1 Restricted Products and the Adèle Ring

The completions K_v of a number field K are locally compact groups, and the ring of integers $\mathcal{O}_v \subseteq K_v$ at each finite place v is a compact open subgroup. The full product $\prod_v K_v$ carries the product topology, but this is too coarse for number-theoretic purposes: it contains elements with unbounded denominators at infinitely many places, and its unit group has the wrong topology. The restricted product resolves both problems by imposing the constraint that almost all components lie in the designated compact open subgroup.

Definition 9.1.1: Restricted Product

Let $\{G_v\}_{v\in V}$ be a family of locally compact topological groups, and for all but finitely many v let $K_v \leq G_v$ be a compact open subgroup. The *restricted product* of (G_v, K_v) is the subgroup:

$$\prod_{v\in V}{}' (G_v, K_v) = \left\{ (g_v)_{v\in V} \in \prod_v G_v \;\middle|\; g_v \in K_v \text{ for all but finitely many } v \right\}$$

equipped with the topology generated by sets of the form $\prod_{v\in S} U_v \times \prod_{v\notin S} K_v$, where $S \subseteq V$ is a finite set containing all v for which K_v is undefined, and each $U_v \subseteq G_v$ is open.

An element of the restricted product is a tuple (g_v) that is "integral at almost all places": only finitely many components are allowed to escape the compact open subgroup K_v. The topology is finer than the subspace topology from $\prod_v G_v$: the basic open sets require the components outside S to lie *exactly* in K_v, not just in any open subset of G_v.

Proposition 9.1.2: Local Compactness of Restricted Products

The restricted product $\prod_v{}' (G_v, K_v)$ is a locally compact Hausdorff topological group.

***Proof*:**
For any finite set S of places (containing all places where K_v is undefined), the subgroup $G_S = \prod_{v\in S} G_v \times \prod_{v\notin S} K_v$ carries the product topology, which is locally compact (a finite product of locally compact groups times a product of compact groups, the latter compact by Tychonoff's theorem). The restricted product is the union $\bigcup_S G_S$ over all finite S, and the topology on the restricted product restricts to the product topology on each G_S. Since G_S is open in $G_{S'}$ for $S \subseteq S'$ (the factor $\prod_{v\in S'\setminus S} K_v$ is open in $\prod_{v\in S'\setminus S} G_v$), the restricted product is locally compact: every point has a compact neighborhood (contained in some G_S). The Hausdorff property follows from the Hausdorff property of each G_v.

The notion of restricted product applies equally to rings: if each G_v is a locally compact ring and each K_v is a compact open subring, the restricted product inherits a ring structure with componentwise operations.

Throughout this chapter, K denotes a number field with ring of integers $\mathcal{O}_K$. The set of places of K is denoted V_K, decomposed as $V_K = V_K^{\text{fin}} \cup V_K^{\infty}$ into finite and archimedean places. For each finite place v, the completion K_v is a non-archimedean local field with ring of integers $\mathcal{O}_v$ and maximal ideal $\mathfrak{m}_v$. For each archimedean place v, the completion K_v is $\mathbb{R}$ or $\mathbb{C}$. The reader is referred to [14] for the construction of completions and their structural properties, and to [6] for the arithmetic of $\mathcal{O}_K$ (prime splitting, class group, unit group).

Definition 9.1.3: Adèle Ring

The *adèle ring* of K is the restricted product:

$$\mathbb{A}_K = \prod_{v \in V_K}{}' (K_v, \mathcal{O}_v)$$

where the restricted product is taken with respect to the compact open subrings $\mathcal{O}_v \subseteq K_v$ at each finite place v, and the archimedean completions K_v ($v \in V_K^\infty$) appear without restriction (i.e., K_v serves as its own "compact open subgroup" for the purpose of the restricted product construction). The ring operations are componentwise addition and multiplication.

An adèle $a = (a_v)_v \in \mathbb{A}_K$ is a tuple giving an element $a_v \in K_v$ at each place v, subject to the constraint $a_v \in \mathcal{O}_v$ for all but finitely many finite places v. The *ring of integral adèles* is the compact open subring $\mathbb{A}_K^\circ = \prod_{v \mid \infty} K_v \times \prod_{v \nmid \infty} \mathcal{O}_v$. The finite adèles are the restricted product over finite places only: $\mathbb{A}_K^{\mathrm{fin}} = \prod_{v \in V_K^{\mathrm{fin}}}{}' (K_v, \mathcal{O}_v)$.

Proposition 9.1.4: The Adèle Ring Is Locally Compact

$\mathbb{A}_K$ is a locally compact Hausdorff topological ring.

***Proof*:**
Immediate from proposition 9.1.2: $\mathbb{A}_K$ is a restricted product of the locally compact fields K_v with respect to the compact open subrings $\mathcal{O}_v$.

The number field K embeds into $\mathbb{A}_K$ diagonally.

Proposition 9.1.5: Diagonal Embedding

The diagonal map $\iota \colon K \hookrightarrow \mathbb{A}_K$, defined by $\iota(a) = (a, a, a, \dots)$ (viewing $a \in K_v$ via the canonical embedding $K \hookrightarrow K_v$ at each place), is a ring homomorphism embedding K as a discrete subring of $\mathbb{A}_K$.

***Proof*:**
The map ι is a ring homomorphism (componentwise operations). The image is well defined as an adèle: for any $a \in K^\times$, the absolute value $|a|_v = 1$ for all but finitely many v (since $a \in \mathcal{O}_v^\times$ for all v not dividing the numerator or denominator ideal of a), so $\iota(a) \in \mathbb{A}_K$. Injectivity is immediate ($a = 0$ if and only if $a_v = 0$ at any place).

For discreteness: the open neighborhood $U = \left\{(a_v) \in \mathbb{A}_K \;\middle|\; |a_v|_v < 1 \text{ for all } v \in V_K^\infty,\ a_v \in \mathcal{O}_v \text{ for all } v \in V_K^{\mathrm{fin}}\right\}$ of $0 \in \mathbb{A}_K$ satisfies $\iota(K) \cap U = \{0\}$: an element $a \in K$ with $|a|_v < 1$ at all archimedean places and $|a|_v \leq 1$ at all finite places satisfies $\prod_v |a|_v < 1$ if $a \neq 0$, contradicting the product formula $\prod_v |a|_v = 1$ (in logarithmic form, [14, Arakelov Degree of Principal Divisors]). So $a = 0$.

From this point forward, K is identified with its image $\iota(K) \subseteq \mathbb{A}_K$ and treated as a discrete subring of $\mathbb{A}_K$.

Theorem 9.1.6: Compactness of $\mathbb{A}_K/K$

The quotient $\mathbb{A}_K/K$ is compact.

***Proof*:**
Step 1: Reduction to a fundamental domain. Let $n = [K : \mathbb{Q}]$. A $\mathbb{Z}$-basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}_K$ determines an isomorphism of topological groups $\mathbb{A}_K \cong \mathbb{A}_{\mathbb{Q}}^n$ (via the basis). Under this isomorphism, K embeds as a discrete subgroup of $\mathbb{A}_{\mathbb{Q}}^n$, and it suffices to show $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is compact (the general case follows by taking the n-fold product).

Step 2: The case $K = \mathbb{Q}$. Here $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^{\mathrm{fin}}$ and $\mathbb{A}_{\mathbb{Q}}^{\mathrm{fin}} = \prod_p{}' (\mathbb{Q}_p, \mathbb{Z}_p)$. The subring of finite integral adèles is $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, which is compact. The claim is that the compact set $[0,1) \times \widehat{\mathbb{Z}}$ maps surjectively onto $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$. Given an adèle $(a_\infty, (a_p)_p)$: first, since the a_p are integral at almost all p, there exists $N \in \mathbb{Z}$ with $Na_p \in \mathbb{Z}_p$ for all p (take N divisible by all primes where $a_p \notin \mathbb{Z}_p$). The Chinese Remainder Theorem and strong approximation for $\mathbb{Q}$ produce $q \in \mathbb{Q}$ with $a_p - q \in \mathbb{Z}_p$ for all p and $a_\infty - q \in [0,1)$ (first approximate the finite components, then adjust the real component by an integer). Then $(a_v)_v - \iota(q) \in [0,1) \times \widehat{\mathbb{Z}}$.

Step 3: Compactness. The image of the compact set $[0,1] \times \widehat{\mathbb{Z}}$ under the continuous projection $\mathbb{A}_{\mathbb{Q}} \to \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is compact. Since this image is all of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ (by Step 2), the quotient is compact.

The compactness of $\mathbb{A}_K/K$ is the adèlic reformulation of two classical finiteness theorems: the finiteness of the class number and the finiteness of the unit group modulo roots of unity. The adèlic language packages these into a single topological statement.

Example 9.1.7: Adèles of $\mathbb{Q}$

1. The integral adèles of $\mathbb{Q}$ are $\mathbb{A}_{\mathbb{Q}}^{\circ} = \mathbb{R} \times \widehat{\mathbb{Z}}$. The diagonal image of $\mathbb{Z}$ in $\mathbb{A}_{\mathbb{Q}}^{\circ}$ is the set $\{(n, (n, n, \dots)) : n \in \mathbb{Z}\}$, which is dense in $\{0\} \times \widehat{\mathbb{Z}}$ (by the density of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$ from exercise 1.1.1 (1)) but discrete in the full adèle topology.

2. Strong approximation for $\mathbb{Q}$ takes the following form: the image of $\mathbb{Q}$ in $\mathbb{A}_{\mathbb{Q}}^{\mathrm{fin}} = \prod_p{}' \mathbb{Q}_p$ is dense. Stated differently: given elements $a_{p_1} \in \mathbb{Q}_{p_1}, \dots, a_{p_r} \in \mathbb{Q}_{p_r}$ and an error bound $\epsilon > 0$, there exists $q \in \mathbb{Q}$ with $|q - a_{p_i}|_{p_i} < \epsilon$ for all i and $|q|_p \leq 1$ for all other primes p. The proof uses the Chinese Remainder Theorem ([9, Chinese Remainder Theorem For Integers]) and is standard.

3. (If the reader has read [4]) The adèle ring is the ring of global sections of a certain sheaf on $\operatorname{Spec} \mathcal{O}_K$ with generic fiber K. More precisely, for a smooth group scheme G over $\mathcal{O}_K$, the restricted product $\prod_v{}' G(K_v)$ (restricted with respect to $G(\mathcal{O}_v)$) is the group of adèlic points $G(\mathbb{A}_K)$. The case $G = \mathbb{G}_a$ recovers $\mathbb{A}_K$, and $G = \mathbb{G}_m$ will produce the idèle group in section 9.2.

Exercise 9.1.1

1. Show that the restricted product topology on $\prod_v'(G_v, K_v)$ makes it a topological group (i.e., the group operations are continuous).
2. Verify the fundamental domain claim for $\mathbb{A}_\mathbb{Q}/\mathbb{Q}$ directly: show that every element of $\mathbb{A}_\mathbb{Q}$ is equivalent modulo $\mathbb{Q}$ to an element of $[0,1) \times \widehat{\mathbb{Z}}$.
3. Let $S \subseteq V_K$ be a finite set of places containing all archimedean places. Show that the S-adèles $\mathbb{A}_{K,S} = \prod_{v\in S} K_v \times \prod_{v\notin S} \mathcal{O}_v$ form an open subring of $\mathbb{A}_K$, and that $\mathbb{A}_K = \bigcup_S \mathbb{A}_{K,S}$ as S ranges over all such finite sets.
4. Let L/K be a finite extension of number fields. Show that $\mathbb{A}_L \cong \mathbb{A}_K \otimes_K L$ as topological rings, where the right side carries the topology making $\mathbb{A}_K \otimes_K L \cong \mathbb{A}_K^{[L:K]}$ (via a K-basis of L) a product of copies of $\mathbb{A}_K$.
5. Use the product formula $\prod_v |a|_v = 1$ for $a \in K^\times$ to show that the diagonal embedding $K \hookrightarrow \mathbb{A}_K$ has discrete image, without appealing to the explicit fundamental domain argument.

9.2 The Idèle Group and Idèle Class Group

The multiplicative group $\mathbb{A}_K^\times$ of the adèle ring, equipped with the subspace topology from $\mathbb{A}_K$, fails to be a topological group: inversion is not continuous (an adèle can have components approaching zero at infinitely many places, making the inverse escape every compact set). The idèle group corrects this by using the restricted product topology on the units, treating the local unit groups $\mathcal{O}_v^\times$ as the compact open subgroups.

Definition 9.2.1: Idèle Group

The *idèle group* of K is the restricted product:

$$\mathbb{I}_K = \prod_{v\in V_K}'(K_v^\times, \mathcal{O}_v^\times)$$

where $\mathcal{O}_v^\times$ is the group of local units at each finite place v, and the archimedean factors $K_v^\times$ ($v \in V_K^\infty$, with $K_v^\times = \mathbb{R}^\times$ or $\mathbb{C}^\times$) appear without restriction. The group operation is componentwise multiplication.

An idèle $\alpha = (\alpha_v)_v$ is a tuple of local elements $\alpha_v \in K_v^\times$ with $\alpha_v \in \mathcal{O}_v^\times$ (i.e., $|\alpha_v|_v = 1$) for all but finitely many finite places. The set of places where $\alpha_v \notin \mathcal{O}_v^\times$ is the *support* of the idèle. The *finite idèles* are $\mathbb{I}_K^{\mathrm{fin}} = \prod_{v\in V_K^{\mathrm{fin}}}'(K_v^\times, \mathcal{O}_v^\times)$.

Proposition 9.2.2: Topology of the Idèle Group

The idèle group $\mathbb{I}_K$ is a locally compact Hausdorff topological group. The natural inclusion $\mathbb{I}_K \hookrightarrow \mathbb{A}_K$ (viewing $K_v^\times \subseteq K_v$) is continuous, but the subspace topology on $\mathbb{I}_K$ inherited from $\mathbb{A}_K$ is strictly coarser than the idèle topology.

***Proof*:**
The first claim follows from proposition 9.1.2: the groups $K_v^\times$ are locally compact and the subgroups $\mathcal{O}_v^\times$ are compact and open in $K_v^\times$ (at a finite place, $\mathcal{O}_v^\times = \{x \in K_v \mid |x|_v = 1\}$ is the preimage of $\{1\} \subseteq \mathbb{R}_{\geq 0}$ under the continuous open map $|\cdot|_v$, hence compact and open).

For the inclusion: the map $\mathbb{I}_K \hookrightarrow \mathbb{A}_K$ sends $(\alpha_v) \mapsto (\alpha_v)$, and a basic open set in $\mathbb{A}_K$ restricts to an open set in $\mathbb{I}_K$ (since $\mathcal{O}_v^\times \subseteq \mathcal{O}_v$ is an intersection of open-in-$\mathbb{I}_K$ with open-in-$\mathbb{A}_K$).

For the strict coarseness: the map $\alpha \mapsto \alpha^{-1}$ is continuous in the restricted product topology on $\mathbb{I}_K$ (since inversion is continuous on each $K_v^\times$ and preserves $\mathcal{O}_v^\times$), but not continuous in the subspace topology from $\mathbb{A}_K$. To see this, consider a sequence of idèles $\alpha^{(n)}$ in $\mathbb{A}_\mathbb{Q}$ with $\alpha_{p_n}^{(n)} = p_n$ (a uniformizer at the nth prime) and $\alpha_v^{(n)} = 1$ for all other v. In the subspace topology from $\mathbb{A}_\mathbb{Q}$, $\alpha^{(n)} \to 1$ (each component eventually equals 1). But $(\alpha^{(n)})_{p_n}^{-1} = p_n^{-1}$, and the sequence $(\alpha^{(n)})^{-1}$ does not converge to 1 in $\mathbb{A}_\mathbb{Q}$ (the p_n-component $p_n^{-1} \notin \mathbb{Z}_{p_n}$ for each n). So inversion is discontinuous in the subspace topology, confirming it is strictly coarser.

The idèle group carries a natural "size" function that measures the total effect of an idèle across all places.

Definition 9.2.3: Idèle Norm

The *idèle norm* (or *content*) is the continuous homomorphism:

$$|\cdot|_K : \mathbb{I}_K \to \mathbb{R}_{>0}, \qquad |(\alpha_v)|_K = \prod_{v \in V_K} |\alpha_v|_v$$

The product converges because $|\alpha_v|_v = 1$ for all but finitely many v, reducing it to a finite product.

Continuity of the idèle norm follows from the fact that on each basic open subgroup $\prod_{v \in S} K_v^\times \times \prod_{v \notin S} \mathcal{O}_v^\times$, the norm is the finite product $\prod_{v \in S} |\alpha_v|_v$, a continuous function of finitely many variables.

The number field K embeds into $\mathbb{I}_K$ diagonally, and the product formula constrains this embedding.

Theorem 9.2.4: Product Formula and Discreteness

The diagonal embedding $K^\times \hookrightarrow \mathbb{I}_K$, sending $a \mapsto (a, a, a, \dots)$, has discrete image. For every $a \in K^\times$:

$$|a|_K = \prod_v |a|_v = 1$$

In particular, $K^\times \subseteq \ker |\cdot|_K$.

***Proof*:**
The product formula is [14, Arakelov Degree of Principal Divisors] in logarithmic form. For discreteness: the open neighborhood $U = \left\{(\alpha_v) \in \mathbb{I}_K \;\middle|\; |\alpha_v - 1|_v < 1 \text{ for all } v \in V_K^\infty,\ \alpha_v \in \mathcal{O}_v^\times \text{ for all } v \in V_K^{\text{fin}}\right\}$ of the identity in $\mathbb{I}_K$ satisfies $K^\times \cap U = \{1\}$. If $a \in K^\times \cap U$, then $|a|_v \leq 1$ at all finite places (so $a \in \mathcal{O}_K$) and $|a - 1|_v < 1$ at all archimedean places. The condition $|a - 1|_v < 1$ at archimedean

places and $a \in \mathcal{O}_K$ leaves only finitely many possibilities for a (the ring $\mathcal{O}_K$ is discrete in the archimedean completions). Combined with $|a|_v = 1$ at all finite v (from $a \in \mathcal{O}_v^\times$), the element a is a unit of $\mathcal{O}_K$ satisfying $|a-1|_v < 1$ at all archimedean places. By Dirichlet's unit theorem (see [6, Dirichlet Unit Theorem]), $\mathcal{O}_K^\times$ is finitely generated, and the archimedean constraint forces $a = 1$ (the only unit in $\mathcal{O}_K$ within distance 1 of 1 at every archimedean place, for the neighborhood U chosen sufficiently small).

From this point forward, $K^\times$ is identified with its image in $\mathbb{I}_K$.

Definition 9.2.5: Idèle Class Group

The *idèle class group* of K is the quotient:

$$C_K = \mathbb{I}_K / K^\times$$

equipped with the quotient topology. The idèle norm descends to a continuous homomorphism $|\cdot|_K \colon C_K \to \mathbb{R}_{>0}$ (since $|a|_K = 1$ for $a \in K^\times$). The *norm-one idèle class group* is:

$$C_K^0 = \ker\bigl(|\cdot|_K \colon C_K \to \mathbb{R}_{>0}\bigr)$$

The idèle class group is the global analogue of $K_v^\times$ for a local field. The local reciprocity map of theorem 8.4.4 sends $K_v^\times \to G_{K_v}^{\mathrm{ab}}$. Global reciprocity (chapter 10) will send $C_K \to G_K^{\mathrm{ab}}$.

The following theorem is the idèlic repackaging of two classical finiteness results: the finiteness of the class group $\mathrm{Cl}\,(K)$ and the Dirichlet unit theorem.

Theorem 9.2.6: Compactness of C_K^0

The norm-one idèle class group C_K^0 is compact.

***Proof*:**

Step 1: The class group. Let $\mathbb{I}_K^\infty = \prod_{v\mid\infty} K_v^\times \times \prod_{v\nmid\infty} \mathcal{O}_v^\times$ be the idèles that are units at every finite place. Sending an idèle $\alpha = (\alpha_v)$ to the class of the fractional ideal $\prod_{v\nmid\infty} \mathfrak{p}_v^{v(\alpha_v)}$ is a surjective homomorphism $\mathbb{I}_K \to \mathrm{Cl}\,(K)$ with kernel $K^\times \mathbb{I}_K^\infty$: if the ideal of α is generated by $a \in K^\times$, then $a^{-1}\alpha$ is a unit at every finite place. The map kills $K^\times$, so it factors through C_K. It stays surjective on C_K^0, because multiplying one archimedean component by a positive real number makes the norm 1 and leaves the ideal unchanged. Since $|a|_K = 1$ for $a \in K^\times$, the kernel of $C_K^0 \to \mathrm{Cl}\,(K)$ is the image H of $\mathbb{I}_K^\infty \cap \ker|\cdot|_K$ in C_K. So $C_K^0/H \cong \mathrm{Cl}\,(K)$, which is finite by [6, Class Number].

Step 2: H is compact. An idèle in $\mathbb{I}_K^\infty$ has $|\alpha_v|_v = 1$ at every finite place, so $\mathbb{I}_K^\infty \cap \ker|\cdot|_K = A \times B$ with

$$A = \left\{ (x_v) \in \prod_{v\mid\infty} K_v^\times \;\middle|\; \prod_{v\mid\infty} |x_v|_v = 1 \right\}, \qquad B = \prod_{v\nmid\infty} \mathcal{O}_v^\times$$

Each $\mathcal{O}_v^\times$ is compact, so B is compact by [17, Tychonoff's Theorem]. Let r_1 and r_2 be the numbers of real and complex places, and let $W \subseteq \mathbb{R}^{r_1+r_2}$ be the hyperplane of vectors with coordinate sum 0. The map $\lambda \colon A \to W$, $\lambda(x) = (\log|x_v|_v)_{v\mid\infty}$, is a continuous surjective

homomorphism. Writing each x_v as its absolute value times a sign or a point of the unit circle identifies A with $\ker\lambda \times W$ as a topological group, and $\ker\lambda$ is a product of the compact groups $\{\pm 1\}$ and S^1. So $\lambda^{-1}(P)$ is compact for every compact $P \subseteq W$.

The units $\mathcal{O}_K^\times = K^\times \cap \mathbb{I}_K^\infty$ sit diagonally in $A \times B$, because $\prod_{v|\infty} |u|_v = |u|_K = 1$. The subgroup $\lambda(\mathcal{O}_K^\times)$ of W is discrete: the units u with $\lambda(u)$ in a bounded set have all their archimedean absolute values bounded, and only finitely many elements of $\mathcal{O}_K$ do, since the archimedean embedding of $\mathcal{O}_K$ is a lattice[a] and a bounded set meets a lattice in finitely many points ([6, Equivalence Of Conditions]). The same finiteness makes the kernel of λ on $\mathcal{O}_K^\times$ a finite group of units, so it consists of roots of unity. It contains $\mu(K)$, whose elements have absolute value 1 everywhere, so it equals $\mu(K)$. By Dirichlet's unit theorem ([6, Dirichlet Unit Theorem]), $\lambda(\mathcal{O}_K^\times) \cong \mathcal{O}_K^\times/\mu(K)$ is free of rank $r_1 + r_2 - 1 = \dim W$. A discrete subgroup of W that is free of rank $\dim W$ is a lattice spanning W ([6, Equivalence Of Conditions]), so the closed fundamental parallelepiped P of a $\mathbb{Z}$-basis is compact and $W = \lambda(\mathcal{O}_K^\times) + P$. Then $F = \lambda^{-1}(P)$ is compact and $A = \mathcal{O}_K^\times F$.

Given $(a, b) \in A \times B$, choose $u \in \mathcal{O}_K^\times$ with $u^{-1}a \in F$. Since u is a unit at every finite place, $u^{-1}(a, b) = (u^{-1}a, u^{-1}b)$ lies in $F \times B$. So H is the image of the compact set $F \times B$ under the continuous projection $\mathbb{I}_K \to C_K$, and H is compact.

Step 3: Assembly. By Step 1, H has finite index in C_K^0, so C_K^0 is the union of finitely many cosets gH. Each is compact by Step 2, so C_K^0 is compact.

[a] [6, Archimedean Embedding of $\mathcal{O}_K$ is Lattice].

The compactness of C_K^0 is the single most important structural fact about the idèle class group. It enters the global theory through the following consequence: the Herbrand quotient of C_L^0 for a cyclic extension L/K is well defined and computable, which is the starting point for the cohomological approach to global class field theory.

Proposition 9.2.7: Structure of C_K

The idèle norm gives a split short exact sequence:

$$1 \to C_K^0 \to C_K \xrightarrow{|\cdot|_K} \mathbb{R}_{>0} \to 1$$

The splitting is given by $t \mapsto (\alpha_v)$ where $\alpha_{v_0} = t$ at a fixed archimedean place v_0 and $\alpha_v = 1$ for $v \neq v_0$. In particular, $C_K \cong C_K^0 \times \mathbb{R}_{>0}$.

***Proof*:**
Surjectivity of $|\cdot|_K$: for any $t \in \mathbb{R}_{>0}$, the idèle with $\alpha_{v_0} = t$ at a real place v_0 and $\alpha_v = 1$ elsewhere has $|(\alpha_v)|_K = t$. Its image in C_K maps to t. The section $s\colon \mathbb{R}_{>0} \to C_K$ is a continuous homomorphism (it is the composition of the inclusion $\mathbb{R}_{>0} \hookrightarrow K_{v_0}^\times$ with the projection $\mathbb{I}_K \to C_K$). The identity $|\cdot|_K \circ s = \mathrm{id}$ gives the splitting.

Example 9.2.8: Idèles of $\mathbb{Q}$

1. The idèle group of $\mathbb{Q}$ is $\mathbb{I}_{\mathbb{Q}} = \mathbb{R}^\times \times \prod_p{}' (\mathbb{Q}_p^\times, \mathbb{Z}_p^\times)$. Since $\mathbb{Q}$ is a PID, the class group $\mathrm{Cl}\,(\mathbb{Q})$ is trivial, so $C_{\mathbb{Q}} = \mathbb{I}_{\mathbb{Q}}/\mathbb{Q}^\times$ has no "class group part." The structure is: $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$. To see this: every idèle $(\alpha_\infty, (\alpha_p)_p)$ is equivalent modulo $\mathbb{Q}^\times$ to one with $\alpha_\infty > 0$ (multiply by -1 if needed) and $\alpha_p \in \mathbb{Z}_p^\times$ for all p (multiply by a suitable power of each bad prime). The residual freedom is multiplication by positive rationals that are p-adic units for all p, i.e., by $\{1\}$. So $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$. The norm-one subgroup is $C_{\mathbb{Q}}^0 \cong \widehat{\mathbb{Z}}^\times$ (compact, as required by theorem 9.2.6).
2. The local-to-global comparison: for each prime p, the inclusion $\mathbb{Q}_p^\times \hookrightarrow \mathbb{I}_{\mathbb{Q}}$ (at the p-component, with 1 elsewhere) followed by the projection $\mathbb{I}_{\mathbb{Q}} \to C_{\mathbb{Q}}$ gives a continuous homomorphism $\mathbb{Q}_p^\times \to C_{\mathbb{Q}}$. These local maps assemble the local arithmetic of each completion into the global idèle class group. The local reciprocity maps rec_{K_v} of chapter 8 will similarly assemble into the global reciprocity map via these inclusions.
3. For comparison with the classical ideal class group: the map $\mathbb{I}_{\mathbb{Q}}^{\mathrm{fin}} \to \mathrm{Div}(\mathbb{Z}) = \bigoplus_p \mathbb{Z}$ sending $(\alpha_p)_p \mapsto \sum_p v_p(\alpha_p) \cdot (p)$ is surjective with kernel $\prod_p \mathbb{Z}_p^\times$. Passing to the quotient by $\mathbb{Q}^\times$ recovers $\mathrm{Cl}\,(\mathbb{Z}) = 0$ (every fractional ideal of $\mathbb{Z}$ is principal). For a general number field, the same construction gives the class group as a quotient of C_K.

Exercise 9.2.1

1. Show that the subspace topology on $\mathbb{I}_K$ from $\mathbb{A}_K$ is strictly coarser than the restricted product topology by verifying that the set $W = \left\{(\alpha_v) \in \mathbb{I}_K \;\middle|\; |\alpha_v|_v = 1 \text{ for all } v \in V_K^{\mathrm{fin}}\right\}$ is open in the restricted product topology but not in the subspace topology.
2. Verify the claim $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$ by constructing an explicit isomorphism and checking it is a homeomorphism.
3. Show that the idèle norm $|\cdot|_K \colon \mathbb{I}_K \to \mathbb{R}_{>0}$ is surjective, and that the composition $K^\times \hookrightarrow \mathbb{I}_K \xrightarrow{|\cdot|_K} \mathbb{R}_{>0}$ has image $\{1\}$.
4. Let S be a finite set of places containing V_K^∞. Define the S-idèle class group $C_{K,S} = \mathbb{I}_K/(K^\times \cdot \prod_{v \notin S} \mathcal{O}_v^\times)$. Show that there is a natural surjection $C_{K,S} \twoheadrightarrow \mathrm{Cl}_S(K)$ to the S-class group, and identify its kernel in terms of the S-unit group $\mathcal{O}_{K,S}^\times$.
5. Let L/K be a finite extension. Show that the norm map $N_{L/K} \colon \mathbb{I}_L \to \mathbb{I}_K$, defined by $N_{L/K}(\alpha)_v = \prod_{w|v} N_{L_w/K_v}(\alpha_w)$, is a continuous homomorphism satisfying $|N_{L/K}(\alpha)|_K = |\alpha|_L$ for all $\alpha \in \mathbb{I}_L$.

9.3 Galois Action on Idèles

The adèle ring and idèle group become Galois modules when a Galois extension L/K is fixed. The action of $G = \mathrm{Gal}(L/K)$ on $\mathbb{I}_L$ permutes the places of L above each place of K and acts on the corresponding completions. The central result of this section is that the G-module $\mathbb{I}_L$ decomposes

as a restricted product of induced modules (proposition 9.3.4), reducing the Galois cohomology of idèles to a direct sum of local cohomology groups. This decomposition is the bridge between the local theory of chapter 8 and the global theory of chapter 10.

Throughout this section, L/K denotes a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$.

Proposition 9.3.1: Galois Action on Adèles and Idèles

The group G acts continuously on $\mathbb{A}_L$ and $\mathbb{I}_L$ by:

$$(\sigma \cdot \alpha)_w = \sigma(\alpha_{\sigma^{-1}w})$$

for $\sigma \in G$, $\alpha = (\alpha_w)_w \in \mathbb{A}_L$ (or $\mathbb{I}_L$), and w a place of L. Here σw is the place $x \mapsto |\sigma^{-1}(x)|_w$, so that $\sigma^{-1}w$ is the place $x \mapsto |\sigma(x)|_w$, and σ acts on $L_{\sigma^{-1}w}$ via the canonical isomorphism $\sigma\colon L_{\sigma^{-1}w} \xrightarrow{\sim} L_w$ induced by $\sigma\colon L \xrightarrow{\sim} L$.

***Proof*:**
The map σ permutes the places of L: if w is a place extending v of K, then σw is another place extending v (since σ fixes K), and $(\tau\sigma)w = \tau(\sigma w)$ because $|x|_{\tau(\sigma w)} = |\tau^{-1}(x)|_{\sigma w} = |\sigma^{-1}\tau^{-1}(x)|_w$. Since $|\sigma(x)|_w = |x|_{\sigma^{-1}w}$ for $x \in L$, the automorphism σ of L is an isometry from $|\cdot|_{\sigma^{-1}w}$ to $|\cdot|_w$, so it extends by continuity to a continuous isomorphism $\sigma\colon L_{\sigma^{-1}w} \to L_w$ of locally compact fields, sending $\mathcal{O}_{\sigma^{-1}w}$ to $\mathcal{O}_w$ (the absolute value is preserved). The extension of $\tau\sigma$ is the composite of the extensions of σ and τ, both being continuous and agreeing on the dense subfield L, so $(\tau \cdot (\sigma \cdot \alpha))_w = \tau(\sigma(\alpha_{\sigma^{-1}\tau^{-1}w})) = ((\tau\sigma) \cdot \alpha)_w$ and the formula is an action. Each σ acts by a ring automorphism of $\mathbb{A}_L$ (componentwise): it permutes the factors and applies a continuous field isomorphism on each. The image of an adèle is an adèle: if $\alpha_w \in \mathcal{O}_w$ for all but finitely many w, then $(\sigma \cdot \alpha)_w = \sigma(\alpha_{\sigma^{-1}w}) \in \mathcal{O}_w$ for all but finitely many w (the set of exceptional places is permuted by σ). Continuity follows from the continuity of each local map $\sigma\colon L_{\sigma^{-1}w} \to L_w$. The same argument applies to $\mathbb{I}_L$ (units are permuted to units, and $\mathcal{O}_w^\times$ maps to $\mathcal{O}_w^\times$).

The Galois action on adèles and idèles recovers the global elements as fixed points.

Theorem 9.3.2: Galois Fixed Points

$\mathbb{A}_L^G = \mathbb{A}_K$ and $\mathbb{I}_L^G = \mathbb{I}_K$.

***Proof*:**
An adèle $\alpha \in \mathbb{A}_L$ is G-fixed if and only if $\sigma(\alpha_{\sigma^{-1}w}) = \alpha_w$ for all $\sigma \in G$ and all places w of L. Fixing a place v of K and choosing a place w_0 of L above v: the places above v are $\{\sigma w_0 : \sigma \in G/G_{w_0}\}$ where $G_{w_0} = \{\sigma \in G \mid \sigma w_0 = w_0\}$ is the decomposition group. The G-fixed condition requires $\alpha_{\sigma w_0} = \sigma(\alpha_{w_0})$ for all σ, and the G_{w_0}-fixed condition on the w_0-component requires $\alpha_{w_0} \in L_{w_0}^{G_{w_0}} = K_v$. So a G-fixed adèle has $\alpha_{w_0} \in K_v$ and all components above v determined by this single value via the Galois action. The resulting tuple is exactly the diagonal embedding of an element of K_v into $\prod_{w \mid v} L_w$. Assembling over all v: $\mathbb{A}_L^G = \mathbb{A}_K$.

The same argument applies to $\mathbb{I}_L$, replacing K_v by $K_v^\times$ and L_w by $L_w^\times$ throughout.

The norm map on idèles assembles the local norms at each place.

Definition 9.3.3: Idèle Norm Map

The *idèle norm map* $N_{L/K}\colon \mathbb{I}_L \to \mathbb{I}_K$ is defined by:

$$N_{L/K}(\alpha)_v = \prod_{w\mid v} N_{L_w/K_v}(\alpha_w)$$

for each place v of K, where w ranges over the places of L above v and N_{L_w/K_v} is the local norm.

The norm map is compatible with the abstract norm $N_G(\alpha) = \prod_{\sigma\in G} \sigma\cdot\alpha$ on the G-module $\mathbb{I}_L$: the product $\prod_\sigma \sigma(\alpha_{\sigma^{-1}w_0})$ at a fixed place $w_0\mid v$ computes $\prod_{w\mid v} N_{L_w/K_v}(\alpha_w)$ after accounting for the decomposition group orbits (see exercise 9.3.1 (1)).

The key structural result decomposes $\mathbb{I}_L$ as a G-module into induced modules from the local fields.

Proposition 9.3.4: Decomposition of Idèles as G-Module

For each place v of K, fix a place w_0 of L above v, and let $G_{w_0} = \mathrm{Gal}(L_{w_0}/K_v) \le G$ be the decomposition group. Then:

$$\mathbb{I}_L \cong \prod_{v\in V_K}{}' \mathrm{Ind}_{G_{w_0}}^G(L_{w_0}^\times)$$

as G-modules, where $\mathrm{Ind}_{G_{w_0}}^G(L_{w_0}^\times) = \mathbb{Z}[G]\otimes_{\mathbb{Z}[G_{w_0}]} L_{w_0}^\times$ is the induced module, and the restricted product is taken with respect to $\mathrm{Ind}_{G_{w_0}}^G(\mathcal{O}_{w_0}^\times)$ at finite places.

***Proof*:**
The places w of L above v are in bijection with G/G_{w_0} via $\sigma\mapsto\sigma w_0$. The v-component of $\mathbb{I}_L$ is $\prod_{w\mid v} L_w^\times = \prod_{\sigma\in G/G_{w_0}} L_{\sigma w_0}^\times$. The G-action permutes these factors: τ sends the σw_0-component to the $\tau\sigma w_0$-component via $\tau\colon L_{\sigma w_0}\xrightarrow{\sim} L_{\tau\sigma w_0}$. This is the defining property of the induced module $\mathrm{Ind}_{G_{w_0}}^G(L_{w_0}^\times) = \bigoplus_{\sigma\in G/G_{w_0}} L_{\sigma w_0}^\times$: the G-module that is freely generated from the G_{w_0}-module $L_{w_0}^\times$ by letting G permute the coset summands.

Taking the restricted product over all v: the integrality condition $\alpha_w\in\mathcal{O}_w^\times$ for all but finitely many w translates to the restricted product being taken with respect to $\mathrm{Ind}_{G_{w_0}}^G(\mathcal{O}_{w_0}^\times) = \bigoplus_{\sigma\in G/G_{w_0}} \mathcal{O}_{\sigma w_0}^\times$ at each finite v. The isomorphism is G-equivariant by construction.

Applying Shapiro's lemma (theorem 3.5.5) to this decomposition gives the fundamental reduction from global to local cohomology.

Theorem 9.3.5: Cohomology of Idèles

For every $n \in \mathbb{Z}$:

$$\hat{H}^n(G, \mathbb{I}_L) \cong \bigoplus_{v \in V_K} \hat{H}^n(G_{w_0}, L_{w_0}^\times)$$

where for each place v of K a place w_0 of L above v is fixed and $G_{w_0} = \mathrm{Gal}(L_{w_0}/K_v)$ is its decomposition group. In particular $H^n(G, \mathbb{I}_L) \cong \bigoplus_{v \in V_K} H^n(G_{w_0}, L_{w_0}^\times)$ for $n \geq 1$.

***Proof*:**

For a place v of K put $F_v = \prod_{w|v} L_w^\times$, and for v finite put $U_v = \prod_{w|v} \mathcal{O}_w^\times$. By proposition 9.3.4, $\mathbb{I}_L$ is the restricted product of the induced modules $\mathrm{Ind}_{G_{w_0}}^G(L_{w_0}^\times)$ with respect to the induced modules $\mathrm{Ind}_{G_{w_0}}^G(\mathcal{O}_{w_0}^\times)$: for each v there is an isomorphism of G-modules $F_v \cong \mathrm{Ind}_{G_{w_0}}^G(L_{w_0}^\times)$, carrying U_v onto $\mathrm{Ind}_{G_{w_0}}^G(\mathcal{O}_{w_0}^\times)$ when v is finite.

Step 1: Products and unions. Let $\hat{P}_\bullet$ be the complete standard resolution, so that $\hat{H}^n(G, M)$ is the n-th cohomology of the complex $\mathrm{Hom}_G(\hat{P}_\bullet, M)$ (definition 4.1.4). For a family (M_i) of G-modules, $\mathrm{Hom}_G(\hat{P}_k, \prod_i M_i) = \prod_i \mathrm{Hom}_G(\hat{P}_k, M_i)$ compatibly with the differentials, and kernels and images in a product of complexes are computed factor by factor. Hence the projections induce an isomorphism $\hat{H}^n(G, \prod_i M_i) \cong \prod_i \hat{H}^n(G, M_i)$, natural in the family. Next let M be the union of a family of G-submodules M_S that is directed: any two members lie in a third. Each $\hat{P}_k$ is a finitely generated abelian group (definition 4.1.3), so every G-homomorphism $\hat{P}_k \to M$ takes its values in a single M_S. Consequently every class in $\hat{H}^n(G, M)$ is the image of a class in some $\hat{H}^n(G, M_S)$. Likewise, if a cocycle z with values in M_S is the coboundary of some $c \in \mathrm{Hom}_G(\hat{P}_{n-1}, M)$, then z is already a coboundary in $\mathrm{Hom}_G(\hat{P}_\bullet, M_{S'})$ for any member $M_{S'}$ containing M_S and the values of c. So a class of $\hat{H}^n(G, M_S)$ that vanishes in $\hat{H}^n(G, M)$ vanishes in some $\hat{H}^n(G, M_{S'})$ with $M_S \subseteq M_{S'}$.

Step 2: Shapiro's lemma in Tate cohomology. Let $H \leq G$ and let A be an H-module. Then $\hat{H}^n(G, \mathrm{Ind}_H^G A) \cong \hat{H}^n(H, A)$ for every $n \in \mathbb{Z}$. For $n \geq 1$ both sides are ordinary cohomology (proposition 4.1.5(1)), and the isomorphism is theorem 3.5.5 for the open subgroup H of the finite group G. For $n \leq 0$, let $J = \mathbb{Z}[H] \otimes_{\mathbb{Z}} A$ with H acting on the left factor, let $\varepsilon \colon J \to A$ be $h \otimes a \mapsto ha$, and put $A_1 = \ker \varepsilon$. The functor Ind_H^G is exact, being a finite direct sum of copies of the identity on underlying abelian groups (definition 3.5.2), and $\mathrm{Ind}_H^G J = \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} (\mathbb{Z}[H] \otimes_{\mathbb{Z}} A) \cong \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$ by $g \otimes (h \otimes a) \mapsto gh \otimes a$, with G acting on the left factor. By proposition 4.1.7, for H and for G, both J and $\mathrm{Ind}_H^G J$ are Tate-acyclic, so the long exact sequences (proposition 4.1.6) of $0 \to A_1 \to J \to A \to 0$ and of the induced sequence give isomorphisms $\hat{H}^m(H, A) \cong \hat{H}^{m+1}(H, A_1)$ and $\hat{H}^m(G, \mathrm{Ind}_H^G A) \cong \hat{H}^{m+1}(G, \mathrm{Ind}_H^G A_1)$ for every m. Repeating the construction with A_1 in place of A, and so on, $1 - n$ times in all, identifies both sides of the claim with the two sides of the case of degree 1, already treated.

Step 3: Units at unramified places. Let v be a finite place of K unramified in L. Then L_{w_0}/K_v is an unramified Galois extension of local fields with group G_{w_0} [14, Global To Local Correspondence], cyclic of order $f = [L_{w_0} : K_v]$ [14, Characterizing Unramified Extension Of Local Fields]. By theorem 8.1.3(1) and proposition 4.1.5(1), $\hat{H}^1(G_{w_0}, \mathcal{O}_{w_0}^\times) = 0$, and $\hat{H}^0(G_{w_0}, \mathcal{O}_{w_0}^\times) = 0$

by theorem 8.1.3(4). By theorem 4.3.4, $\hat{H}^k(G_{w_0}, \mathcal{O}_{w_0}^\times) = 0$ for every $k \in \mathbb{Z}$, and by Step 2, $\hat{H}^k(G, U_v) = 0$ for every $k \in \mathbb{Z}$.

Step 4: The S-idèles. Let S_0 be the set of archimedean places of K together with the finite places ramified in L, a finite set [6, Ramification In Number Fields]. For a finite set $S \supseteq S_0$ of places of K put

$$\mathbb{I}_{L,S} = \prod_{v \in S} F_v \times \prod_{v \notin S} U_v$$

a subgroup of $\mathbb{I}_L$. It is a G-submodule: for $\sigma \in G$ the place $\sigma^{-1}w$ lies above the same place of K as w, since σ fixes K, and $\sigma\colon L_{\sigma^{-1}w} \to L_w$ is an isometry, so G preserves each F_v and each U_v (proposition 9.3.1). By Steps 1 and 3, and since S is finite,

$$\hat{H}^n(G, \mathbb{I}_{L,S}) \cong \prod_{v \in S} \hat{H}^n(G, F_v) \times \prod_{v \notin S} \hat{H}^n(G, U_v) = \bigoplus_{v \in S} \hat{H}^n(G, F_v)$$

For $S \subseteq S'$ the inclusion $\mathbb{I}_{L,S} \to \mathbb{I}_{L,S'}$ is the identity on the factors F_v with $v \in S$ and U_v with $v \notin S'$, and the inclusion $U_v \to F_v$ for $v \in S' \setminus S$. By the naturality in Step 1, and since $\hat{H}^n(G, U_v) = 0$, the induced map is the inclusion of $\bigoplus_{v \in S} \hat{H}^n(G, F_v)$ into $\bigoplus_{v \in S'} \hat{H}^n(G, F_v)$, which is injective.

Step 5: Passage to $\mathbb{I}_L$. An idèle has unit components at all but finitely many places, so $\mathbb{I}_L$ is the union of the $\mathbb{I}_{L,S}$, a directed family since $\mathbb{I}_{L,S}$ and $\mathbb{I}_{L,S'}$ lie in $\mathbb{I}_{L,S\cup S'}$. By Step 1, every class of $\hat{H}^n(G, \mathbb{I}_L)$ comes from some $\mathbb{I}_{L,S}$, and a class of $\hat{H}^n(G, \mathbb{I}_{L,S})$ that vanishes in $\hat{H}^n(G, \mathbb{I}_L)$ vanishes in some $\hat{H}^n(G, \mathbb{I}_{L,S'})$ with $S \subseteq S'$, hence is zero by Step 4. So $\hat{H}^n(G, \mathbb{I}_L)$ is the union of its subgroups $\bigoplus_{v \in S} \hat{H}^n(G, F_v)$, that is, $\bigoplus_{v \in V_K} \hat{H}^n(G, F_v)$. Finally $\hat{H}^n(G, F_v) \cong \hat{H}^n(G_{w_0}, L_{w_0}^\times)$ by Step 2, and Tate cohomology in degrees $n \geq 1$ is ordinary cohomology (proposition 4.1.5(1)), as claimed.

The sum is not finite in general, and only the unit factors are cohomologically trivial at the unramified places. At a finite place v unramified in L the summand in degree 2 is $H^2(G_{w_0}, L_{w_0}^\times) \cong \mathbb{Z}/[L_{w_0} : K_v]\mathbb{Z}$ (theorem 8.1.3(2)), which is nonzero unless v splits completely in L.

The decomposition $\hat{H}^n(G, \mathbb{I}_L) \cong \bigoplus_v \hat{H}^n(G_{w_0}, L_{w_0}^\times)$ is the mechanism by which the local duality theorems of chapter 8 enter the global picture. Each summand on the right is a local cohomology group computed in section 8.1 and section 8.2, and the global theory of chapter 10 will analyze the long exact sequence arising from the fundamental short exact sequence of G-modules.

Proposition 9.3.6: Fundamental Exact Sequence

The short exact sequence of G-modules:

$$1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$$

induces a long exact sequence in cohomology:

$$\cdots \to H^n(G, L^\times) \to H^n(G, \mathbb{I}_L) \to H^n(G, C_L) \xrightarrow{\delta^n} H^{n+1}(G, L^\times) \to \cdots$$

The terms $H^n(G, L^\times)$ are controlled by Hilbert 90 ($H^1 = 0$, theorem 8.1.1) and the Brauer group ($H^2 \cong \mathrm{Br}(L/K)$, theorem 7.2.7). The terms $H^n(G, \mathbb{I}_L)$ with $n \geq 1$ decompose into local cohomology by theorem 9.3.5. The terms $H^n(G, C_L)$ are the object of study in chapter 10.

***Proof*:**
The short exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ is exact: the diagonal embedding $L^\times \hookrightarrow \mathbb{I}_L$ has discrete image (theorem 9.2.4), and the idèle class group is the quotient by definition (definition 9.2.5). All three terms are discrete G-modules (the action factors through the finite group $G = \mathrm{Gal}(L/K)$), so the long exact sequence in group cohomology applies (see [13, Long Exact Sequence From a Right-Derived Functor]).

The beginning of this long exact sequence gives the information that chapter 10 will exploit. At $n = 0$: the sequence $1 \to K^\times \to \mathbb{I}_K \to C_K \to H^1(G, L^\times)$ and Hilbert 90 ($H^1(G, L^\times) = 0$) give the exactness of $1 \to K^\times \to \mathbb{I}_K \to C_K \to 0$, recovering the definition of C_K. At $n = 1$: the connecting map $\delta^0 \colon C_K \to H^1(G, L^\times) = 0$ is trivial, and the sequence continues:

$$0 \to H^1(G, C_L) \to H^2(G, L^\times) \to H^2(G, \mathbb{I}_L)$$

The map $H^2(G, L^\times) \to H^2(G, \mathbb{I}_L) \cong \bigoplus_v H^2(G_{w_0}, L_{w_0}^\times)$ sends a class of $\mathrm{Br}(L/K)$ to its family of local restrictions in the direct sum $\bigoplus_v \mathrm{Br}(L_{w_0}/K_v)$ of local Brauer groups, each identified with $\frac{1}{[L_{w_0}:K_v]}\mathbb{Z}/\mathbb{Z}$ by its local invariant map. The kernel of this map is exactly $H^1(G, C_L)$, which measures the failure of the local-to-global principle for the Brauer group. This is the starting point of the Poitou-Tate sequence in chapter 10.

Example 9.3.7: The Fundamental Sequence for Cyclic Extensions
Let L/K be a cyclic extension of number fields with $G = \langle \sigma \rangle \cong C_n$. The Tate cohomology of $\mathbb{I}_L$ decomposes as:

$$\hat{H}^0(G, \mathbb{I}_L) \cong \bigoplus_v \hat{H}^0(G_{w_0}, L_{w_0}^\times) = \bigoplus_v K_v^\times / N_{L_{w_0}/K_v}(L_{w_0}^\times)$$

At unramified finite places, the local norm is surjective on units and $N(\pi_{w_0}) = \pi_v^{f_v}$ (where f_v is the residue degree), so the local norm index $[K_v^\times : N(L_{w_0}^\times)] = [L_{w_0} : K_v]$ is accounted for by the valuation. At split places ($f_v = 1$, $G_{w_0} = 1$): the local norm is an isomorphism, contributing a trivial factor. The nonzero summands therefore come from the places that do not split completely, ramified or not: at an unramified place with $f_v > 1$ the summand is $\mathbb{Z}/f_v\mathbb{Z}$, generated by the class of π_v.

The Herbrand quotient $h(G, C_L)$ is not obtained from this decomposition. The group $\hat{H}^0(G, \mathbb{I}_L)$ is infinite whenever infinitely many places of K fail to split completely in L, and then $h(G, \mathbb{I}_L)$ is undefined. The computation of $h(G, C_L)$ uses the S-idèles $\mathbb{I}_{L,S}$ of the proof of theorem 9.3.5

for a large finite set S of places, whose Tate cohomology is the finite sum $\bigoplus_{v\in S}\hat{H}^n(G_{w_0}, L_{w_0}^\times)$, together with the S-units of L.

Exercise 9.3.1

1. Let $\alpha \in \mathbb{I}_L$ and $\sigma \in G$. Show that the abstract norm $N_G(\alpha) = \prod_{\sigma\in G}\sigma\cdot\alpha$ (computed in the G-module $\mathbb{I}_L$) agrees with the idèle norm map $N_{L/K}(\alpha) \in \mathbb{I}_K$ defined in definition 9.3.3. (*Hint:* compare the v-components for a fixed place v of K, using the orbit-stabilizer decomposition $G = \bigsqcup_{\sigma\in G/G_{w_0}} \sigma G_{w_0}$.)

2. Verify $\mathbb{A}_L^G = \mathbb{A}_K$ directly for $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, with $G = \{1, \sigma\}$ where σ is complex conjugation. Identify the places of L above each place of K and describe the Galois action explicitly.

3. Show that $H^1(G, \mathbb{I}_L) = 0$ by combining the Shapiro decomposition (theorem 9.3.5) with Hilbert 90 applied locally (theorem 8.1.1).

4. Let L/K be a cyclic extension with group G. Show that the Tate cohomology long exact sequence of $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ and Hilbert 90 identify $C_K/N_{L/K}(C_L)$ with the cokernel of the map $K^\times/N_{L/K}(L^\times) \to \bigoplus_v K_v^\times/N_{L_w/K_v}(L_w^\times)$ induced by the diagonal embedding (use theorem 9.3.5 in degree 0). For $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$, compute the local norm indices $[\mathbb{Q}_v^\times : N(L_w^\times)]$, determine the image of $\mathbb{Q}^\times$, and verify that $[C_\mathbb{Q} : N(C_{\mathbb{Q}(i)})] = 2$.

5. Show that the norm map on idèle class groups $N_{L/K}\colon C_L \to C_K$ is well defined (i.e., $N_{L/K}(L^\times) \subseteq K^\times$, so the map descends to the quotient), and that $|N_{L/K}(\bar{\alpha})|_K = |\bar{\alpha}|_L$ for the induced norm on C_L.

10

Global Fields

The local theory of chapter 8 computed the cohomology of G_K for a local field K, recorded local Tate duality, and constructed the local reciprocity map from the Tate-Nakayama theorem. The global theory faces a fundamentally different challenge: the absolute Galois group of a number field acts on modules that carry information at every place simultaneously, and the relationship between local and global cohomology is governed by nontrivial exact sequences rather than by a single computation.

Early cohomological approaches to global class field theory (Artin-Tate, 1950s) attempted to work directly with the multiplicative group $L^\times$ and the ideal class group, mirroring the ideal-theoretic formulation of Takagi and Artin. The difficulty is that $H^n(G, L^\times)$ captures global information (the Brauer group, Hilbert 90) but offers no access to the individual local completions: there is no natural map from $H^n(G, L^\times)$ to $H^n(G_w, L_w^\times)$ that accounts for all places simultaneously. The idèlic reformulation resolves this. Chapter 9 assembled local data into the idèle group $\mathbb{I}_L$ and established the Shapiro decomposition $\hat{H}^n(G, \mathbb{I}_L) \cong \bigoplus_v \hat{H}^n(G_w, L_w^\times)$ (theorem 9.3.5), making the local cohomology groups (computed in chapter 8) directly visible as summands of a global object. The fundamental exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ (proposition 9.3.6) then provides the exact relationship between the global cohomology $H^n(G, L^\times)$ and the local decomposition $H^n(G, \mathbb{I}_L)$, with the idèle class group C_L measuring the discrepancy. This chapter exploits this structure, culminating in the Poitou-Tate exact sequence, the global Euler-Poincaré characteristic, and the global reciprocity law. The chapter closes with Selmer groups and the Shafarevich-Tate group, connecting the cohomological machinery to the arithmetic of elliptic curves and abelian varieties.

10.1 Idèle Cohomology and the Global Fundamental Class

The fundamental exact sequence $1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$ of G-modules (proposition 9.3.6), combined with the Shapiro decomposition of $H^n(G, \mathbb{I}_L)$ (theorem 9.3.5), reduces the study of $H^n(G, C_L)$ to understanding the relationship between the global cohomology $H^n(G, L^\times)$ (the Brauer group and its

relatives) and the local cohomology groups $H^n(G_w, L_w^\times)$. This section carries out this analysis in low degrees and records the vanishing of $H^1(G, C_L)$ (the Albert-Brauer-Hasse-Noether theorem) and the global fundamental class $u_{L/K} \in H^2(G, C_L)$ that will drive the global reciprocity law in section 10.4. Their proofs rest on the two fundamental inequalities of global class field theory, which are cited from [2, Ch. VII].

Throughout this section, L/K denotes a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$. For each place v of K, a place w of L above v is fixed, with decomposition group $G_w = \mathrm{Gal}(L_w/K_v) \leq G$.

The long exact sequence of the fundamental short exact sequence begins:

$$1 \to K^\times \to \mathbb{I}_K \to C_K \xrightarrow{\delta^0} H^1(G, L^\times) \xrightarrow{\alpha_1} H^1(G, \mathbb{I}_L) \xrightarrow{\beta_1} H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)$$

The first three terms are the definition of C_K (the fixed points $\mathbb{I}_L^G = \mathbb{I}_K$ by theorem 9.3.2 and $C_L^G = C_K$). Hilbert 90 (theorem 8.1.1) gives $H^1(G, L^\times) = 0$, simplifying the sequence to:

$$0 \to H^1(G, \mathbb{I}_L) \to H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L) \to \cdots$$

The Shapiro decomposition (theorem 9.3.5) and local Hilbert 90 (theorem 8.1.1) give $H^1(G, \mathbb{I}_L) \cong \bigoplus_v H^1(G_w, L_w^\times) = 0$. The sequence reduces further:

Proposition 10.1.1: H^1 of the Idèle Class Group

$H^1(G, C_L) \cong \ker\big(H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)\big)$.

***Proof*:**

The vanishing of $H^1(G, L^\times)$ and $H^1(G, \mathbb{I}_L)$ reduces the long exact sequence to $0 \to H^1(G, C_L) \xrightarrow{\delta^1} H^2(G, L^\times) \xrightarrow{\alpha_2} H^2(G, \mathbb{I}_L)$, and δ^1 identifies $H^1(G, C_L)$ with $\ker \alpha_2$.

The map $\alpha_2 \colon H^2(G, L^\times) \to H^2(G, \mathbb{I}_L)$ sends a global Brauer class to its collection of local components. Identifying $H^2(G, L^\times) \cong \mathrm{Br}(L/K)$ (theorem 7.2.7) and $H^2(G, \mathbb{I}_L) \cong \bigoplus_v H^2(G_w, L_w^\times) \cong \bigoplus_v \mathrm{Br}(L_w/K_v)$ (theorem 9.3.5), the map α_2 is the localization map sending a global central simple algebra to its local invariants. The group $H^1(G, C_L)$ therefore measures the failure of the Hasse principle for the Brauer group: it consists of global Brauer classes that are split at every place.

The following theorem, whose proof was deferred from chapter 7, states that this failure does not occur.

Theorem 10.1.2: Albert-Brauer-Hasse-Noether Sequence and $H^1(G, C_L) = 0$

The localization sequence:

$$0 \to \mathrm{Br}(L/K) \xrightarrow{\mathrm{loc}} \bigoplus_v \mathrm{Br}(L_w/K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}$$

is exact. In particular:

1. A central simple algebra over K split by L is trivial if and only if it is locally trivial at every place (Hasse principle for the Brauer group).
2. $H^1(G, C_L) = 0$.

***Proof Key ideas*:**
The theorem rests on the two fundamental inequalities of global class field theory and on the reciprocity law for cyclotomic extensions, which this book does not prove.

Step 1: $H^1(G, C_L) = 0$. For L/K cyclic of degree n the Herbrand quotient is $h(G, C_L) = n$ [2, Ch. VII, 8.3]. Its proof compares the group of S-idèles of L, whose Herbrand quotient is the product of the local degrees, with the group of S-units of L, whose Herbrand quotient is computed from the logarithmic embedding of the unit theorem. For every finite Galois L/K the orders of $\hat{H}^0(G, C_L)$ and $\hat{H}^2(G, C_L)$ divide n, and $H^1(G, C_L) = 0$ [2, Ch. VII, 9.1]. For L/K cyclic of prime degree, after roots of unity are adjoined, the index $[C_K : N_{L/K}(C_L)]$ of a Kummer extension is computed through S-units and shown to divide n. As $h(G, C_L) = n$ and $|\hat{H}^{-1}(G, C_L)| = |H^1(G, C_L)|$ by theorem 4.3.4, the integer $|H^1(G, C_L)| = |\hat{H}^0(G, C_L)|/n$ is then at most 1. The general case follows from this one by the induction of [2, Ch. VI, §1.5, Lemma 4], through the Sylow subgroups of G and their normal subgroups of prime index. This gives (2).

Step 2: The Hasse principle. By proposition 10.1.1, $H^1(G, C_L)$ is the kernel of α_2, and α_2 is the localization map under the identifications recalled before the theorem. So Step 1 makes loc injective, which is (1).

Step 3: The sequence is a complex. The local invariants of a global class sum to zero: $\sum_v \mathrm{inv}_v(a) = 0$ for every $a \in \mathrm{Br}(K)$ [2, Ch. VII, 10.2, Thm. B]. The proof there reduces to classes split by cyclic cyclotomic extensions, for which the identity follows from the reciprocity law for cyclotomic extensions.

Step 4: Exactness at the middle term. By [2, Ch. VII, 11.2 and 11.2 (bis)], $H^2(G, C_L)$ is cyclic of order n and the sum of the local invariants factors as the map $H^2(G, \mathbb{I}_L) \to H^2(G, C_L)$ followed by an injective map $H^2(G, C_L) \to \mathbb{Q}/\mathbb{Z}$, so the sequence $0 \to H^2(G, L^\times) \to H^2(G, \mathbb{I}_L) \to \mathbb{Q}/\mathbb{Z}$ is exact. Under the identifications of Step 2 this is the exactness of the localization sequence at the middle term, completing the sketch.

The ABHN theorem passes to the limit over all finite Galois extensions L/K to give the global Brauer group exact sequence.

Corollary 10.1.3: Global Brauer Group Sequence

The sequence:

$$0 \to \mathrm{Br}(K) \xrightarrow{\mathrm{loc}} \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

is exact.

***Proof*:**
Every class of $\mathrm{Br}(K)$ lies in $\mathrm{Br}(L/K)$ for some finite Galois L/K (proposition 7.1.10), and $\mathrm{Br}(L_w/K_v) \subseteq \mathrm{Br}(K_v)$, so the injectivity of loc and the vanishing of $\sum_v \mathrm{inv}_v \circ \mathrm{loc}$ follow from theorem 10.1.2. The map $\sum_v \mathrm{inv}_v$ is surjective because a single $\mathrm{inv}_v \colon \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$ at a finite place v is already surjective (theorem 8.1.5). Exactness at the middle term, that a family of local classes whose invariants sum to zero comes from a global class, is not proved in this book. It is the exactness of the limit sequence $0 \to \mathrm{Br}(K) \to \bigoplus_v \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$ shown in [2, Ch. VII, 11.2 (bis)], obtained there from cyclic cyclotomic extensions, whose local degrees can be made as large as desired at finitely many given places, completing the proof.

This exact sequence is the definitive statement about the Brauer group of a number field: a central simple algebra over K is determined up to Brauer equivalence by its local invariants, which are rational numbers summing to zero. This fulfills the promise made in section 7.3.

With $H^1(G, C_L) = 0$, the global fundamental class is defined as in the local case.

Definition 10.1.4: Global Invariant Map

The *global invariant map* $\mathrm{inv}_{L/K} \colon H^2(G, C_L) \to \mathbb{Q}/\mathbb{Z}$ is the homomorphism characterized as follows. Let $x \in H^2(G, C_L)$, let L'/K be a finite Galois extension containing L, with group G', and suppose that the inflation of x to $H^2(G', C_{L'})$ is the image of a class $b \in H^2(G', \mathbb{I}_{L'})$. Then $\mathrm{inv}_{L/K}(x) = \sum_v \mathrm{inv}_v(b_v)$, where b_v is the component of b at v under the decomposition $H^2(G', \mathbb{I}_{L'}) \cong \bigoplus_v H^2(G'_{w'}, L'^{\times}_{w'})$ of theorem 9.3.5 and inv_v is the local invariant map.

That such L' and b exist for every x, and that the value does not depend on the choices, is part of the global theory cited in theorem 10.1.2: every class becomes the image of an idèle class after inflation to a field with large local degrees, and the sum of the local invariants of a global Brauer class vanishes [2, Ch. VII, 10.2, 11.2 and 11.2 (bis)]. For a class in the image of $H^2(G, \mathbb{I}_L)$ one may take $L' = L$, so the composite $H^2(G_w, L_w^{\times}) \to H^2(G, \mathbb{I}_L) \to H^2(G, C_L) \xrightarrow{\mathrm{inv}_{L/K}} \mathbb{Q}/\mathbb{Z}$ is the local invariant inv_v.

Theorem 10.1.5: The Global Fundamental Class

The global invariant map $\mathrm{inv}_{L/K}\colon H^2(G,C_L)\to\mathbb{Q}/\mathbb{Z}$ is an isomorphism onto $\frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$, compatible with inflation. The *global fundamental class* is the unique element $u_{L/K}\in H^2(G,C_L)$ with:

$$\mathrm{inv}_{L/K}(u_{L/K})=\frac{1}{[L:K]}$$

For every field K' with $K\subseteq K'\subseteq L$, the restriction of $u_{L/K}$ to $\mathrm{Gal}(L/K')$ is $u_{L/K'}$. In particular, for a place w of L above v, the restriction of $u_{L/K}$ to the decomposition group G_w is the image of the local fundamental class u_{L_w/K_v} (theorem 8.4.3) under the map $H^2(G_w,L_w^\times)\to H^2(G_w,C_L)$ induced by $L_w^\times\to\mathbb{I}_L\to C_L$.

***Proof Key ideas*:**
This book does not prove the structure of $H^2(G,C_L)$. The order of $H^2(G,C_L)$ divides $[L:K]$ by [2, Ch. VII, 9.1], the second inequality recalled in the proof of theorem 10.1.2. By [2, Ch. VII, 11.2 (bis)], the global invariant map is injective on $H^2(G,C_L)$, compatibly with inflation, and its image is $\frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$: the image contains $\frac{1}{n_0}\mathbb{Z}/\mathbb{Z}$, where n_0 is the least common multiple of the local degrees, and the reduction to cyclic cyclotomic extensions of the proof of theorem 10.1.2 yields all of $\frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$. Restricting $u_{L/K}$ to $\mathrm{Gal}(L/K')$ multiplies its invariant by $[K':K]$, as for the local invariant maps of theorem 7.3.1(3), so it gives $u_{L/K'}$ [2, Ch. VII, 11.2].

For the last assertion let $K'=L^{G_w}$ be the decomposition field of w and w' the place of K' below w. Then $G_w=\mathrm{Gal}(L/K')$, the place w is the only place of L above w', and $K'_{w'}=K_v$. By definition 10.1.4 applied to L/K', the image of u_{L_w/K_v} in $H^2(G_w,C_L)$ has global invariant $\mathrm{inv}_{K_v}(u_{L_w/K_v})=\frac{1}{[L_w:K_v]}=\frac{1}{[L:K']}$, so it is $u_{L/K'}$, which is the restriction of $u_{L/K}$, completing the sketch.

The global fundamental class is the central input for the global reciprocity law in section 10.4. Its local-global compatibility, that the restriction of $u_{L/K}$ to a decomposition group G_w is the image of the local fundamental class u_{L_w/K_v}, is the mechanism by which the global Artin map restricts to the local Artin maps at each place.

Example 10.1.6: The Global Fundamental Class for Quadratic Extensions
Let $L=K(\sqrt{d})$ for some non-square $d\in K^\times$, so $G\cong C_2$. The global invariant map gives $H^2(G,C_L)\cong\mathbb{Z}/2\mathbb{Z}$, generated by $u_{L/K}$ with $\mathrm{inv}(u_{L/K})=1/2$, and $[C_K:N_{L/K}(C_L)]=|\hat{H}^0(G,C_L)|=2$ by theorem 4.3.4. The local fundamental class u_{L_w/K_v} has invariant $1/[L_w:K_v]$: it is $1/2$ at every place v that does not split in L, whether v is inert, ramified, or a real place where $d<0$, and 0 at the split places, where $L_w=K_v$. At a non-split place the decomposition group is all of G, so by theorem 10.1.5 the image of u_{L_w/K_v} in $H^2(G,C_L)$ is $u_{L/K}$ itself.

Exercise 10.1.1

1. Let L/K be a cyclic extension of prime degree p with group G. Show that $\hat{H}^0(G,C_L)=C_K/N_{L/K}(C_L)$, and that the second inequality $[C_K:N_{L/K}(C_L)]\leq p$ together with $h(G,C_L)=p$ [2, Ch. VII, 8.3] implies $H^1(G,C_L)=0$ and $[C_K:N_{L/K}(C_L)]=p$.

2. Let $a, b \in \mathbb{Q}^\times$. Show that the local invariants of the quaternion algebra $(a,b)_\mathbb{Q}$ sum to zero: $\sum_v \mathrm{inv}_v((a,b)_{\mathbb{Q}_v}) = 0$ in $\mathbb{Q}/\mathbb{Z}$. (*Hint:* relate each invariant to the Hilbert symbol $(a,b)_v$ and use the product formula [14, The Product Formula].)

3. Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Determine the places v of $\mathbb{Q}$ with $[L_w : \mathbb{Q}_v] = 2$, describe $H^2(G, \mathbb{I}_L)$, and use the global invariant map to compute $H^2(G, C_L)$. Which classes of $H^2(G, \mathbb{I}_L)$ come from $\mathrm{Br}(L/\mathbb{Q})$?

4. Show that the ABHN theorem implies the Hasse norm theorem for cyclic extensions: an element $a \in K^\times$ is a norm from $L^\times$ if and only if $a \in N_{L_w/K_v}(L_w^\times)$ for every place v. (*Hint:* use $\hat{H}^0(G, L^\times) = K^\times/N(L^\times)$ and the Tate cohomology long exact sequence.)

5. Using corollary 10.1.3, compute $\mathrm{Br}(\mathbb{Q})$, and describe the local invariants of a quaternion algebra $(a,b)_\mathbb{Q}$, for $a, b \in \mathbb{Q}^\times$, in terms of the Hilbert symbol.

10.2 The Poitou-Tate Exact Sequence

Local Tate duality (theorem 8.3.2) provides a perfect pairing $H^r(G_v, M) \times H^{2-r}(G_v, M^*) \to \mathbb{Q}/\mathbb{Z}$ for finite G_{K_v}-modules M at each non-archimedean place v of a number field K. The Poitou-Tate exact sequence is the global counterpart: it assembles these local pairings into an exact sequence, nine terms long in full, that controls the relationship between global and local cohomology of finite Galois modules. The sequence is the central result of global Galois cohomology, and Poitou-Tate duality and the finiteness of Selmer groups below follow from it.

Throughout this section, K is a number field, M is a finite G_K-module (a finite abelian group with a continuous action of G_K), and M^* denotes its Cartier dual.

Definition 10.2.1: Cartier Dual over a Number Field

Let M be a finite G_K-module. The *Cartier dual* of M is the finite G_K-module:

$$M^* = \mathrm{Hom}_\mathbb{Z}(M, (K^{\mathrm{sep}})^\times)$$

where G_K acts by $(\sigma \cdot f)(m) = \sigma(f(\sigma^{-1}m))$ for $\sigma \in G_K$, $f \in M^*$, $m \in M$. Since M is finite of exponent n, every homomorphism $M \to (K^{\mathrm{sep}})^\times$ lands in $\mu_n \subseteq (K^{\mathrm{sep}})^\times$, so $M^* = \mathrm{Hom}_\mathbb{Z}(M, \mu_n)$ as a G_K-module.

The Cartier dual satisfies $(M^*)^* \cong M$ (canonically), and for $M = \mathbb{Z}/n\mathbb{Z}$ (with trivial action), $M^* \cong \mu_n$ (with the cyclotomic action). The duality exchanges "constant" and "multiplicative" modules.

Definition 10.2.2: S-Ramified Galois Group

Let S be a finite set of places of K containing all archimedean places and all places where M is ramified (i.e., where the inertia group I_v acts nontrivially on M). The *maximal S-ramified extension* K_S is the maximal algebraic extension of K unramified at all $v \notin S$, and:

$$G_{K,S} = \mathrm{Gal}(K_S/K)$$

The action of G_K on M factors through $G_{K,S}$ (since I_v acts trivially for $v \notin S$). A critical distinction: while the action factors through $G_{K,S}$, cocycles $G_K \to M$ need not factor through $G_{K,S}$, so $H^n(G_K, M) \neq H^n(G_{K,S}, M)$ in general. For instance, $H^1(G_K, \mathbb{Z}/p\mathbb{Z}) = \mathrm{Hom}(G_K, \mathbb{Z}/p\mathbb{Z})$ classifies *all* cyclic p-extensions of K and is infinite, while $H^1(G_{K,S}, \mathbb{Z}/p\mathbb{Z})$ classifies only those unramified outside S and is finite. The Poitou-Tate sequence requires finite cohomology groups and is formulated for $G_{K,S}$.

For each place v of K, the choice of an embedding $\overline{K} \hookrightarrow \overline{K}_v$ determines a continuous homomorphism $G_v = G_{K_v} \to G_K$, well-defined up to conjugacy. The restriction along this map gives the *localization maps*.

Definition 10.2.3: Localization Maps

For each $r \geq 0$ and each place v of K, the *localization map* is the restriction:

$$\mathrm{loc}_v \colon H^r(G_{K,S}, M) \to H^r(G_v, M)$$

These assemble into a single map:

$$\mathrm{loc} \colon H^r(G_{K,S}, M) \to \bigoplus_{v \in S} H^r(G_v, M)$$

The restriction to $v \in S$ is essential: at $v \notin S$, the unramified cohomology $H^1_{\mathrm{nr}}(G_v, M) = H^1(G_v/I_v, M)$ can be nontrivial (e.g., $H^1_{\mathrm{nr}}(G_v, \mathbb{Z}/p\mathbb{Z}) = \mathbb{Z}/p\mathbb{Z}$ for every v), so the sum over all v would be infinite. The localization maps for $v \in S$ factor through $G_{K,S}$ via the decomposition groups at places in S.

The local Tate duality pairings allow the local cohomology groups to be identified with duals of cohomology groups of M^*. The *dual localization map* is the composition:

$$\bigoplus_{v \in S} H^r(G_v, M) \xrightarrow{\bigoplus_v \mathrm{inv}_v} \bigoplus_{v \in S} H^{2-r}(G_v, M^*)^* \xrightarrow{\mathrm{loc}^*} H^{2-r}(G_{K,S}, M^*)^*$$

where inv_v is the local duality isomorphism, given by theorem 8.3.2 at the non-archimedean places of S and by [22, Ch. I, Thm. 2.13(a)] at the archimedean ones, with Tate cohomology in degree 0, and loc^* is the Pontryagin dual of loc for M^*.

Definition 10.2.4: Shafarevich-Tate Groups

For a finite G_K-module M and $r \geq 0$, the *rth Shafarevich-Tate group* is:

$$Ш^r(K, M) = \ker\Big(H^r(G_{K,S}, M) \xrightarrow{\text{loc}} \bigoplus_{v \in S} H^r(G_v, M)\Big)$$

the subgroup of global cohomology classes that are locally trivial at every place of S. The group depends on S: for $M = \mathbb{Z}/n\mathbb{Z}$ it is $\mathrm{Hom}(\mathrm{Cl}_S(K), \mathbb{Z}/n\mathbb{Z})$ (example 10.2.7), which changes as places are added to S.

The groups $Ш^r(K, M)$ measure the failure of the local-global principle at the level of cohomology: a class in $Ш^r$ is "globally nontrivial but locally invisible." For $r = 0$: $Ш^0(K, M) = \ker(M^{G_K} \to \prod_v M^{G_v})$, which is trivial (an element fixed by G_K is fixed by every decomposition group). For $r = 1$ and $r = 2$, these groups are nontrivial in general and are the central objects of the Poitou-Tate sequence.

Theorem 10.2.5: Poitou-Tate Exact Sequence

Let S be a finite set of places of K containing the archimedean places, and let M be a finite $G_{K,S}$-module whose order is invertible at every place outside S, with Cartier dual M^*. Then the following sequence of finite groups is exact:

$$\begin{aligned} 0 \to Ш^1(K, M) \to H^1(G_{K,S}, M) &\xrightarrow{\text{loc}} \bigoplus_{v \in S} H^1(G_v, M) \\ \to H^1(G_{K,S}, M^*)^* \to H^2(G_{K,S}, M) &\xrightarrow{\text{loc}} \bigoplus_{v \in S} H^2(G_v, M) \\ \to H^0(G_{K,S}, M^*)^* &\to 0 \end{aligned} \tag{10.1}$$

The maps from local cohomology to the duals of global cohomology are induced by the local Tate duality pairings and the sum of local invariants $\sum_v \mathrm{inv}_v \colon \bigoplus_v \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$.

***Proof Key ideas*:**
This book does not prove the theorem. The sequence is the part from $H^1(G_{K,S}, M)$ onwards of the nine-term exact sequence of [22, Ch. I, Thm. 4.10], with its first three terms replaced by the kernel $Ш^1(K, M)$ of loc, and the finiteness of the global groups is [22, Ch. I, Cor. 4.15]. That loc followed by the dual localization map is zero comes from the reciprocity law: for global classes α and β, the local invariants of the global class $\alpha \smile \beta$ sum to zero (corollary 10.1.3). Milne proves exactness by identifying the sequence with the long exact sequence of $\mathrm{Ext}_{G_{K,S}}(M^*, -)$ applied to $0 \to E_S \to J_S \to C_S \to 0$, the S-units, S-idèles and S-idèle classes. The global terms are computed by the duality theorem for the S-idèle class group [22, Ch. I, Thm. 4.6] and the local terms by local duality at the places of S (theorem 8.3.2, and [22, Ch. I, Thm. 2.13(a)] at the archimedean places). The argument does not use the global Euler-Poincaré characteristic, completing the sketch.

The proof of the Poitou-Tate sequence is among the most technical in algebraic number theory: the

identification of the connecting maps and the exactness at $H^1(G_{K,S}, M^*)^*$ require careful diagram chases. Besides the account in Milne cited above, a complete proof is given in [26].

The most striking consequence of the Poitou-Tate sequence is the duality between the Shafarevich-Tate groups.

Theorem 10.2.6: Poitou-Tate Duality

For S and M as in theorem 10.2.5:

$$Ш^1(K, M) \cong Ш^2(K, M^*)^*$$

as finite abelian groups, where $(-)^*$ on the right denotes the Pontryagin dual.

***Proof*:**
The module M^* again satisfies the hypotheses of theorem 10.2.5, and $M^{**} \cong M$, so it suffices to prove $Ш^2(K, M) \cong Ш^1(K, M^*)^*$ and apply it to M^*. In the exact sequence of theorem 10.2.5 for M, the group $Ш^2(K, M)$, the kernel of loc on $H^2(G_{K,S}, M)$, is the image of $H^1(G_{K,S}, M^*)^*$, hence the cokernel of the map $\bigoplus_{v\in S} H^1(G_v, M) \to H^1(G_{K,S}, M^*)^*$. This map is the dual localization map: the local duality isomorphisms followed by the Pontryagin dual of loc: $H^1(G_{K,S}, M^*) \to \bigoplus_{v\in S} H^1(G_v, M^*)$. For a homomorphism of finite groups the cokernel of its dual is the dual of its kernel, so $Ш^2(K, M)$ is the dual of $\ker(\text{loc}) = Ш^1(K, M^*)$, as we sought to show.

Poitou-Tate duality swaps the degree and the module: $Ш^1$ for M is dual to $Ш^2$ for the Cartier dual M^*. Since the Cartier dual of $\mathbb{Z}/n\mathbb{Z}$ is μ_n and vice versa, the duality interchanges "additive" and "multiplicative" local-global obstructions.

Example 10.2.7: Poitou-Tate for $\mathbb{Z}/n\mathbb{Z}$ and μ_n
Let S and $M = \mathbb{Z}/n\mathbb{Z}$ satisfy the hypotheses of theorem 10.2.5, so that n is invertible at every place outside S, and let $\text{Cl}_S(K)$ be the S-class group, the ideal class group of the ring $\mathcal{O}_{K,S}$ of S-integers of K.

1. For $M = \mathbb{Z}/n\mathbb{Z}$ with trivial action, $M^* = \mu_n$. The group $H^1(G_{K,S}, \mathbb{Z}/n\mathbb{Z}) = \text{Hom}(G_{K,S}, \mathbb{Z}/n\mathbb{Z})$ classifies cyclic extensions of K of degree dividing n that are unramified outside S, and $Ш^1(K, \mathbb{Z}/n\mathbb{Z})$ consists of those in which every place of S splits completely. By class field theory the maximal abelian extension of K unramified outside S in which every place of S splits completely has Galois group $\text{Cl}_S(K)$, so $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) \cong \text{Hom}(\text{Cl}_S(K), \mathbb{Z}/n\mathbb{Z})$ [26, Ch. VIII, §6, (8.6.3)].

   The dual side recovers this. Let $\mathcal{O}_S^\times$ be the group of S-units of the maximal extension K_S of K unramified outside S. The Kummer sequence $1 \to \mu_n \to \mathcal{O}_S^\times \xrightarrow{n} \mathcal{O}_S^\times \to 1$ is an exact sequence of $G_{K,S}$-modules [26, Ch. VIII, §3, (8.3.4)], and $H^0(G_{K,S}, \mathcal{O}_S^\times) = \mathcal{O}_{K,S}^\times$, $H^1(G_{K,S}, \mathcal{O}_S^\times) \cong \text{Cl}_S(K)$ [26, Ch. VIII, §3, (8.3.11)]. Its long exact sequence gives the exact sequences

$$0 \to \mathcal{O}_{K,S}^\times/(\mathcal{O}_{K,S}^\times)^n \to H^1(G_{K,S}, \mu_n) \to \text{Cl}_S(K)[n] \to 0$$

$$0 \to \text{Cl}_S(K)/n\,\text{Cl}_S(K) \to H^2(G_{K,S}, \mu_n) \to H^2(G_{K,S}, \mathcal{O}_S^\times)[n] \to 0$$

The n-torsion of $H^2(G_{K,S}, \mathcal{O}_S^\times)$ maps injectively to $\bigoplus_{v\in S} \mathrm{Br}(K_v)$ by localization (part (iii) of the same proposition), and the localization of $H^2(G_{K,S}, \mu_n)$ into $\bigoplus_{v\in S} H^2(G_v, \mu_n) \subseteq \bigoplus_{v\in S} \mathrm{Br}(K_v)$ factors through it. So $Ш^2(K, \mu_n) = \mathrm{Cl}_S(K)/n\,\mathrm{Cl}_S(K)$, and Poitou-Tate duality (theorem 10.2.6) gives $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) \cong (\mathrm{Cl}_S(K)/n\,\mathrm{Cl}_S(K))^* \cong \mathrm{Hom}(\mathrm{Cl}_S(K), \mathbb{Z}/n\mathbb{Z})$, as above. In particular $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) = 0$ when n is prime to the order of $\mathrm{Cl}_S(K)$, for instance when the classes of the primes in S generate the class group of K. In general a cyclic extension unramified outside S in which every place of S splits completely need not be trivial.

2. Take $K = \mathbb{Q}$, $M = \mu_p$ for an odd prime p, and S containing p and ∞. Then $\mathrm{Cl}_S(\mathbb{Q}) = 0$, so by the first sequence in (1) $H^1(G_{\mathbb{Q},S}, \mu_p) \cong \mathbb{Z}_S^\times/(\mathbb{Z}_S^\times)^p$, where $\mathbb{Z}_S^\times = \left\{\pm\prod_{\ell\in S} \ell^{a_\ell} \;\middle|\; a_\ell \in \mathbb{Z}\right\}$ is the group of S-units of $\mathbb{Q}$, and $H^1(G_v, \mu_p) \cong \mathbb{Q}_v^\times/(\mathbb{Q}_v^\times)^p$ by Kummer theory and Hilbert's Theorem 90. The localization map is injective: if an S-unit $\pm\prod_\ell \ell^{a_\ell}$ is a p-th power in $\mathbb{Q}_\ell$ for every prime $\ell \in S$, then every a_ℓ is divisible by p, so the S-unit is a p-th power, -1 being one. So $Ш^1(\mathbb{Q}, \mu_p) = 0$ and the sequence of theorem 10.2.5 begins

$$0 \to H^1(G_{\mathbb{Q},S}, \mu_p) \xrightarrow{\mathrm{loc}} \bigoplus_{v\in S} H^1(G_v, \mu_p) \to H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})^* \to \cdots$$

Its exactness identifies the cokernel of the localization map with a subgroup of the dual of $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})$, the group classifying cyclic extensions of $\mathbb{Q}$ of degree p unramified outside S.

Exercise 10.2.1

1. Verify that the Poitou-Tate sequence is a complex at the term $\bigoplus_{v\in S} H^2(G_v, M)$: show that the composition $H^2(G_{K,S}, M) \xrightarrow{\mathrm{loc}} \bigoplus_{v\in S} H^2(G_v, M) \to H^0(G_{K,S}, M^*)^*$ is zero.
2. Using example 10.2.7, show that $Ш^1(K, \mathbb{Z}/n\mathbb{Z}) = 0$ when n is prime to the order of $\mathrm{Cl}_S(K)$, and that $Ш^1(K, \mathbb{Z}/3\mathbb{Z}) \cong \mathbb{Z}/3\mathbb{Z}$ for $K = \mathbb{Q}(\sqrt{-31})$ and $S = \{\infty, (3)\}$.
3. Using the first exact sequence of example 10.2.7(1), show that there is an exact sequence $0 \to E \to Ш^1(K, \mu_n) \to \mathrm{Cl}_S(K)[n]$, where E is the group of classes in $\mathcal{O}_{K,S}^\times/(\mathcal{O}_{K,S}^\times)^n$ of the S-units that are n-th powers in K_v for every $v \in S$.
4. Show that the Poitou-Tate exact sequence is functorial in M: a morphism of finite G_K-modules $f\colon M \to N$ induces a morphism of the exact sequences of theorem 10.2.5 (commutative ladder).
5. (If the reader has read [6]) For $K = \mathbb{Q}$ and $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action), write out the Poitou-Tate sequence explicitly. Identify $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z})$ with the group of quadratic extensions of $\mathbb{Q}$, the local terms with local quadratic extensions, and interpret the exactness in terms of the Hilbert reciprocity law.

10.3 The Global Euler-Poincaré Characteristic

For a finite G_{K_v}-module M at a non-archimedean place v, the local Euler-Poincaré characteristic is $\chi(G_v, M) = |\#M|_v$, the normalized absolute value at v of the order of M (theorem 8.3.4). It is 1 when the order of M is prime to the residue characteristic of v. This section records, with its proof

cited, the global analogue: the alternating product of orders of $H^r(G_{K,S}, M)$ is determined by a product of local terms at the archimedean places alone. The formula is independent of the choice of S (provided S is large enough) and expresses a fundamental balance between the global cohomology groups.

Throughout this section, K is a number field with $[K:\mathbb{Q}] = n$, having r_1 real places and r_2 complex places ($n = r_1 + 2r_2$). The set S is a finite set of places containing all archimedean places and all primes where M is ramified or dividing $|M|$, so that the G_K-action on M factors through $G_{K,S}$ (definition 10.2.2).

Definition 10.3.1: Global Euler-Poincaré Characteristic

For a finite $G_{K,S}$-module M, the *global Euler-Poincaré characteristic* is:

$$\chi(G_{K,S}, M) = \frac{|H^0(G_{K,S}, M)| \cdot |H^2(G_{K,S}, M)|}{|H^1(G_{K,S}, M)|}$$

All three cohomology groups are finite [22, Ch. I, Cor. 4.15].

The global EP characteristic has an unexpectedly simple formula: it depends only on the module M, the degree $[K:\mathbb{Q}]$, and the archimedean places of K.

Theorem 10.3.2: Global Euler-Poincaré Formula

For a finite $G_{K,S}$-module M:

$$\chi(G_{K,S}, M) = \frac{\prod_{v|\infty} |H^0(G_v, M)|}{|M|^{[K:\mathbb{Q}]}}$$

where the product runs over all archimedean places of K. At a real place v, $H^0(G_v, M) = M^{G_v}$ where $G_v \cong \mathbb{Z}/2\mathbb{Z}$ acts via complex conjugation. At a complex place v, $G_v = 1$ and $H^0(G_v, M) = M$, contributing $|M|$ to the product.

The formula can be rewritten: since there are r_1 real places and r_2 complex places,

$$\chi(G_{K,S}, M) = \frac{\prod_{v \text{ real}} |M^{G_v}| \cdot |M|^{r_2}}{|M|^{r_1+2r_2}} = \frac{\prod_{v \text{ real}} |M^{G_v}|}{|M|^{r_1+r_2}}$$

For a totally imaginary field ($r_1 = 0$): $\chi(G_{K,S}, M) = |M|^{-r_2}$, depending only on $|M|$ and the number of complex places.

***Proof Key ideas*:**
This is Tate's global Euler-Poincaré formula, which this book does not prove. It is [22, Ch. I, Thm. 5.1], whose archimedean factor $|\#M|_v$ is $|M|$ at a real place and $|M|^2$ at a complex place, so that the product of these factors is $|M|^{r_1+2r_2} = |M|^{[K:\mathbb{Q}]}$. Writing $\varphi(M)$ for the quotient of the two sides of the formula, the proof there shows three things. First, $\varphi(M)\varphi(M^*) = 1$ for the Cartier dual M^*, by the nine-term Poitou-Tate sequence [22, Ch. I, Thm. 4.10] and the local formula recalled above [22, Ch. I, Rem. 5.2(b)]. Second, φ is multiplicative in short exact sequences [22, Ch. I, Lemma 5.3]. Third, $\varphi(M) = \varphi(M^*)$ for M killed by a prime p, by an argument in the Grothendieck group of $\mathbb{F}_p[G]$-modules for a finite quotient G of $G_{K,S}$ [22, Ch. I,

Lemma 5.4]. Together these give $\varphi(M) = 1$, completing the sketch.

Only the archimedean places appear in the formula. At a non-archimedean place $v \in S$ the local characteristic $|\#M|_v$ depends only on the order of M, and $\#M$ is a unit at every place outside S, so by the product formula the product of these local factors over the non-archimedean $v \in S$ is the inverse of $\prod_{v|\infty} |\#M|_v = |M|^{[K:\mathbb{Q}]}$ [22, Ch. I, Rem. 5.2(b)]. At the archimedean places the factor $|H^0(G_v, M)|$ depends on the action, and this dependence is the only arithmetic content of the global formula.

The contrast between the $\mathbb{Z}/p\mathbb{Z}$ and μ_p examples below illustrates this: both modules have order p, but $|(\mathbb{Z}/p\mathbb{Z})^{G_\mathbb{R}}| = p$ (trivial action, everything fixed) while $|\mu_p^{G_\mathbb{R}}| = 1$ for odd p (complex conjugation inverts roots of unity, leaving only 1 fixed). The global EP characteristics differ by a factor of p, and this difference propagates to the sizes of H^1 and H^2.

Example 10.3.3: Global EP Computations

1. For $K = \mathbb{Q}$ ($r_1 = 1$, $r_2 = 0$) and $M = \mathbb{Z}/p\mathbb{Z}$ with trivial action (p odd): $|H^0(G_\mathbb{R}, \mathbb{Z}/p\mathbb{Z})| = p$ (everything fixed under trivial action), so $\chi = p/p = 1$. This means $|H^0| \cdot |H^2| = |H^1|$, and since $|H^0| = p$: $|H^2(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})| = |H^1(G_{\mathbb{Q},S}, \mathbb{Z}/p\mathbb{Z})|/p$.

2. For $K = \mathbb{Q}$ and $M = \mu_p$ (p odd): $|H^0(G_\mathbb{R}, \mu_p)| = |\mu_p(\mathbb{R})| = 1$ (the only real pth root of unity is 1), so $\chi = 1/p$. With $|H^0(G_\mathbb{Q}, \mu_p)| = |\mu_p(\mathbb{Q})| = 1$: $|H^2(G_{\mathbb{Q},S}, \mu_p)| = |H^1(G_{\mathbb{Q},S}, \mu_p)|/p$. By Kummer theory, $H^1(G_{\mathbb{Q},S}, \mu_p) \cong \mathbb{Q}_S^\times/(\mathbb{Q}_S^\times)^p$ where $\mathbb{Q}_S^\times$ denotes elements of $\mathbb{Q}^\times$ that are S-units. For $S = \{\infty, p\}$: $\mathbb{Q}_S^\times = \langle -1, p\rangle \cdot (\mathbb{Q}^\times)^p$, giving $|H^1| = p$ (generated by the class of p). So $|H^2| = 1$.

3. For a totally imaginary field K ($r_1 = 0$) and any M with trivial action: $\chi = |M|^{r_2}/|M|^{2r_2} = |M|^{-r_2}$. As r_2 grows, the EP characteristic shrinks, reflecting the increasing complexity of the cohomology.

4. For $K = \mathbb{Q}$ and $M = \mathbb{Z}/2\mathbb{Z}$ (trivial action): $|H^0(G_\mathbb{R}, \mathbb{Z}/2\mathbb{Z})| = 2$, so $\chi = 2/2 = 1$. For $S = \{\infty, 2\}$: $H^1(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z}) = \mathrm{Hom}(G_{\mathbb{Q},S}, \mathbb{Z}/2\mathbb{Z})$ classifies quadratic extensions of $\mathbb{Q}$ unramified outside $\{2, \infty\}$, namely $\mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{-2})$ (plus the trivial extension), giving $|H^1| = 4$. Then $|H^2| = |H^1|/|H^0| = 4/2 = 2$.

The global EP formula constrains the sizes of cohomology groups: knowing any two of $|H^0|$, $|H^1|$, $|H^2|$ determines the third. In applications to arithmetic (Selmer groups, descent), the formula relates the "size" of H^1 (which controls the Selmer group) to computable local data.

Corollary 10.3.4: EP for Dual Modules

For a finite $G_{K,S}$-module M with Cartier dual M^*:

$$\chi(G_{K,S}, M) \cdot \chi(G_{K,S}, M^*) = \frac{\prod_{v|\infty} |H^0(G_v, M)| \cdot |H^0(G_v, M^*)|}{|M|^{2[K:\mathbb{Q}]}}$$

For M self-dual ($M \cong M^*$), this gives $\chi(G_{K,S}, M)^2 = \prod_{v|\infty} |H^0(G_v, M)|^2/|M|^{2[K:\mathbb{Q}]}$, recovering the formula from the square root.

***Proof*:**
Apply theorem 10.3.2 to both M and M^* and multiply.

Exercise 10.3.1

1. Verify the global EP formula for $K = \mathbb{Q}$, $M = \mathbb{Z}/2\mathbb{Z}$, $S = \{\infty, 2\}$ by computing all three cohomology groups independently and checking $\chi = 1$.
2. Let K be a totally imaginary number field and M a finite $G_{K,S}$-module with trivial G_K-action. Show that $\chi(G_{K,S}, M) = |M|^{-r_2}$ where $r_2 = [K : \mathbb{Q}]/2$. Conclude that $|H^1(G_{K,S}, M)| = |M|^{r_2} \cdot |H^0| \cdot |H^2|$.
3. Show that $\chi(G_{K,S}, M) \cdot \chi(G_{K,S}, M^*) = |M|^{-2[K:\mathbb{Q}]} \cdot \prod_{v|\infty} |M|^2 = |M|^{-2r_1}$ when the G_K-action on M is trivial (so $|H^0(G_v, M)| = |M|$ at every archimedean place and $M^* = \mathrm{Hom}(M, \mu)$ need not have trivial action).
4. (If the reader has read [11]) Let $E/\mathbb{Q}$ be an elliptic curve and p an odd prime of good reduction for E. The module $M = E[p]$ is a finite $G_{\mathbb{Q}}$-module of order p^2 with Cartier dual $M^* \cong E[p]$ (via the Weil pairing). Compute $\chi(G_{\mathbb{Q},S}, E[p])$ in terms of p and the number of real places, and explain why this constrains the size of the p-Selmer group $\mathrm{Sel}_p(E/\mathbb{Q})$.

10.4 The Cohomological Reciprocity Isomorphism

The local reciprocity law (theorem 8.4.4) was obtained from the local class formation through Tate's theorem on cup products with the fundamental class. The same strategy applies globally: the pair $(G_K, C_{K^{\mathrm{sep}}})$ satisfies the class formation axioms, and cup product with the global fundamental class produces the global Artin map $\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$. The axioms are recorded in section 10.1: the vanishing of H^1 is theorem 10.1.2, and the structure of H^2 is theorem 10.1.5. This section assembles these ingredients into the global reciprocity law and examines its compatibility with the local theory.

As in the local case (section 8.4), what is constructed here is the reciprocity *isomorphism* of the cohomological formalism: the global Artin map rec_K as the degree -2 image of the global fundamental class. Its development into class field theory belongs to *Class Field Theory* [24]: the Artin-Frobenius description of rec_K, the idele-class and ray-class field correspondence, the global existence theorem, conductors, and the arithmetic reciprocity laws.

Definition 10.4.1: Global Class Formation

The pair $(G_K, C_{K^{\mathrm{sep}}})$ is a *class formation*, where $C_{K^{\mathrm{sep}}} = \varinjlim_L C_L$ is the idèle class group of the separable closure (the colimit over all finite extensions L/K of the idèle class groups C_L, with transition maps given by the natural inclusions). The class formation axioms are:

1. For every finite Galois extension L/K with $G = \mathrm{Gal}(L/K)$: $H^1(G, C_L) = 0$.
2. For every finite Galois extension L/K with $G = \mathrm{Gal}(L/K)$: the global invariant map (definition 10.1.4) gives an isomorphism $\mathrm{inv}_{L/K} \colon H^2(G, C_L) \xrightarrow{\sim} \frac{1}{[L:K]}\mathbb{Z}/\mathbb{Z}$, compatible with inflation.
3. For every finite Galois extension L/K, every field K' with $K \subseteq K' \subseteq L$ and every $x \in H^2(\mathrm{Gal}(L/K), C_L)$: $\mathrm{inv}_{L/K'}(\mathrm{Res} x) = [K' : K] \cdot \mathrm{inv}_{L/K}(x)$, where Res is restriction to $\mathrm{Gal}(L/K')$.

These conditions, for K and for each finite extension of K in place of K, are the axioms of definition 8.4.1 for the pair $(G_K, C_{K^{\mathrm{sep}}})$, the invariant map of a pair of open subgroups with fixed fields $L \supseteq K'$ being $\mathrm{inv}_{L/K'}$.

Axiom 1 is theorem 10.1.2(2), and axioms 2 and 3 are theorem 10.1.5. All three rest on the inequalities of global class field theory cited in the proof of theorem 10.1.2. With the axioms in hand, Tate's theorem on cup products applies to any class formation [31, Ch. XI, §3, Thm. 1], as it did for the local class formation in theorem 8.4.3.

Theorem 10.4.2: Global Artin Reciprocity

There exists a unique continuous surjective homomorphism

$$\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$$

(the *global Artin map*) with the following properties:

1. *(Local-global compatibility)* For each place v of K, the diagram

$$\begin{array}{ccccc} K_v^\times & \hookrightarrow & \mathbb{I}_K & \twoheadrightarrow & C_K \\ {\scriptstyle \mathrm{rec}_{K_v}}\downarrow & & & & \downarrow{\scriptstyle \mathrm{rec}_K} \\ G_{K_v}^{\mathrm{ab}} & & \longrightarrow & & G_K^{\mathrm{ab}} \end{array}$$

   commutes, where rec_{K_v} is the local Artin map from theorem 8.4.4 and the bottom arrow is the natural map $G_{K_v}^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ induced by the inclusion of the decomposition group $G_v \hookrightarrow G_K$.

2. *(Norm groups)* For every finite Galois extension L/K: rec_K induces an isomorphism

$$C_K/N_{L/K}(C_L) \xrightarrow{\sim} \mathrm{Gal}(L/K)^{\mathrm{ab}}$$

3. *(Existence)* The open subgroups of C_K of finite index are exactly the norm groups $N_{L/K}(C_L)$ for finite abelian extensions L/K. The kernel of rec_K is the connected component of the identity in C_K.

***Proof Key ideas*:**
This is the main theorem of global class field theory, which this book does not prove. It is [2, Ch. VII, 5.1, with the proof in §§10–12].

Construction. For a finite Galois extension L/K with group G, Tate's theorem for the global class formation (definition 10.4.1) makes cup product with $u_{L/K}$ an isomorphism $\hat{H}^{-2}(G,\mathbb{Z}) \to \hat{H}^0(G, C_L)$ [31, Ch. XI, §3, Thm. 1]. The left side is G^{ab} (proposition 4.1.5, theorem 2.5.2), the right side is $C_K/N_{L/K}(C_L)$ because $C_L^G = C_K$, and inverting gives $\mathrm{rec}_{L/K}$ [2, Ch. VII, 11.3]. These maps are compatible as L grows [31, Ch. XI, §3, diagram (4)], so they define a continuous homomorphism $\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$ whose composite with the projection to $\mathrm{Gal}(L/K)$ is surjective for every finite abelian L/K. Its image is therefore dense. The connected subgroup $\mathbb{R}_{>0}$ of C_K (proposition 9.2.7) maps to the identity of the totally disconnected group G_K^{ab}, so the image of rec_K is the image of the compact group C_K^0 (theorem 9.2.6), which is closed. Hence rec_K is surjective.

1. Property (1) says that rec_K is the product of the local Artin maps, which is [2, Ch. VII, 10.2, Thm. A]. In the cohomological construction it reflects the restriction property of the fundamental class in theorem 10.1.5: on a decomposition group, $u_{L/K}$ restricts to the image of the local fundamental class. Uniqueness follows: the classes of idèles with only finitely many components different from 1 are dense in C_K, so a continuous homomorphism

satisfying (1) is determined by the local Artin maps.

2. For a finite Galois extension L/K, the composite of rec_K with the projection $G_K^{\mathrm{ab}} \to \mathrm{Gal}(L/K)^{\mathrm{ab}}$ is $\mathrm{rec}_{L/K}$, an isomorphism $C_K/N_{L/K}(C_L) \to \mathrm{Gal}(L/K)^{\mathrm{ab}}$ by the construction.

3. The existence theorem is [2, Ch. VII, 5.1(D)], proved in [2, Ch. VII, §12] from the second inequality and Kummer theory. By (2) and the existence theorem, the kernel of rec_K is the intersection of the open subgroups of finite index in C_K. That this intersection is the connected component of the identity in C_K is stated in [2, Ch. VII, 5.6], where the proof is omitted, and this book does not prove it. It rests on the structure of C_K as the product of the compact group C_K^0 and $\mathbb{R}_{>0}$, completing the sketch.

The global Artin map $\mathrm{rec}_K \colon C_K \to G_K^{\mathrm{ab}}$ is the idèlic formulation of the main theorem of class field theory: it classifies all abelian extensions of K in terms of the idèle class group. Property (1) says that the global reciprocity law is assembled from local reciprocity laws, one at each place. Property (2) says that the norm group $N_{L/K}(C_L)$ determines (and is determined by) the abelian extension L/K. Property (3) says that every "admissible" subgroup of C_K (open, finite index) arises from an abelian extension (a strong existence statement that goes beyond the cohomological formalism).

Example 10.4.3: The Artin Map for $\mathbb{Q}$

The idèle class group of $\mathbb{Q}$ is $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}$ (example 9.2.8). The maximal abelian extension of $\mathbb{Q}$ is $\mathbb{Q}^{\mathrm{ab}} = \mathbb{Q}(\zeta_n : n \geq 1)$ (Kronecker-Weber; see [6, Kronecker Weber's Theorem]), with $G_{\mathbb{Q}}^{\mathrm{ab}} = \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \widehat{\mathbb{Z}}^{\times}$ via the cyclotomic character (exercise 1.4.1 (5)). The global Artin map $\mathrm{rec}_{\mathbb{Q}} \colon C_{\mathbb{Q}} \to G_{\mathbb{Q}}^{\mathrm{ab}}$ is trivial on $\mathbb{R}_{>0}$ and sends the class of $(r, u) \in \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}$ to the automorphism acting on every root of unity ζ by $\zeta \mapsto \zeta^{u^{-1}}$ [2, Ch. VII, 5.7]. So $\mathrm{rec}_{\mathbb{Q}}$ is the projection to $\widehat{\mathbb{Z}}^{\times}$ followed by inversion and the identification $\widehat{\mathbb{Z}}^{\times} \cong G_{\mathbb{Q}}^{\mathrm{ab}}$. The kernel is $\mathbb{R}_{>0}$, the connected component of $C_{\mathbb{Q}}$.

The inverse is forced by the local-global compatibility at a prime p. The idèle with p at the place p and 1 elsewhere has class (p^{-1}, u_p) in $\mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}$, where u_p has component p^{-1} in $\mathbb{Z}_{\ell}^{\times}$ for every $\ell \neq p$ and component 1 in $\mathbb{Z}_p^{\times}$: multiply by the principal idèle p^{-1}. By property (1) of theorem 10.4.2, its image under $\mathrm{rec}_{\mathbb{Q}}$ is $\mathrm{rec}_{\mathbb{Q}_p}(p)$, which acts on $\mathbb{Q}(\zeta_m)$ for $p \nmid m$ as the arithmetic Frobenius, $\zeta_m \mapsto \zeta_m^p$ (proposition 8.4.5(1)), and $u_p^{-1} \equiv p \pmod{m}$.

The global reciprocity law, formulated here cohomologically, is equivalent to the classical ideal-theoretic formulation of Takagi and Artin: the Artin symbol $\left(\frac{L/K}{\mathfrak{p}}\right)$ for unramified primes $\mathfrak{p}$ is recovered from rec_K via the local Artin map at $\mathfrak{p}$, and the Artin reciprocity law $\prod_v \left(\frac{L/K}{v}\right) = 1$ (for a principal idèle) is the content of property (2). The cohomological approach replaces the intricate ideal-theoretic arguments with the three axioms of a class formation, recorded in section 10.1.

Proposition 10.4.4: Functoriality of the Artin Map

Let L/K be a finite extension of number fields (not necessarily Galois). The Artin maps for K and L satisfy two compatibilities:

1. *(Norm compatibility)* The diagram

$$\begin{array}{ccc} C_L & \xrightarrow{\mathrm{rec}_L} & G_L^{\mathrm{ab}} \\ {\scriptstyle N_{L/K}}\downarrow & & \downarrow \\ C_K & \xrightarrow{\mathrm{rec}_K} & G_K^{\mathrm{ab}} \end{array}$$

   commutes, where the right vertical arrow is the natural map $G_L^{\mathrm{ab}} \to G_K^{\mathrm{ab}}$ induced by the inclusion $G_L \hookrightarrow G_K$.

2. *(Transfer compatibility)* The diagram

$$\begin{array}{ccc} C_K & \xrightarrow{\mathrm{rec}_K} & G_K^{\mathrm{ab}} \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Ver}} \\ C_L & \xrightarrow{\mathrm{rec}_L} & G_L^{\mathrm{ab}} \end{array}$$

   commutes, where the left vertical arrow is the natural inclusion $C_K \hookrightarrow C_L$ and $\mathrm{Ver}\colon G_K^{\mathrm{ab}} \to G_L^{\mathrm{ab}}$ is the transfer (Verlagerung) homomorphism.

***Proof Key ideas*:**
Both diagrams are the limits $(1)_*$ and $(2)_*$ of the functoriality of the reciprocity isomorphism of a class formation [31, Ch. XI, §3], applied to the global class formation (definition 10.4.1), in which the module attached to a finite extension E of K is C_E. For idèle classes they are also [2, Ch. VII, 11.3, diagrams (13)]. This book does not prove them. Serre deduces the commutativity before the limit from the behaviour of the fundamental classes under restriction and corestriction and the corresponding formulas for cup products [31, Ch. XI, §3].

1. The vertical maps of the finite-level diagram (horizontal in Serre's layout) are corestrictions in degrees 0 and -2. In degree 0, corestriction from $\mathrm{Gal}(F/L)$ to $\mathrm{Gal}(F/K)$ is induced by the norm $C_L \to C_K$, and in degree -2 it is the map $\mathrm{Gal}(F/L)^{\mathrm{ab}} \to \mathrm{Gal}(F/K)^{\mathrm{ab}}$ induced by the inclusion.

2. The vertical maps are restrictions. In degree 0, restriction is induced by the inclusion $C_K \to C_L$, and in degree -2 it is the transfer, which is by definition the restriction on $H_1(-,\mathbb{Z}) = \hat{H}^{-2}(-,\mathbb{Z})$ [31, Ch. VII, §8], completing the sketch.

Exercise 10.4.1

1. Verify the class formation axioms explicitly for the quadratic extension $L = \mathbb{Q}(\sqrt{-1})$, $K = \mathbb{Q}$: check that $H^1(G, C_L) = 0$ (using ABHN) and that $\mathrm{inv}_{L/\mathbb{Q}}\colon H^2(G, C_L) \to \frac{1}{2}\mathbb{Z}/\mathbb{Z}$ is an isomorphism (exercise 10.1.1 (3)). Axiom (3) is vacuous here.

2. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_p)$ for an odd prime p. Show that the global Artin map $\mathrm{rec}_{\mathbb{Q}} \colon C_{\mathbb{Q}} \to \mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ agrees with the cyclotomic character at each prime $\ell \neq p$: the local Artin map sends $\ell \in \mathbb{Q}_\ell^\times$ to $\ell \bmod p \in (\mathbb{Z}/p\mathbb{Z})^\times$.

3. Compute the norm group $N_{\mathbb{Q}(i)/\mathbb{Q}}(C_{\mathbb{Q}(i)}) \subseteq C_{\mathbb{Q}}$ explicitly, and verify that it has index 2 in $C_{\mathbb{Q}}$. (*Hint:* identify $C_{\mathbb{Q}} \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times$ and determine which elements are norms from $C_{\mathbb{Q}(i)}$.)

4. The existence theorem states that every open finite-index subgroup of C_K is a norm group. Show that this implies: for every finite abelian group A and every continuous surjection $\varphi \colon C_K \twoheadrightarrow A$, there exists a unique abelian extension L/K with $\mathrm{Gal}(L/K) \cong A$ and $\ker \varphi = N_{L/K}(C_L)$.

5. Let L/K be a finite abelian extension of number fields and $a \in K^\times$. Show that the local Artin symbols satisfy the product formula $\prod_v \mathrm{rec}_{L_w/K_v}(a) = 1$ in $\mathrm{Gal}(L/K)$, the product running over all places of K and almost all of its factors being 1. (*Hint:* the idèle class of the diagonal element a is trivial in C_K.)

10.5 Selmer Groups and the Shafarevich-Tate Group

The Poitou-Tate sequence (theorem 10.2.5) applies to every finite G_K-module M, with S containing the archimedean places, the places where M is ramified and those dividing $|M|$, but its most consequential applications arise from the p-torsion of abelian varieties over number fields. This section introduces Selmer groups and the Shafarevich-Tate group as instances of the general local-global framework, connecting the cohomological machinery developed in this chapter to the arithmetic of elliptic curves and abelian varieties.

Throughout this section, A denotes an abelian variety over a number field K (in practice, most applications concern elliptic curves E/K), and p is a prime number. The reader is referred to [11] for the theory of elliptic curves and to [3] for abelian varieties. This section requires only the basic properties of the group $A(K)$ of rational points and the p-torsion subgroup $A[p] \subseteq A(\overline{K})$ as a finite G_K-module.

The starting point is the Kummer exact sequence. The multiplication-by-p map $[p] \colon A(\overline{K}) \to A(\overline{K})$ is surjective (over an algebraically closed field), with kernel $A[p]$. This gives a short exact sequence of G_K-modules:

$$0 \to A[p] \to A(\overline{K}) \xrightarrow{[p]} A(\overline{K}) \to 0$$

The associated long exact sequence in Galois cohomology begins:

$$0 \to A(K)[p] \to A(K) \xrightarrow{[p]} A(K) \xrightarrow{\delta} H^1(G_K, A[p]) \to H^1(G_K, A(\overline{K})) \xrightarrow{[p]} \cdots$$

The connecting homomorphism $\delta \colon A(K) \to H^1(G_K, A[p])$ sends a point $P \in A(K)$ to the cocycle $\sigma \mapsto \sigma(Q) - Q$, where $Q \in A(\overline{K})$ satisfies $[p]Q = P$. The image of δ is $A(K)/pA(K)$, which embeds as a subgroup of $H^1(G_K, A[p])$.

The same construction at each place v gives local connecting homomorphisms $\delta_v \colon A(K_v) \to$

$H^1(G_v, A[p])$, and the localization maps (definition 10.2.3) make the following diagram commute:

$$\begin{array}{ccc} A(K)/pA(K) & \xhookrightarrow{\delta} & H^1(G_K, A[p]) \\ \downarrow & & \downarrow{\scriptstyle \mathrm{loc}_v} \\ A(K_v)/pA(K_v) & \xhookrightarrow{\delta_v} & H^1(G_v, A[p]) \end{array}$$

Definition 10.5.1: p-Selmer Group

The *p-Selmer group* of A/K is:

$$\mathrm{Sel}_p(A/K) = \left\{\alpha \in H^1(G_K, A[p]) \;\middle|\; \mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v) \text{ for all } v\right\}$$

the subgroup of global cohomology classes whose local restrictions lie in the image of the Kummer map at every place.

The Selmer group sits between the Mordell-Weil group modulo p and the full cohomology group: $A(K)/pA(K) \subseteq \mathrm{Sel}_p(A/K) \subseteq H^1(G_K, A[p])$. The first inclusion holds because a global point is a local point at every place. The Selmer group is defined by the *local conditions* $\mathrm{im}(\delta_v)$: a global cohomology class belongs to Sel_p if and only if it is "locally a Kummer class" at every place.

Definition 10.5.2: Shafarevich-Tate Group

The *Shafarevich-Tate group* of A/K is:

$$Ш(A/K) = \ker\left(H^1(G_K, A(\overline{K})) \xrightarrow{\mathrm{loc}} \prod_v H^1(G_v, A(\overline{K}_v))\right)$$

the group of locally trivial torsors under A (principal homogeneous spaces for A that have a rational point over every completion K_v but not over K itself).

The Shafarevich-Tate group $Ш(A/K)$ is the obstruction to the Hasse principle for torsors under A: a torsor X under A defines a class $[X] \in H^1(G_K, A)$, and $[X] \in Ш(A/K)$ if and only if $X(K_v) \neq \emptyset$ for all v but $X(K) = \emptyset$. The group Ш is conjectured to be finite for all abelian varieties over number fields (see [11] for the connection to the Birch-Swinnerton-Dyer conjecture).

The three groups $A(K)/pA(K)$, $\mathrm{Sel}_p(A/K)$, and $Ш(A/K)$ are related by a fundamental exact sequence.

Proposition 10.5.3: Selmer Exact Sequence

There is a short exact sequence:

$$0 \to A(K)/pA(K) \xrightarrow{\delta} \mathrm{Sel}_p(A/K) \to Ш(A/K)[p] \to 0$$

***Proof*:**

The Kummer map $\delta\colon A(K)/pA(K) \hookrightarrow H^1(G_K, A[p])$ has image contained in $\mathrm{Sel}_p(A/K)$: a global point is a local point at every place, so its Kummer class lies in $\mathrm{im}(\delta_v)$ for all v. The cokernel of δ

inside Sel_p maps to $H^1(G_K, A)$ via the long exact sequence (the map $H^1(G_K, A[p]) \to H^1(G_K, A)$ sends a cohomology class of $A[p]$ to the corresponding torsor). A class $\alpha \in \mathrm{Sel}_p(A/K)$ maps to $H^1(G_K, A)$, and its localization $\mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v)$ means the local torsor has a K_v-point (it lies in the image of $A(K_v)$). So the image of Sel_p in $H^1(G_K, A)$ lands in $Ш(A/K)$. Since the map $H^1(G_K, A[p]) \to H^1(G_K, A)$ has image killed by p: the image lies in $Ш(A/K)[p]$. The surjectivity onto $Ш(A/K)[p]$ and the identification of the kernel with $\mathrm{im}(\delta)$ follow from the snake lemma applied to the commutative diagram of long exact sequences (global and local Kummer sequences).

The Selmer exact sequence is the primary tool for studying the Mordell-Weil group $A(K)$. The rank of $A(K)$ (as a finitely generated abelian group, by the Mordell-Weil theorem) is constrained by the $\mathbb{F}_p$-dimension of $\mathrm{Sel}_p(A/K)$: the exact sequence gives $\dim_{\mathbb{F}_p} \mathrm{Sel}_p(A/K) = \mathrm{rank}\, A(K) + \dim_{\mathbb{F}_p} A(K)[p] + \dim_{\mathbb{F}_p} Ш(A/K)[p]$. Computing Sel_p is feasible (it is a finite, computable group), while Ш remains mysterious.

Proposition 10.5.4: Finiteness of the Selmer Group

$\mathrm{Sel}_p(A/K)$ is a finite group.

***Proof*:**
Let S be a finite set of places containing all archimedean places, all places of bad reduction for A, and all places dividing p. A class $\alpha \in \mathrm{Sel}_p(A/K) \subseteq H^1(G_K, A[p])$ is unramified at every $v \notin S$: the local condition $\mathrm{loc}_v(\alpha) \in \mathrm{im}(\delta_v)$ implies α is trivial on the inertia group I_v (since A has good reduction at v and the Néron model provides a smooth integral structure). So $\mathrm{Sel}_p(A/K) \subseteq H^1(G_{K,S}, A[p])$. The group $H^1(G_{K,S}, A[p])$ is finite ($A[p]$ is a finite $G_{K,S}$-module and $G_{K,S}$ has finite cohomology; see [26]), so its subgroup $\mathrm{Sel}_p(A/K)$ is finite.

The Selmer group fits into the Poitou-Tate framework as a special case of "cohomology with local conditions." The module $M = A[p]$ is a finite G_K-module with Cartier dual $M^* \cong A^t[p]$ (the p-torsion of the dual abelian variety, identified via the Weil pairing), and the local conditions $\mathrm{im}(\delta_v) \subseteq H^1(G_v, A[p])$ define a "Selmer structure" in the sense of [26]. The Poitou-Tate sequence (theorem 10.2.5) applied to $M = A[p]$ controls the relationship between $\mathrm{Sel}_p(A/K)$ and the dual Selmer group $\mathrm{Sel}_p(A^t/K)$.

Theorem 10.5.5: Cassels-Tate Pairing

There exists a bilinear pairing

$$\langle \cdot, \cdot \rangle \colon Ш(A/K) \times Ш(A^t/K) \to \mathbb{Q}/\mathbb{Z}$$

whose left and right kernels are the maximal divisible subgroups $Ш(A/K)_{\mathrm{div}}$ and $Ш(A^t/K)_{\mathrm{div}}$. It therefore induces a non-degenerate pairing between $Ш(A/K)/Ш(A/K)_{\mathrm{div}}$ and $Ш(A^t/K)/Ш(A^t/K)_{\mathrm{div}}$. When $Ш(A/K)$ is identified with $Ш(A^t/K)$ through a polarization defined by a K-rational divisor, the pairing is alternating. In particular, for an elliptic curve E, identified with E^t by its canonical principal polarization, the pairing on $Ш(E/K)$ is alternating.

***Proof Key ideas*:**
The theorem is due to Cassels for elliptic curves and to Tate in general, and this book does not prove it. For each prime p the pairing on the p-primary parts is assembled from the Poitou-Tate duality of theorem 10.2.6 for the finite modules $A[p^n]$ and $A^t[p^n]$, which the Weil pairing makes Cartier dual, by passing to the limit in n. The description of the kernels is Milne's Theorem I.6.13(a), and the alternating property for a polarization defined by a K-rational divisor is proved in the same section.[a] For elliptic curves both statements are Cassels's theorem.[b] For a principal polarization that does not come from a K-rational divisor the pairing need not be alternating, as Poonen and Stoll showed.[c]

[a] [22, Ch. I, Thm. 6.13(a) and the discussion before Rem. 6.12].
[b] [32, Ch. X, Thm. 4.14].
[c] [22, Ch. I, §6, note to the construction of the pairing].

A consequence of the alternating property: if $Ш(E/K)$ is finite for an elliptic curve E/K, then $|Ш(E/K)|$ is a perfect square.

10.5.6 The Birch-Swinnerton-Dyer Conjecture The Selmer exact sequence and the Cassels-Tate pairing are central ingredients in the Birch-Swinnerton-Dyer (BSD) conjecture, which predicts that the rank of $E(K)$ equals the order of vanishing of the L-function $L(E/K, s)$ at $s = 1$, and gives a precise formula for the leading coefficient involving $|Ш(E/K)|$, the regulator, the real period, and the Tamagawa numbers at primes of bad reduction. The cohomological framework of this chapter provides the tools (Selmer groups, local-global duality, Cassels-Tate pairing) that are used in all known partial results toward BSD. The conjecture and its implications are developed in [11].

Example 10.5.7: 2-Descent for an Elliptic Curve
Let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 - x$ (conductor 32). The 2-torsion is rational: $E[2] = \{O, (0,0), (1,0), (-1,0)\} \cong (\mathbb{Z}/2\mathbb{Z})^2$ with trivial $G_{\mathbb{Q}}$-action. The Kummer map $\delta\colon E(\mathbb{Q})/2E(\mathbb{Q}) \hookrightarrow H^1(G_{\mathbb{Q}}, E[2]) = H^1(G_{\mathbb{Q}}, (\mathbb{Z}/2\mathbb{Z})^2)$ embeds the Mordell-Weil group modulo 2 into the cohomology. The Selmer group $\mathrm{Sel}_2(E/\mathbb{Q})$ is computed by imposing local conditions at each place: at $v = \infty$ (the archimedean place) and $v = 2$ (the unique prime of bad reduction), the local images $\mathrm{im}(\delta_v)$ constrain the global classes. A classical 2-descent computation (see [11]) gives $\mathrm{Sel}_2(E/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^2$ and $Ш(E/\mathbb{Q})[2] = 0$. The Selmer exact sequence then gives $E(\mathbb{Q})/2E(\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^2$, which (combined with the known torsion $E(\mathbb{Q})_{\mathrm{tors}} \cong (\mathbb{Z}/2\mathbb{Z})^2$) shows $\mathrm{rank}\, E(\mathbb{Q}) = 0$.

Exercise 10.5.1

1. Show that $A(K)/pA(K) \subseteq \mathrm{Sel}_p(A/K)$ by verifying that the Kummer image of a global point satisfies the local conditions at every place.
2. Show that the Selmer group $\mathrm{Sel}_p(A/K)$ does not depend on the choice of the finite set S used in its definition (i.e., the local conditions at places of good reduction are automatically satisfied by unramified classes).
3. For an elliptic curve E/K: assuming $Ш(E/K)$ is finite, show that the alternating property of the Cassels-Tate pairing implies $|Ш(E/K)|$ is a perfect square. (*Hint:* an alternating non-degenerate pairing on a finite abelian group forces the group to have square order.)

4. Explain how the p-Selmer group $\mathrm{Sel}_p(E/\mathbb{Q})$ for an elliptic curve $E/\mathbb{Q}$ can be viewed as a subgroup of $H^1(G_{\mathbb{Q},S}, E[p])$ for a suitable S, and describe how the Poitou-Tate exact sequence (theorem 10.2.5) applied to $M = E[p]$ gives an upper bound on $|\,\mathrm{Sel}_p\,|$.

11

Descent and Non-Abelian Cohomology

The preceding chapters developed the cohomology of profinite groups as an algebraic machine and applied it to the arithmetic of local and global fields. The theory was abelian throughout: the coefficient modules were abelian groups, the cohomology groups carried abelian group structures, and the duality theorems exploited this linearity. This chapter moves beyond the abelian setting in two directions. The first (section 11.1 and section 11.2) applies the abelian theory to classify field extensions and algebraic objects via *descent*, the principle that structures over a base field K are controlled by Galois cohomology of their automorphism groups over K^{sep}. The second (sections 11.3 and 11.4) develops the foundations of non-abelian H^1 and H^2, where the coefficient object is a non-abelian group.

11.1 Kummer Theory and Artin-Schreier Theory

The Kummer exact sequence $1 \to \mu_n \to (K^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^n} (K^{\mathrm{sep}})^\times \to 1$ appeared in chapter 10 in the context of Selmer groups and the Shafarevich-Tate group (section 10.5). Over a field K containing the nth roots of unity μ_n, the same sequence controls all abelian extensions of exponent dividing n: the connecting homomorphism in the long exact sequence gives $K^\times/(K^\times)^n \cong H^1(G_K, \mu_n)$, and the class of a gives the cyclic extension $K(\sqrt[n]{a})/K$, of degree the order of the class (classes generating the same subgroup give the same field). This section develops Kummer theory as a classification theorem for abelian extensions and treats its characteristic-p counterpart, the Artin-Schreier classification.

Throughout this section, K denotes a field and $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$ its absolute Galois group.

Theorem 11.1.1: Kummer Theory

Let K be a field and $n \geq 1$ an integer with $\mathrm{char}(K) \nmid n$.

1. The Kummer sequence $1 \to \mu_n \to (K^{\mathrm{sep}})^\times \xrightarrow{(\cdot)^n} (K^{\mathrm{sep}})^\times \to 1$ is an exact sequence of discrete G_K-modules, and its connecting homomorphism induces an isomorphism
$$\delta\colon K^\times/(K^\times)^n \xrightarrow{\sim} H^1(G_K, \mu_n)$$
For $a \in K^\times$ and $\alpha \in K^{\mathrm{sep}}$ with $\alpha^n = a$, the class $\delta(a)$ is represented by the cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$.
2. Suppose $\mu_n \subseteq K$. There is an inclusion-preserving bijection
$$\left\{ \begin{array}{c} \text{abelian extensions } L/K \text{ inside } K^{\mathrm{sep}} \\ \text{of exponent dividing } n \end{array} \right\} \xleftrightarrow{1:1} \left\{ \text{ subgroups } \Delta \leq K^\times/(K^\times)^n \right\}$$
given by $L \mapsto \Delta_L = ((L^\times)^n \cap K^\times)/(K^\times)^n$ and $\Delta \mapsto L_\Delta = K(\{\sqrt[n]{a} \mid a \in \Delta\})$.
3. Under the bijection of (2), the pairing $\langle \sigma, a\rangle = \sigma(\alpha)/\alpha$, for $\sigma \in \mathrm{Gal}(L/K)$, $a \in \Delta_L$ and $\alpha \in L$ with $\alpha^n = a$, induces an isomorphism of profinite groups $\mathrm{Gal}(L/K) \cong \mathrm{Hom}(\Delta_L, \mu_n)$, the right side with the topology of pointwise convergence, as a closed subgroup of $\mu_n^{\Delta_L}$. The extension L/K is finite if and only if Δ_L is finite, and then $[L : K] = |\Delta_L|$.

Here μ_n is the group of nth roots of unity in K^{sep}, and an abelian extension of exponent dividing n is a Galois extension whose Galois group is abelian and killed by n. When $\mu_n \subseteq K$, all nth roots of an element of $K^\times$ differ by elements of K, so L_Δ does not depend on the choice of the roots $\sqrt[n]{a}$.

***Proof*:**

Step 1: Part (1). The kernel of $x \mapsto x^n$ on $(K^{\mathrm{sep}})^\times$ is μ_n. For $a \in (K^{\mathrm{sep}})^\times$, the polynomial $X^n - a$ is separable, as its derivative nX^{n-1} vanishes only at 0, which is not a root, because $\mathrm{char}(K) \nmid n$. So it has a root in K^{sep}, and $x \mapsto x^n$ is onto. The maps are G_K-equivariant, so the long exact sequence of proposition 3.1.6, with $((K^{\mathrm{sep}})^\times)^{G_K} = K^\times$ (theorem 1.4.1), contains the exact segment
$$K^\times \xrightarrow{(\cdot)^n} K^\times \xrightarrow{\delta} H^1(G_K, \mu_n) \to H^1(G_K, (K^{\mathrm{sep}})^\times)$$
whose last term vanishes by Hilbert's Theorem 90 (theorem 8.1.1(2)). So δ induces the isomorphism of (1). Lifting a to α and computing the connecting homomorphism on cochains as in proposition 2.4.1 gives the cocycle $\sigma \mapsto \sigma(\alpha)/\alpha$.

From now on $\mu_n \subseteq K$, so G_K acts trivially on μ_n and $H^1(G_K, \mu_n) = \mathrm{Hom}_{\mathrm{cts}}(G_K, \mu_n)$, the group of continuous homomorphisms (proposition 3.1.7). Under this identification the class $\delta(a)$ is the homomorphism $\sigma \mapsto \sigma(\alpha)/\alpha$, as $B^1_{\mathrm{cts}}(G_K, \mu_n) = 0$. The nth roots of $a \in K^\times$ in K^{sep} are the $\zeta\alpha$ with $\zeta \in \mu_n$, for any one root α.

Step 2: The pairing. Let L/K be a Galois extension inside K^{sep}, and $\Delta \leq K^\times/(K^\times)^n$ a subgroup such that every $a \in \Delta$ has an nth root $\alpha \in L$. For $\sigma \in \mathrm{Gal}(L/K)$, put $\langle \sigma, a\rangle = \sigma(\alpha)/\alpha$.

It lies in μ_n, as $\sigma(\alpha)^n = a$. Another root $\zeta\alpha$, or the root αb of ab^n with $b \in K^\times$, gives the same value, since $\zeta, b \in K$ are fixed by σ. The value at ab is the product of the values at a and b, as $\alpha\beta$ is a root of ab. Since $\tau(\alpha)/\alpha \in \mu_n \subseteq K$ is fixed by σ,

$$\frac{\sigma\tau(\alpha)}{\alpha} = \sigma\Big(\frac{\tau(\alpha)}{\alpha}\Big)\frac{\sigma(\alpha)}{\alpha} = \frac{\tau(\alpha)}{\alpha} \cdot \frac{\sigma(\alpha)}{\alpha}$$

So $\sigma \mapsto \langle\sigma, \cdot\rangle$ is a homomorphism $\mathrm{Gal}(L/K) \to \mathrm{Hom}(\Delta, \mu_n)$. It is continuous, since $\langle\sigma, a\rangle$ depends only on the restriction of σ to $K(\alpha)$. If $L = L_\Delta$ it is injective, since an element pairing trivially with Δ fixes every α and so all of L_Δ. If $\langle\sigma, a\rangle = 1$ for all $\sigma \in \mathrm{Gal}(L/K)$, then $\alpha \in L^{\mathrm{Gal}(L/K)} = K$ and a is trivial in $K^\times/(K^\times)^n$.

Step 3: L_Δ is abelian of exponent dividing n, and $\Delta_{L_\Delta} = \Delta$. Let Δ be a subgroup and $L = L_\Delta$. The field L is generated by roots of the separable polynomials $X^n - a$, a representing elements of Δ, and contains all their roots $\zeta\alpha$, so L/K is Galois. By Step 2, $\mathrm{Gal}(L/K)$ embeds in the abelian group $\mathrm{Hom}(\Delta, \mu_n)$ of exponent dividing n. Every element of Δ has an nth root in L, so $\Delta \subseteq \Delta_L$. Suppose first that Δ is finite. Then L/K is finite, and

$$[L : K] = |\mathrm{Gal}(L/K)| \leq |\mathrm{Hom}(\Delta, \mu_n)| = |\Delta|$$

the last equality because Δ is a finite abelian group of exponent dividing n and μ_n is cyclic of order n. Step 2 applied to Δ_L shows that $\Delta_L \to \mathrm{Hom}(\mathrm{Gal}(L/K), \mu_n)$ is injective, and a finite group Γ has $|\mathrm{Hom}(\Gamma, \mu_n)| \leq |\Gamma|$, since a homomorphism factors through the abelian group $\Gamma^{\mathrm{ab}}/n\Gamma^{\mathrm{ab}}$, whose dual has its order. So $|\Delta_L| \leq |\mathrm{Gal}(L/K)| \leq |\Delta|$. With $\Delta \subseteq \Delta_L$ this gives $\Delta_L = \Delta$ and $[L : K] = |\Delta|$. For arbitrary Δ, let $[b] \in \Delta_L$ with $\beta \in L$, $\beta^n = b$. The element β lies in $L_{\Delta'}$ for the subgroup Δ' generated by finitely many elements of Δ, which is finite because Δ is abelian of exponent dividing n. So $[b] \in \Delta_{L_{\Delta'}} = \Delta' \subseteq \Delta$, and $\Delta_L = \Delta$.

Step 4: $L_{\Delta_L} = L$. Let L/K be abelian of exponent dividing n. Every $a \in \Delta_L$ has an nth root in L, so $L_{\Delta_L} \subseteq L$. Let $\sigma \in \mathrm{Gal}(L/K)$ be nontrivial. There is an open normal subgroup $U \leq \mathrm{Gal}(L/K)$ with $\sigma \notin U$, and the finite abelian group $\mathrm{Gal}(L/K)/U$ has exponent dividing n, so it is a product of cyclic groups of orders dividing n, each of which embeds in the cyclic group μ_n of order n. Hence there is a continuous homomorphism $\chi\colon \mathrm{Gal}(L/K) \to \mu_n$ with $\chi(\sigma) \neq 1$. Composed with the restriction $G_K \to \mathrm{Gal}(L/K)$, it is an element of $\mathrm{Hom}_{\mathrm{cts}}(G_K, \mu_n)$, so by Step 1 there are $a \in K^\times$ and $\alpha \in K^{\mathrm{sep}}$ with $\alpha^n = a$ and $\chi(g) = g(\alpha)/\alpha$ for all $g \in G_K$. Every $g \in G_L$ fixes α, so $\alpha \in L$ (theorem 1.4.1), $a \in \Delta_L$, and $\langle\sigma, a\rangle = \chi(\sigma) \neq 1$. Thus only the identity of $\mathrm{Gal}(L/K)$ pairs trivially with Δ_L. If $L_{\Delta_L} \neq L$, a nontrivial element of $\mathrm{Gal}(L/L_{\Delta_L})$ would fix every nth root of every $a \in \Delta_L$, a contradiction. So $L_{\Delta_L} = L$.

Step 5: Parts (2) and (3). By Steps 3 and 4 the two maps are mutually inverse, and they preserve inclusions. Let L correspond to $\Delta = \Delta_L$. By Step 2 the continuous homomorphism $\mathrm{Gal}(L/K) \to \mathrm{Hom}(\Delta, \mu_n)$ is injective. For a finite subgroup $\Delta' \leq \Delta$, the restriction $\mathrm{Gal}(L/K) \to \mathrm{Gal}(L_{\Delta'}/K)$ is surjective, and $\mathrm{Gal}(L_{\Delta'}/K) \to \mathrm{Hom}(\Delta', \mu_n)$ is an injective map between groups of the same order $|\Delta'|$ by Step 3, hence bijective. So every homomorphism $\Delta \to \mu_n$ agrees on each finite subgroup with some $\langle\sigma, \cdot\rangle$, the image of $\mathrm{Gal}(L/K)$ is dense in the profinite group $\mathrm{Hom}(\Delta, \mu_n)$, and it is closed, being compact. The map is therefore a continuous bijection from a compact space to a Hausdorff space, hence an isomorphism of profinite groups. If Δ is finite, $[L : K] = |\Delta|$ by Step 3. If Δ is infinite, it has finite subgroups Δ' of arbitrarily large order, and $\mathrm{Gal}(L/K)$ maps onto groups $\mathrm{Gal}(L_{\Delta'}/K)$ of order $|\Delta'|$, so L/K is infinite.

Example 11.1.2: Kummer Theory in Practice

1. For $n = 2$ and any field K with $\mathrm{char}(K) \neq 2$: $\mu_2 = \{\pm 1\} \subseteq K$ always, so Kummer theory applies. The group $K^\times/(K^\times)^2$ classifies quadratic extensions of K: each $a \in K^\times \setminus (K^\times)^2$ gives $K(\sqrt{a})/K$. For $K = \mathbb{Q}$: the squarefree integers $d \neq 1$ parametrize $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ (each class contains a unique squarefree representative), recovering the classical classification of quadratic fields.

2. For $K = \mathbb{Q}(\zeta_5)$ and $n = 5$: $\mu_5 \subseteq K$, so Kummer theory classifies all abelian extensions of K with exponent dividing 5. The group $K^\times/(K^\times)^5$ is infinite: the class number of K is 1, so it is $\mathcal{O}_K^\times/(\mathcal{O}_K^\times)^5 \oplus \bigoplus_{\mathfrak{p}} \mathbb{Z}/5\mathbb{Z}$, one factor for each prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$. The *unramified* abelian extensions of exponent dividing 5 correspond, by class field theory, to $\mathrm{Hom}(\mathrm{Cl}\,(K), \mathbb{Z}/5\mathbb{Z})$, the Hilbert class field of K having Galois group $\mathrm{Cl}\,(K)$ [25, Ch. VI, (6.9)]. Since $\mathrm{Cl}\,(\mathbb{Q}(\zeta_5)) = 1$ (the class number is 1), there are no unramified abelian 5-extensions of $\mathbb{Q}(\zeta_5)$.

3. (If the reader has read [14]) For $K = \mathbb{Q}_p$ with $p \equiv 1 \pmod{n}$: $\mu_n \subseteq \mathbb{Q}_p$ (Hensel's lemma), and $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ (one factor from the valuation $v(a) \bmod n$ and one from the unit group $\mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^n \cong \mathbb{Z}/n\mathbb{Z}$). The abelian extensions of $\mathbb{Q}_p$ of exponent dividing n correspond to the subgroups of this group, and the cyclic ones to its cyclic subgroups.

In positive characteristic, the pth power map on $(K^{\mathrm{sep}})^\times$ is no longer surjective (the equation $x^p = a$ may have no solution when $\mathrm{char}(K) = p$), and Kummer theory fails for p-extensions. The Artin-Schreier exact sequence provides the replacement.

Theorem 11.1.3: Artin-Schreier Theory

Let K be a field with $\mathrm{char}(K) = p > 0$, and define the Artin-Schreier map $\wp\colon K^{\mathrm{sep}} \to K^{\mathrm{sep}}$ by $\wp(x) = x^p - x$.

1. There is a short exact sequence of discrete G_K-modules

$$0 \to \mathbb{F}_p \to K^{\mathrm{sep}} \xrightarrow{\wp} K^{\mathrm{sep}} \to 0$$

and its connecting homomorphism induces an isomorphism of $\mathbb{F}_p$-vector spaces onto the group of continuous homomorphisms $G_K \to \mathbb{F}_p$

$$\delta\colon K/\wp(K) \xrightarrow{\sim} H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}_{\mathrm{cts}}(G_K, \mathbb{F}_p)$$

with $\delta(a)(\sigma) = \sigma(\alpha) - \alpha$ for any root $\alpha \in K^{\mathrm{sep}}$ of $x^p - x - a$.

2. For $a \in K \setminus \wp(K)$ and a root α of $x^p - x - a$, the extension $K(\alpha)/K$ is cyclic of degree p, and $G_{K(\alpha)}$ is the kernel of $\delta(a)$.

3. The assignment $a \mapsto K(\alpha)$ induces a bijection between the one-dimensional $\mathbb{F}_p$-subspaces of $K/\wp(K)$ and the cyclic extensions of K of degree p inside K^{sep}. Two classes $[a], [b] \neq 0$ give the same extension if and only if $[b] = i[a]$ for some $i \in \mathbb{F}_p^\times$.

***Proof*:**
Part (1). The map $\wp$ is additive, since $(x+y)^p = x^p + y^p$ in characteristic p, and G_K-equivariant. Its kernel consists of the roots of $x^p - x$, which are the p elements of the prime field $\mathbb{F}_p$, fixed by G_K. For $a \in K^{\mathrm{sep}}$, the polynomial $x^p - x - a$ has derivative -1, so it is separable and has a root $\alpha \in K^{\mathrm{sep}}$, with $\wp(\alpha) = a$. The long exact sequence (proposition 3.1.6), with $(K^{\mathrm{sep}})^{G_K} = K$ (theorem 1.4.1), contains the exact segment

$$K \xrightarrow{\wp} K \xrightarrow{\delta} H^1(G_K, \mathbb{F}_p) \to H^1(G_K, K^{\mathrm{sep}})$$

and the additive form of Hilbert's Theorem 90 (theorem 8.1.2(2)) gives $H^1(G_K, K^{\mathrm{sep}}) = 0$. So δ induces an isomorphism of groups $K/\wp(K) \to H^1(G_K, \mathbb{F}_p)$, which is $\mathbb{F}_p$-linear since every additive map between $\mathbb{F}_p$-vector spaces is. As G_K acts trivially on $\mathbb{F}_p$, $H^1(G_K, \mathbb{F}_p) = \mathrm{Hom}_{\mathrm{cts}}(G_K, \mathbb{F}_p)$ (proposition 3.1.7), and the formula for $\delta(a)$ is the cochain computation of proposition 2.4.1.

Part (2). The roots of $x^p - x - a$ are the $\alpha + i$ with $i \in \mathbb{F}_p$, since $\wp(\alpha + i) = \wp(\alpha) = a$, so $K(\alpha)$ contains all of them and $K(\alpha)/K$ is Galois. The homomorphism $\chi = \delta(a) \colon G_K \to \mathbb{F}_p$ is nonzero, because $[a] \neq 0$ and δ is injective, so it is surjective onto the group $\mathbb{F}_p$ of prime order. An element $\sigma \in G_K$ lies in $\ker \chi$ if and only if $\sigma(\alpha) = \alpha$, that is, if and only if $\sigma \in G_{K(\alpha)}$. So $G_{K(\alpha)} = \ker \chi$, and $\mathrm{Gal}(K(\alpha)/K) \cong G_K/\ker \chi \cong \mathbb{F}_p$ by theorem 1.4.1.

Part (3). Let $b = ia + \wp(c)$ with $i \in \mathbb{F}_p^\times$ and $c \in K$. Then $\wp(i\alpha + c) = i\wp(\alpha) + \wp(c) = b$, as $i^p = i$, so $i\alpha + c$ is a root of $x^p - x - b$, and $K(i\alpha + c) = K(\alpha)$. So the extension depends only on the line $\mathbb{F}_p \cdot [a]$. If roots α of $x^p - x - a$ and β of $x^p - x - b$ generate the same field M, then by part (2) the surjections $\delta(a), \delta(b) \colon G_K \to \mathbb{F}_p$ have the same kernel G_M. They induce two isomorphisms $G_K/G_M \to \mathbb{F}_p$, which differ by an automorphism of $\mathbb{F}_p$, that is, by multiplication by some $i \in \mathbb{F}_p^\times$, so $\delta(b) = i\,\delta(a) = \delta(ia)$ and $[b] = i[a]$. This proves injectivity on lines. For surjectivity, let M/K be cyclic of degree p inside K^{sep}, and choose an isomorphism $\mathrm{Gal}(M/K) \cong \mathbb{F}_p$. Its composite with the restriction $G_K \to \mathrm{Gal}(M/K)$ is a nonzero element of $\mathrm{Hom}_{\mathrm{cts}}(G_K, \mathbb{F}_p)$ with kernel G_M, equal to $\delta(a)$ for some $[a] \neq 0$ by part (1). By part (2), $G_{K(\alpha)} = \ker \delta(a) = G_M$, so $K(\alpha) = M$ (theorem 1.4.1).

The parallel between Kummer and Artin-Schreier is structural: both use a short exact sequence involving a "power map" ($(\cdot)^n$ or $\wp$), Hilbert 90 to kill H^1 of the ambient group, and the connecting homomorphism to identify a quotient of $K^\times$ (or K) with the group classifying cyclic extensions. The Artin-Schreier-Witt generalization (replacing $\mathbb{F}_p$ by $\mathbb{Z}/p^n\mathbb{Z}$ and $\wp$ by $F - 1$ on Witt vectors, with F the Witt vector Frobenius) extends the theory to all cyclic p-power extensions in characteristic p. See [26] for the full treatment.

Example 11.1.4: Artin-Schreier Extensions of $\mathbb{F}_p(t)$
For $K = \mathbb{F}_p(t)$, the one-dimensional $\mathbb{F}_p$-subspaces of $K/\wp(K)$ classify the cyclic extensions of degree p (theorem 11.1.3(3)). The element t does not lie in $\wp(K)$: a constant $x \in \mathbb{F}_p$ has $\wp(x) = 0 \neq t$, and a nonconstant $x \in K$ has a pole of some order $m \geq 1$, where $\wp(x)$ has a pole of order pm, while t has a single simple pole, at $t = \infty$. So t gives the cyclic extension $L = K(\alpha)$ of degree p, where $\alpha^p - \alpha = t$. This extension is totally ramified at the place $t = \infty$ (the pole of t) and unramified elsewhere. For $p = 2$: $L = \mathbb{F}_2(t)(\alpha)$ with $\alpha^2 + \alpha = t$ is an Artin-Schreier extension of degree 2. In characteristic 2 the extensions $K(\sqrt{a})$ with $a \notin K^2$ are inseparable, and the separable quadratic extensions are exactly these Artin-Schreier extensions.

Exercise 11.1.1

1. Show that Kummer theory for $n = 2$ recovers the classical bijection between quadratic extensions of K (with $\mathrm{char}(K) \neq 2$) and elements of $K^\times/(K^\times)^2$, with $a \mapsto K(\sqrt{a})$.
2. Let ℓ be a prime and p a prime with $p \equiv 1 \pmod{\ell}$. Using Kummer theory and $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^\ell \cong (\mathbb{Z}/\ell\mathbb{Z})^2$ (example 11.1.2(3)), show that $\mathbb{Q}_p$ has exactly $\ell + 1$ cyclic extensions of degree ℓ inside a fixed separable closure.
3. For $K = \mathbb{F}_3(t)$, show that the classes of the powers t^i with $i \geq 1$ and $3 \nmid i$ are linearly independent over $\mathbb{F}_3$ in $K/\wp(K)$, and deduce that K has infinitely many cyclic extensions of degree 3. *Hint:* compare pole orders, as in example 11.1.4.
4. Let K be a number field with $\mu_n \subseteq K$. Show that the Kummer isomorphism $K^\times/(K^\times)^n \cong H^1(G_K, \mu_n) = \mathrm{Hom}(G_K, \mu_n)$ is compatible with the Artin map from theorem 10.4.2: for $a \in K^\times$, the Kummer character $\chi_a \colon G_K \to \mu_n$ satisfies $\chi_a(\mathrm{rec}_K(\alpha)) = \prod_v (a, \alpha_v)_{n,v}$ (the product of local nth power residue symbols).
5. (If the reader has read [26]) State the Artin-Schreier-Witt generalization: for $\mathrm{char}(K) = p$ and the ring of Witt vectors $W_m(K)$, the sequence $0 \to \mathbb{Z}/p^m\mathbb{Z} \to W_m(K^{\mathrm{sep}}) \xrightarrow{\wp} W_m(K^{\mathrm{sep}}) \to 0$ classifies the cyclic extensions of degree dividing p^m via $W_m(K)/\wp(W_m(K)) \cong H^1(G_K, \mathbb{Z}/p^m\mathbb{Z})$, those of degree exactly p^m coming from the surjections $G_K \to \mathbb{Z}/p^m\mathbb{Z}$ up to automorphisms of $\mathbb{Z}/p^m\mathbb{Z}$.

11.2 Galois Descent

Kummer theory (theorem 11.1.1) classified abelian extensions of K using H^1 with abelian coefficients μ_n. A far broader principle underlies this classification: given any algebraic structure X_0 over K (a vector space, an algebra, a quadratic form, a variety), the structures over K that become isomorphic to X_0 after extending scalars to K^{sep} are classified by the cohomology $H^1(G_K, \mathrm{Aut}(X_0 \otimes_K K^{\mathrm{sep}}))$. When the automorphism group is abelian, this is the ordinary H^1 of chapters 3 and 4. When it is non-abelian, the cohomology set H^1 is a pointed set, which in general carries no group structure (the subject of section 11.3). This section develops the descent principle through concrete examples, deferring the formal non-abelian theory to the next section.

Throughout this section, K denotes a field with absolute Galois group $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$.

Definition 11.2.1: Twisted Forms

Let X_0 be an algebraic object defined over K (a K-algebra, a quadratic form over K, a K-variety, etc.). A *twisted form* (or *K-form*) of X_0 is an object X of the same type over K such that:

$$X \otimes_K K^{\mathrm{sep}} \cong X_0 \otimes_K K^{\mathrm{sep}}$$

(isomorphism of the corresponding structures over K^{sep}). The set of K-isomorphism classes of twisted forms of X_0 is denoted $\mathrm{Twist}(X_0/K)$.

The question "classify all objects of a given type over K" often reduces to classifying twisted forms of a single "split" object X_0: an object that is as simple as possible over K^{sep}. The descent principle

provides the answer.

A twisted form of X_0 is recovered from $X_0 \otimes_K K^{\mathrm{sep}}$ as the fixed points of a modified Galois action, and that recovery rests on one fact about vector spaces: a K^{sep}-vector space with a continuous semilinear action of G_K is spanned by its fixed vectors.

Lemma 11.2.2: Speiser's Lemma

Let W be a K^{sep}-vector space with an action of G_K that is semilinear, $\sigma(\lambda w) = \sigma(\lambda)\sigma(w)$ for $\lambda \in K^{\mathrm{sep}}$ and $w \in W$, and continuous, meaning every $w \in W$ is fixed by an open subgroup of G_K. Then the natural map $W^{G_K} \otimes_K K^{\mathrm{sep}} \to W$ is an isomorphism.

***Proof*:**

Step 1: Injectivity. Let $w_1, \dots, w_m \in W^{G_K}$ be linearly independent over K and suppose they are dependent over K^{sep}. Choose a nontrivial relation $\sum_i \lambda_i w_i = 0$ with the fewest nonzero coefficients, scaled so that $\lambda_1 = 1$. Applying $\sigma \in G_K$ gives $\sum_i \sigma(\lambda_i) w_i = 0$, and subtracting the two relations gives one in which the coefficient of w_1 is 0. By minimality that relation is trivial, so $\sigma(\lambda_i) = \lambda_i$ for every i and every σ. Hence every λ_i lies in $(K^{\mathrm{sep}})^{G_K} = K$ (theorem 1.4.1(2)), which contradicts the independence over K.

Step 2: Surjectivity. Let $w \in W$. The open subgroups G_L for finite Galois extensions L/K inside K^{sep} form a neighbourhood basis of the identity (theorem 1.4.1(1)), so w is fixed by G_L for such an L. Let $\sigma_1 = 1, \sigma_2, \dots, \sigma_n$ be representatives of the cosets of G_L in G_K, where $n = [L : K]$, so that their restrictions are the distinct elements of $\mathrm{Gal}(L/K)$. Let $a_1, \dots, a_n$ be a K-basis of L, and put

$$t_i = \sum_{j=1}^{n} \sigma_j(a_i w) = \sum_{j=1}^{n} \sigma_j(a_i)\, \sigma_j(w).$$

Each t_i is fixed by G_K: an element $\tau \in G_K$ permutes the cosets $\sigma_j G_L$, and $a_i w$ is fixed by G_L. The matrix $(\sigma_j(a_i))_{i,j}$ is invertible, because a vector (x_j) with $\sum_j x_j \sigma_j(a_i) = 0$ for every i gives $\sum_j x_j \sigma_j = 0$ on L, so $x = 0$ by Dedekind's lemma from *Core Algebra*.[a] Solving the linear system for the unknowns $\sigma_j(w)$ expresses $w = \sigma_1(w)$ as an L-linear combination of $t_1, \dots, t_n \in W^{G_K}$.

[a] [9, linear independence of Automorphisms (Dedekind's Lemma)].

The lemma says that Galois descent for vector spaces is effective: every continuous semilinear G_K-action on W comes from the K-form W^{G_K}. In cohomological terms it is the following vanishing statement, whose case $n = 1$ is Hilbert's Theorem 90 (theorem 8.1.1(2)).

Corollary 11.2.3: Generalized Hilbert's Theorem 90

For every $n \geq 1$, the pointed set $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}}))$ is trivial.

***Proof*:**

Let $c\colon G_K \to \mathrm{GL}_n(K^{\mathrm{sep}})$ be a continuous 1-cocycle, and let G_K act on $W = (K^{\mathrm{sep}})^n$ by $\sigma *_c v = c(\sigma)\, \sigma(v)$. This is an action, since $(\sigma\tau) *_c v = c(\sigma)\, \sigma(c(\tau))\, \sigma\tau(v) = \sigma *_c (\tau *_c v)$ by the cocycle condition. It is semilinear. It is continuous: the cocycle condition gives $c(1) = 1$, so $c^{-1}(1)$

is an open neighbourhood of 1 and contains an open subgroup, and each v is fixed by an open subgroup under the standard action. By lemma 11.2.2 the fixed space has a K-basis $e_1, \dots, e_n$ that is a K^{sep}-basis of W. Let $a \in \mathrm{GL}_n(K^{\mathrm{sep}})$ have columns $e_1, \dots, e_n$. Then $c(\sigma)\,\sigma(a) = a$ for every σ, so $c(\sigma) = a\,\sigma(a)^{-1}$ and c is cohomologous to the trivial cocycle.

Theorem 11.2.4: Descent Principle

Let V_0 be a finite-dimensional K-vector space with a finite family t of tensors, each an element of some $V_0^{\otimes p} \otimes_K (V_0^\vee)^{\otimes q}$, and write $X_0 = (V_0, t)$. Its twisted forms are the pairs (V, s) of the same type over K with $(V, s) \otimes_K K^{\mathrm{sep}} \cong X_0 \otimes_K K^{\mathrm{sep}}$. There is a natural bijection of pointed sets

$$\mathrm{Twist}(X_0/K) \cong H^1(G_K, \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)),$$

where $\mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$ is the group of K^{sep}-linear automorphisms of $V_0 \otimes_K K^{\mathrm{sep}}$ preserving t, with the G_K-action $(\sigma \cdot \varphi)(x) = \sigma(\varphi(\sigma^{-1}(x)))$, and the distinguished point corresponds to X_0 itself.

An algebra structure on V_0 is its multiplication, a tensor in $(V_0^\vee)^{\otimes 2} \otimes V_0$, and a bilinear form is a tensor in $(V_0^\vee)^{\otimes 2}$. So the theorem covers central simple algebras, quadratic forms and étale algebras, three of the four families of examples below. The fourth, elliptic curves, is a variety.

***Proof*:**

Step 1: From twisted forms to cocycles. Let X be a twisted form of X_0, and choose an isomorphism $\varphi\colon X \otimes_K K^{\mathrm{sep}} \xrightarrow{\sim} X_0 \otimes_K K^{\mathrm{sep}}$. For each $\sigma \in G_K$, the map:

$$c(\sigma) = \varphi \circ \sigma(\varphi)^{-1} = \varphi \circ \sigma \circ \varphi^{-1} \circ \sigma^{-1} \in \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$$

measures the failure of φ to be G_K-equivariant. The map $c\colon G_K \to \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$ satisfies the cocycle condition: $c(\sigma\tau) = c(\sigma) \cdot \sigma(c(\tau))$ (direct computation from the definition). It is continuous: the matrix of φ in bases of V and V_0 has finitely many entries, all in some finite Galois extension L/K, so $c(\sigma)$ depends only on the restriction of σ to L. A different choice of isomorphism $\varphi' = a \circ \varphi$ (with $a \in \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$) replaces c by the cohomologous cocycle $c'(\sigma) = a \cdot c(\sigma) \cdot \sigma(a)^{-1}$. So X determines a well-defined class $[c] \in H^1(G_K, \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0))$.

Step 2: From cocycles to twisted forms. Given a cocycle $c\colon G_K \to \mathrm{Aut}_{K^{\mathrm{sep}}}(X_0)$, define a new G_K-action on $X_0 \otimes_K K^{\mathrm{sep}}$ by $\sigma *_c x = c(\sigma) \cdot \sigma(x)$ ("twisting" the standard action by the cocycle). The cocycle condition ensures this is a group action: $(\sigma\tau) *_c x = c(\sigma\tau) \cdot \sigma\tau(x) = c(\sigma) \cdot \sigma(c(\tau)) \cdot \sigma\tau(x) = c(\sigma) \cdot \sigma(c(\tau) \cdot \tau(x)) = \sigma *_c (\tau *_c x)$. The twisted action is semilinear and continuous, so by lemma 11.2.2 its fixed space V_c is a K-vector space with $V_c \otimes_K K^{\mathrm{sep}} \cong V_0 \otimes_K K^{\mathrm{sep}}$. The tensors t are fixed by the twisted action, since σ fixes them (they are defined over K) and $c(\sigma)$ preserves them, so they lie in the tensor spaces of V_c. Hence $X_c = (V_c, t)$ is a twisted form of X_0. If $c'(\sigma) = a\,c(\sigma)\,\sigma(a)^{-1}$, then a carries V_c onto $V_{c'}$ and preserves t, so cohomologous cocycles give K-isomorphic twisted forms.

Step 3: The bijection. Starting from X and φ, every $x \in V$ satisfies $c(\sigma)\,\sigma(\varphi(x)) = \varphi(\sigma(x)) = \varphi(x)$, so φ maps V into V_c, and both have dimension $\dim V_0$, so φ is a K-isomorphism $X \cong X_c$. Starting from c, the identification $V_c \otimes_K K^{\mathrm{sep}} \cong V_0 \otimes_K K^{\mathrm{sep}}$ of Step 2 has cocycle c. So the maps $X \mapsto [c]$ and $[c] \mapsto X_c$ are mutual inverses. The identity cocycle $c = 1$ corresponds to X_0 itself (the untwisted form), so X_0 maps to the distinguished point of H^1.

The descent principle reduces the classification of algebraic structures to a cohomological computation. The difficulty shifts to computing H^1 of the automorphism group, which depends on the specific type of structure.

The same bijection holds for a quasi-projective variety X_0 over K, with $\mathrm{Aut}(X_0 \otimes_K K^{\mathrm{sep}})$ its group of automorphisms as a variety [28, Thm. 4.5.2]. The proof follows Steps 1 to 3, with the twisted Galois action now a descent datum on a variety, and quasi-projectivity is what makes such a datum come from a variety over K. Descent of schemes can fail: *Moduli, Deformation Theory, and Stacks* [16, Non-Effective Descent] gives a descent datum along an étale cover that comes from no scheme. The argument is unchanged when the variety carries a marked K-point that the automorphisms must fix, and the elliptic-curve example below is that case.

The following examples illustrate the principle for several fundamental classes of objects.

Example 11.2.5: Central Simple Algebras

The "split" central simple algebra of degree n over K is $M_n(K)$ (the matrix algebra). Every CSA of degree n becomes isomorphic to $M_n(K^{\mathrm{sep}})$ after extending scalars (by the Artin-Wedderburn theorem, see [10, Artin-Wedderburn For k-Algebras]). The automorphism group is $\mathrm{Aut}(M_n(K^{\mathrm{sep}})) = \mathrm{PGL}_n(K^{\mathrm{sep}})$ (the Skolem-Noether theorem: every automorphism is inner, and the center acts trivially). The descent principle gives:

$$\{\text{CSAs of degree } n \text{ over } K \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}}))$$

This recovers the identification of the Brauer group in terms of cohomology from theorem 7.2.7: taking the union over all n, the exact sequence $1 \to K^{\mathrm{sep},\times} \to \mathrm{GL}_n(K^{\mathrm{sep}}) \to \mathrm{PGL}_n(K^{\mathrm{sep}}) \to 1$ and the vanishing $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = 1$ (generalized Hilbert 90, corollary 11.2.3) give a map $H^1(G_K, \mathrm{PGL}_n) \to H^2(G_K, K^{\mathrm{sep},\times}) = \mathrm{Br}(K)$ with trivial kernel. It sends a central simple algebra of degree n to its Brauer class, and it is injective: two central simple algebras of the same degree with the same Brauer class are isomorphic, by theorem 7.1.2.

Example 11.2.6: Quadratic Forms

The "split" quadratic form of rank n over K ($\mathrm{char}(K) \neq 2$) is $q_0 = \langle 1, 1, \ldots, 1\rangle$ (the sum of n squares). Every non-degenerate quadratic form of rank n becomes isomorphic to q_0 over K^{sep} (diagonalize and scale). The orthogonal group $O_n(K^{\mathrm{sep}})$ acts as automorphisms of q_0, and the descent principle gives:

$$\{\text{non-deg. quadratic forms of rank } n \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, O_n(K^{\mathrm{sep}}))$$

For $n = 1$: $O_1 = \{\pm 1\} = \mu_2$, so $H^1(G_K, O_1) = K^\times/(K^\times)^2$, classifying quadratic forms $\langle a\rangle$ up to isometry (i.e., elements of $K^\times$ up to squares). For $n = 2$: the exact sequence $1 \to SO_2 \to O_2 \to \mathbb{Z}/2\mathbb{Z} \to 1$ and the identification $SO_2 \cong \mathbb{G}_m$ (rotation group) produce a connecting map $H^1(G_K, O_2) \to H^1(G_K, \mathbb{Z}/2\mathbb{Z})$ that sends a binary form to its discriminant class.

Example 11.2.7: Elliptic Curves

(If the reader has read [11]) For an elliptic curve E/K: the twisted forms of E are elliptic curves E'/K isomorphic to E over K^{sep}. The automorphism group $\mathrm{Aut}(E)$ depends on the j-invariant:

1. For $j(E) \neq 0, 1728$: $\mathrm{Aut}(E) = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$ (only the identity and negation). The descent principle gives $\mathrm{Twist}(E/K) \cong H^1(G_K, \mathbb{Z}/2\mathbb{Z}) \cong K^\times/(K^\times)^2$. Each twisted form is a

quadratic twist E^d: $dy^2 = f(x)$ for $d \in K^\times/(K^\times)^2$.

2. For $j(E) = 1728$ (e.g., E: $y^2 = x^3 - x$ over $\mathbb{Q}$): $\mathrm{Aut}(E) \cong \mathbb{Z}/4\mathbb{Z}$ (generated by $(x, y) \mapsto (-x, iy)$, with $i = \sqrt{-1}$). Over a field containing μ_4: $\mathrm{Twist}(E/K) \cong H^1(G_K, \mathbb{Z}/4\mathbb{Z}) \cong K^\times/(K^\times)^4$ (by Kummer theory). The twisted forms include quartic twists E^d: $dy^2 = x^3 - x$.

3. For $j(E) = 0$ (e.g., E: $y^2 = x^3+1$ over $\mathbb{Q}$): $\mathrm{Aut}(E) \cong \mathbb{Z}/6\mathbb{Z}$ (generated by $(x, y) \mapsto (\zeta_3 x, -y)$). Over a field containing μ_6: the twisted forms include sextic twists.

Example 11.2.8: Étale Algebras and Galois Extensions

The "split" étale algebra of degree n is $K^n = K \times K \times \cdots \times K$. An étale K-algebra of degree n is a product of separable field extensions $L_1 \times \cdots \times L_r$ with $\sum[L_i : K] = n$. The automorphism group of $(K^{\mathrm{sep}})^n$ is the symmetric group S_n (permuting the factors), with G_K acting trivially. The descent principle gives:

$$\{\text{étale } K\text{-algebras of degree } n \mid \text{up to } K\text{-isom.}\} \cong H^1(G_K, S_n) = \mathrm{Hom}(G_K, S_n)/\text{conj.}$$

(since the G_K-action on S_n is trivial, H^1 is the set of conjugacy classes of continuous homomorphisms $G_K \to S_n$). Such a homomorphism is a continuous action of G_K on $\{1, \ldots, n\}$, and its orbits correspond to the factors of the étale algebra: an orbit of size d with stabilizer G_{L_i} corresponds to a separable field extension L_i/K of degree d. The transitive actions are exactly those giving fields, the separable extensions of degree n up to isomorphism, while the other actions give products $L_1 \times \cdots \times L_r$ with $r \geq 2$. The distinguished point is K^n itself (the split algebra), corresponding to the trivial homomorphism, whose n orbits are points.

A common thread in these examples: the generalized Hilbert 90 theorem states that $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = 1$ (every twisted form of K^n as a *vector space* is trivially isomorphic to K^n, by corollary 11.2.3). This is the descent-theoretic content of Hilbert 90: there are no nontrivial vector bundles over $\mathrm{Spec}\, K$ in the Galois topology. The nontrivial cohomology arises when additional structure (algebra, quadratic form, group law) constrains the automorphism group to a proper subgroup $A \leq \mathrm{GL}_n$.

The vanishing $H^1(G_K, \mathrm{GL}_n) = 1$ has a computational consequence: for any subgroup $A \leq \mathrm{GL}_n(K^{\mathrm{sep}})$ stabilized by G_K, the exact sequence of pointed sets ending in $H^1(G_K, A) \to H^1(G_K, \mathrm{GL}_n) = 1$ from the inclusion $A \hookrightarrow \mathrm{GL}_n$ [30, Ch. I, §5.4, Prop. 36 and Cor. 1] identifies $H^1(G_K, A)$ with the set of orbits of $\mathrm{GL}_n(K)$ acting on $(\mathrm{GL}_n(K^{\mathrm{sep}})/A)^{G_K}$, so it can be analyzed via the quotient GL_n/A. When A is normal (as for $A = \mathrm{SL}_n \trianglelefteq \mathrm{GL}_n$), the long exact sequence in non-abelian H^1 (section 11.3) provides a systematic approach to computing $H^1(G_K, A)$ from the simpler group GL_n/A.

Exercise 11.2.1

1. Using the descent principle, show that the quaternion algebras $(a, b)_{\mathbb{Q}}$ (see example 7.1.4) are exactly the nontrivial elements of $H^1(G_{\mathbb{Q}}, \mathrm{PGL}_2(\overline{\mathbb{Q}}))$, and identify the connecting map to $H^2(G_{\mathbb{Q}}, \overline{\mathbb{Q}}^\times) = \mathrm{Br}(\mathbb{Q})$.

2. For an elliptic curve E/K with $j(E) \neq 0, 1728$: show that the quadratic twist E^d (for $d \in K^\times/(K^\times)^2$) has the property $E^d(K^{\mathrm{sep}}) \cong E(K^{\mathrm{sep}})$ as G_K-sets, but $E^d(K)$ can differ from $E(K)$ (the Mordell-Weil groups of twists are related but not isomorphic in general).

3. Show that the degree-3 étale $\mathbb{Q}$-algebras are: $\mathbb{Q}^3$ (split), $\mathbb{Q} \times L$ for quadratic fields L, and cubic fields K'. Identify each with a conjugacy class of homomorphisms $G_{\mathbb{Q}} \to S_3$.

4. (If the reader has read [4]) A *Severi-Brauer variety* of dimension $n-1$ over K is a twisted form of $\mathbb{P}^{n-1}_K$. Show that $\mathrm{Twist}(\mathbb{P}^{n-1}/K) \cong H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}}))$, and that the Brauer class of a Severi-Brauer variety X is trivial if and only if $X(K) \neq \emptyset$.

11.3 Non-Abelian H^1

The descent principle (theorem 11.2.4) used $H^1(G_K, \mathrm{Aut}(X_0))$ without requiring $\mathrm{Aut}(X_0)$ to be abelian: the automorphism groups PGL_n, O_n, and S_n from the examples in section 11.2 are all non-abelian (for $n \geq 2$, $n \geq 3$, and $n \geq 3$ respectively). This section develops the formal theory of H^1 for non-abelian coefficient groups. The key difference from the abelian theory is that H^1 is a *pointed set* rather than a group: it has a distinguished element (the trivial class) but no group operation in general.

Throughout this section, G denotes a profinite group and A a topological group with a continuous G-action by group automorphisms (a *G-group*).

Definition 11.3.1: Non-Abelian H^1

Let A be a G-group (not necessarily abelian). A 1-*cocycle* is a continuous map $c\colon G \to A$ satisfying the cocycle condition:

$$c(\sigma\tau) = c(\sigma) \cdot \sigma(c(\tau)) \qquad \text{for all } \sigma, \tau \in G$$

Two cocycles c, c' are *cohomologous* if there exists $a \in A$ with:

$$c'(\sigma) = a^{-1} \cdot c(\sigma) \cdot \sigma(a) \qquad \text{for all } \sigma \in G$$

The *non-abelian* H^1 is the quotient:

$$H^1(G, A) = Z^1(G, A)/\sim$$

where $Z^1(G, A)$ is the set of 1-cocycles and $\sim$ is the cohomologous relation. This is a *pointed set* with distinguished element $[1]$ (the class of the trivial cocycle $c(\sigma) = 1$ for all σ).

When A is abelian, the relation $c' \sim c$ becomes $c'(\sigma) = c(\sigma) \cdot \sigma(a) \cdot a^{-1}$, and the set $H^1(G, A)$ carries an abelian group structure (the product of cocycles is a cocycle). When A is non-abelian, the product of two cocycles need not satisfy the cocycle condition, so $H^1(G, A)$ has no natural group structure. The correct categorical framework is that of pointed sets and morphisms preserving the basepoint.

Proposition 11.3.2: Functoriality

The assignment $A \mapsto H^1(G, A)$ is a functor from the category of G-groups (with G-equivariant group homomorphisms) to the category of pointed sets (with basepoint-preserving maps). For a continuous homomorphism $\varphi\colon G' \to G$ of profinite groups: the restriction $\mathrm{res}\colon H^1(G, A) \to H^1(G', A)$ sending $c \mapsto c \circ \varphi$ is a morphism of pointed sets.

***Proof*:**
A G-equivariant homomorphism $f\colon A \to B$ sends cocycles to cocycles ($f(c(\sigma\tau)) = f(c(\sigma)) \cdot \sigma(f(c(\tau)))$ using G-equivariance) and preserves the cohomologous relation ($f(a^{-1} \cdot c(\sigma) \cdot \sigma(a)) = f(a)^{-1} \cdot f(c(\sigma)) \cdot \sigma(f(a))$). The trivial cocycle maps to the trivial cocycle. For restriction: the composition $c \circ \varphi$ satisfies the cocycle condition for G' whenever c satisfies it for G.

The central structural result for non-abelian H^1 is the long exact sequence in pointed sets, which generalizes the long exact sequence of abelian cohomology.

Theorem 11.3.3: Exact Sequence of Pointed Sets

Let $1 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 1$ be an exact sequence of G-groups (with i injective, π surjective, and $\ker \pi = i(A)$). There is an exact sequence of pointed sets:

$$1 \to A^G \xrightarrow{i} B^G \xrightarrow{\pi} C^G \xrightarrow{\delta} H^1(G, A) \xrightarrow{i_*} H^1(G, B) \xrightarrow{\pi_*} H^1(G, C)$$

where "exactness" means: the fiber of each map over the distinguished point equals the image of the previous map.

***Proof*:**
The sequence at A^G, B^G, C^G is the usual exact sequence of fixed points, using the left-exactness of the fixed-point functor.

The connecting map δ. For $\gamma \in C^G$: choose a lift $b \in B$ with $\pi(b) = \gamma$ (possible by surjectivity of π). For each $\sigma \in G$: $\pi(\sigma(b)) = \sigma(\pi(b)) = \sigma(\gamma) = \gamma = \pi(b)$, so $b^{-1} \cdot \sigma(b) \in \ker \pi = i(A)$. Define $a(\sigma) = i^{-1}(b^{-1} \cdot \sigma(b)) \in A$. This is a 1-cocycle: $a(\sigma\tau) = i^{-1}(b^{-1} \cdot \sigma\tau(b)) = i^{-1}(b^{-1} \cdot \sigma(b) \cdot \sigma(b^{-1}) \cdot \sigma\tau(b)) = i^{-1}(b^{-1} \cdot \sigma(b)) \cdot \sigma(i^{-1}(b^{-1} \cdot \tau(b))) = a(\sigma) \cdot \sigma(a(\tau))$ (using that $i(A)$ is normal in B and G-stable). A different lift $b' = b \cdot i(a_0)$ replaces a by the cohomologous cocycle $a'(\sigma) = a_0^{-1} \cdot a(\sigma) \cdot \sigma(a_0)$. So $\delta(\gamma) = [a] \in H^1(G, A)$ is well defined.

Exactness at C^G. The image of $\pi\colon B^G \to C^G$ consists of those $\gamma \in C^G$ that lift to a G-fixed element $b \in B^G$. For such γ: $\sigma(b) = b$ for all σ, giving $a(\sigma) = i^{-1}(b^{-1} \cdot b) = 1$, so $\delta(\gamma) = [1]$. Conversely, if $\delta(\gamma) = [1]$: the cocycle a is a coboundary, $a(\sigma) = a_0^{-1} \cdot \sigma(a_0)$ for some $a_0 \in A$. Then $b' = b \cdot i(a_0)^{-1}$ satisfies $\sigma(b') = \sigma(b) \cdot \sigma(i(a_0))^{-1} = b \cdot i(a(\sigma)) \cdot i(\sigma(a_0))^{-1} = b \cdot i(a_0^{-1} \cdot \sigma(a_0) \cdot \sigma(a_0)^{-1}) = b \cdot i(a_0)^{-1} = b'$, so $b' \in B^G$ lifts γ.

Exactness at $H^1(G, A)$. The image of δ consists of classes $[a]$ arising from some $\gamma \in C^G$. The map $i_*\colon H^1(G, A) \to H^1(G, B)$ sends $[a]$ to $[i \circ a]$. For $[a] \in \mathrm{im}(\delta)$: $(i \circ a)(\sigma) = b^{-1} \cdot \sigma(b)$ is a coboundary in B, so $i_*([a]) = [1]$. Conversely, if $i_*([a]) = [1]$: the cocycle $i \circ a$ is a coboundary in B, i.e., $(i \circ a)(\sigma) = b^{-1} \cdot \sigma(b)$ for some $b \in B$. Then $\gamma = \pi(b) \in C^G$ (since $\pi(b^{-1} \cdot \sigma(b)) = \pi(i(a(\sigma))) = 1$, giving $\sigma(\pi(b)) = \pi(b)$), and $\delta(\gamma) = [a]$.

Exactness at $H^1(G, B)$. The image of i_* consists of classes $[c] \in H^1(G, B)$ with $c(\sigma) \in i(A)$ (up to B-coboundary). The map π_* sends $[c]$ to $[\pi \circ c]$. For $[c] \in \mathrm{im}(i_*)$: $(\pi \circ c)(\sigma) = \pi(i(a(\sigma))) = 1$, so $\pi_*([c]) = [1]$. Conversely, if $\pi_*([c]) = [1]$: the cocycle $\pi \circ c$ is a coboundary in C, i.e.,

$\pi(c(\sigma)) = \gamma^{-1} \cdot \sigma(\gamma)$ for some $\gamma \in C$. Lifting γ to $b \in B$: the cocycle $c'(\sigma) = b \cdot c(\sigma) \cdot \sigma(b)^{-1}$ is cohomologous to c and satisfies $\pi(c'(\sigma)) = 1$, so $c'(\sigma) \in i(A)$. Thus $[c] = [c'] \in \mathrm{im}(i_*)$.

The exact sequence terminates at $H^1(G, C)$: there is no natural continuation to H^2 in the non-abelian setting (without additional structure). The obstruction to extending the sequence to H^2 is the subject of section 11.4. When A lies in the center of B, the sequence does extend to the abelian $H^2(G, A)$.

Definition 11.3.4: Twisting

Let $c \in Z^1(G, A)$ be a 1-cocycle. The *twisted G-group* ${}_cA$ is the group A equipped with the twisted G-action:

$$\sigma *_c a = c(\sigma) \cdot \sigma(a) \cdot c(\sigma)^{-1}$$

The cocycle condition ensures this is a group action. The *twisting bijection* is:

$$H^1(G, A) \cong H^1(G, {}_cA)$$

sending a class $[c']$ to $[c^{-1} \cdot c']$ (with the cocycle condition verified using the twisted action). Under this bijection, $[c] \in H^1(G, A)$ maps to $[1] \in H^1(G, {}_cA)$: twisting shifts the basepoint from $[1]$ to $[c]$.

Twisting is the mechanism that controls fibers in the non-abelian exact sequence. For the map $\pi_*\colon H^1(G, B) \to H^1(G, C)$: the fiber $\pi_*^{-1}([c_C])$ over a class $[c_C] \in H^1(G, C)$ is not a subgroup of $H^1(G, B)$ (since $H^1(G, B)$ is not a group), but a torsor under $H^1(G, {}_{c_B}A)$ for any c_B lifting c_C. The "size" of the fiber depends on the twisted group, which varies with the basepoint.

Example 11.3.5: The Brauer Group via Non-Abelian H^1

The exact sequence of G_K-groups:

$$1 \to (K^{\mathrm{sep}})^\times \xrightarrow{\cdot \mathrm{Id}} \mathrm{GL}_n(K^{\mathrm{sep}}) \xrightarrow{\pi} \mathrm{PGL}_n(K^{\mathrm{sep}}) \to 1$$

(where $(K^{\mathrm{sep}})^\times$ embeds as scalar matrices) induces the exact sequence of pointed sets:

$$\begin{aligned} K^\times \to \mathrm{GL}_n(K) \to \mathrm{PGL}_n(K) &\xrightarrow{\delta} H^1(G_K, (K^{\mathrm{sep}})^\times) \\ \to H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) \to H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}})) &\xrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) \end{aligned}$$

Hilbert 90 (theorem 8.1.1) gives $H^1(G_K, (K^{\mathrm{sep}})^\times) = 0$, and generalized Hilbert 90 (corollary 11.2.3) gives $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = 1$, so δ' has trivial kernel:

$$1 \to H^1(G_K, \mathrm{PGL}_n(K^{\mathrm{sep}})) \xrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$$

The connecting map δ' sends a CSA of degree n (viewed as a twisted form of $M_n(K)$ via descent, example 11.2.5) to its Brauer class. It is injective, since two CSAs of the same degree with the same Brauer class are isomorphic (theorem 7.1.2). The image of δ' is $\mathrm{Br}(K)[n]$ (the n-torsion), and the injectivity of δ' says: a CSA of degree n is determined up to isomorphism by its Brauer class (the Brauer class plus the degree determines the algebra). Passing to the colimit over n: $\mathrm{Br}(K) = \varinjlim_n H^1(G_K, \mathrm{PGL}_n)$, recovering the cohomological description of the Brauer group from theorem 7.2.7.

Exercise 11.3.1

1. Show that $H^1(G_K, \mathrm{GL}_n(K^{\mathrm{sep}})) = \{*\}$ implies, via the exact sequence of pointed sets for $1 \to \mathrm{SL}_n \to \mathrm{GL}_n \xrightarrow{\det} (K^{\mathrm{sep}})^\times \to 1$, that every class in $H^1(G_K, \mathrm{SL}_n(K^{\mathrm{sep}}))$ is $\delta(a)$ for some $a \in K^\times$, where δ is the connecting map. Describe $\delta(a)$ by an explicit cocycle, and deduce that $H^1(G_K, \mathrm{SL}_n(K^{\mathrm{sep}})) = \{*\}$.
2. For $K = \mathbb{R}$: compute $H^1(G_{\mathbb{R}}, O_n(\mathbb{C}))$ by classifying non-degenerate real quadratic forms of rank n up to isometry. (*Hint:* the classification is by signature (p, q) with $p + q = n$.)
3. Let $c \in Z^1(G_K, \mathrm{PGL}_2(K^{\mathrm{sep}}))$ be the cocycle corresponding to a quaternion algebra D over K. Describe the twisted group ${}_c\mathrm{GL}_2(K^{\mathrm{sep}})$ explicitly as the unit group $D^\times$ of D (viewed as a G_K-group), and verify that $H^1(G_K, {}_c\mathrm{GL}_2) = \{*\}$ (Hilbert 90 for $D^\times$).
4. For the connecting map $\delta' \colon H^1(G_K, \mathrm{PGL}_n) \to \mathrm{Br}(K)$: show that the fiber $(\delta')^{-1}([\alpha])$ over a Brauer class $[\alpha]$ is either empty (if $[\alpha]$ has order not dividing n) or a single point (if $[\alpha] \in \mathrm{Br}(K)[n]$).

11.4 Non-Abelian H^2 and Gerbes

The exact sequence of theorem 11.3.3 terminates at $H^1(G, C)$: the non-abelian H^1 has no natural continuation to H^2 for arbitrary coefficient groups. A partial extension is possible when the kernel A is *central* in B (i.e., $A \subseteq Z(B)$), in which case the connecting map extends one step further into the abelian $H^2(G, A)$. This section describes this extension, introduces the non-abelian H^2 in its limited form, and connects the theory to the broader categorical framework of gerbes.

Theorem 11.4.1: Obstruction Exact Sequence

Let $1 \to A \xrightarrow{i} B \xrightarrow{\pi} C \to 1$ be an exact sequence of G-groups with $A \subseteq Z(B)$ (i.e., A is central in B, and in particular A is abelian). The exact sequence of theorem 11.3.3 extends by one term:

$$\cdots \to H^1(G, B) \xrightarrow{\pi_*} H^1(G, C) \xrightarrow{\delta'} H^2(G, A)$$

where $H^2(G, A)$ is the ordinary (abelian) H^2. The map δ' is the *obstruction*: a class $[c] \in H^1(G, C)$ lifts to $H^1(G, B)$ if and only if $\delta'([c]) = 0$ in $H^2(G, A)$.

***Proof*:**
Construction of δ'. Let $c \colon G \to C$ be a 1-cocycle. For each $\sigma \in G$, choose a lift $\tilde{c}(\sigma) \in B$ with $\pi(\tilde{c}(\sigma)) = c(\sigma)$ (possible by surjectivity). The map $\tilde{c}$ need not satisfy the cocycle condition in B. The failure is measured by:

$$\alpha(\sigma, \tau) = \tilde{c}(\sigma) \cdot \sigma(\tilde{c}(\tau)) \cdot \tilde{c}(\sigma\tau)^{-1}$$

Since $\pi(\alpha(\sigma, \tau)) = c(\sigma) \cdot \sigma(c(\tau)) \cdot c(\sigma\tau)^{-1} = 1$ (the cocycle condition for c), the element $\alpha(\sigma, \tau)$ lies in $\ker \pi = i(A)$. The map $\alpha \colon G \times G \to A$ is a 2-cocycle: the associativity of the group operation in B gives the cocycle identity $\sigma(\alpha(\tau, \rho)) \cdot \alpha(\sigma, \tau\rho) = \alpha(\sigma, \tau) \cdot \alpha(\sigma\tau, \rho)$ (using $A \subseteq Z(B)$ to commute α past elements of B). The centrality of A is essential here: without it, α would depend on the conjugation action of $\tilde{c}$ on A, and the cocycle identity would fail.

Well-definedness. A different choice of lifts $\tilde{c}'(\sigma) = \tilde{c}(\sigma) \cdot i(\beta(\sigma))$ (with $\beta\colon G \to A$) modifies α by a coboundary: $\alpha'(\sigma,\tau) = \alpha(\sigma,\tau) \cdot \beta(\sigma) \cdot \sigma(\beta(\tau)) \cdot \beta(\sigma\tau)^{-1} = \alpha(\sigma,\tau) \cdot (d\beta)(\sigma,\tau)$. A cohomologous cocycle $c'(\sigma) = \gamma^{-1} \cdot c(\sigma) \cdot \sigma(\gamma)$ (with $\gamma \in C$) also modifies α by a coboundary (lift γ to $\tilde{\gamma} \in B$ and adjust). So $\delta'([c]) = [\alpha] \in H^2(G, A)$ is well defined.

Exactness. The class $[c] \in H^1(G, C)$ lies in $\mathrm{im}(\pi_*)$ if and only if $c = \pi \circ b$ for some 1-cocycle $b\colon G \to B$ (up to coboundary). Taking $\tilde{c} = b$: the obstruction $\alpha(\sigma,\tau) = b(\sigma) \cdot \sigma(b(\tau)) \cdot b(\sigma\tau)^{-1} = 1$ (the cocycle condition for b). So $\delta'([c]) = 0$. Conversely, if $\delta'([c]) = 0$: the 2-cocycle α is a coboundary, $\alpha(\sigma,\tau) = \beta(\sigma) \cdot \sigma(\beta(\tau)) \cdot \beta(\sigma\tau)^{-1}$ for some $\beta\colon G \to A$. Replacing $\tilde{c}(\sigma)$ by $\tilde{c}(\sigma) \cdot i(\beta(\sigma))^{-1}$: the new lift satisfies the cocycle condition in B (the obstruction vanishes), providing a lift $b \in Z^1(G, B)$ with $\pi \circ b = c$.

Example 11.4.2: The Brauer Group as an Obstruction

The central extension $1 \to (K^{\mathrm{sep}})^\times \to \mathrm{GL}_n(K^{\mathrm{sep}}) \to \mathrm{PGL}_n(K^{\mathrm{sep}}) \to 1$ has $(K^{\mathrm{sep}})^\times$ central in $\mathrm{GL}_n(K^{\mathrm{sep}})$, so the obstruction exact sequence applies:

$$H^1(G_K, \mathrm{GL}_n) \to H^1(G_K, \mathrm{PGL}_n) \xrightarrow{\delta'} H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$$

The obstruction class $\delta'([A]) \in \mathrm{Br}(K)$ of a CSA A (viewed as a class in $H^1(G_K, \mathrm{PGL}_n)$) is its Brauer class. Since $H^1(G_K, \mathrm{GL}_n) = 1$ (corollary 11.2.3), the connecting map δ' has trivial kernel, and it is injective because two CSAs of the same degree with the same Brauer class are isomorphic (theorem 7.1.2). The content of the obstruction: A splits (i.e., $A \cong M_n(K)$) if and only if its Brauer class is trivial.

The 2-cocycle $\alpha(\sigma,\tau)$ from the proof has a concrete description: choose a K^{sep}-basis of $A \otimes_K K^{\mathrm{sep}} \cong M_n(K^{\mathrm{sep}})$ and express the G_K-action on A as $\sigma \mapsto \tilde{c}(\sigma) \in \mathrm{GL}_n(K^{\mathrm{sep}})$ (the semilinear action). The failure of $\tilde{c}$ to be a homomorphism is measured by $\alpha(\sigma,\tau) = \tilde{c}(\sigma) \cdot \sigma(\tilde{c}(\tau)) \cdot \tilde{c}(\sigma\tau)^{-1} \in (K^{\mathrm{sep}})^\times$, which is exactly the 2-cocycle representing the Brauer class of A in $H^2(G_K, (K^{\mathrm{sep}})^\times)$. This recovers the crossed product construction of definition 7.2.1.

Example 11.4.3: Lifting Projective Representations

A *projective representation* of G_K of dimension n is a continuous homomorphism $\bar{\rho}\colon G_K \to \mathrm{PGL}_n(K^{\mathrm{sep}})$ (equivalently, a 1-cocycle with values in PGL_n). The obstruction to lifting $\bar{\rho}$ to a linear representation $\rho\colon G_K \to \mathrm{GL}_n(K^{\mathrm{sep}})$ is the class $\delta'([\bar{\rho}]) \in H^2(G_K, (K^{\mathrm{sep}})^\times) = \mathrm{Br}(K)$. This class is the *Schur obstruction*: it measures whether the projective representation arises from a central simple algebra that splits. For K a local field: $\mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}$ (corollary 8.1.4 from chapter 8), so the obstruction is a rational number modulo 1. For $K = \mathbb{Q}$: the Brauer group exact sequence (corollary 10.1.3 from chapter 10) constrains which projective representations can be lifted, and the obstruction vanishes if and only if the associated CSA splits at every place.

The obstruction exact sequence handles the case where A is central in B. The centrality condition is essential: without it, the "2-cocycle" $\alpha(\sigma,\tau) = \tilde{c}(\sigma) \cdot \sigma(\tilde{c}(\tau)) \cdot \tilde{c}(\sigma\tau)^{-1}$ does not commute with elements of B, and the cocycle identity fails. For a non-central kernel, the obstruction takes values not in an abelian H^2 but in a more general object: the non-abelian H^2 of Giraud.

Definition 11.4.4: Non-Abelian H^2 (Giraud)

For a G-group A (possibly non-abelian), the *non-abelian* $H^2(G, A)$ is the set of equivalence classes of *G-gerbes bound by A*: categorical objects that generalize torsors from groups to groupoids. A G-gerbe bound by A is a category $\mathcal{G}$ equipped with a G-action such that: (i) $\mathcal{G}$ is locally non-empty (it has objects over some open cover in the G-topology), and (ii) the automorphism group of any object is isomorphic to A (with the isomorphism compatible with the G-action up to inner automorphisms). Two gerbes are equivalent if they are connected by a G-equivariant equivalence of categories. The set $H^2(G, A)$ is pointed (the trivial gerbe is the category of A-torsors with the standard G-action).

When A is abelian, this construction recovers the ordinary $H^2(G, A)$ (an abelian gerbe bound by A is equivalent to a 2-cocycle $\alpha\colon G \times G \to A$). When A is non-abelian, $H^2(G, A)$ depends on the "band" (or "lien") of A, the outer automorphism class of the G-action on A, and does not admit a group structure. The dependence on the band is a genuinely non-abelian phenomenon: two G-groups with the same underlying group but different G-actions can have different H^2, even though their underlying abstract H^2 (ignoring the G-action) would be the same. The full theory is developed in Giraud [20] using the language of stacks and fibered categories. The construction above gives the essential idea without the full categorical apparatus.

The relationship between the obstruction exact sequence and the gerbe framework is as follows. For a central extension $1 \to A \to B \to C \to 1$: the obstruction $\delta'([c]) \in H^2(G, A)$ of a class $[c] \in H^1(G, C)$ corresponds, under the Giraud construction, to the gerbe of "liftings of c to B." This gerbe has objects (over K^{sep}) that are 1-cocycles $b\colon G \to B$ with $\pi \circ b = c$, and the automorphism group of any such object is A (acting by the coboundary relation). The gerbe is trivial (has a global object) if and only if the lifting exists, i.e., $\delta'([c]) = 0$.

The non-abelian H^2 has no further extension: there is no general "non-abelian H^3." The obstruction theory stops at H^2, reflecting the structural fact that torsors (classified by H^1) and gerbes (classified by H^2) exhaust the naturally arising "non-abelian" cohomological objects in classical algebraic geometry. Higher cohomology with non-abelian coefficients requires the theory of higher stacks and ∞-categories, which lies beyond the scope of this book.

11.4.5 Connections to the Langlands Program The non-abelian $H^1(G_K, \hat{G})$ for the Langlands dual group $\hat{G}$ of a reductive group G parametrizes *L-parameters*: continuous homomorphisms from the Weil group W_K (or the conjectural Langlands group L_K) to $\hat{G}(\mathbb{C})$, up to conjugacy. The local Langlands correspondence predicts a bijection between L-parameters and "L-packets" of irreducible representations of $G(K_v)$ for local fields K_v. The global Langlands correspondence extends this to automorphic representations. The cohomological framework of this chapter (non-abelian H^1, twisting, descent, and obstruction classes in H^2) provides the language for formulating and studying these correspondences. The Langlands program remains one of the central research programs in modern number theory. See [26] for the connection between Galois representations and automorphic forms.

The non-abelian H^1 and H^2 developed in this section and section 11.3 provide a classification framework that extends far beyond abelian Galois cohomology: H^1 classifies twisted forms and torsors (theorem 11.2.4), the obstruction map in H^2 controls lifting problems (theorem 11.4.1), and the gerbe framework gives a categorical interpretation of the obstructions.

Exercise 11.4.1

1. Verify the obstruction cocycle identity: for the 2-cocycle $\alpha(\sigma,\tau) = \tilde{c}(\sigma)\cdot\sigma(\tilde{c}(\tau))\cdot\tilde{c}(\sigma\tau)^{-1}$ (with $\tilde{c}$ a set-theoretic lift of a 1-cocycle $c\colon G \to C$), show that $\sigma(\alpha(\tau,\rho))\cdot\alpha(\sigma,\tau\rho) = \alpha(\sigma,\tau)\cdot\alpha(\sigma\tau,\rho)$ using the centrality of A in B.

2. Show that when A is abelian, the non-abelian $H^2(G,A)$ (in the sense of definition 11.4.4) reduces to the ordinary $H^2(G,A)$: a gerbe bound by an abelian group A is determined up to equivalence by a 2-cocycle $\alpha\colon G\times G\to A$, and the equivalence relation on gerbes corresponds to the coboundary relation on 2-cocycles.

3. For $K=\mathbb{Q}$ and the exact sequence $1\to\mu_2\to\mathrm{SL}_2\to\mathrm{PGL}_2\to 1$: the obstruction map $\delta'\colon H^1(G_{\mathbb{Q}},\mathrm{PGL}_2)\to H^2(G_{\mathbb{Q}},\mu_2)=\mathrm{Br}(\mathbb{Q})[2]$ sends a quaternion algebra D to its Brauer class. Show that D lifts to $H^1(G_{\mathbb{Q}},\mathrm{SL}_2)$ (i.e., D has a "reduced norm 1 form") if and only if $[D]=0\in\mathrm{Br}(\mathbb{Q})$ (i.e., $D\cong M_2(\mathbb{Q})$).

4. Describe the gerbe associated to a Brauer class $[\alpha]\in\mathrm{Br}(K)$: it is the category whose objects over K^{sep} are K^{sep}-vector spaces V of dimension n with an isomorphism $\mathrm{End}(V)\cong A\otimes_K K^{\mathrm{sep}}$ (where A is the CSA representing $[\alpha]$), and whose morphisms are isomorphisms of vector spaces compatible with the endomorphism structure. Show that this category is non-empty over K^{sep} but has no global object over K (unless $[\alpha]=0$).

12

What Next?

Class field theory. The cohomological reciprocity isomorphism of chapters 8 and 10 identifies $\mathrm{Gal}(L/K)^{\mathrm{ab}}$ with a norm quotient of the multiplicative or idèle class group, but it does not say which subgroups are norm groups, or how a prime decomposes in a given abelian extension. The existence theorem, the Frobenius description of the Artin map, conductors and the explicit reciprocity laws answer those questions. *Class Field Theory* [24] develops them, and Neukirch's *Algebraic Number Theory* and Cassels and Fröhlich's *Algebraic Number Theory* are the classical references.

From a field to a scheme. A discrete G_K-module is the same thing as a sheaf on the étale site of $\operatorname{Spec} K$, and the cohomology of G_K is the étale cohomology of that one-point scheme. Replacing $\operatorname{Spec} K$ by a scheme X gives the étale cohomology of X. Its analogue of the absolute Galois group is the étale fundamental group, and that group controls the first cohomology with finite locally constant coefficients. *Étale Cohomology* [12] builds the theory, from the cohomology of a point and of curves to Deligne's proof of the Riemann hypothesis for varieties over finite fields. Milne's *Étale Cohomology* and Freitag and Kiehl's *Étale Cohomology and the Weil Conjecture* are the standard texts.

Descent and gerbes over a base. The Galois descent of section 11.2 and the non-abelian H^2 of section 11.4 are the field case of descent along faithfully flat covers and of gerbes on a site, and the Brauer group of a field is the field case of the group of $\mathbb{G}_m$-gerbes on a scheme. *Moduli, Deformation Theory, and Stacks* [16] develops descent, stacks and gerbes, with twisted sheaves as an application. Olsson's *Algebraic Spaces and Stacks* and Giraud's *Cohomologie non abélienne* are the references.

Elliptic curves. The Selmer and Shafarevich-Tate groups of section 10.5 are where local and global duality meet the arithmetic of a single curve. *Elliptic Curves* [11] carries out descent on elliptic curves, with the Mordell-Weil theorem as its first payoff. Silverman's *The Arithmetic of Elliptic Curves* is the standard text.

Nonabelian reciprocity. Local reciprocity describes the one-dimensional representations of the Weil group of a local field by characters of $K^\times$. The local Langlands correspondence extends this to a bijection between n-dimensional Frobenius-semisimple Weil-Deligne representations and irreducible

smooth representations of $\mathrm{GL}_n(K)$, and *Local Langlands for* $GL_2(F)$ [15] treats the case $n = 2$. Bushnell and Henniart's *The Local Langlands Conjecture for* $\mathrm{GL}(2)$ is the reference.

Arithmetic duality. The duality theorems of chapters 8 and 10 have counterparts for the étale and flat cohomology of rings of integers and of curves over finite fields, Artin-Verdier duality among them. Milne's *Arithmetic Duality Theorems* develops them.

The cohomology of groups. The discrete and finite group cohomology of chapters 2 and 4 is a subject in its own right, with its own geometry: classifying spaces, Euler characteristics and the cohomology rings of finite groups. Brown's *Cohomology of Groups* and Evens' *The Cohomology of Groups* develop it.

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